

CH32V407 / V467

Application Manual

Original: CH32V407、V467 应用手册 V1.1

Translation: English Technical Translation

Source: WCH (wch.cn)

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CH32V407 / V467 Application Manual

V1.1

Description

The CH32V407 is an industrial-grade interconnect microcontroller based on the QingKe RISC-V core, supporting zero-wait operation at up to 200 MHz, and supporting the RVV vector instruction set for accelerating computation or parallel processing applications. The CH32V407 integrates 2 sets of USB 2.0 high-speed PHY transceivers (480 Mbps), an Ethernet MAC controller and 10/100 Mbps physical-layer transceiver; built-in dual ADC units, dual DAC units, OPA operational amplifiers and other analog resources; provides 2 sets of 24-channel DMA controllers, SDIO, DVP digital video interface, ARGB, LTDC, FSMC, multiple timer groups, 10 USART serial ports, I2C interfaces, I3C interfaces, 3 SPI groups, 2 I2S groups, CAN interfaces and other rich peripheral resources.

The CH32V467 adds 4 MB or 8 MB of on-chip extended PSRAM memory (with DMA support) on top of the CH32V407.

This manual provides detailed usage information for the CH32V407 and V467 chips for user application development. For device characteristics of the CH32V407 series, please refer to the CH32V407DS0 manual.

RISC-V Core Version Overview

Core	Version	Instruction Set	Hardware Stack Levels	Interrupt Nesting Levels	Fast Interrupt Channels	Pipeline Stages	Vector Table Mode	Extension Interface	Debug
QingKe V3V	IMACV-X2	24	3	address or instruction	supports single/dual-wire	—	—	—	—

For information about the RISC-V core, please refer to the QingKe V3 microprocessor manual: *QingKeV3_Processor_Manual*.

Register Bit Attribute Abbreviations

Attribute	Description
RF	Read-only, fixed value.
RO	Read-only, changed by hardware.
RZ	Read-only, auto-cleared after read.

WO	Write-only (not readable, read value is indeterminate).
WA	Write-only, writable in security mode.
WZ	Write-only, auto-cleared after write.
RW	Readable, writable.
RWA	Readable, writable in security mode.
RW1	Readable, write 1 effective, write 0 ineffective.
RW0	Readable, write 0 effective, write 1 ineffective.
RW1T	Readable, write 0 ineffective, write 1 toggles.
RW1Z	Readable, write 1 clears the bit, write 0 ineffective.
SC	Self-clearing.

Chapter 1: Memory and Bus Architecture

1.1 Bus Architecture

This series of chips is an industrial-grade general-purpose microcontroller based on the QingKe RISC-V3V core. The core, arbitration unit, DMA module, SRAM memory and other parts in its architecture interact through multiple bus groups. The system block diagrams are shown in Figure 1-1-1 and Figure 1-1-2.

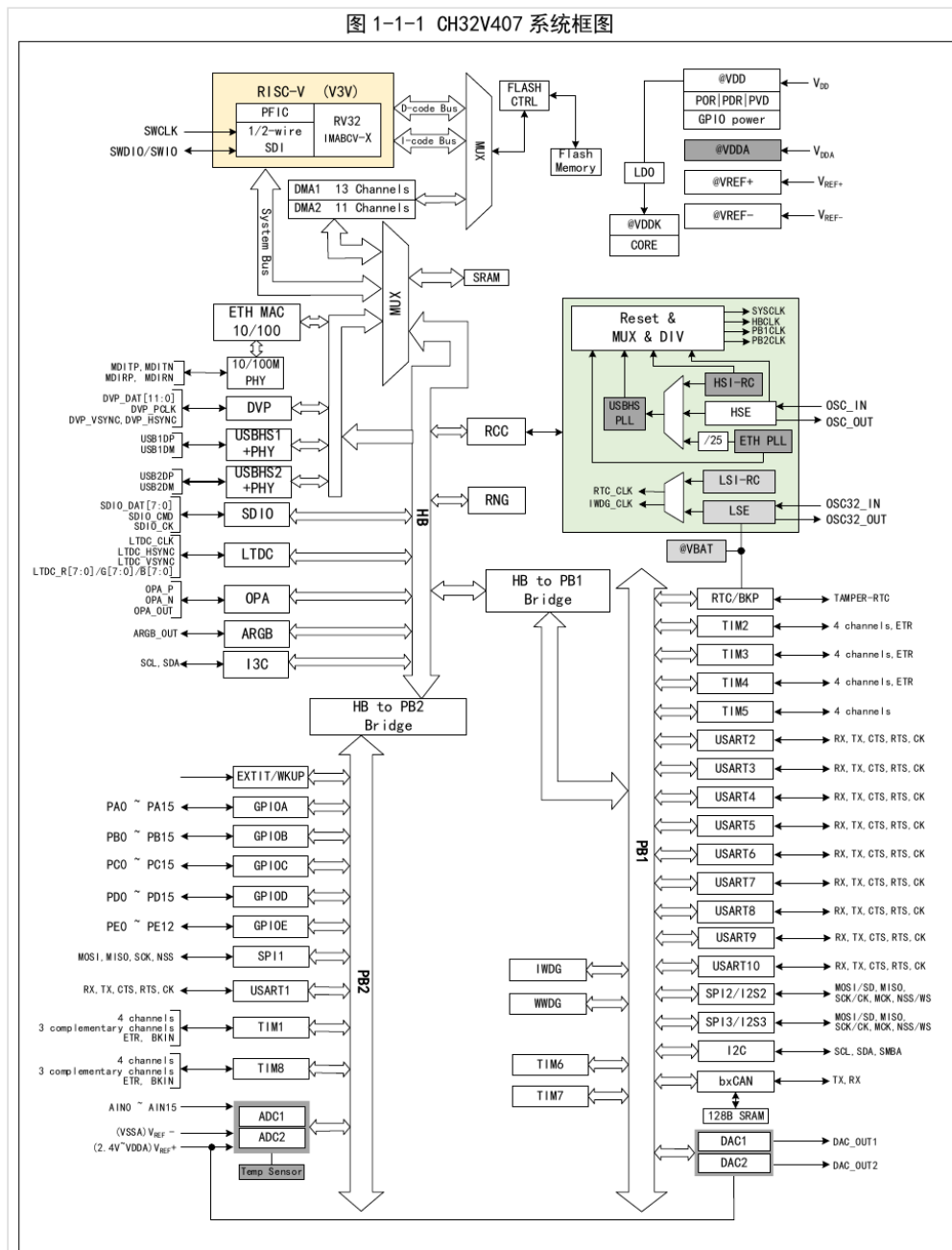


Figure 1-1-1: CH32V407 System Block Diagram

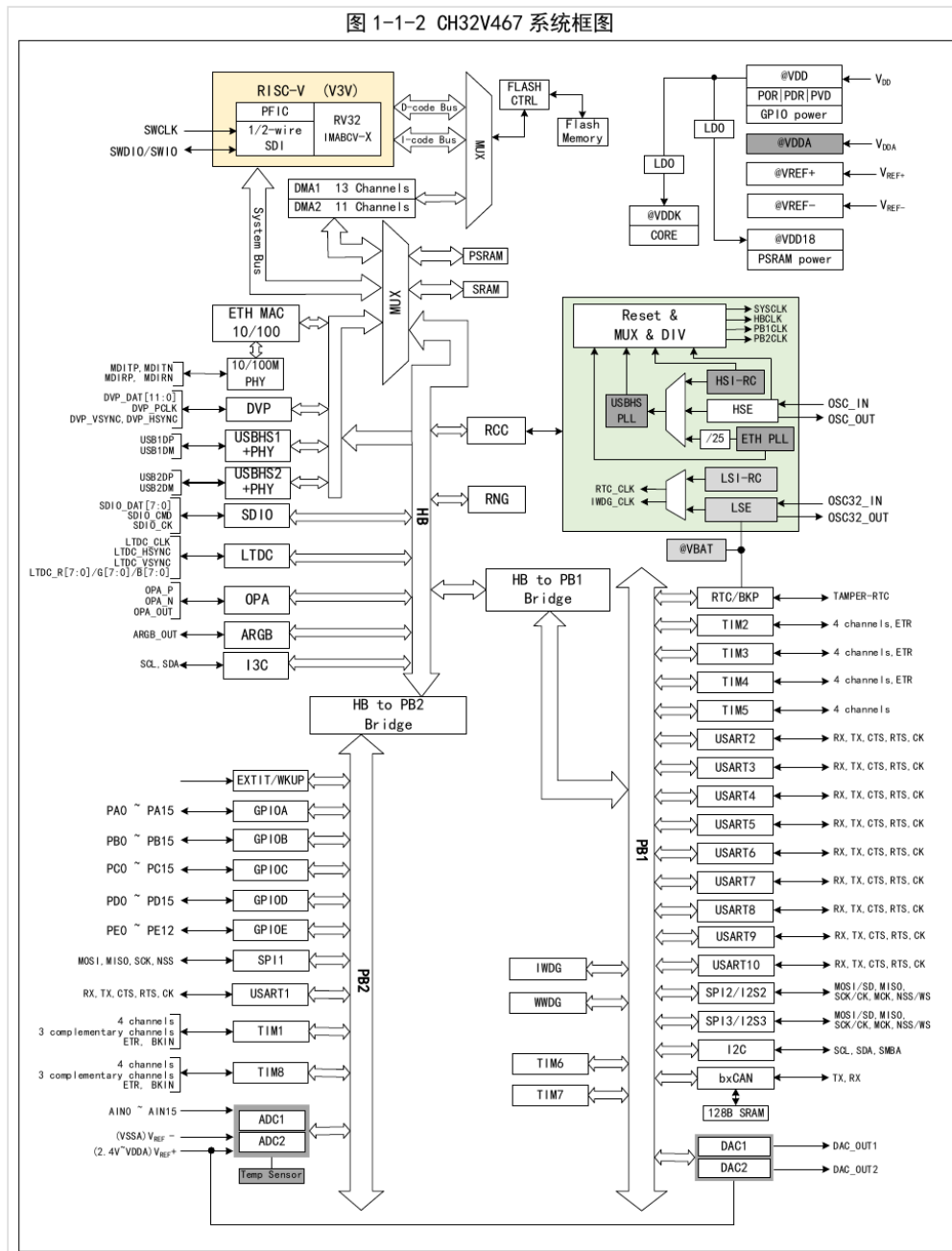


Figure 1-1-2: CH32V467 System Block Diagram

The system includes: a general-purpose DMA controller to reduce CPU burden and improve efficiency; a hierarchical clock tree management to reduce the total operating power consumption of peripherals, along with data protection mechanisms, clock safety system protection mechanisms and other measures to increase system stability.

- The instruction bus (I-Code) connects the core to the FLASH instruction interface; prefetch is completed on this bus.
- The data bus (D-Code) connects the core to the FLASH data interface, used for constant loading and debugging.
- The system bus connects the core to the bus matrix, used to coordinate access between the core, DMA, SRAM and peripherals.
- The DMA bus connects the DMA HB master interface to the bus matrix. This bus accesses FLASH data, SRAM and peripherals.

- The bus matrix is responsible for access coordination between the system bus, data bus, DMA bus, SRAM and HB/PB bridges.
- The HB/PB bridge provides synchronous connection between the HB bus and two PB buses. Different peripherals are attached to different PB buses, allowing different bus clocks to be configured as needed to optimize performance.

1.2 Memory Map

The chip contains program memory, data memory, core registers and peripheral registers, all addressed in a 4 GB linear space.

System memory stores data in little-endian format, i.e., the low byte is stored at the low address and the high byte is stored at the high address.

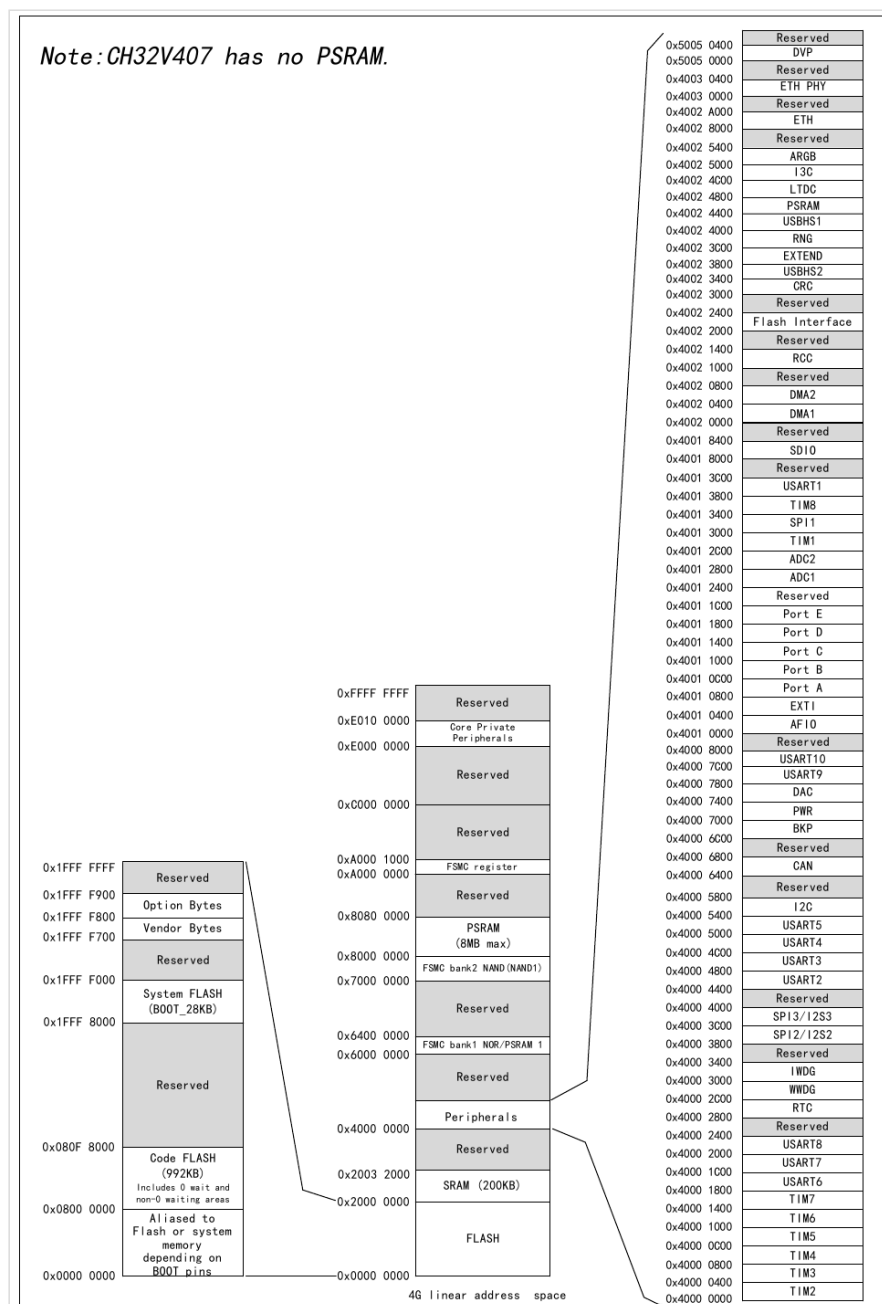


Figure 1-2: Memory Map

Memory Map Address Table

Address Range	Region	Peripheral
0x5005_0000 – 0x5005_03FF	Reserved	DVP
0x4003_0000 – 0x4003_03FF	Reserved	ETH PHY
0x4002_8000 – 0x4002_9FFF	Reserved	ETH
0x4002_5000 – 0x4002_53FF	Reserved	ARGB
0x4002_4C00 – 0x4002_4FFF	I3C	
0x4002_4800 – 0x4002_4BFF	LTDC	
0x4002_4400 – 0x4002_47FF	PSRAM	
0x4002_4000 – 0x4002_43FF	USBHS1	
0x4002_3C00 – 0x4002_3FFF	RNG	
0x4002_3800 – 0x4002_3BFF	EXTEND	
0x4002_3400 – 0x4002_37FF	USBHS2	
0x4002_3000 – 0x4002_33FF	CRC	
0x4002_2000 – 0x4002_23FF	Reserved	Flash Interface
0x4002_1000 – 0x4002_13FF	Reserved	RCC
0x4002_0400 – 0x4002_07FF	Reserved	DMA2
0x4002_0000 – 0x4002_03FF	Reserved	DMA1
0x4001_8000 – 0x4001_83FF	Reserved	SDIO
0x4001_3800 – 0x4001_3BFF	Reserved	USART1
0x4001_3400 – 0x4001_37FF	TIM8	
0x4001_3000 – 0x4001_33FF	SPI1	
0x4001_2C00 – 0x4001_2FFF	TIM1	
0x4001_2800 – 0x4001_2BFF	ADC2	
0x4001_2400 – 0x4001_27FF	ADC1	
0x4001_1800 – 0x4001_1BFF	Reserved	Port E
0x4001_1400 – 0x4001_17FF	Port D	
0x4001_1000 – 0x4001_13FF	Port C	
0x4001_0C00 – 0x4001_0FFF	Port B	
0x4001_0800 – 0x4001_0BFF	Port A	
0x4001_0400 – 0x4001_07FF	EXTI	
0x4001_0000 – 0x4001_03FF	AFIO	
0x4000_7C00 – 0x4000_7FFF	Reserved	USART10
0x4000_7800 – 0x4000_7BFF	USART9	
0x4000_7400 – 0x4000_77FF	DAC	
0x4000_7000 – 0x4000_73FF	PWR	
0x4000_6C00 – 0x4000_6FFF	BKP	
0x4000_6400 – 0x4000_67FF	Reserved	CAN

0x4000_5400 – 0x4000_57FF	Reserved	I2C
0x4000_5000 – 0x4000_53FF	USART5	
0x4000_4C00 – 0x4000_4FFF	USART4	
0x4000_4800 – 0x4000_4BFF	USART3	
0x4000_4400 – 0x4000_47FF	Reserved	USART2
0x4000_3C00 – 0x4000_3FFF	SPI3/I2S3	
0x4000_3800 – 0x4000_3BFF	SPI2/I2S2	
0x4000_3000 – 0x4000_33FF	Reserved	IWDG
0x4000_2C00 – 0x4000_2FFF	WWDG	
0x4000_2800 – 0x4000_2BFF	RTC	
0x4000_2000 – 0x4000_23FF	Reserved	USART8
0x4000_1C00 – 0x4000_1FFF	USART7	
0x4000_1800 – 0x4000_1BFF	USART6	
0x4000_1400 – 0x4000_17FF	TIM7	
0x4000_1000 – 0x4000_13FF	TIM6	
0x4000_0C00 – 0x4000_0FFF	TIM5	
0x4000_0800 – 0x4000_0BFF	TIM4	
0x4000_0400 – 0x4000_07FF	TIM3	
0x4000_0000 – 0x4000_03FF	TIM2	

Note: CH32V407 has no PSRAM.

Other memory regions:

Address Range	Region
0xE000_0000 – 0xE010_0000	Core Private Peripherals
0xA000_0000 – 0xA000_0FFF	FSMC register
0x8080_0000 – 0x807F_FFFF	PSRAM (8 MB max, CH32V467 only)
0x7000_0000 – 0x7000_FFFF	FSMC bank2 NAND (NAND1)
0x6000_0000 – 0x63FF_FFFF	FSMC bank1 NOR/PSRAM 1
0x2000_0000 – 0x2003_1FFF	SRAM (200 KB)
0x0800_0000 – 0x080F_7FFF	Code FLASH (992 KB, includes 0-wait and non-0-wait areas)
0x1FFF_F000 – 0x1FFF_7FFF	System FLASH (BOOT, 28 KB)
0x1FFF_F700 – 0x1FFF_F7FF	Vendor Bytes (128 B)
0x1FFF_F800 – 0x1FFF_F8FF	Option Bytes (128 B)
0x1FFF_F900 – 0x1FFF_FFFF	Reserved
0x0000_0000	Aliased to Flash or system memory depending on BOOT pins

1.2.1 Memory Allocation

Built-in zero-wait 200 KB SRAM for data storage; data is lost on power-down. Starting address 0x20000000, supports byte, half-word (2 bytes), and word (4 bytes) access.

Built-in 992 KB program flash memory area (Code FLASH), i.e., the user area, for user application programs and constant data storage. Includes 480 KB of non-zero-wait program area and 512 KB of zero-wait program execution area.

64 KB of the zero-wait data SRAM can be configured for zero-wait program execution, resulting in 416 KB of non-zero-wait program area, 576 KB of zero-wait program execution area, and 136 KB of zero-wait data SRAM area.

Built-in 28 KB system storage area (System FLASH), i.e., the BOOT area, for system boot program storage (factory-fixed bootloader).

128 bytes for system non-volatile configuration information storage, used for vendor configuration word storage, factory-fixed, user cannot modify.

128 bytes for user-defined information storage, used for user option byte storage.

Note: For chips whose lot number's fifth digit from the end is greater than 0, the non-zero-wait area supports both RVV instruction and DMA 64-bit access.

1.3 Boot Configuration

The system can select three different boot modes through the BOOT0 and BOOT1 pins.

Table 1-1: Boot Modes

BOOT0	BOOT1	Boot Mode
0	X	Boot from program flash memory
1	0	Boot from system memory
1	1	Boot from internal SRAM

The user selects the boot mode after reset by setting the state of the BOOT pins. Both system reset and power reset cause the BOOT pin values to be re-latched.

Depending on the boot mode, program flash memory, system memory, and internal SRAM have different access methods:

- When booting from program flash memory, the program flash memory address is mapped to the 0x00000000 address area, and can also be accessed at the original address area 0x08000000.
- When booting from system memory, the system memory address is mapped to the 0x00000000 address area, and can also be accessed at the original address area 0x1FFF0000.
- When booting from internal SRAM, it can only be accessed from the 0x20000000 address area.

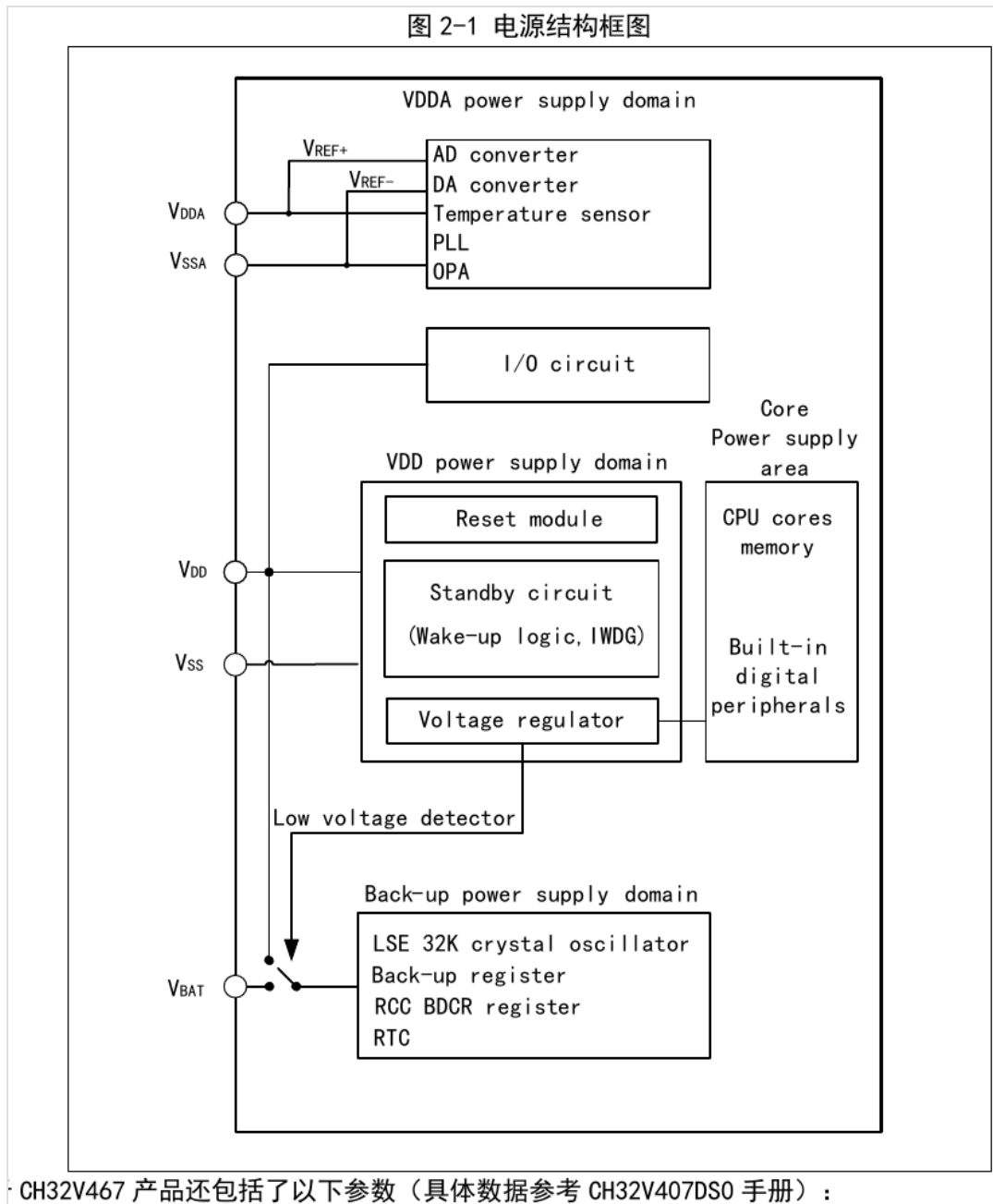
Chapter 2: Power Control (PWR)

2.1 Overview

The system operating voltage VDD ranges from 2.9 to 3.6V. A built-in voltage regulator provides the low-voltage supply required by the core. When the main supply VDD is powered down, a backup supply such as a battery can power the Real-Time Clock (RTC) and backup registers via the VBAT pin. If no backup supply is needed, it is recommended to connect VDD directly to the VBAT pin.

The VDDA and VSSA pins are dedicated to supplying power to the analog-related circuits in the system, including the ADC, DAC, temperature sensor, etc. VREF+ and VREF- serve as the reference points for some analog circuits and are internally equal to VDDA and VSSA.

图 2-1 电源结构框图



CH32V467 产品还包括了以下参数（具体数据参考 CH32V407DS0 手册）：

Figure 2-1: Power supply block diagram

For the CH32V467 product, the following parameter is also included (refer to the CH32V407DS0 manual for specific data):

VDD18: Supplies power to the on-chip PSRAM module.

After the main supply VDD is powered down, the analog switch switches to VBAT, and the backup domain is powered by the VBAT pin. In this case, PC13-PC15 cannot be used as GPIO; only the following functions are available:

- PC13 can be used as the TAMPER pin, RTC alarm or second output.
- PC14 and PC15 can only be used as LSE pins.

When the main supply VDD is powered on and stable, the system automatically switches the backup domain to be powered by VDD, and PC13-PC15 can be used as GPIO functions.

When PC13–PC15 pins are used as GPIO outputs, the speed must be limited to below 2 MHz, the maximum load capacitance is 30 pF, and they must not be used in applications requiring continuous current output or sinking, such as LED driving.

Note: During the process of the main supply VDD recovering power, the internal VBAT supply is still connected to the external backup supply via the corresponding VBAT pin. If VDD reaches a stable level within less than the reset lag time $t_{RSTTEMPO}$ and is higher than VBAT by more than 0.6V, there may be a brief moment where current flows into VBAT through the diode between VDD and VBAT, and then injects into the backup supply (such as a battery) via the VBAT pin. If the backup supply cannot withstand such transient injection current, it is recommended to add a forward-conducting low-drop diode between the backup supply and the VBAT pin.

2.2 Power Management

2.2.1 Power-on Reset and Power-down Reset

The system integrates Power-on Reset (POR) and Power-down Reset (PDR) circuits. When the chip supply voltage VDD and VDDA fall below the corresponding threshold voltages, the system is reset by the relevant circuitry, without the need for an external reset circuit. For the parameters of the power-on threshold voltage V_{POR} and the power-down threshold voltage V_{PDR} , please refer to the corresponding datasheet.

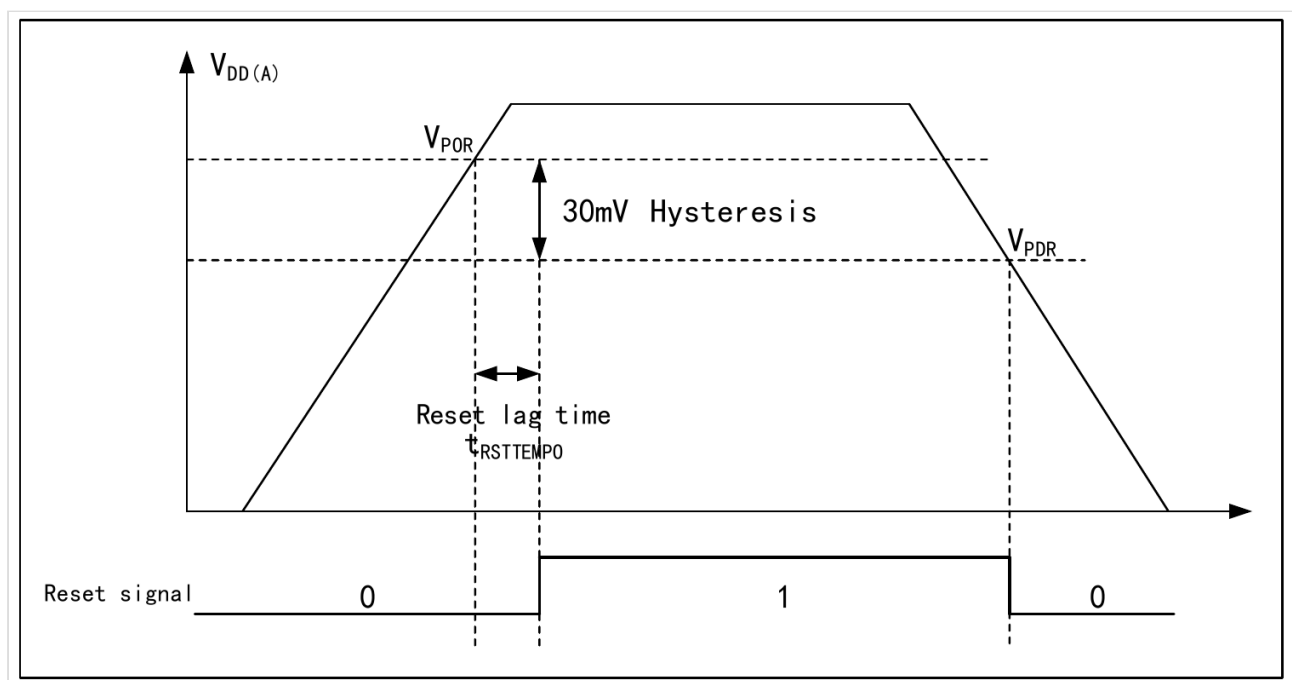


Figure 2-2: POR and PDR operation diagram

2.2.2 Programmable Voltage Detector

The Programmable Voltage Detector (PVD) is mainly used to monitor changes in the system main supply. It compares with the threshold voltage set by the PLS[1:0] field of the power control register PWR_CTLR. Together with the external interrupt register (EXTI) configuration, it can generate a relevant interrupt to promptly notify the system to perform pre-power-loss operations such as data saving.

The specific configuration is as follows:

1. Set the PLS[1:0] field of the PWR_CTLR register to select the voltage threshold to be monitored.
2. Optional interrupt handling. The PVD function is internally connected to line 16 of the EXTI module for rising/falling edge trigger configuration. Enable this interrupt (configure EXTI); when VDD falls below or rises above the PVD threshold, a PVD interrupt will be generated.
3. Set the PVDE bit of the PWR_CTLR register to enable the PVD function.
4. Reading the PVD0 bit of the PWR_CSR status register provides the current relationship between the system main supply and the threshold set by PLS[1:0], allowing corresponding software processing. When the VDD voltage is higher than the threshold set by PLS[1:0], the PVD0 bit is set to 0; when the VDD voltage is lower than the threshold set by PLS[1:0], the PVD0 bit is set to 1.

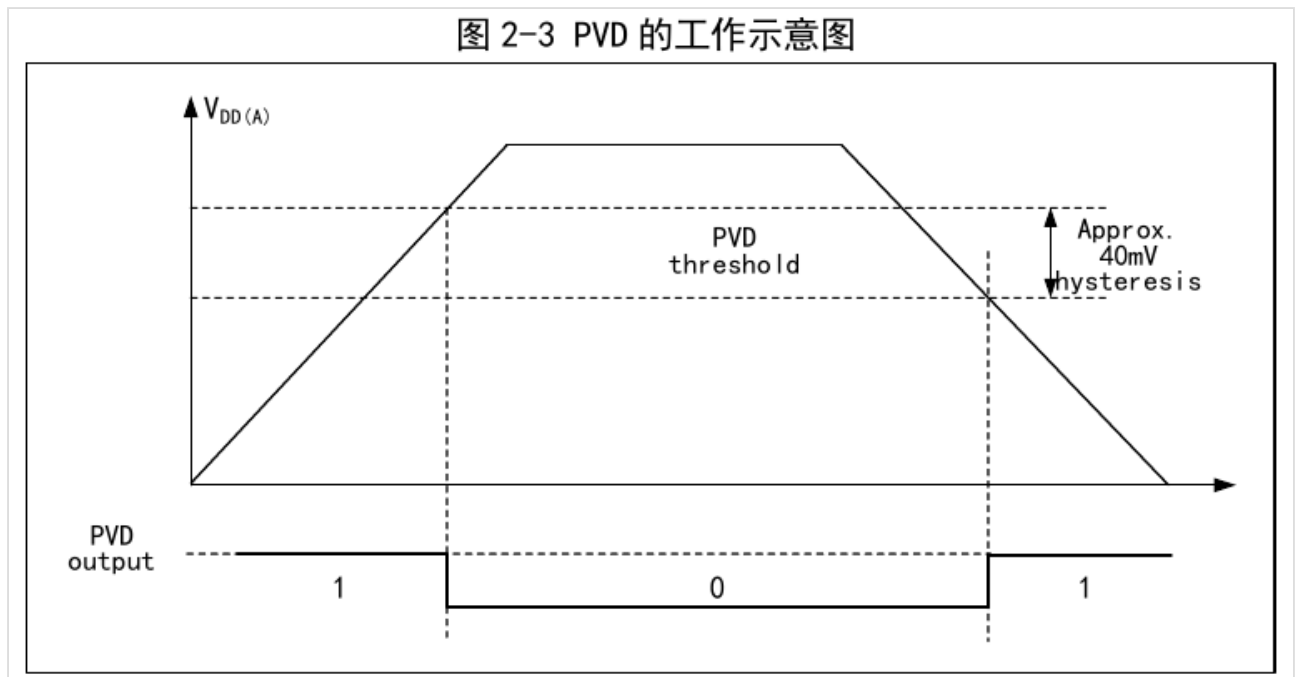


Figure 2-3: PVD operation diagram

2.3 Low-Power Modes

After system reset, the microcontroller is in normal operating state (run mode). At this point, system power consumption can be reduced by lowering the main system frequency, turning off clocks of unused peripherals, or reducing the clock of working peripherals. If the system does not need to operate, it can be set to enter a low-power mode and be brought out of this state by a specific event.

The microcontroller currently provides 3 low-power modes, which are distinguished by the differences in the operation of the processor, peripherals, voltage regulator, etc.:

- Sleep mode: The core stops running, but all peripherals (including core-private peripherals) continue to run.
- Stop mode: All clocks are stopped; after wakeup, the system continues to run.
- Standby mode: All clocks are stopped; after wakeup, the microcontroller is reset (power reset).

Table 2-1: Low-power mode summary

Mode	Entry	Wakeup source	Effect on clocks	Voltage regulator
Sleep (SLEEP)	WFI / WFE	Any interrupt wakeup / wakeup event	Core clock off, other clocks unaffected	Normal
Stop (STOP)	SLEEPDEEP set to 1, PDDDS cleared to 0, WFI or WFE	Any external interrupt/event (EXTI signal), external reset signal on NRST, IWDG reset. EXTI signals include any of 77 external I/O ports, PVD output, RTC alarm, I3C wakeup signal, USB wakeup signal.	HSE, HSI, PLL and peripheral clocks off	Normal: LPDS=0, or Low power: LPDS=1
Standby (STANDBY)	SLEEPDEEP set to 1, PDDDS set to 1, WFI or WFE	Any external event (EXTI signal), external reset signal on NRST, IWDG reset, a rising edge on the WKUP pin. EXTI signals include any of 77 external I/O ports, RTC alarm.	HSE, HSI, PLL and peripheral clocks off	Off

Note: The SLEEPDEEP bit is a core-private peripheral control bit; refer to the PFIC_SCTLR register.

2.3.1 Low-Power Configuration Options

• WFI and WFE methods

WFI: The microcontroller is woken up by an interrupt source responded to by the interrupt controller. After wakeup, the interrupt service function will be executed first (except for microcontroller reset).

WFE: A wakeup event triggers the microcontroller to exit the low-power mode. Wakeup events include:

1. Configure an external or internal EXTI line as event mode; in this case, there is no need to configure the interrupt controller;
2. Or configure an interrupt source, equivalent to WFI wakeup; the system will execute the interrupt service function first;
3. Or configure the SEVONPEND bit, enable the peripheral interrupt enable, but do not enable the interrupt enable in the interrupt controller; after the system wakes up, the interrupt pending bit needs to be cleared.

• SLEEPONEXIT

Enabled: After executing the WFI or WFE instruction, the microcontroller ensures that it enters the low-power mode after all pending interrupt services have exited.

Not enabled: After executing the WFI or WFE instruction, the microcontroller immediately enters the low-power mode.

• SEVONPEND

Enabled: All interrupts or wakeup events can wake up the low-power mode entered via executing WFE.

Not enabled: Only interrupts or wakeup events enabled in the interrupt controller can wake up the low-power mode entered via executing WFE.

2.3.2 Sleep Mode (SLEEP)

In this mode, all I/O pins retain their states from run mode, and all peripheral clocks are normal. Therefore, before entering sleep mode, try to turn off unused peripheral clocks to reduce power consumption. This mode has the shortest wakeup time.

Entry: Configure the core register control bit SLEEPDEEP=0, power control register PDDS=0; LPDS determines the state of the internal regulator. Execute WFI or WFE; optionally SEVONPEND and SLEEPONEXIT.

Exit: Any interrupt or wakeup event.

2.3.3 Stop Mode (STOP)

Stop mode is based on the core's deep sleep mode (SLEEPDEEP) combined with the peripheral clock control mechanism, and puts the voltage regulator into a lower power consumption state. In this mode, the high-frequency clock (HSE/HSI/PLL) domain is turned off, SRAM and register contents are retained, and I/O pin states are maintained. After wakeup from this mode, the system can continue to run, with HSI as the default system clock.

If flash programming is in progress, the system will not enter stop mode until the memory access is complete; if access to PB is in progress, the system will not enter stop mode until the PB access is complete.

Modules that can operate in stop mode: Independent Watchdog (IWDG), Real-Time Clock (RTC), Low-Speed Clock (LSI/LSE).

Entry: Configure the core register control bit SLEEPDEEP=1, power control register PDDS=0, optionally the LPDS bit. Execute WFI or WFE; optionally SEVONPEND and SLEEPONEXIT.

Exit: Any external interrupt/event (EXTI signal), external reset signal on NRST, IWDG reset. EXTI signals include any of 77 external I/O ports, PVD output, RTC alarm, I3C wakeup signal, USB wakeup signal.

2.3.4 Standby Mode (STANDBY)

In standby mode, the main system LDO is turned off, and a low-power LDO powers the wakeup circuit. All other digital circuits are powered down, and FLASH is in a powered-down state. Waking up the system from standby mode generates a reset, and SBF (PWR_CSR) is set. After wakeup, querying the SBF status indicates the low-power mode before wakeup; SBF is cleared by the CSBF (PWR_CR) bit. In standby mode, the contents of the 32 KB SRAM can be retained (depending on the configuration before sleep), and the backup register contents are retained.

Modules that can operate in standby mode: Independent Watchdog (IWDG), Real-Time Clock (RTC), Low-Speed Clock (LSI/LSE).

Entry: Configure the core register control bit SLEEPDEEP=1, power control register PDDS=1. Execute WFI or WFE; optionally SEVONPEND and SLEEPONEXIT.

Exit: Any external event (EXTI signal), external reset signal on NRST, IWDG reset, a rising edge on the WKUP pin. EXTI signals include any of 77 external I/O ports, RTC alarm.

In standby mode, when powered normally, configuring R32KSTY=1 in the PWR_CTLR register controls the 32K-byte RAM to remain powered; when powered by VBAT, configuring R32KVBAT=1 in the PWR_CTLR register controls the 32K-byte RAM to remain powered. On this basis, by configuring RAMLV=1 in the PWR_CTLR register, the RAM low-voltage mode is enabled, achieving the lowest power consumption.

Note: (1) In debug mode, putting the microprocessor into stop or standby mode will lose the debug connection.
 (2) For chips whose fifth digit from the end of the lot number is 0, before entering standby mode, I/O pins not used for wakeup must be configured as analog input.
 (3) R32KSTY=1 controls the address range of the 32K-byte RAM: 0x20000000—0x20000000+32K.

2.3.5 RTC Auto Wakeup

The RTC can achieve auto wakeup without an external interrupt. By programming the time base, the system can be periodically woken up from stop or standby mode.

The precise external low-frequency 32.768 kHz crystal oscillator LSE can be selected as the RTC clock source, or the internal LSI oscillator can be selected as the RTC clock source. The accuracy and power consumption specifications of the LSI are inferior to those of the LSE.

The RTC alarm event can wake the MCU up from stop mode. To implement this function, external interrupt line 17 must be configured, and the RTC must be set to generate an alarm event. To wake up from standby mode, only the RTC needs to be set to generate an alarm event.

2.4 Register Description

Table 2-2: PWR register list

Name	Address	Description	Reset Value
R32_PWR_CTLR	0x40007000	Power control register	0x1B000200
R32_PWR_CSR	0x40007004	Power control/status register	0x00000000

2.4.1 Power Control Register (PWR_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	Reserved	LDO_VDD12A[2:0]	Reserved	Reserved	LDO_VDDK[2:0]	Reserved	Reserved	Reserved	Reserved	RAMLV	Reser- ved	Reserved	R32K VBAT	Reser- ved	R32K STY
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	LDO18_EN	DBP	Reser- ved	PLS[1:0]	PVDE	CSBF	CWUF	PDDS	LPDS	

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
[29:27]	LDO_VDD12A[2:0]	RW	Adjusts the internal analog supply voltage.	0x3
[26:24]	LDO_VDDK[2:0]	RW	Adjusts the digital core voltage, VDDK voltage value.	0x3
[23:21]	Reserved	RO	Reserved.	0
20	RAMLV	RW	RAM low-voltage mode enable control bit (relatively lower power consumption): 1: Enabled; 0: Disabled. Note: Valid when the LPDS bit of the PWR_CTLR register is 1.	0
19	Reserved	RO	Reserved.	0
18	R32KVBAT	RW	Control bit for whether the 32K RAM is powered in Standby mode when powered by VBAT: 1: Powered; 0: Not powered.	0
17	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
16	R32KSTY	RW	Control bit for whether the 32K RAM is powered in Standby mode: 1: Powered; 0: Not powered.	0
[15:10]	Reserved	RO	Reserved.	0
9	LDO18_EN	RW	VDD18 power enable bit: 1: Turn on PSRAM dedicated power; 0: Turn off PSRAM dedicated power.	1
8	DBP	RW	Write enable for the backup domain: In the reset state, the registers in the backup domain are write-protected. This bit must be set to 1 to perform write access to these registers. 1: Enable access to backup domain registers; 0: Disable access to backup domain registers.	0
7	Reserved	RO	Reserved.	0
[6:5]	PLS[1:0]	RW	PVD voltage monitoring threshold setting. See the electrical characteristics section in the datasheet for details. 00: Rising edge 2.85V / Falling edge 2.81V; 01: Rising edge 2.94V / Falling edge 2.90V; 10: Rising edge 3.04V / Falling edge 3.00V; 11: Rising edge 3.14V / Falling edge 3.10V.	0
4	PVDE	RW	Power voltage monitoring function enable flag bit: 1: Enable power voltage monitoring function; 0: Disable power voltage monitoring function.	0
3	CSBF	RW1	Clear standby status flag bit; always reads as 0. 1: Set to 1 to clear the SBF standby status flag; 0: Clearing to 0 has no effect.	0
2	CWUF	RW1	Clear wakeup status flag bit; always reads as 0. 1: After being set to 1, the WUF flag bit is cleared after 2 system clock cycles; 0: Clearing to 0 has no effect.	0
1	PDDS	RW	Standby/Stop mode selection bit in power-down deep sleep scenario. 1: Enter standby mode; 0: Enter stop mode, voltage regulator state controlled by LPDS.	0
0	LPDS	RW	Voltage regulator operating mode selection bit in stop mode. Valid when PDDS=0. 1: Voltage regulator operates in low-power mode; 0: Voltage regulator operates in normal mode.	0

Note: BIT16, BIT18, and BIT20 of this register can only be reset by backup reset; other bits are reset when waking up from standby mode.

2.4.2 Power Control/Status Register (PWR_CSR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							EWUP	Reserved				PVDO	SBF	WUF	

Bits	Name	Access	Description	Reset
[31:9]	Reserved	RO	Reserved.	0
8	EWUP	RW		0

Bits	Name	Access	Description	Reset
			WKUP pin enable bit: 1: WKUP is forced into input pull-down state, used to wake the MCU up from standby; 0: WKUP pin can be used as general-purpose IO, with no standby wakeup function.	
[7:3]	Reserved	RO	Reserved.	0
2	PVD0	RO	PVD output status flag bit. Valid when PVDE of the PWR_CTLR register is 1. 1: VDD and VDDA are below the PVD threshold set by PLS[1:0]; 0: VDD and VDDA are above the PVD threshold set by PLS[1:0].	0
1	SBF	RO	Standby status flag bit, can be cleared by setting CSBF to 1. 1: MCU is in standby mode; 0: MCU is not in standby mode.	0
0	WUF	RO	Wakeup event status flag bit, can be cleared by setting CWUF to 1. 1: A wakeup event or RTC alarm event is detected on the WKUP pin; 0: No wakeup event has occurred.	0

Note: This register remains unchanged after wakeup from standby mode.

Chapter 3: Reset and Clock Control (RCC)

The controller provides different reset forms and a configurable clock tree structure based on the division of power domains and peripheral power consumption management considerations in applications. This chapter describes the scope of each clock in the system.

3.1 Main Features

- Multiple reset forms
- Multiple clock sources, bus clock management
- Built-in external crystal oscillator monitoring and clock security system
- Independent clock management for each peripheral: reset, enable, disable
- Supports internal clock output

3.2 Reset

The controller provides 3 reset forms: power reset, system reset, and backup domain reset.

3.2.1 Power Reset

When a power reset occurs, all registers except those in the backup domain are reset (the backup domain is powered by VBAT). Its trigger conditions include:

- Power-on/power-down reset (POR/PDR reset)
- Wakeup from standby mode

3.2.2 System Reset

When a system reset occurs, all registers except the reset flags in the control/status register `RCC_RSTSCKR` and the backup domain are reset. The reset event source can be identified by checking the reset status flag bits in the `RCC_RSTSCKR` register. Its trigger conditions include:

- Low-level signal on the NRST pin (external reset)
- Window watchdog count termination (WWDG reset)
- Independent watchdog count termination (IWDG reset)
- Software reset (SW reset)
- Low-power management reset
- Core deadlock reset

Window/independent watchdog reset: triggered by the overflow of the window/independent watchdog peripheral timer count period. For detailed description, refer to the corresponding chapters.

Software reset: The chip resets the system by setting the `SYSRST` bit in the interrupt configuration register `PFIC_CFGR` of the programmable interrupt controller `PFIC`, or by setting the `SYSRST` bit in the configuration register `PFIC_SCTLR`. For details, refer to the corresponding chapter.

Low-power management reset: By setting the STANBYRST bit in the user option bytes to 0, standby mode reset is enabled. In this case, after executing the procedure to enter standby mode, a system reset will be executed instead of entering standby mode. By setting the STOPRST bit in the user option bytes to 0, stop mode reset is enabled. In this case, after executing the procedure to enter stop mode, a system reset will be executed instead of entering stop mode.

Core deadlock reset: When the LKUPEN bit of the EXTEN_CTR register is enabled, if the core runs abnormally and cannot recover, resulting in a deadlock, the system will perform a reset.

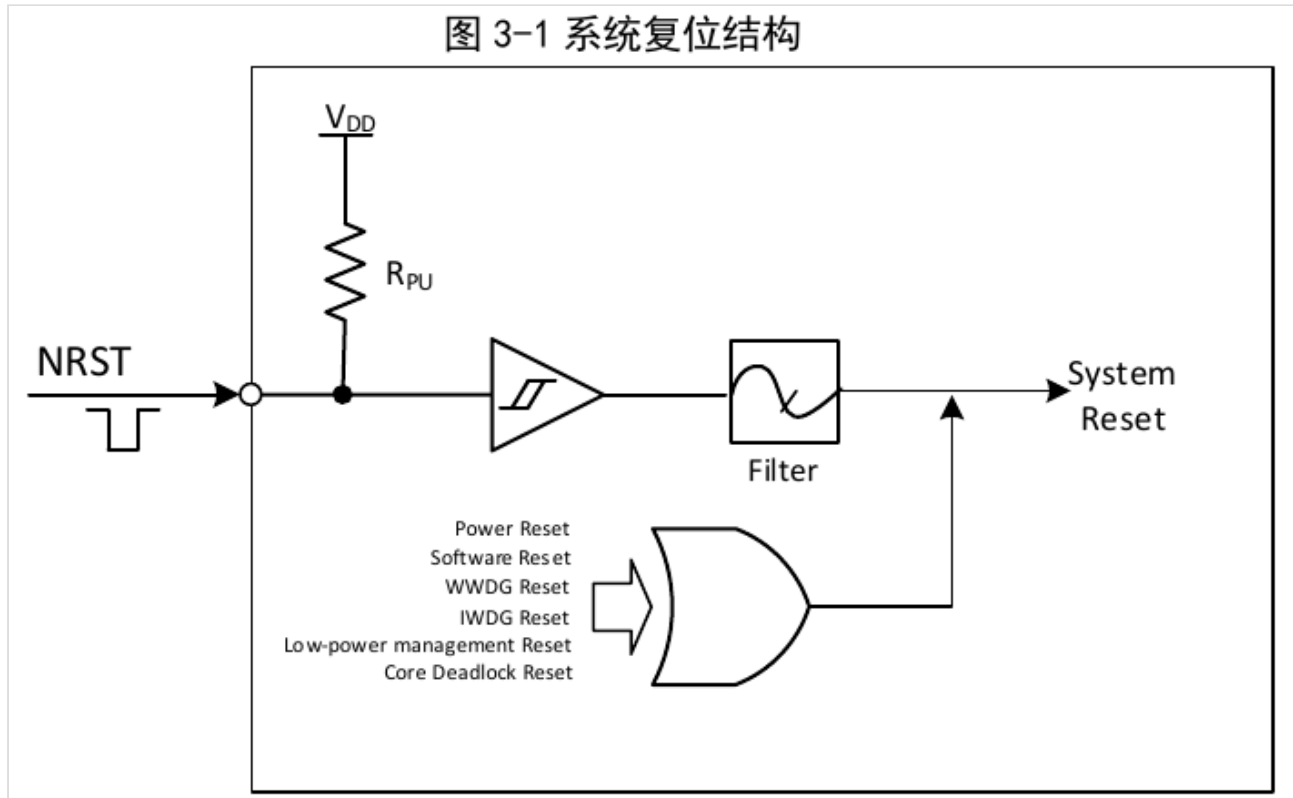


Figure 3-1: System reset structure

3.2.3 Backup Domain Reset

When a backup domain reset occurs, only the backup domain registers are reset, including the backup registers and the RCC_BDCTL register (RTC enable and LSE oscillator). Its trigger conditions include:

- Caused by power-up of VDD or VBAT under the premise that both VDD and VBAT are powered down
- Setting the BDRST bit in the RCC_BDCTL register to 1
- Setting the BKPRST bit in the RCC_PB1PRSTR register to 1

3.3 Clock

3.3.1 System Clock Structure

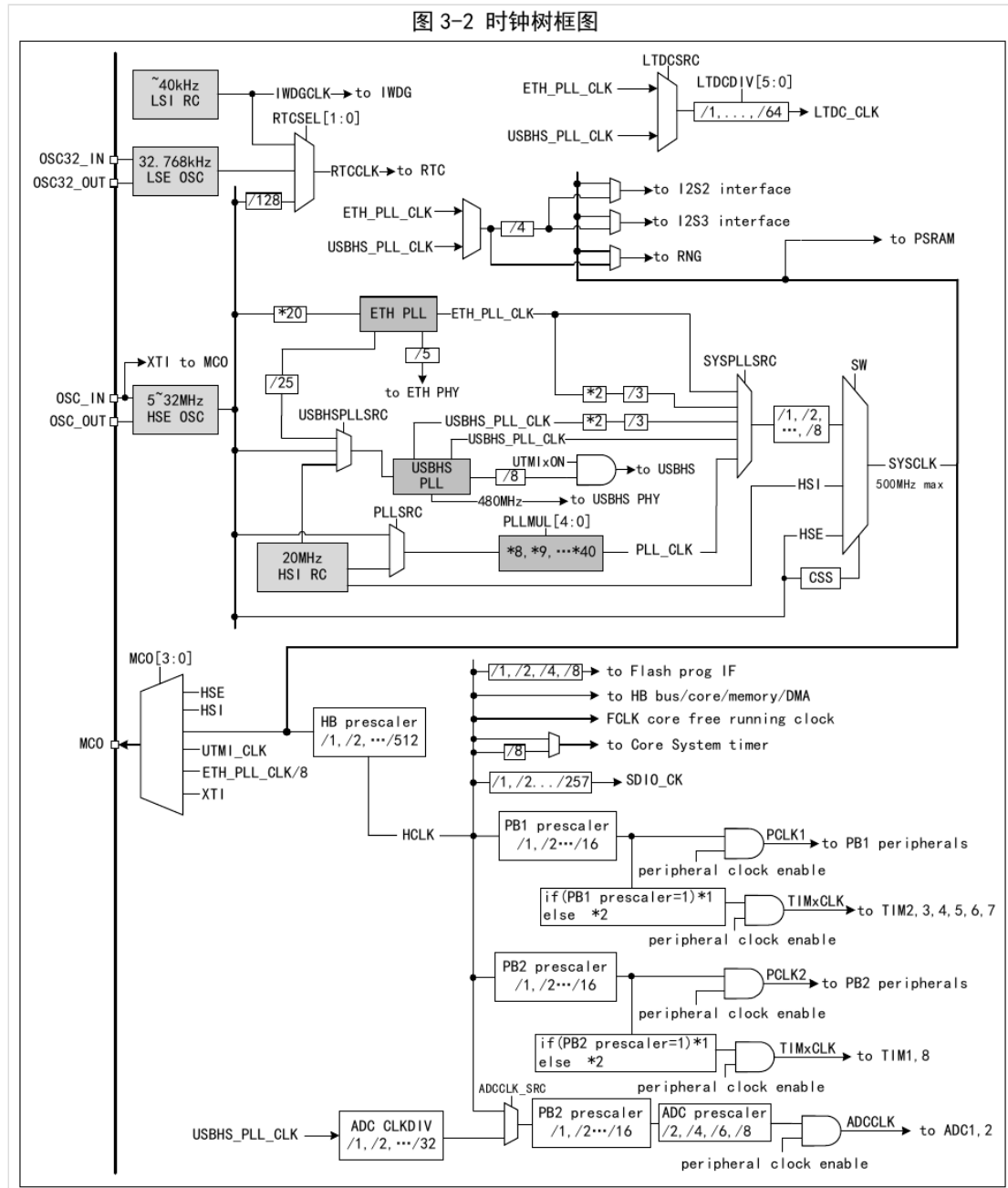


Figure 3-2: Clock tree block diagram

3.3.2 High-Speed Clocks (HSI/HSE)

HSI is a high-speed clock signal generated by the internal 20MHz RC oscillator of the system. The HSI RC oscillator can provide a system clock without requiring any external components. Its startup time is very short, but the clock frequency accuracy is relatively poor. HSI is enabled and disabled by setting the HSION bit in the RCC_CTLR register. The HSIRDY bit indicates whether the HSI RC oscillator is

stable. The system defaults to HSION and HSIRDY set to 1 (it is recommended not to disable it). If the HSIRDYIE bit of the RCC_INTR register is set, a corresponding interrupt will be generated.

- Factory calibration: Manufacturing process variations cause each chip to have a different RC oscillation frequency, so before the chip leaves the factory, HSI calibration is performed for each chip. After system reset, the factory calibration value is loaded into HSICAL[7:0] of the RCC_CTLR register.
- User adjustment: Based on different voltages or ambient temperatures, the application can adjust the HSI frequency through the HSITRIM[4:0] bits in the RCC_CTLR register.

Note: If the HSE crystal oscillator fails, the HSI clock will be used as a backup clock source (clock security system).

HSE is an external high-speed clock signal, generated by an external crystal/ceramic resonator or fed in from an external high-speed clock source.

- External crystal/ceramic resonator (HSE crystal): An external 3–32MHz oscillator provides a more precise clock source for the system. For further information, refer to the electrical characteristics section of the datasheet. The HSE crystal can be enabled and disabled by setting the HSEON bit in the RCC_CTLR register. The HSERDY bit indicates whether the HSE crystal oscillator is stable. The hardware only feeds the clock into the system after HSERDY is set to 1. If the HSERDYIE bit of the RCC_INTR register is set, a corresponding interrupt will be generated.

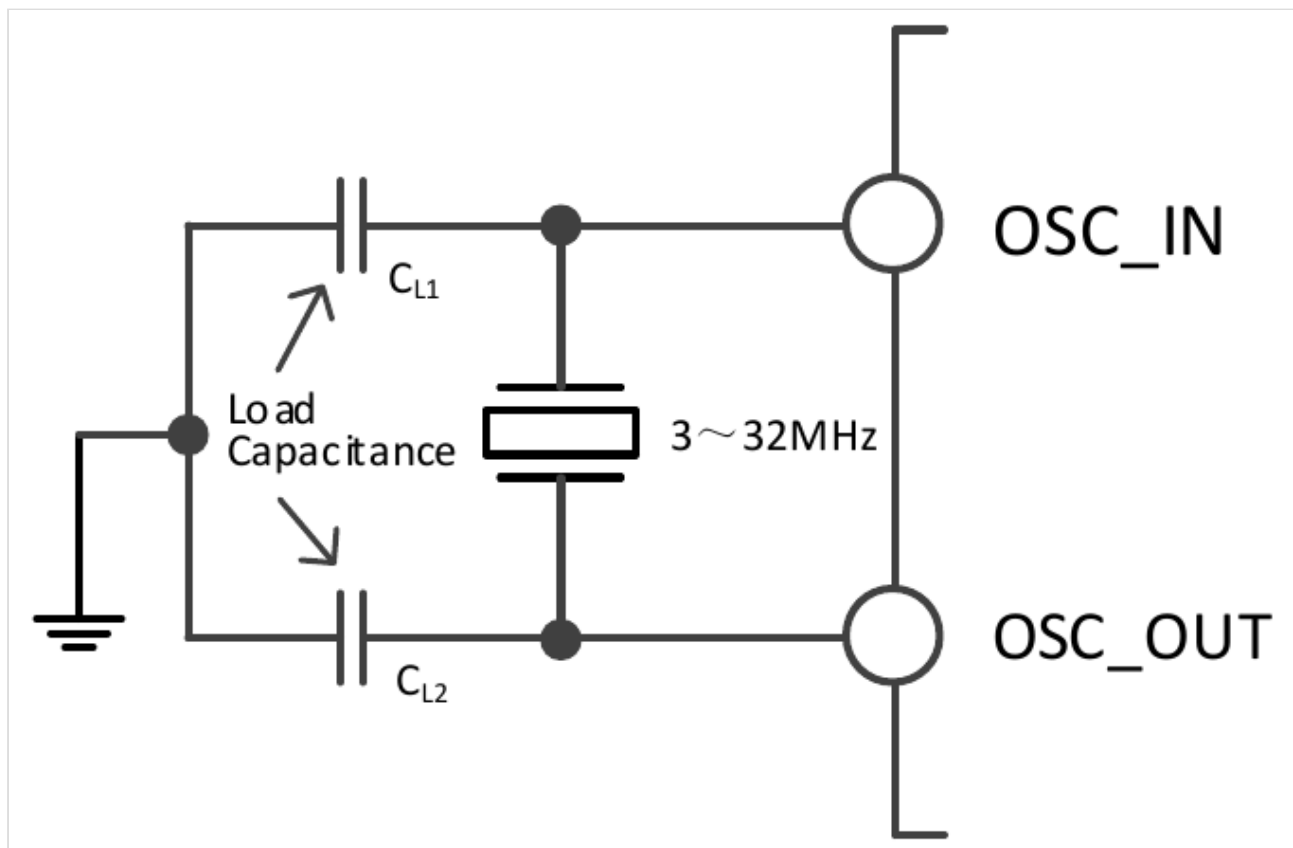


Figure 3-3: High-speed external crystal circuit

Note: The load capacitors need to be placed as close to the oscillator pins as possible, and the capacitance values should be selected according to the crystal manufacturer's parameters.

- External high-speed clock source (HSE bypass): In this mode, an external clock source is fed directly to the OSC_IN pin, and the OSC_OUT pin is left floating. A maximum frequency of 32MHz is supported. The application needs to set the HSEBYP bit while the HSEON bit is 0 to enable the HSE bypass function, and then set the HSEON bit.

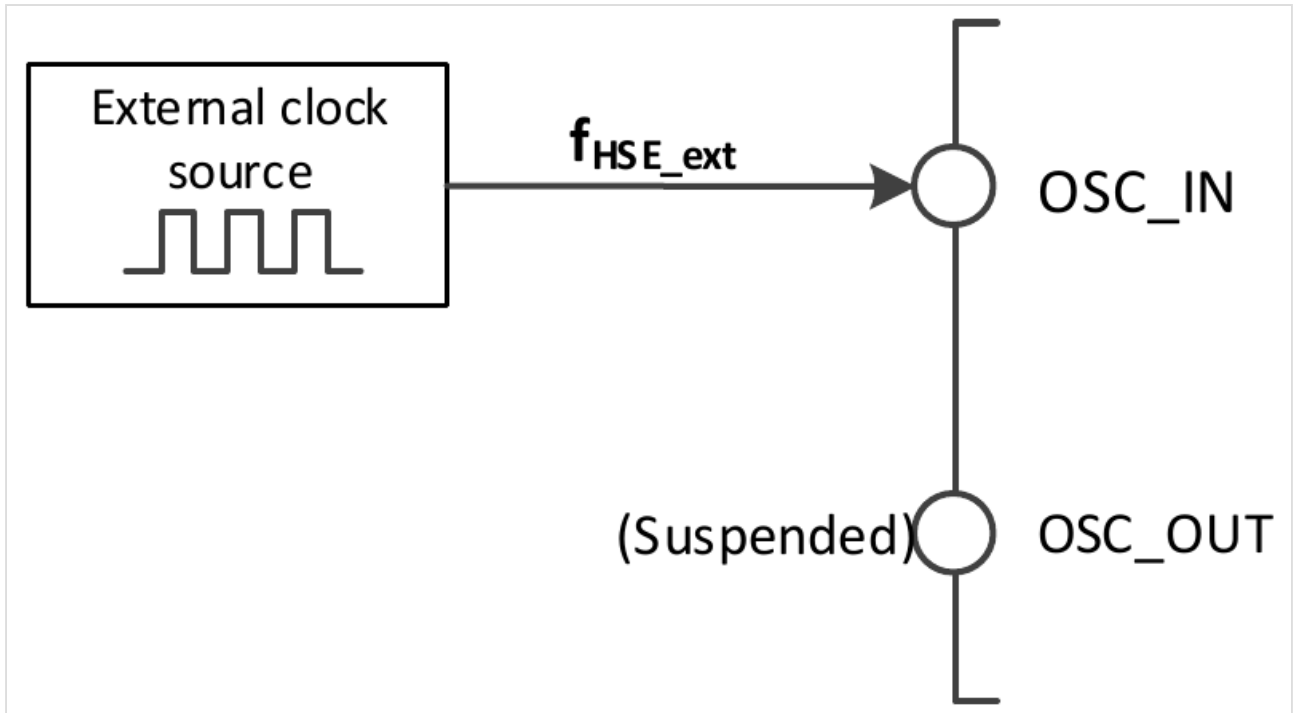


Figure 3-4: High-speed clock source circuit

3.3.3 Low-Speed Clocks (LSI/LSE)

LSI is a low-speed clock signal generated by the internal approximately 40kHz RC oscillator of the system. It can keep running in stop and standby modes, providing a clock reference for the RTC clock, independent watchdog, and wakeup unit. For further information, refer to the electrical characteristics section of the datasheet. LSI can be enabled and disabled by setting the LSION bit in the RCC_RSTSKR register, and then the LSIRDY bit can be queried to detect whether the LSI RC oscillator is stable. The hardware only feeds the clock in after the LSIRDY bit is set to 1. If the LSIRDYIE bit of the RCC_INTR register is set, a corresponding interrupt will be generated.

LSE is an external low-speed clock signal, generated by an external crystal/ceramic resonator or fed in from an external low-speed clock source. It provides a low-power and precise clock source for the RTC clock or other timing functions.

- External crystal/ceramic resonator (LSE crystal): An external 32.768kHz low-speed oscillator is connected. LSE is enabled and disabled by setting the LSEON bit in the RCC_BDCTLR register. The LSERDY bit indicates whether the LSE crystal oscillator is stable. The hardware only feeds the clock into the system after the LSERDY bit is set to 1. If the LSERDYIE bit of the RCC_INTR register is set, a corresponding interrupt will be generated.

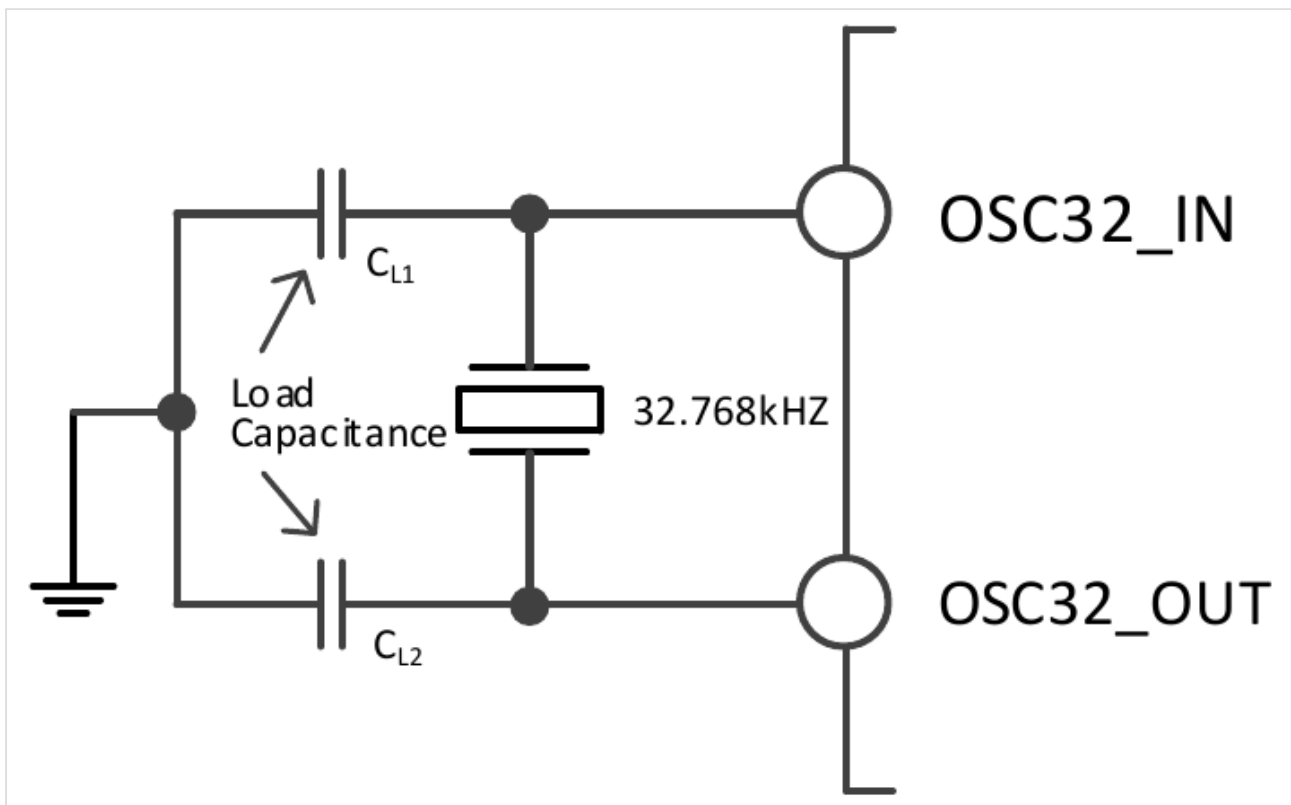


Figure 3-5: Low-speed external crystal circuit

- External low-speed clock source (LSE bypass): In this mode, an external clock source is fed directly to the OSC32_IN pin, and the OSC32_OUT pin is left floating. The application needs to set the LSEBYP bit while the LSEON bit is 0 to enable the LSE bypass function, and then set the LSEON bit.

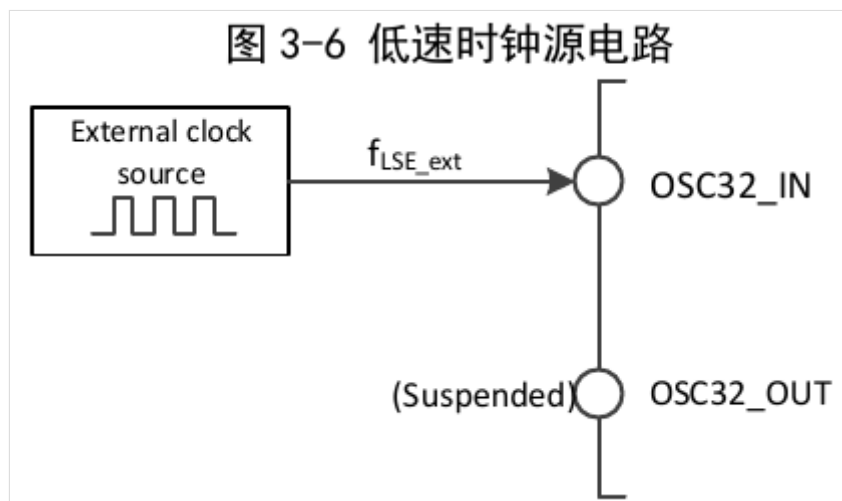


Figure 3-6: Low-speed clock source circuit

3.3.4 System PLL Clock (SYSPLL)

By configuring the RCC_CFGR0 register and RCC_CFGR2 register, the system PLL clock can select 5 clock sources and division ratios. These settings must be completed before each PLL is enabled. Once the PLL is started, these parameters cannot be changed.

The PLL is enabled and disabled by setting the PLLON bit in the RCC_CTLR register. The PLLRDY bit indicates whether the PLL clock is stable. The hardware only feeds the clock into the system after the PLLRDY bit is set to 1.

The ETH_PLL is enabled and disabled by setting the ETH_PLLON bit in the RCC_CTLR register. The ETH_PLLRDY bit indicates whether the ETH_PLL clock is stable. The hardware only feeds the clock into the system after the ETH_PLLRDY bit is set to 1.

The USBHS_PLL is enabled and disabled by setting the USBHS_PLLON bit in the RCC_CTLR register. The USBHS_PLLRDY bit indicates whether the USBHS_PLL clock is stable. The hardware only feeds the clock into the system after the USBHS_PLLRDY bit is set to 1.

SYSPLL clock sources:

- PLL_CLK clock input
- USBHS_PLL (480MHz) clock input
- ETH_PLL (500MHz) clock input
- USBHS_PLL (320MHz) divided clock input
- ETH_PLL (333MHz) divided clock input

3.3.5 Bus/Peripheral Clocks

3.3.5.1 System Clock (SYSCLK)

The system clock source is configured through the SW[1:0] bits of the RCC_CFGR0 register. The SWS[1:0] bits indicate the current system clock source.

- HSI as system clock
- HSE as system clock
- SYSPLL clock as system clock

After controller reset, the HSI clock is selected as the system clock source by default. Switching between clock sources only occurs after the target clock source is ready.

3.3.5.2 HB/PB1/PB2 Bus Peripheral Clocks (HCLK/PCLK1/PCLK2)

HB is the High Performance Bus, and its bus peripheral clock is HCLK; PB1 is Peripheral Bus 1, and its bus peripheral clock is PCLK1; PB2 is Peripheral Bus 2, and its bus peripheral clock is PCLK2.

By configuring the HPRE[3:0], PPRE1[2:0], and PPRE2[2:0] bits of the RCC_CFGR0 register, the clocks of the HB, PB1, and PB2 buses can be configured respectively. These bus clocks determine the access clock reference for the peripheral interfaces mounted under them. The application can adjust different values to reduce the power consumption of some peripherals during operation.

Through the individual bits in the RCC_HBRSTR, RCC_PB1PRSTR, and RCC_PB2PRSTR registers, different peripheral modules can be reset to their initial states.

Through the individual bits in the RCC_HBPCENR, RCC_PB1PCENR, and RCC_PB2PCENR registers, the communication clock interfaces of different peripheral modules can be individually enabled or disabled. When using a peripheral, its clock enable bit must first be set before its registers can be accessed.

3.3.5.3 RTC Clock (RTCCLK)

By setting the RTCSEL[1:0] bits of the RCC_BDCTLR register, the RTCCLK clock source can be provided by HSE/128, LSE, or LSI clock. Before modifying this bit, ensure that the DBP bit in the power control register (PWR_CTLR) is set to 1. Only a backup domain reset can reset this bit.

- LSE as RTC clock: Since the LSE is in the backup domain powered by VBAT, as long as VBAT maintains power, the RTC continues to operate even if the VDD supply is cut off.
- LSI as RTC clock: If the VDD supply is cut off, RTC auto-wakeup cannot be guaranteed.
- HSE/128 as RTC clock: If the VDD supply is cut off or the internal voltage regulator is turned off (the 1.8V domain supply is cut off), the RTC state is undefined.

3.3.5.4 Independent Watchdog Clock

If the independent watchdog has been configured by hardware or started by software, the LSI oscillator will be forcibly turned on and cannot be turned off. After the LSI oscillator stabilizes, the clock is supplied to the IWDG.

3.3.5.5 Clock Output (MCO)

The microcontroller allows outputting a clock signal to the MCO pin. By configuring the corresponding GPIO port register for alternate function push-pull output mode, and through configuring the MCO[3:0] bits of the RCC_CFGR0 register, the following 6 clock signals can be selected as the MCO clock output:

- System clock (SYSCLK) output
- HSI clock output
- HSE clock output
- UTMI clock output
- ETH_PLL clock divided by 8 output
- XT1 external oscillator clock output

3.3.6 Clock Security System

The clock security system is an operational protection mechanism of the controller. It can switch to the HSI clock in the event of an HSE clock failure and generate an interrupt notification, allowing the application software to perform rescue operations.

By setting the CSSON bit of the RCC_CTLR register to 1, the clock security system is activated. At this point, the clock monitor will be enabled after the HSE oscillator starts (HSERDY=1) with a delay, and will be disabled after the HSE clock is turned off. Once the HSE clock fails during system operation, the HSE oscillator will be turned off, the clock failure event will be sent to the brake input of the advanced timer (TIM1), and a clock security interrupt will be generated with the CSSF bit set to 1. The application then enters the NMI non-maskable interrupt. By setting the CSSC bit to 1, the CSSF flag can be cleared and the NMI interrupt pending bit can be withdrawn.

If the HSE is currently the system clock, or the HSE is currently the PLL input clock and the PLL is the system clock, the clock security system will automatically switch the system clock to the HSI oscillator when an HSE fault occurs, and turn off the HSE oscillator and PLL.

3.4 Register Description

Table 3-1: RCC-related register list

Name	Address	Description	Reset Value
R32_RCC_CTLR	0x40021000	Clock control register	0x0000xx83
R32_RCC_CFGR0	0x40021004	Clock configuration register 0	0x20000000
R32_RCC_INTR	0x40021008	Clock interrupt register	0x00000000
R32_RCC_PB2PRSTR	0x4002100C	PB2 peripheral reset register	0x00000000
R32_RCC_PB1PRSTR	0x40021010	PB1 peripheral reset register	0x00000000
R32_RCC_HBPCENR	0x40021014	HB peripheral clock enable register	0x00000004
R32_RCC_PB2PCENR	0x40021018	PB2 peripheral clock enable register	0x00000000
R32_RCC_PB1PCENR	0x4002101C	PB1 peripheral clock enable register	0x00000000
R32_RCC_BDCTLR	0x40021020	Backup domain control register	0x00000000
R32_RCC_RSTSCKR	0x40021024	Control/status register	0x0C000000
R32_RCC_HBRSTR	0x40021028	HB peripheral reset register	0x00000000
R32_RCC_CFGR2	0x4002102C	Clock configuration register 2	0x00000000
R32_RCC_CFGR3	0x40021030	Clock configuration register 3	0x00000000

3.4.1 Clock Control Register (RCC_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved						PLLRDY	PLLON	ETH_PLLRDY	ETH_PLLON	USBHS_PLLRDY	USBHS_PLLON	CSSON	HSEBYP	HSERDY	HSEON

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]				Reserved		HSIRDY	HSION

Bits	Name	Access	Description	Reset
[31:26]	Reserved	RO	Reserved.	0
25	PLLRDY	RO	PLL clock ready locked flag: 1: PLL clock locked; 0: PLL clock not locked.	0
24	PLLON	RW	PLL clock enable control bit: 1: Enable PLL clock; 0: Disable PLL clock. Note: After entering stop or standby low-power mode, this bit is cleared to 0 by hardware.	0
23	ETH_PLLRDY	RO	ETH_PLL clock ready locked flag: 1: ETH_PLL clock locked; 0: ETH_PLL clock not locked.	0
22	ETH_PLLON	RW	ETH_PLL enable control bit: 1: Enable ETH_PLL clock; 0: Disable ETH_PLL clock.	0
21	USBHS_PLLRDY	RO		0

Bits	Name	Access	Description	Reset
			USBHS_PLL clock ready locked flag: 1: USBHS_PLL clock locked; 0: USBHS_PLL clock not locked.	
20	USBHS_PLLON	RW	USBHS_PLL enable control bit: 1: Enable USBHS_PLL clock; 0: Disable USBHS_PLL clock.	0
19	CSSON	RW	Clock security system enable control bit: 1: Enable clock security system. When HSE is ready (HSERDY set to 1), hardware enables the HSE clock monitoring function. If an HSE abnormality is detected, the CSSF flag and NMI interrupt are triggered. When HSE is not ready, hardware disables the HSE clock monitoring function. 0: Disable clock security system.	0
18	HSEBYP	RW	External high-speed crystal bypass control bit: 1: Bypass external high-speed crystal/ceramic resonator (use external clock source); 0: Do not bypass high-speed external crystal/ceramic resonator. Note: This bit must be written while HSEON is 0.	0
17	HSERDY	RO	External high-speed crystal oscillator stable ready flag (set by hardware): 1: External high-speed crystal oscillator is stable; 0: External high-speed crystal oscillator is not stable. Note: After the HSEON bit is cleared to 0, this bit requires 6 HSE cycles to clear.	0
16	HSEON	RW	External high-speed crystal oscillator enable control bit: 1: Enable HSE oscillator; 0: Disable HSE oscillator. Note: After entering stop or standby low-power mode, this bit is cleared to 0 by hardware.	0
[15:8]	HSICAL[7:0]	RO	Internal high-speed clock calibration value, automatically initialized at system startup.	x
[7:3]	HSITRIM[4:0]	RW	Internal high-speed clock trim value: The user can input a trim value to add to the HSICAL[7:0] value, adjusting the internal HSI RC oscillator frequency according to voltage and temperature changes. The default value is 16, which can adjust HSI to 20MHz±0.25%; each step of HSICAL change adjusts approximately 40kHz.	10000b
2	Reserved	RO	Reserved.	0
1	HSIRDY	RO	Internal high-speed clock (20MHz) stable ready flag (set by hardware): 1: Internal high-speed clock (20MHz) is stable; 0: Internal high-speed clock (20MHz) is not stable. Note: After the HSION bit is cleared to 0, this bit requires 6 HSI cycles to clear.	1
0	HSION	RW	Internal high-speed clock (20MHz) enable control bit: 1: Enable HSI oscillator; 0: Disable HSI oscillator. Note: When returning from standby and stop modes or when the	1

Bits	Name	Access	Description	Reset
			external oscillator HSE used as the system clock fails, this bit is set to 1 by hardware to start the internal 20MHz RC oscillator.	

3.4.2 Clock Configuration Register 0 (RCC_CFGR0)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	ADC_DUTY_SEL[1:0]			ADCSRC	MCO[3:0]			Reserved			PLLMUL[4:0]			PLLSRC	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADCPRE[1:0]		PPRE2[2:0]			PPRE1[2:0]			HPRE[3:0]				SWS[1:0]		SW[1:0]	

Bits	Name	Access	Description	Reset
31	Reserved	RO	Reserved.	0
[30:29]	ADC_DUTY_SEL[1:0]	RW	ADC clock duty cycle selection: 00: ADC clock duty cycle is 50%; 01: ADC clock duty cycle is 62.5%; 10: ADC clock duty cycle is 75%; 11: ADC clock duty cycle is 62.5%.	01b
28	ADCSRC	RW	ADC input clock source selection: 1: USBHS_PLL divided clock; 0: HCLK.	0
[27:24]	MCO[3:0]	RW	Microcontroller MCO pin clock output control: 00xx: No clock output; 0100: System clock (SYSCLK) output; 0101: Internal 20MHz RC oscillator clock (HSI) output; 0110: External oscillator clock (HSE) output; 0111: UTMI clock output; 1000: ETH_PLL clock divided by 8 output; 1010: XT1 external oscillator clock output; Others: Reserved. Note: There may be a few cycles of clock loss when starting or switching the MCO clock.	0000b
[23:22]	Reserved	RO	Reserved.	0
[21:17]	PLLMUL[4:0]	RW	System PLL multiplication factor: 00000: ×8; 00001: ×9; 00010: ×10; 00011: ×11; 00100: ×12; 00101: ×12.5; 00110: ×13; 00111: ×13.5; 01000: ×14; 01001: ×14.5; 01010: ×15; 01011: ×15.5; 01100: ×16; 01101: ×16.5; 01110: ×17; 01111: ×17.5; 10000: ×18; 10001: ×18.5; 10010: ×19; 10011: ×19.5; 10100: ×20; 10101: ×21; 10110: ×22; 10111: ×24; 11000: ×26; 11001: ×28;	0

Bits	Name	Access	Description	Reset
			11010: ×30; 11011: ×32; 11100: ×34; 11101: ×36; 11110: ×38; 11111: ×40.	
16	PLLSRC	RW	System PLL source: 1: HSE; 0: HSI.	0
[15:14]	ADCPRE[1:0]	RW	ADC clock source prescaler control: 00: PCLK2 divided by 2 as ADC clock; 01: PCLK2 divided by 4 as ADC clock; 10: PCLK2 divided by 6 as ADC clock; 11: PCLK2 divided by 8 as ADC clock. Note: ADC clock must not exceed 30MHz.	00b
[13:11]	PPRE2[2:0]	RW	PB2 clock source prescaler control: 0xx: HCLK not divided; 100: HCLK divided by 2; 101: HCLK divided by 4; 110: HCLK divided by 8; 111: HCLK divided by 16.	000b
[10:8]	PPRE1[2:0]	RW	PB1 clock source prescaler control: 0xx: HCLK not divided; 100: HCLK divided by 2; 101: HCLK divided by 4; 110: HCLK divided by 8; 111: HCLK divided by 16.	000b
[7:4]	HPRE[3:0]	RW	HB clock source prescaler control: 0xxx: SYSCLK not divided; 1000: SYSCLK divided by 2; 1001: SYSCLK divided by 4; 1010: SYSCLK divided by 8; 1011: SYSCLK divided by 16; 1100: SYSCLK divided by 64; 1101: SYSCLK divided by 128; 1110: SYSCLK divided by 256; 1111: SYSCLK divided by 512.	0000b
[3:2]	SWS[1:0]	RO	System clock (SYSCLK) status (set by hardware): 00: System clock source is HSI; 01: System clock source is HSE; 10: System clock source is SYSPLL; 11: Not available.	00b
[1:0]	SW[1:0]	RW	System clock source selection: 00: HSI as system clock; 01: HSE as system clock; 10: SYSPLL output as system clock; 11: Not available. Note: When the clock security system is enabled (CSSON=1), when returning from standby and stop modes or when the external oscillator HSE used as the system clock fails, the hardware forces HSI to be selected as the system clock.	00b

3.4.3 Clock Interrupt Register (RCC_INTR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved						CSSC	Reserved				HSERDYC	HSIRDYC	LSERDYC	LSIRDYC	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				HSERDYIE	HSIRDYIE	LSERDYIE	LSIRDYIE	CSSF	Reserved			HSERDYF	HSIRDYF	LSERDYF	LSIRDYF

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
23	CSSC	WO	Clear clock security system interrupt flag (CSSF): 1: Clear CSSF interrupt flag; 0: No action.	0
[22:20]	Reserved	RO	Reserved.	0
19	HSERDYC	WO	Clear HSE oscillator ready interrupt flag: 1: Clear HSERDYF interrupt flag; 0: No action.	0
18	HSIRDYC	WO	Clear HSI oscillator ready interrupt flag: 1: Clear HSIRDYF interrupt flag; 0: No action.	0
17	LSERDYC	WO	Clear LSE oscillator ready interrupt flag: 1: Clear LSERDYF interrupt flag; 0: No action.	0
16	LSIRDYC	WO	Clear LSI oscillator ready interrupt flag: 1: Clear LSIRDYF interrupt flag; 0: No action.	0
[15:12]	Reserved	RO	Reserved.	0
11	HSERDYIE	RW	HSE ready interrupt enable bit: 1: Enable HSE ready interrupt; 0: Disable HSE ready interrupt.	0
10	HSIRDYIE	RW	HSI ready interrupt enable bit: 1: Enable HSI ready interrupt; 0: Disable HSI ready interrupt.	0
9	LSERDYIE	RW	LSE ready interrupt enable bit: 1: Enable LSE ready interrupt; 0: Disable LSE ready interrupt.	0
8	LSIRDYIE	RW	LSI ready interrupt enable bit: 1: Enable LSI ready interrupt; 0: Disable LSI ready interrupt.	0
7	CSSF	RO	Clock security system interrupt flag: 1: HSE clock failure, clock security interrupt CSSI generated; 0: No clock security system interrupt. Set by hardware, cleared by software writing CSSC bit to 1.	0
[6:4]	Reserved	RO	Reserved.	0
3	HSERDYF	RO	HSE clock ready interrupt flag: 1: HSE clock ready interrupt generated;	0

Bits	Name	Access	Description	Reset
			0: No HSE clock ready interrupt. Set by hardware, cleared by software writing HSERDYC bit to 1.	
2	HSIRDYF	RO	HSI clock ready interrupt flag: 1: HSI clock ready interrupt generated; 0: No HSI clock ready interrupt. Set by hardware, cleared by software writing HSIRDYC bit to 1.	0
1	LSERDYF	RO	LSE clock ready interrupt flag: 1: LSE clock ready interrupt generated; 0: No LSE clock ready interrupt. Set by hardware, cleared by software writing LSERDYC bit to 1.	0
0	LSIRDYF	RO	LSI clock ready interrupt flag: 1: LSI clock ready interrupt generated; 0: No LSI clock ready interrupt. Set by hardware, cleared by software writing LSIRDYC bit to 1.	0

3.4.4 PB2 Peripheral Reset Register (RCC_PB2PRSTR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	USART1RST	TIM8RST	SPI1RST	TIM1RST	ADC2RST	ADC1RST	Reserved	IOPERST	IOPDRST	IOPCRST	IOPBRST	IOPARST	Reserved	AFIORST	

Bits	Name	Access	Description	Reset
[31:15]	Reserved	RO	Reserved.	0
14	USART1RST	RW	USART1 module reset control: 1: Reset module; 0: No effect.	0
13	TIM8RST	RW	TIM8 module reset control: 1: Reset module; 0: No effect.	0
12	SPI1RST	RW	SPI1 module reset control: 1: Reset module; 0: No effect.	0
11	TIM1RST	RW	TIM1 module reset control: 1: Reset module; 0: No effect.	0
10	ADC2RST	RW	ADC2 module reset control: 1: Reset module; 0: No effect.	0
9	ADC1RST	RW	ADC1 module reset control: 1: Reset module; 0: No effect.	0
[8:7]	Reserved	RO	Reserved.	0
6	IOPERST	RW	IO port E module reset control: 1: Reset module; 0: No effect.	0
5	IOPDRST	RW	IO port D module reset control: 1: Reset module; 0: No effect.	0
4	IOPCRST	RW	IO port C module reset control: 1: Reset module; 0: No effect.	0

Bits	Name	Access	Description	Reset
3	IOPBRST	RW	IO port B module reset control: 1: Reset module; 0: No effect.	0
2	IOPARST	RW	IO port A module reset control: 1: Reset module; 0: No effect.	0
1	Reserved	RO	Reserved.	0
0	AFIORST	RW	IO auxiliary function module reset control: 1: Reset module; 0: No effect.	0

3.4.5 PB1 Peripheral Reset Register (RCC_PB1PRSTR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	DACRST	PWRRST	BKPRST	Reserved	CAN1RST	Reserved	I2CRST	USART5RST	USART4RST	USART3RST	USART2RST	Reserved			

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
SPI3RST	SPI2RST	Reserved	WWDGRST	USART10RST	USART9RST	USART8RST	USART7RST	USART6RST	TIM7RST	TIM6RST	TIM5RST	TIM4RST	TIM3RST	

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
29	DACRST	RW	DAC module reset control: 1: Reset module; 0: No effect.	0
28	PWRRST	RW	Power interface module reset control: 1: Reset module; 0: No effect.	0
27	BKPRST	RW	Backup unit reset control: 1: Reset module; 0: No effect.	0
26	Reserved	RO	Reserved.	0
25	CAN1RST	RW	CAN1 module reset control: 1: Reset module; 0: No effect.	0
[24:22]	Reserved	RO	Reserved.	0
21	I2CRST	RW	I2C module reset control: 1: Reset module; 0: No effect.	0
20	USART5RST	RW	USART5 module reset control: 1: Reset module; 0: No effect.	0
19	USART4RST	RW	USART4 module reset control: 1: Reset module; 0: No effect.	0
18	USART3RST	RW	USART3 module reset control: 1: Reset module; 0: No effect.	0
17	USART2RST	RW	USART2 module reset control: 1: Reset module; 0: No effect.	0
16	Reserved	RO	Reserved.	0
15	SPI3RST	RW	SPI3 module reset control: 1: Reset module; 0: No effect.	0
14	SPI2RST	RW		0

Bits	Name	Access	Description	Reset
			SPI2 module reset control: 1: Reset module; 0: No effect.	
[13:12]	Reserved	RO	Reserved.	0
11	WWDGRST	RW	Window watchdog reset control: 1: Reset module; 0: No effect.	0
10	USART10RST	RW	USART10 module reset control: 1: Reset module; 0: No effect.	0
9	USART9RST	RW	USART9 module reset control: 1: Reset module; 0: No effect.	0
8	USART8RST	RW	USART8 module reset control: 1: Reset module; 0: No effect.	0
7	USART7RST	RW	USART7 module reset control: 1: Reset module; 0: No effect.	0
6	USART6RST	RW	USART6 module reset control: 1: Reset module; 0: No effect.	0
5	TIM7RST	RW	Timer 7 module reset control: 1: Reset module; 0: No effect.	0
4	TIM6RST	RW	Timer 6 module reset control: 1: Reset module; 0: No effect.	0
3	TIM5RST	RW	Timer 5 module reset control: 1: Reset module; 0: No effect.	0
2	TIM4RST	RW	Timer 4 module reset control: 1: Reset module; 0: No effect.	0
1	TIM3RST	RW	Timer 3 module reset control: 1: Reset module; 0: No effect.	0
0	TIM2RST	RW	Timer 2 module reset control: 1: Reset module; 0: No effect.	0

3.4.6 HB Peripheral Clock Enable Register (RCC_HBPCENR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	ETHMACEN	DVPEN	USBHS1EN	USBHS2EN	SDIOEN	RNGEN	FSMCEN	ARGBEN	CRCEN	I3CEN	PSRAMEN	LTDCEN	SRAMEN	DMA2EN	DMA1EN

Bits	Name	Access	Description	Reset
[31:15]	Reserved	RO	Reserved.	0
14	ETHMACEN	RW	Ethernet MAC clock enable: 1: Ethernet MAC clock enabled; 0: Ethernet MAC clock disabled.	0
13	DVPEN	RW	DVP module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0

Bits	Name	Access	Description	Reset
12	USBHS1EN	RW	USBHS1 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
11	USBHS2EN	RW	USBHS2 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
10	SDIOEN	RW	SDIO module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
9	RNGEN	RW	RNG module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
8	FSMCEN	RW	FSMC module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
7	ARGBEN	RW	ARGB module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
6	CRCEN	RW	CRC module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
5	I3CEN	RW	I3C module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
4	PSRAMEN	RW	PSRAM module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
3	LTDCEN	RW	LTDC module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
2	SRAMEN	RW	SRAM interface module clock enable bit: 1: SRAM interface module clock enabled in sleep mode; 0: SRAM interface module clock disabled in sleep mode.	1
1	DMA2EN	RW	DMA2 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
0	DMA1EN	RW	DMA1 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0

3.4.7 PB2 Peripheral Clock Enable Register (RCC_PB2PCENR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	USART1EN	TIM8EN	SPI1EN	TIM1EN	ADC2EN	ADC1EN	Reserved	IOPEEN	IOPDEN	IOPCEN	IOPBEN	IOPAEN	Reserved	Reserved	AFIOEN

Bits	Name	Access	Description	Reset
[31:15]	Reserved	RO	Reserved.	0
14	USART1EN	RW	USART1 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
13	TIM8EN	RW	TIM8 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
12	SPI1EN	RW		0

Bits	Name	Access	Description	Reset
			SPI1 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	
11	TIM1EN	RW	TIM1 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
10	ADC2EN	RW	ADC2 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
9	ADC1EN	RW	ADC1 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
[8:7]	Reserved	RO	Reserved.	0
6	IOPEEN	RW	IO port E module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
5	IOPDEN	RW	IO port D module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
4	IOPCEN	RW	IO port C module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
3	IOPBEN	RW	IO port B module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
2	IOPAEN	RW	IO port A module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
1	Reserved	RO	Reserved.	0
0	AFIOEN	RW	IO auxiliary function module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0

3.4.8 PB1 Peripheral Clock Enable Register (RCC_PB1PCENR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	DACEN	PWREN	BKPEN	Reserved	CAN1EN	Reserved	I2CEN	USART5EN	USART4EN	USART3EN	USART2EN	Reserved			

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPI3EN	SPI2EN	Reserved	WWDGEN	USART10EN	USART9EN	USART8EN	USART7EN	USART6EN	TIM7EN	TIM6EN	TIM5EN	TIM4EN	TIM3EN	TIM2EN	

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
29	DACEN	RW	DAC module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
28	PWREN	RW	Power interface module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
27	BKPEN	RW	Backup unit clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
26	Reserved	RO	Reserved.	0
25	CAN1EN	RW	CAN1 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
[24:22]	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
21	I2CEN	RW	I2C module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
20	USART5EN	RW	USART5 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
19	USART4EN	RW	USART4 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
18	USART3EN	RW	USART3 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
17	USART2EN	RW	USART2 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
16	Reserved	RO	Reserved.	0
15	SPI3EN	RW	SPI3 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
14	SPI2EN	RW	SPI2 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
[13:12]	Reserved	RO	Reserved.	0
11	WWDGEN	RW	Window watchdog clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
10	USART10EN	RW	USART10 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
9	USART9EN	RW	USART9 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
8	USART8EN	RW	USART8 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
7	USART7EN	RW	USART7 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
6	USART6EN	RW	USART6 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
5	TIM7EN	RW	Timer 7 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
4	TIM6EN	RW	Timer 6 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
3	TIM5EN	RW	Timer 5 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
2	TIM4EN	RW	Timer 4 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
1	TIM3EN	RW	Timer 3 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0
0	TIM2EN	RW	Timer 2 module clock enable bit: 1: Module clock enabled; 0: Module clock disabled.	0

3.4.9 Backup Domain Control Register (RCC_BDCTL)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Reserved																BDRST
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
RTCEN	Reserved					RTCSEL[1:0]		Reserved					LSEBYP	LSERDY	LSEON	

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
16	BDRST	RW	Backup domain software reset control: 1: Reset the entire backup domain; 0: Release reset.	0
15	RTCEN	RW	RTC clock enable control: 1: Enable RTC clock; 0: Disable RTC clock. Note: RTC clock can only be enabled when RTCSEL $\neq$ 0, otherwise hardware forces it to 0.	0
[14:10]	Reserved	RO	Reserved.	0
[9:8]	RTCSEL[1:0]	RW	RTC clock source selection: 00: No clock; 01: LSE oscillator as RTC clock; 10: LSI oscillator as RTC clock; 11: HSE oscillator divided by 128 as RTC clock. Note: Once the RTC clock source is selected (RTCEN=1), it cannot be changed until the next backup domain reset. It can be restored to default by setting the BDRST bit.	0
[7:3]	Reserved	RO	Reserved.	0
2	LSEBYP	RW	External low-speed crystal (LSE) bypass control bit: 1: Bypass external low-speed crystal/ceramic resonator (use external clock source); 0: Do not bypass low-speed external crystal/ceramic resonator. Note: This bit must be written while LSEON is 0.	0
1	LSERDY	RO	External low-speed crystal oscillator stable ready flag (set by hardware): 1: External low-speed crystal oscillator is stable; 0: External low-speed crystal oscillator is not stable. Note: After the LSEON bit is cleared to 0, this bit requires 6 LSE cycles to clear.	0
0	LSEON	RW	External low-speed crystal oscillator enable control bit: 1: Enable LSE oscillator; 0: Disable LSE oscillator.	0

Note: The LSEON, LSEBYP, RTCSEL, and RTCEN bits in the backup domain control register (RCC_BDCTL) are in the backup domain. Therefore, these bits are write-protected after reset and can only be modified after the DBP bit in the power control register (PWR_CTLR) is set to 1. These bits can only be cleared by a backup domain reset. No internal or external reset affects these bits.

3.4.10 Control/Status Register (RCC_RSTSCKR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LPWRRSTF	WWDGRSTF	IWDGRSTF	SFTRSTF	PORRSTF	PINRSTF	Reserved	RMVF	Reserved							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved															LSIRDY	LSION

Bits	Name	Access	Description	Reset
31	LPWRRSTF	RO	Low-power reset flag: 1: Low-power reset occurred; 0: No low-power reset occurred. Set to 1 by hardware when a low-power management reset occurs; cleared by software writing RMVF bit. Note: This function only applies to chips where the fifth-from-last digit of the lot number is greater than 0.	0
30	WWDGRSTF	RO	Window watchdog reset flag: 1: Window watchdog reset occurred; 0: No window watchdog reset occurred. Set to 1 by hardware when a window watchdog reset occurs; cleared by software writing RMVF bit.	0
29	IWDGRSTF	RO	Independent watchdog reset flag: 1: Independent watchdog reset occurred; 0: No independent watchdog reset occurred. Set to 1 by hardware when an independent watchdog reset occurs; cleared by software writing RMVF bit.	0
28	SFTRSTF	RO	Software reset flag: 1: Software reset occurred; 0: No software reset occurred. Set to 1 by hardware when a software reset occurs; cleared by software writing RMVF bit.	0
27	PORRSTF	RO	Power-on/power-down reset flag: 1: Power-on/power-down reset occurred; 0: No power-on/power-down reset occurred. Set to 1 by hardware when a power-on/power-down reset occurs; cleared by software writing RMVF bit.	1
26	PINRSTF	RO	External manual reset (NRST pin) flag: 1: NRST pin reset occurred; 0: No NRST pin reset occurred. Set to 1 by hardware when an NRST pin reset occurs; cleared by software writing RMVF bit.	0
25	Reserved	RO	Reserved.	0
24	RMVF	RW	Clear reset flag control: 1: Clear reset flags; 0: No effect.	0
[23:2]	Reserved	RO	Reserved.	0
1	LSIRDY	RO	Internal low-speed clock (LSI) stable ready flag (set by hardware): 1: Internal low-speed clock (40kHz) is stable; 0: Internal low-speed clock (40kHz) is not stable. Note: After the LSION bit is cleared to 0, this bit requires 3 LSI cycles to clear.	0

Bits	Name	Access	Description	Reset
0	LSION	RW	Internal low-speed clock (LSI) enable control bit: 1: Enable LSI (40kHz) oscillator; 0: Disable LSI (40kHz) oscillator.	0

Note: Except for bit 1 which is cleared by power-on reset, the other write-to-clear reset flags can be cleared by writing the RMVF bit.

3.4.11 HB Peripheral Reset Register (RCC_HBRSTR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	ETHMACRST	DVPRST	USBHS1RST	USBHS2RST	Reserved	ARGBRST	Reserved	I3CRST	PSRAMRST	LTDCRST	Reserved				

Bits	Name	Access	Description	Reset
[31:15]	Reserved	RO	Reserved.	0
14	ETHMACRST	RW	Ethernet MAC reset control: 1: Reset module; 0: No effect.	0
13	DVPRST	RW	DVP reset control: 1: Reset module; 0: No effect.	0
12	USBHS1RST	RW	USBHS1 module reset control: 1: Reset module; 0: No effect.	0
11	USBHS2RST	RW	USBHS2 module reset control: 1: Reset module; 0: No effect.	0
[10:8]	Reserved	RO	Reserved.	0
7	ARGBRST	RW	ARGB module reset control: 1: Reset module; 0: No effect.	0
6	Reserved	RO	Reserved.	0
5	I3CRST	RW	I3C module reset control: 1: Reset module; 0: No effect.	0
4	PSRAMRST	RW	PSRAM module reset control: 1: Reset module; 0: No effect.	0
3	LTDCRST	RW	LTDC module reset control: 1: Reset module; 0: No effect.	0
[2:0]	Reserved	RO	Reserved.	0

3.4.12 Clock Configuration Register 2 (RCC_CFGR2)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
UTMI2ON	UTMI1ON	USBHSPLLCLK[1:0]		USBHSPLLSRC[1:0]		Reserved			RNGSRC	I2S3SRC	I2S2SRC	Reserved			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

SYSPLLSRC[2:0]		SYSPLL_GATE	Reserved	SYSPLLDIV[2:0]	I2SPLLSRC	LTDCSRC	LTDCDIV[5:0]
Bits	Name	Access	Description			Reset	
31	UTMI2ON	RW	UTMI2 clock gate enable: 1: Enable; 0: Disable.			0	
30	UTMI1ON	RW	UTMI1 clock gate enable: 1: Enable; 0: Disable.			0	
[29:28]	USBHSPLLCLK[1:0]	RW	USBHSPLL reference clock frequency selection: 00: 25MHz; 01: 20MHz; 10: 24MHz; 11: 32MHz; Only writable when USBHS_PLLON is 0.			01b	
[27:26]	USBHSPLLSRC[1:0]	RW	USBHSPLL reference source selection: 00: HSE; 01: HSI; 10: ETHCLK_20M. Only writable when USBHS_PLLON is 0.			0	
[25:20]	Reserved	RO	Reserved.			0	
19	RNGSRC	RW	RNG clock source selection: 1: PLL clock selected via I2SPLLSRC bit; 0: System clock (SYSCLK).			0	
18	I2S3SRC	RW	I2S3 clock source: 1: PLL clock selected via I2SPLLSRC bit; 0: System clock (SYSCLK).			0	
17	I2S2SRC	RW	I2S2 clock source: 1: PLL clock selected via I2SPLLSRC bit; 0: System clock (SYSCLK).			0	
16	Reserved	RO	Reserved.			0	
[15:13]	SYSPLLSRC[2:0]	RW	System clock switch to PLL selection, only writable when SYSPLL_GATE is 0: 000: USBHS_PLL (480MHz); 001: ETH_PLL (500MHz); 010: USBHS_PLL×2/3 (320MHz); 011: ETH_PLL×2/3 (333MHz); 110: PLL_CLK; Others: Reserved.			0	
12	SYSPLL_GATE	RW	Before switching the system clock to PLL, this bit must be set.			0	
11	Reserved	RO	Reserved.			0	
[10:8]	SYSPLLDIV[2:0]	RW	System PLL divider configuration: 000: No division; 001: Divide by 2; ... 111: Divide by 8.			0	
7	I2SPLLSRC	RW				0	

Bits	Name	Access	Description	Reset
			I2S/RNG PLL source selection: 1: ETH_PLL; 0: USBHS_PLL.	
6	LTDCSRC	RW	LTDC clock source selection: 1: ETH_PLL clock; 0: USBHS_PLL clock.	0
[5:0]	LTDCDIV[5:0]	RW	LTDC clock source prescaler control: 000000: No division; 000001: Divide by 2; 000010: Divide by 3; 000011: Divide by 4; 000100: Divide by 5; ... 111110: Divide by 63; 111111: Divide by 64.	0

3.4.13 Clock Configuration Register 3 (RCC_CFGR3)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											ADC_CLKDIV[4:0]				

Bits	Name	Access	Description	Reset
[31:5]	Reserved	RO	Reserved.	0
[4:0]	ADC_CLKDIV[4:0]	RW	ADC clock source USBHS_PLL divider configuration: 00000: No division; 00001: Divide by 2; 00010: Divide by 3; ... 11111: Divide by 32.	0000b

Chapter 4: Backup Registers (BKP)

The Backup Registers (BKP) provide 42 16-bit backup data registers, which can be used to store 84 bytes of user data. After the main power supply (VDD) is powered off, this data can still be maintained by VBAT power, unaffected by standby state, system reset, or power reset. In addition, the BKP unit also provides tamper detection management, RTC clock calibration, and pulse output functions.

4.1 Main Features

- Provides 84 bytes of backup data registers
- RTC clock calibration function
- Outputs RTC clock divided by 64, alarm pulse, or second pulse on the PC13 pin
- Tamper detection (TAMPER) function

4.2 Functional Description

After the microcontroller is reset, access to the backup registers and RTC is disabled. Access to the backup registers must be enabled through the following operations:

1. Set the PWREN and BKPEN bits of the RCC_PB1PCENR register to enable the operating clocks for power and backup interface;
2. Set the DBP bit of the power control register PWR_CTLR to enable access to the backup registers and RTC registers.

4.2.1 Backup Data Registers

The backup data registers can be used as general-purpose data storage. Since they retain data under VBAT power when VDD is lost, they can be used to store important or sensitive data. However, this data will be entirely cleared after a tamper event occurs.

4.2.2 Tamper Detection

Tamper detection means that when an external signal is provided (rising edge or falling edge), it indicates that a "tamper event" has occurred, and the hardware will automatically clear important information retained in the current system. This method can increase the security of system information.

When a transition edge appears on the tamper detection pin (depending on the TPAL bit), a tamper event is generated. If the tamper detection interrupt is enabled, a tamper detection interrupt will also be generated at the same time. Whenever a tamper event occurs, all backup data registers will be cleared. In addition, the hardware detection uses a latching method; even when the tamper detection function is not enabled (TPE=0), the system will sample whether a transition edge has occurred, and under the condition selected by the TPAL bit, it will pre-lock the tamper event and trigger the tamper event when the TPE bit is set to 1.

For example: When TPAL=0, if TPE=0 means the function is not enabled, but the TAMPER pin is already at a high level, once TPE=1, an additional tamper event will be generated (the system pre-locked the

rising edge). When TPAL=1, if TPE=0 means the function is not enabled, but the TAMPER pin is already at a low level, once TPE=1, an additional tamper event will be generated (the system pre-locked the falling edge).

Therefore, to prevent unnecessary tamper events that would clear the backup registers, it is recommended that at the moment when hardware detection of the tamper pin is desired to start, by writing 1 to the CTE bit of the BKP_TPCSR register, first clear any tamper event that the hardware may have memorized, and ensure that the current state of the tamper detection pin is inactive.

Note: When the VDD power is disconnected, the tamper detection function remains active. To avoid unnecessary resetting of the backup data registers, the TAMPER pin should be connected to the correct logic level externally.

4.2.3 RTC Calibration

This function must be configured to use the tamper detection pin as a regular I/O port. Clear the TPE bit of the BKP_TPCTLR register.

- Pulse output: Configure the ASOE bit of the BKP_OCTLR register to enable RTC pulse output; set the ASOS bit to select between second pulse output and alarm pulse output.
- RTC calibration: After configuring the CCO bit of the BKP_OCTLR register, the internal RTC clock will be output to the tamper detection pin (TAMPER) after being divided by 64. Through actual testing, the software cooperates with modifying the CAL[6:0] bits to adjust the clock and calibrate the RTC.

4.2.4 BKP Interface Reset

The BKP domain can be independently powered by VBAT when the VDD main power is off. During application code control of BKP domain register reset, the backup data registers BKP_DATAR1-10, ASOS bit, and ASOE bit are reset under software configuration of the BDRST bit of the RCC_BDCTLR register, and are not affected by the RCC peripheral interface control BKPRST bit.

4.3 Register Description

Table 4-1: BKP-related register list

Name	Address	Description	Reset Value
R16_BKP_DATAR1	0x40006C04	Backup data register 1	0x0000
R16_BKP_DATAR2	0x40006C08	Backup data register 2	0x0000
R16_BKP_DATAR3	0x40006C0C	Backup data register 3	0x0000
R16_BKP_DATAR4	0x40006C10	Backup data register 4	0x0000
R16_BKP_DATAR5	0x40006C14	Backup data register 5	0x0000
R16_BKP_DATAR6	0x40006C18	Backup data register 6	0x0000
R16_BKP_DATAR7	0x40006C1C	Backup data register 7	0x0000
R16_BKP_DATAR8	0x40006C20	Backup data register 8	0x0000
R16_BKP_DATAR9	0x40006C24	Backup data register 9	0x0000
R16_BKP_DATAR10	0x40006C28	Backup data register 10	0x0000
R16_BKP_OCTLR	0x40006C2C	RTC calibration register	0x0000

Name	Address	Description	Reset Value
R16_BKP_TPCTLR	0x40006C30	Tamper detection control register	0x0000
R16_BKP_TPCSR	0x40006C34	Tamper detection status register	0x0000
R16_BKP_DATAR11	0x40006C40	Backup data register 11	0x0000
R16_BKP_DATAR12	0x40006C44	Backup data register 12	0x0000
R16_BKP_DATAR13	0x40006C48	Backup data register 13	0x0000
R16_BKP_DATAR14	0x40006C4C	Backup data register 14	0x0000
R16_BKP_DATAR15	0x40006C50	Backup data register 15	0x0000
R16_BKP_DATAR16	0x40006C54	Backup data register 16	0x0000
R16_BKP_DATAR17	0x40006C58	Backup data register 17	0x0000
R16_BKP_DATAR18	0x40006C5C	Backup data register 18	0x0000
R16_BKP_DATAR19	0x40006C60	Backup data register 19	0x0000
R16_BKP_DATAR20	0x40006C64	Backup data register 20	0x0000
R16_BKP_DATAR21	0x40006C68	Backup data register 21	0x0000
R16_BKP_DATAR22	0x40006C6C	Backup data register 22	0x0000
R16_BKP_DATAR23	0x40006C70	Backup data register 23	0x0000
R16_BKP_DATAR24	0x40006C74	Backup data register 24	0x0000
R16_BKP_DATAR25	0x40006C78	Backup data register 25	0x0000
R16_BKP_DATAR26	0x40006C7C	Backup data register 26	0x0000
R16_BKP_DATAR27	0x40006C80	Backup data register 27	0x0000
R16_BKP_DATAR28	0x40006C84	Backup data register 28	0x0000
R16_BKP_DATAR29	0x40006C88	Backup data register 29	0x0000
R16_BKP_DATAR30	0x40006C8C	Backup data register 30	0x0000
R16_BKP_DATAR31	0x40006C90	Backup data register 31	0x0000
R16_BKP_DATAR32	0x40006C94	Backup data register 32	0x0000
R16_BKP_DATAR33	0x40006C98	Backup data register 33	0x0000
R16_BKP_DATAR34	0x40006C9C	Backup data register 34	0x0000
R16_BKP_DATAR35	0x40006CA0	Backup data register 35	0x0000
R16_BKP_DATAR36	0x40006CA4	Backup data register 36	0x0000
R16_BKP_DATAR37	0x40006CA8	Backup data register 37	0x0000
R16_BKP_DATAR38	0x40006CAC	Backup data register 38	0x0000
R16_BKP_DATAR39	0x40006CB0	Backup data register 39	0x0000
R16_BKP_DATAR40	0x40006CB4	Backup data register 40	0x0000
R16_BKP_DATAR41	0x40006CB8	Backup data register 41	0x0000
R16_BKP_DATAR42	0x40006CBC	Backup data register 42	0x0000

4.3.1 Backup Data Register (BKP_DATARx) (x=1-42)

Offset address: 0x04-0x28, 0x40-0xBC

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
D[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	D[15:0]	RW	Backup data, can be accessed by the user program. Note: They are only reset by a backup domain reset (BDRST) or (if the tamper detection pin TAMPER function is enabled) by a tamper pin event.	0

4.3.2 RTC Calibration Register (BKP_OCTLR)

Offset address: 0x2C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						ASOS	ASOE	CCO	CAL[6:0]						

Bits	Name	Access	Description	Reset
[15:10]	Reserved	RO	Reserved.	0
9	ASOS	RW	TAMPER pin alarm/second pulse output selection. 1: Output second pulse; 0: Output alarm pulse. Note: This bit is only reset by backup domain reset (BDRST).	0
8	ASOE	RW	TAMPER pin pulse output enable bit. 1: Enable output of alarm pulse or second pulse; 0: Disable output of alarm pulse or second pulse. Note: This bit is only reset by backup domain reset (BDRST).	0
7	CCO	RW	Calibration clock output selection bit. 1: The TAMPER pin outputs the RTC clock divided by 64; 0: No calibration clock output. Note 1: To enable this function, the tamper detection function must be disabled. Note 2: When VDD power is disconnected, this bit is cleared.	0
[6:0]	CAL[6:0]	RW	Calibration value register; the value of this register indicates how many out of every 2 clock pulses are skipped. This function is used to calibrate the RTC clock. The RTC clock can be slowed down by 0 to 121 ppm.	0

4.3.3 Tamper Detection Control Register (BKP_TPCTLR)

Offset address: 0x30

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														TPAL	TPE

Bits	Name	Access	Description	Reset
[15:2]	Reserved	RO	Reserved.	0
1	TPAL	RW	Tamper detection pin (TAMPER pin) active level setting. 1: A low level on the tamper detection pin clears all backup data registers (hardware latches falling edge); 0: A high level on the tamper detection pin clears all backup data registers (hardware latches rising edge).	0
0	TPE	RW	Tamper detection pin enable bit: 1: TAMPER pin is used for tamper detection; 0: TAMPER pin is used as a regular I/O port.	0

Note: Clearing both the TPAL and TPE bits at the same time will generate a false tamper event. It is recommended to change the TPAL bit state only when TPE is 0.

4.3.4 Tamper Detection Status Register (BKP_TPCSR)

Offset address: 0x34

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					TIF	TEF	Reserved				TPIE	CTI	CTE		

Bits	Name	Access	Description	Reset
[15:10]	Reserved	RO	Reserved.	0
9	TIF	RO	Tamper interrupt flag bit; this bit is set when a tamper event is detected and the TPIE bit is set to 1. This flag bit is cleared by writing 1 to the CTI bit. If the TPIE bit is reset, this bit will also be reset. Note: This bit is only reset upon a system reset or wakeup from standby mode.	0
8	TEF	RO	Tamper event flag bit; this bit is set when a tamper event is detected. This bit is cleared by writing 1 to the CTE bit. Note: When this bit is 1, the values of all BKP_DATARx registers are cleared, and until this bit is reset, all write operations to the BKP_DATARx registers are invalid.	0
[7:3]	Reserved	RO	Reserved.	0
2	TPIE	RW	Tamper interrupt enable bit: 1: Enable tamper detection interrupt (TPE must be set to 1); 0: Disable tamper detection interrupt. Note 1: The tamper interrupt cannot wake up the core from low-power mode. Note 2: This bit is only reset upon a system reset or wakeup from standby mode.	0
1	CTI	WO	Tamper detection interrupt clear bit; write 1 to clear, read is invalid.	0
0	CTE	WO	Tamper detection event clear bit; write 1 to clear, read is invalid.	0

Chapter 5: Cyclic Redundancy Check (CRC)

The Cyclic Redundancy Check (CRC) computation unit produces a CRC calculation result for any 32-bit data based on a fixed generator polynomial. It is generally used in data storage and data communication fields to verify data integrity. The hardware CRC computation unit provided by the system can greatly save CPU and RAM resources and improve efficiency.

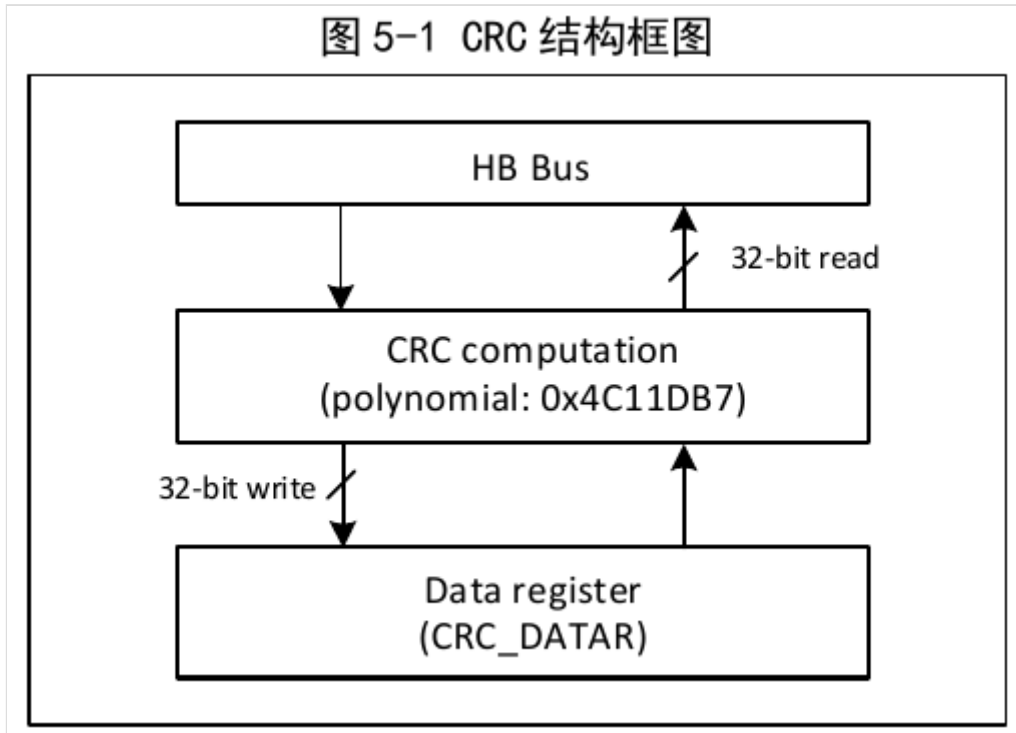


Figure 5-1: CRC Block Diagram

5.1 Main Features

- Uses the CRC32 polynomial (0x4C11DB7):

$$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$$
- The same 32-bit register serves as both data input and CRC32 calculation output
- Single conversion time: 4 HB clock cycles (HCLK)

5.2 Function Description

CRC Unit Reset

To start a CRC calculation for a new set of data, the CRC computation unit must be reset. Writing 1 to the RST bit of the control register CRC_CTLR will cause the hardware to reset the data register, restoring the initial value 0xFFFFFFFF.

CRC Calculation

The CRC unit's calculation result is the CRC of the previous CRC calculation result combined with the newly participating data. For the CRC_DATAR data register, a write operation feeds new data into the hardware computation unit, while a read operation returns the latest CRC calculation value. During hardware computation, write operations to the system are stalled, so new values can be written continuously.

Note: The CRC unit computes on the entire 32-bit data, not byte-by-byte.

Independent Data Buffer

The CRC unit provides an 8-bit independent data register CRC_IDATAR for application code to temporarily store 1 byte of data, which is not affected by CRC unit reset.

5.3 Register Description

Table 5-1: CRC-related register list

Name	Address	Description	Reset Value
R32_CRC_DATAR	0x40023000	Data register	0xFFFFFFFF
R8_CRC_IDATAR	0x40023004	Independent data buffer	0x00
R32_CRC_CTLR	0x40023008	Control register	0x00000000

5.3.1 Data Register (CRC_DATAR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	DR[31:0]	RW	Write raw data; read the calculation result.	0xFFFFFFFF

5.3.2 Independent Data Buffer (CRC_IDATAR)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								IDR[7:0]							

Bits	Name	Access	Description	Reset
[7:0]	IDR[7:0]	RW	8-bit general-purpose register that can be used as a data buffer. This register is not affected by the RST bit of the control register.	0

5.3.3 Control Register (CRC_CTLR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															RST

Bits	Name	Access	Description	Reset
[31:1]	Reserved	RO	Reserved.	0
0	RST	WO	CRC computation unit reset control. Writing 1 triggers the reset; the hardware automatically clears this bit. After execution, the data register becomes 0xFFFFFFFF.	0

Chapter 6: Real-Time Clock (RTC)

The Real-Time Clock (RTC) is an independent timer module. Its programmable counter can be up to 32 bits. Combined with software, it can implement real-time clock functions, and the counter value can be modified to reconfigure the system's current time and date. The RTC module is located in the backup power domain; system reset and wake-up from Standby mode do not affect it.

6.1 Main Features

- Prescaler factor up to 2^{20}
- 32-bit programmable counter
- Multiple clock sources, interrupts
- Independent reset

6.2 Functional Description

6.2.1 Overview

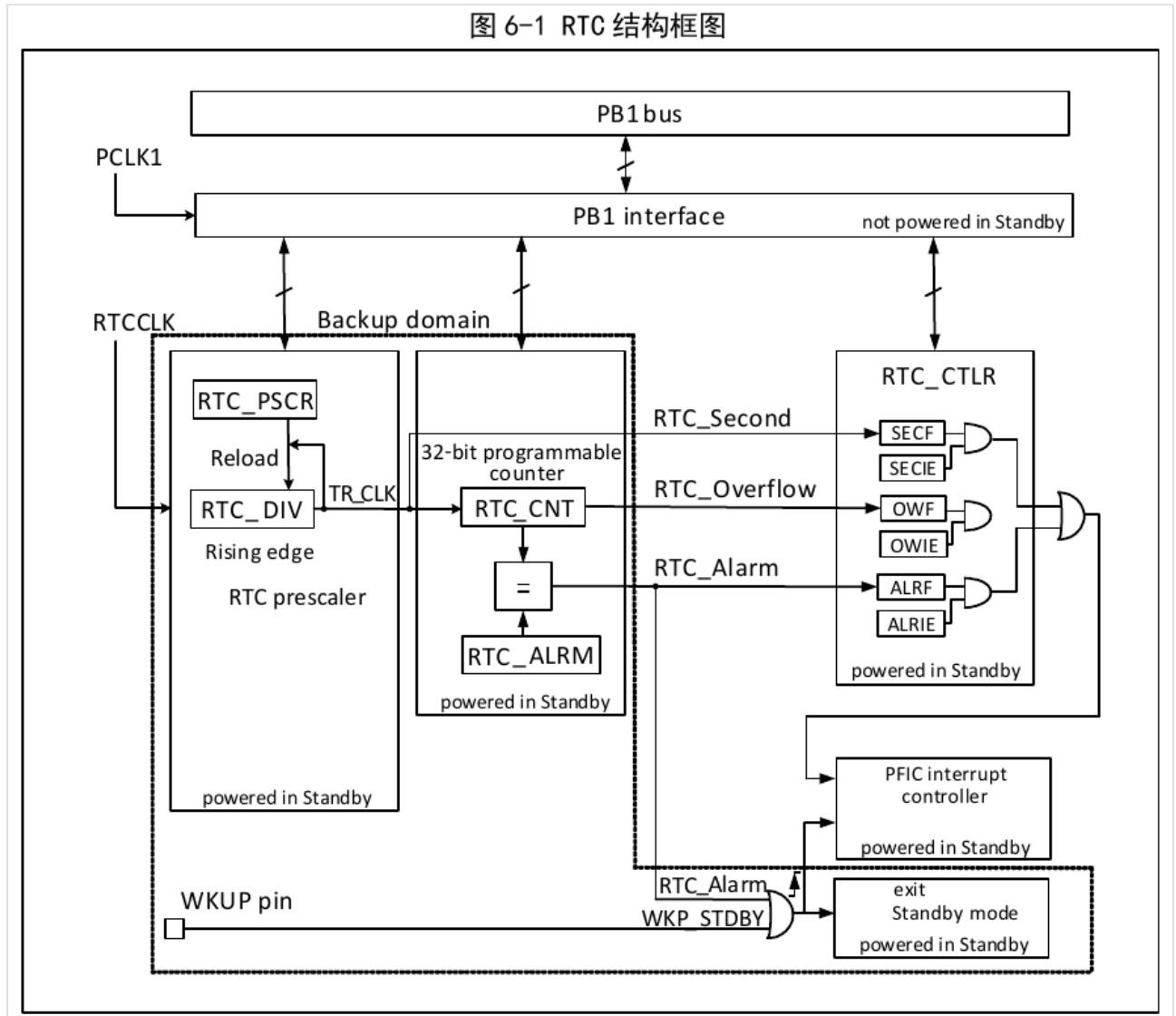


Figure 6-1: RTC block diagram

As shown in Figure 6-1, the RTC module consists mainly of three parts: the PB1 bus interface, the divider and counter, and the control and status registers. The divider and counter are located in the backup domain and can be powered by VBAT. After RTCCLK enters the divider (RTC_DIV), it is divided to produce TR_CLK. Note that the divider (RTC_DIV) internally is a down-counter; when it underflows, it outputs one TR_CLK and then reloads the preset value from the reload value register (RTC_PSCR). Reading the divider actually reads its real-time value (read only); the prescaler factor should be written to the reload value register (RTC_PSCR). Generally, the TR_CLK period is set to 1 second. TR_CLK triggers a second event and simultaneously increments the main counter (RTC_CNT) by 1. When the main counter increments to match the alarm register value, an alarm event is triggered. When the main counter increments to overflow, an overflow event is triggered. All three events can trigger interrupts, each controlled by a corresponding interrupt enable bit.

6.2.2 Reset

Due to the special purpose of the real-time clock, its four register groups in the backup domain — prescaler, prescaler reload value, main counter, and alarm — can only be reset by the backup domain reset signal; refer to the RCC backup domain reset section. The control register of the real-time clock is controlled by system reset or power reset.

6.2.3 Special Register Read/Write Operations

Due to the special purpose of the real-time clock, the RTC is independent from the PB1 bus. PB1 reads of the RTC are not necessarily real-time. Reading RTC registers through PB1 must occur after PB1 is started and after one RTC rising edge. This situation may occur after system reset and power reset, or after wake-up from Standby or Stop mode. A convenient approach is to wait for the RSF bit in the control register (CTLR) to be set. Writing to the RTC requires waiting for the previous write operation to finish, and configuration mode must be entered. The specific steps are:

1. Query the RTOFF bit until it becomes 1;
2. Set the CNF bit to enter configuration mode;
3. Perform write operations on one or more RTC registers;
4. Clear the CNF bit to exit configuration mode; the PB1 interface begins writing to the RTC registers;
5. Query the RTOFF bit until it becomes 1, indicating the write is complete.

6.3 Register Description

Table 6-1: RTC-related register list

Name	Address	Description	Reset Value
R16_RTC_CTLRH	0x40002800	RTC control register high	0x0000
R16_RTC_CTLRL	0x40002804	RTC control register low	0x0020
R16_RTC_PSCRH	0x40002808	Prescaler reload value register high	0x000X
R16_RTC_PSCRL	0x4000280C	Prescaler reload value register low	0xFFFF
R16_RTC_DIVH	0x40002810	Divider register high	0x000X
R16_RTC_DIVL	0x40002814	Divider register low	0xFFFF
R16_RTC_CNTH	0x40002818	RTC counter register high	0xFFFF
R16_RTC_CNTL	0x4000281C	RTC counter register low	0xFFFF
R16_RTC_ALRMH	0x40002820	Alarm register high	0xFFFF
R16_RTC_ALRML	0x40002824	Alarm register low	0xFFFF

6.3.1 RTC Control Register High (RTC_CTLRH)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													OWIE	ALRIE	SECE

Bits	Name	Access	Description	Reset
[15:3]	Reserved	RO	Reserved.	0
2	OWIE	RW	Overflow interrupt enable: 1: Enable; 0: Disable.	0
1	ALRIE	RW	Alarm interrupt enable: 1: Enable; 0: Disable.	0
0	SECIE	RW	Second interrupt enable: 1: Enable; 0: Disable.	0

6.3.2 RTC Control Register Low (RTC_CTLRL)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RTOFF	CNF	RSF	OWF	ALRF	SECF

Bits	Name	Access	Description	Reset
[15:6]	Reserved	RO	Reserved.	0
5	RTOFF	RO	RTC operation status indicator, indicating the execution status of the last operation on the RTC. RTC operations must wait for this bit to be 1. 1: The last RTC operation has completed; 0: The last RTC operation is still in progress.	1
4	CNF	RW	Configuration flag. Writing 1 to this bit enters configuration mode, thereby allowing values to be written to the counter (R16_RTC_CNTx), alarm register (R16_RTC_ALRMx), and prescaler reload value register (R16_RTC_PSCRx). The write operation is only executed after this bit is written 1 and then cleared by software: 1: Enter configuration mode; 0: Exit configuration mode, start updating RTC registers.	0
3	RSF	RW0	Register synchronization flag. Before reading or writing the RTC module's prescaler (PSCRx), alarm (ALRMx), and counter (CNTx) registers, this bit must be set by hardware to confirm these registers have been synchronized. When reading/writing these registers, or after PB1 reset or PB1 clock stop, this bit should be cleared first. 1: Registers have been synchronized; 0: Registers not synchronized.	0
2	OWF	RW0	Counter overflow flag. When the 32-bit counter overflows, this bit is set by hardware. If the OWIE bit is set, an overflow interrupt is also generated. This bit can only be cleared by software and cannot be set by software.	0
1	ALRF	RW0	Alarm flag. When the counter value reaches the alarm register (ALRMx) value, this bit is set by hardware. If the alarm interrupt enable bit (ALRIE) is set, an alarm interrupt is also generated. This bit can only be cleared by software and cannot be set by software.	0
0	SECF	RW0	Second event flag. After the clock passes through the prescaler, each falling edge generated increments the counter by 1 and simultaneously generates a second event; this bit is set. If the second interrupt is enabled	0

Bits	Name	Access	Description	Reset
			(SECIE set), a second interrupt is also generated. This bit can only be cleared by software and cannot be set by software.	

6.3.3 Prescaler Reload Value Register High (RTC_PSCRH)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												PRL[19:16]			

Bits	Name	Access	Description	Reset
[15:4]	Reserved	RO	Reserved.	0
[3:0]	PRL[19:16]	WO	Reload value high.	x

6.3.4 Prescaler Reload Value Register Low (RTC_PSCRL)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRL[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	PRL[15:0]	WO	Reload value low. The actual prescaler factor is (PRL[15:0]+1). For example, if the RTC input frequency is 32768 Hz, setting this value to 0x7fff produces a signal with a 1-second period.	x

6.3.5 Divider Register High (RTC_DIVH)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												DIV[19:16]			

Bits	Name	Access	Description	Reset
[15:4]	Reserved	RO	Reserved.	0
[3:0]	DIV[19:16]	RO	Divider register high.	x

6.3.6 Divider Register Low (RTC_DIVL)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	DIV[15:0]	RO	Divider register low. DIV is actually a down-counter; each RTC_CLK clock decrements the DIV counter by 1. On underflow, a TR_CLK is output and the value is reloaded from PSCR. DIV is read-only; reading it returns the remaining value of the current divider counter.	x

6.3.7 RTC Counter Register High (RTC_CNTH)

Offset address: 0x18

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[31:16]															
Bits		Name		Access		Description				Reset					
[15:0]		CNT[31:16]		RW		Counter high.				x					

6.3.8 RTC Counter Register Low (RTC_CNTL)

Offset address: 0x1C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	CNT[15:0]	RW	Counter low. The core device of the RTC timer, clocked by TR_CLK (period generally set to 1 second). The current time is calculated by reading CNT[31:0]. Writing this value requires entering configuration mode.	x											

6.3.9 Alarm Register High (RTC_ALRMH)

Offset address: 0x20

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ALR[31:16]															
Bits		Name	Access	Description									Reset		
[15:0]		ALR[31:16]	WO	Alarm register high.									x		

6.3.10 Alarm Register Low (RTC_ALRML)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ALR[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	ALR[15:0]	WO	Alarm register low. When the alarm register ALRM[31:0] value matches the counter CNT[31:0] value, an alarm event is generated. Changing this value requires entering configuration mode.	x											

Chapter 7: Independent Watchdog (IWDG)

The system includes an Independent Watchdog (IWDG) to detect software faults caused by logic errors and external environmental interference. The IWDG clock source is derived from the LSI and can operate independently of the main program, making it suitable for applications with low precision requirements.

7.1 Main Features

- 12-bit down-counter
- Clock source derived from LSI divider; can run in low-power modes
- Reset condition: counter value decrements to 0

7.2 Functional Description

7.2.1 Principle and Usage

The Independent Watchdog is clocked by the LSI clock, and its function continues to operate normally in Stop and Standby modes. When the watchdog counter decrements to 0, a system reset is generated, so the timeout period is (reload value + 1) clock cycles.

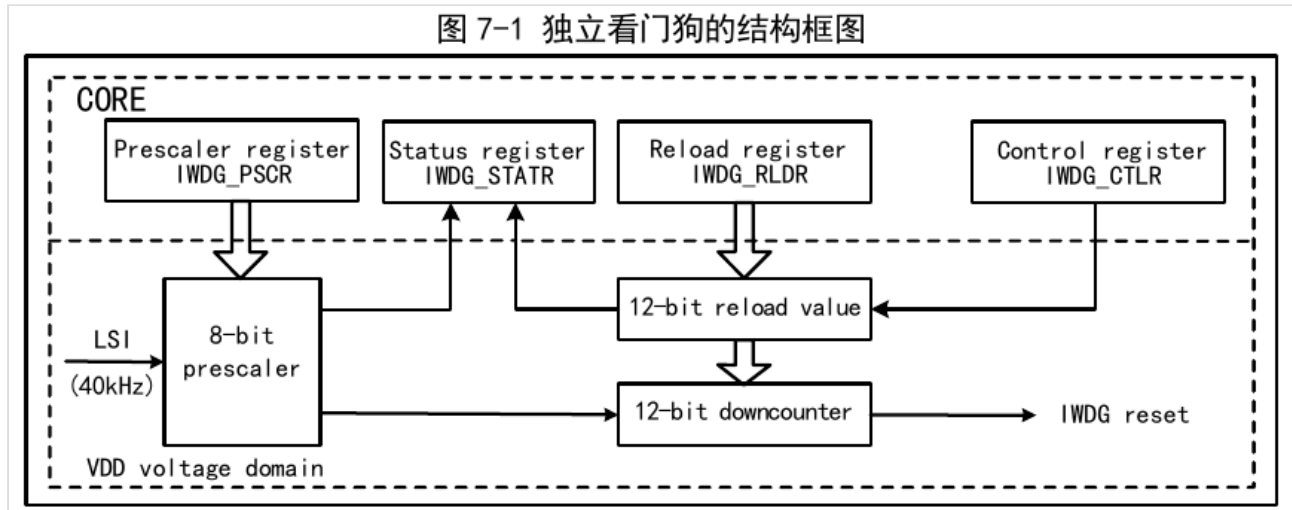


Figure 7-1: Independent Watchdog block diagram

• Starting the Independent Watchdog

After system reset, the watchdog is disabled. Writing 0xCCCC to the IWDG_CTLR register starts the watchdog; it cannot be disabled afterwards unless a reset occurs.

If the hardware Independent Watchdog enable bit (IWDGSW) is enabled in the user option bytes, the IWDG will be permanently enabled after microcontroller reset.

• Watchdog Configuration

The watchdog internally contains a 12-bit down-counter. When the counter value reaches 0, a system reset occurs. To enable the IWDG function, the following operations must be performed:

1. **Counting time base:** The IWDG clock is sourced from the LSI. Set the LSI divider value via the IWDG_PSCR register to configure the IWDG counting time base. The operation method is to first write 0x5555 to the IWDG_CTLR register, then modify the divider value in the IWDG_PSCR register. The PVU bit in the IWDG_STATR register indicates the divider value update status; the divider value can only be modified and read when the update is complete.
2. **Reload value:** Used to update the current counter value of the Independent Watchdog, from which the counter decrements. The operation method is to first write 0x5555 to the IWDG_CTLR register, then modify the IWDG_RLDR register to set the target reload value. The RUV bit in the IWDG_STATR register indicates the reload value update status; the IWDG_RLDR register can only be modified and read when the update is complete.
3. **Watchdog enable:** Write 0xCCCC to the IWDG_CTLR register to enable the watchdog function.
4. **Kick (feed) the watchdog:** That is, refresh the current counter value before the watchdog counter decrements to 0 to prevent a system reset. Write 0xAAAA to the IWDG_CTLR register to have the hardware load the IWDG_RLDR register value into the watchdog counter. This action must be executed periodically after the watchdog function is enabled, otherwise a watchdog reset will occur.

7.2.2 Debug Mode

When the system enters debug mode, the IWDG counter can be configured to continue running or stop via the debug module registers.

7.3 Register Description

Table 7-1: IWDG related register list

Name	Address	Description	Reset Value
R16_IWDG_CTLR	0x40003000	Control register	0x0000
R16_IWDG_PSCR	0x40003004	Prescaler register	0x0000
R16_IWDG_RLDR	0x40003008	Reload register	0x0FFF
R16_IWDG_STATR	0x4000300C	Status register	0x0000

7.3.1 IWDG Control Register (IWDG_CTLR)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEY[15:0]															
Bits	Name	Access	Description												Reset
[15:0]	KEY[15:0]	WO	Operation key value lock. 0xAAAA: Kick (feed) the watchdog. Load the IWDG_RLDR register value into the Independent Watchdog counter; 0x5555: Allow modification of R16_IWDG_PSCR and R16_IWDG_RLDR registers;												0

Bits	Name	Access	Description	Reset
			0xCCCC: Start the watchdog. If the hardware watchdog is enabled (configured via user option bytes), this restriction does not apply.	

7.3.2 Prescaler Register (IWDG_PSCR)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													PR[2:0]		

Bits	Name	Access	Description	Reset
[15:3]	Reserved	RO	Reserved.	0
[2:0]	PR[2:0]	RW	IWDG clock prescaler coefficient. Write 0x5555 to KEY before modifying this field. 000: divide by 4; 001: divide by 8; 010: divide by 16; 011: divide by 32; 100: divide by 64; 101: divide by 128; 110: divide by 256; 111: reserved. IWDG counting time base = LSI / prescaler coefficient. Note: Before reading this field, ensure that the PVU bit in the IWDG_STATR register is 0, otherwise the read value is invalid.	000b

7.3.3 Reload Register (IWDG_RLDR)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						RL[11:0]									

Bits	Name	Access	Description	Reset
[15:12]	Reserved	RO	Reserved.	0
[11:0]	RL[11:0]	RW	Counter reload value. Write 0x5555 to KEY before modifying this field. After writing 0xAAAA to KEY, the value of this field is loaded into the counter by hardware, and the counter then starts decrementing from this value. Note: Before reading or writing this field, ensure that the RUV bit in the IWDG_STATR register is 0, otherwise read/write access to this field is invalid.	0xFFFF

Note: This register is reset in Standby mode.

7.3.4 Status Register (IWDG_STATR)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													RVU		PVU

Bits	Name	Access	Description	Reset
[15:2]	Reserved	RO	Reserved.	0
1	RVU	RO	Reload value update flag. Set or cleared by hardware. 1: Reload value update in progress; 0: Reload value update complete (up to 5 LSI cycles). Note: The reload register IWDG_RLDR can only be read/written after the RVU bit is cleared to 0.	0
0	PVU	RO	Clock prescaler coefficient update flag. Set or cleared by hardware. 1: Clock prescaler value update in progress; 0: Clock prescaler value update complete (up to 5 LSI cycles). Note: The prescaler register IWDG_PSCR can only be read/written after the PVU bit is cleared to 0.	0

Note: After updating the prescaler or reload value, there is no need to wait for RVU or PVU to reset; code execution can continue. (Even in low-power mode, this write operation will still be completed.)

Chapter 8: Window Watchdog (WWDG)

The Window Watchdog is generally used to monitor software faults during system operation, such as external interference, unpredictable logic errors, and other conditions. It requires refreshing the counter (kicking the dog) within a specific window time (with upper and lower limits); otherwise, a watchdog circuit reset of the system will be generated if the refresh occurs earlier or later than this window time.

8.1 Main Features

- Programmable 7-bit down-counter
- Dual-condition reset: the current counter value is less than 0x40, or the counter value is reloaded outside the window time
- Early wakeup notification function (EWI), used to perform a timely kick action to prevent system reset

8.2 Functional Description

8.2.1 Principle and Usage

The Window Watchdog operates based on a 7-bit down-counter, which is attached to the PB1 bus. The counting time base WWDG_CLK is derived from the (PCLK1/4096) clock divider, and the divider coefficient is set in the WDGTB[1:0] field of the configuration register WWDG_CFGR. The down-counter is in a free-running state; whether the watchdog function is enabled or not, the counter continuously decrements in a loop. As shown in Figure 8-1, the internal structure block diagram of the Window Watchdog.

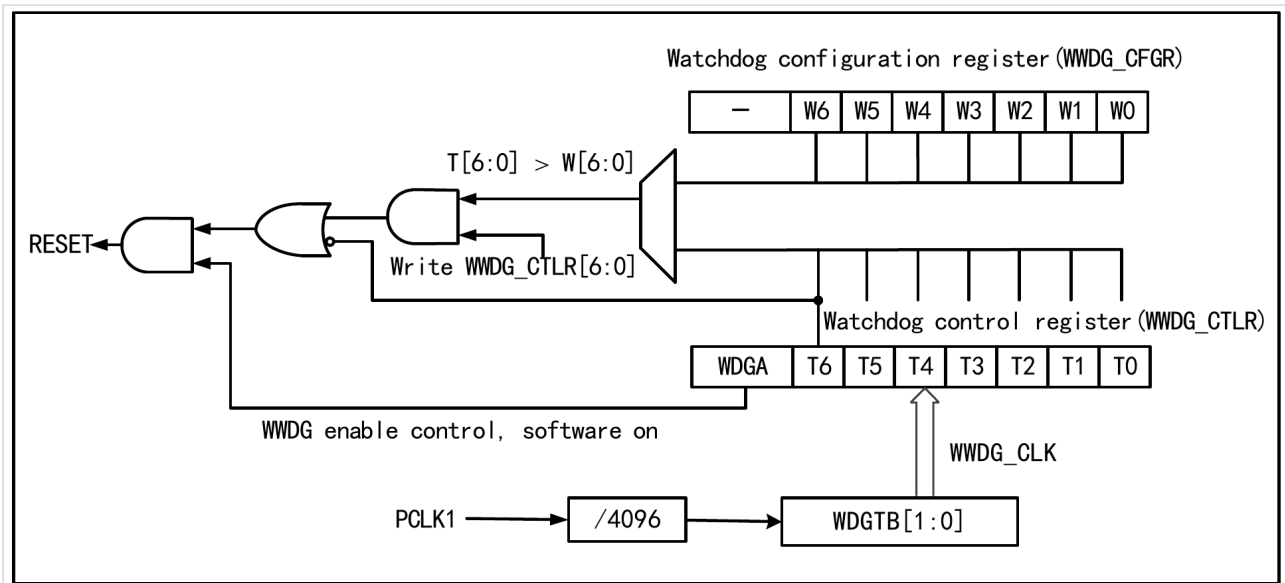


Figure 8-1: Window Watchdog block diagram

• Starting the Window Watchdog

After system reset, the watchdog is in the disabled state. Setting the WDGA bit of the WWDG_CTLR register can enable the watchdog; afterwards it cannot be disabled unless a reset occurs.

Note: The WWDG clock source can be disabled by setting the RCC_PB1PCENR register, pausing the WWDG_CLK counting and indirectly stopping the watchdog function; alternatively, the WWDG module can be reset by setting the RCC_PB1PRSTR register, which has the same effect as a reset.

• Watchdog Configuration

The watchdog internally contains a 7-bit counter that continuously decrements in a loop, supporting read/write access. To use the watchdog reset function, the following operations must be performed:

1. **Counting time base:** Via the WDG TB[1:0] field of the WWDG_CFGR register. Note that the WWDG module clock of the RCC unit must be enabled.
2. **Window counter:** Set the W[6:0] field of the WWDG_CFGR register. This counter is used by hardware for comparison with the current counter. The value is configured by user software and does not change. It serves as the upper limit value of the window time.
3. **Watchdog enable:** Software sets the WDGA bit of the WWDG_CTLR register to 1 to enable the watchdog function, which can then generate a system reset.
4. **Kick (feed) the watchdog:** That is, refresh the current counter value by configuring the T[6:0] field of the WWDG_CTLR register. This action must be executed within the periodic window time after the watchdog function is enabled, otherwise a watchdog reset will occur.

• Kick window time

As shown in Figure 8-2, the gray area is the monitoring window area of the Window Watchdog. Its upper limit time t_2 corresponds to the time point when the current counter value reaches the window value W[6:0]; its lower limit time t_3 corresponds to the time point when the current counter value reaches 0x3F. Within this area, where $t_2 < t < t_3$, a kick operation (writing T[6:0]) can be performed to refresh the current counter value.

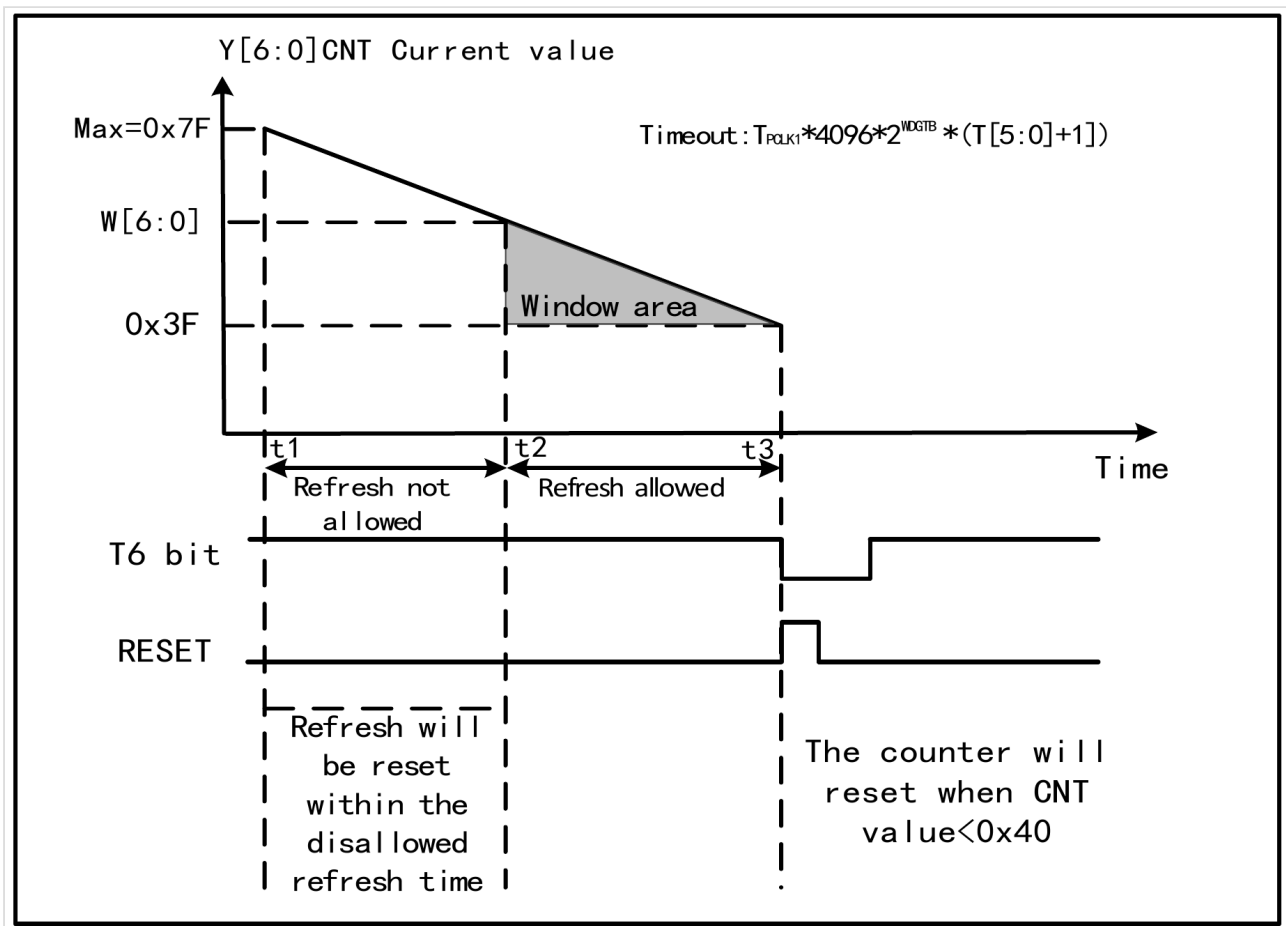


Figure 8-2: Counting mode of the Window Watchdog

- **Watchdog reset:**

1. When a kick operation is not performed in time, causing the value of the T[6:0] counter to change from 0x40 to 0x3F, a "window watchdog reset" will occur, generating a system reset. That is, when T6-bit is detected by hardware as 0, a system reset will occur.

Note: The application can write T6-bit to 0 via software to generate a system reset, equivalent to a software reset function.

1. When a counter refresh action is performed during the time when kicking is not allowed, i.e., writing to the T[6:0] field during the time $t1 \leq t \leq t2$, a "window watchdog reset" will occur, generating a system reset.

- **Early wakeup**

To prevent system reset caused by not refreshing the counter in time, the watchdog module provides an Early Wakeup Interrupt (EWI) notification. When the counter decrements to 0x40, an early wakeup signal is generated and the WEIF flag is set to 1. If the EWI bit is set, a window watchdog interrupt is triggered at the same time. At this point, there is 1 counter clock cycle remaining before hardware reset (decrementing to 0x3F), during which the application can immediately perform a kick operation.

8.2.2 Debug Mode

When the system enters debug mode, the WWDG counter can be configured to continue running or stop via the debug module registers.

8.3 Register Description

Table 8-1: WWDG related register list

Name	Address	Description	Reset Value
R16_WWDG_CTLR	0x40002C00	Control register	0x007F
R16_WWDG_CFGR	0x40002C04	Configuration register	0x007F
R16_WWDG_STATR	0x40002C08	Status register	0x0000

8.3.1 WWDG Control Register (WWDG_CTLR)

Offset address: 0x00

15	14	13	12	11	10	9	8	7		6	5	4	3	2	1	0
Reserved								WDGA		T[6:0]						

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
7	WDGA	RW1	Window watchdog reset enable bit. 1: Enable watchdog function (can generate reset signal); 0: Disable watchdog function. Software writes 1 to enable, but it can only be cleared by hardware after reset.	0
[6:0]	T[6:0]	RW	7-bit down-counter, which decrements by 1 every 4096×2^{WDGTB} PCLK1 cycles. When the counter decrements from 0x40 to 0x3F, i.e., when T6 toggles to 0, a watchdog reset is generated.	0x7F

8.3.2 WWDG Configuration Register (WWDG_CFGR)

Offset address: 0x04

15	14	13	12	11	10	9	8	7		6	5	4	3	2	1	0
Reserved						EWI		WDGTB[1:0]		W[6:0]						

Bits	Name	Access	Description	Reset
[15:10]	Reserved	RO	Reserved.	0
9	EWI	RW1	Early wakeup interrupt enable bit. If this bit is set to 1, an interrupt is generated when the counter value reaches 0x40. This bit can only be cleared by hardware after reset.	0
[8:7]	WDGTB[1:0]	RW	Window watchdog clock divider selection: 00: divide by 1, counting time base = PCLK1/4096; 01: divide by 2, counting time base = PCLK1/4096/2; 10: divide by 4, counting time base = PCLK1/4096/4; 11: divide by 8, counting time base = PCLK1/4096/8.	0
[6:0]	W[6:0]	RW	7-bit window value of the window watchdog. Used for comparison with the counter value. The kick operation can only be performed when the counter value is less than the window value and greater than 0x3F.	0x7F

8.3.3 WWDG Status Register (WWDG_STATR)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														EWIF	
Bits	Name	Access	Description	Reset											
[15:1]	Reserved	WO	Reserved.	0											
0	EWIF	RW0	Early wakeup interrupt flag bit. When the counter reaches 0x40, this bit is set by hardware and must be cleared by software; setting it by the user has no effect. Even if EWI is not set, this bit will still be set normally when the event occurs.	0											

Chapter 9: Central Processing Unit (CPU)

The CH32V407 uses the Qingke V3V core. RISC-V3V supports the IMACV-X instruction set, with an internal 2-3 stage pipeline and in-order execution architecture. The core supports the RISC-V V1.0 vector instruction set and the zvbb subset, with a maximum EEW width of 64 bits.

Internally, it is managed in a modular fashion, including a Fast Programmable Interrupt Controller (PFIC), memory protection, branch prediction mode, extended instruction support, and other units. Multiple sets of internal buses connect to external module units, enabling interaction between external functional modules and the core.

9.1 Main Features

- In-order single-issue architecture
- 2-3 stage pipeline
- Provides one non-maskable interrupt (NMI)
- 2-level interrupt nesting
- Fast Programmable Interrupt Controller (PFIC)
- Custom extended instructions
- Hardware stacking
- Vector operations / parallel processing

9.2 Functional Description

9.2.1 Instruction Set

The Qingke V3V processor supports vector integer and fixed-point arithmetic instructions from the RISC-V V1.0 instruction set, but does not support vector floating-point instructions. Vector operations use 32 groups of 64-bit-wide vector registers, with a maximum vector instruction EEW of 64 bits. The configuration conforms to the vector embedded extension Zve64x extension set, supporting vector instruction encoding checks and instruction execution exception checks.

9.2.1.1 RISC-V V1.0

Vector operations use 32 groups of 64-bit-wide vector registers as operand registers, fully supporting the vector embedded extension Zve64x extension set.

Note:

1. The zve64x extension instruction set consists of instructions defined in the RVV1.0 manual. For specific instruction functions, refer to "RISC-V "V" Vector Extension 1.0".
2. All memory access instructions support eew=64 and iew=64 operations, but the scalar address width XLEN=32 bits, which does not support 64-bit addressing, conforming to the Zve64x specification.

9.2.1.2 RISC-V zvbb

zvbb is a basic operation subset of the vector instruction set extension. It fully supports the Zve64x extension set.

9.2.1.3 Custom Delay Instruction

The custom delay instruction is a precise delay instruction at the cost of a single instruction overhead. It can delay a specified matching instruction precisely to clock cycle precision, avoiding the uncertainty of instruction jump execution in timer and delay operations. The instruction can automatically match instruction encoding, configure delay cycles and clock division coefficients, and is flexible to use, offering significant advantages in precise control, flow management, and low-overhead delay applications.

Custom Delay Instruction Encoding

Bits	Name	Description
[31:20]	imm	Delay immediate value. Depending on configuration, it represents the number of clock cycles or divided cycles to delay.
[19:15]	rs1	Matching rs1 register encoding.
[14:12]	func3	Fixed value 001b.
[11:9]	match	Matching instruction type. Instructions matching this type cannot execute while the delay is not complete. 000: Reserved; 001: Match load instruction; 010: Match store instruction; 011: Match load/store instruction; 100: Match delay instruction; 101: Reserved; 110: Match all instructions (pipeline stall); 111: Match instruction with specified func3 and opcode (match value configured by CSR register).
8	div	Clock selection for counting: 1: Use divided clock cycle count (division factor configured by CSR register); 0: Use main clock cycle count.
7	sel	Delay count selection: 1: rs1 + rs2; 0: rs1 + imm.
[6:0]	func7	Fixed value 0001011b.

Note: Read/write instructions using the sp register as base address are not subject to the match load/store instruction restriction (match=001/010/011).

9.2.1.4 Custom mcpy Instruction

The mcpy instruction is used to replace the memcpy function, implementing continuous memory copy functionality. The instruction registers hold the start address, target address, and end address information. All addresses have no alignment requirements.

Custom mcpy Instruction Encoding

Bits	Name	Description
[31:27]	rs3	End address register encoding.
[26:25]	Reserved	Fixed value 00b.
[24:20]	rs2	Start address register encoding.
[19:15]	rs1	Target address register encoding.
[14:12]	func3	Fixed value 111b.
[11:7]	func5	Fixed value 00000b.
[6:0]	func7	Fixed value 0001111b.

9.2.2 Vector Privilege Registers (CSR)

Table 9-1: CSR Related Register List

Name	Access Address	Description	Reset Value
VXSAT	0x009	Fixed-point saturation status register	0x00000000
VXRM	0x00A	Fixed-point rounding register	0x00000000
VCSR	0x00F	Vector control and status register	0x00000000
VL	0xC20	Element body length register	0x00000000
VTTYPE	0xC21	Vector data type register	0x00000000
VLENB	0xC22	Vector register byte length register	0x00000000
VCONTROL	0x805	Custom vector control register	0x00000X0C
VPPADDR	0x806	Custom vector register stacking base address register	0x00000000
VCAUSE	0x808	Custom vector exception source register	0x00000000
VTVAL	0x809	Custom vector exception trap information register	0x00000000

9.2.2.1 Fixed-Point Saturation Status Register (VXSAT)

CSR address: 0x009

Bits	Name	Access	Description	Reset
[31:1]	Reserved	URW	Reserved.	0
0	VXSAT	URW	Fixed-point saturation status register. Set to 1 when the fixed-point operation result must output a saturated value.	0

9.2.2.2 Fixed-Point Rounding Register (VXRM)

CSR address: 0x00A

Bits	Name	Access	Description	Reset
[31:2]	Reserved	URW	Reserved.	0
[1:0]	VXRM	URW	Fixed-point operation rounding register, used to control the rounding mode of fixed-point operations: 00: Round to nearest (round half up); 01: Round to nearest (round half to even);	0

Bits	Name	Access	Description	Reset
			10: Round down; 11: Round to odd.	

9.2.2.3 Vector Control and Status Register (VCSR)

CSR address: 0x00F

Bits	Name	Access	Description	Reset
[31:3]	Reserved	URW	Reserved.	0
[2:1]	VXRM	URW	Fixed-point operation rounding register, used to control the rounding mode of fixed-point operations: 00: Round to nearest (round half up); 01: Round to nearest (round half to even); 10: Round down; 11: Round to odd.	0
0	VXSAT	URW	Fixed-point saturation status register. Set to 1 when the fixed-point operation result must output a saturated value.	0

9.2.2.4 Element Body Length Register (VL)

CSR address: 0xC20

Bits	Name	Access	Description	Reset
[31:7]	Reserved	URW	Reserved.	0
[6:0]	VL	URO	Vector body length register, used to store the element length used in operations. The register value can only be modified by vset instructions. For the assignment rules, see "RISC-V "V" Vector Extension 1.0" manual.	0

9.2.2.5 Vector Data Type Register (VTYPE)

CSR address: 0xC21

Bits	Name	Access	Description	Reset
[31:6]	Reserved	URW	Reserved.	0
[5:3]	VSEW	URO	Vector width register, used to select the width of vectors and elements. Only VSET instruction can configure: 000: 8bit; 001: 16bit; 010: 32bit; 011: 64bit; 1xx: Reserved. Note: Configuring eww to 64bit requires enabling the VCONTROL related configuration.	0
[2:0]	VLMUL	URO	Vector group allocation register, used to select grouping information in vector operations. Only VSET instruction can configure: 000: 1; 001: 2; 010: 4; 011: 8;	0

Bits	Name	Access	Description	Reset
			100: Reserved; 101: 1/8; 110: 1/4; 111: 1/2.	

9.2.2.6 Vector Register Byte Length Register (VLENB)

CSR address: 0xC22

Bits	Name	Access	Description	Reset
[31:0]	VLENB	URO	Vector register group byte length register. This core has a vector register length of 64bit, byte length of 8, fixed value.	0x8

9.2.2.7 Custom Vector Control Register (VCONTROL)

CSR address: 0x805

Bits	Name	Access	Description	Reset
31	VEC_ILL	URW1C	Vector instruction exception flag. Set when a vector instruction generates an exception. When the NMI_EN register is enabled, vector exceptions will enter NMI exception.	0
[30:24]	Reserved	URO	Reserved.	0
[23:22]	NEST_INT_PTR	URO	Interrupt nesting depth: 00: No interrupt; 01: Level 1 interrupt nesting; 10: Level 2 interrupt nesting; 11: Level 3 interrupt nesting.	0
[21:20]	VEC_INT_PTR	URW	Vector pipeline interrupt nesting depth: 00: No interrupt; 01: Level 1 interrupt nesting; 10: Level 2 interrupt nesting; 11: Level 3 interrupt nesting. When vector instructions are not used in the interrupt service function, the vector pipeline interrupt nesting depth retains its previous value.	0
[19:16]	VEC_REG_STATE	URW	Vector register state register. 4 bits correspond to a maximum of four levels of vector register state depth. A value of 1 indicates that the corresponding interrupt depth vector register state is dirty, meaning vector instructions have been executed causing vector register value changes.	0
[15:10]	Reserved	URO	Reserved.	0
9	VEC_ILL_INFO	URW	Vector exception information save enable. When set to 1, VCAUSE and VTVAL registers will save vector exception information.	1
8	Reserved	URO	Reserved, cannot write 0.	x
[7:4]	Reserved	URO	Reserved.	0
3	LSU_WAX_ON	URW	Force write ordering enable. Forces write operations to execute in instruction order.	1

Bits	Name	Access	Description	Reset
2	LSU_RAW_ON	URW	Force read ordering enable. Forces read operations to execute in instruction order.	1
1	VEC_HW_PP_ABI	URW	Vector register hardware stack protection ABI configuration register: 1: Variant ABI call, hardware stack protects V0, V8-V23 registers; 0: Standard ABI call, hardware stack protects V0-V31 registers. Note: For ABI calls, refer to "RISC-V ABIS SPECIFICATION V1.1".	0
0	VEC_HW_PP_EN	URW	Vector register hardware stack protection enable: 1: Enable vector stack protection function; 0: Disable vector stack protection function.	0

9.2.2.8 Custom Vector Register Stacking Base Address Register (VPPADDR)

CSR address: 0x806

Bits	Name	Access	Description	Reset
[31:0]	VPPADDR	URW	Vector register stack protection memory address register, used to configure the vector register stack protection address. The lower 10 bits are fixed to 0, 1KB aligned in memory.	0x20000000

9.2.2.9 Custom Vector Exception Source Register (VCAUSE)

CSR address: 0x808

Bits	Name	Access	Description	Reset
[31:0]	VCAUSE	URO	Vector instruction exception cause register. When VEC_ILL_INFO is enabled, it saves the exception cause of the vector exception. Note: Debug mode will cause exception cause information to be lost and cannot be used in debug mode.	0

9.2.2.10 Custom Vector Exception Trap Information Register (VTVAL)

CSR address: 0x809

Bits	Name	Access	Description	Reset
[31:0]	VTVAL	URO	Vector instruction trap information register. When VEC_ILL_INFO is enabled, it saves the PC value of the vector exception. Note: Debug mode will cause exception cause information to be lost and cannot be used in debug mode.	0

9.2.3 General Privilege Registers

In the RISC-V architecture, some Control and Status Registers (CSR) are defined for configuring, identifying, or recording running states. CSR registers belong to registers internal to the core and use a dedicated 12-bit address space.

Note: Registers marked with "MRW, MRO, MRW1" attributes require system access in machine mode.

Table 9-2: General Privilege Register List

Name	CSR Address	Description	Reset Value
MVENDORID	0xF11	Chip manufacturer ID register	0x00000000
MARCHID	0xF12	Chip architecture ID register	0xDC68D876
MIMPID	0xF13	Core ID register	0xDC688003
MSTATUS	0x300	Machine status register	0x00000000
MISA	0x301	Machine instruction set register	0x40B01107
MTVEC	0x305	Machine mode exception base address register	0x00000000
MSCRATCH	0x340	Machine mode temporary register	X
MEPC	0x341	Machine exception program counter register	X
MCAUSE	0x342	Machine exception register	0xX00000XX
MTVAL	0x343	Machine trap value register	X
TINFO	0x7A4	Trigger information register	0x00000004
DPC	0x7B1	Debug PC address register	X
DSCRATCH0	0x7B2	Debug mode temporary register 0	X
DSCRATCH1	0x7B3	Debug mode temporary register 1	X
DBGMCU_0	0x7C0	Debug register 0	0x00000000
UACCES_MSTATUS	0x800	User access machine status register	0x00000000
HW_POPDM_CTLR	0x804	Hardware stacking control register	0x00000004
TSELECT	0x7A0	Trigger channel selection register	0x0000000X
TDATA1	0x7A1	Trace trigger data register	0x23E00048
TDATA2	0x7A2	Save trigger data register	X
MCOUNT_INHIBIT	0x320	Counter inhibit register	0x00000000
MCYCLE	0xB00	Machine cycle counter register	0x00000000
MINSTRET	0xB02	Machine instruction counter register	0x00000000
UCYCLE	0xC00	User mode cycle counter register	0x00000000
UINSTRET	0xC02	User mode instruction counter register	0x00000000

9.2.3.1 Chip Manufacturer ID Register (MVENDORID)

CSR address: 0xF11

Bits	Name	Access	Description	Reset
[31:0]	mvendorid[31:0]	MRO	Chip manufacturer ID, fixed value of 0.	0

9.2.3.2 Chip Architecture ID Register (MARCHID)

CSR address: 0xF12

Bits	Name	Access	Description	Reset
[31:0]	marchid[31:0]	MRO	Chip architecture ID, fixed value, cannot be modified.	0xDC68D876

9.2.3.3 Machine Implementation ID Register (MIMPID)

CSR address: 0xF13

Bits	Name	Access	Description	Reset
[31:0]	mimpid[31:0]	MRO	Machine implementation ID, fixed value, cannot be modified.	0xDC688003

9.2.3.4 Machine Status Register (MSTATUS)

CSR address: 0x300

Bits	Name	Access	Description	Reset
[31:13]	Reserved	MRO	Reserved.	0
[12:11]	MPP[1:0]	MRW	Saves/restores the machine state before entering an exception. Updated to the pre-exception machine state when entering an exception: 00: Machine state set to U mode on exception exit; 01: Reserved; 10: Reserved; 11: Machine state set to M mode on exception exit.	0
[10:9]	VS[1:0]	MRW	Vector instruction set state. When the value is not 0, vector instructions (V) are allowed to execute: 00: OFF; 01: Initial; 10: Clean; 11: Dirty.	0
8	Reserved	MRW	Reserved.	0
7	MPIE	MRW	Saves the interrupt enable state when entering an interrupt: 1: Enable global interrupt after exiting interrupt; 0: Disable global interrupt after exiting interrupt.	0
[6:4]	Reserved	MRO	Reserved.	0
3	MIE	MRW	Global interrupt enable. Updated to MPIE value on interrupt exit: 1: Global interrupt enabled, allows the processor to respond to interrupt requests; 0: Global interrupt disabled, processor does not respond to interrupt requests.	0
[2:0]	Reserved	MRO	Reserved.	0

9.2.3.5 Machine Instruction Set Register (MISA)

CSR address: 0x301

Bits	Name	Access	Description	Reset
[31:30]	XLEN[1:0]	MRO	Machine basic integer width and basic instruction width information. 32-bit wide core, fixed value of 1.	01b
[29:24]	Reserved	MRO	Reserved.	0
23	X	MRO	This core supports execution of custom instruction sets.	1
[22:21]	Reserved	MRO	Reserved.	01b

Bits	Name	Access	Description	Reset
20	U	MRO	Supports user mode operation.	1
[19:13]	Reserved	MRO	Reserved.	0
12	M	MRO	This core supports execution of the multiply instruction set.	0
[11:9]	Reserved	MRO	Reserved.	0
8	I	MRO	This core supports execution of the basic integer instruction set.	1
[7:6]	Reserved	MRO	Reserved.	0
5	F	MRO	This core supports execution of the single-precision floating-point instruction set.	0
[4:3]	Reserved	MRO	Reserved.	0
2	C	MRO	This core supports execution of the compressed instruction set.	1
1	B	MRO	This core supports execution of the bit manipulation instruction set.	1
0	A	MRO	This core supports execution of the atomic instruction set.	1

9.2.3.6 Machine Mode Exception Base Address Register (MTVEC)

CSR address: 0x305

Bits	Name	Access	Description	Reset
[31:10]	mtvec_base[21:0]	MRW	Base address high bits for processor exception jump. 1KB aligned. The actual 32-bit base address $BASE = \{mtvec_base, 10'h0\}$.	0
[9:2]	Reserved	MRO	Reserved.	0
1	mode1	RW	Vector table mode 1: 1: Interrupt vector table stores jump addresses; 0: Interrupt vector table stores jump instructions.	0
0	mode0	RW	Vector table mode 0: 1: All interrupts/exceptions have independent entries, jump address is $BASE+4*IRQ/EXC\ ID$; 0: All interrupts/exceptions share a unified entry, all jump to the exception jump base address.	0

9.2.3.7 Machine Mode Temporary Register (MSCRATCH)

CSR address: 0x340

Bits	Name	Access	Description	Reset
[31:0]	mscratch[31:0]	MRW	Machine mode temporary register.	x

9.2.3.8 Machine Exception Program Counter Register (MEPC)

CSR address: 0x341

Bits	Name	Access	Description	Reset
[31:0]	mepc[31:0]	MRW	Saves the PC address to be restored when exiting interrupt/exception. mepc[0] is fixed to 0.	x

9.2.3.9 Machine Exception Register (MCAUSE)

CSR address: 0x342

Bits	Name	Access	Description	Reset
31	Interrupt	RW	Interrupt/exception flag bit: 1: Exception/interrupt generated externally; 0: Exception/interrupt generated by the core.	x
[30:8]	Reserved	RO	Reserved.	0
[7:0]	Exception Code[7:0]	RW	Saves the source information of the exception or interrupt.	x

Interrupt/Exception Source Encoding

Interrupt/Exception	Exception Code	Interrupt/Exception Source
1	2	Reserved NMI
1	3	Machine mode software interrupt
1	4	User mode timer interrupt
1	5	Monitor mode timer interrupt
1	6	Reserved
1	7	Machine mode timer interrupt
1	8	User mode external interrupt
1	11	Machine mode external interrupt
1	12	System timer interrupt
1	≥ 16	Reserved platform interrupt ID
0	0	Instruction address misaligned
0	1	Instruction access fault
0	2	Illegal instruction
0	3	Breakpoint
0	4	Load address misaligned
0	5	Load access fault
0	6	Store/AMO address misaligned
0	7	Store/AMO access fault
0	8	User mode ecall
0	11	Machine mode ecall
0	≥ 16	Reserved

9.2.3.10 Machine Trap Value Register (MTVAL)

CSR address: 0x343

Bits	Name	Access	Description	Reset
[31:0]	mtval[31:0]	RW	Saves additional information for traps.	x

mtval Register Information Saved in Trap

Exception Type	Mcause	Saved Information
Instruction address misaligned	0	Instruction PC
Instruction access fault	1	Instruction PC
Illegal instruction	2	Instruction machine code
Load address misaligned	4	Load data address
Load access fault	5	Load data address
Store/AMO access fault	6	Store data address
Store/AMO address misaligned	7	Store data address

9.2.3.11 Trigger Information Register (TINFO)

CSR address: 0x7A4

Bits	Name	Access	Description	Reset
[31:0]	tinfo[31:0]	MRO	Trigger information register. Corresponds to the TDATA1 register type value. The type value is 2, so bit 2 of tinfo is set to 1.	0x4

9.2.3.12 Debug PC Address Register (DPC)

CSR address: 0x7B1

Bits	Name	Access	Description	Reset
[31:0]	dpc[31:0]	RW	Saves the PC address to be restored when exiting debug mode.	x

The addresses saved in dpc for different debug request sources that cause a jump to debug mode are as follows:

DPC Saved Address by Debug Cause

Cause	Saved Address
ebreak	Address of the ebreak instruction.
Single step	Address of the next instruction that should be executed in non-debug mode.
Trigger module	tdata1.timing=0: Address of the instruction that triggered the breakpoint.
Halt request	Address of the next instruction that should be executed when entering debug mode.

9.2.3.13 Debug Mode Temporary Register 0 (DSCRATCH0)

CSR address: 0x7B2

Bits	Name	Access	Description	Reset
[31:0]	Dscratch0[31:0]	RW	Debug mode temporary register 0.	x

9.2.3.14 Debug Mode Temporary Register 1 (DSCRATCH1)

CSR address: 0x7B3

Bits	Name	Access	Description	Reset
[31:0]	Dscratch1[31:0]	RW	Debug mode temporary register 1.	x

9.2.3.15 Debug Register 0 (DBGMCU_0)

CSR address: 0x7C0

Bits	Name	Access	Description	Reset
[31:8]	dbg_mode_stop	MRW	Registered module stops working when entering debug mode: 1: Registered peripherals stop working when the core enters debug mode; 0: Registered peripherals continue working normally when the core enters debug mode.	0
[7:0]	Reserved	MRO	Reserved.	0

9.2.3.16 User Access Machine Status Register (UACCES_MSTATUS)

CSR address: 0x800

Bits	Name	Access	Description	Reset
31	SD	RO	Used to mark whether floating-point instructions are in dirty state.	0
[30:15]	Reserved	RO	Reserved.	0
[14:13]	FS[1:0]	RO	Floating-point instruction set state. When the value is not 0, single-precision floating-point instructions (F) are allowed to execute: 00: OFF; 01: Initial; 10: Clean; 11: Dirty.	0
[12:11]	MPP[1:0]	RO	Saves/restores the machine state before entering an exception. Updated to the pre-exception machine state when entering an exception: 00: Machine state set to U mode on exception exit; 01: Reserved; 10: Reserved; 11: Machine state set to M mode on exception exit.	0
[10:8]	Reserved	RO	Reserved.	0
7	MPIE	RW	Saves the interrupt enable state when entering an interrupt: 1: Enable global interrupt after exiting interrupt; 0: Disable global interrupt after exiting interrupt.	0
[6:4]	Reserved	RO	Reserved.	0
3	MIE	RW	Global interrupt enable. Updated to MPIE value on interrupt exit: 1: Global interrupt enabled, allows the processor to respond to interrupt requests; 0: Global interrupt disabled, processor does not respond to interrupt requests.	0
[2:0]	Reserved	RO	Reserved.	0

Note: This register is used to configure the MSTATUS register when the core is running in user mode. When the CPU_RUN_CTLR register's non_usta bit is enabled, this address becomes available. The register settings are the same as the MSTATUS (0x300) register.

9.2.3.17 Hardware Stacking Control Register (HW_POPDM_CTLR)

CSR address: 0x804

Bits	Name	Access	Description	Reset
31	lock	RW	User mode lock flag. When locked, user mode cannot modify this group of register configurations: 1: Locked, only machine mode can configure; 0: Unlocked, both machine/user mode can configure.	0
[30:6]	Reserved	RO	Reserved.	0
5	hw_pop_off	RW	One-time hardware pop disable. Automatically reset after exiting interrupt: 1: Mask hardware pop on next interrupt exit; 0: Do not mask hardware pop.	0
4	Reserved	RO	Reserved.	0
[3:2]	preempt[1:0]	RW	Preemption priority width configuration register, used to configure the preemption priority width in interrupt priority: 00: Preemption width is 0, interrupts of any priority cannot nest; 01: Preemption width is 1, i.e. bit [7] of the interrupt priority register is the preemption priority; 10: Preemption width is 2, i.e. bits [7:6] of the interrupt priority register are the preemption priority; 11: Preemption width is 3, i.e. bits [7:5] of the interrupt priority register are the preemption priority.	0x1
1	nest_en	RW	Interrupt nesting enable register: 1: Allow interrupt nesting; 0: Do not allow interrupt nesting.	0
0	hw_stk_en	RW	Hardware stack protection enable: 1: Use hardware stack protection when entering/exiting interrupts; 0: Do not use hardware stack protection when entering/exiting interrupts.	0

9.2.3.18 Trigger Channel Selection Register (TSELECT)

CSR address: 0x7A0

Bits	Name	Access	Description	Reset
[31:2]	Reserved	MRO	Reserved.	0
[1:0]	mscratch[1:0]	MRW	Trigger channel selection register.	x

Note: (1) The core implements four trigger channels. The trigger channel is selected for configuration through the TSELECT register. The lower 2 bits of the TSELECT trigger are valid, and the remaining bits are always 0.

9.2.3.19 Trace Trigger Data Register (TDATA1)

CSR address: 0x7A1

Bits	Name	Access	Description	Reset
[31:28]	type[3:0]	MRO	Trigger type definition.	0x2

Bits	Name	Access	Description	Reset
27	dmode	MRO	Both machine mode and debug mode can modify trigger-related registers.	0
[26:21]	maskmax	MRO	Maximum range of matching exponent powers allowed when match=1.	0x1F
[20:19]	Reserved	MRO	Reserved.	0
18	timing	MRO	Trigger before instruction/operation execution.	0
[17:13]	Reserved	MRO	Reserved.	0
12	action	MRW	Set the processing mode when triggered: 1: Enter debug mode on trigger, dpc saves the instruction address to be restored on debug exit; 0: Generate an exception that directs the thread to M mode on trigger.	0
[11:8]	Reserved	MRO	Reserved.	0
7	match	MRO	Match strategy configuration. Allows triggering when the comparison value equals the tdata2 value.	0
6	m	MRO	Trigger enable in M mode: 1: Enable; 0: Disable.	1
[5:4]	Reserved	MRO	Reserved.	0
3	u	MRO	Trigger enable in U mode: 1: Enable; 0: Disable.	1
2	execute	MRW	Trigger based on virtual address of executed instruction: 1: Trigger compares instruction address; 0: Trigger does not compare instruction address.	0
1	store	MRW	Trigger based on virtual address of store data: 1: Trigger compares store data address; 0: Trigger does not compare store data address.	0
0	load	MRW	Trigger based on virtual address of load data: 1: Trigger compares load data address; 0: Trigger does not compare load data address.	0

Note: (1) The processor trigger configuration conforms to the RISC-V debug standard. The trigger configuration type is 2, matching based on the lowest bit of the accessed virtual address, and executing the trigger operation before the matched instruction or read/write data.

9.2.3.20 Save Trigger Data Register (TDATA2)

CSR address: 0x7A2

Bits	Name	Access	Description	Reset
[31:0]	tdata2[31:0]	MRW	Used to save trigger data (instruction PC or data address).	x

9.2.3.21 Counter Inhibit Register (MCOUNT_INHIBIT)

CSR address: 0x320

Bits	Name	Access	Description	Reset
[31:3]	Reserved	MRO	Reserved.	0
2	IR	MRW	Instruction counter inhibit register: 1: Inhibit instruction counter, MINSTRET and UNSTRET stop counting; 0: Do not inhibit instruction counter, MINSTRET and UNSTRET count normally.	0
1	Reserved	MRO	Reserved.	0
0	CY	MRW	Cycle counter inhibit register: 1: Inhibit cycle counter, MCYCLE and UCYCLE stop counting; 0: Do not inhibit cycle counter, MCYCLE and UCYCLE count normally.	0

9.2.3.22 Machine Cycle Counter Register (MCYCLE)

CSR address: 0xB00

Bits	Name	Access	Description	Reset
[31:0]	MCYCLE	MRW	Machine cycle counter, used to count the clock cycles of machine operation. Counts when the MCOUNT_INHIBIT register CY bit is 0. Does not count when the counter is stopped in debug mode.	0

9.2.3.23 Machine Instruction Counter Register (MINSTRET)

CSR address: 0xB02

Bits	Name	Access	Description	Reset
[31:0]	MINSTRET	MRW	Instruction counter, used to count the number of instructions correctly executed by the machine. Counts when the MCOUNT_INHIBIT register IR bit is 0. Does not count when the counter is stopped in debug mode.	0

9.2.3.24 User Mode Cycle Counter Register (UCYCLE)

CSR address: 0xC00

Bits	Name	Access	Description	Reset
[31:0]	UCYCLE	RO	User mode cycle counter, used to count the clock cycles of machine operation. Counts when the MCOUNT_INHIBIT register CY bit is 0. Does not count when the counter is stopped in debug mode.	0

Note: User mode cycle counter, function is the same as the MCYCLE register, used to read machine run cycle count in user mode.

9.2.3.25 User Mode Instruction Counter Register (UINSTRET)

CSR address: 0xC02

Bits	Name	Access	Description	Reset
[31:0]	UINSTRET	RO	Instruction counter, used to count the number of instructions correctly executed by the machine. Counts when the MCOUNT_INHIBIT register IR bit is 0. Does not count when the counter is stopped in debug mode.	0

Note: User mode instruction counter, function is the same as the MINSTRET register, used to read the number of correctly executed machine instructions in user mode.

9.2.4 RISC-V3V Core Custom Registers

Table 9-3: RISC-V3V Core Custom Register List

Name	CSR Address	Description	Reset Value
CPU_RUN_CTLR	0xBC0	Processor run control register	0x00000000
INEST_CTLR	0xBC1	Interrupt nesting control register	0x00000000
MIE	0xBC8	Machine interrupt register	0x00000000
DCSR	0x7B0	Debug control and status register	0x40000000
U_NONS_DLY_0	0x8C0	Delay instruction control register	0xFFFF0XXX

9.2.4.1 Processor Run Control Register (CPU_RUN_CTLR)

CSR address: 0xBC0

Bits	Name	Access	Description	Reset
[31:8]	Reserved	MRO	Reserved.	0
7	nmi_ie	MRW	Non-maskable interrupt enable: 1: Enter NMI interrupt when an exception causes nesting overflow; 0: Do not enter NMI interrupt when an exception causes nesting overflow.	0
6	int_fence	MRW	Fence interrupt mask enable: 1: Clear interrupt request when executing fence instruction; 0: Do not clear interrupt request when executing fence instruction.	0
5	non_usta	MRW	UACCES_MSTATUS register enable: 1: Bit 3 and bit 7 of CSR address 0x800 are mapped to MSTATUS register bit MIE and MSTATUS register bit MPIE, respectively; 0: CSR address 0x800 is a read-only register, returning the value of MSTATUS.	0
[4:2]	Reserved	MRO	Reserved.	0
[1:0]	pipe_acc[1:0]	MRW	Fetch mode: 00: Prefetch disabled. Instruction prefetch function is off, avoiding invalid fetch operations. At most 1 valid instruction exists in the CPU pipeline. This mode has the lowest power consumption, with performance degradation of approximately 2-3x. 01: Prefetch enabled. Instruction prefetch function is on. The CPU will continuously access instruction memory until the number of instructions waiting for execution in the internal instruction buffer exceeds a certain amount, or the instruction buffer is full, then instruction fetching is paused. This mode has high power consumption and strong performance. (CPU prediction failure will introduce redundant fetch operations. In some cases, the execution unit will introduce an additional 0-2 cycles of bubble. Performance degradation is not significant for most programs.) Others: Reserved.	0

9.2.4.2 Interrupt Nesting Control Register (INEST_CTLR)

CSR address: 0xBC1

Bits	Name	Access	Description	Reset
31	Reserved	RO	Reserved.	0
30	nest_ovr	MRW	Interrupt/exception nesting overflow flag bit. Set to 1 when the nesting depth exceeds the maximum depth. Write 1 to clear. Note: Interrupt overflow only occurs when an instruction exception and NMI interrupt are generated during execution of a level 2 interrupt service function. In this case, the exception and NMI interrupt enter normally, but the CPU stack overflows and cannot exit from this exception and NMI interrupt.	0
[29:12]	Reserved	MRO	Reserved.	0
[11:8]	nest_sta[3:0]	MRO	Nesting status flag bits: 0000: No interrupt; 0001: Level 1 interrupt; 0011: Level 2 interrupt nesting; 0111: Level 3 interrupt nesting, overflow; Others: Reserved.	0
[7:2]	Reserved	MRO	Reserved.	0
[1:0]	nest_max[1:0]	MRW	Maximum nesting level: 00: Nesting disabled (nesting function off); 01: Level 2 nesting (nesting function on); 10: Cannot be written; 11: Cannot be written. Note: (1) Writing 10b or 11b to this field sets the register to 01b. (2) When writing 11b, reading this register returns the chip's maximum nesting level.	0

9.2.4.3 Machine Interrupt Register (MIE)

CSR address: 0xBC8

Bits	Name	Access	Description	Reset
[31:5]	Reserved	RO	Reserved.	0
[4:0]	nest_mie[4:0]	RW	Machine interrupt register. Saves the mie information for each interrupt level in multi-level interrupt nesting: status.mie=nest_mie[0]; status.mpie=nest_mie[1].	0

9.2.4.4 Debug Control and Status Register (DCSR)

CSR address: 0x7B0

Bits	Name	Access	Description	Reset
[31:28]	exdebug[3:0]	MRO	An external debugger conforming to the debugger protocol is present.	0x4
[27:16]	Reserved	MRO	Reserved.	0

Bits	Name	Access	Description	Reset
15	ebreakm	MRW	Machine mode ebreak instruction enter debug mode enable: 1: When ebreak instruction is executed in machine mode, the processor enters debug mode; 0: When ebreak instruction is executed in machine mode, the processor enters exception.	0
[14:13]	Reserved	MRO	Reserved.	0
12	ebreaku	MRW	User mode ebreak instruction enter debug mode enable: 1: When ebreak instruction is executed in user mode, the processor enters debug mode; 0: When ebreak instruction is executed in user mode, the processor enters exception.	0
[11:10]	Reserved	MRO	Reserved.	0
9	stoptime	MRW	Timer pause enable: 1: Counter stops counting in debug mode; 0: Counter continues normally in debug mode.	0
[8:6]	dcause[2:0]	MRW	Debug source register, used to mark the specific event that caused entry into debug mode: 000: Reset value; 001: ebreak instruction in program triggered debug (priority 3); 010: Trigger module caused a breakpoint exception (priority 4); 011: Debugger generated halt request (priority 1); 100: Single-step mode halt (priority 0, lowest); 101: Not implemented; 110: Not implemented; 111: Reserved.	0
[5:4]	Reserved	MRO	Reserved.	0
3	nmip	MRO	Core NMI interrupt pending flag.	0
2	step	MRW	Single-step mode: 1: Single-step mode; 0: Non-single-step mode.	0
[1:0]	prv[1:0]	MRW	Records the machine state before entering debug mode: 00: Entered debug mode from user mode, processor restores to user mode; 01, 10: Reserved; 11: Entered debug mode from machine mode, processor restores to machine mode.	0

9.2.4.5 Delay Instruction Control Register (U_NONS_DLY_0)

CSR address: 0x8C0

Bits	Name	Access	Description	Reset
[31:24]	dly_mask	URW	Instruction parameter mask for dly instruction match=111.	x
[23:16]	dly_data	URW	Matching {func, opcode} for dly instruction match=111.	x
15	CPU_dly_busy_sta	URO	Core delay state busy flag.	0
14	dly_tout	RW1Z		0

Bits	Name	Access	Description	Reset
			Timeout information. No matching instruction request within the delay range.	
13	dly_ie	URW	Delay interrupt request. When enabled, if a timeout condition occurs, a software interrupt request is generated.	0
12	dly_priv	URW	1: Both machine mode and user mode are allowed to match; 0: Delay operations executed in user mode do not match instructions in machine mode.	0
[11:10]	Reserved	URO	Reserved.	0
[9:0]	dly_freq	URW	Delay count division factor configuration. If configured to the main clock frequency, microsecond-based delay can be achieved.	x

9.3 System Counter (SysTick)

Table 9-4: STK Related Register List

Name	Access Address	Description	Reset Value
R32_STK_CTLR	0xE000F000	System counter control register	0x00000000
R32_STK_SR	0xE000F004	System counter status register	0x00000000
R32_STK_CNTL	0xE000F008	System counter low register	0xFFFFFFFF
R32_STK_CMPLR	0xE000F010	Counter compare low register	0xFFFFFFFF

Note: Applicable to general-purpose microcontrollers based on 32-bit RISC-V instruction set and architecture design.

9.3.1 System Counter Control Register (STK_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										INIT	MODE	STRE	STCLK	STIE	STE

Bits	Name	Access	Description	Reset
[31:6]	Reserved	RO	Reserved.	0
5	INIT	RW1	Counter initial value update: 1: When counting up, update to 0; when counting down, update to the compare value; 0: No effect.	0
4	MODE	RW	Count mode: 1: Count down; 0: Count up.	0
3	STRE	RW	Auto-reload count enable bit: 1: When counting up reaches the compare value, restart from 0; when	0

Bits	Name	Access	Description	Reset
			counting down reaches 0, restart from the compare value; 0: Continue counting up/down.	
2	STCLK	RW	Counter clock source selection bit: 1: HCLK as time base; 0: HCLK/8 as time base.	0
1	STIE	RW	Counter interrupt enable control bit: 1: Enable counter interrupt; 0: Disable counter interrupt.	0
0	STE	RW	System counter enable control bit: 1: Start system counter STK; 0: Disable system counter STK, counter stops counting.	0

9.3.2 System Counter Status Register (STK_SR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SWIE	Reserved														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													CNTIF		

Bits	Name	Access	Description	Reset
31	SWIE	RW	Software interrupt trigger enable (SW): 1: Trigger software interrupt; 0: Disable trigger. Note: After entering software interrupt, this bit must be cleared to 0, otherwise it will keep triggering.	0
[30:1]	Reserved	RO	Reserved.	0
0	CNTIF	RW	Count value compare flag. Write 0 to clear, write 1 has no effect: 1: Counting up reached the compare value, or counting down reached 0; 0: Compare value not reached.	0

9.3.3 System Counter Low Register (STK_CNTL)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	CNT[31:0]	RW	Low 32 bits of the current counter value.	X

9.3.4 Counter Compare Low Register (STK_CMPLR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CMP[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	CMP[31:0]	RW	Sets the low 32 bits of the compare counter value. When counting up, if the CMP register value equals the CNT counter value, or when counting down, if the CNT counter value reaches 0, an STK interrupt will be triggered.	X

9.4 Interrupts and Events (PFIC)

9.4.1 PFIC Controller

- Each interrupt request has independent trigger and mask control bits, with dedicated status bits
- Supports configurable up to 2-level interrupt nesting
- Unique fast interrupt entry/exit mechanism, hardware automatic stacking and restoration, maximum hardware stacking depth of 2 levels, no instruction overhead required
- Unique Vector Table Free (VTF) interrupt response mechanism, 4-channel programmable direct interrupt vector address

9.4.2 Vector Table for Interrupts and Exceptions

Table 9-5: Vector Table

Number	Priority	Type	Name	Description	Entry Address
0	-6	Fixed	RESET	Reset interrupt	0x00000000
1	-	-	-	-	0x00000004
2	-5	Fixed	NMI	Non-maskable interrupt	0x00000008
3	-4	Fixed	HardFault	Exception interrupt	0x0000000C
4	-	-	-	-	0x00000010
5	-3	Fixed	Ecall-M	Machine mode ecall interrupt	0x00000014
6-7	-	-	-	-	0x00000018-0x0000001C
8	-2	Fixed	Ecall-U	User mode ecall interrupt	0x00000020
9	-1	Fixed	BreakPoint	Breakpoint interrupt	0x00000024
10-11	-	-	-	-	0x00000028-0x0000002C
12	0	Programmable	SysTick0	System timer interrupt 0	0x00000030
13	-	-	-	-	0x00000034
14	1	Programmable	SW	Software interrupt	0x00000038
15	-	-	-	-	0x0000003C
16	2	Programmable	WWDG		0x00000040

Number	Priority	Type	Name	Description	Entry Address
				Window watchdog interrupt	
17	3	Programmable	PVD	Power voltage detector interrupt (EXTI)	0x00000044
18	4	Programmable	TAMPER	Tamper detection interrupt	0x00000048
19	5	Programmable	RTC	Real-time clock interrupt	0x0000004C
20	6	Programmable	FLASH	Flash global interrupt	0x00000050
21	7	Programmable	RCC	Reset and clock control interrupt	0x00000054
22	8	Programmable	EXTI0	EXTI line 0 interrupt	0x00000058
23	9	Programmable	EXTI1	EXTI line 1 interrupt	0x0000005C
24	10	Programmable	EXTI2	EXTI line 2 interrupt	0x00000060
25	11	Programmable	EXTI3	EXTI line 3 interrupt	0x00000064
26	12	Programmable	EXTI4	EXTI line 4 interrupt	0x0000006C
27	13	Programmable	DMA1_CH1	DMA channel 1 global interrupt	0x00000070
28	14	Programmable	DMA1_CH2	DMA channel 2 global interrupt	0x00000074
29	15	Programmable	DMA1_CH3	DMA channel 3 global interrupt	0x00000078
30	16	Programmable	DMA1_CH4	DMA channel 4 global interrupt	0x0000007C
31	17	Programmable	DMA1_CH5	DMA channel 5 global interrupt	0x00000080
32	18	Programmable	DMA1_CH6	DMA channel 6 global interrupt	0x00000084
33	19	Programmable	DMA1_CH7	DMA channel 7 global interrupt	0x00000088
34	20	Programmable	ADC1_2	ADC1_2 global interrupt	0x0000008C
35	21	Programmable	CAN1_TX	CAN1_TX global interrupt	0x00000090
36	22	Programmable	CAN1_RX0	CAN1_RX0 global interrupt	0x00000094
37	23	Programmable	CAN1_RX1	CAN1_RX1 global interrupt	0x00000098
38	24	Programmable	CAN1_SCE	CAN1_SCE global interrupt	0x0000009C
39	25	Programmable	EXTI9_5	EXTI line[9:5] interrupt	0x000000A0
40	26	Programmable	TIM1_BRK	TIM1 break interrupt	0x000000A4
41	27	Programmable	TIM1_UP	TIM1 update interrupt	0x000000A8
42	28	Programmable	TIM1_TRG_COM	TIM1 trigger and communication interrupt	0x000000AC
43	29	Programmable	TIM1_CC	TIM1 capture/compare interrupt	0x000000B0

Number	Priority	Type	Name	Description	Entry Address
44	30	Programmable	TIM2	TIM2 global interrupt	0x000000B4
45	31	Programmable	TIM3	TIM3 global interrupt	0x000000B8
46	32	Programmable	TIM4	TIM4 global interrupt	0x000000BC
47	33	Programmable	I2C_EV	I2C event interrupt	0x000000C0
48	34	Programmable	I2C_ER	I2C error interrupt	0x000000C4
49	35	Programmable	I3C_EV	I3C event interrupt	0x000000C8
50	36	Programmable	I3C_ER	I3C error interrupt	0x000000CC
51	37	Programmable	SPI1	SPI1 global interrupt	0x000000D0
52	38	Programmable	SPI2	SPI2 global interrupt	0x000000D4
53	39	Programmable	USART1	USART1 global interrupt	0x000000D8
54	40	Programmable	USART2	USART2 global interrupt	0x000000DC
55	41	Programmable	USART3	USART3 global interrupt	0x000000E0
56	42	Programmable	EXTI15_10	EXTI line[15:10] interrupt	0x000000E4
57	43	Programmable	RTCAlarm	RTC alarm interrupt (EXTI)	0x000000E8
58	44	Programmable	USBHS1WakeUP	USBHS1 wake-up interrupt	0x000000EC
59	45	Programmable	TIM8_BRK	TIM8 break interrupt	0x000000F0
60	46	Programmable	TIM8_UP	TIM8 update interrupt	0x000000F4
61	47	Programmable	TIM8_TRG_COM	TIM8 trigger and communication interrupt	0x000000F8
62	48	Programmable	TIM8_CC	TIM8 capture/compare interrupt	0x000000FC
63	49	Programmable	RNG	RNG global interrupt	0x00000100
64	-	-	-	-	0x00000104
65	50	Programmable	SDIO	SDIO global interrupt	0x00000108
66	51	Programmable	TIM5	TIM5 global interrupt	0x0000010C
67	52	Programmable	SPI3	SPI3 global interrupt	0x00000110
68	53	Programmable	USART4	USART4 global interrupt	0x00000114
69	54	Programmable	USART5	USART5 global interrupt	0x00000118
70	55	Programmable	TIM6	TIM6 global interrupt	0x0000011C
71	56	Programmable	TIM7	TIM7 global interrupt	0x00000120
72	57	Programmable	DMA2_CH1	DMA2 channel 1 global interrupt	0x00000124
73	58	Programmable	DMA2_CH2	DMA2 channel 2 global interrupt	0x00000128
74	59	Programmable	DMA2_CH3	DMA2 channel 3 global interrupt	0x0000012C
75	60	Programmable	DMA2_CH4	DMA2 channel 4 global interrupt	0x00000130

Number	Priority	Type	Name	Description	Entry Address
76	61	Programmable	DMA2_CH5	DMA2 channel 5 global interrupt	0x00000134
77	62	Programmable	ETH	ETH global interrupt	0x00000138
78	63	Programmable	ETH_WKUP	ETH wake-up interrupt	0x0000013C
79	64	Programmable	LTDC	LTDC global interrupt	0x00000140
80	65	Programmable	PSRAM	PSRAM global interrupt	0x00000144
81	66	Programmable	USART9	USART9 global interrupt	0x00000148
82	67	Programmable	USART10	USART10 global interrupt	0x0000014C
83	68	Programmable	USBHS1	USBHS1 global interrupt	0x00000150
84	69	Programmable	USBHS2WakeUP	USBHS2 wake-up interrupt	0x00000154
85	70	Programmable	USBHS2	USBHS2 global interrupt	0x00000158
86	71	Programmable	DVP	DVP global interrupt	0x0000015C
87	72	Programmable	USART6	USART6 global interrupt	0x00000160
88	73	Programmable	USART7	USART7 global interrupt	0x00000164
89	74	Programmable	USART8	USART8 global interrupt	0x00000168
90	75	Programmable	DMA1_CH8	DMA1 channel 8 global interrupt	0x0000016C
91	76	Programmable	DMA1_CH9	DMA1 channel 9 global interrupt	0x00000170
92	77	Programmable	DMA1_CH10	DMA1 channel 10 global interrupt	0x00000174
93	78	Programmable	DMA1_CH11	DMA1 channel 11 global interrupt	0x00000178
94	79	Programmable	DMA1_CH12	DMA1 channel 12 global interrupt	0x0000017C
95	80	Programmable	DMA1_CH13	DMA1 channel 13 global interrupt	0x00000180
96	81	Programmable	I3CWakeUP	I3C wake-up interrupt	0x00000184
97	82	Programmable	ARGB	ARGB global interrupt	0x00000188
98	83	Programmable	DMA2_CH6	DMA2 channel 6 global interrupt	0x0000018C
99	84	Programmable	DMA2_CH7	DMA2 channel 7 global interrupt	0x00000190
100	85	Programmable	DMA2_CH8	DMA2 channel 8 global interrupt	0x00000194
101	86	Programmable	DMA2_CH9	DMA2 channel 9 global interrupt	0x00000198
102	87	Programmable	DMA2_CH10	DMA2 channel 10 global interrupt	0x0000019C
103	88	Programmable	DMA2_CH11		0x000001A0

Number	Priority	Type	Name	Description	Entry Address
				DMA2 channel 11 global interrupt	

9.4.3 External Interrupt and Event Controller (EXTI)

9.4.3.1 Overview

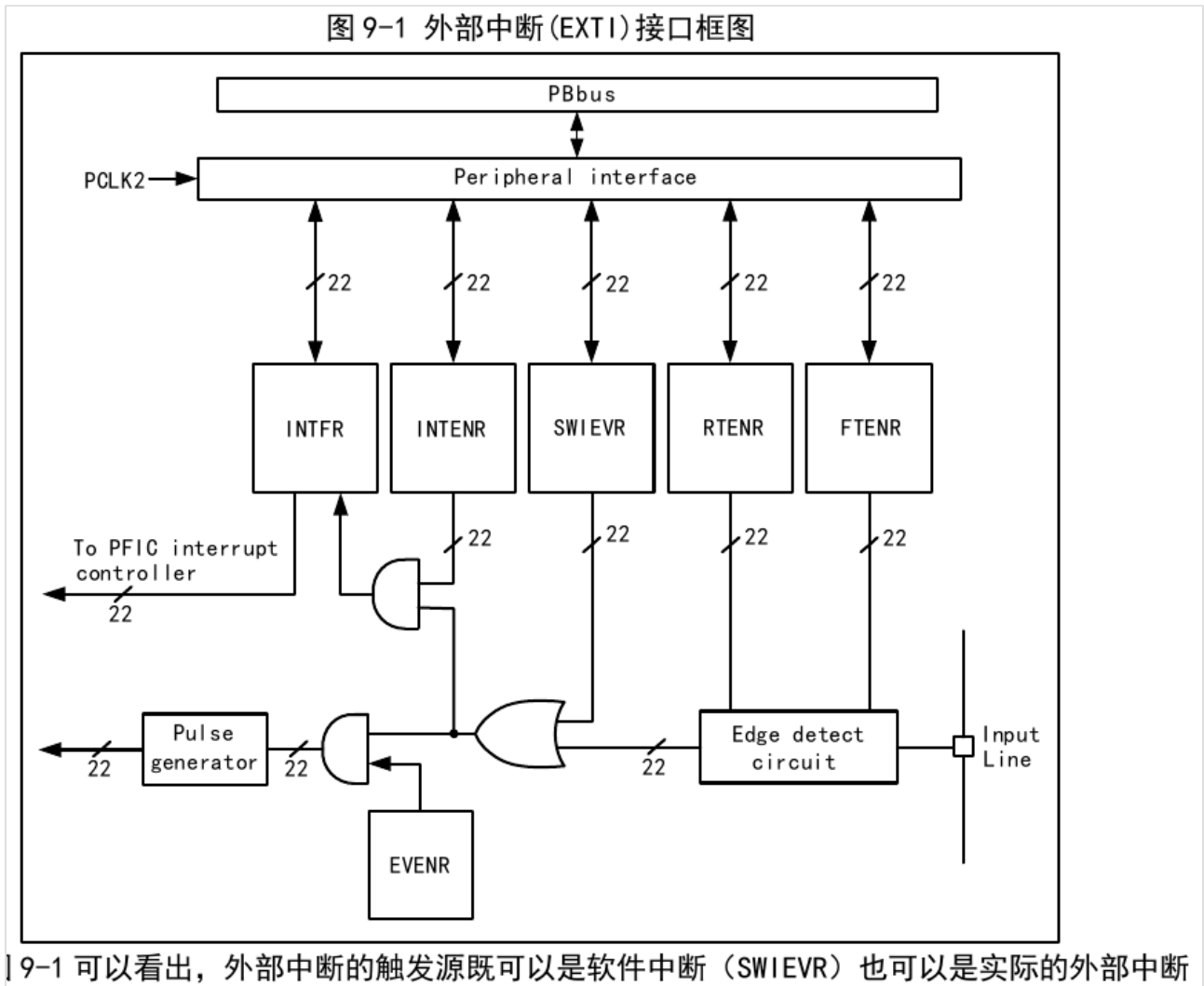


Figure 9-1: External interrupt (EXTI) interface block diagram

As shown in Figure 9-1, the trigger source of an external interrupt can be either a software interrupt (SWIEVR) or an actual external interrupt channel. The signal of the external interrupt channel first passes through the edge detect circuit for filtering. As long as either a software interrupt or an external interrupt signal is generated, it will be output through the OR gate in the diagram to the event enable and interrupt enable AND gate circuits. As long as an interrupt or event is enabled, an interrupt or event will be generated. The six EXTI registers are accessed by the processor through the PB2 interface.

9.4.3.2 Wake-up Event Description

The system can be woken up from sleep mode caused by the WFE instruction through wake-up events. Wake-up events are generated through the following two configurations:

- Enable an interrupt in a peripheral's register, but do not enable this interrupt in the core's PFIC, and enable the SEVONPEND bit in the core. For EXTI, this means enabling the EXTI interrupt, but not enabling the EXTI interrupt in the PFIC, while enabling the SEVONPEND bit. After the CPU wakes up from WFE, the EXTI interrupt flag bit and PFIC pending bit need to be cleared.
- Enable an EXTI channel as an event channel. After the CPU wakes up from WFE, there is no need to clear the interrupt flag bit and PFIC pending bit.

9.4.3.3 Notes

To use external interrupts, the corresponding external interrupt channel must be configured, i.e., select the trigger edge and enable the corresponding interrupt. When the configured trigger edge appears on the external interrupt channel, an interrupt request will be generated, and the corresponding interrupt flag bit will be set. Writing 1 to the flag bit can clear it.

Steps for using external hardware interrupts:

1. Configure GPIO operation;
2. Configure the interrupt enable bit of the corresponding external interrupt channel (EXTI_INTENR);
3. Configure the trigger edge (EXTI_RTENR or EXTI_FTENR), selecting rising edge trigger, falling edge trigger, or both edge trigger;
4. Configure the EXTI interrupt in the core's PFIC to ensure it can be correctly responded to.

Steps for using external hardware events:

1. Configure GPIO operation;
2. Configure the event enable bit of the corresponding external interrupt channel (EXTI_EVENTR);
3. Configure the trigger edge (EXTI_RTENR or EXTI_FTENR), selecting rising edge trigger, falling edge trigger, or both edge trigger.

Steps for using software interrupt/event:

1. Enable external interrupt (EXTI_INTENR) or external event (EXTI_EVENTR);
2. If using an interrupt service function, configure the EXTI interrupt in the core's PFIC;
3. Set the software interrupt trigger (EXTI_SWIEVR), which will generate an interrupt.

9.4.3.4 External Event Mapping

Table 9-6: EXTI Interrupt Mapping

External Interrupt/ Event Line	Mapped Event Description
EXTI0-EXTI15	Px0-Px15 (x=A/B/C/D/E). Any IO port can enable the external interrupt/event function, configured by the AFIO_EXTICRx register.
EXTI16	PVD event: voltage monitoring threshold exceeded
EXTI17	RTC alarm event

External Interrupt/ Event Line	Mapped Event Description
EXTI18	USBHS1 wake-up event
EXTI19	ETH wake-up event
EXTI20	USBHS2 wake-up event
EXTI21	I3C wake-up event

9.4.4 EXTI Register Description

Table 9-7: EXTI Related Register List

Name	Access Address	Description	Reset Value
R32_EXTI_INTENR	0x40010400	Interrupt enable register	0x00000000
R32_EXTI_EVENTR	0x40010404	Event enable register	0x00000000
R32_EXTI_RTENR	0x40010408	Rising edge trigger enable register	0x00000000
R32_EXTI_FTENR	0x4001040C	Falling edge trigger enable register	0x00000000
R32_EXTI_SWIEVR	0x40010410	Software interrupt event register	0x00000000
R32_EXTI_INTFR	0x40010414	Interrupt flag register	0x0000XXXX

9.4.4.1 Interrupt Enable Register (EXTI_INTENR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										MR21	MR20	MR19	MR18	MR17	MR16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
[21:0]	MRx	RW	Enable the interrupt request signal of external interrupt channel x: 1: Enable the interrupt of this channel; 0: Mask the interrupt of this channel.	0

9.4.4.2 Event Enable Register (EXTI_EVENTR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										MR21	MR20	MR19	MR18	MR17	MR16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MR15	MR14	MR13	MR12	MR11	MR10	MR9	MR8	MR7	MR6	MR5	MR4	MR3	MR2	MR1	MR0

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
[21:0]	MRx	RW		0

Bits	Name	Access	Description	Reset
			Enable the event request signal of external interrupt channel x: 1: Enable the event of this channel; 0: Mask the event of this channel.	

9.4.4.3 Rising Edge Trigger Enable Register (EXTI_RTENR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TR21	TR20	TR19	TR18	TR17	TR16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
[21:0]	TRx	RW	Enable rising edge trigger of external interrupt channel x: 1: Enable rising edge trigger of this channel; 0: Disable rising edge trigger of this channel.	0

9.4.4.4 Falling Edge Trigger Enable Register (EXTI_FTENR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										TR21	TR20	TR19	TR18	TR17	TR16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TR15	TR14	TR13	TR12	TR11	TR10	TR9	TR8	TR7	TR6	TR5	TR4	TR3	TR2	TR1	TR0

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
[21:0]	TRx	RW	Enable falling edge trigger of external interrupt channel x: 1: Enable falling edge trigger of this channel; 0: Disable falling edge trigger of this channel.	0

9.4.4.5 Software Interrupt Event Register (EXTI_SWIEVR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										SWIER21	SWIER20	SWIER19	SWIER18	SWIER17	SWIER16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWIER15	SWIER14	SWIER13	SWIER12	SWIER11	SWIER10	SWIER9	SWIER8	SWIER7	SWIER6	SWIER5	SWIER4	SWIER3	SWIER2	SWIER1	SWIER0

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
[21:0]	SWIERx	RW	Set a software interrupt on the corresponding external trigger interrupt channel. Setting this bit will set the corresponding bit in the interrupt	0

Bits	Name	Access	Description	Reset
			flag register (EXTI_INTFR). If interrupt enable (EXTI_INTENR) or event enable (EXTI_EVENTR) is on, an interrupt or event will be generated.	

9.4.4.6 Interrupt Flag Register (EXTI_INTFR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										IF21	IF20	IF19	IF18	IF17	IF16
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IF15	IF14	IF13	IF12	IF11	IF10	IF9	IF8	IF7	IF6	IF5	IF4	IF3	IF2	IF1	IF0

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
[21:0]	IFx	RW1	Interrupt flag bit. This bit being set indicates that the corresponding external interrupt has occurred. Writing 1 can clear this bit.	0

9.4.5 PFIC Registers

Table 9-8: PFIC Related Register List

Name	Access Address	Description	Reset Value
R32_PFIC_ISR1	0xE000E000	PFIC interrupt enable status register 1	0x0000032C
R32_PFIC_ISR2	0xE000E004	PFIC interrupt enable status register 2	0x00000000
R32_PFIC_ISR3	0xE000E008	PFIC interrupt enable status register 3	0x00000000
R32_PFIC_ISR4	0xE000E00C	PFIC interrupt enable status register 4	0x00000000
R32_PFIC_IPR1	0xE000E020	PFIC interrupt pending status register 1	0x00000000
R32_PFIC_IPR2	0xE000E024	PFIC interrupt pending status register 2	0x00000000
R32_PFIC_IPR3	0xE000E028	PFIC interrupt pending status register 3	0x00000000
R32_PFIC_IPR4	0xE000E02C	PFIC interrupt pending status register 4	0x00000000
R32_PFIC_ITHRESDR	0xE000E040	PFIC interrupt priority threshold configuration register	0x00000000
R32_PFIC_CFGR	0xE000E048	PFIC interrupt configuration register	0x00000000
R32_PFIC_GISR	0xE000E04C	PFIC interrupt global status register	0x00000000
R32_PFIC_VTFIDR	0xE000E050	PFIC VTF interrupt ID configuration register	0xFFFFFFFF
R32_PFIC_VTFADDRR0	0xE000E060	PFIC VTF interrupt 0 offset address register	0xFFFFFFFF
R32_PFIC_VTFADDRR1	0xE000E064	PFIC VTF interrupt 1 offset address register	0xFFFFFFFF
R32_PFIC_VTFADDRR2	0xE000E068	PFIC VTF interrupt 2 offset address register	0xFFFFFFFF
R32_PFIC_VTFADDRR3	0xE000E06C	PFIC VTF interrupt 3 offset address register	0xFFFFFFFF
R32_PFIC_IENR1	0xE000E100	PFIC interrupt enable set register 1	0x00000000
R32_PFIC_IENR2	0xE000E104	PFIC interrupt enable set register 2	0x00000000
R32_PFIC_IENR3	0xE000E108	PFIC interrupt enable set register 3	0x00000000

Name	Access Address	Description	Reset Value
R32_PFIC_IENR4	0xE000E10C	PFIC interrupt enable set register 4	0x00000000
R32_PFIC_IRER1	0xE000E180	PFIC interrupt enable clear register 1	0x00000000
R32_PFIC_IRER2	0xE000E184	PFIC interrupt enable clear register 2	0x00000000
R32_PFIC_IRER3	0xE000E188	PFIC interrupt enable clear register 3	0x00000000
R32_PFIC_IRER4	0xE000E18C	PFIC interrupt enable clear register 4	0x00000000
R32_PFIC_IPSR1	0xE000E200	PFIC interrupt pending set register 1	0x00000000
R32_PFIC_IPSR2	0xE000E204	PFIC interrupt pending set register 2	0x00000000
R32_PFIC_IPSR3	0xE000E208	PFIC interrupt pending set register 3	0x00000000
R32_PFIC_IPSR4	0xE000E20C	PFIC interrupt pending set register 4	0x00000000
R32_PFIC_IPRR1	0xE000E280	PFIC interrupt pending clear register 1	0x00000000
R32_PFIC_IPRR2	0xE000E284	PFIC interrupt pending clear register 2	0x00000000
R32_PFIC_IPRR3	0xE000E288	PFIC interrupt pending clear register 3	0x00000000
R32_PFIC_IPRR4	0xE000E28C	PFIC interrupt pending clear register 4	0x00000000
R32_PFIC_IACTR1	0xE000E300	PFIC interrupt active status register 1	0x00000000
R32_PFIC_IACTR2	0xE000E304	PFIC interrupt active status register 2	0x00000000
R32_PFIC_IACTR3	0xE000E308	PFIC interrupt active status register 3	0x00000000
R32_PFIC_IACTR4	0xE000E30C	PFIC interrupt active status register 4	0x00000000
R32_PFIC_IPRIORx	0xE000E400	PFIC interrupt priority configuration register	0x00000000
R32_PFIC_WAKEIP0	0xE000E720	PFIC wake-up instruction pointer register 0	0x00000001
R32_PFIC_CSTAR0	0xE000E780	PFIC core status register 0	0x00000000
R32_PFIC_EENR	0xE000EC80	PFIC event enable register	0xFFFFFFFF
R32_PFIC_EPR	0xE000EC84	PFIC event pending register	0x00000000
R32_PFIC_EWUPR	0xE000EC88	PFIC event wake-up register	0x00000000
R32_PFIC_SCTLR	0xE000ED10	PFIC system control register	0x00000000

Note:

1. NMI, HardFault, ECALL-M, ECALL-U, BREAKPOINT interrupts are always enabled by default.
2. ECALL-M, ECALL-U, BREAKPOINT are all cases of EXC, the status is indicated by EXC status bit bit3.
3. NMI and EXC support interrupt pending clear and set operations, but do not support interrupt enable clear and set operations.
4. ECALL-M, ECALL-U, BREAKPOINT do not support interrupt pending clear and set, or interrupt enable clear and set operations.

9.4.5.1 PFIC Interrupt Enable Status Register 1 (PFIC_ISR1)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	INTENSTA14	Reserved	INTENSTA12	Reserved	INTENSTA10	INTENSTA9	INTENSTA8	Reserved	INTENSTA5	Reserved	INTENSTA3	INTENSTA2	Reserved		

Bits	Name	Access	Description	Reset
[31:16]	INTENSTA	RO	Current enable status of interrupt #16-#31. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	0
15	Reserved	RO	Reserved.	0
14	INTENSTA	RO	Current enable status of interrupt #14. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	0
13	Reserved	RO	Reserved.	0
12	INTENSTA	RO	Current enable status of interrupt #12. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	0
[11:10]	Reserved	RO	Reserved.	0
[9:8]	INTENSTA	RO	Current enable status of interrupt #8-#9. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	0x3
[7:6]	Reserved	RO	Reserved.	0
5	INTENSTA	RO	Current enable status of interrupt #5. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	1
4	Reserved	RO	Reserved.	0
[3:2]	INTENSTA	RO	Current enable status of interrupt #2-#3. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	0x3
[1:0]	Reserved	RO	Reserved.	0

9.4.5.2 PFIC Interrupt Enable Status Register 2 (PFIC_ISR2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTENSTA[47:32]															

Bits	Name	Access	Description	Reset
[31:0]	INTENSTA	RO	Current enable status of interrupt #32-#63. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	0

9.4.5.3 PFIC Interrupt Enable Status Register 3 (PFIC_ISR3)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTENSTA[95:80]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTENSTA[79:64]															
Bits	Name	Access	Description	Reset											
[31:0]	INTENSTA	RO	Current enable status of interrupt #64-#95. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	0											

9.4.5.4 PFIC Interrupt Enable Status Register 4 (PFIC_ISR4)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								INTENSTA[103:96]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	INTENSTA	RO	Current enable status of interrupt #96-#103. 1: The interrupt of the current number is enabled; 0: The interrupt of the current number is not enabled.	0

9.4.5.5 PFIC Interrupt Pending Status Register 1 (PFIC_IPR1)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSTA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PENDSTA14	Reserved	PENDSTA12	Reserved	PENDSTA9	PENDSTA8	Reserved	PENDSTA5	Reserved	PENDSTA3	PENDSTA2	Reserved			

Bits	Name	Access	Description	Reset
[31:16]	PENDSTA	RO	Current pending status of interrupt #16-#31. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0
15	Reserved	RO	Reserved.	0
14	PENDSTA	RO	Current pending status of interrupt #14. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0
13	Reserved	RO	Reserved.	0
12	PENDSTA	RO	Current pending status of interrupt #12. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0
[11:10]	Reserved	RO	Reserved.	0
[9:8]	PENDSTA	RO	Current pending status of interrupt #8-#9. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0

Bits	Name	Access	Description	Reset
[7:6]	Reserved	RO	Reserved.	0
5	PENDSTA	RO	Current pending status of interrupt #5. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0
4	Reserved	RO	Reserved.	0
[3:2]	PENDSTA	RO	Current pending status of interrupt #2-#3. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0
[1:0]	Reserved	RO	Reserved.	0

9.4.5.6 PFIC Interrupt Pending Status Register 2 (PFIC_IPR2)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSTA[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSTA[47:32]															

Bits	Name	Access	Description	Reset
[31:0]	PENDSTA	RO	Current pending status of interrupt #32-#63. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0

9.4.5.7 PFIC Interrupt Pending Status Register 3 (PFIC_IPR3)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSTA[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSTA[79:64]															

Bits	Name	Access	Description	Reset
[31:0]	PENDSTA	RO	Current pending status of interrupt #64-#95. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0

9.4.5.8 PFIC Interrupt Pending Status Register 4 (PFIC_IPR4)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								PENDSTA[103:96]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	PENDSTA	RO	Current pending status of interrupt #96-#103. 1: The interrupt of the current number is pending; 0: The interrupt of the current number is not pending.	0

9.4.5.9 PFIC Interrupt Priority Threshold Configuration Register (PFIC_ITHRESDR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								THRESHOLD[7:0]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	THRESHOLD	RW	Interrupt priority threshold setting value. Interrupts with priority lower than the set threshold will not execute interrupt service when pending. When this register is 0, the threshold register function is invalid. [7:4]: Priority threshold; [3:0]: Reserved, fixed to 0.	0

9.4.5.10 PFIC Interrupt Configuration Register (PFIC_CFGR)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16			
KEYCODE[15:0]																		
15	14	13	12	11	10	9	8	7				6	5	4	3	2	1	0
Reserved								SYSRST			Reserved							

Bits	Name	Access	Description	Reset
[31:16]	KEYCODE[15:0]	WO	Key control keyword. Effective when written as 0xBEEF, otherwise the operation is invalid.	0
[15:8]	Reserved	RO	Reserved.	0
7	SYSRST	WO	Reset register. Writing 0xBEEF0080 triggers a system reset, other values are invalid. Note: Same function as the PFIC_SCTLR register SYSRST bit.	0
[6:0]	Reserved	RO	Reserved.	0

9.4.5.11 PFIC Interrupt Global Status Register (PFIC_GISR)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EX_STATE[1:0]		LOCK_UP	DBG_MODE	GLOBL_IE	Reserved	GPENDSTA	GACTSTA	NESTSTA[7:0]							

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:14]	EX_STATE[1:0]	RO	Core status register, used to obtain the running state of the core: 00: Running; 01: Light sleep; 10: Deep sleep; 11: Deep sleep (locked).	0
13	LOCK_UP	RO	Core lock status register, used to query whether the core is in a locked state. 1: Locked; 0: Not locked.	0
12	DBG_MODE	RO	Core debug mode register, used to query whether the core is in debug mode: 1: Core is in debug mode; 0: Core is not in debug mode.	0
11	GLOBL_IE	RO	Core global interrupt enable register, used to query whether the core global interrupt is enabled: 1: Core global interrupt enable is on; 0: Core global interrupt enable is off.	0
10	Reserved	RO	Reserved.	0
9	GPENDSTA	RO	Whether any interrupt is currently pending: 1: Yes; 0: No.	0
8	GACTSTA	RO	Whether any interrupt is currently being executed: 1: Yes; 0: No.	0
[7:0]	NESTSTA	RO	Current interrupt nesting status, used to query the nesting status of core interrupts.	0

9.4.5.12 PFIC VTF Interrupt ID Configuration Register (PFIC_VTFIDR)

Offset address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
VTFID3															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VTFID1								VTFID0							

Note: VTFID2 occupies bits [23:16].

Bits	Name	Access	Description	Reset
[31:24]	VTFID3	RW	Configure the interrupt number for VTF interrupt 3.	X
[23:16]	VTFID2	RW	Configure the interrupt number for VTF interrupt 2.	X
[15:8]	VTFID1	RW	Configure the interrupt number for VTF interrupt 1.	X
[7:0]	VTFID0	RW	Configure the interrupt number for VTF interrupt 0.	X

9.4.5.13 PFIC VTF Interrupt 0 Address Register (PFIC_VTFADDR0)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR0[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR0[15:1]														VTF0EN	
Bits	Name	Access	Description											Reset	
[31:1]	ADDR0	RW	VTF interrupt 0 service program address bit[31:1].											X	
0	VTF0EN	RW	VTF interrupt 0 enable bit: 1: Enable VTF interrupt 0 channel; 0: Disable.											0	

9.4.5.14 PFIC VTF Interrupt 1 Address Register (PFIC_VTFADDR1)

Offset address: 0x64

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR1[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR1[15:1]														VTF1EN	
Bits	Name	Access	Description											Reset	
[31:1]	ADDR1	RW	VTF interrupt 1 service program address bit[31:1].											X	
0	VTF1EN	RW	VTF interrupt 1 enable bit: 1: Enable VTF interrupt 1 channel; 0: Disable.											0	

9.4.5.15 PFIC VTF Interrupt 2 Address Register (PFIC_VTFADDR2)

Offset address: 0x68

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR2[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR2[15:1]														VTF2EN	
Bits	Name	Access	Description											Reset	
[31:1]	ADDR2	RW	VTF interrupt 2 service program address bit[31:1].											X	
0	VTF2EN	RW	VTF interrupt 2 enable bit: 1: Enable VTF interrupt 2 channel; 0: Disable.											0	

9.4.5.16 PFIC VTF Interrupt 3 Address Register (PFIC_VTFADDR3)

Offset address: 0x6C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADDR3[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR3[15:1]													VTF3EN		

Bits	Name	Access	Description	Reset
[31:1]	ADDR3	RW	VTF interrupt 3 service program address bit[31:1].	X
0	VTF3EN	RW	VTF interrupt 3 enable bit: 1: Enable VTF interrupt 3 channel; 0: Disable.	0

9.4.5.17 PFIC Interrupt Enable Set Register 1 (PFIC_IENR1)

Offset address: 0x100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		INTEN14		Reserved		INTEN12		Reserved							

Bits	Name	Access	Description	Reset
[31:16]	INTEN	WO	Interrupt #16-#31 enable control. 1: Enable the interrupt of the current number; 0: No effect.	0
15	Reserved	RO	Reserved.	0
14	INTEN	WO	Interrupt #14 enable control. 1: Enable the interrupt of the current number; 0: No effect.	0
13	Reserved	RO	Reserved.	0
12	INTEN	WO	Interrupt #12 enable control. 1: Enable the interrupt of the current number; 0: No effect.	0
[11:0]	Reserved	RO	Reserved.	0

9.4.5.18 PFIC Interrupt Enable Set Register 2 (PFIC_IENR2)

Offset address: 0x104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTEN[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTEN[47:32]															

Bits	Name	Access	Description	Reset
[31:0]	INTEN	WO	Interrupt #32-#63 enable control. 1: Enable the interrupt of the current number; 0: No effect.	0

9.4.5.19 PFIC Interrupt Enable Set Register 3 (PFIC_IENR3)

Offset address: 0x108

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
INTEN[95:80]																
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
INTEN[79:64]																
Bits	Name	Access	Description													Reset
[31:0]	INTEN	WO	Interrupt #64-#95 enable control. 1: Enable the interrupt of the current number; 0: No effect.													0

9.4.5.20 PFIC Interrupt Enable Set Register 4 (PFIC_IENR4)

Offset address: 0x10C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									INTEN[103:96]						

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	INTEN	WO	Interrupt #96-#103 enable control. 1: Enable the interrupt of the current number; 0: No effect.	0

9.4.5.21 PFIC Interrupt Enable Clear Register 1 (PFIC_IRER1)

Offset address: 0x180

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16								
INTRSET[31:16]																							
15	14			13			12			11			10	9	8	7	6	5	4	3	2	1	0
Reserved			INTRSET14			Reserved			INTRSET12			Reserved											

Bits	Name	Access	Description	Reset
[31:16]	INTRSET	WO	Interrupt #16-#31 disable control. 1: Disable the interrupt of the current number; 0: No effect.	0
15	Reserved	RO	Reserved.	0
14	INTRSET	WO	Interrupt #14 disable control. 1: Disable the interrupt of the current number; 0: No effect.	0
13	Reserved	RO	Reserved.	0
12	INTRSET	WO		0

Bits	Name	Access	Description	Reset
			Interrupt #12 disable control. 1: Disable the interrupt of the current number; 0: No effect.	
[11:0]	Reserved	RO	Reserved.	0

9.4.5.22 PFIC Interrupt Enable Clear Register 2 (PFIC_IRER2)

Offset address: 0x184

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET[47:32]															

Bits	Name	Access	Description	Reset
[31:0]	INTRSET	WO	Interrupt #32-#63 disable control. 1: Disable the interrupt of the current number; 0: No effect.	0

9.4.5.23 PFIC Interrupt Enable Clear Register 3 (PFIC_IRER3)

Offset address: 0x188

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
INTRSET[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTRSET[79:64]															

Bits	Name	Access	Description	Reset
[31:0]	INTRSET	WO	Interrupt #64-#95 disable control. 1: Disable the interrupt of the current number; 0: No effect.	0

9.4.5.24 PFIC Interrupt Enable Clear Register 4 (PFIC_IRER4)

Offset address: 0x18C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								INTRSET[103:96]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	INTRSET	WO	Interrupt #96-#103 disable control. 1: Disable the interrupt of the current number; 0: No effect.	0

9.4.5.25 PFIC Interrupt Pending Set Register 1 (PFIC_IPSR1)

Offset address: 0x200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PENDSET14	Reserved	PENDSET12	Reserved	PENDSET9	PENDSET8	Reserved	PENDSET5	Reserved	PENDSET3	PENDSET2	Reserved			

Bits	Name	Access	Description	Reset
[31:16]	PENDSET	WO	Interrupt #16-#31 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	0
15	Reserved	RO	Reserved.	0
14	PENDSET	WO	Interrupt #14 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	0
13	Reserved	RO	Reserved.	0
12	PENDSET	WO	Interrupt #12 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	0
[11:10]	Reserved	RO	Reserved.	0
[9:8]	PENDSET	RO	Interrupt #8-#9 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	0
[7:6]	Reserved	RO	Reserved.	0
5	PENDSET	RO	Interrupt #5 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	0
4	Reserved	RO	Reserved.	0
[3:2]	PENDSET	WO	Interrupt #2-#3 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	0
[1:0]	Reserved	RO	Reserved.	0

9.4.5.26 PFIC Interrupt Pending Set Register 2 (PFIC_IPSR2)

Offset address: 0x204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSET[47:32]															
Bits	Name	Access	Description											Reset	
[31:0]	PENDSET	WO												0	

Bits	Name	Access	Description	Reset
			Interrupt #32-#63 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	

9.4.5.27 PFIC Interrupt Pending Set Register 3 (PFIC_IPSR3)

Offset address: 0x208

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDSET[95:80]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDSET[79:64]															

Bits	Name	Access	Description	Reset
[31:0]	PENDSET	WO	Interrupt #64-#95 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	0

9.4.5.28 PFIC Interrupt Pending Set Register 4 (PFIC_IPSR4)

Offset address: 0x20C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								PENDSET[103:96]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	PENDSET	WO	Interrupt #96-#103 pending set. 1: Set the interrupt of the current number as pending; 0: No effect.	0

9.4.5.29 PFIC Interrupt Pending Clear Register 1 (PFIC_IPRR1)

Offset address: 0x280

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	PENDRST14	Reserved	PENDRST12	Reserved	PENDRST9	PENDRST8	Reserved	PENDRST5	Reserved	PENDRST3	PENDRST2	Reserved			

Bits	Name	Access	Description	Reset
[31:16]	PENDRST	WO	Interrupt #16-#31 pending clear. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	0
15	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
14	PENDRST	WO	Interrupt #14 pending clear, #13 and #15 are reserved. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	0
13	Reserved	RO	Reserved.	0
12	PENDRST	WO	Interrupt #12 pending clear. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	0
[11:10]	Reserved	RO	Reserved.	0
[9:8]	PENDRST	RO	Interrupt #8-#9 pending clear. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	0
[7:6]	Reserved	RO	Reserved.	0
5	PENDRST	RO	Interrupt #5 pending clear. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	0
4	Reserved	RO	Reserved.	0
[3:2]	PENDRST	WO	Interrupt #2-#3 pending clear. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	0
[1:0]	Reserved	RO	Reserved.	0

9.4.5.30 PFIC Interrupt Pending Clear Register 2 (PFIC_IPRR2)

Offset address: 0x284

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST[47:32]															
Bits	Name	Access	Description	Reset											
[31:0]	PENDRST	WO	Interrupt #32-#63 pending clear. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	0											

9.4.5.31 PFIC Interrupt Pending Clear Register 3 (PFIC_IPRR3)

Offset address: 0x288

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PENDRST[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PENDRST[79:64]															
Bits	Name	Access	Description												Reset
[31:0]	PENDRST	WO													0

Bits	Name	Access	Description	Reset
			Interrupt #64-#95 pending clear. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	

9.4.5.32 PFIC Interrupt Pending Clear Register 4 (PFIC_IPRR4)

Offset address: 0x28C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								PENDRST[103:96]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	PENDSET	WO	Interrupt #96-#103 pending clear. 1: Clear the pending status of the interrupt of the current number; 0: No effect.	0

9.4.5.33 PFIC Interrupt Active Status Register 1 (PFIC_IACTR1)

Offset address: 0x300

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	IACTS14	Reserved	IACTS12	Reserved	IACTS3	IACTS2	Reserved								

Bits	Name	Access	Description	Reset
[31:16]	IACTS	RO	Interrupt #16-#31 active status. 1: The interrupt of the current number is executing; 0: The interrupt of the current number is not executing.	0
15	Reserved	RO	Reserved.	0
14	IACTS	RO	Interrupt #14 active status. 1: The interrupt of the current number is executing; 0: The interrupt of the current number is not executing.	0
13	Reserved	RO	Reserved.	0
12	IACTS	RO	Interrupt #12 active status. 1: The interrupt of the current number is executing; 0: The interrupt of the current number is not executing.	0
[11:4]	Reserved	RO	Reserved.	0
[3:2]	IACTS	RO	Interrupt #2-#3 active status. 1: The interrupt of the current number is executing; 0: The interrupt of the current number is not executing.	0
[1:0]	Reserved	RO	Reserved.	0

9.4.5.34 PFIC Interrupt Active Status Register 2 (PFIC_IACTR2)

Offset address: 0x304

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS[47:32]															
Bits	Name	Access	Description												Reset
[31:0]	IACTS	RO	Interrupt #32-#63 active status. 1: The interrupt of the current number is executing; 0: The interrupt of the current number is not executing.												0

9.4.5.35 PFIC Interrupt Active Status Register 3 (PFIC_IACTR3)

Offset address: 0x308

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IACTS[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IACTS[79:64]															
Bits	Name	Access	Description												Reset
[31:0]	IACTS	RO	Interrupt #64-#95 active status. 1: The interrupt of the current number is executing; 0: The interrupt of the current number is not executing.												0

9.4.5.36 PFIC Interrupt Active Status Register 4 (PFIC_IACTR4)

Offset address: 0x30C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									IACTS[103:96]						

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	IACTS	WO	Interrupt #96-#103 active status. 1: The interrupt of the current number is executing; 0: The interrupt of the current number is not executing.	0

9.4.5.37 PFIC Interrupt Priority Configuration Register (PFIC_IPRIORx) (x=0-63)

Offset address: 0x400-0x4FF

The controller supports 256 interrupts (0-255), each using 8 bits to set the control priority.

Bits	Name	Access	Description	Reset
[2047:2040]	IP_255	RW	Same as IP_0 description.	0
...	...	...	...	...
[31:24]	IP_3	RW	Same as IP_0 description.	0
[23:16]	IP_2	RW	Same as IP_0 description.	0
[15:8]	IP_1	RW	Same as IP_0 description.	0
[7:0]	IP_0	RW	Interrupt #0 priority configuration: [7:4]: Priority control bits. The smaller the value, the higher the priority. The preemption bits are controlled by the CSR register preempt. When the preempt register value is 0, there are no preemption bits. When the preempt register value is 1, bit7 is the preemption bit. When the preempt register value is 2, bits7-6 are the preemption bits. When the preempt register value is 3, bits7-5 are the preemption bits. The core supports a maximum of 2 levels of nesting. When nesting is full, further preemption is not possible. [3:0] bits are fixed to 0.	0

9.4.5.38 PFIC Wake-up Instruction Pointer Register 0 (PFIC_WAKEIP0)

Offset address: 0x720

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IP_RELOAD0[30:15]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IP_RELOAD0[14:0]												SHUTDOWN0			

Bits	Name	Access	Description	Reset
[31:1]	IP_RELOAD0	RW	Core wake-up PC address register, used for PC value reload after power-on wake-up and reset wake-up.	0
0	SHUTDOWN0	RW	Core deep sleep (locked) cancel register, used to cancel the locked state of the core that entered deep sleep (locked) after power-on, allowing wake-up by time: 1: Locked; 0: Unlock the lock.	1

9.4.5.39 PFIC Core Status Register 0 (PFIC_CSTAR0)

Offset address: 0x780

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CPU_EX_STATE_0	CPU_LOCK_UP_0	CPU_DBG_MODE_0	CPU_GLOBL_IE_0	RESERVED	CPU_IRQ_PEND_0	CPU_IRQ_ACTIVE_0	CPU_NEST_STA_0								

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:14]	CPU_EX_STATE_0	RO	Core status register, used to obtain the running state of the core:	0

Bits	Name	Access	Description	Reset
			00: Running; 01: Light sleep; 10: Deep sleep; 11: Deep sleep (locked).	
13	CPU_LOCK_UP_0	RO	Core lock status register, used to query whether the core is in a locked state: 1: Locked; 0: Not locked.	0
12	CPU_DBG_MODE_0	RO	Core debug mode register, used to query whether the core is in debug mode: 1: Core is in debug mode; 0: Core is not in debug mode.	0
11	CPU_GLOBL_IE_0	RO	Core global interrupt enable register, used to query whether the core global interrupt is enabled: 1: Core global interrupt enable is on; 0: Core global interrupt enable is off.	0
10	Reserved	RO	Reserved.	0
9	CPU_IRQ_PEND_0	RO	Core interrupt pending flag register, used to query whether the core has unprocessed interrupts: 1: Core has unprocessed interrupts; 0: Core has no unprocessed interrupts.	0
8	CPU_IRQ_ACTIVE_0	RO	Core interrupt active flag register, used to query whether the core is currently processing an interrupt: 1: Core is processing an interrupt; 0: Core is not processing an interrupt.	0
[7:0]	CPU_NEST_STA_0	RO	Core interrupt nesting status register, used to query the nesting status of core interrupts.	0

9.4.5.40 PFIC Event Enable Register (PFIC_EENR)

Offset address: 0xC80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVENTEN[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVENTEN[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	EVENTEN	RW	Event #31-#0 wake-up enable: 1: Allow the corresponding event to wake up the core during sleep; 0: Mask the corresponding event, this event is not allowed to wake up the core. Supports up to 32 event sources, of which 0-7 are level triggered (active high), 8-31 are pulse triggered: #0: Core halt request; #1: Core interrupt request; #7-#2: Reserved; #8: Core enters interrupt;	0xFFFFFFFF

Bits	Name	Access	Description	Reset
			#9: Core exits interrupt; #10: New interrupt pending; (for shared interrupts, does not distinguish whether the interrupt is routed to the local core) #11: Core sends event; (send event of the local core) #12: Core receives event; (send event of other cores) #13: System receives event; (system configured events, e.g. GPIO toggle events...) #31-#14: Reserved.	

9.4.5.41 PFIC Event Pending Register (PFIC_EPR)

Offset address: 0xC84

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVENT_PEND[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVENT_PEND[15:8]								EVENT_PEND[7:0]							

Bits	Name	Access	Description	Reset
[31:8]	EVENT_PEND	RW1Z	Event #31-#8 pending status, write 1 to clear: 1: The corresponding event occurred and has not been processed; 0: The corresponding event did not occur.	0
[7:0]	EVENT_PEND	RO	Event #7-#0 pending status, cannot be cleared: 1: The corresponding event occurred and has not been processed; 0: The corresponding event did not occur.	0

9.4.5.42 PFIC Event Wake-up Register (PFIC_EWUPR)

Offset address: 0xC88

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
EVENT_WUP[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVENT_WUP[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	EVENT_WUP	RO	Event #31-#0 wake-up register: 1: The corresponding event woke up the core from sleep; 0: The corresponding event is not the event that woke up the core this time.	0

9.4.5.43 PFIC System Control Register (PFIC_SCTLR)

Offset address: 0xD10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SYSRST		Reserved						HART_ID							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved						SENDEVENT		SEVONPEND		WFIWFE		SLEEPDEEP		SLEEPONEXIT		Reserved

Bits	Name	Access	Description	Reset
31	SYSRST	RW1	System reset, auto-clears to 0. Write 1 is effective, write 0 has no effect. Same function as PFIC_CFGR register.	0
[30:24]	Reserved	RO	Reserved.	0
[23:16]	HART_ID	RO	Core ID register, used to obtain the core ID that reads this register.	0
[15:6]	Reserved	RO	Reserved.	0
5	SENDEVENT	WO	Send event, generates a send event, which can be used to wake up other cores.	0
4	SEVONPEND	RW	When an event or interrupt pending status occurs, the system can be woken up from after the WFE instruction. If the WFE instruction has not been executed, the system will wake up immediately after executing the instruction next time. 1: Enabled events and all interrupts (including those with interrupts not enabled) can wake up the system; 0: Only enabled events and enabled interrupts can wake up.	0
3	WFIWFE	RW	WFI command executes as WFE. WFE is a custom instruction. When this register is set to 1, the core enters low-power state after executing the WFI instruction, and can be woken up after an external event occurs. 1: Wait for event to wake up the core after WFI instruction; 0: Wait for interrupt to wake up the core after WFI instruction.	0
2	SLEEPDEEP	RW	Control the system's low-power mode: 1: deep sleep; 0: sleep.	0
1	SLEEPONEXIT	RW	Control the system state after leaving the interrupt service program: 1: System enters low-power mode; 0: System enters main program.	0
0	Reserved	RO	Reserved.	0

Chapter 10: GPIO and Alternate Functions (GPIO/AFIO)

GPIO ports can be configured into various input or output modes, with built-in, disableable pull-up or pull-down resistors, and can be configured for push-pull or open-drain operation. GPIO ports can also be multiplexed to other functions.

10.1 Main Features

Each pin of a port can be configured into one of the following modes:

- Floating input
- Pull-up input
- Pull-down input
- Analog input
- Open-drain output
- Push-pull output
- Alternate function input and output

Many pins have alternate functions; many other peripherals map their output and input channels onto these pins. The specific usage of these multiplexed pins should be referred to in each peripheral's documentation, while whether these pins are multiplexed or remapped is described in this chapter.

10.2 Functional Description

10.2.1 Overview

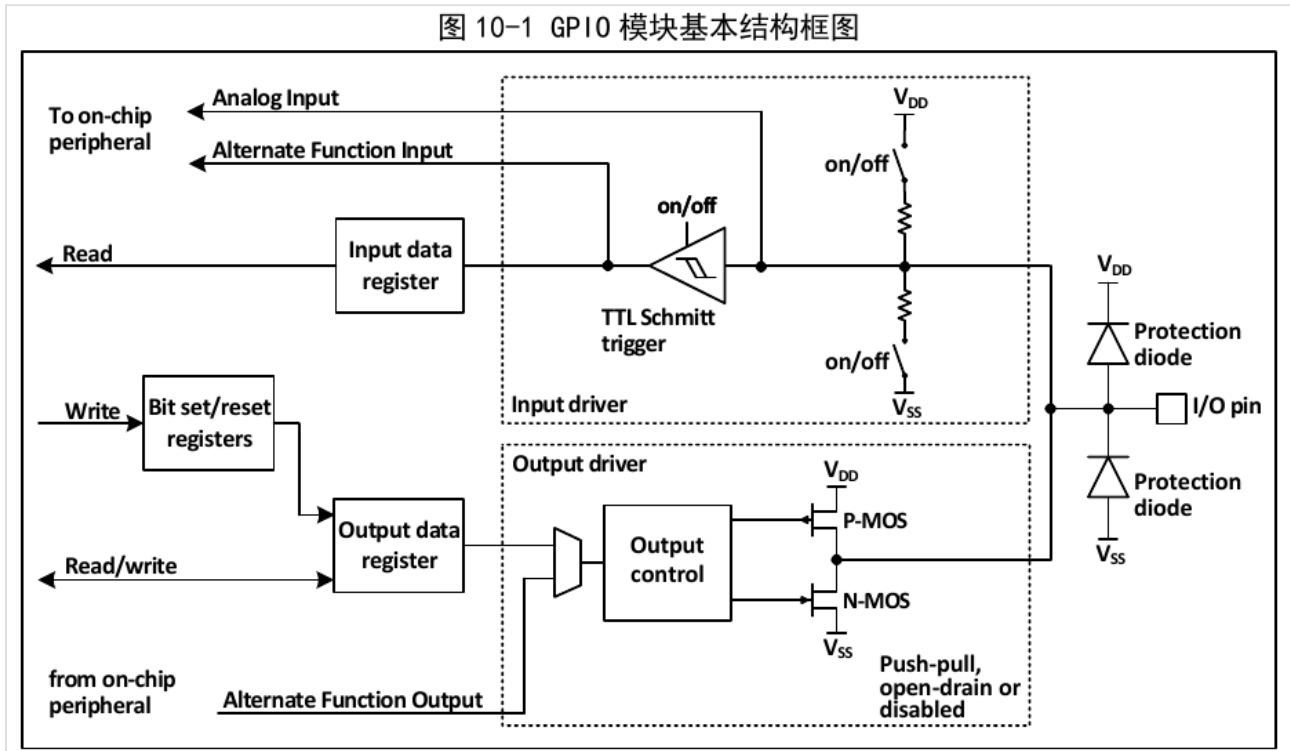


Figure 10-1: GPIO module basic structure block diagram

As shown in the IO port structure in Figure 10-1, each pin has two protection diodes inside the chip, and the IO port is internally divided into input and output driver modules. The input driver has optional weak pull-up/pull-down resistors and can be connected to analog input peripherals such as AD. If the input goes to a digital peripheral, it must pass through a TTL Schmitt trigger before being connected to the GPIO input data register or other alternate function peripherals. The output driver has a pair of MOS transistors; by configuring whether the upper and lower MOS transistors are enabled, the IO port can be configured as open-drain or push-pull output. The output driver can also be configured to be controlled by GPIO output or by another multiplexed peripheral.

10.2.2 GPIO Initialization Function

Immediately after reset, the GPIO port runs in its initial state. At this point, most IO ports are in floating input mode, but some peripheral-related pins such as HSE run in their peripheral alternate function mode. For specific initialization functions, please refer to the pin description related chapters.

10.2.3 External Interrupts

All GPIO ports can be configured as external interrupt input channels, but one external interrupt input channel can be mapped to at most one GPIO pin, and the external interrupt channel number must match the GPIO port bit number. For example, PA1 (or PB1, PC1, PD1, PE1, etc.) can only be mapped to EXTI1, and EXTI1 can only accept one of PA1, PB1, PC1, PD1, or PE1 — the mapping is one-to-one in both directions.

10.2.4 Alternate Functions

When using alternate functions, note the following:

- To use an input-direction alternate function, the port must be configured as alternate function input mode; the pull-up/pull-down settings can be configured as needed.
- To use an output-direction alternate function, the port must be configured as alternate function output mode; push-pull or open-drain can be configured as needed.
- For bidirectional alternate functions, the port must be configured as alternate function output mode, and the input driver is configured as floating input mode.

The same IO port may have multiple peripherals multiplexed onto it. To give each peripheral maximum flexibility, the alternate function pins — in addition to the default alternate function pins — can be remapped to other pins to avoid occupied pins.

10.2.5 Lock Mechanism

The lock mechanism can lock the IO port configuration. After a specific write sequence, the selected IO pin configuration is locked and cannot be changed until the next reset.

10.2.6 Input Configuration

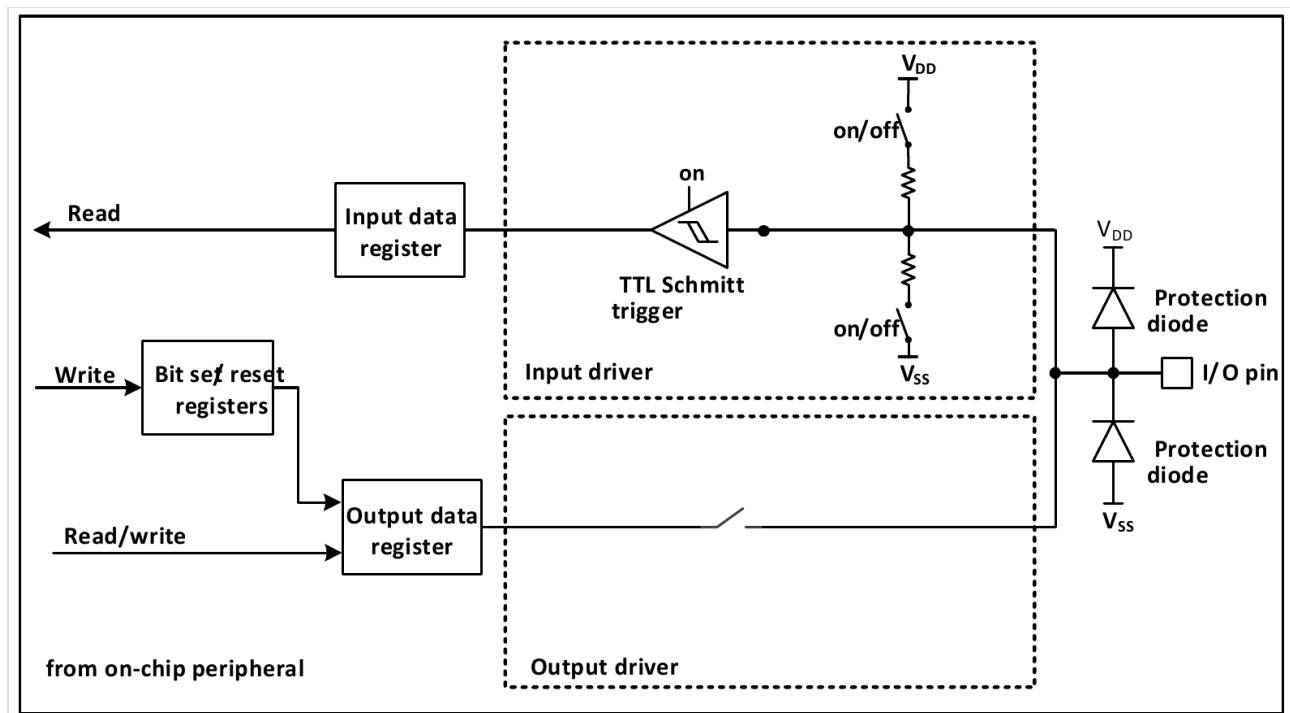


Figure 10-2: GPIO module input configuration block diagram

When the IO port is configured as input mode, the output driver is disconnected, the input pull-up/pull-down is optional, and the alternate function and analog input are not connected. The data on each IO port is sampled into the input data register at each PB2 clock; reading the corresponding bit of the input data register obtains the logic level of the corresponding pin.

10.2.7 Output Configuration

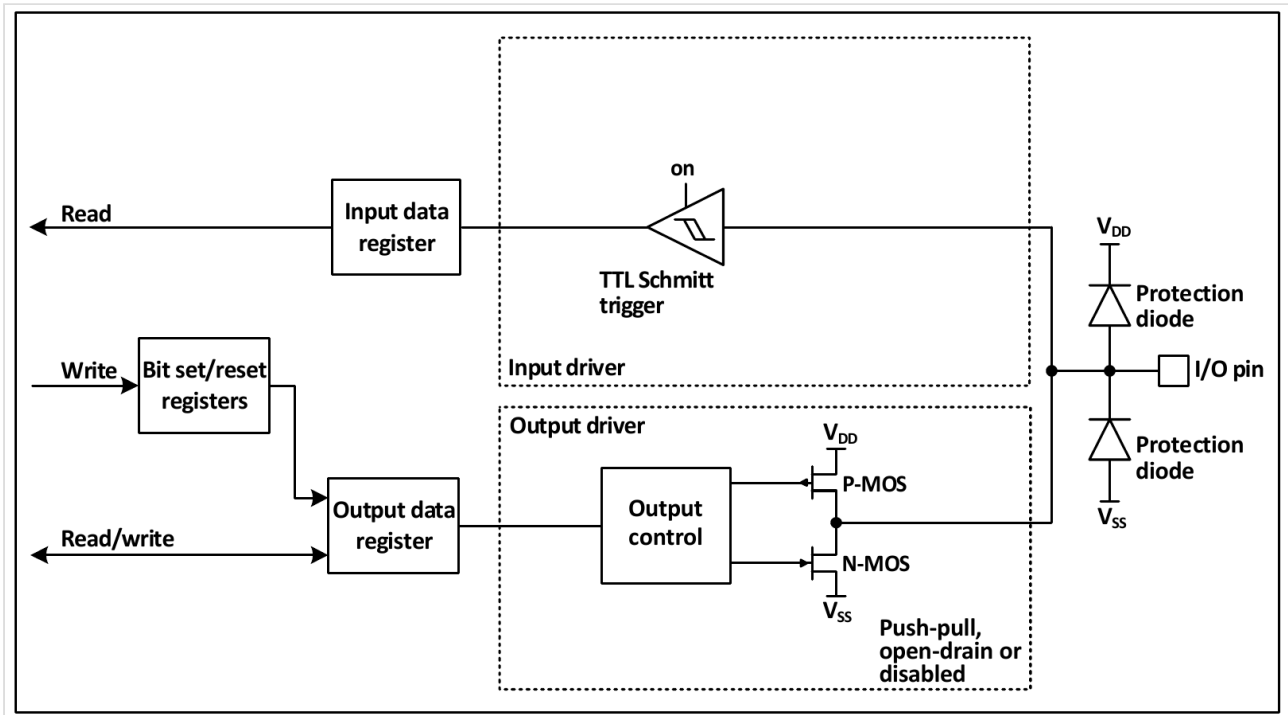


Figure 10-3: GPIO module output configuration block diagram

When the IO port is configured as output mode, the pair of MOS transistors in the output driver can be configured as push-pull or open-drain mode as needed, without using the alternate function. The input driver's pull-up/pull-down resistors are disabled, and the TTL Schmitt trigger is activated. The level appearing on the IO pin will be sampled into the input data register at each PB2 clock, so reading the input data register will return the IO status. In push-pull output mode, accessing the output data register will return the last written value.

10.2.8 Alternate Function Configuration

图 10-4 GPIO 模块被其他外设复用时的结构框图

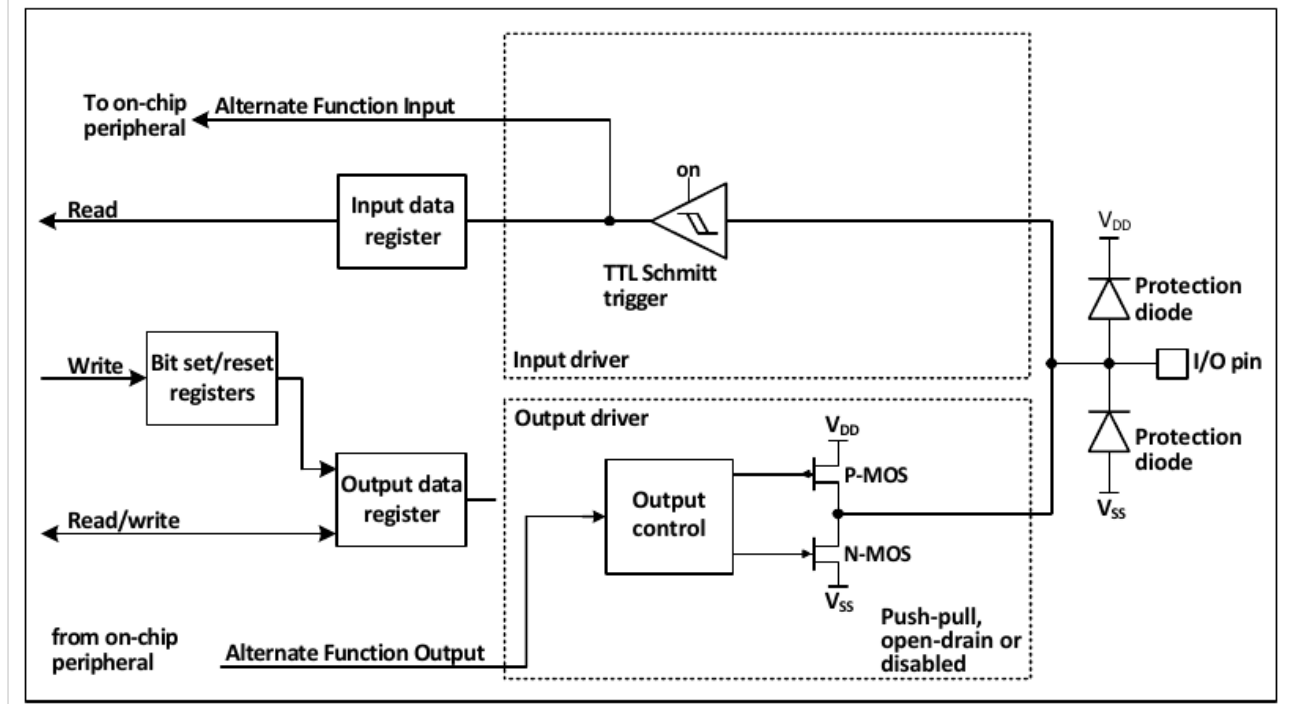


Figure 10-4: GPIO module structure when used by other peripherals (alternate function)

When the alternate function is enabled, the output driver is enabled and can be configured as open-drain or push-pull mode as needed. The Schmitt trigger is also turned on, and the alternate function input and output lines are connected, but the output data register is disconnected. The level appearing on the IO pin will be sampled into the input data register at each PB2 clock. In open-drain mode, reading the input data register will return the current IO port status; in push-pull mode, reading the output data register will return the last written value.

10.2.9 Analog Input Configuration

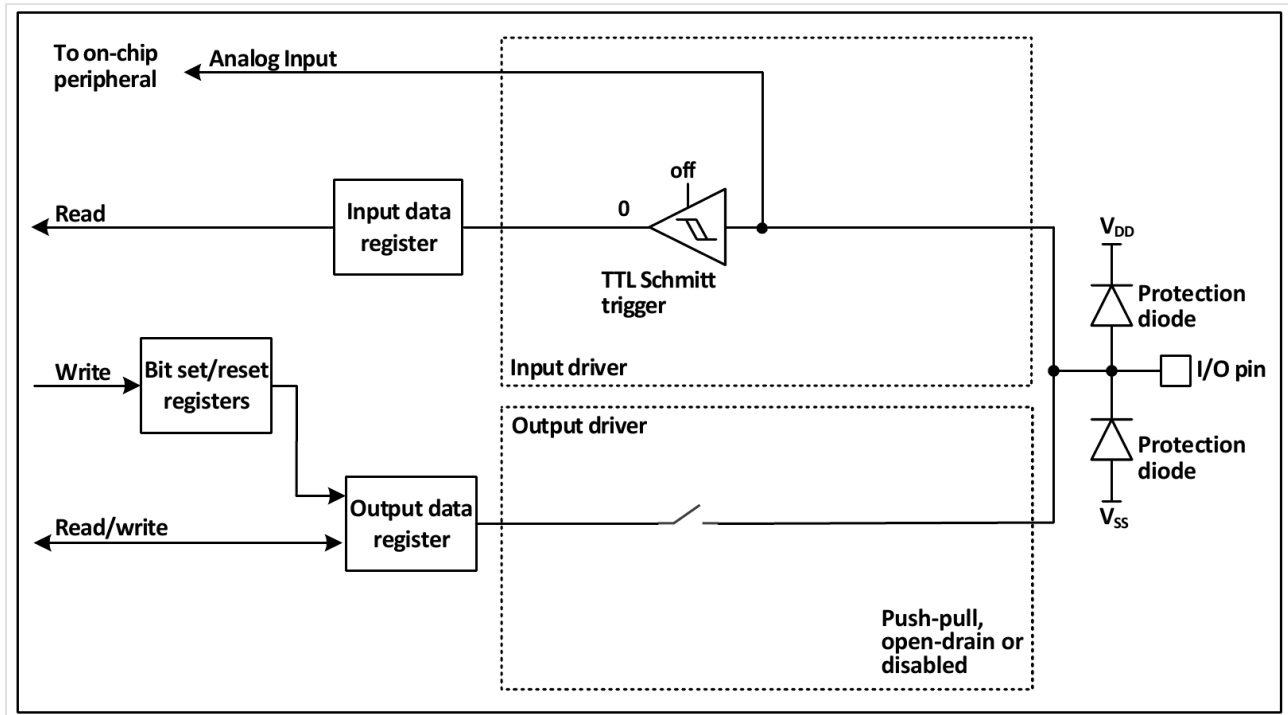


Figure 10-5: GPIO module configuration as analog input

When analog input is enabled, the output buffer is disconnected, the Schmitt trigger input of the input driver is disabled to prevent current consumption on the IO port, the pull-up/pull-down resistors are disabled, and reading the input data register will always return 0.

10.2.10 GPIO Settings for Peripherals

The following tables recommend the GPIO port configuration for the pins of each peripheral.

Table 10-1: Advanced Timer (TIM1/8)

TIM1/8	Configuration	GPIO Configuration
TIM1/8_CHx	Input capture channel x	Floating input
	Output compare channel x	Push-pull alternate output
TIM1/8_CHxN	Complementary output channel x	Push-pull alternate output
TIM1/8_BKIN	Break input	Floating input
TIM1/8_ETR	External trigger clock input	Floating input

Table 10-2: General-Purpose Timer (TIM2/3/4/5)

TIM2/3/4/5 Pin	Configuration	GPIO Configuration
TIM2/3/4/5_CHx	Input capture channel x	Floating input
	Output compare channel x	Push-pull alternate output
TIM2/3/4_ETR	External trigger clock input	Floating input

Table 10-3: Universal Synchronous/Asynchronous Receiver Transmitter (USART1/2/3/4/5/6/7/8/9/10)

USART Pin	Configuration	GPIO Configuration
USARTx_TX	Full-duplex mode	Push-pull alternate output
	Half-duplex synchronous mode	Open-drain alternate output
USARTx_RX	Full-duplex mode	Floating input or pull-up input
	Half-duplex synchronous mode	Not used
USARTx_CK	Synchronous mode	Push-pull alternate output
USARTx_RTS	Hardware flow control	Push-pull alternate output
USARTx_CTS	Hardware flow control	Floating input or pull-up input

Table 10-4: Serial Peripheral Interface (SPI1/2/3) Module

SPI Pin	Configuration	GPIO Configuration
SPIx_SCK	Master mode	Push-pull alternate output
	Slave mode	Floating input
SPIx_MOSI	Full-duplex master mode	Push-pull alternate output
	Full-duplex slave mode	Floating input or pull-up input
SPIx_MOSI	Simple bidirectional data line / master mode	Push-pull alternate output
SPIx_MOSI	Simple bidirectional data line / slave mode	Not used
SPIx_MISO	Full-duplex master mode	Floating input or pull-up input
	Full-duplex slave mode	Push-pull alternate output
SPIx_MISO	Simple bidirectional data line / master mode	Not used
SPIx_MISO	Simple bidirectional data line / slave mode	Push-pull alternate output
SPIx_NSS	Hardware master or slave mode	Floating, pull-up, or pull-down input
	Hardware master mode / NSS output enable	Push-pull alternate output
	Software mode	Not used

Table 10-5: Built-in Audio Bus (I2S2/3) Module

I2S Pin	Configuration	GPIO Configuration
I2Sx_WS	Master mode	Push-pull alternate output
	Slave mode	Floating input
I2Sx_CK	Master mode	Push-pull alternate output
	Slave mode	Floating input
I2Sx_SD	Transmitter	Push-pull alternate output
	Receiver	Floating, pull-up, or pull-down input
I2Sx_MCK	Master mode	Push-pull alternate output
	Slave mode	Not used

Table 10-6: Inter-Integrated Circuit Bus (I2C) Module

I2C Pin	Configuration	GPIO Configuration
I2C_SCL	I2C clock	Open-drain alternate output
I2C_SDA	I2C data	Open-drain alternate output

Table 10-7: I3C Bus (I3C) Module

I3C Pin	Configuration	GPIO Configuration
I3C_SCL	I3C clock	Push-pull alternate output
I3C_SDA	I3C data	Push-pull alternate output

Table 10-8: Controller Area Network (CAN) Module

CAN Pin	GPIO Configuration
CAN1_TX	Push-pull alternate output
CAN1_RX	Floating input or pull-up input

Table 10-9: Digital Video Port (DVP) Module

DVP Pin	Configuration	GPIO Configuration
PCLK	Pixel clock input	Floating input or pull-up input
DATA[11:0]	Pixel data input	Floating input or pull-up input
VSYNC	Frame sync signal input	Floating input or pull-up input
HSYNC	Line sync signal input	Floating input or pull-up input

Table 10-10: LCD-TFT Display Controller (LTDC) Module

LTDC Pin	Configuration	GPIO Configuration
LCD_CLK	Clock output	Alternate push-pull output
LCD_VSYNC	Vertical sync output	Alternate push-pull output
LCD_HSYNC	Horizontal sync output	Alternate push-pull output
LCD_DE	Data enable output	Alternate push-pull output
LCD_R[7:0]	Data: 8-bit red data output	Alternate push-pull output
LCD_G[7:0]	Data: 8-bit green data output	Alternate push-pull output
LCD_B[7:0]	Data: 8-bit blue data output	Alternate push-pull output

Note: The LCD-TFT controller pins must be configured through the user program. Unused pins can be used for other functions. For LTDC output up to 24 bits (RGB888), if using a pixel depth lower than 8bpp to output RGB565 or RGB666 to a 16-bit or 18-bit display, the RGB display data lines must be connected to the MSBs of the LCD-TFT controller RGB data lines. For example, when the LCD-TFT controller is connected to an RGB565 16-bit display, the LCD display's R[4:0], G[4:0], B[4:0] data line pins must be connected to the LCD-TFT controller's LCD_R[7:3], LCD_G[7:3], LCD_B[7:3].

Table 10-11: Secure Digital Input Output (SDIO) Module

SDIO Pin	Configuration	GPIO Configuration
SDIO_CK	Clock	Push-pull alternate output
SDIO_CMD	Command	Push-pull alternate output
SDIO[D7:D0]	Data	Push-pull alternate output

Table 10-12: Flexible Static Memory Controller (FSMC)

FSMC Pin	GPIO Configuration
FSMC_A[23:16]	Push-pull alternate output
FSMC_D[15:0]	Push-pull alternate output
FSMC_CK	Push-pull alternate output
FSMC_NOE	Push-pull alternate output
FSMC_NWE	Push-pull alternate output
FSMC_NE1	Push-pull alternate output
FSMC_NCE2	Push-pull alternate output
FSMC_NWAIT	Floating input or pull-up input
FSMC_NBL[1:0]	Push-pull alternate output

Table 10-13: Analog-to-Digital Converter (ADC) and Digital-to-Analog Converter (DAC)

ADC/DAC Pin	GPIO Configuration
ADC/DAC	Analog input

Table 10-14: Other IO Function Settings

Pin	Function	GPIO Configuration
TAMPER_RTC	RTC output	Hardware auto-configured
	Tamper event input	Hardware auto-configured
MCO	Clock output	Push-pull alternate output
EXTI	External interrupt input	Floating, pull-up, or pull-down input

10.2.11 Alternate Function Remapping GPIO Settings

10.2.11.1 OSC32_IN/OSC32_OUT as GPIO Port PC14/PC15

When LSEON=0, the LSE oscillator pins OSC32_IN/OSC32_OUT can be used as GPIO PC14/PC15, respectively. When LSEON=1, they serve as LSE pins.

10.2.11.2 OSC_IN/OSC_OUT Pins as GPIO Port PD0/PD1

OSC_IN/OSC_OUT can be used as GPIO PD0/PD1, configured via remap register 1 (AFIO_PCFR1).

10.2.11.3 Timer Alternate Function Remapping

Table 10-15: TIM1 Alternate Function Remapping

AF	TIM1_RM=00 Default	TIM1_RM=01 Partial	TIM1_RM=11 Full
TIM1_ETR	PA12	PA12	PE7
TIM1_CH1	PA8	PA8	PE9
TIM1_CH2	PA9	PA9	PE11
TIM1_CH3	PA10	PA10	PB8
TIM1_CH4	PA11	PA11	PA7
TIM1_BKIN	PB12	PA6	PB10

AF	TIM1_RM=00 Default	TIM1_RM=01 Partial	TIM1_RM=11 Full
TIM1_CH1N	PB13	PA7	PE8
TIM1_CH2N	PB14	PB0	PE10
TIM1_CH3N	PB15	PB1	PE12

Table 10-16: TIM2 Alternate Function Remapping

AF	TIM2_RM=00 Default	TIM2_RM=01 Partial	TIM2_RM=10 Partial	TIM2_RM=11 Full
TIM2_ETR	PA0	PA15	PA0	PA15
TIM2_CH1	PA0	PA15	PA0	PA15
TIM2_CH2	PA1	PB3	PA1	PB3
TIM2_CH3	PA2	PA2	PB10	PB10
TIM2_CH4	PA3	PA3	PB11	PB11

Table 10-17: TIM3 Alternate Function Remapping

AF	TIM3_RM=00 Default	TIM3_RM=10 Partial	TIM3_RM=11 Full
TIM3_ETR	PD2	PD2	PD2
TIM3_CH1	PA6	PB4	PC6
TIM3_CH2	PA7	PB5	PC7
TIM3_CH3	PB0	PB0	PC8
TIM3_CH4	PB1	PB1	PC9

Table 10-18: TIM4 Alternate Function Remapping

AF	TIM4_RM=0 Default	TIM4_RM=1 Remap
TIM4_ETR	PE0	PE0
TIM4_CH1	PB6	PD12
TIM4_CH2	PB7	PD13
TIM4_CH3	PB8	PD14
TIM4_CH4	PB9	PD15

Table 10-19: TIM5 Alternate Function Remapping

AF	Default
TIM5_CH1	PA0
TIM5_CH2	PA1
TIM5_CH3	PA2

AF	TIM5CH4_RM=0 Default	TIM5CH4_RM=1 Remap
TIM5_CH4	PA3	LSI internal clock

Table 10-20: TIM8 Alternate Function Remapping

AF	TIM8_RM=0 Default	TIM8_RM=1 Remap
TIM8_ETR	PA0	PA0

AF	TIM8_RM=0 Default	TIM8_RM=1 Remap
TIM8_CH1	PC6	PB6
TIM8_CH2	PC7	PB7
TIM8_CH3	PC8	PB8
TIM8_CH4	PC9	PC13
TIM8_BKIN	PA6	PB9
TIM8_CH1N	PA7	PA13
TIM8_CH2N	PB0	PA14
TIM8_CH3N	PB1	PA15

10.2.11.4 USART Alternate Function Remapping

Table 10-21: USART1 Alternate Function Remapping

AF	USART1_RM1=0, USART1_RM=0 Default	USART1_RM1=0, USART1_RM=1 Remap	USART1_RM1=1, USART1_RM=0 Remap	USART1_RM1=1, USART1_RM=1 Remap
USART1_CK	PA8	PA8	PA10	PA5
USART1_TX	PA9	PB6	PB15	PA6
USART1_RX	PA10	PB7	PA8	PA7
USART1_CTS	PA11	PA11	PA5	PC4
USART1_RTS	PA12	PA12	PA9	PC5

Note: (1) USART1_RM is AFIO_PCFR1 register bit2, the mapping configuration low bit. (2) USART1_RM1 is AFIO_PCFR2 register bit26, the mapping configuration high bit.

Table 10-22: USART2 Alternate Function Remapping

AF	USART2_RM=0 Default	USART2_RM=1 Remap
USART2_CTS	PA0	PD3
USART2_RTS	PA1	PD4
USART2_TX	PA2	PD5
USART2_RX	PA3	PD6
USART2_CK	PA4	PD7

Table 10-23: USART3 Alternate Function Remapping

AF	USART3_RM=00 Default	USART3_RM=01 Partial	USART3_RM=10 Partial	USART3_RM=11 Full
USART3_TX	PB10	PC10	PA13	PD8
USART3_RX	PB11	PC11	PA14	PD9
USART3_CK	PB12	PC12	PD10	PD10
USART3_CTS	PB13	PB13	PD11	PD11
USART3_RTS	PB14	PB14	PD12	PD12

Table 10-24: USART4 Alternate Function Remapping

AF	USART4_RM=00 Default	USART4_RM=01 Remap	USART4_RM=1x Remap
USART4_TX	PC10	PB0	PE0
USART4_RX	PC11	PB1	PE1
USART4_CK	PC12	PB2	PE12
USART4_CTS	PA13	PC4	PB8
USART4_RTS	PD2	PC5	PB9

Table 10-25: USART5 Alternate Function Remapping

AF	USART5_RM=00 Default	USART5_RM=01 Remap	USART5_RM=1x Remap
USART5_TX	PC12	PB4	PE8
USART5_RX	PD2	PB5	PE9
USART5_CK	PB5	PB3	PB2
USART5_CTS	PB3	PB8	PB0
USART5_RTS	PB4	PB9	PB1

Table 10-26: USART6 Alternate Function Remapping

AF	USART6_RM=00 Default	USART6_RM=01 Remap	USART6_RM=10 Remap	USART6_RM=11 Remap
USART6_TX	PC0	PB8	PE10	PC0
USART6_RX	PC1	PB9	PE11	PC1
USART6_CK	PB0	PB5	PA13	PA5
USART6_CTS	PC2	PE0	PA14	PC2
USART6_RTS	PC3	PE1	PA15	PC3

Table 10-27: USART7 Alternate Function Remapping

AF	USART7_RM=00 Default	USART7_RM=01 Remap	USART7_RM=10 Remap	USART7_RM=11 Remap
USART7_TX	PC2	PA6	PA9	PA9
USART7_RX	PC3	PA7	PA10	PA10
USART7_CK	PC1	PB0	PA8	PA8
USART7_CTS	PC4	PC4	PB6	PC8
USART7_RTS	PC5	PC5	PB7	PC9

Table 10-28: USART8 Alternate Function Remapping

AF	USART8_RM=00 Default	USART8_RM=01 Remap	USART8_RM=1x Remap
USART8_TX	PC4	PA14	PB15
USART8_RX	PC5	PA15	PB14
USART8_CK	PA7	PB15	PB11
USART8_CTS	PB0	PE10	PB13
USART8_RTS	PB1	PE11	PB12

Table 10-29: USART9 Alternate Function Remapping

AF	USART9_RM=00 Default	USART9_RM=01 Remap	USART9_RM=1x Remap
USART9_TX	PB1	PA4	PE3
USART9_RX	PB2	PA5	PE4
USART9_CK	PD8	PD8	PE2
USART9_CTS	PE8	PE8	PE5
USART9_RTS	PE9	PE9	PE6

Table 10-30: USART10 Alternate Function Remapping

AF	USART10_RM=00 Default	USART10_RM=01 Remap	USART10_RM=1x Remap
USART10_TX	PE0	PB8	PD11
USART10_RX	PE1	PB9	PD12
USART10_CK	PE12	PE12	PD10
USART10_CTS	PC14	PC14	PD13
USART10_RTS	PC15	PC15	PD14

10.2.11.5 SPI Alternate Function Remapping

Table 10-31: SPI1 Alternate Function Remapping

AF	SPI1_RM=0 Default	SPI1_RM=1 Remap
SPI1_NSS	PA4	PA15
SPI1_SCK	PA5	PB3
SPI1_MISO	PA6	PB4
SPI1_MOSI	PA7	PB5

Table 10-32: SPI3 Alternate Function Remapping

AF	SPI3_RM=0 Default	SPI3_RM=1 Remap
SPI3_NSS	PA15	PA4
SPI3_SCK	PB3	PC10
SPI3_MISO	PB4	PC11
SPI3_MOSI	PB5	PC12

Table 10-33: I2S3 Alternate Function Remapping

AF	SPI3_RM=0 Default	SPI3_RM=1 Remap
I2S3_WS	PA15	PA4
I2S3_CK	PB3	PC10
I2S3_MCK	PC7	PC7
I2S3_SD	PB5	PC12

Note: When the I2SMOD bit of the SPI_I2S_CFGR register is set to 1, I2S mode is selected.

10.2.11.6 I2C Alternate Function Remapping

Table 10-34: I2C Alternate Function Remapping

AF	I2C_RM=0 Default	I2C_RM=1 Remap
I2C_SCL	PB6	PB8
I2C_SDA	PB7	PB9

10.2.11.7 CAN Alternate Function Remapping

Table 10-35: CAN1 Alternate Function Remapping

AF	CAN1_RM=00 Default	CAN1_RM=10 Remap	CAN1_RM=11 Remap
CAN1_RX	PA11	PB8	PD0
CAN1_TX	PA12	PB9	PD1

10.2.11.8 ADC Alternate Function Remapping

Table 10-36: ADC1 External Trigger Injected Conversion Alternate Function Remapping

AF	ADC1_ETRGINJ_RM=0 Default	ADC1_ETRGINJ_RM=1 Remap
ADC1 external trigger injected conversion	ADC1 external trigger injected conversion connected to EXTI15	ADC1 external trigger injected conversion connected to TIM8_CH4

Table 10-37: ADC1 External Trigger Regular Conversion Alternate Function Remapping

AF	ADC1_ETRGREG_RM=0 Default	ADC1_ETRGREG_RM=1 Remap
ADC1 external trigger regular conversion	ADC1 external trigger regular conversion connected to EXTI11	ADC1 external trigger regular conversion connected to TIM8_TRGO

Table 10-38: ADC2 External Trigger Injected Conversion Alternate Function Remapping

AF	ADC2_ETRGINJ_RM=0 Default	ADC2_ETRGINJ_RM=1 Remap
ADC2 external trigger injected conversion	ADC2 external trigger injected conversion connected to EXTI15	ADC2 external trigger injected conversion connected to TIM8_CH4

Table 10-39: ADC2 External Trigger Regular Conversion Alternate Function Remapping

AF	ADC2_ETRGREG_RM=0 Default	ADC2_ETRGREG_RM=1 Remap
ADC2 external trigger regular conversion	ADC2 external trigger regular conversion connected to EXTI11	ADC2 external trigger regular conversion connected to TIM8_TRGO

Note: ADC1 remapping is only supported on products that have TIM8. For details, see the relevant datasheet.

10.2.11.9 FSMC Alternate Function Remapping

Table 10-40: FSMC Alternate Function Remapping

AF	FSMCEN=0 Default	FSMCEN=1 Remap
NBL[0]	PE0	PE0
NBL[1]	PE1	PE1

AF	FSMCEN=0 Default	FSMCEN=1 Remap
CLK	PD3	PD3
NOE	PD4	PD4
NWE	PD5	PD5
NWAIT	PD6	PD6
NE1/NCE2	PD7	PD7
NADV	PB7	PB7
A16	PD11	PD11
A17	PD12	PD12
A18	PD13	PD13
A19	PE3	PE3
A20	PE4	PE4
A21	PE5	PE5
A22	PE6	PE6
A23	PE2	PE2
D0	PD14	PD14
D1	PD15	PD15
D2	PD0	PD0
D3	PD1	PD1
D4	PE7	PC4
D5	PE8	PE8
D6	PE9	PE9
D7	PE10	PE10
D8	PE11	PE11
D9	PB11	PB11
D10	PB12	PB12
D11	PB13	PB13
D12	PB14	PB14
D13	PD8	PD8
D14	PD9	PD9
D15	PD10	PD10

10.2.11.10 FSMC_NADV Alternate Function Remapping

Table 10-41: FSMC_NADV Alternate Function Remapping

AF	FSMCEN=1 Default	FSMCEN=1 & USBHS2EN=1 & RB_UD_RST_SIE=0
FSMC_NADV	PB7	PD2

Note: (1) When FSMC_NADV=1, the FSMC_NADV output is disabled.

10.2.11.11 DVP Alternate Function Remapping

Table 10-42: DVP Alternate Function Remapping

AF	DVPEN=1, DVP_RM=0 Default	DVPEN=1 & USBHS1EN=1 & RB_UD_RST_SIE=0	DVP_RM=1 Remap
PCLK	PA6	PA6	PA6
HSYNC	PA4	PA4	PA4
VSYNC	PA5	PA5	PA5
D0	PA9	PA9	PA9
D1	PA10	PA10	PA10
D2	PC8	PC8	PA8
D3	PC9	PC9	PB4
D4	PC11	PC11	PC11
D5	PB6	PB3	PB3
D6	PB8	PB8	PB8
D7	PB9	PB9	PB9
D8	PC10	PC10	PC10
D9	PC12	PC12	PC12
D10	PD6	PD6	PB5
D11	PD2	PD2	PD2

10.2.11.12 SDIO Alternate Function Remapping

Table 10-43: SDIO Alternate Function Remapping

AF	SDIO_RM=0 Default	SDIOEN=1 & ETHMACEN=1	SDIO_RM=1 Remap
CK	PC12	PC12	PC12
CMD	PD2	PD2	PD2
SD0	PC8	PB14	PB14
SD1	PC9	PB15	PB15
SD2	PC10	PC10	PC10
SD3	PC11	PC11	PC11
SD4	PB8	PB8	PB8
SD5	PB9	PB9	PB9
SD6	PC6	PC6	PA9
SD7	PC7	PC7	PA10

10.2.11.13 I3C Alternate Function Remapping

Table 10-44: I3C Alternate Function Remapping

AF	I3C_RM=0 Default	I3C_RM=1 Default
I3C_SCL	PB10	PB4

AF	I3C_RM=0 Default	I3C_RM=1 Default
I3C_SDA	PB11	PB5

10.2.11.14 LTDC Alternate Function Remapping

Table 10-45: LTDC Alternate Function Remapping

AF	LTDC_RM=0 Default	LTDC_RM=1 Remap
CLK	PE1	PA10
HSYNC	PB4	PB8
VSYNC	PB5	PB9
DE	PE0	PA9
R[0]	PC0	PC0
R[1]	PC1	PC1
R[2]	PC2	PC2
R[3]	PC3	PC3
R[4]	PA1	PA1
R[5]	PA2	PA2
R[6]	PA3	PA3
R[7]	PA7	PA7
G[0]	PC4	PC4
G[1]	PC5	PC5
G[2]	PB0	PB0
G[3]	PB1	PB1
G[4]	PB2	PB3
G[5]	PD8	PA4
G[6]	PB12	PB12
G[7]	PB13	PB13
B[0]	PB14	PB14
B[1]	PB15	PB15
B[2]	PB11	PB11
B[3]	PB10	PB10
B[4]	PA8	PA8
B[5]	PA13	PA13
B[6]	PA14	PA14
B[7]	PA15	PA15

10.3 Register Description

10.3.1 GPIO Register Description

Unless otherwise specified, GPIO registers must be accessed as word operations (operating on these registers as 32-bit).

Table 10-46: GPIO-related register list

Name	Address	Description	Reset Value
R32_GPIOA_CFGLR	0x40010800	PA port configuration register low	0x44444444
R32_GPIOB_CFGLR	0x40010C00	PB port configuration register low	0x44444444
R32_GPIOC_CFGLR	0x40011000	PC port configuration register low	0x44444444
R32_GPIOD_CFGLR	0x40011400	PD port configuration register low	0x44444444
R32_GPIOE_CFGLR	0x40011800	PE port configuration register low	0x44444444
R32_GPIOA_CFGHR	0x40010804	PA port configuration register high	0x44444444
R32_GPIOB_CFGHR	0x40010C04	PB port configuration register high	0x44444444
R32_GPIOC_CFGHR	0x40011004	PC port configuration register high	0x44444444
R32_GPIOD_CFGHR	0x40011404	PD port configuration register high	0x44444444
R32_GPIOE_CFGHR	0x40011804	PE port configuration register high	0x44444444
R32_GPIOA_INDR	0x40010808	PA port input data register	0x0000XXXX
R32_GPIOB_INDR	0x40010C08	PB port input data register	0x0000XXXX
R32_GPIOC_INDR	0x40011008	PC port input data register	0x0000XXXX
R32_GPIOD_INDR	0x40011408	PD port input data register	0x0000XXXX
R32_GPIOE_INDR	0x40011808	PE port input data register	0x0000XXXX
R32_GPIOA_OUTDR	0x4001080C	PA port output data register	0x00000000
R32_GPIOB_OUTDR	0x40010C0C	PB port output data register	0x00000000
R32_GPIOC_OUTDR	0x4001100C	PC port output data register	0x00000000
R32_GPIOD_OUTDR	0x4001140C	PD port output data register	0x00000000
R32_GPIOE_OUTDR	0x4001180C	PE port output data register	0x00000000
R32_GPIOA_BSHR	0x40010810	PA port bit set/reset register	0x00000000
R32_GPIOB_BSHR	0x40010C10	PB port bit set/reset register	0x00000000
R32_GPIOC_BSHR	0x40011010	PC port bit set/reset register	0x00000000
R32_GPIOD_BSHR	0x40011410	PD port bit set/reset register	0x00000000
R32_GPIOE_BSHR	0x40011810	PE port bit set/reset register	0x00000000
R32_GPIOA_BCR	0x40010814	PA port bit reset register	0x00000000
R32_GPIOB_BCR	0x40010C14	PB port bit reset register	0x00000000
R32_GPIOC_BCR	0x40011014	PC port bit reset register	0x00000000
R32_GPIOD_BCR	0x40011414	PD port bit reset register	0x00000000
R32_GPIOE_BCR	0x40011814	PE port bit reset register	0x00000000
R32_GPIOA_LCKR	0x40010818	PA port lock configuration register	0x00000000

Name	Address	Description	Reset Value
R32_GPIOB_LCKR	0x40010C18	PB port lock configuration register	0x00000000
R32_GPIOC_LCKR	0x40011018	PC port lock configuration register	0x00000000
R32_GPIOD_LCKR	0x40011418	PD port lock configuration register	0x00000000
R32_GPIOE_LCKR	0x40011818	PE port lock configuration register	0x00000000

10.3.1.1 GPIO Configuration Register Low (GPIOx_CFGLR) (x=A/B/C/D/E)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF7[1:0]	MODE7[1:0]	CNF6[1:0]	MODE6[1:0]	CNF5[1:0]	MODE5[1:0]	CNF4[1:0]	MODE4[1:0]								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF3[1:0]	MODE3[1:0]	CNF2[1:0]	MODE2[1:0]	CNF1[1:0]	MODE1[1:0]	CNF0[1:0]	MODE0[1:0]								

Bits	Name	Access	Description	Reset
[31:30] [27:26] [23:22] [19:18] [15:14] [11:10] [7:6] [3:2]	CNFy[1:0]	RW	(y=0–7), port x configuration bits, used to configure the corresponding port. In input mode (MODE=00b): 00: Analog input mode; 01: Floating input mode; 10: Input with pull-up/pull-down. 11: Reserved. In output mode (MODE>00b): 00: General-purpose push-pull output mode; 01: General-purpose open-drain output mode; 10: Alternate function push-pull output mode; 11: Alternate function open-drain output mode.	01b
[29:28] [25:24] [21:20] [17:16] [13:12] [9:8] [5:4] [1:0]	MODEy[1:0]	RW	(y=0–7), port x mode selection bits, used to configure the corresponding port. 00: Input mode; 01: Output mode, max speed 10 MHz; 10: Output mode, max speed 2 MHz; 11: Output mode, max speed 50 MHz.	00b

10.3.1.2 GPIO Configuration Register High (GPIOx_CFGHR) (x=A/B/C/D/E)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CNF15[1:0]	MODE15[1:0]	CNF14[1:0]	MODE14[1:0]	CNF13[1:0]	MODE13[1:0]	CNF12[1:0]	MODE12[1:0]								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNF11[1:0]	MODE11[1:0]	CNF10[1:0]	MODE10[1:0]	CNF9[1:0]	MODE9[1:0]	CNF8[1:0]	MODE8[1:0]								

Bits	Name	Access	Description	Reset
[31:30] [27:26]	CNFy[1:0]	RW	(y=8–15), port x configuration bits, used to configure the corresponding port.	01b

Bits	Name	Access	Description	Reset
[23:22] [19:18] [15:14] [11:10] [7:6] [3:2]			In input mode (MODE=00b): 00: Analog input mode; 01: Floating input mode; 10: Input with pull-up/pull-down. 11: Reserved. In output mode (MODE>00b): 00: General-purpose push-pull output mode; 01: General-purpose open-drain output mode; 10: Alternate function push-pull output mode; 11: Alternate function open-drain output mode.	
[29:28] [25:24] [21:20] [17:16] [13:12] [9:8] [5:4] [1:0]	MODEy[1:0]	RW	(y=8–15), port x mode bits, used to configure the corresponding port. 00: Input mode; 01: Output mode, max speed 10 MHz; 10: Output mode, max speed 2 MHz; 11: Output mode, max speed 50 MHz.	00b

10.3.1.3 Port Input Data Register (GPIOx_INDR) (x=A/B/C/D/E)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IDR15	IDR14	IDR13	IDR12	IDR11	IDR10	IDR9	IDR8	IDR7	IDR6	IDR5	IDR4	IDR3	IDR2	IDR1	IDR0

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	IDRy	RO	(y=0–15), port input data. These bits are read-only and can only be read as 16-bit values. The read value is the high/low state of the corresponding pin.	X

10.3.1.4 Port Output Data Register (GPIOx_OUTDR) (x=A/B/C/D/E)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ODR15	ODR14	ODR13	ODR12	ODR11	ODR10	ODR9	ODR8	ODR7	ODR6	ODR5	ODR4	ODR3	ODR2	ODR1	ODR0

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	ODRy	RW	For output mode: (y=0–15), port output data. This data can only be accessed as 16-bit. The IO port outputs the value of this register.	0

Bits	Name	Access	Description	Reset
			For input mode with pull-up/pull-down: 1: Pull-up input; 0: Pull-down input.	

10.3.1.5 Port Bit Set/Reset Register (GPIOx_BSHR) (x=A/B/C/D/E)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0

Bits	Name	Access	Description	Reset
[31:16]	BRy	WO	(y=0–15), setting these bits clears the corresponding OUTDR bit; writing 0 has no effect. These bits can only be accessed as 16-bit. If both BR and BS bits are set simultaneously, the BS bit takes effect.	0
[15:0]	BSy	WO	(y=0–15), setting these bits sets the corresponding OUTDR bit; writing 0 has no effect. These bits can only be accessed as 16-bit. If both BR and BS bits are set simultaneously, the BS bit takes effect.	0

10.3.1.6 Port Bit Reset Register (GPIOx_BCR) (x=A/B/C/D/E)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	BRy	WO	(y=0–15), setting these bits clears the corresponding OUTDR bit; writing 0 has no effect. These bits can only be accessed as 16-bit.	0

10.3.1.7 Lock Configuration Register (GPIOx_LCKR) (x=A/B/C/D/E)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														LCKK	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LCK15	LCK14	LCK13	LCK12	LCK11	LCK10	LCK9	LCK8	LCK7	LCK6	LCK5	LCK4	LCK3	LCK2	LCK1	LCK0

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
16	LCKK	RW	Lock key. It can be written via a specific sequence to activate the lock, but can be read at any time. When read as 0, the lock is not active; when read	0

Bits	Name	Access	Description	Reset
			as 1, the lock is active. The lock key write sequence is: write 1 – write 0 – write 1 – read 0 – read 1. The last step is not required but can be used to confirm the lock key has been activated. Any error during the write sequence will not activate the lock, and the LCK[15:0] value cannot be changed during the write sequence. Once the lock is active, the port configuration can only be changed after the next reset.	
[15:0]	LCKy	RW	(y=0–15), when these bits are 1, the corresponding port configuration is locked. These bits can only be changed before LCKK is locked. The locked configuration refers to the configuration registers GPIOx_CFGLR and GPIOx_CFGHR.	0

Note: After a LOCK sequence has been executed on the corresponding port bits, the port bit configuration cannot be changed until the next system reset.

10.3.2 AFIO Registers

Unless otherwise specified, AFIO registers must be accessed as word operations (operating on these registers as 32-bit).

Table 10-47: AFIO-related register list

Name	Address	Description	Reset Value
R32_AFIO_ECR	0x40010000	Event control register	0x00000000
R32_AFIO_PCFR1	0x40010004	Remap register 1	0x00000000
R32_AFIO_EXTICR1	0x40010008	External interrupt configuration register 1	0x00000000
R32_AFIO_EXTICR2	0x4001000C	External interrupt configuration register 2	0x00000000
R32_AFIO_EXTICR3	0x40010010	External interrupt configuration register 3	0x00000000
R32_AFIO_EXTICR4	0x40010014	External interrupt configuration register 4	0x00000000
R32_AFIO_PCFR2	0x4001001C	Remap register 2	0x00000000

10.3.2.1 Event Control Register (AFIO_ECR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						EVOE		PORT[2:0]			PIN[3:0]				

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
7	EVOE	RW	Event output enable. Setting this bit connects the core's EVENTOUT to the IO port selected by PORT and PIN.	0
[6:4]	PORT[2:0]	RW	Used to select the port for the core's EVENTOUT output: 000: Select PA port; 001: Select PB port; 010: Select PC port; 011: Select PD port; 100: Select PE port; Others: Reserved.	000b

Bits	Name	Access	Description	Reset
[3:0]	PIN[3:0]	RW	The value of these bits determines the specific pin number of the port to which the core outputs EVENTOUT. Values 0–15 correspond to pins 0–15 of the selected PORT (Px).	0

10.3.2.2 Remap Register 1 (AFIO_PCFR1)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18
ETHPHY_LED_RM	PTP_PPS_RM	TIM2ITR1_RM	SPI3_RM	Reserved	SW_CFG[2:0]	Reserved	ADC2_ETRGREG_RM	ADC2_ETRGINJ_RM	ADC1_ETRGREG_RM				

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PDOPD1_RM	CAN1_RM[1:0]	TIM4_RM	TIM3_RM[1:0]	TIM2_RM[1:0]	TIM1_RM[1:0]	USART3_RM[1:0]	USART2_RM	USART1_RM	I2C_RM	SPI1_RM					

Bits	Name	Access	Description	Reset
31	ETHPHY_LED_RM	RW	ETHPHY LED remapping: 1: LED0:PD14, LED1:PD15; 0: LED0:PE8, LED1:PE9.	0
30	PTP_PPS_RM	RW	Ethernet PTP PPS remapping: 1: PTP PPS output to PB5 pin; 0: PTP PPS not output to PB5 pin.	0
29	TIM2ITR1_RM	RW	TIM2 internal trigger 1 remapping: 1: Reserved; 0: Internally connect TIM2_ITR1 to Ethernet PTP output.	0
28	SPI3_RM	RW	SPI3 remapping: 1: Remap (NSS/PA4, SCK/PC10, MISO/PC11, MOSI/PC12); 0: Default (NSS/PA15, SCK/PB3, MISO/PB4, MOSI/PB5).	0
27	Reserved	RO	Reserved.	0
[26:24]	SW_CFG[2:0]	RW	These bits configure the IO port for SW function and trace function. SWD (SDI) is the debug interface to access the core. After system reset, it always serves as the SWD port. 0xx: Enable SWD (SDI); 100: Disable SWD (SDI), use as GPIO function; Others: Invalid.	0
[23:21]	Reserved	RO	Reserved.	0
20	ADC2_ETRGREG_RM	RW	ADC2 external trigger regular conversion remapping bit. 1: ADC2 external trigger regular conversion connected to TIM8_TRGO; 0: ADC2 external trigger regular conversion connected to EXTI11.	0
19	ADC2_ETRGINJ_RM	RW	ADC2 external trigger injected conversion remapping bit. 1: ADC2 external trigger injected conversion connected to TIM8_CH4; 0: ADC2 external trigger injected conversion connected to EXTI15.	0
18	ADC1_ETRGREG_RM	RW	ADC1 external trigger regular conversion remapping bit. 1: ADC1 external trigger regular conversion connected to TIM8_TRGO;	0

Bits	Name	Access	Description	Reset
			0: ADC1 external trigger regular conversion connected to EXTI11.	
17	ADC1_ETRGINJ_RM	RW	ADC1 external trigger injected conversion remapping bit. 1: ADC1 external trigger injected conversion connected to TIM8_CH4; 0: ADC1 external trigger injected conversion connected to EXTI15.	0
16	TIM5CH4_RM	RW	Timer 5 channel 4 remapping: 1: Remap, Timer 5 channel 4 mapped to LSI internal clock; 0: Default, Timer 5 channel 4 default mapped to PA3.	0
15	PD0PD1_RM	RW	Pin PD0 & PD1 remapping bit; this bit can be read/written by the user. It controls whether the GPIO function of PD0 and PD1 is remapped, i.e., PD0 & PD1 mapped to OSC_IN & OSC_OUT. 1: Pins used as GPIO; 0: Pins used as crystal oscillator pins. Note: For LQFP100 package, since PD0 and PD1 are inherently function pins, there is no need for software remapping.	0
[14:13]	CAN1_RM[1:0]	RW	CAN1 alternate function remapping bits; these bits can be read/written by the user. Controls the remapping of CAN1_RX and CAN1_TX: 00: CAN1_RX mapped to PA11, CAN1_TX mapped to PA12; 10: CAN1_RX mapped to PB8, CAN1_TX mapped to PB9; 01: Reserved; 11: CAN1_RX mapped to PD0, CAN1_TX mapped to PD1.	0
12	TIM4_RM	RW	Timer 4 remapping bit; this bit can be read/written by the user. It controls the remapping of Timer 4 channels 1–4 on the GPIO port: 1: Remap (ETR/PE0, CH1/PD12, CH2/PD13, CH3/PD14, CH4/PD15); 0: Default (ETR/PE0, CH1/PB6, CH2/PB7, CH3/PB8, CH4/PB9).	0
[11:10]	TIM3_RM[1:0]	RW	Timer 3 remapping bits; these bits can be read/written by the user. They control the remapping of Timer 3 channels 1–4 on the GPIO port: 00: Default (ETR/PD2, CH1/PA6, CH2/PA7, CH3/PB0, CH4/PB1); 01: Reserved; 10: Partial remap (ETR/PD2, CH1/PB4, CH2/PB5, CH3/PB0, CH4/PB1); 11: Full remap (ETR/PD2, CH1/PC6, CH2/PC7, CH3/PC8, CH4/PC9). Note: Remapping does not affect TIM3_ETR on PD2.	0
[9:8]	TIM2_RM[1:0]	RW	Timer 2 remapping bits. These bits can be read/written by the user. They control the mapping of Timer 2 channels 1–4 and external trigger (ETR) on the GPIO port: 00: Default (ETR/PA0, CH1/PA0, CH2/PA1, CH3/PA2, CH4/PA3); 01: Partial remap (ETR/PA15, CH1/PA15, CH2/PB3, CH3/PA2, CH4/PA3); 10: Partial remap (ETR/PA0, CH1/PA0, CH2/PA1, CH3/PB10, CH4/PB11);	0

Bits	Name	Access	Description	Reset
			11: Full remap (ETR/PA15, CH1/PA15, CH2/PB3, CH3/PB10, CH4/PB11).	
[7:6]	TIM1_RM[1:0]	RW	Timer 1 remapping bits. These bits can be read/written by the user. They control the mapping of Timer 1 channels 1–4, 1N–3N, external trigger (ETR), and break input (BKIN) on the GPIO port: 00: Default (ETR/PA12, CH1/PA8, CH2/PA9, CH3/PA10, CH4/PA11, BKIN/PB12, CH1N/PB13, CH2N/PB14, CH3N/PB15); 01: Partial remap (ETR/PA12, CH1/PA8, CH2/PA9, CH3/PA10, CH4/PA11, BKIN/PA6, CH1N/PA7, CH2N/PB0, CH3N/PB1); 10: Reserved; 11: Full remap (ETR/PE7, CH1/PE9, CH2/PE11, CH3/PB8, CH4/PA7, BKIN/PB10, CH1N/PE8, CH2N/PE10, CH3N/PE12).	0
[5:4]	USART3_RM[1:0]	RW	USART3 remapping bits; these bits can be read/written by the user. They control the mapping of USART3 CTS, RTS, CK, TX, and RX alternate functions on the GPIO port: 00: Default (TX/PB10, RX/PB11, CK/PB12, CTS/PB13, RTS/PB14); 01: Partial remap (TX/PC10, RX/PC11, CK/PC12, CTS/PB13, RTS/PB14); 10: Partial remap (TX/PA13, RX/PA14, CK/PD10, CTS/PD11, RTS/PD12); 11: Full remap (TX/PD8, RX/PD9, CK/PD10, CTS/PD11, RTS/PD12).	0
3	USART2_RM	RW	USART2 remapping bit. This bit can be read/written by the user. It controls the mapping of USART2 CTS, RTS, CK, TX, and RX alternate functions on the GPIO port: 1: Remap (CTS/PD3, RTS/PD4, TX/PD5, RX/PD6, CK/PD7); 0: Default (CTS/PA0, RTS/PA1, TX/PA2, RX/PA3, CK/PA4).	0
2	USART1_RM	RW	USART1 mapping configuration low bit (used together with AFIO_PCFR2 register bit26 USART1_RM1). 00: Default (CK/PA8, TX/PA9, RX/PA10, CTS/PA11, RTS/PA12); 01: Remap (CK/PA8, TX/PB6, RX/PB7, CTS/PA11, RTS/PA12); 10: Remap (CK/PA10, TX/PB15, RX/PA8, CTS/PA5, RTS/PA9); 11: Remap (CK/PA5, TX/PA6, RX/PA7, CTS/PC4, RTS/PC5).	0
1	I2C_RM	RW	I2C remapping. This bit can be read/written by the user. It controls the mapping of I2C SCL and SDA alternate functions on the GPIO port: 1: Remap (SCL/PB8, SDA/PB9); 0: Default (SCL/PB6, SDA/PB7).	0
0	SPI1_RM	RW	SPI1 remapping. This bit can be read/written by the user. It controls the mapping of SPI1 NSS, SCK, MISO, and MOSI alternate functions on the GPIO port: 1: Remap (NSS/PA15, SCK/PB3, MISO/PB4, MOSI/PB5); 0: Default (NSS/PA4, SCK/PA5, MISO/PA6, MOSI/PA7).	0

10.3.2.3 External Interrupt Configuration Register 1 (AFIO_EXTICR1)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI3[3:0]				EXTI2[3:0]				EXTI1[3:0]				EXTI0[3:0]			

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=0–3), external interrupt input pin configuration bits. Used to determine which port pin the external interrupt pin is mapped to: 0000: PA port pin x; 0001: PB port pin x; 0010: PC port pin x; 0011: PD port pin x; 0100: PE port pin x; Others: Reserved.	0000b

10.3.2.4 External Interrupt Configuration Register 2 (AFIO_EXTICR2)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI7[3:0]				EXTI6[3:0]				EXTI5[3:0]				EXTI4[3:0]			

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=4–7), external interrupt input pin configuration bits. Used to determine which port pin the external interrupt pin is mapped to: 0000: PA port pin x; 0001: PB port pin x; 0010: PC port pin x; 0011: PD port pin x; 0100: PE port pin x; Others: Reserved.	0000b

10.3.2.5 External Interrupt Configuration Register 3 (AFIO_EXTICR3)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI11[3:0]				EXTI10[3:0]				EXTI9[3:0]				EXTI8[3:0]			

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:12] [11:8]	EXTIx[3:0]	RW	(x=8–11), external interrupt input pin configuration bits. Used to determine which port pin the external interrupt pin is mapped to:	0000b

Bits	Name	Access	Description	Reset
[7:4] [3:0]			0000: PA port pin x; 0001: PB port pin x; 0010: PC port pin x; 0011: PD port pin x; 0100: PE port pin x; Others: Reserved.	

10.3.2.6 External Interrupt Configuration Register 4 (AFIO_EXTICR4)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXTI15[3:0]				EXTI14[3:0]				EXTI13[3:0]				EXTI12[3:0]			

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:12] [11:8] [7:4] [3:0]	EXTIx[3:0]	RW	(x=12–15), external interrupt input pin configuration bits. Used to determine which port pin the external interrupt pin is mapped to: 0000: PA port pin x; 0001: PB port pin x; 0010: PC port pin x; 0011: PD port pin x; 0100: PE port pin x; Others: Reserved.	0000b

10.3.2.7 Remap Register 2 (AFIO_PCFR2)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
USART10_RM	USART9_RM	Reserved	USART1_RM1	USART8_RM[1:0]	USART7_RM[1:0]	USART6_RM[1:0]	USART5_RM[1:0]	USART4_RM[1:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	I3C_RM	DVP_RM	SDIO_RM	LTDC_RM	FSMC_NADV	Reserved	TIM8_RM	Reserved							

Bits	Name	Access	Description	Reset
[31:30]	USART10_RM	RW	USART10 remapping: 00: Default (CK/PE12, TX/PE0, RX/PE1, CTS/PC14, RTS/PC15); 01: Remap (CK/PE12, TX/PB8, RX/PB9, CTS/PC14, RTS/PC15); 1x: Remap (CK/PD10, TX/PD11, RX/PD12, CTS/PD13, RTS/PD14).	0
[29:28]	USART9_RM	RW	USART9 remapping: 00: Default (CK/PD8, TX/PB1, RX/PB2, CTS/PE8, RTS/PE9); 01: Remap (CK/PD8, TX/PA4, RX/PA5, CTS/PE8, RTS/PE9); 1x: Remap (CK/PE2, TX/PE3, RX/PE4, CTS/PE5, RTS/PE6).	0
27	Reserved	RO	Reserved.	0
26	USART1_RM1	RW	USART1 mapping configuration high bit (used together with AFIO_PCFR1 register bit2 USART1_RM).	0

Bits	Name	Access	Description	Reset
			00: Default (CK/PA8, TX/PA9, RX/PA10, CTS/PA11, RTS/PA12); 01: Remap (CK/PA8, TX/PB6, RX/PB7, CTS/PA11, RTS/PA12); 10: Remap (CK/PA10, TX/PB15, RX/PA8, CTS/PA5, RTS/PA9); 11: Remap (CK/PA5, TX/PA6, RX/PA7, CTS/PC4, RTS/PC5).	
[25:24]	USART8_RM[1:0]	RW	USART8 remapping: 00: Default (TX/PC4, RX/PC5, CK/PA7, CTS/PB0, RTS/PB1); 01: Remap (TX/PA14, RX/PA15, CK/PB15, CTS/PE10, RTS/PE11); 1x: Remap (TX/PB15, RX/PB14, CK/PB11, CTS/PB13, RTS/PB12).	0
[23:22]	USART7_RM[1:0]	RW	USART7 remapping: 00: Default (TX/PC2, RX/PC3, CK/PC1, CTS/PC4, RTS/PC5); 01: Remap (TX/PA6, RX/PA7, CK/PB0, CTS/PC4, RTS/PC5); 10: Remap (TX/PA9, RX/PA10, CK/PA8, CTS/PB6, RTS/PB7); 11: Remap (TX/PA9, RX/PA10, CK/PA8, CTS/PC8, RTS/PC9).	0
[21:20]	USART6_RM[1:0]	RW	USART6 remapping: 00: Default (TX/PC0, RX/PC1, CK/PB0, CTS/PC2, RTS/PC3); 01: Remap (TX/PB8, RX/PB9, CK/PB5, CTS/PE0, RTS/PE1); 10: Remap (TX/PE10, RX/PE11, CK/PA13, CTS/PA14, RTS/PA15); 11: Remap (TX/PC0, RX/PC1, CK/PA5, CTS/PC2, RTS/PC3).	0
[19:18]	USART5_RM[1:0]	RW	USART5 remapping: 00: Default (TX/PC12, RX/PD2, CK/PB5, CTS/PB3, RTS/PB4); 01: Remap (TX/PB4, RX/PB5, CK/PB3, CTS/PB8, RTS/PB9); 1x: Remap (TX/PE8, RX/PE9, CK/PB2, CTS/PB0, RTS/PB1).	0
[17:16]	USART4_RM[1:0]	RW	UART4 remapping: 00: Default (TX/PC10, RX/PC11, CK/PC12, CTS/PA13, RTS/PD2); 01: Remap (TX/PB0, RX/PB1, CK/PB2, CTS/PC4, RTS/PC5); 1x: Remap (TX/PE0, RX/PE1, CK/PE12, CTS/PB8, RTS/PB9).	0
15	Reserved	RW	Reserved.	0
14	I3C_RM	RW	I3C remapping: 1: Remap (SCL/PB4, SDA/PB5); 0: Default (SCL/PB10, SDA/PB11).	0
13	DVP_RM	RW	DVP remapping: 1: Remap (PCLK/PE0, HSYNC/PE1, VSYNC/PD3, D0/PD4, D1/PD5, D2/PD6, D3/PD7, D4/PB7, D5/PD11, D6/PD12, D7/PD13, D8/PE3, D9/PE4, D10/PE5, D11/PE6); 0: Default (PCLK/PA6, HSYNC/PA4, VSYNC/PA5, D0/PA9, D1/PA10, D2/PC8, D3/PC9, D4/PC11, D5/PB6, D6/PB8, D7/PB9, D8/PC10, D9/PC12, D10/PD6, D11/PD2).	0
12	SDIO_RM	RW	SDIO remapping: 1: Remap (CK/PC12, CMD/PD2, SD0/PB14, SD1/PB15, SD2/PC10, SD3/PC11, SD4/PB8, SD5/PB9, SD6/PA9, SD7/PA10); 0: Default (CK/PC12, CMD/PD2, SD0/PC8, SD1/PC9, SD2/PC10, SD3/PC11, SD4/PB8, SD5/PB9, SD6/PC6, SD7/PC7).	0
11	LTDC_RM	RW	LTDC remapping: 1: Remap (CLK/PA10, HSYNC/PB8, VSYNC/PB9, DE/PA9, R[0]/PC0, R[1]/PC1, R[2]/PC2, R[3]/PC3, R[4]/PA1, R[5]/PA2, R[6]/PA3, R[7]/PA7, G[0]/PC4, G[1]/PC5, G[2]/PB0, G[3]/PB1, G[4]/PB3, G[5]/PA4, G[6]/PB12, G[7]/PB13, B[0]/PB14, B[1]/PB15, B[2]/PB11, B[3]/PB10, B[4]/PA8, B[5]/PA13, B[6]/PA14, B[7]/PA15);	0

Bits	Name	Access	Description	Reset
			0: Default (CLK/PE1, HSYNC/PB4, VSYNC/PB5, DE/PE0, R[0]/PC0, R[1]/PC1, R[2]/PC2, R[3]/PC3, R[4]/PA1, R[5]/PA2, R[6]/PA3, R[7]/PA7, G[0]/PC4, G[1]/PC5, G[2]/PB0, G[3]/PB1, G[4]/PB2, G[5]/PD8, G[6]/PB12, G[7]/PB13, B[0]/PB14, B[1]/PB15, B[2]/PB11, B[3]/PB10, B[4]/PA8, B[5]/PA13, B[6]/PA14, B[7]/PA15).	
10	FSMC_NADV	RW	FSMC_NADV remapping: 1: Disable FSMC NADV output; 0: FSMC NADV mapped to PB7.	0
[9:3]	Reserved	RO	Reserved.	0
2	TIM8_RM	RW	TIM8 remapping bit. 1: Remap (ETR/PA0, CH1/PB6, CH2/PB7, CH3/PB8, CH4/PC13, BKIN/PB9, CH1N/PA13, CH2N/PA14, CH3N/PA15); 0: Default (ETR/PA0, CH1/PC6, CH2/PC7, CH3/PC8, CH4/PC9, BKIN/PA6, CH1N/PA7, CH2N/PB0, CH3N/PB1).	0
1	Reserved	RW	Reserved.	0
0	FSMC_RM	RW	FSMC remapping: 1: Remap (NBL[0]/PE0, NBL[1]/PE1, CLK/PD3, NOE/PD4, NWE/PD5, NWAIT/PD6, NE1/NCE2/PD7, NADV/PB7, A16/PD11, A17/PD12, A18/PD13, A19/PE3, A20/PE4, A21/PE5, A22/PE6, A23/PE2, D0/PD14, D1/PD15, D2/PD0, D3/PD1, D4/PC4, D5/PE8, D6/PE9, D7/PE10, D8/PE11, D9/PB11, D10/PB12, D11/PB13, D12/PB14, D13/PD8, D14/PD9, D15/PD10); 0: Default (NBL[0]/PE0, NBL[1]/PE1, CLK/PD3, NOE/PD4, NWE/PD5, NWAIT/PD6, NE1/NCE2/PD7, NADV/PB7, A16/PD11, A17/PD12, A18/PD13, A19/PE3, A20/PE4, A21/PE5, A22/PE6, A23/PE2, D0/PD14, D1/PD15, D2/PD0, D3/PD1, D4/PE7, D5/PE8, D6/PE9, D7/PE10, D8/PE11, D9/PB11, D10/PB12, D11/PB13, D12/PB14, D13/PD8, D14/PD9, D15/PD10).	0

Chapter 11: Direct Memory Access Control (DMA)

The Direct Memory Access controller (DMA) provides high-speed data transfer between peripherals and memory or between memory and memory, without CPU intervention. Data can be moved quickly via DMA, freeing up CPU resources for other operations.

Each channel of the DMA controller is dedicated to managing memory access requests from one or more peripherals. An arbiter coordinates the priority among the channels.

11.1 Main Features

- Multiple independent configurable channels
- Each channel is directly connected to a dedicated hardware DMA request and supports software triggering
- Supports circular buffer management
- The request priority among multiple channels can be programmed via software (highest, high, medium, and low). When the priority settings are equal, it is determined by the channel number (the lower the channel number, the higher the priority)
- Supports peripheral-to-memory, memory-to-peripheral, and memory-to-memory transfers
- Flash, SRAM, peripheral SRAM, PB1, PB2, and HB peripherals can all serve as source and destination of access
- Programmable number of data transfers: up to 65535

11.2 Functional Description

11.2.1 DMA Channel Processing

1) Arbitration Priority

DMA requests generated by multiple independent channels are input to the DMA controller via a logical-OR structure; only one channel's request can be serviced at a time. The internal arbiter selects the peripheral/memory access to initiate based on the priority of the channel requests.

In software management, the application can independently configure the priority level for each channel through the PL[1:0] bits of the DMAy_CFGRx register, including four levels: highest, high, medium, and low. When the software-set levels among channels are equal, the module selects according to a fixed hardware priority — a lower channel number has higher priority than a higher one.

2) DMA Configuration

When the DMA controller receives a request signal, it accesses the requesting peripheral or memory and establishes data transfer between the peripheral or memory and memory. This mainly involves the following three steps:

1. Fetch data from the peripheral data register or the memory address indicated by the current peripheral/memory address register. The starting address for the first transfer is the peripheral base address or memory address specified by the DMAy_PADDRx or DMAy_MADDRx register.
2. Store data to the peripheral data register or the memory address indicated by the current peripheral/memory address register. The starting address for the first transfer is the peripheral base address or memory address specified by the DMAy_PADDRx or DMAy_MADDRx register.
3. Perform a decrement of the value in the DMAy_CNTRx register, which indicates the number of remaining outstanding transfers.

Each channel includes three DMA data transfer modes:

- Peripheral to memory (MEM2MEM=0, DIR=0)
- Memory to peripheral (MEM2MEM=0, DIR=1)
- Memory to memory (MEM2MEM=1)

Note: Memory-to-memory mode does not require a peripheral request signal. Once configured for this mode (MEM2MEM=1), enabling the channel (EN=1) starts the data transfer. This mode does not support circular mode.

The configuration procedure is as follows:

1. Set the base address of the peripheral register in the DMAy_PADDRx register, or in memory-to-memory mode (MEM2MEM=1) set the memory data address. When a DMA request occurs, this address will be the source or destination address of the data transfer.
2. Set the memory data address in the DMAy_MADDRx register. When a DMA request occurs, the transfer data will be read from or written to this address.
3. Set the number of data to be transferred in the DMAy_CNTRx register. This value is decremented after each data transfer.
4. Set the channel priority in the PL[1:0] bits of the DMAy_CFGRx register.
5. Set the data transfer direction, circular mode, peripheral and memory increment mode, peripheral and memory data width, transfer half, transfer complete, and transfer error interrupt enable bits in the DMAy_CFGRx register.
6. Set the EN bit of the DMAy_CFGRx register to enable channel x.

Note: The DMAy_PADDRx/DMAy_MADDRx/DMAy_CNTRx registers and the control bits in the DMAy_CFGRx register for data transfer direction (DIR), circular mode (CIRC), peripheral and memory increment mode (MINC/PINC), etc., can only be configured when the DMA channel is disabled.

3) Circular Mode

Setting the CIRC bit of the DMAy_CFGRx register to 1 enables the circular mode function for channel data transfer. In circular mode, when the number of data transfers becomes 0, the contents of the DMAy_CNTRx register are automatically reloaded to their initial value, and the internal peripheral and memory address registers are also reloaded to the initial address values set in the DMAy_PADDRx and DMAy_MADDRx registers. The DMA operation continues until the channel is disabled or DMA mode is turned off.

4) DMA Processing Status

- **Transfer half:** Corresponds to the HTIFx bit in the DMAy_INTFR register, set by hardware. When the DMA transfer count decreases to below half of the initial set value, a DMA transfer half flag is generated. If HTIE is set in the DMAy_CFGRx register, an interrupt will be generated. Hardware uses this flag to remind the application that it can prepare for a new round of data transfer.
- **Transfer complete:** Corresponds to the TCIFx bit in the DMAy_INTFR register, set by hardware. When the DMA transfer count decreases to 0, a DMA transfer complete flag is generated. If TCIE is set in the DMAy_CFGRx register, an interrupt will be generated.
- **Transfer error:** Corresponds to the TEIFx bit in the DMAy_INTFR register, set by hardware. Reading or writing a reserved address area generates a DMA transfer error. At the same time, the module hardware automatically clears the EN bit of the DMAy_CFGRx register for the channel where the error occurred, disabling that channel. If TEIE is set in the DMAy_CFGRx register, an interrupt will be generated.

When the application queries the DMA channel status, it can first access the GIFx bit of the DMAy_INTFR register to determine which channel currently had a DMA event, and then handle the specific DMA event content for that channel.

11.2.2 Programmable Data Transfer Total Size / Data Width / Alignment

The total amount of data transferred per round by each DMA channel is programmable, up to 65535 transfers. The DMAy_CNTRx register indicates the number of transfers to be performed. When EN=0, the set value is written; when EN=1 enables the DMA transfer channel, this register becomes read-only and decrements after each transfer.

The peripheral and memory transfer data supports an automatic address pointer increment function, and the pointer increment is programmable. The address of the first data accessed for transfer is stored in the DMAy_PADDRx and DMAy_MADDRx registers. By setting the PINC bit or MINC bit of the DMAy_CFGRx register to 1, the peripheral address auto-increment mode or memory address auto-increment mode can be enabled respectively. PSIZE[1:0] sets the peripheral address data access size and address increment size, and MSIZE[1:0] sets the memory address data access size and address increment size, with four options: 8-bit, 16-bit, 32-bit, and 64-bit. The specific data transfer methods are shown in the table below:

Note: For chips whose fifth-from-last digit of the lot number is greater than 0, the non-zero-wait region supports 64-bit data width DMA access.

Table 11-1: DMA transfer under different data widths (PINC=MINC=1)

Source width	Target width	Transfer count	Source: address/data	Target: address/data	Transfer operation
8	8	4	0x00/B0	0x00/B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage

Source width	Target width	Transfer count	Source: address/data	Target: address/data	Transfer operation
			0x01/B1	0x01/B1	method: little-endian mode, low address stores low byte, high address stores high byte.
			0x02/B2	0x02/B2	
			0x03/B3	0x03/B3	
8	16	4	0x00/B0	0x00/00B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x01/B1	0x02/00B1	
			0x02/B2	0x04/00B2	
			0x03/B3	0x06/00B3	
8	32	4	0x00/B0	0x00/000000B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x01/B1	0x04/000000B1	
			0x02/B2	0x08/000000B2	
			0x03/B3	0x0C/000000B3	
8	64	4	0x00/B0	0x00/00...0000B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x01/B1	0x08/00...0000B1	
			0x02/B2	0x10/00...0000B2	
			0x03/B3	0x18/00...0000B3	
16	8	4	0x00/B1B0	0x00/B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x02/B3B2	0x01/B2	
			0x04/B5B4	0x02/B4	
			0x06/B7B6	0x03/B6	

Source width	Target width	Transfer count	Source: address/data	Target: address/data	Transfer operation
16	16	4	0x00/B1B0	0x00/B1B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x02/B3B2	0x02/B3B2	
			0x04/B5B4	0x04/B5B4	
			0x06/B7B6	0x06/B7B6	
16	32	4	0x00/B1B0	0x00/0000B1B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x02/B3B2	0x04/0000B3B2	
			0x04/B5B4	0x08/0000B5B4	
			0x06/B7B6	0x0C/0000B7B6	
16	64	4	0x00/B1B0	0x00/00...00B1B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x02/B3B2	0x08/00...00B3B2	
			0x04/B5B4	0x10/00...00B5B4	
			0x06/B7B6	0x18/00...00B7B6	
32	8	4	0x00/B3B2B1B0	0x00/B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x04/B7B6B5B4	0x01/B4	
			0x08/BBBAB9B8	0x02/B8	
			0x0C/BFBEBDBC	0x03/BC	
32	16	4	0x00/B3B2B1B0	0x00/B1B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data

Source width	Target width	Transfer count	Source: address/data	Target: address/data	Transfer operation
			0x04/B7B6B5B4	0x02/B5B4	width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x08/BBBAB9B8	0x04/B9B8	
			0x0C/BFBEBDBC	0x06/BDBC	
32	32	4	0x00/B3B2B1B0	0x00/B3B2B1B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x04/B7B6B5B4	0x04/B7B6B5B4	
			0x08/BBBAB9B8	0x08/BBBAB9B8	
			0x0C/BFBEBDBC	0x0C/BFBEBDBC	
32	64	4	0x00/B3B2B1B0	0x00/00...B3B2B1B0	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x04/B7B6B5B4	0x08/00...B7B6B5B4	
			0x08/BBBAB9B8	0x10/00...BBBAB9B8	
			0x0C/BFBEBDBC	0x18/00...BFBEBDBC	
64	8	4	0x00/1716...10	0x00/10	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x08/1F1E...18	0x01/18	
			0x10/2726...20	0x02/20	
			0x18/2F2E...28	0x03/28	
64	16	4	0x00/1716...10	0x00/1110	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size

Source width	Target width	Transfer count	Source: address/data	Target: address/data	Transfer operation
			0x08/1F1E...18	0x02/1918	overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x10/2726...20	0x04/2120	
			0x18/2F2E...28	0x06/2928	
64	32	4	0x00/1716...10	0x00/13121110	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x08/1F1E...18	0x04/1B1A1918	
			0x10/2726...20	0x08/23222120	
			0x18/2F2E...28	0x0C/2B2A2928	
64	64	4	0x00/1716...10	0x00/1716...10	Source address increment is aligned with the source-set data width, and the access size equals the source data width. Target address increment is aligned with the target-set data width, and the access size equals the target data width. Data sent to the target follows the principle: if the data size is insufficient, the high bits are padded with 0; if the data size overflows, the high bits are discarded. Storage method: little-endian mode, low address stores low byte, high address stores high byte.
			0x08/1F1E...18	0x08/1F1E...18	
			0x10/2726...20	0x10/2726...20	
			0x18/2F2E...28	0x18/2F2E...28	

11.2.3 DMA Request Mapping

The DMA controller provides 24 channels, of which DMA1 contains 13 channels and DMA2 contains 11 channels. Each channel corresponds to multiple peripheral requests. By setting the corresponding DMA control bits in the respective peripheral registers, the DMA function of each peripheral can be independently enabled or disabled. The specific correspondences are as follows.

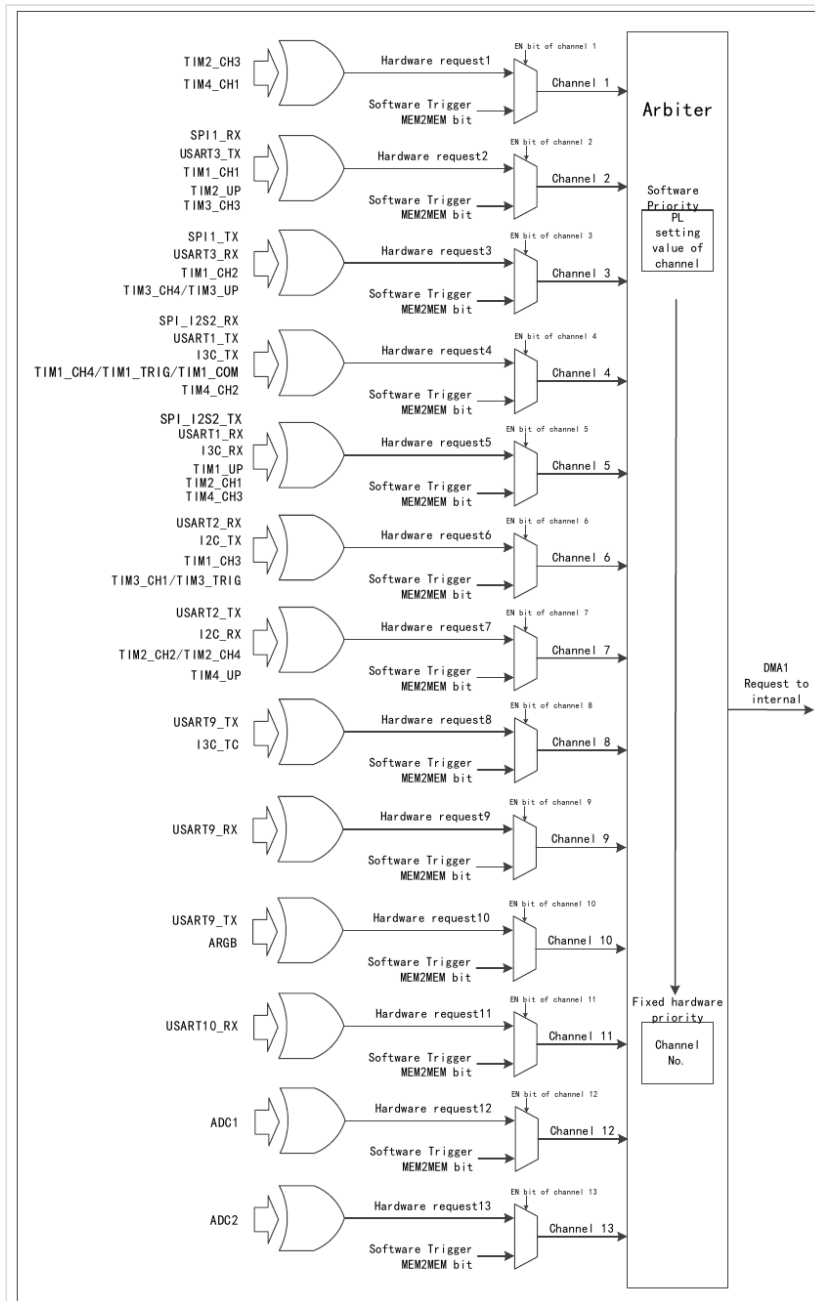


Figure 11-1: DMA1 request mapping

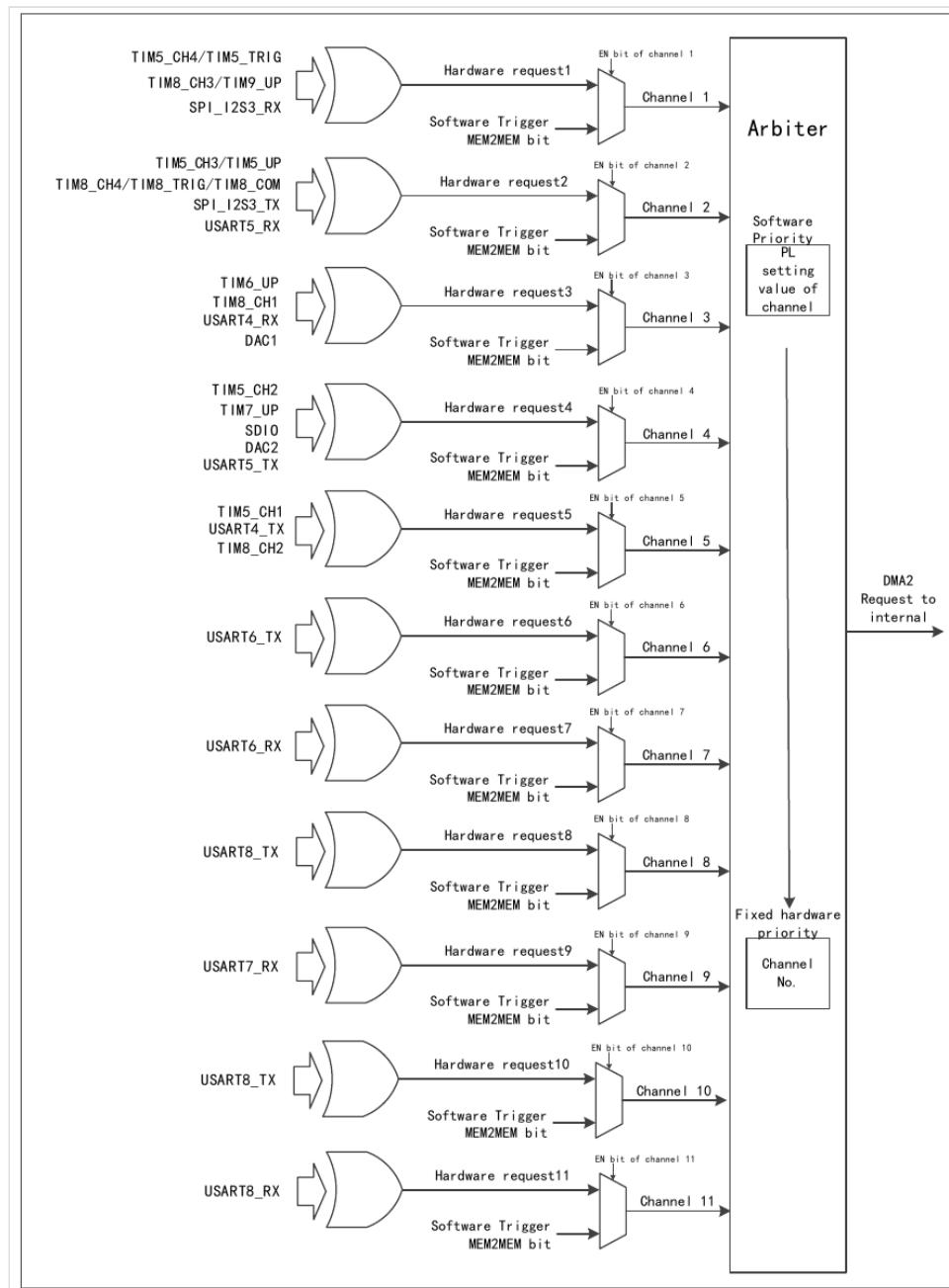


Figure 11-2: DMA2 request mapping

Table 11-2: DMA1 channel peripheral mapping table

Peripheral	Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7
SPI1		SPI1_RX	SPI1_TX				
SPI/I2S2				SPI_I2S2_RX	SPI_I2S2_TX		
USART1				USART1_TX	USART1_RX		
USART2						USART2_RX	USART2_TX
USART3		USART3_TX	USART3_RX				
I2C						I2C_TX	I2C_RX
I3C				I3C_TX	I3C_RX		
TIM1		TIM1_CH1	TIM1_CH2		TIM1_UP	TIM1_CH3	

Peripheral	Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7
				TIM1_CH4 / TIM1_TRIG / TIM1_COM			
TIM2	TIM2_CH3	TIM2_UP			TIM2_CH1		TIM2_CH2 / TIM2_CH4
TIM3		TIM3_CH3	TIM3_UP / TIM3_CH4		TIM3_CH1 / TIM3_TRIG		
TIM4	TIM4_CH1			TIM4_CH2	TIM4_CH3		TIM4_UP

Table 11-2 (continued): DMA1 channel peripheral mapping table

Peripheral	Channel 8	Channel 9	Channel 10	Channel 11	Channel 12	Channel 13
ADC1					ADC1	
ADC2						ADC2
USART9	USART9_TX	USART9_RX				
USART10			USART10_TX	USART10_RX		
I3C	I3C_TC					
ARGB			ARGB			

Table 11-3: DMA2 channel peripheral mapping table

Peripheral	Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7
TIM5	TIM5_CH4 / TIM5_TRIG	TIM5_CH3 / TIM5_UP		TIM5_CH2	TIM5_CH1		
TIM6			TIM6_UP				
TIM7				TIM7_UP			
TIM8	TIM8_CH3 / TIM8_UP	TIM8_CH4 / TIM8_TRIG / TIM8_COM	TIM8_CH1		TIM8_CH2		
SPI/I2S3	SPI_I2S3_RX	SPI_I2S3_TX					
SDIO				SDIO			
DAC1			DAC1				
DAC2				DAC2			
USART4			USART4_RX		USART4_TX		
USART5		USART5_RX		USART5_TX			
USART6						USART6_TX	USART6_RX

Table 11-3 (continued): DMA2 channel peripheral mapping table

Peripheral	Channel 8	Channel 9	Channel 10	Channel 11
USART7	USART7_TX	USART7_RX		
USART8			USART8_TX	USART8_RX

11.3 Register Description

Table 11-4: DMA1 related register list

Name	Address	Description	Reset Value
R32_DMA1_INTFR	0x40020000	DMA1 interrupt status register	0x00000000
R32_DMA1_INTFCR	0x40020004	DMA1 interrupt flag clear register	0x00000000
R32_DMA1_CFGR1	0x40020008	DMA1 channel 1 configuration register	0x00000000
R32_DMA1_CNTR1	0x4002000C	DMA1 channel 1 transfer data count register	0x00000000
R32_DMA1_PADDR1	0x40020010	DMA1 channel 1 peripheral address register	0x00000000
R32_DMA1_MADDR1	0x40020014	DMA1 channel 1 memory address register	0x00000000
R32_DMA1_CFGR2	0x4002001C	DMA1 channel 2 configuration register	0x00000000
R32_DMA1_CNTR2	0x40020020	DMA1 channel 2 transfer data count register	0x00000000
R32_DMA1_PADDR2	0x40020024	DMA1 channel 2 peripheral address register	0x00000000
R32_DMA1_MADDR2	0x40020028	DMA1 channel 2 memory address register	0x00000000
R32_DMA1_CFGR3	0x40020030	DMA1 channel 3 configuration register	0x00000000
R32_DMA1_CNTR3	0x40020034	DMA1 channel 3 transfer data count register	0x00000000
R32_DMA1_PADDR3	0x40020038	DMA1 channel 3 peripheral address register	0x00000000
R32_DMA1_MADDR3	0x4002003C	DMA1 channel 3 memory address register	0x00000000
R32_DMA1_CFGR4	0x40020044	DMA1 channel 4 configuration register	0x00000000
R32_DMA1_CNTR4	0x40020048	DMA1 channel 4 transfer data count register	0x00000000
R32_DMA1_PADDR4	0x4002004C	DMA1 channel 4 peripheral address register	0x00000000
R32_DMA1_MADDR4	0x40020050	DMA1 channel 4 memory address register	0x00000000
R32_DMA1_CFGR5	0x40020058	DMA1 channel 5 configuration register	0x00000000
R32_DMA1_CNTR5	0x4002005C	DMA1 channel 5 transfer data count register	0x00000000
R32_DMA1_PADDR5	0x40020060	DMA1 channel 5 peripheral address register	0x00000000
R32_DMA1_MADDR5	0x40020064	DMA1 channel 5 memory address register	0x00000000
R32_DMA1_CFGR6	0x4002006C	DMA1 channel 6 configuration register	0x00000000
R32_DMA1_CNTR6	0x40020070	DMA1 channel 6 transfer data count register	0x00000000
R32_DMA1_PADDR6	0x40020074	DMA1 channel 6 peripheral address register	0x00000000
R32_DMA1_MADDR6	0x40020078	DMA1 channel 6 memory address register	0x00000000
R32_DMA1_CFGR7	0x40020080	DMA1 channel 7 configuration register	0x00000000
R32_DMA1_CNTR7	0x40020084	DMA1 channel 7 transfer data count register	0x00000000
R32_DMA1_PADDR7	0x40020088	DMA1 channel 7 peripheral address register	0x00000000
R32_DMA1_MADDR7	0x4002008C	DMA1 channel 7 memory address register	0x00000000
R32_DMA1_CFGR8	0x40020090	DMA1 channel 8 configuration register	0x00000000
R32_DMA1_CNTR8	0x40020094	DMA1 channel 8 transfer data count register	0x00000000
R32_DMA1_PADDR8	0x40020098	DMA1 channel 8 peripheral address register	0x00000000
R32_DMA1_MADDR8	0x4002009C	DMA1 channel 8 memory address register	0x00000000
R32_DMA1_CFGR9	0x400200A0	DMA1 channel 9 configuration register	0x00000000

Name	Address	Description	Reset Value
R32_DMA1_CNTR9	0x400200A4	DMA1 channel 9 transfer data count register	0x00000000
R32_DMA1_PADDR9	0x400200A8	DMA1 channel 9 peripheral address register	0x00000000
R32_DMA1_MADDR9	0x400200AC	DMA1 channel 9 memory address register	0x00000000
R32_DMA1_CFGR10	0x400200B0	DMA1 channel 10 configuration register	0x00000000
R32_DMA1_CNTR10	0x400200B4	DMA1 channel 10 transfer data count register	0x00000000
R32_DMA1_PADDR10	0x400200B8	DMA1 channel 10 peripheral address register	0x00000000
R32_DMA1_MADDR10	0x400200BC	DMA1 channel 10 memory address register	0x00000000
R32_DMA1_CFGR11	0x400200C0	DMA1 channel 11 configuration register	0x00000000
R32_DMA1_CNTR11	0x400200C4	DMA1 channel 11 transfer data count register	0x00000000
R32_DMA1_PADDR11	0x400200C8	DMA1 channel 11 peripheral address register	0x00000000
R32_DMA1_MADDR11	0x400200CC	DMA1 channel 11 memory address register	0x00000000
R32_DMA1_CFGR12	0x400200D0	DMA1 channel 12 configuration register	0x00000000
R32_DMA1_CNTR12	0x400200D4	DMA1 channel 12 transfer data count register	0x00000000
R32_DMA1_PADDR12	0x400200D8	DMA1 channel 12 peripheral address register	0x00000000
R32_DMA1_MADDR12	0x400200DC	DMA1 channel 12 memory address register	0x00000000
R32_DMA1_CFGR13	0x400200E0	DMA1 channel 13 configuration register	0x00000000
R32_DMA1_CNTR13	0x400200E4	DMA1 channel 13 transfer data count register	0x00000000
R32_DMA1_PADDR13	0x400200E8	DMA1 channel 13 peripheral address register	0x00000000
R32_DMA1_MADDR13	0x400200EC	DMA1 channel 13 memory address register	0x00000000
R32_DMA1_EXTEN_INTFR	0x400200F0	DMA1 extended interrupt status register	0x00000000
R32_DMA1_EXTEN_INTFCR	0x400200F4	DMA1 extended interrupt flag clear register	0x00000000

Table 11-5: DMA2 related register list

Name	Address	Description	Reset Value
R32_DMA2_INTFR	0x40020400	DMA2 interrupt status register	0x00000000
R32_DMA2_INTFCR	0x40020404	DMA2 interrupt flag clear register	0x00000000
R32_DMA2_CFGR1	0x40020408	DMA2 channel 1 configuration register	0x00000000
R32_DMA2_CNTR1	0x4002040C	DMA2 channel 1 transfer data count register	0x00000000
R32_DMA2_PADDR1	0x40020410	DMA2 channel 1 peripheral address register	0x00000000
R32_DMA2_MADDR1	0x40020414	DMA2 channel 1 memory address register	0x00000000
R32_DMA2_CFGR2	0x4002041C	DMA2 channel 2 configuration register	0x00000000
R32_DMA2_CNTR2	0x40020420	DMA2 channel 2 transfer data count register	0x00000000
R32_DMA2_PADDR2	0x40020424	DMA2 channel 2 peripheral address register	0x00000000
R32_DMA2_MADDR2	0x40020428	DMA2 channel 2 memory address register	0x00000000
R32_DMA2_CFGR3	0x40020430	DMA2 channel 3 configuration register	0x00000000
R32_DMA2_CNTR3	0x40020434	DMA2 channel 3 transfer data count register	0x00000000
R32_DMA2_PADDR3	0x40020438	DMA2 channel 3 peripheral address register	0x00000000
R32_DMA2_MADDR3	0x4002043C	DMA2 channel 3 memory address register	0x00000000

Name	Address	Description	Reset Value
R32_DMA2_CFGR4	0x40020444	DMA2 channel 4 configuration register	0x00000000
R32_DMA2_CNTR4	0x40020448	DMA2 channel 4 transfer data count register	0x00000000
R32_DMA2_PADDR4	0x4002044C	DMA2 channel 4 peripheral address register	0x00000000
R32_DMA2_MADDR4	0x40020450	DMA2 channel 4 memory address register	0x00000000
R32_DMA2_CFGR5	0x40020458	DMA2 channel 5 configuration register	0x00000000
R32_DMA2_CNTR5	0x4002045C	DMA2 channel 5 transfer data count register	0x00000000
R32_DMA2_PADDR5	0x40020460	DMA2 channel 5 peripheral address register	0x00000000
R32_DMA2_MADDR5	0x40020464	DMA2 channel 5 memory address register	0x00000000
R32_DMA2_CFGR6	0x4002046C	DMA2 channel 6 configuration register	0x00000000
R32_DMA2_CNTR6	0x40020470	DMA2 channel 6 transfer data count register	0x00000000
R32_DMA2_PADDR6	0x40020474	DMA2 channel 6 peripheral address register	0x00000000
R32_DMA2_MADDR6	0x40020478	DMA2 channel 6 memory address register	0x00000000
R32_DMA2_CFGR7	0x40020480	DMA2 channel 7 configuration register	0x00000000
R32_DMA2_CNTR7	0x40020484	DMA2 channel 7 transfer data count register	0x00000000
R32_DMA2_PADDR7	0x40020488	DMA2 channel 7 peripheral address register	0x00000000
R32_DMA2_MADDR7	0x4002048C	DMA2 channel 7 memory address register	0x00000000
R32_DMA2_CFGR8	0x40020490	DMA2 channel 8 configuration register	0x00000000
R32_DMA2_CNTR8	0x40020494	DMA2 channel 8 transfer data count register	0x00000000
R32_DMA2_PADDR8	0x40020498	DMA2 channel 8 peripheral address register	0x00000000
R32_DMA2_MADDR8	0x4002049C	DMA2 channel 8 memory address register	0x00000000
R32_DMA2_CFGR9	0x400204A0	DMA2 channel 9 configuration register	0x00000000
R32_DMA2_CNTR9	0x400204A4	DMA2 channel 9 transfer data count register	0x00000000
R32_DMA2_PADDR9	0x400204A8	DMA2 channel 9 peripheral address register	0x00000000
R32_DMA2_MADDR9	0x400204AC	DMA2 channel 9 memory address register	0x00000000
R32_DMA2_CFGR10	0x400204B0	DMA2 channel 10 configuration register	0x00000000
R32_DMA2_CNTR10	0x400204B4	DMA2 channel 10 transfer data count register	0x00000000
R32_DMA2_PADDR10	0x400204B8	DMA2 channel 10 peripheral address register	0x00000000
R32_DMA2_MADDR10	0x400204BC	DMA2 channel 10 memory address register	0x00000000
R32_DMA2_CFGR11	0x400204C0	DMA2 channel 11 configuration register	0x00000000
R32_DMA2_CNTR11	0x400204C4	DMA2 channel 11 transfer data count register	0x00000000
R32_DMA2_PADDR11	0x400204C8	DMA2 channel 11 peripheral address register	0x00000000
R32_DMA2_MADDR11	0x400204CC	DMA2 channel 11 memory address register	0x00000000
R32_DMA2_EXTM_INTFR	0x400204D0	DMA2 extended interrupt status register	0x00000000
R32_DMA2_EXTM_INTFCR	0x400204D4	DMA2 extended interrupt flag clear register	0x00000000

11.3.1 DMAy Interrupt Status Register (DMAy_INTFR) (y=1/2)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TEIF7	HTIF7	TCIF7	GIF7	TEIF6	HTIF6	TCIF6	GIF6	TEIF5	HTIF5	TCIF5	GIF5
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TEIF4	HTIF4	TCIF4	GIF4	TEIF3	HTIF3	TCIF3	GIF3	TEIF2	HTIF2	TCIF2	GIF2	TEIF1	HTIF1	TCIF1	GIF1

Bits	Name	Access	Description	Reset
[31:28]	Reserved	RO	Reserved.	0
27 / 23 / 19 / 15 / 11 / 7 / 3	TEIFx	RO	Transfer error flag of channel x (x=1/2/3/4/5/6/7): 1 = a transfer error occurred on channel x; 0 = no transfer error on channel x. Set by hardware; software writes the CTEIFx bit to clear this flag.	0
26 / 22 / 18 / 14 / 10 / 6 / 2	HTIFx	RO	Transfer half flag of channel x (x=1/2/3/4/5/6/7): 1 = a transfer half event occurred on channel x; 0 = no transfer half on channel x. Set by hardware; software writes the CHTIFx bit to clear this flag.	0
25 / 21 / 17 / 13 / 9 / 5 / 1	TCIFx	RO	Transfer complete flag of channel x (x=1/2/3/4/5/6/7): 1 = a transfer complete event occurred on channel x; 0 = no transfer complete event on channel x. Set by hardware; software writes the CTCIFx bit to clear this flag.	0
24 / 20 / 16 / 12 / 8 / 4 / 0	GIFx	RO	Global interrupt flag of channel x (x=1/2/3/4/5/6/7): 1 = TEIFx or HTIFx or TCIFx occurred on channel x; 0 = no TEIFx or HTIFx or TCIFx occurred on channel x. Set by hardware; software writes the CGIFx bit to clear this flag.	0

11.3.2 DMAy Interrupt Flag Clear Register (DMAy_INTFCR) (y=1/2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				CTEIF7	CHTIF7	CTCIF7	CGIF7	CTEIF6	CHTIF6	CTCIF6	CGIF6	CTEIF5	CHTIF5	CTCIF5	CGIF5
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CTEIF4	CHTIF4	CTCIF4	CGIF4	CTEIF3	CHTIF3	CTCIF3	CGIF3	CTEIF2	CHTIF2	CTCIF2	CGIF2	CTEIF1	CHTIF1	CTCIF1	CGIF1

Bits	Name	Access	Description	Reset
[31:28]	Reserved	RO	Reserved.	0
27 / 23 / 19 / 15 / 11 / 7 / 3	CTEIFx	WO	Clear transfer error flag of channel x (x=1/2/3/4/5/6/7): 1 = clears the TEIFx flag in the DMAy_INTFR register; 0 = no effect.	0
26 / 22 / 18 / 14 / 10 / 6 / 2	CHTIFx	WO	Clear transfer half flag of channel x (x=1/2/3/4/5/6/7): 1 = clears the HTIFx flag in the DMAy_INTFR register; 0 = no effect.	0
25 / 21 / 17 / 13 / 9 / 5 / 1	CTCIFx	WO	Clear transfer complete flag of channel x (x=1/2/3/4/5/6/7): 1 = clears the TCIFx flag in the DMAy_INTFR register; 0 = no effect.	0
24 / 20 / 16 / 12 / 8 / 4 / 0	CGIFx	WO	Clear global interrupt flag of channel x (x=1/2/3/4/5/6/7): 1 = clears the TEIFx/HTIFx/TCIFx/GIFx flags in the DMAy_INTFR register; 0 = no effect.	0

11.3.3 DMAy Channel x Configuration Register (DMAy_CFGRx) (x=1-13, y=1; x=1-11, y=2)

Offset address: 0x08 + (x-1)*20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	MEM2MEM	PL[1:0]	MSIZE[1:0]	PSIZE[1:0]	MINC	PINC	CIRC	DIR	TEIE	HTIE	TCIE	EN			

Bits	Name	Access	Description	Reset
[31:15]	Reserved	RO	Reserved.	0
14	MEM2MEM	RW	Memory-to-memory mode enable: 1 = enable memory-to-memory data transfer mode; 0 = not memory-to-memory data transfer.	0
[13:12]	PL[1:0]	RW	Channel priority setting: 00 = low; 01 = medium; 10 = high; 11 = highest.	00b
[11:10]	MSIZE[1:0]	RW	Memory address data width setting: 00 = 8-bit; 01 = 16-bit; 10 = 32-bit; 11 = 64-bit. Note: For chips whose fifth-from-last digit of the lot number is greater than 0, the non-zero-wait region supports 64-bit data width DMA access.	00b
[9:8]	PSIZE[1:0]	RW	Peripheral address data width setting: 00 = 8-bit; 01 = 16-bit; 10 = 32-bit; 11 = 64-bit. Note: For chips whose fifth-from-last digit of the lot number is greater than 0, the non-zero-wait region supports 64-bit data width DMA access.	00b
7	MINC	RW	Memory address increment mode enable: 1 = enable memory address increment operation; 0 = memory address remains unchanged.	0
6	PINC	RW	Peripheral address increment mode enable: 1 = enable peripheral address increment operation; 0 = peripheral address remains unchanged.	0
5	CIRC	RW	DMA channel circular mode enable: 1 = enable circular operation; 0 = perform single operation.	0
4	DIR	RW	Data transfer direction: 1 = read from memory; 0 = read from peripheral.	0
3	TEIE	RW	Transfer error interrupt enable control: 1 = enable transfer error interrupt; 0 = disable transfer error interrupt.	0
2	HTIE	RW	Transfer half interrupt enable control: 1 = enable transfer half interrupt; 0 = disable transfer half interrupt.	0
1	TCIE	RW	Transfer complete interrupt enable control: 1 = enable transfer complete interrupt; 0 = disable transfer complete interrupt.	0
0	EN	RW	Channel enable control: 1 = channel enabled; 0 = channel disabled. When a DMA transfer error occurs, hardware automatically clears this bit to 0, disabling the channel.	0

11.3.4 DMAy Channel x Transfer Data Count Register (DMAy_CNTRx) (x=1-13, y=1; x=1-11, y=2)

Offset address: 0x0C + (x-1)*20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NDT[15:0]															

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	NDT[15:0]	RW	Number of data transfers, range 0-65535. Indicates the number of remaining transfers (the register content is decremented after each DMA transfer). In circular mode, the register content is automatically reloaded to the previously configured value.	0

Note: Indicates the current number of transfers remaining. When the register content is 0, no data transfer will occur regardless of whether the channel is enabled.

11.3.5 DMAy Channel x Peripheral Address Register (DMAy_PADDRx) (x=1-13, y=1; x=1-11, y=2)

Offset address: $0x10 + (x-1)*20$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PA[31:0]																															

Bits	Name	Access	Description	Reset
[31:0]	PA[31:0]	RW	Peripheral base address, used as the source or destination address for peripheral data transfer. When PSIZE[1:0]='01' (16-bit), the module automatically ignores bit0 and the operation address is auto-aligned to 2 bytes; when PSIZE[1:0]='10' (32-bit), the module automatically ignores bit[1:0] and the operation address is auto-aligned to 4 bytes.	0

Note: This register can only be changed when EN=0; it is not writable when EN=1.

11.3.6 DMAy Channel x Memory Address Register (DMAy_MADDRx) (x=1-13, y=1; x=1-11, y=2)

Offset address: $0x14 + (x-1)*20$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MA[31:0]																															

Bits	Name	Access	Description	Reset
[31:0]	MA[31:0]	RW	Memory data address, used as the source or destination address for data transfer. When MSIZE[1:0]='01' (16-bit), the module automatically ignores bit0 and the operation address is auto-aligned to 2 bytes; when MSIZE[1:0]='10' (32-bit), the module automatically ignores bit[1:0] and the operation address is auto-aligned to 4 bytes.	0

Note: This register can only be changed when EN=0; it is not writable when EN=1.

11.3.7 DMAy Extended Interrupt Status Register (DMAy_EXTM_INTFR) (y=1/2)

Offset address: DMA1: 0xF0; DMA2: 0xD0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								TEIF13	HTIF13	TCIF13	GIF13	TEIF12	HTIF12	TCIF12	GIF12

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TEIF11	HTIF11	TCIF11	GIF11	TEIF10	HTIF10	TCIF10	GIF10	TEIF9	HTIF9	TCIF9	GIF9	TEIF8	HTIF8	TCIF8	GIF8

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
23 / 19 / 15 / 11 / 7 / 3	TEIFx	RO	Transfer error flag of channel x (x=8/9/10/11/12/13): 1 = a transfer error occurred on channel x; 0 = no transfer error on channel x. Set by hardware; software writes the CTEIFx bit to clear this flag.	0
22 / 18 / 14 / 10 / 6 / 2	HTIFx	RO	Transfer half flag of channel x (x=8/9/10/11/12/13): 0 = no transfer half on channel x; 1 = a transfer half event occurred on channel x. Set by hardware; software writes the CHTIFx bit to clear this flag.	0
21 / 17 / 13 / 9 / 5 / 1	TCIFx	RO	Transfer complete flag of channel x (x=8/9/10/11/12/13): 1 = a transfer complete event occurred on channel x; 0 = no transfer complete event on channel x. Set by hardware; software writes the CTCIFx bit to clear this flag.	0
20 / 16 / 12 / 8 / 4 / 0	GIFx	RO	Global interrupt flag of channel x (x=8/9/10/11/12/13): 1 = TEIFx or HTIFx or TCIFx occurred on channel x; 0 = no TEIFx or HTIFx or TCIFx occurred on channel x. Set by hardware; software writes the CGIFx bit to clear this flag.	0

11.3.8 DMAy Extended Interrupt Flag Clear Register (DMAy_EXTM_INTFCR) (y=1/2)

Offset address: DMA1: 0xF4; DMA2: 0xD4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CTEIF13	CHTIF13	CTCIF13	CGIF13	CTEIF12	CHTIF12	CTCIF12	CGIF12
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CTEIF11	CHTIF11	CTCIF11	CGIF11	CTEIF10	CHTIF10	CTCIF10	CGIF10	CTEIF9	CHTIF9	CTCIF9	CGIF9	CTEIF8	CHTIF8	CTCIF8	CGIF8

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
23 / 19 / 15 / 11 / 7 / 3	CTEIFx	WO	Clear transfer error flag of channel x (x=8/9/10/11/12/13): 1 = clears the TEIFx flag in the DMA_INTFR register; 0 = no effect.	0
22 / 18 / 14 / 10 / 6 / 2	CHTIFx	WO	Clear transfer half flag of channel x (x=8/9/10/11/12/13): 1 = clears the HTIFx flag in the DMA_INTFR register; 0 = no effect.	0
21 / 17 / 13 / 9 / 5 / 1	CTCIFx	WO	Clear transfer complete flag of channel x (x=8/9/10/11/12/13): 1 = clears the TCIFx flag in the DMA_INTFR register; 0 = no effect.	0
20 / 16 / 12 / 8 / 4 / 0	CGIFx	WO	Clear global interrupt flag of channel x (x=8/9/10/11/12/13): 1 = clears the TEIFx/HTIFx/TCIFx/GIFx flags in the DMA_INTFR register; 0 = no effect.	0

Chapter 12: Analog-to-Digital Conversion (ADC)

The ADC module contains two 12-bit successive approximation analog-to-digital converters with a maximum input clock of 30 MHz. It supports 16 external channels and 2 internal signal sources as sampling sources. It can perform single conversion, continuous conversion, automatic scan mode between channels, discontinuous mode, external trigger mode, and other functions on channels. The analog watchdog function can be used to monitor whether the channel voltage is within the threshold range.

12.1 Main Features

- 12-bit resolution
- Supports 16 external channels and 2 internal signal source sampling
- Multiple sampling conversion methods for multi-channel: single, continuous, scan, trigger, discontinuous, etc.
- Data alignment modes: left-aligned, right-aligned
- Sampling time is individually programmable per channel
- Both regular conversion and injected conversion support external trigger
- Analog watchdog monitors channel voltage
- ADC channel input range: $0 \leq V_{IN} \leq V_{DDA}$
- Adjustable input gain for small signal amplification sampling

12.2 Functional Description

12.2.1 Module Structure

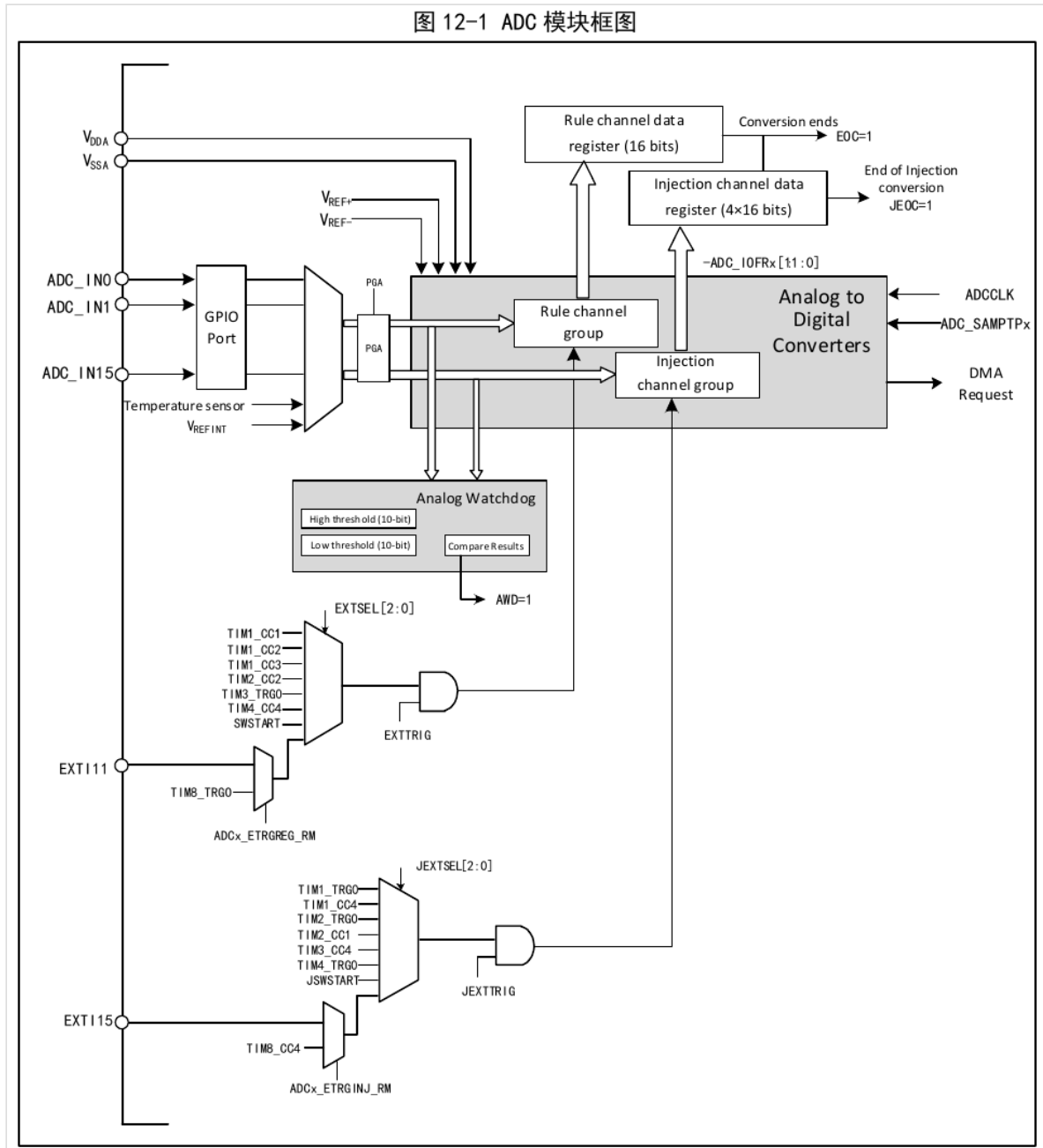


Figure 12-1: ADC module block diagram

12.2.2 ADC Configuration

1) Module power-up

The ADON bit in the ADCx_CTLR2 register is set to 1 to power up the ADC module. When the ADC module transitions from power-down mode (ADON=0) to power-up state (ADON=1), a delay of t_{STAB} is required for module stabilization. After that, writing ADON to 1 again serves as the software start signal to initiate ADC conversion. Clearing the ADON bit to 0 terminates the current conversion and puts the ADC module into power-down mode, in which state the ADC consumes almost no power.

2) Sampling clock

The register operations of the module are based on the PCLK2 (PB2 bus) clock. Its conversion unit clock reference ADCCLK is synchronized with PCLK2 and is configured by the ADCPRE[1:0] field in the RCC_CFGR0 register, with a maximum of 30 MHz.

3) Channel configuration

The ADC module provides 18 channel sampling sources, including 16 external channels and 2 internal channels. They can be configured into two conversion groups: the regular group and the injected group. This enables group conversions consisting of a series of conversions on any number of channels in any order.

Conversion groups:

- **Regular group:** Consists of up to 16 conversions. The regular channels and their conversion order are set in the ADCx_RSQRx registers. The total number of conversions in the regular group is written to L[3:0] in the ADCx_RSQR1 register.
- **Injected group:** Consists of up to 4 conversions. The injected channels and their conversion order are set in the ADCx_ISQR register. The total number of conversions in the injected group is written to JL[1:0] in the ADCx_ISQR register.

Note: If the ADCx_RSQRx or ADCx_ISQR register is modified during a conversion, the current conversion is terminated and a new start signal is sent to the ADC to convert the newly selected group.

The 2 internal channels:

- **Temperature sensor:** Connected to the ADC_IN16 channel, used to measure the internal chip temperature.
- **VREFINT internal reference voltage:** Connected to the ADC_IN17 channel.

4) Calibration

Writing the RSTCAL bit in the ADCx_CTLR2 register to 1 initializes the calibration registers; wait for RSTCAL to be hardware-cleared to 0, indicating initialization is complete. Set the CAL bit to start the calibration function; once calibration is complete, the hardware automatically clears the CAL bit. Offset calibration is performed. After that, normal conversion functions can begin, with the offset value added after each ADC conversion. It is recommended to perform an ADC calibration once when the ADC module is powered up.

Note: Before starting calibration, the ADC module must be in the power-up state (ADON=1) for at least two ADC clock cycles.

5) Programmable sampling time

The ADC uses a number of ADCCLK cycles to sample the input voltage. The number of sampling cycles for a channel can be changed via the SMPx[2:0] bits in the ADCx_SAMPTR1 and ADCx_SAMPTR2 registers. Each channel can be sampled using a different time individually.

The total conversion time is calculated as follows:

$$T_{\text{CONV}} = \text{Sampling time} + 12.5 \times T_{\text{ADCCLK}}$$

The sampling time is determined by SMPx[2:0].

The ADC regular channel conversion supports the DMA function. The regular channel conversion value is stored in a single data register, ADCx_RDATAR. To prevent the data in the ADCx_RDATAR register from not being read in time during continuous conversion of multiple regular channels, the ADC DMA

function can be enabled. The hardware generates a DMA request at the end of the regular channel conversion (EOC set) and transfers the converted data from the ADCx_RDATAR register to the user-specified destination address.

After configuring the channel of the DMA controller module, write the DMA bit in the ADCx_CTLR2 register to 1 to enable the ADC DMA function.

Note: The injected group conversion does not support the DMA function.

6) Data alignment

The ALIGN bit in the ADCx_CTLR2 register selects the storage alignment mode of the ADC converted data. The 12-bit data supports left-aligned and right-aligned modes.

The data register of the regular group channel, ADCx_RDATAR, stores the actual converted 12-bit digital value; while the data register of the injected group channel, ADCx_IDATARx, stores the value written after subtracting the offset defined in the ADCx_IOFRx register from the actual converted data. Since positive and negative values can occur, there is a sign bit (SIGNB).

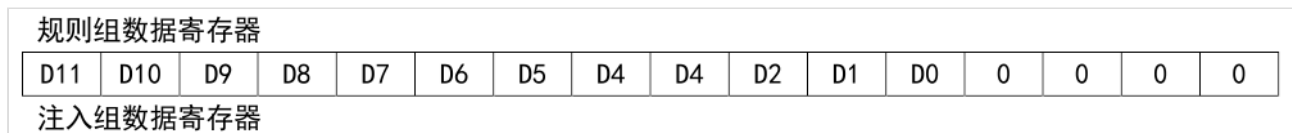


Figure 12-2: Data left alignment

Regular group data register: D11 D10 D9 D8 D7 D6 D5 D4 D3 D2 D1 D0 0 0 0 0

Injected group data register: SIGNB D11 D10 D9 D8 D7 D6 D5 D4 D3 D2 D1 D0 0 0 0

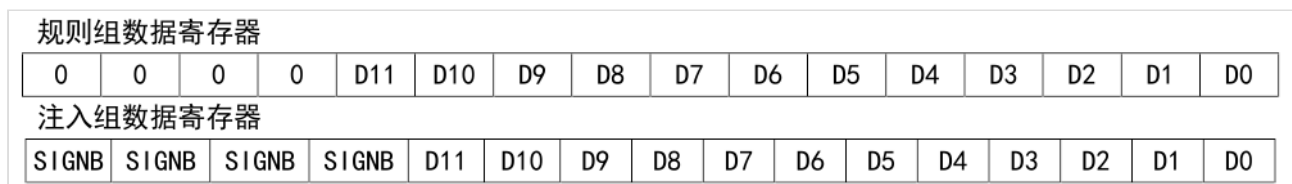


Figure 12-3: Data right alignment

Regular group data register: 0 0 0 0 D11 D10 D9 D8 D7 D6 D5 D4 D3 D2 D1 D0

Injected group data register: SIGNB SIGNB SIGNB SIGNB D11 D10 D9 D8 D7 D6 D5 D4 D3 D2 D1 D0

12.2.3 External Trigger Sources

The start event of an ADC conversion can be triggered by external events. If the EXTTRIG or JEXTTRIG bit in the ADCx_CTLR2 register is set, the conversion of the regular group or injected group channel can be triggered by an external event, respectively. In this case, the EXTSEL[2:0] and JEXTSEL[2:0] bit configurations determine the external event sources for the regular group and injected group.

Table 12-1: External trigger sources for regular group channels

EXTSEL[2:0]	Trigger source	Type
000	Timer 1 CC1 event	Internal signal from on-chip timer
001	Timer 1 CC2 event	
010	Timer 1 CC3 event	
011	Timer 2 CC2 event	

EXTSEL[2:0]	Trigger source	Type
100	Timer 3 TRGO event	
101	Timer 4 CC4 event	
110	EXTI line 11 / TIM8_TRGO	External pin / internal timer signal
111	SWSTART bit set to 1 software trigger	Software control bit

Table 12-2: External trigger sources for injected group channels

JEXTSEL[2:0]	Trigger source	Type
000	Timer 1 TRGO event	Internal signal from on-chip timer
001	Timer 1 CC4 event	
010	Timer 2 TRGO event	
011	Timer 2 CC1 event	
100	Timer 3 CC4 event	
101	Timer 4 TRGO event	
110	EXTI line 15 / TIM8_CC4	External pin / internal timer signal
111	JSWSTART bit set to 1 software trigger	Software control bit

12.2.4 Conversion Modes

Table 12-3: Conversion mode combinations

CONT	SCAN	DISCEN/ JDISCEN	IAUTO	Start event	ADC conversion mode
0	0	0	0	ADON bit set to 1	Single single-channel mode: A single conversion is performed on one regular channel.
0	0	0	0	External trigger	Single single-channel mode: A single conversion is performed on one channel of a regular channel or injected channel.
0	1	0	0	ADON bit set to 1 or external trigger	Single scan mode: All selected regular group channels (ADCx_RSQRx) or all injected group channels (ADCx_ISQR) are sequentially converted one by one in a single pass.
0	1	0	1	ADON bit set to 1 or external trigger	Single scan mode: All selected regular group channels (ADCx_RSQRx) or all injected group channels (ADCx_ISQR) are sequentially converted one by one in a single pass. Auto-injection mode: After the regular group channel conversion is complete, the injected group channels are automatically converted. Note: External trigger signals for injected channels are not allowed during conversion.
0	0	1 (DISCEN and JDISCEN cannot both be 1)	0	External trigger	Single discontinuous mode: Each start event executes a short sequence (number defined by DISCNUM[2:0]) of channel conversions, until all selected channels are converted, then it restarts from the beginning. Note: The control bits for selecting this mode for the

CONT	SCAN	DISCEN/ JDISCEN	IAUTO	Start event	ADC conversion mode
					regular group and injected group are DISCEN and JDISCEN, respectively. Discontinuous mode cannot be configured for both the regular group and the injected group at the same time; discontinuous mode can only be used for one group of conversions.
0	0	1	-	-	This mode is disabled.
1	1	X	-	-	No such mode.
1	0	0	0	ADON bit set to 1 or external trigger	Continuous single-channel/scan mode: After each round completes, a new round of conversion repeats until CONT is cleared to 0 to terminate.

Note: The external trigger events for the regular group and injected group are different, and the ADON bit can only start regular group channel conversion. Therefore, the start events for regular group and injected group channel conversions are independent.

1) Single single-channel conversion mode

In this mode, only one conversion is performed on the current channel. This mode performs a conversion on the channel ranked first in the regular group or injected group. It can be started by setting the ADON bit in the ADCx_CTLR2 register to 1 (only applicable to regular channels) or by an external trigger (applicable to regular or injected channels). Once the selected channel conversion is complete:

If a regular group channel is converted, the conversion data is stored in the 16-bit ADCx_RDATAR register, the EOC flag is set, and if the EOCIE bit is set, an ADC interrupt is triggered.

If an injected group channel is converted, the conversion data is stored in the 16-bit ADCx_IDATAR1 register, both EOC and JEOC flags are set, and if the JEOCIE or EOCIE bit is set, an ADC interrupt is triggered.

2) Single scan mode conversion

The ADC scan mode is entered by setting the SCAN bit in the ADCx_CTLR1 register to 1. This mode is used to scan a group of analog channels, performing a single conversion on each of all channels selected by the ADCx_RSQRx register (for regular channels) or ADCx_ISQR (for injected channels). When the current channel conversion ends, the next channel in the same group is automatically converted.

In scan mode, depending on the state of the IAUTO bit, it is further divided into triggered injection mode and auto-injection mode.

Triggered injection

When IAUTO is 0, if a trigger event for the injected group channel conversion occurs during the scanning of regular group channels, the current conversion is reset and the injected channel sequence is executed in a single scan mode. After all selected injected group channels have been scanned and converted, the interrupted regular group channel conversion resumes.

If a regular channel start event occurs while scanning the injected group channel sequence, the injected group conversion is not interrupted; instead, the regular sequence conversion is executed after the injected sequence conversion is complete.

Note: When using triggered injection conversion, the interval between trigger events must be longer than the injected sequence. For example, if completing the injected sequence conversion requires a total time of 28 ADCCLK cycles, the minimum interval between trigger events for the injected channel is 29 ADCCLK cycles.

Auto-injection

When IAUTO is 1, after scanning all selected regular group channels, the selected injected group channels are automatically converted. This method can be used to convert a sequence of up to 20 conversions in the ADCx_RSQRx and ADCx_ISQR registers.

In this mode, the external trigger of the injected channel must be disabled (JEXTTRIG=0).

Note: For ADC clock prescaler coefficients (ADCPRE[1:0]) of 4 to 8, when switching from regular conversion to injected sequence or from injected conversion to regular sequence, 1 ADCCLK interval is automatically inserted; when the ADC clock prescaler coefficient is 2, there is a delay of 2 ADCCLK intervals.

3) Single discontinuous mode conversion

The discontinuous mode for the regular group or injected group is entered by setting the DISCEN or JDISCEN bit in the ADCx_CTLR1 register to 1. This mode differs from scan mode in that instead of scanning a complete group of channels, it divides a group of channels into multiple short sequences. Each external trigger event executes a short sequence of scan conversions.

The length n ($n \leq 8$) of the short sequence is defined in DISCNUM[2:0] in the ADCx_CTLR1 register. When DISCEN is 1, it is the discontinuous mode for the regular group, and the total length to be converted is defined in L[3:0] in the ADCx_RSQR1 register; when JDISCEN is 1, it is the discontinuous mode for the injected group, and the total length to be converted is defined in JL[1:0] in the ADCx_ISQR register. The regular group and injected group cannot be set to discontinuous mode at the same time.

Example of regular group discontinuous mode:

DISCEN=1, DISCNUM[2:0]=3, L[3:0]=8, channels to be converted = 1, 3, 2, 5, 8, 4, 7, 6

- 1st external trigger: conversion sequence: 1, 3, 2
- 2nd external trigger: conversion sequence: 5, 8, 4
- 3rd external trigger: conversion sequence: 7, 6, and EOC event is generated
- 4th external trigger: conversion sequence: 1, 3, 2

Example of injected group discontinuous mode:

JDISCEN=1, DISCNUM[2:0]=1, JL[1:0]=3, channels to be converted = 1, 3, 2

- 1st external trigger: conversion sequence: 1
- 2nd external trigger: conversion sequence: 3
- 3rd external trigger: conversion sequence: 2, and EOC and JEOC events are generated
- 4th external trigger: conversion sequence: 1

Note:

1. When converting a regular group or injected group in discontinuous mode, the conversion sequence does not automatically restart from the beginning after it ends. When all sub-groups have been converted, the next trigger event starts the conversion of the first sub-group.
2. Auto-injection (IAUTO=1) and discontinuous mode cannot be used simultaneously.
3. Discontinuous mode cannot be set for both the regular group and the injected group at the same time; discontinuous mode can only be used for one group of conversions.
4. In the discontinuous mode of the injected group, the number of injected channels to be converted after an external trigger is 1.

4) Continuous conversion

In continuous conversion mode, as soon as the previous ADC conversion ends, another conversion is started immediately. The conversion does not stop on the last channel of the selected group but continues from the first channel of the selected group again. The start events for this mode include external trigger events and setting the ADON bit to 1. After starting, the CONT bit must be set to 1.

If a regular channel is converted, the conversion data is stored in the ADCx_RDATAR register, the EOC end-of-conversion flag is set, and if EOCIE is set, an interrupt is generated.

If an injected channel is converted, the conversion data is stored in the ADCx_IDATARx register, the JEOC injected end-of-conversion flag is set, and if JEOCIE is set, an interrupt is generated.

12.2.5 Analog Watchdog

If the analog voltage converted by the ADC is below the low threshold or above the high threshold, the AWD analog watchdog status bit is set. The threshold settings are located in the lowest 12 significant bits of the ADCx_WDHTR and ADCx_WDLTR registers. Setting the AWDIE bit in the ADCx_CTLR1 register allows the corresponding interrupt to be generated. Setting the AWRST_EN bit in the ADCx_CFG register enables the analog watchdog reset.

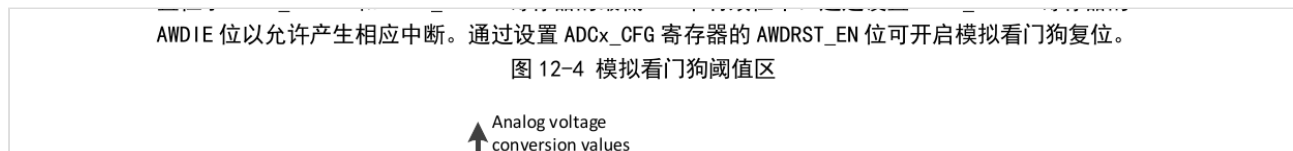


Figure 12-4: Analog watchdog threshold region

Configure the AWDSGL, AWDEN, JAWDEN, and AWDCH[4:0] bits in the ADCx_CTLR1 register to select the channel(s) guarded by the analog watchdog. The specific relationships are shown in the table below:

Table 12-4: Analog watchdog channel selection

Analog watchdog guarded channel	AWDSGL	AWDEN	JAWDEN	AWDCH[4:0]
No guarding	X	0	0	X
All injected channels	0	0	1	X
All regular channels	0	1	0	X
All injected and regular channels	0	1	1	X
Single injected channel	1	0	1	Determines channel number
Single regular channel	1	1	0	Determines channel number
Single injected and regular channel	1	1	1	Determines channel number

12.2.6 Temperature Sensor

The chip has a built-in temperature sensor connected to the ADC_INT16 channel. The ADC converts the sensor output voltage to a digital value to reflect the internal chip temperature. The recommended sampling time setting is 17.1 μ s. The voltage output by the temperature sensor varies linearly with temperature. Due to manufacturing dispersion, the slope and offset of the linear variation curve differ; therefore, the internal temperature sensor is more suitable for detecting

temperature changes rather than measuring absolute temperature. If precise temperature measurement is required, an external temperature sensor should be used.

Setting the TSVREFE bit in the ADCx_CTLR2 register to 1 wakes up the ADC internal sampling channel. Software start or external trigger starts the ADC temperature sensor channel conversion, and the data result (mV) is read. The conversion formula between the digital value and temperature (°C) is as follows:

$$\text{Temperature (}^{\circ}\text{C)} = ((V_{\text{SENSE}} - V_{25}) / \text{Avg_Slope}) + 25$$

- V_{25} : Voltage value of the temperature sensor at 25 °C
- Avg_Slope: Average slope of the temperature vs. V_{SENSE} curve (mV/°C)

Refer to the electrical characteristics section of the datasheet for the actual values of V_{25} and Avg_Slope.

Note: The internal temperature sensor power-up (TSVREFE bit changed from 0 to 1) requires a settling time, and the ADC module power-up (ADON bit changed from 0 to 1) also requires a settling time. To shorten the wait time, the ADON and TSVREFE bits can be set simultaneously.

12.2.7 Dual ADC Mode

In products with two ADC modules, the two ADCs can be used together to realize dual ADC mode. In dual ADC mode, ADC1 is the master ADC and ADC2 is the slave ADC. By configuring DUALMODE[3:0] in ADC1_CTLR1, the mode is selected to achieve alternating or synchronous trigger conversion of ADC1 and ADC2.

Note: In dual ADC mode, when selecting external trigger events for triggering, the user must enable the external trigger of both the master and slave ADCs, and set the master ADC to the corresponding trigger and the slave ADC to software trigger, to prevent unnecessary triggers from causing the slave ADC to perform conversions.

The following modes can be configured:

- Independent mode
- Synchronous injection mode
- Synchronous regular mode
- Fast alternate mode
- Slow alternate mode
- Alternating trigger mode
- Synchronous regular mode + synchronous injection mode
- Synchronous regular mode + alternating trigger mode
- Synchronous injection mode + fast alternate mode
- Synchronous injection mode + slow alternate mode

Note: In dual ADC mode, to read the slave ADC conversion data from the master data register, the DMA bit must be enabled.

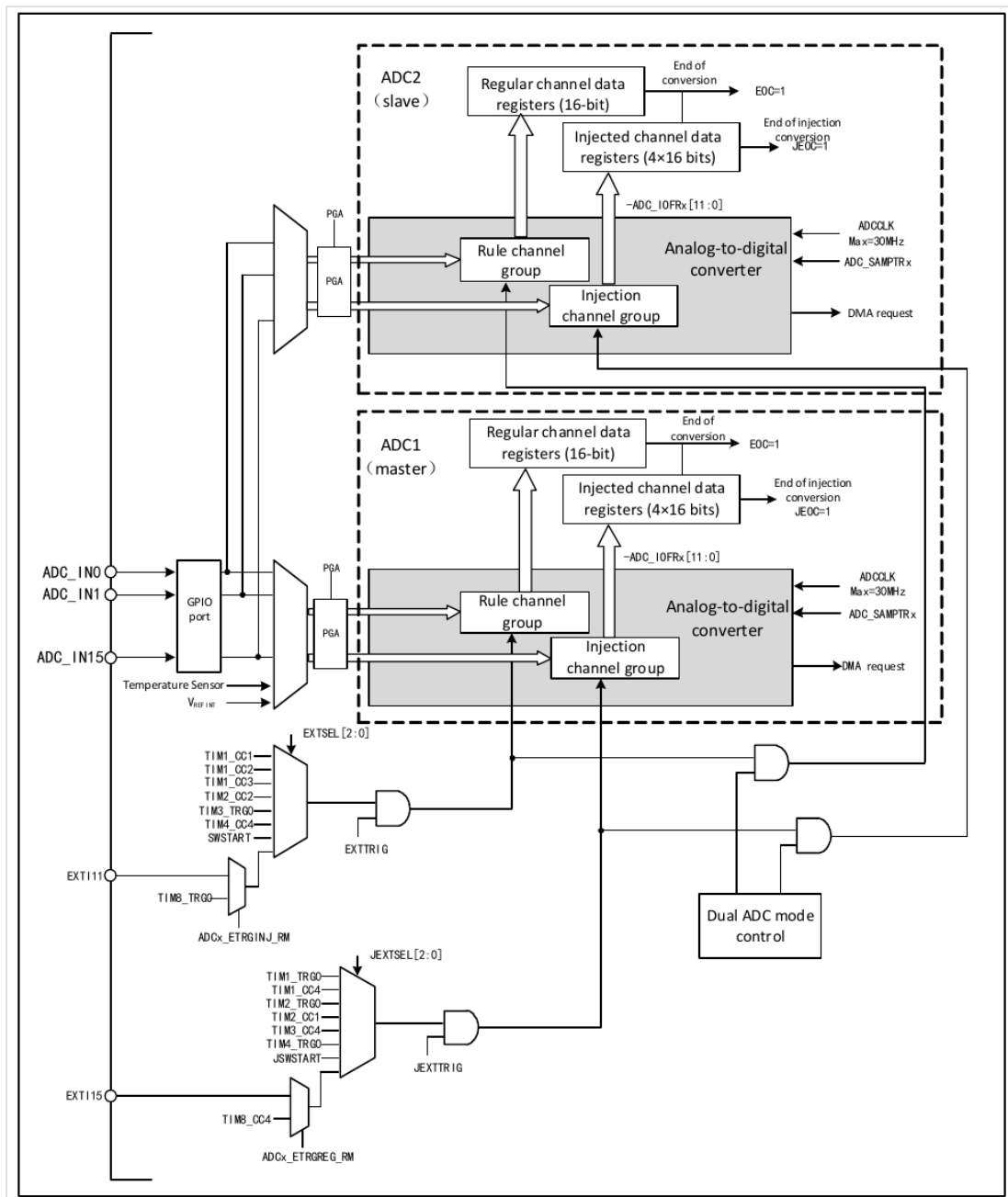


Figure 12-5: Dual ADC block diagram

1) Independent mode

In this mode, the dual ADCs do not operate synchronously; each works independently.

2) Synchronous injection mode

This mode is used to convert an injected channel group. Set JEXTSEL[2:0] in ADC1_CTLR2 to select the trigger source, which is also used to synchronously trigger ADC2. When the conversion is complete, the converted data is stored in ADCx_IDATARx of each ADC, and if any ADC interrupt is enabled, a JEOP interrupt is generated after the conversion ends.

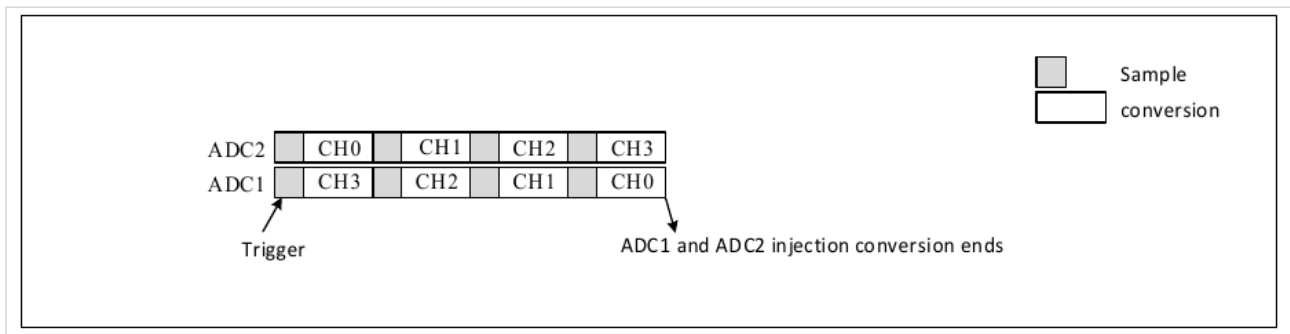


Figure 12-6: 4-channel synchronous injection conversion

Note:

1. The conversion channels of ADC1 and ADC2 should not overlap at the same time.
2. In synchronous mode, ADC1 and ADC2 should have conversion sequences of the same time length, or the time length of the longer conversion sequence should be less than the trigger time interval, to ensure that both sequences can complete conversion on each trigger.

3) Synchronous regular mode

This mode is used to convert a regular channel sequence. Set EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source, which is also used to synchronously trigger ADC2. When the conversion is complete, a 32-bit DMA transfer request is generated to transfer the contents of the data register ADC1_RDATAR to SRAM. The upper 16 bits contain the ADC2 conversion data, and the lower 16 bits contain the ADC1 conversion data. If any ADC interrupt is enabled, an EOC interrupt is generated.

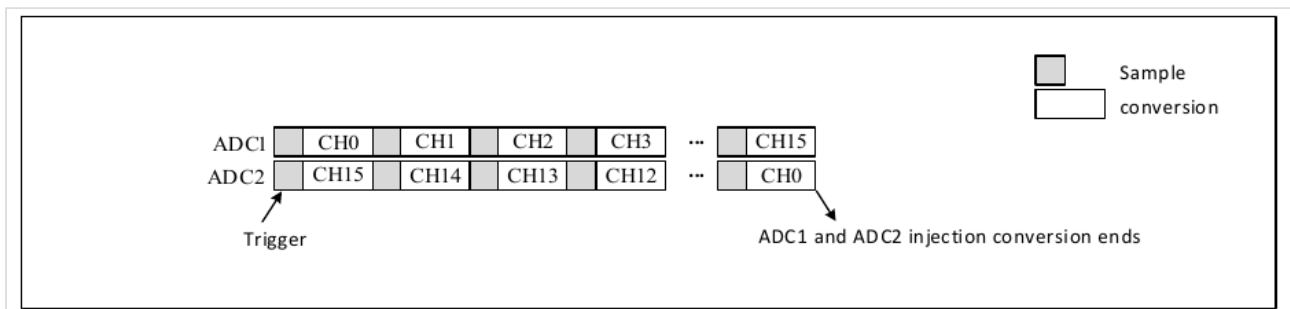


Figure 12-7: 16-channel synchronous regular conversion

Note:

1. The conversion channels of ADC1 and ADC2 should not overlap at the same time.
2. In synchronous mode, ADC1 and ADC2 should have conversion sequences of the same time length, or the time length of the longer conversion sequence should be less than the trigger time interval, to ensure that both sequences can complete conversion on each trigger.

4) Fast alternate mode

This mode is only applicable to regular channels (typically only one channel). Set EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. When a trigger is generated, ADC2 starts conversion immediately, and ADC1 starts conversion after a delay of 7 ADC clock cycles. If continuous mode is enabled for both (CONT set), the two ADCs will continuously alternate conversion on the regular channel. If interrupts are enabled, ADC1 generates an EOC interrupt. If DMA is also enabled, a 32-bit DMA transfer request is generated to transfer the contents of the data register ADC1_RDATAR to SRAM. The upper 16 bits contain the ADC2 conversion data, and the lower 16 bits contain the ADC1 conversion data.

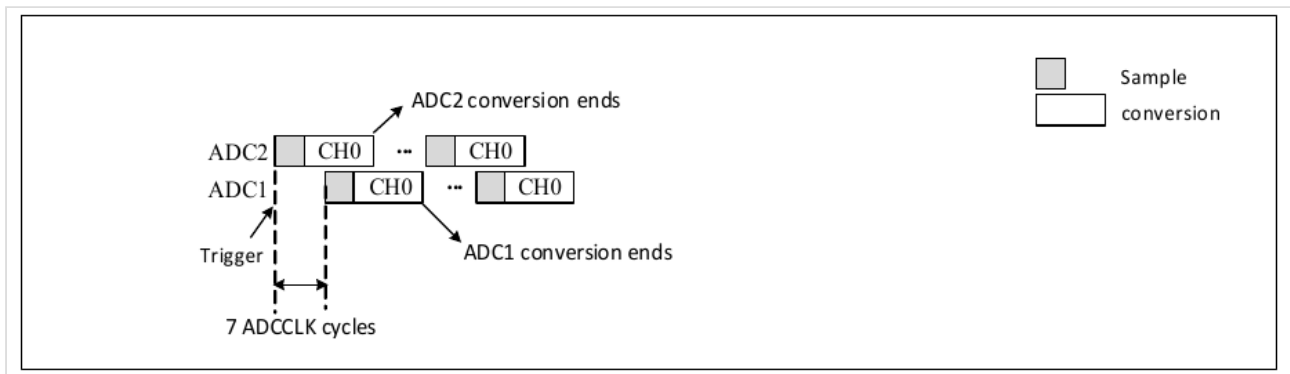


Figure 12-8: Single-channel fast alternate continuous conversion

Note: The sampling time should be less than 7 ADC clock cycles to avoid overlap of sampling periods when ADC1 and ADC2 sample the same channel.

5) Slow alternate mode

This mode is only applicable to regular channels and only one channel. Set EXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. When a trigger is generated, ADC2 starts conversion immediately, and ADC1 starts conversion after a delay of 14 ADC clock cycles. After another delay of 14 ADC clock cycles, ADC2 starts again, and so on in a cycle. If interrupts are enabled, ADC1 generates an EOC interrupt. If DMA is also enabled, a 32-bit DMA transfer request is generated to transfer the contents of the data register ADC1_RDATAR to SRAM. The upper 16 bits contain the ADC2 conversion data, and the lower 16 bits contain the ADC1 conversion data.

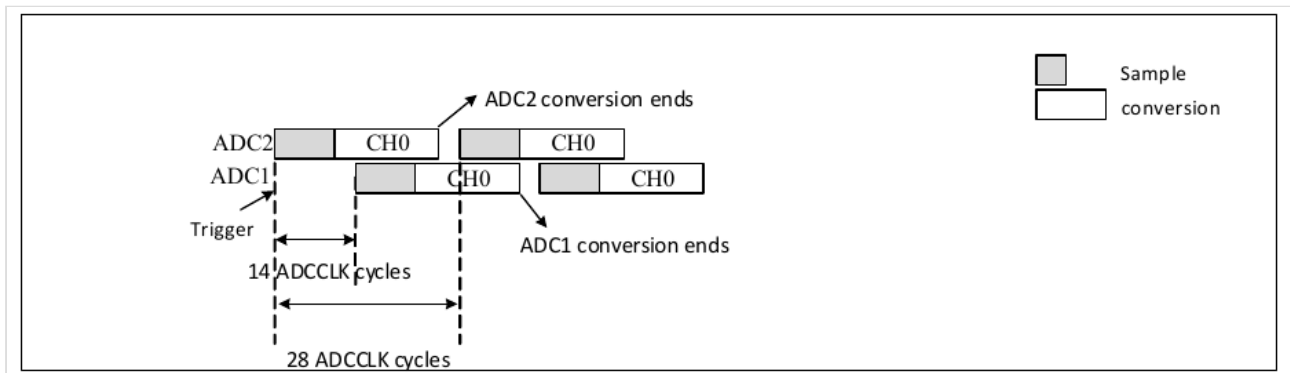


Figure 12-9: Single-channel slow alternate conversion

Note:

1. The sampling time should be less than 14 ADC clock cycles to avoid overlap with the next sampling period.
2. A new ADC2 conversion is automatically started after 28 ADC clock cycles.
3. The CONT bit does not need to be set.

6) Alternating trigger mode

This mode is only applicable to injected channel groups. Set JEXTSEL[2:0] in ADC1_CTLR2 to select the trigger source. When the first trigger event occurs, all injected channels on ADC1 are converted. When the second trigger event occurs, all injected channels on ADC2 are converted. This alternates in a cycle. If interrupts are enabled, a JEOC interrupt is generated when all injected channels of ADC1 have completed conversion, and a JEOC interrupt is generated when all injected channels of ADC2 have completed conversion.

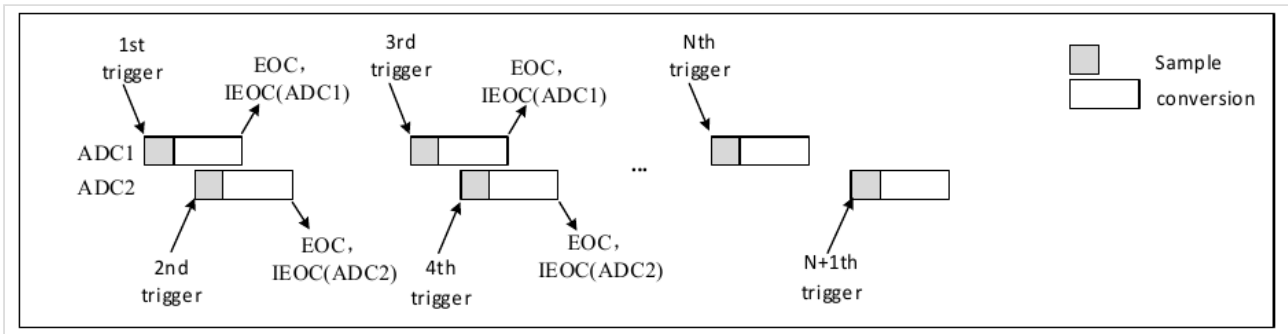


Figure 12-10: Alternating trigger conversion of each ADC injected channel group

If injected discontinuous mode is used simultaneously, when the first trigger event occurs, the first injected channel on ADC1 is converted. When the second trigger event occurs, the first injected channel on ADC2 is converted. This alternates in a cycle. If interrupts are enabled, a JEOC interrupt is generated when all injected channels of ADC1 have completed conversion, and a JEOC interrupt is generated when all injected channels of ADC2 have completed conversion.

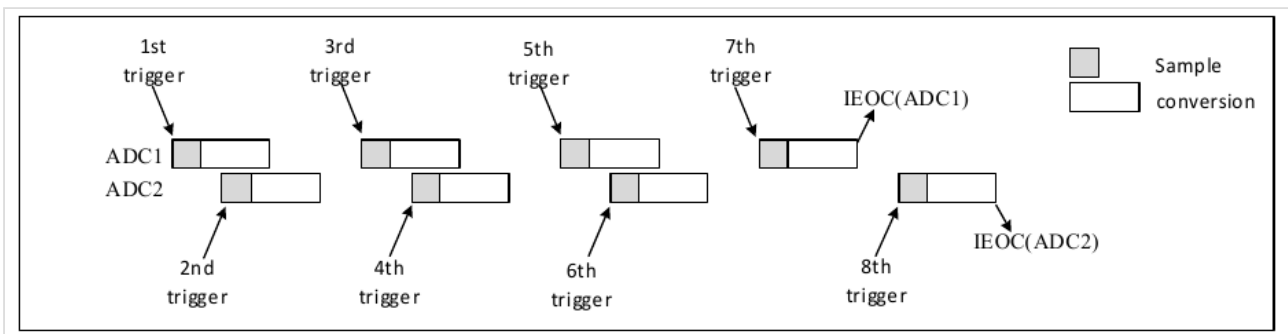


Figure 12-11: Alternating trigger conversion of each ADC injected channel in discontinuous mode

7) Synchronous regular mode + synchronous injection mode

In this mode, the synchronous conversion of the regular group can be interrupted to start the synchronous conversion of the injected group. This mode requires that the conversions have sequences of the same time length, or that the trigger time interval is longer than the time of the longer sequence of the two.

8) Synchronous regular mode + alternating trigger mode

In this mode, the synchronous conversion of the regular group can be interrupted to start the alternating trigger conversion of the injected group. When an injection event occurs, the alternating trigger conversion starts immediately. If synchronous regular conversion is in progress, the regular conversion of all ADCs is stopped and synchronously resumes after the injected conversion ends.

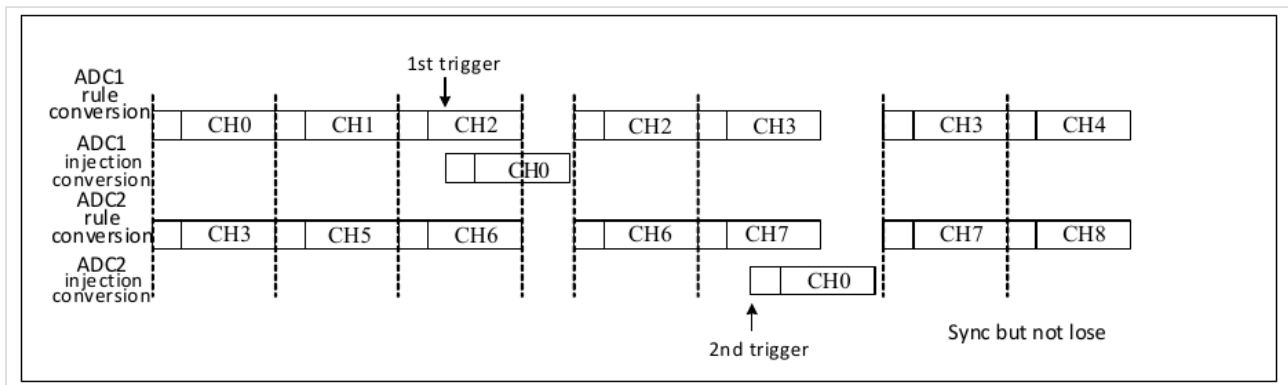


Figure 12-12: Alternating trigger injected channel conversion under synchronous regular mode

Note: This mode requires that the conversions have sequences of the same time length, or that the trigger time interval is longer than the time of the longer sequence of the two.

If an injected trigger event occurs during an injected conversion that has already interrupted a regular conversion, the trigger event is ignored, as shown in the figure below for the 2nd trigger.

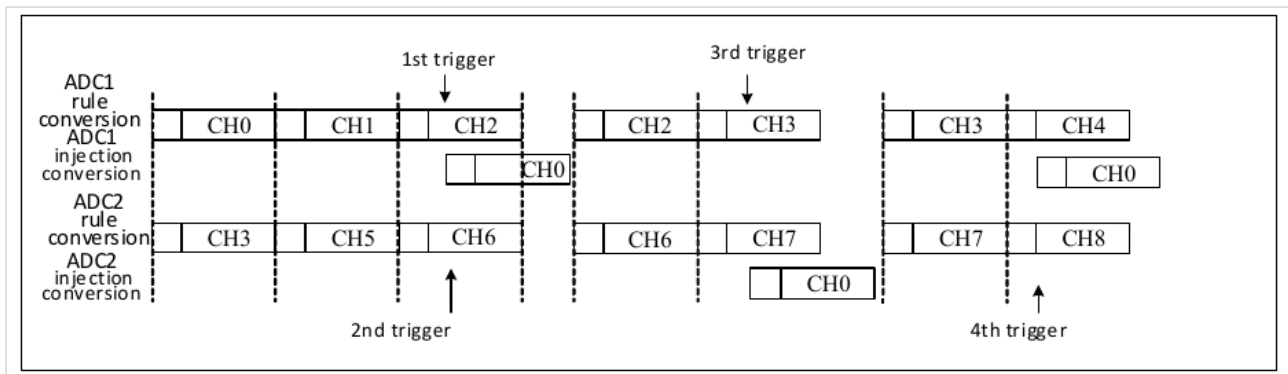


Figure 12-13: Trigger event occurring during injected conversion

9) Synchronous injection mode + alternate mode

In this mode, the injected conversion can interrupt the alternate conversion. When an injection event occurs, the alternate conversion is interrupted and the injected conversion starts. After the injected conversion ends, the alternate conversion is resumed.

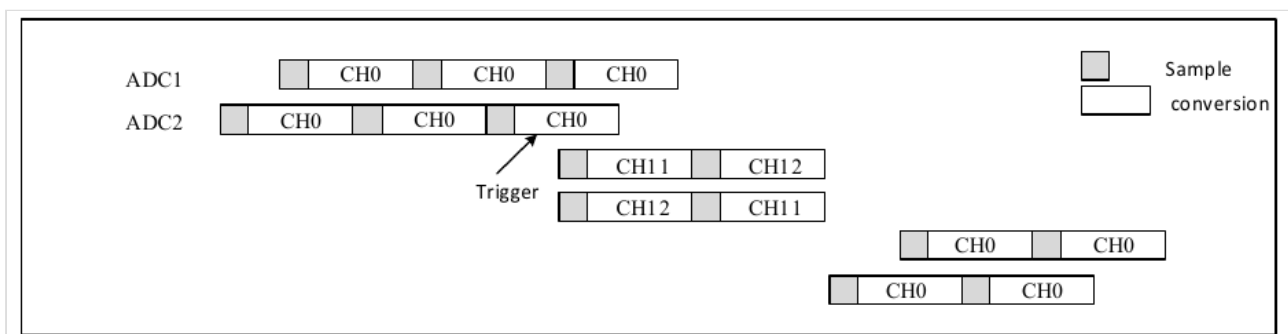


Figure 12-14: Triggering injected group conversion during alternate conversion

Note: When the ADC prescaler coefficient is 4, after the alternate conversion is resumed, the sampling interval is no longer a uniform 7 ADC clocks; it alternates between 8 and 6 clock cycles.

12.3 Register Description

Table 12-5: ADC1 related register list

Name	Address	Description	Reset Value
R32_ADC1_STATR	0x40012400	ADC1 status register	0x00000000
R32_ADC1_CTLR1	0x40012404	ADC1 control register 1	0x00000000
R32_ADC1_CTLR2	0x40012408	ADC1 control register 2	0x00000000
R32_ADC1_SAMPTR1	0x4001240C	ADC1 sampling time configuration register 1	0x00000000
R32_ADC1_SAMPTR2	0x40012410	ADC1 sampling time configuration register 2	0x00000000
R32_ADC1_IOFR1	0x40012414	ADC1 injected channel data offset register 1	0x00000000
R32_ADC1_IOFR2	0x40012418	ADC1 injected channel data offset register 2	0x00000000
R32_ADC1_IOFR3	0x4001241C	ADC1 injected channel data offset register 3	0x00000000
R32_ADC1_IOFR4	0x40012420	ADC1 injected channel data offset register 4	0x00000000
R32_ADC1_WDHTR	0x40012424	ADC1 watchdog high threshold register	0x00000FFF
R32_ADC1_WDLTR	0x40012428	ADC1 watchdog low threshold register	0x00000000
R32_ADC1_RSQR1	0x4001242C	ADC1 regular channel sequence register 1	0x00000000
R32_ADC1_RSQR2	0x40012430	ADC1 regular channel sequence register 2	0x00000000
R32_ADC1_RSQR3	0x40012434	ADC1 regular channel sequence register 3	0x00000000
R32_ADC1_ISQR	0x40012438	ADC1 injected channel sequence register	0x00000000
R32_ADC1_IDATAR1	0x4001243C	ADC1 injected data register 1	0x00000000
R32_ADC1_IDATAR2	0x40012440	ADC1 injected data register 2	0x00000000
R32_ADC1_IDATAR3	0x40012444	ADC1 injected data register 3	0x00000000
R32_ADC1_IDATAR4	0x40012448	ADC1 injected data register 4	0x00000000
R32_ADC1_RDATAR	0x4001244C	ADC1 regular data register	0x00000000
R32_ADC1_AUX	0x40012454	ADC1 sampling time register	0x00000000

Table 12-6: ADC2 related register list

Name	Address	Description	Reset Value
R32_ADC2_STATR	0x40012800	ADC2 status register	0x00000000
R32_ADC2_CTLR1	0x40012804	ADC2 control register 1	0x00000000
R32_ADC2_CTLR2	0x40012808	ADC2 control register 2	0x00000000
R32_ADC2_SAMPTR1	0x4001280C	ADC2 sampling time configuration register 1	0x00000000
R32_ADC2_SAMPTR2	0x40012810	ADC2 sampling time configuration register 2	0x00000000
R32_ADC2_IOFR1	0x40012814	ADC2 injected channel data offset register 1	0x00000000
R32_ADC2_IOFR2	0x40012818	ADC2 injected channel data offset register 2	0x00000000
R32_ADC2_IOFR3	0x4001281C	ADC2 injected channel data offset register 3	0x00000000
R32_ADC2_IOFR4	0x40012820	ADC2 injected channel data offset register 4	0x00000000
R32_ADC2_WDHTR	0x40012824	ADC2 watchdog high threshold register	0x00000FFF
R32_ADC2_WDLTR	0x40012828	ADC2 watchdog low threshold register	0x00000000

Name	Address	Description	Reset Value
R32_ADC2_RSQR1	0x4001282C	ADC2 regular channel sequence register 1	0x00000000
R32_ADC2_RSQR2	0x40012830	ADC2 regular channel sequence register 2	0x00000000
R32_ADC2_RSQR3	0x40012834	ADC2 regular channel sequence register 3	0x00000000
R32_ADC2_ISQR	0x40012838	ADC2 injected channel sequence register	0x00000000
R32_ADC2_IDATAR1	0x4001283C	ADC2 injected data register 1	0x00000000
R32_ADC2_IDATAR2	0x40012840	ADC2 injected data register 2	0x00000000
R32_ADC2_IDATAR3	0x40012844	ADC2 injected data register 3	0x00000000
R32_ADC2_IDATAR4	0x40012848	ADC2 injected data register 4	0x00000000
R32_ADC2_RDATAR	0x4001284C	ADC2 regular data register	0x00000000
R32_ADC2_AUX	0x40012854	ADC2 sampling time register	0x00000000

12.3.1 ADCx Status Register (ADCx_STATR) (x=1/2)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RSTN	Reserved														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					STRT	JSTRT	JEOC	EOC	AWD						

Bits	Name	Access	Description	Reset
31	RSTN	RO	ADC module reset flag: 1: ADC module is in reset released state; 0: ADC module is in reset state.	0
[30:5]	Reserved	RO	Reserved.	0
4	STRT	RW0	Regular channel conversion start status: 1: Regular channel conversion has started; 0: Regular channel conversion has not started. This bit is set to 1 by hardware and cleared to 0 by software (writing 1 has no effect).	0
3	JSTRT	RW0	Injected channel conversion start status: 1: Injected channel conversion has started; 0: Injected channel conversion has not started. This bit is set to 1 by hardware and cleared to 0 by software (writing 1 has no effect).	0
2	JEOC	RW0	Injected channel group conversion end status: 1: Conversion complete; 0: Conversion not complete. This bit is set to 1 by hardware (when all injected channels have been converted) and cleared to 0 by software (writing 1 has no effect).	0
1	EOC	RW0	Conversion end status: 1: Conversion complete; 0: Conversion not complete. This bit is set to 1 by hardware (when regular or injected channel group	0

Bits	Name	Access	Description	Reset
			conversion ends) and cleared to 0 by software (writing 1 has no effect) or by reading ADCx_RDATAR.	
0	AWD	RW0	Analog watchdog flag bit: 1: Analog watchdog event occurred; 0: No analog watchdog event occurred. This bit is set to 1 by hardware (when the converted value exceeds the range of the ADCx_WDHTR and ADCx_WDLTR registers) and cleared to 0 by software (writing 1 has no effect).	0

12.3.2 ADCx Control Register 1 (ADCx_CTLR1) (x=1/2)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SW_PRE	ANA_RST	Reserved						AWDEN	JAWDEN	Reserved		DUALMOD[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DISCNUM[2:0]			JDISCEN	DISCEN	JAUTO	AWDSGL	SCAN	JEOCIE	AWDIE	EOCIE	AWDCH[4:0]				

Bits	Name	Access	Description	Reset
31	SW_PRE	RW	Channel pre-switching off: 1: Channel pre-switching off; 0: Channel pre-switching on.	0
30	ANA_RST	RW	Analog module reset enable: 1: Enable; 0: Disable. Note: Resetting the ADC analog circuit during ADC conversion will corrupt the conversion results. This bit can be used together with RCC's ADCxRST.	0
[29:24]	Reserved	RO	Reserved.	0
23	AWDEN	RW	Analog watchdog enable on regular channels: 1: Enable analog watchdog on regular channels; 0: Disable analog watchdog on regular channels.	0
22	JAWDEN	RW	Analog watchdog enable on injected channels: 1: Enable analog watchdog on injected channels; 0: Disable analog watchdog on injected channels.	0
[21:20]	Reserved	RO	Reserved.	0
[19:16]	DUALMOD[3:0]	RW	Dual mode selection. 0000: Independent mode 0001: Synchronous regular + synchronous injection mode 0010: Synchronous regular + alternating trigger mode 0011: Synchronous injection + fast alternate mode 0100: Synchronous injection + slow alternate mode 0101: Synchronous injection mode 0110: Synchronous regular mode 0111: Fast alternate mode 1000: Slow alternate mode 1001: Alternating trigger mode	0000b

Bits	Name	Access	Description	Reset
			Note: These bits are reserved in ADC2. Any configuration bit modifications should be performed while dual mode is disabled.	
[15:13]	DISCNUM[2:0]	RW	Number of regular channels to be converted after an external trigger in discontinuous mode: 000: 1 channel; ... 111: 8 channels.	000b
12	JDISCEN	RW	Discontinuous mode enable on injected channels: 1: Enable discontinuous mode on injected channels; 0: Disable discontinuous mode on injected channels.	0
11	DISCEN	RW	Discontinuous mode enable on regular channels: 1: Enable discontinuous mode on regular channels; 0: Disable discontinuous mode on regular channels.	0
10	JAUTO	RW	Automatic injected channel group conversion enable after regular channel conversion completes: 1: Enable automatic injected channel group conversion; 0: Disable automatic injected channel group conversion. Note: This mode requires disabling the external trigger function of injected channels.	0
9	AWDSGL	RW	Analog watchdog enable on a single channel in scan mode: 1: Use analog watchdog on a single channel (selected by AWDCH[4:0]); 0: Use analog watchdog on all channels.	0
8	SCAN	RW	Scan mode enable: 1: Enable scan mode (continuously convert all channels selected by ADCx_IOFRx and ADCx_RSQRx); 0: Disable scan mode.	0
7	JEOCIE	RW	Injected channel group conversion end interrupt enable: 1: Enable injected channel group conversion complete interrupt (JEOC flag); 0: Disable injected channel group conversion complete interrupt.	0
6	AWDIE	RW	Analog watchdog interrupt enable: 1: Enable analog watchdog interrupt; 0: Disable analog watchdog interrupt. Note: In scan mode, if this interrupt occurs, the scan will be aborted.	0
5	EOCIE	RW	Conversion end (regular or injected channel group) interrupt enable: 1: Enable conversion end interrupt (EOC flag); 0: Disable conversion end interrupt.	0
[4:0]	AWDCH[4:0]	RW	Analog watchdog channel selection: 00000: Analog input channel 0; 00001: Analog input channel 1; ... 10000: Analog input channel 16; 10001: Analog input channel 17; Others: Reserved.	00000b

12.3.3 ADCx Control Register 2 (ADCx_CTLR2) (x=1/2)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								TSVREFE	RSWSTART	JSWSTART	EXTTRIG	EXTSEL[2:0]			Reserved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JEXTTRIG	JEXTSEL[2:0]			ALIGN	Reserved			DMA	Reserved			RSTCAL	CAL	CONT	ADON

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
23	TSVREFE	RW	Temperature sensor and internal voltage (VREFINT) channel enable: 1: Enable temperature sensor and VREFINT channels; 0: Disable temperature sensor and VREFINT channels. Note: This bit is only applicable to ADC1.	0
22	RSWSTART	RW	Start a regular channel conversion, requires setting the software trigger: 1: Start regular channel conversion; 0: Reset state. This bit is set by software and cleared by hardware after conversion starts.	0
21	JSWSTART	RW	Start an injected channel conversion, requires setting the software trigger: 1: Start injected channel conversion; 0: Reset state. This bit is set by software and cleared by hardware after conversion starts or by software.	0
20	EXTTRIG	RW	External trigger conversion mode enable for regular channels: 1: Use external event to start conversion; 0: Disable external event start function.	0
[19:17]	EXTSEL[2:0]	RW	External trigger event selection for starting regular channel conversion: 000: Timer 1 CC1 event; 001: Timer 1 CC2 event; 010: Timer 1 CC3 event; 011: Timer 2 CC2 event; 100: Timer 3 TRGO event; 101: Timer 4 CC4 event; 110: EXTI line 11 / Timer 8 TRGO event; 111: SWSTART software trigger.	000b
16	Reserved	RO	Reserved.	0
15	JEXTTRIG	RW	External trigger conversion mode enable for injected channels: 1: Use external event to start conversion; 0: Disable external event start function.	0
[14:12]	JEXTSEL[2:0]	RW	External trigger event selection for starting injected channel conversion: 000: Timer 1 TRGO event; 001: Timer 1 CC4 event; 010: Timer 2 TRGO event;	000b

Bits	Name	Access	Description	Reset
			011: Timer 2 CC1 event; 100: Timer 3 CC4 event; 101: Timer 4 TRGO event; 110: EXTI line 15 / Timer 8 CC4 event; 111: JSWSTART software trigger.	
11	ALIGN	RW	Data alignment: 1: Left-aligned; 0: Right-aligned.	0
[10:9]	Reserved	RO	Reserved.	0
8	DMA	RW	Direct memory access (DMA) mode enable: 1: Enable DMA mode; 0: Disable DMA mode.	0
[7:4]	Reserved	RO	Reserved.	0
3	RSTCAL	RW	Reset calibration. This bit is set by software and cleared by hardware when reset is complete: 1: Initialize calibration registers; 0: Calibration registers initialized. Note: If RSTCAL is set while a conversion is in progress, additional cycles are required to clear the calibration registers.	0
2	CAL	RW	A/D calibration. This bit is set by software and cleared by hardware when calibration ends. 1: Start calibration; 0: Calibration complete.	0
1	CONT	RW	Continuous conversion enable: 1: Continuous conversion mode; 0: Single conversion mode. If this bit is set, conversion will continue until this bit is cleared.	0
0	ADON	RW	On/off A/D converter. When this bit is 0, writing 1 wakes up the ADC from power-down mode; When this bit is 1, writing 1 starts conversion. If any other bits change, no new conversion is started. 1: Turn on ADC and start conversion; 0: Turn off ADC conversion/calibration and enter power-down mode.	0

12.3.4 ADCx Sampling Time Configuration Register 1 (ADCx_SAMPTR1) (x=1/2)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					SMP18[2:0]			SMP17[2:0]			SMP16[2:0]			SMP15[2:1]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMP15[0]		SMP14[2:0]			SMP13[2:0]			SMP12[2:0]			SMP11[2:0]			SMP10[2:0]	

Bits	Name	Access	Description	Reset
[31:27]	Reserved	RO	Reserved.	0
[26:0]	SMPx[2:0]	RW	SMPx[2:0]: Sampling time configuration for channel x: 000: 1.5 cycles; 001: 7.5 cycles;	000b

Bits	Name	Access	Description	Reset
			010: 13.5 cycles; 011: 28.5 cycles; 100: 41.5 cycles; 101: 55.5 cycles; 110: 71.5 cycles; 111: 239.5 cycles; SMPx[2:0]: Sampling time configuration for channel x when ADC_SMP_SELx bit is set: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 2.5 cycles; 101: 3.5 cycles; 110: 4.5 cycles; 111: 5.5 cycles; These bits are used to independently select the sampling time for each channel. The channel configuration value must remain unchanged during the sampling period.	

12.3.5 ADCx Sampling Time Configuration Register 2 (ADCx_SAMPTR2) (x=1/2)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SMP9[2:0]			SMP8[2:0]			SMP7[2:0]			SMP6[2:0]			SMP5[2:1]	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMP5[0]		SMP4[2:0]			SMP3[2:0]			SMP2[2:0]			SMP1[2:0]			SMP0[2:0]	

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
[29:0]	SMPx[2:0]	RW	SMPx[2:0]: Sampling time configuration for channel x: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 41.5 cycles; 101: 55.5 cycles; 110: 71.5 cycles; 111: 239.5 cycles; SMPx[2:0]: Sampling time configuration for channel x when ADC_SMP_SELx bit is set: 000: 1.5 cycles; 001: 7.5 cycles; 010: 13.5 cycles; 011: 28.5 cycles; 100: 2.5 cycles; 101: 3.5 cycles; 110: 4.5 cycles; 111: 5.5 cycles; These bits are used to independently select the sampling time for each channel. The channel configuration value must remain unchanged during the sampling period.	000b

12.3.6 ADCy Injected Channel Data Offset Register x (ADCy_IOFRx) (x=1/2/3/4, y=1/2)

Offset address: $0x14 + (x-1) \times 4$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				JOFFSETx[11:0]											

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
[11:0]	JOFFSETx[11:0]	RW	Data offset value for injected channel x. When converting an injected channel, this value defines the value to be subtracted from the raw conversion data. The conversion result can be read from the ADCx_IDATARx register.	0

12.3.7 ADCx Watchdog High Threshold Register (ADCx_WDHTR) (x=1/2)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				HT[11:0]											

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
[11:0]	HT[11:0]	RW	Analog watchdog high threshold setting value.	0xFFF

Note: The values of WDHTR and LTR can be changed during conversion, but they will take effect at the next conversion.

12.3.8 ADCx Watchdog Low Threshold Register (ADCx_WDLTR) (x=1/2)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				LT[11:0]											

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
[11:0]	LT[11:0]	RW	Analog watchdog low threshold setting value.	0

Note: The values of WDHTR and LTR can be changed during conversion, but they will take effect at the next conversion.

12.3.9 ADCx Regular Channel Sequence Register 1 (ADCx_RSQR1) (x=1/2)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								L[3:0]				SQ16[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ16[0]		SQ15[4:0]				SQ14[4:0]				SQ13[4:0]					

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
[23:20]	L[3:0]	RW	Number of channels to be converted in the regular channel conversion sequence: 0000–1111: 1–16 conversions.	0
[19:15]	SQ16[4:0]	RW	Channel number (0–17) for the 16th conversion in the regular sequence.	0
[14:10]	SQ15[4:0]	RW	Channel number (0–17) for the 15th conversion in the regular sequence.	0
[9:5]	SQ14[4:0]	RW	Channel number (0–17) for the 14th conversion in the regular sequence.	0
[4:0]	SQ13[4:0]	RW	Channel number (0–17) for the 13th conversion in the regular sequence.	0

12.3.10 ADCx Regular Channel Sequence Register 2 (ADCx_RSQR2) (x=1/2)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SQ12[4:0]						SQ11[4:0]				SQ10[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ10[0]		SQ9[4:0]					SQ8[4:0]					SQ7[4:0]			

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
[29:25]	SQ12[4:0]	RW	Channel number (0–17) for the 12th conversion in the regular sequence.	0
[24:20]	SQ11[4:0]	RW	Channel number (0–17) for the 11th conversion in the regular sequence.	0
[19:15]	SQ10[4:0]	RW	Channel number (0–17) for the 10th conversion in the regular sequence.	0
[14:10]	SQ9[4:0]	RW	Channel number (0–17) for the 9th conversion in the regular sequence.	0
[9:5]	SQ8[4:0]	RW	Channel number (0–17) for the 8th conversion in the regular sequence.	0
[4:0]	SQ7[4:0]	RW	Channel number (0–17) for the 7th conversion in the regular sequence.	0

12.3.11 ADCx Regular Channel Sequence Register 3 (ADCx_RSQR3) (x=1/2)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		SQ6[4:0]						SQ5[4:0]				SQ4[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SQ4[0]		SQ3[4:0]					SQ2[4:0]					SQ1[4:0]			

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
[29:25]	SQ6[4:0]	RW	Channel number (0–17) for the 6th conversion in the regular sequence.	0
[24:20]	SQ5[4:0]	RW	Channel number (0–17) for the 5th conversion in the regular sequence.	0
[19:15]	SQ4[4:0]	RW	Channel number (0–17) for the 4th conversion in the regular sequence.	0
[14:10]	SQ3[4:0]	RW	Channel number (0–17) for the 3rd conversion in the regular sequence.	0
[9:5]	SQ2[4:0]	RW	Channel number (0–17) for the 2nd conversion in the regular sequence.	0

Bits	Name	Access	Description	Reset
[4:0]	SQ1[4:0]	RW	Channel number (0–17) for the 1st conversion in the regular sequence.	0

12.3.12 ADCx Injected Channel Sequence Register (ADCx_ISQR) (x=1/2)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										JL[1:0]		JSQ4[4:1]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JSQ4[0]		JSQ3[4:0]				JSQ2[4:0]				JSQ1[4:0]					

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
[21:20]	JL[1:0]	RW	Number of channels to be converted in the injected channel conversion sequence: 00–11: 1–4 conversions.	0
[19:15]	JSQ4[4:0]	RW	Channel number (0–17) for the 4th conversion in the injected sequence. Note: Written by software to assign channel number (0–17) as the 4th in the sequence to be converted.	0
[14:10]	JSQ3[4:0]	RW	Channel number (0–17) for the 3rd conversion in the injected sequence.	0
[9:5]	JSQ2[4:0]	RW	Channel number (0–17) for the 2nd conversion in the injected sequence.	0
[4:0]	JSQ1[4:0]	RW	Channel number (0–17) for the 1st conversion in the injected sequence.	0

Note: Unlike the regular conversion sequence, if the length of JL[1:0] is less than 4, the conversion sequence order starts from (4–JL). For example, when JL[1:0]=3 (4 injected conversions in the sequencer), the ADC converts channels in the following order: JSQ1[4:0], JSQ2[4:0], JSQ3[4:0], and JSQ4[4:0]; when JL[1:0]=2 (3 injected conversions in the sequencer), the ADC converts channels in the following order: JSQ2[4:0], JSQ3[4:0], and JSQ4[4:0]; when JL[1:0]=1 (2 injected conversions in the sequencer), the ADC converts channels in the order: JSQ3[4:0] first, then JSQ4[4:0]; when JL[1:0]=0 (1 injected conversion in the sequencer), the ADC converts only the JSQ4[4:0] channel. If ADCx_ISQR[21:0]=10 00111 00011 00111 00010, the ADC converts channels in the following order: JSQ2[4:0], JSQ3[4:0], and JSQ4[4:0], meaning the scan conversion proceeds in the channel order: 7, 3, 7.

12.3.13 ADCy Injected Data Register x (ADCy_IDATARx) (x=1/2/3/4, y=1/2)

Offset address: 0x3C + (x–1) × 4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
JDATA[15:0]															

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	JDATA[15:0]	RO	Injected channel conversion data (data left-aligned or right-aligned).	0

12.3.14 ADCx Regular Data Register (ADCx_RDATAR) (x=1/2)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADC2DATA[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA[15:0]															

Bits	Name	Access	Description	Reset
[31:16]	ADC2DATA[15:0]	RO	ADC2 converted data: In ADC1: In dual mode, these bits contain the regular channel data converted by ADC2. In ADC2: These bits are not used.	0
[15:0]	DATA[15:0]	RO	Regular channel conversion data (data left-aligned or right-aligned).	0

12.3.15 ADCx Sampling Time Register (ADCx_AUX) (x=1/2)

Offset address: 0x54

31	30	29	28	27	26	25	24	23
Reserved								
15	14	13	12	11	10	9	8	7
ADC_SMP_SEL15	ADC_SMP_SEL14	ADC_SMP_SEL13	ADC_SMP_SEL12	ADC_SMP_SEL11	ADC_SMP_SEL10	ADC_SMP_SEL9	ADC_SMP_SEL8	ADC_SMP_SEL7

Bits	Name	Access	Description	Reset
[31:18]	Reserved	RO	Reserved.	0
[17:0]	ADC_SMP_SELx	RW	x=0-17, sampling time optional enable bit for channel x (only when SMPx is 100-111): 1: SMPx=100: sampling time 2.5 cycles; SMPx=101: sampling time 3.5 cycles; SMPx=110: sampling time 4.5 cycles; SMPx=111: sampling time 5.5 cycles; 0: Sampling time is the period corresponding to the SMPx configuration.	0

Chapter 13: Digital-to-Analog Conversion (DAC)

The Digital-to-Analog Conversion module (DAC) contains two 12-bit voltage-output digital-to-analog converters (DAC) that convert two digital signals into two analog voltage signals and output them. It supports independent or simultaneous conversion of dual DAC channels, 12-bit data left-aligned or right-aligned, 12-bit or 8-bit data, and external event-triggered conversion. It can generate triangle waves and noise, and supports DMA functionality.

13.1 Main Features

- 2 DAC converters, each corresponding to one output channel
- Triangle wave and noise waveform generators
- Configurable 8-bit or 12-bit output
- 12-bit data left-aligned or right-aligned
- Simultaneous or independent conversion of dual DAC channels
- DMA support
- Multiple trigger events

13.2 Functional Description

13.2.1 DAC Module Structure

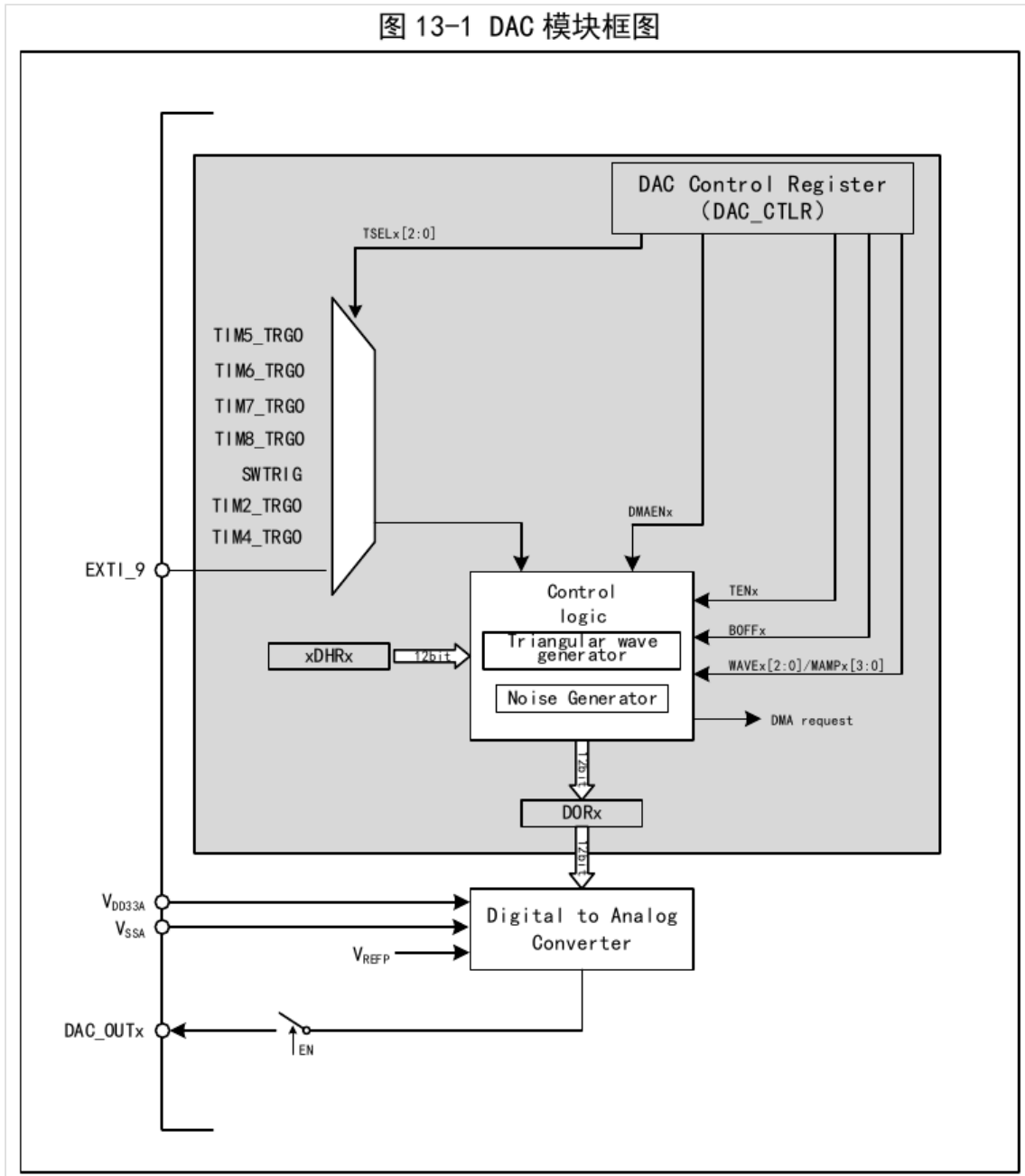


Figure 13-1: DAC module block diagram

13.2.2 DAC Channel Configuration

13.2.2.1 Enabling the DAC

Setting the ENx bit in the DAC_CTLR register to 1 enables the analog power supply to DAC channel x. After a startup time, DAC channel x is enabled. The DAC contains two analog output channels, which can output simultaneously or independently.

Note: To avoid parasitic interference and additional power consumption, the pin corresponding to the DAC channel should be configured as analog input (AIN) mode in advance.

13.2.2.2 Enabling the Output Buffer

The DAC integrates an output buffer that can be used to reduce output impedance and increase driving capability to directly drive external loads. Each DAC channel output buffer can be enabled or disabled by setting the BOFFx bit in the DAC_CTLR register.

13.2.2.3 Data Format

In single DAC channel mode, the data formats include 8-bit right-aligned, 12-bit left-aligned, and 12-bit right-aligned.

In 8-bit right-aligned mode, data is written to DAC_R8BDHRx[7:0], and the module loads (after 1 PB1 clock cycle) the left-shifted data into the data output register DAC_DORx[11:4].

In 12-bit right-aligned mode, data is written to DAC_R12BDHRx[11:0], and the module loads (after 1 PB1 clock cycle) the right-aligned data into the data output register DAC_DORx[11:0].

In 12-bit left-aligned mode, data is written to DAC_L12BDHRx[15:4], and after corresponding shifting, the module loads (after 1 PB1 clock cycle) the left-aligned data into the data output register DAC_DORx[11:0].

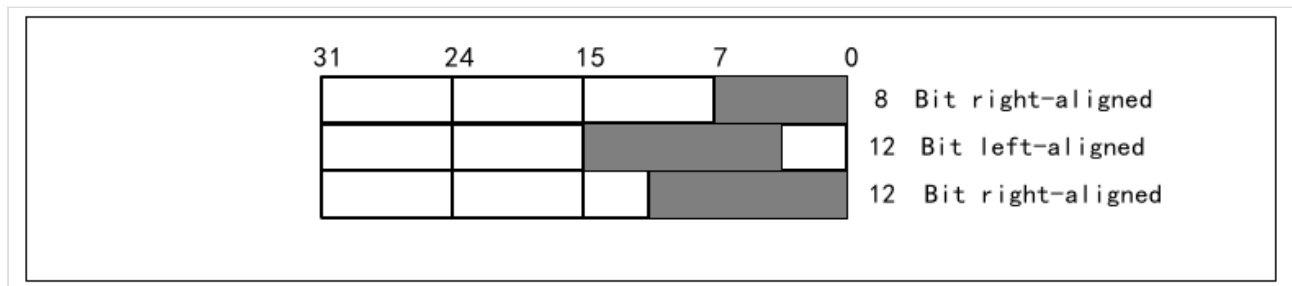


Figure 13-2: Single-channel data format

In dual DAC channel mode, there are also three formats: 8-bit right-aligned, 12-bit left-aligned, and 12-bit right-aligned.

In 8-bit right-aligned mode, data is written to DAC_RD8BDHR[7:0], and the module loads (after 1 PB1 clock cycle) bits[7:0] shifted into DAC_DOR1[11:4], and bits[15:8] shifted into DAC_DOR2[11:4].

In 12-bit left-aligned mode, data is written to DAC_LD12BDHR[31:0], and the module loads (after 1 PB1 clock cycle) bits[15:4] shifted into DAC_DOR1[11:0], and bits[31:20] shifted into DAC_DOR2[11:0].

In 12-bit right-aligned mode, data is written to DAC_RD12BDHR[31:0], and the module loads (after 1 PB1 clock cycle) bits[11:0] into DAC_DOR1[11:0], and bits[27:16] into DAC_DOR2[11:0].

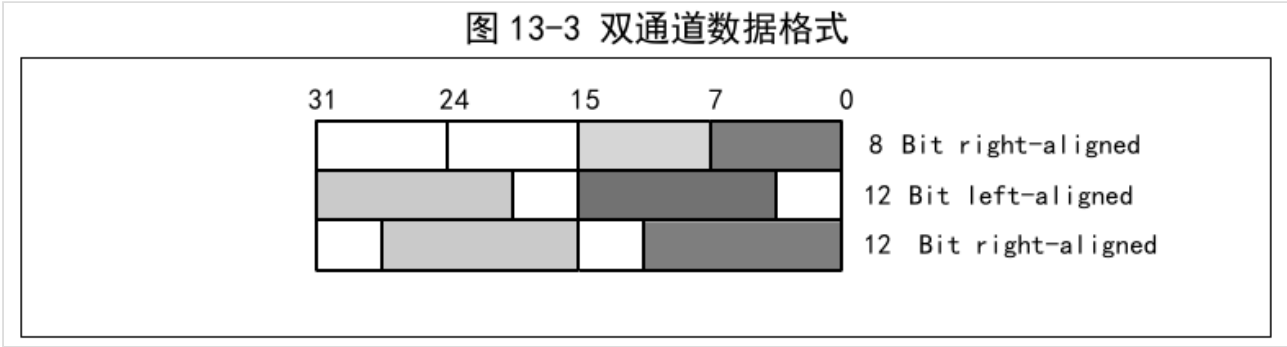


Figure 13-3: Dual-channel data format

13.2.2.4 DMA Function

The DAC channel has DMA functionality. Setting the DMAENx bit in the DAC_CTLR register to 1 enables the DMA function for the corresponding channel. When a trigger event (excluding software trigger) occurs, a DMA request is generated, and then the data in the DAC_DORx register is updated.

13.2.2.5 Trigger Event Selection

DAC conversion can be triggered by the following events. When the TENx bit in the DAC_CTLR register is set to 1, the TSELx[2:0] control bits select a specific trigger event to trigger DAC conversion.

Table 13-1: Trigger events

Trigger source	Type	TSELx[2:0]
Timer 6 TRGO event	Internal signal from on-chip timer	000
Timer 8 TRGO event		001
Timer 7 TRGO event		010
Timer 5 TRGO event		011
Timer 2 TRGO event		100
Timer 4 TRGO event		101
EXTI line 9	External pin	110
SWTRIG (software trigger)	Software control bit	111

The DAC interface detects the rising edge from the selected timer TRGO output or external interrupt line 9, and updates the DAC_DORx register with the new value 3 PB1 clock cycles after the trigger.

If software trigger mode is configured, once the SWTRIG bit is set to 1, a conversion is initiated. The DAC_DORx register is updated with the new value 1 PB1 clock cycle after the trigger, and the hardware automatically clears the SWTRIG bit.

Note: The TSELx[2:0] bits cannot be changed while ENx is 1.

13.2.3 DAC Conversion

The data for the DAC channel comes from the DAC_DORx register, but data cannot be written directly to the DAC_DORx register. Any data output to DAC channel x must be written to the DAC_R12BDHR1, DAC_L12BDHR1, DAC_R12BDHR2, DAC_L12BDHR2, DAC_RD12BDHR, DAC_LD12BDHR, or

DAC_RD8BDHR registers. The internal holding register DAC_DHRx obtains the values from the above registers and transfers them to the DAC_DORx register after the corresponding time.

In non-trigger mode, data written to the DAC_xDHRx register is shifted into the DAC_DORx register after 1 PB1 clock cycle.

In software trigger mode, the DAC_DORx register is automatically updated 1 PB1 clock cycle after the trigger event rising edge.

In hardware trigger mode (timer TRGO event or external interrupt line 9 rising edge), the DAC_DORx register is automatically updated 3 PB1 clock cycles after the trigger event.

After data is loaded into the DAC_DORx register, the output becomes valid after a settling time $t_{SETTLING}$. The duration of this time varies depending on the power supply voltage and the analog output load.

The digital input is linearly converted by the DAC to an analog voltage output in the range of 0 to VDD33A. The output voltage on any DAC channel pin satisfies the following relationship:

DAC output voltage = $VDD33A \times (DAC_DORx / 4096)$.

13.2.4 DAC Triangle Wave Generator

The module has a built-in triangle wave generator that can superimpose a small-amplitude triangle wave on the base signal. Set the WAVEx[1:0] bits to 10b to select the DAC triangle wave generation function. Set the MAMPx[3:0] bits in the DAC_CTLR register to select the triangle wave amplitude.

Internally, the system contains a triangle wave counter starting from 0, which increments by 1 after 3 PB1 clock cycles following each trigger event. The counter value is added to the DAC_DHRx register value, overflow bits are discarded, and the result is written to the DAC_DORx register. When the value loaded into the DAC_DORx register is less than the maximum amplitude defined by the MAMPx[3:0] bits, the triangle wave counter increments step by step. Once the set maximum amplitude is reached, the counter begins to decrement. After reaching 0, it begins incrementing again, repeating the cycle. Setting the WAVEx[1:0] bits to '00' resets the triangle wave generation.

Note:

1. To generate a triangle wave, the DAC trigger must be enabled, i.e., the TENx bit in the DAC_CTLR register must be set to 1.
2. The MAMPx[3:0] bits must be set before the DAC is enabled; otherwise, their values cannot be modified.

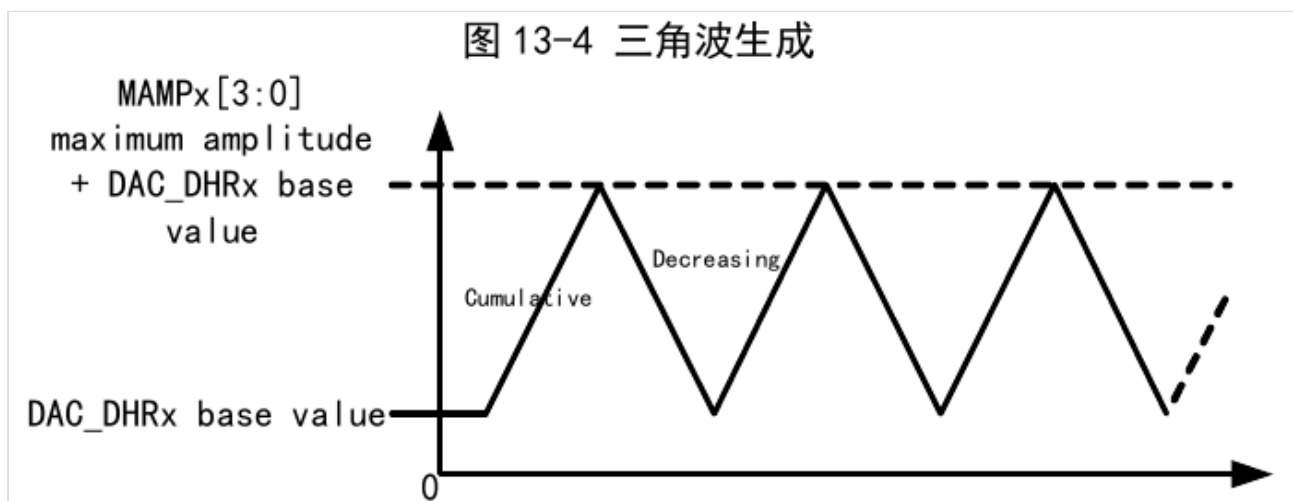


Figure 13-4: Triangle wave generation

13.2.5 DAC Noise Generator

The module has a built-in noise generator that uses a Linear Feedback Shift Register (LFSR) to generate pseudo-noise with varying amplitude. Set the WAVE[1:0] bits to 01b to select the DAC noise generation function. Set the MAMPx[3:0] bits in the DAC_CTLR register to select which part of the LFSR data to mask.

The LFSR register is preloaded with 0xAAA. According to a specific algorithm, the register value is updated after 3 PB1 clock cycles following each trigger event. Setting the MAMPx[3:0] bits in the DAC_CR register can mask part or all of the LFSR data. The resulting LFSR value is added to the DAC_DHRx value, overflow bits are discarded, and the result is written to the DAC_DORx register. If the LFSR register value is 0x000, a 1 is injected (anti-lock mechanism). Setting the WAVEx[1:0] bits to 00b resets the LFSR waveform generation algorithm.

Note: To generate noise, the DAC trigger must be enabled, i.e., the TENx bit in the DAC_CTLR register must be set to 1.

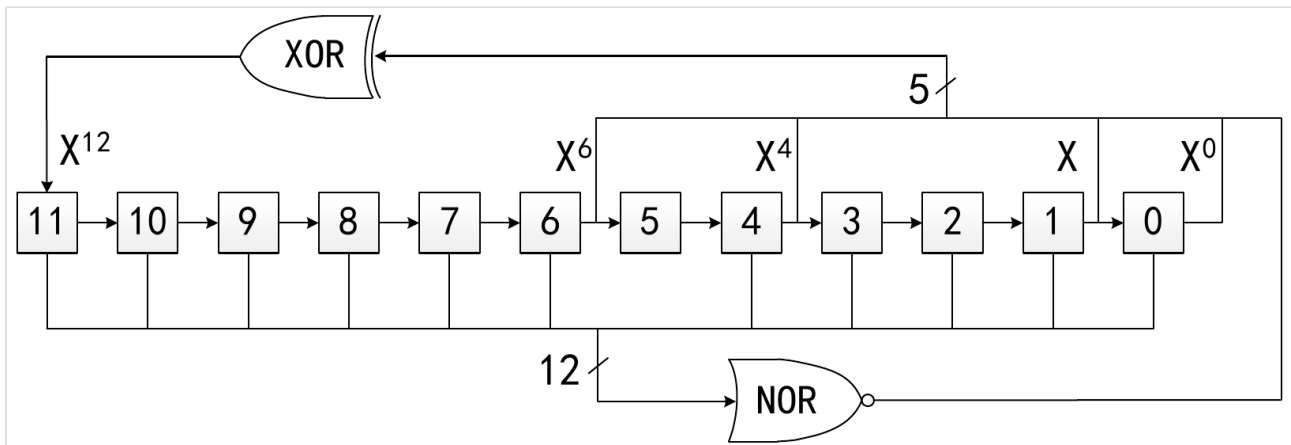


Figure 13-5: LFSR register algorithm

13.3 Dual DAC Conversion

When simultaneous conversion of two DACs is required, for more convenient and efficient operation of the DAC module, the module integrates three dual-DAC-mode data registers: DAC_RD8BDHR, DAC_LD12BDHR, and DAC_RD12BDHR. Operating any one of these registers updates the conversion values of both DACs.

Dual DAC conversion can be combined with other module registers to achieve 11 different combinations of conversion modes. The values to be converted for both channels must be written into one of the three dual-channel data registers.

13.3.1 Same LFSR with Different Triggers

Set TENx, TSELx to different values, WAVEx to 0b01, and MAMPx to the same LFSR mask value. When channel 1 trigger event occurs, the channel 1 data register DAC_DHR1 value is added to the LFSR1 counter value with the same mask, delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion, and LFSR1 is updated. When channel 2 trigger event occurs, the channel 2 data register DAC_DHR2 value is added to the LFSR2 counter value with the same mask, delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion, and LFSR2 is updated.

13.3.2 Different LFSRs with Different Triggers

Set TENx, TSELx to different values, WAVEx to 0b01, and MAMPx to different LFSR mask values. When channel 1 trigger event occurs, the channel 1 data register DAC_DHR1 value is added to the LFSR1 counter value masked by MAMP1[3:0], delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion, and LFSR1 is updated. When channel 2 trigger event occurs, the channel 2 data register DAC_DHR2 value is added to the LFSR2 counter value masked by MAMP2[3:0], delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion, and LFSR2 is updated.

13.3.3 Same Triangle Wave with Different Triggers

Set TENx, TSELx to different values, WAVEx to 0b1x, and MAMPx to the same triangle wave amplitude. When channel 1 trigger event occurs, the channel 1 data register DAC_DHR1 value is added to the triangle wave counter value with the same amplitude set by MAMPx, delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion, and channel 1 triangle wave counter is updated. When channel 2 trigger event occurs, the channel 2 data register DAC_DHR2 value is added to the triangle wave counter value with the same amplitude set by MAMPx, delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion, and channel 2 triangle wave counter is updated.

13.3.4 Different Triangle Waves with Different Triggers

Set TENx, TSELx to different values, WAVEx to 0b1x, and MAMPx to different triangle wave amplitudes. When channel 1 trigger event occurs, the channel 1 data register DAC_DHR1 value is added to the triangle wave counter value with the amplitude set by MAMP1, delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion, and channel 1 triangle wave counter is updated. When channel 2 trigger event occurs, the channel 2 data register DAC_DHR2 value is added to the triangle wave counter value with the amplitude set by MAMP2, delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion, and channel 2 triangle wave counter is updated.

13.3.5 No Waveform Generator with Different Triggers

Set TENx, TSELx to different values to select different trigger sources. When channel 1 trigger event occurs, the channel 1 data register DAC_DHR1 value is delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion. When channel 2 trigger event occurs, the channel 2 data register DAC_DHR2 value is delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion.

13.3.6 Both Using Software Trigger

In this configuration, write the conversion values to the dual-channel data registers. After 1 PB1 clock cycle, the data of DAC_DHR1 and DAC_DHR2 are sent to DAC_DOR1 and DAC_DOR2 respectively for conversion.

13.3.7 Same LFSR with Same Trigger

Set TENx, TSELx to the same value, WAVEx to 0b01, and MAMPx to the same LFSR mask value. When the trigger event occurs, the register DAC_DHR1 value is added to the LFSR1 counter value with the same mask, delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion, and LFSR1 is updated. Simultaneously, the register DAC_DHR2 value is added to the LFSR2 counter value with the same mask, delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion, and LFSR2 is updated.

13.3.8 Different LFSRs with Same Trigger

Set TENx, TSELx to the same value, WAVEx to 0b01, and MAMPx to different LFSR mask values. When the trigger event occurs, the register DAC_DHR1 value is added to the LFSR1 counter value with different mask values, delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion, and LFSR1 is updated. Simultaneously, the register DAC_DHR2 value is added to the LFSR2 counter value with different mask values, delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion, and LFSR2 is updated.

13.3.9 Same Triangle Wave with Same Trigger

Set TENx, TSELx to the same value, WAVEx to 0b1x, and MAMPx to the same triangle wave amplitude. When the trigger event occurs, the register DAC_DHR1 value is added to the counter value with the same triangle wave amplitude, delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion, and channel 1 triangle wave counter is updated. Simultaneously, the register DAC_DHR2 value is added to the counter value with the same triangle wave amplitude, delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion, and channel 2 triangle wave counter is updated.

13.3.10 Different Triangle Waves with Same Trigger

Set TENx, TSELx to the same value, WAVEx to 0b1x, and MAMPx to different triangle wave amplitudes. When the trigger event occurs, the register DAC_DHR1 value is added to the triangle wave amplitude counter value set by MAMP1[3:0], delayed 3 PB1 clock cycles, then sent to DAC_DOR1 for conversion, and channel 1 triangle wave counter is updated. Simultaneously, the register DAC_DHR2 value is added to the triangle wave amplitude counter value set by MAMP2[3:0], delayed 3 PB1 clock cycles, then sent to DAC_DOR2 for conversion, and channel 2 triangle wave counter is updated.

13.3.11 No Waveform Generator with Same Trigger

Set TENx, TSELx to the same value. In this configuration, when the trigger event occurs, the register DAC_DHR1 and DAC_DHR2 values are sent to DAC_DOR1 and DAC_DOR2 respectively for DAC conversion after a delay of 3 PB1 clock cycles.

13.4 Register Description

Table 13-2: DAC related register list

Name	Address	Description	Reset Value
R32_DAC_CTLR	0x40007400	DAC configuration register	0x00000000
R32_DAC_SWTR	0x40007404	DAC software trigger register	0x00000000
R32_DAC_R12BDHR1	0x40007408	DAC channel 1 right-aligned 12-bit data holding register	0x00000000
R32_DAC_L12BDHR1	0x4000740C	DAC channel 1 left-aligned 12-bit data holding register	0x00000000
R32_DAC_R8BDHR1	0x40007410	DAC channel 1 right-aligned 8-bit data holding register	0x00000000
R32_DAC_R12BDHR2	0x40007414	DAC channel 2 right-aligned 12-bit data holding register	0x00000000
R32_DAC_L12BDHR2	0x40007418	DAC channel 2 left-aligned 12-bit data holding register	0x00000000
R32_DAC_R8BDHR2	0x4000741C	DAC channel 2 right-aligned 8-bit data holding register	0x00000000
R32_DAC_RD12BDHR	0x40007420	Dual-channel right-aligned 12-bit data holding register	0x00000000

Name	Address	Description	Reset Value
R32_DAC_LD12BDHR	0x40007424	Dual-channel left-aligned 12-bit data holding register	0x00000000
R32_DAC_RD8BDHR	0x40007428	Dual-channel right-aligned 8-bit data holding register	0x00000000
R32_DAC_DOR1	0x4000742C	DAC channel 1 data output register	0x00000000
R32_DAC_DOR2	0x40007430	DAC channel 2 data output register	0x00000000

13.4.1 DAC Configuration Register (DAC_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			DMAEN2				MAMP2[3:0]		WAVE2[1:0]		TSEL2[2:0]		TEN2	BOFF2	EN2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DMAEN1				MAMP1[3:0]		WAVE1[1:0]		TSEL1[2:0]		TEN1	BOFF1	EN1

Bits	Name	Access	Description	Reset
[31:29]	Reserved	RO	Reserved.	0
28	DMAEN2	RW	DAC channel 2 DMA enable: 1: Enable DAC channel 2 DMA function; 0: Disable DAC channel 2 DMA function.	0
[27:24]	MAMP2[3:0]	RW	DAC channel 2 mask/amplitude setting. Software sets this field to select the LFSR data mask bits in noise generation mode, or to select the waveform amplitude in triangle wave generation mode: 0000: Unmask LFSR bit 0 / triangle wave amplitude 1; 0001: Unmask LFSR bits[1:0] / triangle wave amplitude 3; 0010: Unmask LFSR bits[2:0] / triangle wave amplitude 7; 0011: Unmask LFSR bits[3:0] / triangle wave amplitude 15; 0100: Unmask LFSR bits[4:0] / triangle wave amplitude 31; 0101: Unmask LFSR bits[5:0] / triangle wave amplitude 63; 0110: Unmask LFSR bits[6:0] / triangle wave amplitude 127; 0111: Unmask LFSR bits[7:0] / triangle wave amplitude 255; 1000: Unmask LFSR bits[8:0] / triangle wave amplitude 511; 1001: Unmask LFSR bits[9:0] / triangle wave amplitude 1023; 1010: Unmask LFSR bits[10:0] / triangle wave amplitude 2047; ≥1011: Unmask LFSR bits[11:0] / triangle wave amplitude 4095.	0000b
[23:22]	WAVE2[1:0]	RW	DAC channel 2 noise/triangle wave generation enable: 00: Disable waveform generator; 01: Enable noise waveform generator; 1x: Enable triangle wave waveform generator.	00b
[21:19]	TSEL2[2:0]	RW	DAC channel 2 trigger event selection: 000: TIM6 TRGO event; 001: TIM8 TRGO event; 010: TIM7 TRGO event; 011: TIM5 TRGO event; 100: TIM2 TRGO event; 101: TIM4 TRGO event; 110: External interrupt line 9; 111: Software trigger; Other: Reserved.	000b

Bits	Name	Access	Description	Reset
18	TEN2	RW	DAC channel 2 external trigger mode enable: 1: Enable DAC channel 2 trigger function. Data written to the DAC_xDHR register is transferred to the DAC_DOR2 register after 3 PB1 clock cycles. 0: Disable DAC channel 2 trigger function. Data written to the DAC_xDHR register is transferred to the DAC_DOR2 register after 1 PB1 clock cycle. Note: If software trigger is selected, the data in DAC_xDHR is transferred to the DAC_DOR2 register after only 1 PB1 clock cycle.	0
17	BOFF2	RW	DAC channel 2 output buffer disable control (recommended to enable): 1: Disable DAC channel 2 output buffer; 0: Enable DAC channel 2 output buffer.	0
16	EN2	RW	DAC channel 2 enable: 1: Enable DAC channel 2; 0: Disable DAC channel 2.	0
[15:13]	Reserved	RO	Reserved.	0
12	DMAEN1	RW	DAC channel 1 DMA enable: 1: Enable DAC channel 1 DMA function; 0: Disable DAC channel 1 DMA function.	0
[11:8]	MAMP1[3:0]	RW	DAC channel 1 mask/amplitude setting. Software sets this field to select the LFSR data mask bits in noise generation mode, or to select the waveform amplitude in triangle wave generation mode: 0000: Unmask LFSR bit 0 / triangle wave amplitude 1; 0001: Unmask LFSR bits[1:0] / triangle wave amplitude 3; 0010: Unmask LFSR bits[2:0] / triangle wave amplitude 7; 0011: Unmask LFSR bits[3:0] / triangle wave amplitude 15; 0100: Unmask LFSR bits[4:0] / triangle wave amplitude 31; 0101: Unmask LFSR bits[5:0] / triangle wave amplitude 63; 0110: Unmask LFSR bits[6:0] / triangle wave amplitude 127; 0111: Unmask LFSR bits[7:0] / triangle wave amplitude 255; 1000: Unmask LFSR bits[8:0] / triangle wave amplitude 511; 1001: Unmask LFSR bits[9:0] / triangle wave amplitude 1023; 1010: Unmask LFSR bits[10:0] / triangle wave amplitude 2047; ≥1011: Unmask LFSR bits[11:0] / triangle wave amplitude 4095.	0000b
[7:6]	WAVE1[1:0]	RW	DAC channel 1 noise/triangle wave generation enable: 00: Disable waveform generator; 01: Enable noise waveform generator; 1x: Enable triangle wave waveform generator.	00b
[5:3]	TSEL1[2:0]	RW	DAC channel 1 trigger event selection: 000: TIM6 TRGO event; 001: TIM8 TRGO event; 010: TIM7 TRGO event; 011: TIM5 TRGO event; 100: TIM2 TRGO event; 101: TIM4 TRGO event; 110: External interrupt line 9; 111: Software trigger; Other: Reserved.	000b

Bits	Name	Access	Description	Reset
2	TEN1	RW	DAC channel 1 external trigger mode enable: 1: Enable DAC channel 1 trigger function. Data written to the DAC_xDHR register is transferred to the DAC_DOR1 register after 3 PB1 clock cycles. 0: Disable DAC channel 1 trigger function. Data written to the DAC_xDHR register is transferred to the DAC_DOR1 register after 1 PB1 clock cycle. Note: If software trigger is selected, the data in DAC_xDHR is transferred to the DAC_DOR1 register after only 1 PB1 clock cycle.	0
1	BOFF1	RW	DAC channel 1 output buffer disable control (recommended to enable): 1: Disable DAC channel 1 output buffer; 0: Enable DAC channel 1 output buffer.	0
0	EN1	RW	DAC channel 1 enable: 1: Enable DAC channel 1; 0: Disable DAC channel 1.	0

13.4.2 DAC Software Trigger Register (DAC_SWTR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										SWTRIG2			SWTRIG1		

Bits	Name	Access	Description	Reset
[31:2]	Reserved	RO	Reserved.	0
1	SWTRIG2	WO	DAC channel 2 software trigger control bit: 1: Enable DAC channel 2 software trigger; 0: Disable DAC channel 2 software trigger. Note: Once the data in DAC_xDHR is transferred to the DAC_DOR2 register (after 1 PB1 clock cycle), this bit is hardware-cleared.	0
0	SWTRIG1	WO	DAC channel 1 software trigger control bit: 1: Enable DAC channel 1 software trigger; 0: Disable DAC channel 1 software trigger. Note: Once the data in DAC_xDHR is transferred to the DAC_DOR1 register (after 1 PB1 clock cycle), this bit is hardware-cleared.	0

13.4.3 DAC Channel 1 Right-Aligned 12-Bit Data Holding Register (DAC_R12BDHR1)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						DACC1DHR[11:0]									

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC1DHR[11:0]	RW	12-bit right-aligned data for DAC channel 1.	0

13.4.4 DAC Channel 1 Left-Aligned 12-Bit Data Holding Register (DAC_L12BDHR1)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC1DHR[11:0]												Reserved			

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:4]	DACC1DHR[11:0]	RW	12-bit left-aligned data for DAC channel 1.	0
[3:0]	Reserved	RO	Reserved.	0

13.4.5 DAC Channel 1 Right-Aligned 8-Bit Data Holding Register (DAC_R8BDHR1)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DACC1DHR[7:0]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	DACC1DHR[7:0]	RW	8-bit right-aligned data for DAC channel 1.	0

13.4.6 DAC Channel 2 Right-Aligned 12-Bit Data Holding Register (DAC_R12BDHR2)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DACC2DHR[11:0]										

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC2DHR[11:0]	RW	12-bit right-aligned data for DAC channel 2.	0

13.4.7 DAC Channel 2 Left-Aligned 12-Bit Data Holding Register (DAC_L12BDHR2)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC2DHR[11:0]												Reserved			

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:4]	DACC2DHR[11:0]	RW	12-bit left-aligned data for DAC channel 2.	0
[3:0]	Reserved	RO	Reserved.	0

13.4.8 DAC Channel 2 Right-Aligned 8-Bit Data Holding Register (DAC_R8BDHR2)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DACC2DHR[7:0]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	DACC2DHR[7:0]	RW	8-bit right-aligned data for DAC channel 2.	0

13.4.9 DAC Dual-Channel Right-Aligned 12-Bit Data Holding Register (DAC_RD12BDHR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				DACC2DHR[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DACC1DHR[11:0]											

Bits	Name	Access	Description	Reset
[31:28]	Reserved	RO	Reserved.	0
[27:16]	DACC2DHR[11:0]	RW	12-bit right-aligned data for DAC channel 2.	0
[15:12]	Reserved	RO	Reserved.	0
[11:0]	DACC1DHR[11:0]	RW	12-bit right-aligned data for DAC channel 1.	0

13.4.10 DAC Dual-Channel Left-Aligned 12-Bit Data Holding Register (DAC_LD12BDHR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DACC2DHR[11:0]												Reserved			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC1DHR[11:0]												Reserved			

Bits	Name	Access	Description	Reset
[31:20]	DACC2DHR[11:0]	RW	12-bit left-aligned data for DAC channel 2.	0
[19:16]	Reserved	RO	Reserved.	0
[15:4]	DACC1DHR[11:0]	RW	12-bit left-aligned data for DAC channel 1.	0
[3:0]	Reserved	RO	Reserved.	0

13.4.11 DAC Dual-Channel Right-Aligned 8-Bit Data Holding Register (DAC_RD8BDHR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DACC2DHR[7:0]								DACC1DHR[7:0]							

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:8]	DACC2DHR[7:0]	RW	8-bit right-aligned data for DAC channel 2.	0
[7:0]	DACC1DHR[7:0]	RW	8-bit right-aligned data for DAC channel 1.	0

13.4.12 DAC Channel 1 Data Output Register (DAC_DOR1)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					DACC1DOR[11:0]										

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC1DOR[11:0]	RO	DAC channel 1 output data.	0

13.4.13 DAC Channel 2 Data Output Register (DAC_DOR2)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						DACC2DOR[11:0]									

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
[11:0]	DACC2DOR[11:0]	RO	DAC channel 2 output data.	0

Chapter 14: Advanced Timer (ADTM)

The advanced timer module contains a powerful 16-bit auto-reload timer (TIM1, TIM8) that can be used to measure pulse widths or generate pulses, PWM waves, etc. It is used in motor control, power supply, and other fields.

14.1 Main Features

The main features of the advanced timer (TIM1/8) include:

- 16-bit auto-reload counter, supporting up-counting mode, down-counting mode, and up-down counting mode
- 16-bit prescaler with programmable divide ratio from 1 to 65536
- Four independent capture/compare channels
- Each capture/compare channel supports multiple working modes, such as: input capture, output compare, PWM generation, and single-pulse output
- Supports complementary outputs with programmable dead-time
- Supports external signal control of the timer
- Supports updating the timer after a defined number of periods using the repetition counter
- Supports using the break signal to reset the timer or set it to a defined state
- Supports DMA in multiple modes
- Supports incremental (quadrature) encoder interface
- Supports cascading and synchronization between timers

14.2 Principle and Structure

This section mainly discusses the internal structure of the advanced timer.

14.2.1 Overview

As shown in Figure 14-1, the structure of the advanced timer can be roughly divided into three parts: the input clock section, the core counter section, and the capture/compare channel section.

The clock of the advanced timer can come from the PB bus clock (CK_INT), from the external clock input pin (TIMx_ETR), from other timers with clock output capability (ITRx), or from the input pins of the capture/compare channels (TIMx_CHx). These input clock signals become CK_PSC after various configured filtering and prescaling operations, and are output to the core counter section. In addition, these complex clock sources can also be output as TRGO to other timers, ADCs, and other peripherals.

The core of the advanced timer is a 16-bit counter (CNT). CK_PSC is divided by the prescaler (PSC) to become CK_CNT and output to CNT. CNT supports up-counting mode, down-counting mode, and up-down counting mode, and has an auto-reload register (ATRLR) that reloads the initial value for CNT at the end of each counting cycle. There is also an auxiliary counter that counts the number of times ATRLR reloads the initial value for CNT. When the count reaches the value set in the repetition counter register (RPTCR), a specific event can be generated.

The advanced timer has four sets of capture/compare channels. Each capture/compare channel can input pulses from its dedicated pin or output waveforms to a pin, i.e., the capture/compare channel supports both input and output modes. The input of each capture/compare register channel supports filtering, prescaling, and edge detection operations, and supports inter-channel cross-triggering, and can also provide the clock for the core counter CNT. Each capture/compare channel has a set of capture/compare registers (CHxCVR), which support comparison with the main counter (CNT) to output pulses.

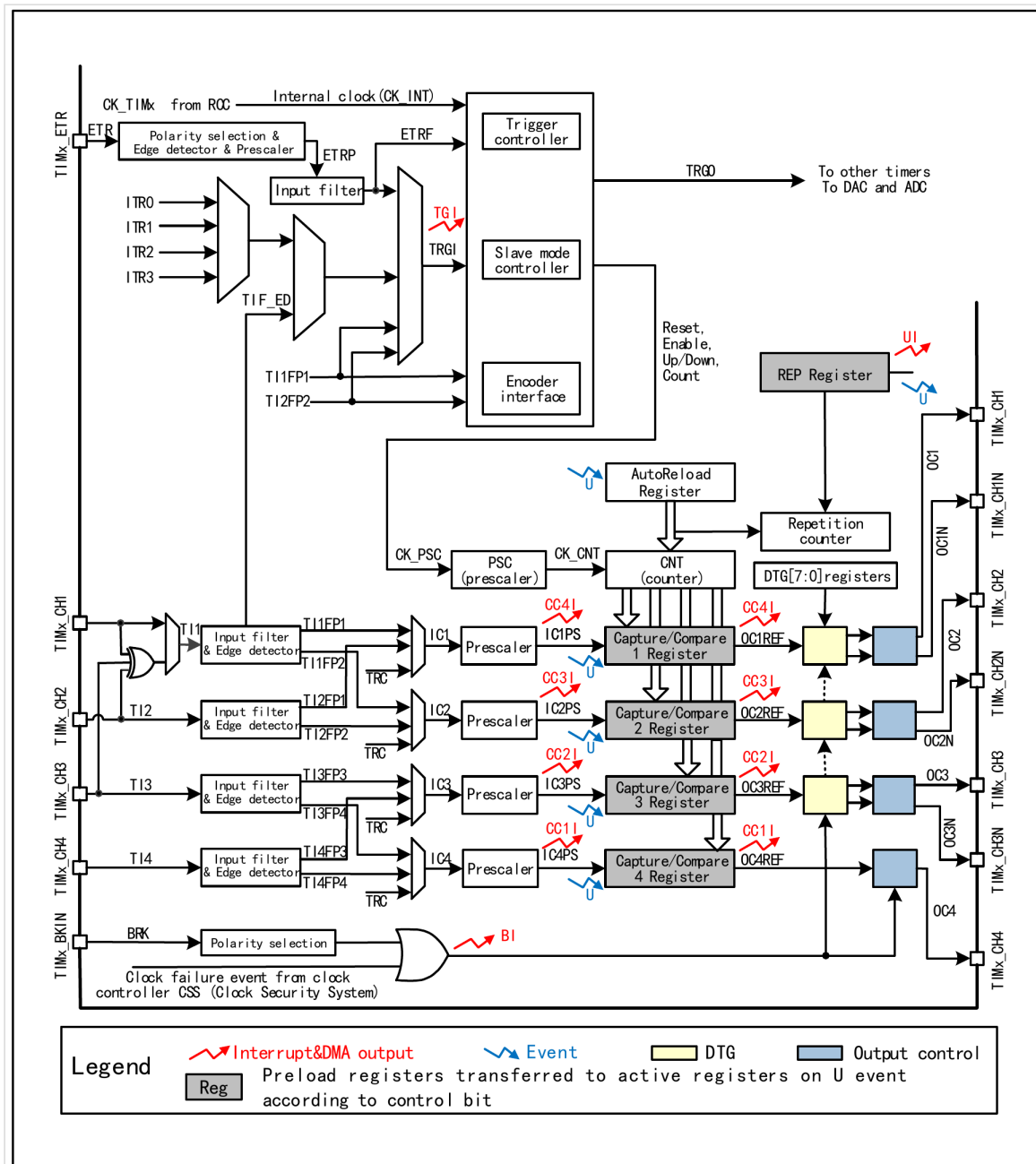


Figure 14-1: Advanced timer block diagram

14.2.2 Clock Input

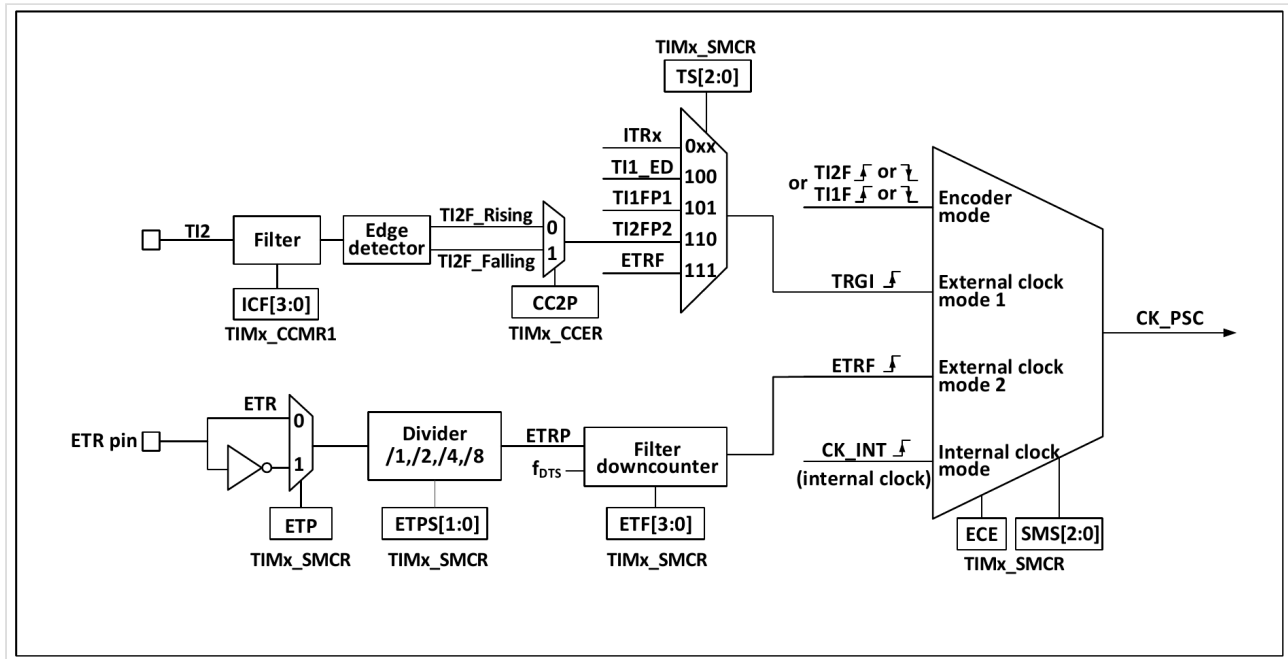


Figure 14-2: Block diagram of CK_PSC sources for the advanced timer

The advanced timer CK_PSC has many clock sources, which can be divided into 4 categories:

1. External clock pin (ETR) input clock route: ETR → ETRP → ETRF;
2. Internal PB clock input route: CK_INT;
3. Route from capture/compare channel pins (TIMx_CHx): TIMx_CHx → TIx → TIxFPx; this route is also used in encoder mode;
4. Input from other internal timers: ITRx;

By selecting the SMS input pulse that determines the CK_PSC source, the actual operations can be divided into 4 categories:

1. Selecting the internal clock source (CK_INT);
2. External clock source mode 1;
3. External clock source mode 2;
4. Encoder mode;

All four clock sources mentioned above can be selected through these four operations.

14.2.2.1 Internal Clock Source (CK_INT)

If the timer is started with the SMS field kept at 000b, the internal clock source (CK_INT) is selected as the clock. In this case, CK_INT is CK_PSC.

14.2.2.2 External Clock Source Mode 1

If the SMS field is set to 111b, external clock source mode 1 is enabled. When external clock source mode 1 is enabled, TRGI is selected as the source of CK_PSC. Note that the TS field must also be configured to select the source of TRGI. The TS field can select the following pulses as clock sources:

1. Internal triggers (ITRx, where x is 0, 1, 2, 3);

2. The signal from capture/compare channel 1 after the edge detector (TI1F_ED);
3. Capture/compare channel signals TI1FP1, TI2FP2;
4. The signal ETRF from the external clock pin input.

14.2.2.3 External Clock Source Mode 2

Using external trigger mode 2 allows counting on each rising or falling edge of the external clock pin input. When the ECE bit is set, external clock source mode 2 is used. When using external clock source mode 2, ETRF is selected as CK_PSC. The ETR pin passes through an optional inverter (ETP), a divider (ETPS) to become ETRP, and then through a filter (ETF) to become ETRF.

When the ECE bit is set and SMS is set to 111b, it is equivalent to selecting ETRF as the input via TS.

14.2.2.4 Encoder Mode

Setting SMS to 001b, 010b, or 011b will enable encoder mode. Enabling encoder mode allows selecting a specific level of TI1FP1 or TI2FP2 and using the transition edge of the other signal as the signal for output. This mode is used when an external encoder is connected. For specific functionality, refer to Section 14.3.10.

14.2.3 Counter and Peripherals

CK_PSC is input to the prescaler (PSC) for division. The PSC is 16-bit, and the actual division factor is equal to the value of R16_TIMx_PSC + 1. CK_PSC becomes CK_CNT after passing through PSC. Changing the value of R16_TIM1_PSC does not take effect immediately; it is updated to PSC after an update event. Update events include clearing the UG bit and reset. The core of the timer is a 16-bit counter (CNT). CK_CNT is finally input to CNT. CNT supports up-counting mode, down-counting mode, and up-down counting mode, and has an auto-reload register (ATRLR) that reloads the initial value for CNT at the end of each counting cycle. There is also an auxiliary counter that records the number of times ATRLR reloads the initial value for CNT. When it reaches the value set in the repetition counter register (RPTCR), a specific event can be generated.

14.2.4 Capture/Compare Channel and Peripherals

The capture/compare channel is the main component of the timer for implementing complex functions. Its core is the capture/compare register, supplemented by digital filtering of the peripheral input section, prescaling and inter-channel multiplexing, and a comparator and output control in the output section.

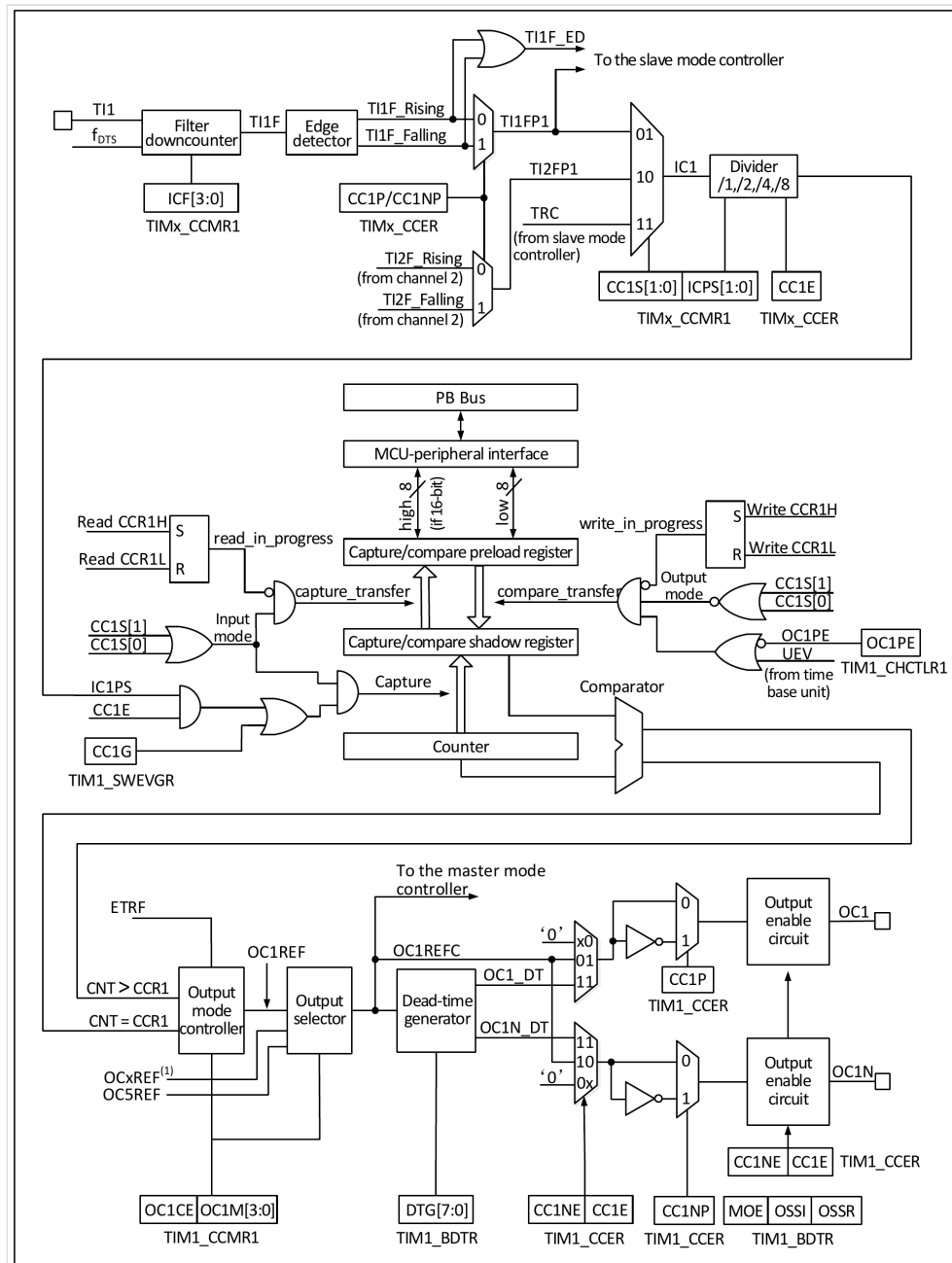


Figure 14-3: Capture/compare channel block diagram

The block diagram of the capture/compare channel is shown in Figure 14-3. After the signal is input from channel x pin, it can be selected as TIx (the source of TI1 may not only be CH1; see the timer block diagram in Figure 14-1). TI1 passes through the filter (ICF[3:0]) to generate TI1F, and then through the edge detector to produce TI1F_Rising and TI1F_Falling. These two signals are selected (CC1P) to generate TI1FP1. TI1FP1 and TI2FP1 from channel 2 are sent to CC1S for selection to become IC1, which is then prescaled by ICPS and sent to the capture/compare register.

The capture/compare register consists of a preload register and a shadow register. The read/write process only operates on the preload register. In capture mode, the capture occurs on the shadow register, and then the value is copied to the preload register. In compare mode, the content of the preload register is copied to the shadow register, and then the content of the shadow register is compared with the core counter (CNT).

14.3 Functions and Implementation

The implementation of complex functions of the advanced timer is achieved through operations on the timer's capture/compare channels, clock input circuit, and counter and peripheral sections. The timer's clock input can come from multiple clock sources, including the input of capture/compare channels. The operations on the capture/compare channels and clock source selection directly determine its function. The capture/compare channel is bidirectional and can work in both input and output modes.

14.3.1 Counter Modes

Up-counting mode

In up-counting mode, the counter counts from 0 to the auto-reload value (the content of the R16_TIMx_ATRLR register), then restarts counting from 0 and generates a counter overflow event.

If the repetition counter is used, an update event (UEV) is generated when the number of up-counting repetitions reaches the number programmed in the repetition counter register plus one (R16_TIMx_RPTCR + 1). Otherwise, an update event is generated on each counter overflow.

Setting the UG bit in the TIMx_SWEVGR register (by software or using the slave mode controller) also generates an update event.

Setting the UDIS bit in the R16_TIMx_CTLR1 register to 1 by software disables the UEV event. This prevents updating the shadow registers when new values are written to the preload registers. No update events are generated until the UDIS bit is written to 0. However, both the counter and the prescaler counter restart counting from 0 (while the prescaler ratio remains unchanged). In addition, if the URS bit (update request selection) in the R16_TIMx_CTLR1 register is set to 1, setting the UG bit generates an update event UEV but does not set the UIF flag (therefore, no interrupt or DMA request is sent). In this way, if the counter is cleared when a capture event occurs, both an update interrupt and a capture interrupt will not be generated simultaneously.

When an update event occurs, all registers are updated and the update flag (UIF bit in the R16_TIMx_INTFR register) is set to 1 (depending on the URS bit):

1. The repetition counter is reloaded with the content of the R16_TIMx_RPTCR register.
2. The auto-reload shadow register is updated with the preload value (R16_TIMx_ATRLR).
3. The prescaler buffer is reloaded with the preload value (content of the R16_TIMx_PSC register).

Down-counting mode

In down-counting mode, the counter counts down from the auto-reload value (the content of the R16_TIMx_ATRLR register) to 0, then restarts counting from the auto-reload value and generates a counter underflow event.

If the repetition counter is used, an update event (UEV) is generated when the number of down-counting repetitions reaches the number programmed in the repetition counter register plus one (R16_TIMx_RPTCR + 1). Otherwise, an update event is generated on each counter underflow.

Setting the UG bit in the R16_TIMx_EGR register (by software or using the slave mode controller) also generates an update event.

Setting the UDIS bit in the R16_TIMx_CTLR1 register to 1 by software disables the UEV update event. This prevents updating the shadow registers when new values are written to the preload registers. No update events are generated until the UDIS bit is written to 0. However, the counter restarts counting from the current auto-reload value, while the prescaler counter restarts from 0 (but the prescaler ratio remains unchanged).

In addition, if the URS bit (update request selection) in the R16_TIMx_CTLR1 register is set to 1, setting the UG bit generates an update event UEV but does not set the UIF flag (therefore, no interrupt or DMA request is sent). In this way, if the counter is cleared when a capture event occurs, both an update interrupt and a capture interrupt will not be generated simultaneously.

When an update event occurs, all registers are updated and the update flag (UIF bit in the R16_TIMx_INTFR register) is set to 1 (depending on the URS bit):

1. The repetition counter is reloaded with the content of the R16_TIMx_RPTCR register.
2. The prescaler buffer is reloaded with the preload value (content of the R16_TIMx_PSC register).
3. The auto-reload active register is updated with the preload value (content of the R16_TIMx_ATRLR register).

Note: The auto-reload register is updated before the counter is reloaded, so the next counting cycle is the desired new cycle length.

Center-aligned mode (up/down counting)

In center-aligned mode, the counter counts from 0 to the auto-reload value (content of the R16_TIMx_ATRLR register) - 1, generating a counter overflow event; then counts down from the auto-reload value to 1 and generates a counter underflow event. It then restarts counting from 0.

Center-aligned mode is active when the CMS bits in the R16_TIMx_CTLR1 register are not "00". When the channel is configured in output mode, its output compare interrupt flag is set in the following modes: counter down-counting (center-aligned mode 1, CMS="01"), counter up-counting (center-aligned mode 2, CMS="10"), and counter up/down counting (center-aligned mode 3, CMS="11").

In this mode, the DIR direction bit in the R16_TIMx_CTLR1 register cannot be written; it is updated by hardware and indicates the current counter direction.

An update event is generated on each counter overflow and underflow, or by setting the UG bit in the R16_TIMx_SWEVGR register (by software or using the slave mode controller). In this case, the counter and the prescaler counter restart counting from 0.

Setting the UDIS bit in the R16_TIMx_CTLR1 register to 1 by software disables the UEV update event. This prevents updating the shadow registers when new values are written to the preload registers. No update events are generated until the UDIS bit is written to 0. However, the counter continues to count up and down according to the current auto-reload value.

In addition, if the URS bit (update request selection) in the R16_TIMx_CTLR1 register is set to 1, setting the UG bit generates a UEV update event but does not set the UIF flag (therefore, no interrupt or DMA request is sent). In this way, if the counter is cleared when a capture event occurs, both an update interrupt and a capture interrupt will not be generated simultaneously.

When an update event occurs, all registers are updated and the update flag (UIF bit in the R16_TIMx_INTFR register) is set to 1 (depending on the URS bit):

1. The repetition counter is reloaded with the content of the R16_TIMx_RPTCR register.
2. The prescaler buffer is reloaded with the preload value (content of the R16_TIMx_PSC register).

3. The auto-reload active register is updated with the preload value (content of the R16_TIMx_ATRLR register). Note that if the update operation is triggered by a counter overflow, the auto-reload register is updated before the counter is reloaded, so the next counting cycle is the desired new cycle length (the counter is reloaded with the new value).

14.3.2 Input Capture Mode

Input capture mode is one of the basic functions of the timer. The principle of input capture mode is that when a defined edge is detected on the ICxPS signal, a capture event occurs, and the current value of the counter is latched into the capture/compare register (R16_TIMx_CHCTLRx). When a capture event occurs, CCxIF (in R16_TIMx_INTFR) is set, and if interrupt or DMA is enabled, a corresponding interrupt or DMA is generated. If CCxIF is already set when a capture event occurs, the CCxOF bit is set. CCxIF can be cleared by software or by hardware through reading the capture/compare register. CCxOF is cleared by software.

Using channel 1 as an example to illustrate the steps of using input capture mode:

1. Configure the CC1S field to select the source of the IC1 signal. For example, set it to 10b to select TI1FP1 as the source of IC1. The default setting cannot be used; the CC1S field defaults to configuring the capture/compare module as an output channel.
2. Configure the IC1F field. These bits set the sampling frequency and digital filter length of the TI1 input. The digital filter consists of an event counter that generates an output transition after recording N events.
3. Configure the CC1P bit to set the polarity of TI1FP1. For example, keep the CC1P bit low to select rising edge transition.
4. Configure the IC1PS field to set the division ratio between the IC1 signal and IC1PS. For example, keep IC1PS at 00b for no division.
5. Configure the CC1E bit to enable capturing the value of the core counter (CNT) into the capture/compare register.
6. Configure the CCxIE and CCxDE bits according to whether interrupt or DMA is needed.

When a captured pulse is input on TI1, the value of the core counter (CNT) is recorded into the capture/compare register, CC1IF is set, and when CC1IF has already been set previously, the CC1OF bit is also set. If the CC1IE bit is set, an interrupt is generated; if the CC1DE bit is set, a DMA request is generated. An input capture event can be generated by software by writing to the event generation register (TIMx_SWEVGR).

14.3.3 Output Compare Mode

Output compare mode is one of the basic functions of the timer. The principle of output compare mode is that when the value of the core counter (CNT) matches the value in the capture/compare register, a specific change or waveform is output. The OCxM field (in R16_TIMx_CHCTLRx) and the CCxP bit (in R16_TIMx_CCER) determine whether the output is a defined high/low level or a level toggle. When a compare match event occurs, the CCxIF bit is also set. If the CCxIE bit has been set beforehand, an interrupt is generated; if the CCxDE bit has been set beforehand, a DMA request is generated.

The steps to configure output compare mode are as follows:

1. Configure the clock source and auto-reload value of the core counter (CNT);
2. Set the desired compare count value into the capture/compare register (R16_TIMx_CHxCVR);

3. If an interrupt is needed, set the CCxIE bit;
4. Keep OCxPE at 0 to disable the preload register of the compare register;
5. Set the output mode, configuring the OCxM field and CCxP bit;
6. Enable the output by setting the CCxE bit;
7. Set the CEN bit to start the timer.

14.3.4 Forced Output Mode

The output mode of the timer's capture/compare channel can be forced by software to output a defined level, without depending on the comparison between the shadow register of the capture/compare register and the core counter.

The specific method is to set OCxM to 100b, which forces OCxREF to low; or set OCxM to 101b, which forces OCxREF to high.

Note that when OCxM is forced to 100b or 101b, the comparison between the internal core counter and the capture/compare register is still ongoing, the corresponding flags are still being set, and interrupts and DMA requests are still being generated.

14.3.5 PWM Input Mode

PWM input mode is used to measure the duty cycle and frequency of a PWM signal, and is a special case of input capture mode. Except for the following differences, the operation is the same as input capture mode: PWM uses two capture/compare channels, and the input polarities of the two channels are set to opposite. One of the signals is set as the trigger input, and SMS is set to reset mode.

For example, to measure the period and frequency of a PWM wave input from TI1, the following operations are needed:

1. Set TI1 (TI1FP1) as the input of IC1 signal. Set CC1S to 01b;
2. Set TI1FP1 to rising edge active. Keep CC1P at 0;
3. Set TI1 (TI1FP2) as the input of IC2 signal. Set CC2S to 10b;
4. Set TI1FP2 to falling edge active. Set CC2P to 1;
5. Select TI1FP1 as the clock source. Set TS to 101b;
6. Set SMS to reset mode, i.e., 100b;
7. Enable input capture. Set CC1E and CC2E;

With this configuration, the value in capture/compare register 1 is the PWM period, and the value in capture/compare register 2 is its duty cycle.

Note: Since only TI1FP1 and TI2FP2 are connected to the slave mode controller, PWM input mode can only use TIM1_CH1/TIM1_CH2.

14.3.6 PWM Output Mode

PWM output mode is one of the basic functions of the timer. The most common approach in PWM output mode is to use the auto-reload value to determine the PWM frequency and the capture/compare register to determine the duty cycle. Set OCxM to 110b or 111b to use PWM mode 1 or mode 2, set the OCxPE bit to enable the preload register, and finally set the ARPE bit to enable auto-reload of the preload register. Since the value of the preload register is transferred to the shadow register only when an update event occurs, the UG bit needs to be set to initialize all registers before the core counter starts counting. In PWM mode, the core counter and the capture/compare register are

continuously compared, and based on the CMS bits, the timer can output edge-aligned or center-aligned PWM signals.

Edge-aligned

When using edge-aligned mode, the core counter counts up or down. In PWM mode 1, when the value of the core counter is greater than the capture/compare register, OCxREF is high; when the value of the core counter is less than the capture/compare register (e.g., when the core counter reaches the value of R16_TIMx_ATRLR and resets to all 0), OCxREF is low.

Center-aligned

When using center-aligned mode, the core counter operates in an alternating up-counting and down-counting mode. OCxREF transitions on rising and falling edges when the value of the core counter matches the capture/compare register. However, the timing of when the compare flag is set differs among the three center-aligned modes. When using center-aligned mode, it is best to generate a software update flag (set the UG bit) before starting the core counter.

14.3.7 Complementary Output and Dead-Time

Each capture/compare channel generally has two output pins (capture/compare channel 4 has only one output pin), which can output two complementary signals (OCx and OCxN). OCx and OCxN can have their polarity set independently through the CCxP and CCxNP bits, their output enable set independently through CCxE and CCxNE, and dead-time and other controls through the MOE, OIS, OISN, OSSI, and OSSR bits. Enabling both OCx and OCxN outputs simultaneously will insert dead-time. Each channel has a 10-bit dead-time generator. If a break circuit is present, the MOE bit must also be set. OCx and OCxN are generated in association with OCxREF. If both OCx and OCxN are active high, then OCx is the same as OCxREF, except that the rising edge of OCx has a delay relative to OCxREF. OCxN is the inverse of OCxREF, and its rising edge has a delay relative to the falling edge of the reference signal. If the delay is greater than the effective output width, the corresponding pulse is not generated.

Figure 14-4 shows the relationship between OCx, OCxN, and OCxREF, and illustrates the dead-time.

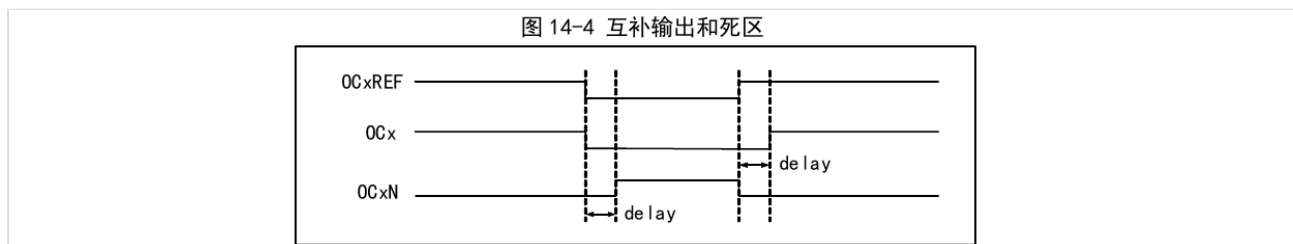


Figure 14-4: Complementary output and dead-time

14.3.8 Break Signal

When a break signal is generated, the output enable signal and inactive level are modified according to the MOE, OIS, OISN, OSSI, and OSSR bits. However, OCx and OCxN do not always stay at the active level. The break event source can come from the break input pin or from a clock failure event, which is generated by the CSS (Clock Security System).

After system reset, the break function is disabled by default (MOE bit is low). Setting the BKE bit enables the break function. The polarity of the input break signal can be set by configuring BKP. The BKE and BKP signals can be written simultaneously, but there is a PB clock delay before the actual write takes effect, so one PB cycle must be waited before the written value can be correctly read back.

When the selected level appears on the break pin, the system performs the following actions:

1. The MOE bit is asynchronously cleared, and the output is set to inactive state, idle state, or reset state depending on the OSS1 bit setting;
2. After MOE is cleared, each output channel outputs the level determined by OISx;
3. When using complementary outputs: the output is set to inactive state, depending on the polarity;
4. If BIE is set, when BIF is set, an interrupt is generated; if BDE is set, a DMA request is generated;
5. If AOE is set, the MOE bit is automatically set on the next update event UEV.

14.3.9 Single Pulse Mode

Single pulse mode can be used to have the microcontroller respond to a specific event and generate a pulse after a delay. Both the delay and pulse width are programmable. Setting the OPM bit causes the core counter to stop when the next update event UEV is generated (counter rolls over to 0).

As shown in Figure 14-5, starting from detecting a rising edge on the TI2 input pin, after a delay T_{delay} , a positive pulse of length T_{pulse} is generated on OC1:

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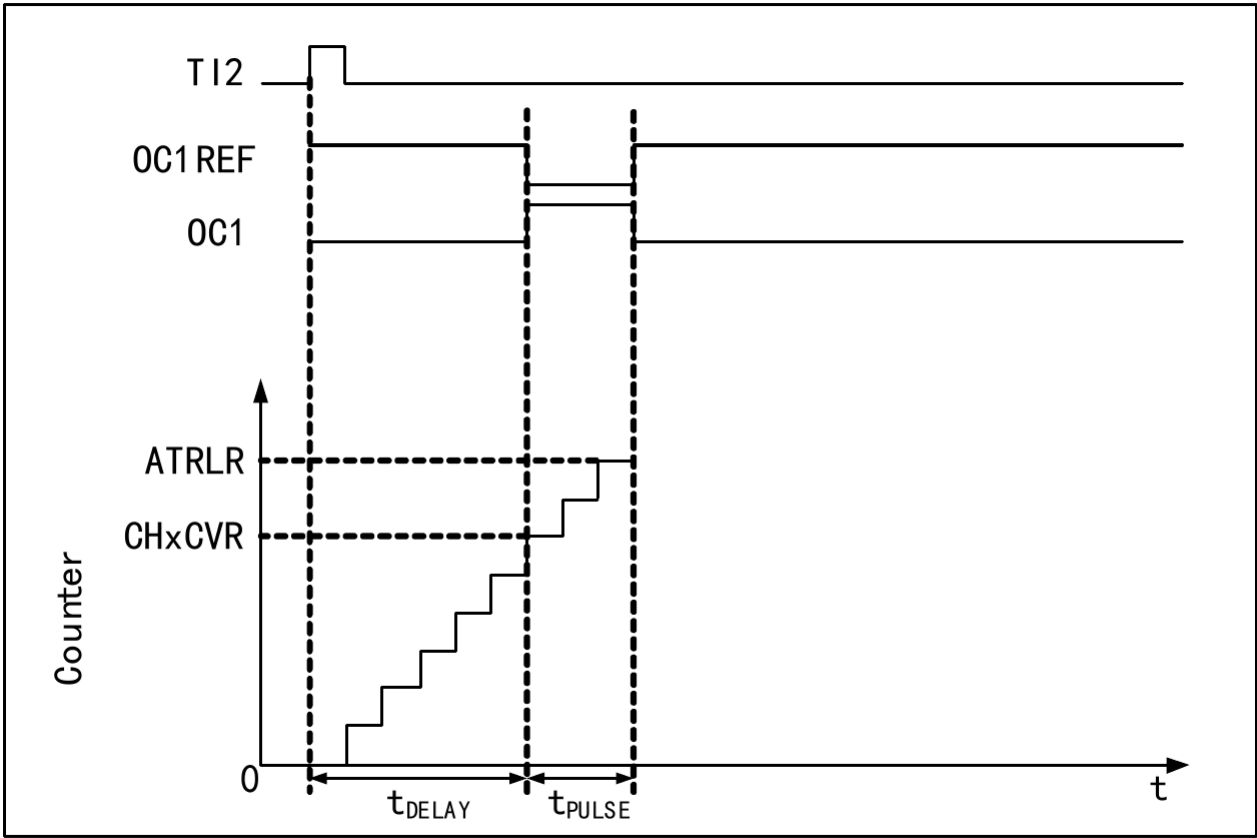


Figure 14-5: Single pulse generation

1. Set TI2 as the trigger. Set the CC2S field to 01b to map TI2FP2 to TI2; set the CC2P bit to 0b, TI2FP2 is set to rising edge detection; set the TS field to 110b, TI2FP2 is set as the trigger source; set the SMS field to 110b, TI2FP2 is used to start the counter.
2. Tdelay is determined by the value of the capture/compare register, and Tpulse is determined by the value of the auto-reload register and the value of the capture/compare register.

14.3.10 Encoder Mode

Encoder mode is a typical application of the timer. It can be used to interface with the dual-phase output of an encoder. The counting direction of the core counter is synchronized with the rotation direction of the encoder shaft. Each pulse output by the encoder causes the core counter to increment or decrement by one. The steps to use the encoder are: set the SMS field to 001b (count on TI2 edges only), 010b (count on TI1 edges only), or 011b (count on both TI1 and TI2 edges), connect the encoder to the inputs of capture/compare channels 1 and 2, and set a value in the auto-reload register. This value can be set relatively large. In encoder mode, the timer's internal capture/compare registers, prescaler, repetition counter register, etc. all work normally. The following table shows the relationship between the counting direction and the encoder signals.

Table 14-1: Relationship between counting direction and encoder signals in timer encoder mode

Active counting edge	Level of the opposite signal	TI1FP1 signal edge		TI2FP2 signal edge	
		Rising	Falling	Rising	Falling
Count on TI1 edges only	High	Down-count	Up-count	No count	
	Low	Up-count	Down-count		
Count on TI2 edges only	High	No count		Up-count	Down-count
	Low			Down-count	Up-count
Count on both TI1 and TI2 edges	High	Down-count	Up-count	Up-count	Down-count
	Low	Up-count	Down-count	Down-count	Up-count

14.3.11 Timer Synchronization Mode

TIMx timers are internally connected together to achieve timer synchronization or cascading. When a timer is configured as master mode, it can perform reset, start, stop operations on the counter of another timer configured as slave mode, or provide its clock.

Using one timer as a prescaler for another timer

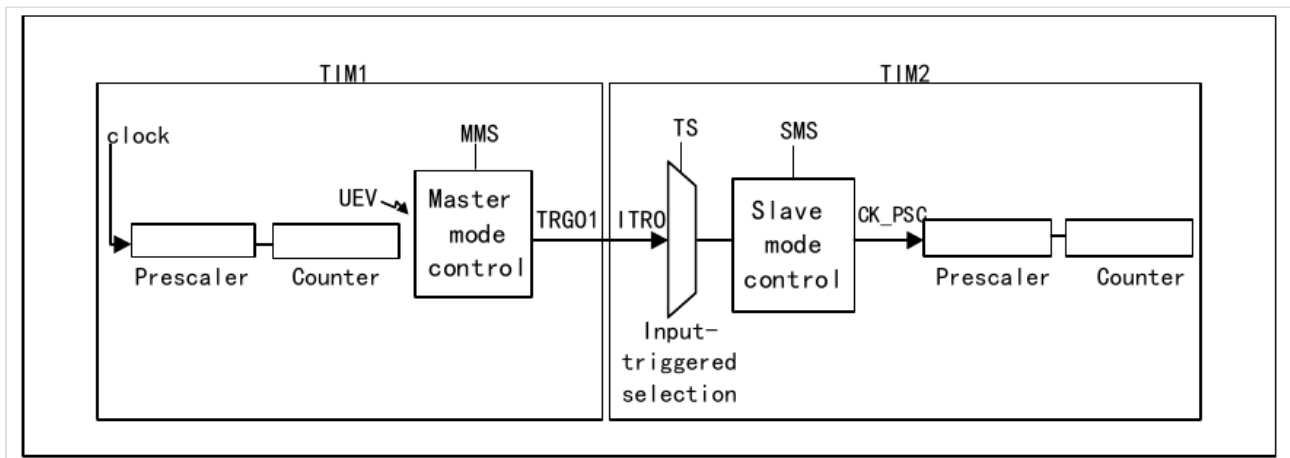


Figure 14-10: Master/slave timer example

For example, timer 1 can be configured as a prescaler for timer 2. To do this:

1. Configure timer 1 as master mode to output a periodic trigger signal each time an update event UEV occurs. If MMS=010 is written to R16_TIM1_CTLR2, then TRGO1 will output a rising edge whenever an update event is generated.
2. To connect the TRGO1 output of timer 1 to timer 2, timer 2 must be configured as slave mode using ITR0 as the internal trigger. This can be selected through the TS bits in the R16_TIM2_SMCFGR register (write TS=000).
3. Then set the slave mode controller to external clock mode 1 (write SMS=111 in the R16_TIM2_SMCFGR register). In this way, the clock of timer 2 is provided by the rising edge of timer 1's periodic trigger signal (corresponding to the counter overflow of timer 1).

4. Finally, both timers must be enabled simultaneously by setting the corresponding CEN bits (R16_TIMx_CTLR1 register) of both timers to 1.

Note: If the OCx signal of timer 1 is selected as the trigger output (MMS=1xx), the rising edge of this signal will be used to drive the counter of timer 2.

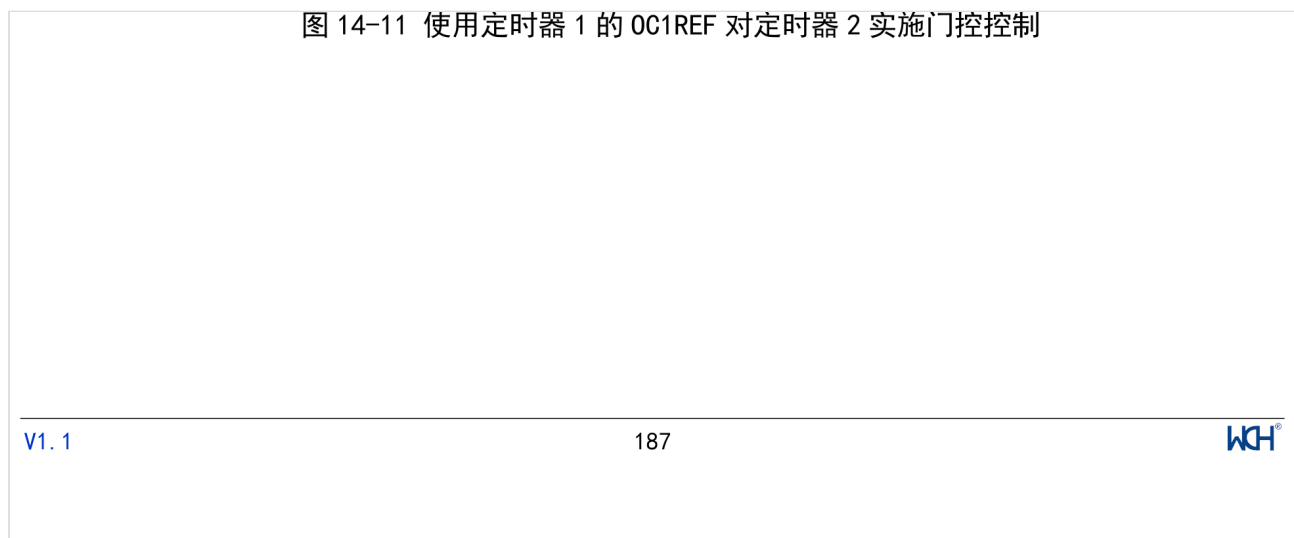
Using one timer to enable another timer

In this example, timer 2 is enabled by the output compare 1 of timer 1. Timer 2 counts according to the divided internal clock only when OC1REF of timer 1 is high. The clock frequency of both counters is based on CK_INT divided by 3 through the prescaler ($f_{CK_CNT} = f_{CK_INT}/3$).

1. Configure timer 1 as master mode, sending its output compare 1 reference signal (OC1REF) as trigger output (MMS=100 in the R16_TIM1_CTLR2 register).
2. Configure the OC1REF waveform of timer 1 (R16_TIM1_CCMR1 register).
3. Configure timer 2 to receive the input trigger from timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
4. Configure timer 2 as gated mode (SMS=101 in the R16_TIM2_SMCFGR register).
5. Enable timer 2 by writing "1" to the CEN bit (R16_TIM2_CTLR1 register).
6. Start timer 1 by writing "1" to the CEN bit (R16_TIM1_CTLR1 register).

Note: The clock of counter 2 is not synchronized with counter 1; this mode only affects the counter enable signal of timer 2.

图 14-11 使用定时器 1 的 OC1REF 对定时器 2 实施门控控制



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Figure 14-11: Gating control of timer 2 using OC1REF of timer 1

The counter and prescaler of timer 2 are not initialized before starting. Therefore, they start counting from their respective current values. Before starting timer 1, both timers can be reset to start counting from specified values. This allows any desired value to be written into the timer counters. Both timers can be easily reset by software using the UG bit in the R16_TIMx_SWEVGR register.

In the next example, timer 1 is synchronized with timer 2. Timer 1 is in master mode, counting from 0. Timer 2 is in slave mode, counting from 0xE7. Both timers have the same prescaler ratio. When timer 1 is disabled by writing "0" to the CEN bit in the R16_TIM1_CTLR1 register, timer 2 will stop:

1. Configure timer 1 as master mode, sending its output compare 1 reference signal (OC1REF) as trigger output (MMS=100 in the R16_TIM1_CTLR2 register).
2. Configure the OC1REF waveform of timer 1 (R16_TIM1_CHCTL1 register).
3. Configure timer 2 to receive the input trigger from timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
4. Configure timer 2 as gated mode (SMS=101 in the R16_TIM2_SMCFGR register).
5. Reset timer 1 by writing "1" to the UG bit (R16_TIM1_SWEVGR register).
6. Reset timer 2 by writing "1" to the UG bit (R16_TIM2_SWEVGR register).
7. Initialize timer 2 to 0xE7 by writing "0xE7" into the counter of timer 2 (R16_TIM2_CNTL).
8. Enable timer 2 by writing "1" to the CEN bit (R16_TIM2_CTLR1 register).
9. Start timer 1 by writing "1" to the CEN bit (R16_TIM1_CTLR1 register).
10. Stop timer 1 by writing "0" to the CEN bit (R16_TIM1_CTLR1 register).

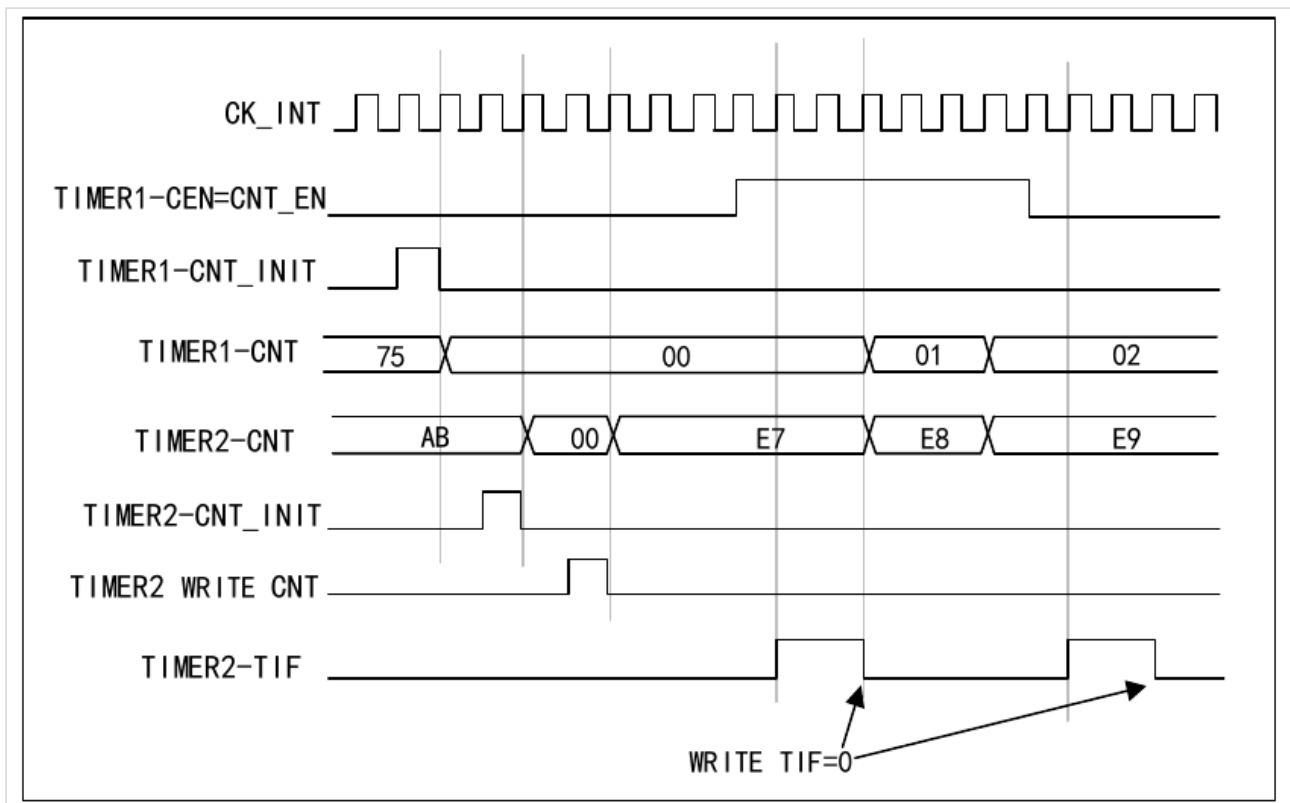


Figure 14-12: Gating control of timer 2 using the enable signal of timer 1

Using one timer to start another timer

In this example, the update event of timer 1 is used to enable timer 2. Whenever timer 1 generates an update event, timer 2 starts counting from its current value (which may not be 0) according to the divided internal clock. When timer 2 receives the trigger signal, its CEN bit is automatically set to 1, and the counter starts counting until "0" is written to the CEN bit of the R16_TIM2_CTLR1 register. The

clock frequency of both counters is based on CK_INT divided by 3 through the prescaler ($f_{CK_CNT} = f_{CK_INT}/3$).

1. Configure timer 1 as master mode, sending its update event (UEV) as trigger output (MMS=010 in the R16_TIM1_CTLR2 register).
2. Configure the period of timer 1 (R16_TIM1_ATRLR register).
3. Configure timer 2 to receive the input trigger from timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
4. Configure timer 2 as trigger mode (SMS=110 in the R16_TIM2_SMCFGR register).
5. Start timer 1 by writing "1" to the CEN bit (R16_TIM1_CTLR1 register).

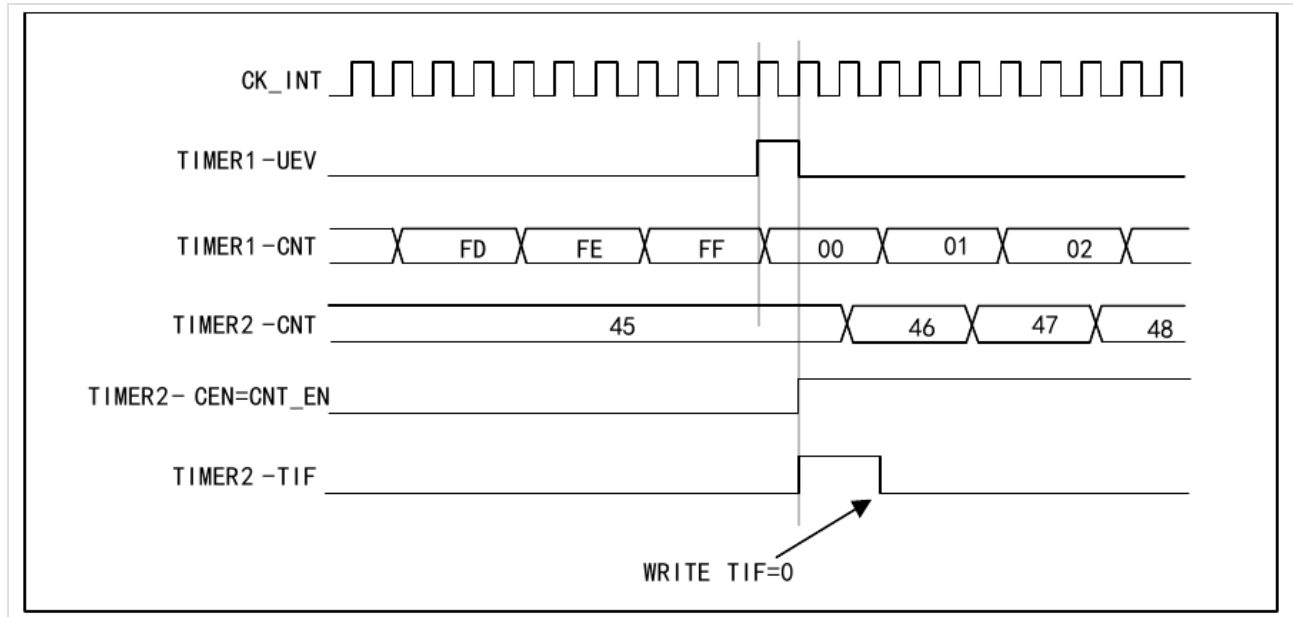


Figure 14-13: Triggering timer 2 using the update event of timer 1

As shown in the above example, the user can initialize both counters before starting to count. Figure 14-14 shows the counting behavior with the same configuration as Figure 14-13, except in trigger mode (SMS=110 in the R16_TIM2_SMCFGR register) instead of gated mode.

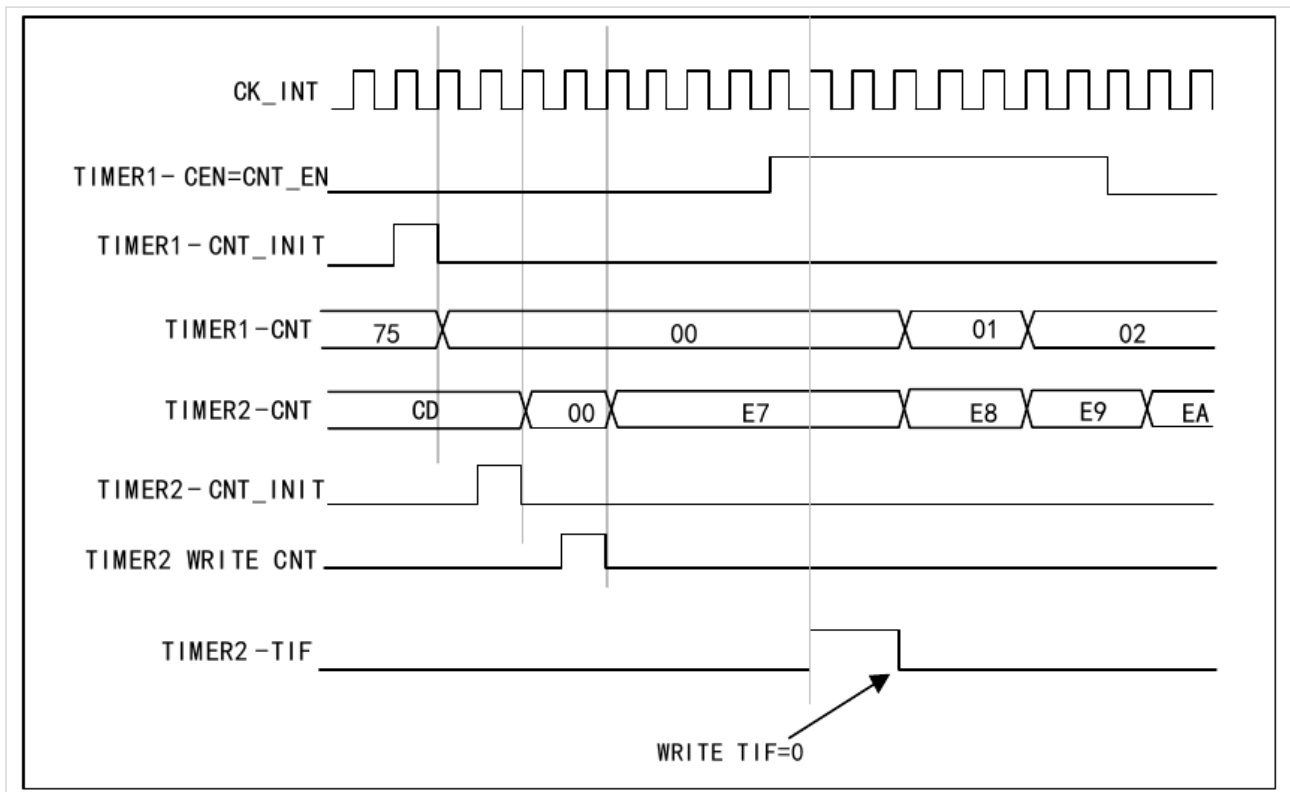


Figure 14-14: Triggering timer 2 using the enable signal of timer 1

Using one timer as a prescaler for another timer

For example, timer 1 can be configured as a prescaler for timer 2. To do this:

1. Configure timer 1 as master mode, sending its update event (UEV) as trigger output (MMS=010 in the R16_TIM1_CTLR2 register). This outputs a periodic signal on each counter overflow.
2. Configure the period of timer 1 (R16_TIM1_ATRLR register).
3. Configure timer 2 to receive the input trigger from timer 1 (TS=000 in the R16_TIM2_SMCFGR register).
4. Configure timer 2 as external clock mode (SMS=111 in the R16_TIM2_SMCFGR register).
5. Start timer 2 by writing "1" to the CEN bit (R16_TIM2_CTLR1 register).
6. Start timer 1 by writing "1" to the CEN bit (R16_TIM1_CTLR1 register).

Starting two timers synchronized by an external trigger

In this example, when a rising edge appears on the TI1 input of timer 1, timer 1 is enabled, and simultaneously timer 2 is enabled. To ensure alignment of both counters, timer 1 must be configured as master/slave mode (corresponding TI1 as slave, and corresponding to timer 2 as master):

1. Configure timer 1 as master mode, sending its enable signal as trigger output (MMS=001 in the R16_TIM1_CTLR2 register).
2. Configure timer 1 as slave mode to receive the input trigger from TI1 (TS=100 in the R16_TIM1_SMCFGR register).
3. Configure timer 1 as trigger mode (SMS=110 in the R16_TIM1_SMCFGR register).
4. Configure timer 1 as master/slave mode by writing MSM=1 (R16_TIMx_SMCR register).
5. Configure timer 2 to receive the input trigger from timer 1 (TS=000 in the R16_TIM2_SMCFGR register).

6. Configure timer 2 as trigger mode (SMS=110 in the R16_TIM2_SMCFGR register).

When a rising edge appears on TI1 (timer 1), both counters start counting synchronously according to the internal clock, and both TIF flags are set to 1.

Note: In this example, both timers are initialized before starting (by setting the respective UG bit to 1). Both counters start counting from 0, but an offset can be easily inserted between them by writing to either counter register (R16_TIMx_CNT). Note that the master/slave mode produces a delay between CNT_EN and CK_PSC of timer 1.

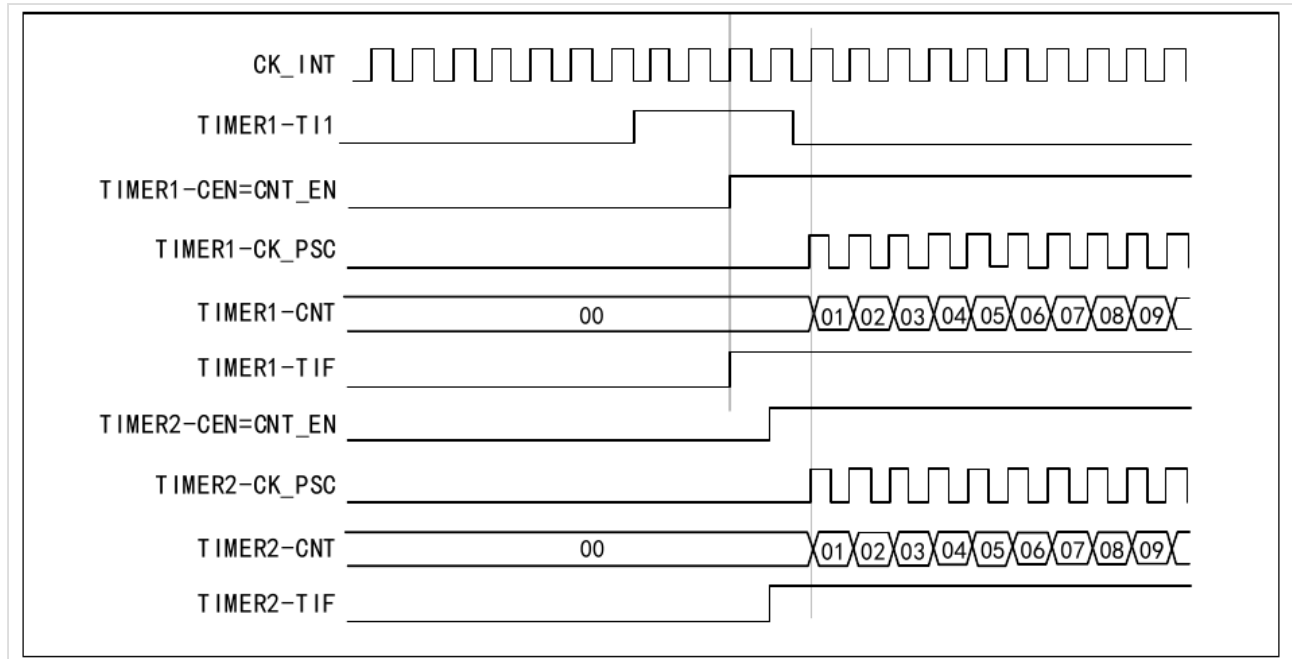


Figure 14-15: Triggering timer 1 and timer 2 using the TI1 input of timer 1

The timer can output clock pulses (TRGO) and can also receive inputs from other timers (ITRx). The sources of ITRx (the TRGO of other timers) are different for different timers. The timer internal trigger connections are shown in Table 14-2.

Table 14-2: TIMx internal trigger connections

Slave timer	ITR0 (TS=000)	ITR1 (TS=001)	ITR2 (TS=010)	ITR3 (TS=011)
TIM1	-	TIM2	TIM3	TIM4

14.3.12 Debug Mode

When the system enters debug mode, the timer continues running or stops according to the DBG module settings.

14.3.13 Dual-edge Capture Mode

Pulse measurement can be performed through the TIMx_AUX register.

For example: to capture pulse width through channel 2, in the capture function configuration, select the source of the IC2 signal (CC2S bits of the TIMx_CHCTLR1 register set to 11b), and enable the dual-edge capture function of channel 2 (CAP_ED_CH2 bit of the TIMx_AUX register set to 1). The dual-edge capture is now configured.

The captured dual-edge pulse width value can be read through the CCR2 bits of the TIMx_CH2CVR register.

14.4 Register Description

Table 14-3: TIM1 related register list

Name	Address	Description	Reset Value
R16_TIM1_CTLR1	0x40012C00	Control register 1	0x0000
R16_TIM1_CTLR2	0x40012C04	Control register 2	0x0000
R16_TIM1_SMCFR	0x40012C08	Slave mode control register	0x0000
R16_TIM1_DMAINTENR	0x40012C0C	DMA/interrupt enable register	0x0000
R16_TIM1_INTFR	0x40012C10	Interrupt status register	0x0000
R16_TIM1_SWEVGR	0x40012C14	Event generation register	0x0000
R16_TIM1_CHCTLR1	0x40012C18	Capture/compare control register 1	0x0000
R16_TIM1_CHCTLR2	0x40012C1C	Capture/compare control register 2	0x0000
R16_TIM1_CCER	0x40012C20	Capture/compare enable register	0x0000
R16_TIM1_CNT	0x40012C24	Counter	0x0000
R16_TIM1_PSC	0x40012C28	Counter clock prescaler	0x0000
R16_TIM1_ATRLR	0x40012C2C	Auto-reload register	0xFFFF
R16_TIM1_RPTCR	0x40012C30	Repetition counter register	0x0000
R16_TIM1_CH1CVR	0x40012C34	Capture/compare register 1	0x0000
R16_TIM1_CH2CVR	0x40012C38	Capture/compare register 2	0x0000
R16_TIM1_CH3CVR	0x40012C3C	Capture/compare register 3	0x0000
R16_TIM1_CH4CVR	0x40012C40	Capture/compare register 4	0x0000
R16_TIM1_BDTR	0x40012C44	Break and dead-time register	0x0000
R16_TIM1_DMACFR	0x40012C48	DMA control register	0x0000
R16_TIM1_DMAADR	0x40012C4C	DMA address for continuous mode register	0x0000
R16_TIM1_AUX	0x40012C50	Dual-edge capture register	0x0000

Table 14-4: TIM8 related register list

Name	Address	Description	Reset Value
R16_TIM8_CTLR1	0x40013400	Control register 1	0x0000
R16_TIM8_CTLR2	0x40013404	Control register 2	0x0000
R16_TIM8_SMCFR	0x40013408	Slave mode control register	0x0000
R16_TIM8_DMAINTENR	0x4001340C	DMA/interrupt enable register	0x0000
R16_TIM8_INTFR	0x40013410	Interrupt status register	0x0000
R16_TIM8_SWEVGR	0x40013414	Event generation register	0x0000
R16_TIM8_CHCTLR1	0x40013418	Capture/compare control register 1	0x0000
R16_TIM8_CHCTLR2	0x4001341C	Capture/compare control register 2	0x0000
R16_TIM8_CCER	0x40013420	Capture/compare enable register	0x0000

Name	Address	Description	Reset Value
R16_TIM8_CNT	0x40013424	Counter	0x0000
R16_TIM8_PSC	0x40013428	Counter clock prescaler	0x0000
R16_TIM8_ATRLR	0x4001342C	Auto-reload register	0xFFFF
R16_TIM8_RPTCR	0x40013430	Repetition counter register	0x0000
R16_TIM8_CH1CVR	0x40013434	Capture/compare register 1	0x0000
R16_TIM8_CH2CVR	0x40013438	Capture/compare register 2	0x0000
R16_TIM8_CH3CVR	0x4001343C	Capture/compare register 3	0x0000
R16_TIM8_CH4CVR	0x40013440	Capture/compare register 4	0x0000
R16_TIM8_BDTR	0x40013444	Break and dead-time register	0x0000
R16_TIM8_DMACHFR	0x40013448	DMA control register	0x0000
R16_TIM8_DMAADR	0x4001344C	DMA address for continuous mode register	0x0000
R16_TIM8_AUX	0x40013450	Dual-edge capture register	0x0000

14.4.1 Control Register 1 (TIMx_CTLR1) (x=1/8)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CKD[1:0]		ARPE	CMS[1:0]		DIR	OPM	URS	UDIS	CEN

Bits	Name	Access	Description	Reset
[15:10]	Reserved	RO	Reserved.	0
[9:8]	CKD[1:0]	RW	These 2 bits define the division ratio between the timer clock (CK_INT) frequency, the dead-time, and the sampling clock used by the dead-time generator and digital filter (ETR, TIX): 00: Tdts = Tck_int; 01: Tdts = 2 × Tck_int; 10: Tdts = 4 × Tck_int; 11: Reserved.	00b
7	ARPE	RW	Auto-reload preload enable: 1: Auto-reload register (ATRLR) is enabled; 0: Auto-reload register (ATRLR) is disabled.	0
[6:5]	CMS[1:0]	RW	Center-aligned mode selection: 00: Edge-aligned mode. The counter counts up or down according to the direction bit (DIR). 01: Center-aligned mode 1. The counter counts up and down alternately. The output compare interrupt flags of channels configured as output (CCxS=00 in CHCTLRx register) are set only when the counter is counting down. 10: Center-aligned mode 2. The counter counts up and down alternately. The output compare interrupt flags of channels configured as output (CCxS=00 in CHCTLRx register) are set only when the counter is counting up. 11: Center-aligned mode 3. The counter counts up and down alternately. The output compare interrupt flags of channels configured as output (CCxS=00 in CHCTLRx register) are set both when the counter is counting	00b

Bits	Name	Access	Description	Reset
			up and down. Note: It is not allowed to switch from edge-aligned mode to center-aligned mode while the counter is enabled (CEN=1).	
4	DIR	RW	Counter direction: 1: The counter counts down; 0: The counter counts up. Note: This bit is invalid when the counter is configured in center-aligned mode or encoder mode.	0
3	OPM	RW	One-pulse mode: 1: The counter stops on the next update event (CEN bit is cleared); 0: The counter does not stop on the next update event.	0
2	URS	RW	Update request source. Software selects the source of UEV events through this bit. 1: If update interrupt or DMA request is enabled, only counter overflow/underflow generates update interrupt or DMA request; 0: If update interrupt or DMA request is enabled, any of the following events generates update interrupt or DMA request: - Counter overflow/underflow; - Setting the UG bit; - Update generated by the slave mode controller.	0
1	UDIS	RW	Update disable. Software enables/disables UEV events through this bit. 1: UEV disabled. No update events are generated, and registers (ARR, PSC, CCRx) keep their values. If the UG bit is set or a hardware reset is issued by the slave mode controller, the counter and prescaler are reinitialized; 0: UEV enabled. Update (UEV) events are generated by any of the following events: - Counter overflow/underflow; - Setting the UG bit; - Update from the slave mode controller. Buffered registers are loaded with their preload values.	0
0	CEN	RW	Counter enable: 1: Counter enabled; 0: Counter disabled. Note: After the software sets the CEN bit, external clock, gated mode, and encoder mode can work. Trigger mode can automatically set the CEN bit through hardware.	0

14.4.2 Control Register 2 (TIMx_CTLR2) (x=1/8)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	TI1S	MMS[2:0]	CCDS	CCUS	Reserved	Reserved	Reserved	CCPC

Bits	Name	Access	Description	Reset
15	Reserved	RO	Reserved.	0
14	OIS4	RW	Output idle state 4: 1: When MOE=0, if OC4N is implemented, OC4=1 after dead-time;	0

Bits	Name	Access	Description	Reset
			0: When MOE=0, if OC4N is implemented, OC4=0 after dead-time. Note: This bit cannot be modified after LOCK level 1, 2, or 3 has been set (TIMx_BDTR register).	
13	OIS3N	RW	Output idle state 3: 1: When MOE=0, OC3N=1 after dead-time; 0: When MOE=0, OC3N=0 after dead-time. Note: This bit cannot be modified after LOCK level 1, 2, or 3 has been set (TIMx_BDTR register).	0
12	OIS3	RW	Output idle state 3, see OIS4.	0
11	OIS2N	RW	Output idle state 2, see OIS3N.	0
10	OIS2	RW	Output idle state 2, see OIS4.	0
9	OIS1N	RW	Output idle state 1, see OIS3N.	0
8	OIS1	RW	Output idle state 1, see OIS4.	0
7	TI1S	RW	TI1 selection: 1: TIMx_CH1, TIMx_CH2, and TIMx_CH3 pins are XOR'ed and connected to TI1 input; 0: TIMx_CH1 pin is directly connected to TI1 input.	0
[6:4]	MMS[2:0]	RW	Master mode selection. These 3 bits select the synchronization information (TRGO) sent to the slave timer in master mode. The possible combinations are: 000: Reset – The UG bit of the TIMx_EGR register is used as trigger output (TRGO). If the reset is generated by the trigger input (slave mode controller in reset mode), the signal on TRGO is delayed relative to the actual reset; 001: Enable – The counter enable signal CNT_EN is used as trigger output (TRGO). This is useful when starting multiple timers simultaneously or controlling the enable of a slave timer for a certain period. The counter enable signal is generated by the logical OR of the CEN control bit and the trigger input signal in gated mode. When the counter enable signal is controlled by the trigger input, there is a delay on TRGO, unless master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCR register); 010: Update – The update event is selected as trigger input (TRGO). For example, the clock of a master timer can be used as a prescaler for a slave timer; 011: Compare pulse – The trigger output sends a positive pulse (TRGO) when a capture or a compare match occurs and the CC1IF flag is to be set (even if it is already high); 100: Compare – OC1REF signal is used as trigger output (TRGO); 101: Compare – OC2REF signal is used as trigger output (TRGO); 110: Compare – OC3REF signal is used as trigger output (TRGO); 111: Compare – OC4REF signal is used as trigger output (TRGO).	000b
3	CCDS	RW	Capture/compare DMA selection. 1: DMA request of CHxCVR is sent when an update event occurs; 0: DMA request of CHxCVR is generated when a CHxCVR event occurs.	0
2	CCUS	RW	Capture/compare control update selection. 1: If CCPC is set, they can be updated by setting the COM bit or by a rising edge on TRGI; 0: If CCPC is not set, they can be updated only by setting the COM bit.	0

Bits	Name	Access	Description	Reset
			0: If CCPC is set, they can only be updated by setting the COM bit. Note: This bit only applies to channels with complementary outputs.	
1	Reserved	RO	Reserved.	0
0	CCPC	RW	Capture/compare preload control bit. 1: CCxE, CCxNE, and OCxM bits are preloaded; after setting this bit, they are updated only after the COM bit is set; 0: CCxE, CCxNE, and OCxM bits are not preloaded. Note: This bit only applies to channels with complementary outputs.	0

14.4.3 Slave Mode Control Register (TIMx_SMCFGR) (x=1/8)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS[1:0]		ETF[3:0]		MSM			TS[2:0]			Reserved		SMS[2:0]	

Bits	Name	Access	Description	Reset
15	ETP	RO	ETR trigger polarity selection. This bit selects whether the ETR is input directly or the inverted ETR is input. 1: ETR is inverted, active low level or falling edge; 0: ETR, active high level or rising edge.	0
14	ECE	RW	External clock mode 2 enable selection: 1: External clock mode 2 enabled; 0: External clock mode 2 disabled. Note 1: The slave mode can be used simultaneously with external clock mode 2: reset mode, gated mode, and trigger mode; however, in this case TRGI cannot be connected to ETRF (TS bits cannot be '111'). Note 2: When both external clock mode 1 and external clock mode 2 are enabled simultaneously, the external clock input is ETRF.	0
[13:12]	ETPS[1:0]	RW	External trigger signal (ETRP) prescaler. The maximum frequency of this signal cannot exceed 1/4 of the TIMxCLK frequency. This field can be used to reduce the frequency: 00: Prescaler disabled; 01: ETRP frequency divided by 2; 10: ETRP frequency divided by 4; 11: ETRP frequency divided by 8.	00b
[11:8]	ETF[3:0]	RW	External trigger filter. In practice, the digital filter is an event counter that uses a certain sampling frequency and produces an output transition after recording N events. 0000: No filter, sampling at Fdts; 0001: Sampling frequency F _{sampling} =F _{ck_int} , N=2; 0010: Sampling frequency F _{sampling} =F _{ck_int} , N=4; 0011: Sampling frequency F _{sampling} =F _{ck_int} , N=8; 0100: Sampling frequency F _{sampling} =F _{dts} /2, N=6; 0101: Sampling frequency F _{sampling} =F _{dts} /2, N=8; 0110: Sampling frequency F _{sampling} =F _{dts} /4, N=6; 0111: Sampling frequency F _{sampling} =F _{dts} /4, N=8; 1000: Sampling frequency F _{sampling} =F _{dts} /8, N=6; 1001: Sampling frequency F _{sampling} =F _{dts} /8, N=8; 1010: Sampling frequency F _{sampling} =F _{dts} /16, N=5;	0000b

Bits	Name	Access	Description	Reset
			1011: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, $N=6$; 1100: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, $N=8$; 1101: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N=5$; 1110: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N=6$; 1111: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, $N=8$.	
7	MSM	RW	Master/slave mode selection: 1: Events on the trigger input (TRGI) are delayed to allow perfect synchronization between the current timer (via TRGO) and its slave timer. This is very useful when several timers need to be synchronized to a single external event; 0: No effect.	0
[6:4]	TS[2:0]	RW	Trigger selection field. These 3 bits select the trigger input source used for synchronizing the counter: 000: Internal trigger 0 (ITR0); 001: Internal trigger 1 (ITR1); 010: Internal trigger 2 (ITR2); 011: Internal trigger 3 (ITR3); 100: TI1 edge detector (TI1F_ED); 101: Filtered timer input 1 (TI1FP1); 110: Filtered timer input 2 (TI2FP2); 111: External trigger input (ETRF); The above can only be changed when SMS is 0. Note: See Table 14-2 for details.	000b
3	Reserved	RO	Reserved.	0
[2:0]	SMS[2:0]	RW	Input mode selection field. Selects the clock and trigger mode of the core counter. 000: Driven by internal clock CK_INT; 001: Encoder mode 1, the core counter counts up/down on TI2FP2 edges depending on the level of TI1FP1; 010: Encoder mode 2, the core counter counts up/down on TI1FP1 edges depending on the level of TI2FP2; 011: Encoder mode 3, the core counter counts up/down on both TI1FP1 and TI2FP2 edges depending on the level of the other input signal; 100: Reset mode, the rising edge of the trigger input (TRGI) initializes the counter and generates a signal to update the registers; 101: Gated mode, the counter clock is enabled when the trigger input (TRGI) is high; when the trigger input goes low, the counter stops. Both the start and stop of the counter are controlled; 110: Trigger mode, the counter starts on the rising edge of the trigger input TRGI. Only the start of the counter is controlled; 111: External clock mode 1, the rising edge of the selected trigger input (TRGI) drives the counter.	000b

14.4.4 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=1/8)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	TDE	COMDE	CC4DE	CC3DE	CC2DE	CC1DE	UDE	BIE	TIE	COMIE	CC4IE	CC3IE	CC2IE	CC1IE	UIE

Bits	Name	Access	Description	Reset
15	Reserved	RO	Reserved.	0
14	TDE	RW	Trigger DMA request enable bit. 1: Trigger DMA request enabled; 0: Trigger DMA request disabled.	0
13	COMDE	RW	COM DMA request enable bit. 1: COM DMA request enabled; 0: COM DMA request disabled.	0
12	CC4DE	RW	Capture/compare channel 4 DMA request enable bit. 1: Capture/compare channel 4 DMA request enabled; 0: Capture/compare channel 4 DMA request disabled.	0
11	CC3DE	RW	Capture/compare channel 3 DMA request enable bit. 1: Capture/compare channel 3 DMA request enabled; 0: Capture/compare channel 3 DMA request disabled.	0
10	CC2DE	RW	Capture/compare channel 2 DMA request enable bit. 1: Capture/compare channel 2 DMA request enabled; 0: Capture/compare channel 2 DMA request disabled.	0
9	CC1DE	RW	Capture/compare channel 1 DMA request enable bit. 1: Capture/compare channel 1 DMA request enabled; 0: Capture/compare channel 1 DMA request disabled.	0
8	UDE	RW	Update DMA request enable bit. 1: Update DMA request enabled; 0: Update DMA request disabled.	0
7	BIE	RW	Break interrupt enable bit. 1: Break interrupt enabled; 0: Break interrupt disabled.	0
6	TIE	RW	Trigger interrupt enable bit. 1: Trigger interrupt enabled; 0: Trigger interrupt disabled.	0
5	COMIE	RW	COM interrupt enable bit. 1: COM interrupt enabled; 0: COM interrupt disabled.	0
4	CC4IE	RW	Capture/compare channel 4 interrupt enable bit. 1: Capture/compare channel 4 interrupt enabled; 0: Capture/compare channel 4 interrupt disabled.	0
3	CC3IE	RW	Capture/compare channel 3 interrupt enable bit. 1: Capture/compare channel 3 interrupt enabled; 0: Capture/compare channel 3 interrupt disabled.	0
2	CC2IE	RW	Capture/compare channel 2 interrupt enable bit. 1: Capture/compare channel 2 interrupt enabled; 0: Capture/compare channel 2 interrupt disabled.	0
1	CC1IE	RW	Capture/compare channel 1 interrupt enable bit. 1: Capture/compare channel 1 interrupt enabled; 0: Capture/compare channel 1 interrupt disabled.	0
0	UIE	RW	Update interrupt enable bit. 1: Update interrupt enabled; 0: Update interrupt disabled.	0

14.4.5 Interrupt Status Register (TIMx_INTFR) (x=1/8)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CC4OF	CC3OF	CC2OF	CC1OF	Reserved	BIF	TIF	COMIF	CC4IF	CC3IF	CC2IF	CC1IF	UIF		

Bits	Name	Access	Description	Reset
[15:13]	Reserved	RO	Reserved.	0
12	CC4OF	RW0	Capture/compare channel 4 over-capture flag.	0
11	CC3OF	RW0	Capture/compare channel 3 over-capture flag.	0
10	CC2OF	RW0	Capture/compare channel 2 over-capture flag.	0
9	CC1OF	RW0	Capture/compare channel 1 over-capture flag, only used when the capture/compare channel is configured in input capture mode. This flag is set by hardware and can be cleared by software writing 0. 1: The counter value was captured to the capture/compare register while CC1IF was already set; 0: No over-capture occurred.	0
8	Reserved	RO	Reserved.	0
7	BIF	RW0	Break interrupt flag. Set by hardware when the break input becomes active; can be cleared by software. 1: Break input active level detected on the break pin; 0: No break event generated.	0
6	TIF	RW0	Trigger interrupt flag. Set by hardware when a trigger event occurs; can be cleared by software. Trigger events include detecting a valid edge on the TRGI input in modes other than gated mode, or any edge in gated mode. 1: Trigger event generated; 0: No trigger event generated.	0
5	COMIF	RW0	COM interrupt flag. Set by hardware when a COM event is generated; can be cleared by software. A COM event includes the update of CCxE, CCxNE, and OCxM. 1: COM event generated; 0: No COM event generated.	0
4	CC4IF	RW0	Capture/compare channel 4 interrupt flag.	0
3	CC3IF	RW0	Capture/compare channel 3 interrupt flag.	0
2	CC2IF	RW0	Capture/compare channel 2 interrupt flag.	0
1	CC1IF	RW0	Capture/compare channel 1 interrupt flag. If the capture/compare channel is configured in output mode: This bit is set by hardware when the counter value matches the compare value, except in center-aligned mode. This bit is cleared by software. 1: The core counter value matches capture/compare register 1; 0: No match occurred. If capture/compare channel 1 is configured in input mode: This bit is set by hardware when a capture event occurs; it is cleared by software or by reading the capture/compare register. 1: The counter value has been captured to capture/compare register 1; 0: No input capture occurred.	0

Bits	Name	Access	Description	Reset
0	UIF	RW0	Update interrupt flag. Set by hardware when an update event is generated; cleared by software. 1: Update interrupt generated; 0: No update event generated. Update events are generated in the following situations: If UDIS=0, when the repetition counter overflows or underflows; If URS=0 and UDIS=0, when the UG bit is set, or when the core counter is reinitialized by software; If URS=0 and UDIS=0, when the counter CNT is reinitialized by a trigger event.	0

14.4.6 Event Generation Register (TIMx_SWEVGR) (x=1/8)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								BG	TG	COMG	CC4G	CC3G	CC2G	CC1G	UG

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
7	BG	WO	Break event generation bit. This bit is set and cleared by software to generate a break event. 1: A break event is generated. MOE=0, BIF=1. If the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated; 0: No action.	0
6	TG	WO	Trigger event generation bit. This bit is set by software and cleared by hardware to generate a trigger event. 1: A trigger event is generated. TIF is set. If the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated; 0: No action.	0
5	COMG	WO	Capture/compare control update generation bit. Generates a capture/compare control update event. This bit is set by software and automatically cleared by hardware. 1: When CCPC=1, allows updating CCxE, CCxNE, and OCxM bits; 0: No action. Note: This bit is only valid for channels with complementary outputs (channels 1, 2, 3).	0
4	CC4G	WO	Capture/compare event generation bit 4. Generates capture/compare event 4.	0
3	CC3G	WO	Capture/compare event generation bit 3. Generates capture/compare event 3.	0
2	CC2G	WO	Capture/compare event generation bit 2. Generates capture/compare event 2.	0
1	CC1G	WO	Capture/compare event generation bit 1. Generates capture/compare event 1. This bit is set by software and cleared by hardware. Used to generate a capture/compare event. 1: A capture/compare event is generated on capture/compare channel 1:	0

Bits	Name	Access	Description	Reset
			If capture/compare channel 1 is configured as output: Sets the CC1IF bit. If the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated; If capture/compare channel 1 is configured as input: The current core counter value is captured to capture/compare register 1. Sets the CC1IF bit; if the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated. If CC1IF is already set, the CC1OF bit is set; 0: No action.	
0	UG	WO	Update event generation bit. Generates an update event. This bit is set by software and automatically cleared by hardware. 1: Initializes the counter and generates an update event; 0: No action. Note: The prescaler counter is also cleared, but the prescaler ratio remains unchanged. In center-aligned mode or up-counting mode, the core counter is cleared; in down-counting mode, the core counter takes the value of the auto-reload register.	0

14.4.7 Capture/Compare Control Register 1 (TIMx_CHCTLR1) (x=1/8)

Offset address: 0x18

The channel can be used for input (capture mode) or output (compare mode). The direction of the channel is defined by the corresponding CCxS bits. The function of the other bits in this register differs between input and output modes. OCxx describes the function of the channel in output mode, and ICxx describes the function of the channel in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
OC2CE	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		OC1CE		OC1M[2:0]		OC1PE	OC1FE	CC1S[1:0]		
IC2F[3:0]				IC2PSC[1:0]					IC1F[3:0]			IC1PSC[1:0]				

Compare mode (pin direction is output):

Bits	Name	Access	Description	Reset
15	OC2CE	RW	Capture/compare channel 2 clear enable bit. 1: OC2REF is cleared to 0 as soon as a high level is detected on the ETRF input; 0: OC2REF is not affected by the ETRF input.	0
[14:12]	OC2M[2:0]	RW	Capture/compare channel 2 mode setting field. These 3 bits define the action of the output reference signal OC2REF, which determines the values of OC2 and OC2N. OC2REF is active high, while the active level of OC2 and OC2N depends on the CC2P and CC2NP bits. 000: Frozen. The comparison between the capture/compare register value and the core counter has no effect on OC2REF; 001: Force active level. When the core counter matches the value of capture/compare register 2, OC2REF is forced high; 010: Force inactive level. When the core counter value matches capture/compare register 2, OC2REF is forced low; 011: Toggle. When the core counter matches the value of capture/	000b

Bits	Name	Access	Description	Reset
			compare register 2, OC2REF toggles; 100: Force inactive level. OC2REF is forced low; 101: Force active level. OC2REF is forced high; 110: PWM mode 1. In up-counting, channel 2 is active as long as the core counter is less than the capture/compare register value, otherwise inactive. In down-counting, channel 2 is inactive (OC2REF=0) as long as the core counter is greater than the capture/compare register value, otherwise active (OC2REF=1); 111: PWM mode 2. In up-counting, channel 2 is inactive as long as the core counter is less than the capture/compare register value, otherwise active. In down-counting, channel 2 is active (OC2REF=1) as long as the core counter is greater than the capture/compare register value, otherwise inactive (OC2REF=0). Note: Once LOCK level is set to 3 and CC1S=00b, this bit cannot be modified. In PWM mode 1 or PWM mode 2, OC2REF level changes only when the comparison result changes or when switching from freeze mode to PWM mode in output compare mode.	
11	OC2PE	RW	Capture/compare register 2 preload enable bit. 1: Preload function of capture/compare register enabled. Read/write operations only operate on the preload register. The preload value of capture/compare register 1 is loaded into the current shadow register on an update event; 0: Preload function of capture/compare register 2 disabled. Capture/compare register 2 can be written at any time, and the newly written value takes effect immediately. Note: Once LOCK level is set to 3 and CC1S=00, this bit cannot be modified. In single-pulse mode (OPM=1), PWM mode can be used without confirming the preload register, otherwise its behavior is undefined.	0
10	OC2FE	RW	Capture/compare channel 2 fast enable bit. This bit is used to accelerate the response of the capture/compare channel output to trigger input events. 1: The effect of the active edge on the trigger is like a compare match has occurred. Therefore, OC is set to the compare level regardless of the comparison result. The delay between the active edge of the trigger and the capture/compare channel 2 output is reduced to 3 clock cycles; 0: The capture/compare channel 2 operates normally according to the counter and capture/compare register 2 values, even when the trigger is enabled. The minimum delay to activate the capture/compare channel 2 output when there is an active edge on the trigger input is 5 clock cycles. OC2FE only takes effect when the channel is configured in PWM1 or PWM2 mode.	0
[9:8]	CC2S[1:0]	RW	Capture/compare channel 2 input selection field. 00: Capture/compare channel 2 configured as output; 01: Capture/compare channel 2 configured as input, IC2 mapped on TI2; 10: Capture/compare channel 2 configured as input, IC2 mapped on TI1; 11: Capture/compare channel 2 configured as input, IC2 mapped on TRC. This mode only works when the internal trigger input is selected	00b

Bits	Name	Access	Description	Reset
			(by TS bits). Note: Capture/compare channel 2 is writable only when the channel is off (CC2E is 0).	
7	OC1CE	RW	Capture/compare channel 1 clear enable bit.	0
[6:4]	OC1M[2:0]	RW	Capture/compare channel 1 mode setting field.	0
3	OC1PE	RW	Capture/compare register 1 preload enable bit.	0
2	OC1FE	RW	Capture/compare channel 1 fast enable bit.	0
[1:0]	CC1S[1:0]	RW	Capture/compare channel 1 input selection field.	0

Capture mode (pin direction is input):

Bits	Name	Access	Description	Reset
[15:12]	IC2F[3:0]	RW	Input capture filter 2 configuration field. These bits set the sampling frequency and digital filter length of the TI1 input. The digital filter consists of an event counter that produces an output transition after recording N events. 0000: No filter, sampling at fDTS; 0001: Sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, N=2; 0010: Sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, N=4; 0011: Sampling frequency $F_{\text{sampling}} = F_{\text{ck_int}}$, N=8; 0100: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/2$, N=6; 0101: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/2$, N=8; 0110: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/4$, N=6; 0111: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/4$, N=8; 1000: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/8$, N=6; 1001: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/8$, N=8; 1010: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, N=5; 1011: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, N=6; 1100: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/16$, N=8; 1101: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, N=5; 1110: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, N=6; 1111: Sampling frequency $F_{\text{sampling}} = F_{\text{dts}}/32$, N=8.	0000b
[11:10]	IC2PSC[1:0]	RW	Capture/compare channel 2 prescaler configuration field. These 2 bits define the prescaler ratio of capture/compare channel 2. The prescaler is reset as soon as CC1E=0. 00: No prescaler, capture is triggered on every edge detected on the input; 01: Capture is triggered every 2 events; 10: Capture is triggered every 4 events; 11: Capture is triggered every 8 events.	00b
[9:8]	CC2S[1:0]	RW	Capture/compare channel 2 input selection field. These 2 bits define the direction (input/output) of the channel and the selection of the input pin. 00: Capture/compare channel 1 configured as output; 01: Capture/compare channel 1 configured as input, IC1 mapped on TI1; 10: Capture/compare channel 1 configured as input, IC1 mapped on TI2;	00b

Bits	Name	Access	Description	Reset
			11: Capture/compare channel 1 configured as input, IC1 mapped on TRC. This mode only works when the internal trigger input is selected (by TS bits). Note: CC1S is writable only when the channel is off (CC1E is 0).	
[7:4]	IC1F[3:0]	RW	Input capture filter 1 configuration field.	0
[3:2]	IC1PSC[1:0]	RW	Capture/compare channel 1 prescaler configuration field.	0
[1:0]	CC1S[1:0]	RW	Capture/compare channel 1 input selection field.	0

14.4.8 Capture/Compare Control Register 2 (TIMx_CHCTLR2) (x=1/8)

Offset address: 0x1C

The channel can be used for input (capture mode) or output (compare mode). The direction of the channel is defined by the corresponding CCxS bits. The function of the other bits in this register differs between input and output modes. OCxx describes the function of the channel in output mode, and ICxx describes the function of the channel in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4CE	OC4M[2:0]			OC4PE	OC4FE	CC4S[1:0]		OC3CE	OC3M[2:0]			OC3PE	OC3FE	CC3S[1:0]	
IC4F[3:0]				IC4PSC[1:0]				IC3F[3:0]				IC3PSC[1:0]			

Compare mode (pin direction is output):

Bits	Name	Access	Description	Reset
15	OC4CE	RW	Capture/compare channel 4 clear enable bit.	0
[14:12]	OC4M[2:0]	RW	Capture/compare channel 4 mode setting field.	0
11	OC4PE	RW	Capture/compare register 4 preload enable bit.	0
10	OC4FE	RW	Capture/compare channel 4 fast enable bit.	0
[9:8]	CC4S[1:0]	RW	Capture/compare channel 4 input selection field.	0
7	OC3CE	RW	Capture/compare channel 3 clear enable bit.	0
[6:4]	OC3M[2:0]	RW	Capture/compare channel 3 mode setting field.	0
3	OC3PE	RW	Capture/compare register 3 preload enable bit.	0
2	OC3FE	RW	Capture/compare channel 3 fast enable bit.	0
[1:0]	CC3S[1:0]	RW	Capture/compare channel 3 input selection field.	0

Capture mode (pin direction is input):

Bits	Name	Access	Description	Reset
[15:12]	IC4F[3:0]	RW	Input capture filter 4 configuration field.	0
[11:10]	IC4PSC[1:0]	RW	Capture/compare channel 4 prescaler configuration field.	0
[9:8]	CC4S[1:0]	RW	Capture/compare channel 4 input selection field.	0
[7:4]	IC3F[3:0]	RW	Input capture filter 3 configuration field.	0
[3:2]	IC3PSC[1:0]	RW	Capture/compare channel 3 prescaler configuration field.	0

Bits	Name	Access	Description	Reset
[1:0]	CC3S[1:0]	RW	Capture/compare channel 3 input selection field.	0

14.4.9 Capture/Compare Enable Register (TIMx_CCER) (x=1/8)

Offset address: 0x20

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CC4P	CC4E	CC3NP	CC3NE	CC3P	CC3E	CC2NP	CC2NE	CC2P	CC2E	CC1NP	CC1NE	CC1P	CC1E	

Bits	Name	Access	Description	Reset
[15:14]	Reserved	RO	Reserved.	0
13	CC4P	RW	Capture/compare channel 4 output polarity setting bit.	0
12	CC4E	RW	Capture/compare channel 4 output enable bit.	0
11	CC3NP	RW	Capture/compare channel 3 complementary output polarity setting bit.	0
10	CC3NE	RW	Capture/compare channel 3 complementary output enable bit.	0
9	CC3P	RW	Capture/compare channel 3 output polarity setting bit.	0
8	CC3E	RW	Capture/compare channel 3 output enable bit.	0
7	CC2NP	RW	Capture/compare channel 2 complementary output polarity setting bit.	0
6	CC2NE	RW	Capture/compare channel 2 complementary output enable bit.	0
5	CC2P	RW	Capture/compare channel 2 output polarity setting bit.	0
4	CC2E	RW	Capture/compare channel 2 output enable bit.	0
3	CC1NP	RW	Capture/compare channel 1 complementary output polarity setting bit.	0
2	CC1NE	RW	Capture/compare channel 1 complementary output enable bit.	0
1	CC1P	RW	Capture/compare channel 1 output polarity setting bit. CC1 channel configured as output: 1: OC1 is active low; 0: OC1 is active high. CC1 channel configured as input: This bit selects whether IC1 or the inverted IC1 is used as trigger or capture signal. 1: Inverted: capture occurs on the falling edge of IC1; when used as external trigger, IC1 is inverted; 0: Not inverted: capture occurs on the rising edge of IC1; when used as external trigger, IC1 is not inverted. Note: Once LOCK level (LOCK bits in TIMx_BDTR register) is set to 3 or 2, this bit cannot be modified.	0
0	CC1E	RW	Capture/compare channel 1 output enable bit. CC1 channel configured as output: 1: On. The OC1 signal is output to the corresponding output pin, and its output level depends on the values of MOE, OSS1, OSSR, OIS1, OIS1N, and CC1NE bits; 0: Off. OC1 output is disabled, so the output level of OC1 depends on the values of MOE, OSS1, OSSR, OIS1, OIS1N, and CC1NE bits. CC1 channel configured as input: This bit determines whether the counter value can be captured into the TIMx_CCR1 register.	0

Bits	Name	Access	Description	Reset
			1: Capture enabled; 0: Capture disabled.	

14.4.10 Advanced Timer Counter (TIMx_CNT) (x=1/8)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	CNT[15:0]	RW	Real-time value of the timer counter.	0

14.4.11 Counter Clock Prescaler (TIMx_PSC) (x=1/8)

Offset address: 0x28

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	PSC[15:0]	RW	Division ratio of the timer prescaler. The counter clock frequency equals the prescaler input frequency / (PSC + 1).	0

14.4.12 Auto-Reload Register (TIMx_ATRLR) (x=1/8)

Offset address: 0x2C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	ARR[15:0]	RW	The value of this field is loaded into the counter. When ATRLR is active and how it is updated is described in Section 14.2.3. When ATRLR is empty, the counter stops.	0xFFFF

14.4.13 Repetition Counter Register (TIMx_RPTCR) (x=1/8)

Offset address: 0x30

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								REP[7:0]							

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
[7:0]	REP[7:0]	RW	Repetition counter value.	0

14.4.14 Capture/Compare Register 1 (TIMx_CH1CVR) (x=1/8)

Offset address: 0x34

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	CCR1[15:0]	RW	Value of capture/compare register channel 1.	0

14.4.15 Capture/Compare Register 2 (TIMx_CH2CVR) (x=1/8)

Offset address: 0x38

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	CCR2[15:0]	RW	Value of capture/compare register channel 2.	0

14.4.16 Capture/Compare Register 3 (TIMx_CH3CVR) (x=1/8)

Offset address: 0x3C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	CCR3[15:0]	RW	Value of capture/compare register channel 3.	0

14.4.17 Capture/Compare Register 4 (TIMx_CH4CVR) (x=1/8)

Offset address: 0x40

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	CCR4[15:0]	RW	Value of capture/compare register channel 4.	0

14.4.18 Break and Dead-Time Register (TIMx_BDTR) (x=1/8)

Offset address: 0x44

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE		AOE		BKP	BKE	OSSR		OSSI	LOCK[1:0]		DTG[7:0]				

Bits	Name	Access	Description	Reset
15	MOE	RW	Main output enable bit. Asynchronously cleared as soon as the break signal is active.	0

Bits	Name	Access	Description	Reset
			1: OCx and OCxN outputs are enabled; 0: OCx and OCxN outputs are disabled or forced to idle state.	
14	AOE	RW	Automatic output enable. 1: MOE can be set by software or automatically set on the next update event; 0: MOE can only be set by software.	0
13	BKP	RW	Break input polarity setting bit. 1: Break input is active high; 0: Break input is active low. Note: After LOCK level 1 is set, this bit cannot be modified. Writing to this bit requires one PB clock cycle to take effect.	0
12	BKE	RW	Break function enable bit. 1: Break input enabled; 0: Break input disabled. Note: After LOCK level 1 is set, this bit cannot be modified. Writing to this bit requires one PB clock cycle to take effect.	0
11	OSSR	RW	1: When the timer is not running, if CCxE=1 or CCxNE=1, the OC/OCN output is first enabled and outputs the inactive level, then the OCx/OCxN enable output signal=1; 0: When the timer is not running, OC/OCN output is disabled. Note: After LOCK level 1 is set, this bit cannot be modified.	0
10	OSSI	RW	1: When the timer is not running, if CCxE=1 or CCxNE=1, the OC/OCN first outputs its idle level, then the OCx/OCxN enable output signal=1; 0: When the timer is not running, OC/OCN output is disabled. Note: After LOCK level 1 is set, this bit cannot be modified.	0
[9:8]	LOCK[1:0]	RW	Lock function setting field. 00: Lock function off; 01: Lock level 1, cannot write DTG, BKE, BKP, AOE, OISx, and OISxN bits; 10: Lock level 2, cannot write the bits locked in level 1, nor the CC polarity bits and OSSR, OSSI bits; 11: Lock level 3, cannot write the bits locked in level 2, nor the CC control bits. Note: After system reset, the LOCK bits can only be written once and cannot be modified again until reset.	00b
[7:0]	DTG[7:0]	RW	Dead-time setting bits. These bits define the dead-time duration between complementary outputs. Let DT denote the duration: DTG[7:5]=0xx $\Rightarrow$ DT = DTG[7:0] $\times$ Tdtg, Tdtg = TDTS; DTG[7:5]=10x $\Rightarrow$ DT = (64 + DTG[5:0]) $\times$ Tdtg, Tdtg = 2 $\times$ TDTS; DTG[7:5]=110 $\Rightarrow$ DT = (32 + DTG[4:0]) $\times$ Tdtg, Tdtg = 8 $\times$ TDTS; DTG[7:5]=111 $\Rightarrow$ DT = (32 + DTG[4:0]) $\times$ Tdtg, Tdtg = 16 $\times$ TDTS. Note: Once LOCK level (LOCK[1:0] bits in TIM1_BDTR register) is set to 1, 2, or 3, these bits cannot be modified.	0

14.4.19 DMA Control Register (TIMx_DMCFGR) (x=1/8)

Offset address: 0x48

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DBL[4:0]			Reserved			Reserved			DBA[4:0]			

Bits	Name	Access	Description	Reset
[15:13]	Reserved	RO	Reserved.	0
[12:8]	DBL[4:0]	RW	DMA burst transfer length. The actual value is this field's value + 1. When a read or write is performed on the TIMx_DMAADR register, the timer performs a burst transfer, i.e., defining the number of transfers. Transfers can be half-word (2 bytes) or byte: 00000: 1 transfer; 00001: 2 transfers; 00010: 3 transfers; ... 10001: 18 transfers. For example, if we perform such a transfer: DBL=7, DBA=TIM2_CTLR1. If DBL=7, DBA=TIM2_CTLR1 indicates the address of the data to be transferred, then the transfer address is given by the following formula: (address of TIMx_CTLR1) + DBA + (DMA index), where DMA index = DBL. Here, (address of TIMx_CTLR1) + DBA + 7 gives the address where data will be written or read. Thus, data transfer will occur in the 7 registers starting from the address (address of TIMx_CTLR1) + DBA. Depending on the DMA data length setting, the following may occur: 1. If the data is set to half-word (16-bit), the data will be transferred to all 7 registers. 2. If the data is set to byte, the data will still be transferred to all 7 registers: the first register contains the first MSB byte, the second register contains the first LSB byte, and so on. Therefore, for the timer, the user must specify the data width transferred by DMA.	0
[7:5]	Reserved	RO	Reserved.	0
[4:0]	DBA[4:0]	RW	These bits define the offset of the DMA in continuous mode from the address of control register 1. 00000: TIMx_CTLR1; 00001: TIMx_CTLR2; 00010: TIMx_SMCFGFR; ...	0

14.4.20 DMA Address for Continuous Mode Register (TIMx_DMAADR) (x=1/8)

Offset address: 0x4C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	DMAB[15:0]	RW	DMA address in continuous mode. A read or write to the TIMx_DMAADR register causes an access to the register at the following address: TIMx_CTLR1 address + DBA + DMA index, where "TIMx_CTLR1 address" is the address of control register 1 (TIMx_CTLR1); "DBA" is the base address defined in the TIMx_DMACFGR register; "DMA index" is the offset automatically controlled by DMA, which depends on the DBL defined in the TIMx_DMACFGR register.	0

14.4.21 Dual-edge Capture Register (TIMx_AUX) (x=1/8)

Offset address: 0x50

15	14	13	12	11	10	9	8	7	6	5	4	3	2		1		0
Reserved												CAP_ED_CH4			CAP_ED_CH3		CAP_ED_CH2

Bits	Name	Access	Description	Reset
[15:3]	Reserved	R0	Reserved.	0
2	CAP_ED_CH4	RW	Dual-edge capture enable for channel 4: 0: Disable dual-edge capture for channel 4; 1: Enable dual-edge capture for channel 4.	0
1	CAP_ED_CH3	RW	Dual-edge capture enable for channel 3: 0: Disable dual-edge capture for channel 3; 1: Enable dual-edge capture for channel 3.	0
0	CAP_ED_CH2	RW	Dual-edge capture enable for channel 2: 0: Disable dual-edge capture for channel 2; 1: Enable dual-edge capture for channel 2.	0

Chapter 15: General Purpose Timer (GPTM)

The general purpose timer module contains four 16-bit auto-reload timers (TIM2, TIM3, TIM4, TIM5), used to measure pulse widths or generate pulses of specific frequencies, PWM waves, etc. They can be used in automation control, power supply, and other fields.

15.1 Main Features

The main features of the 16-bit general purpose timer include:

- 16-bit auto-reload counter, supporting up-counting mode, down-counting mode, and up-down counting mode
- 16-bit prescaler, with division factor dynamically adjustable from 1 to 65536
- Supports four independent compare/capture channels
- Each compare/capture channel supports multiple working modes, such as: input capture, output compare, PWM generation, and single-pulse output
- Supports external signal control of the timer
- Supports DMA in multiple modes
- Supports incremental encoder, cascading and synchronization between timers

15.2 Principles and Structure

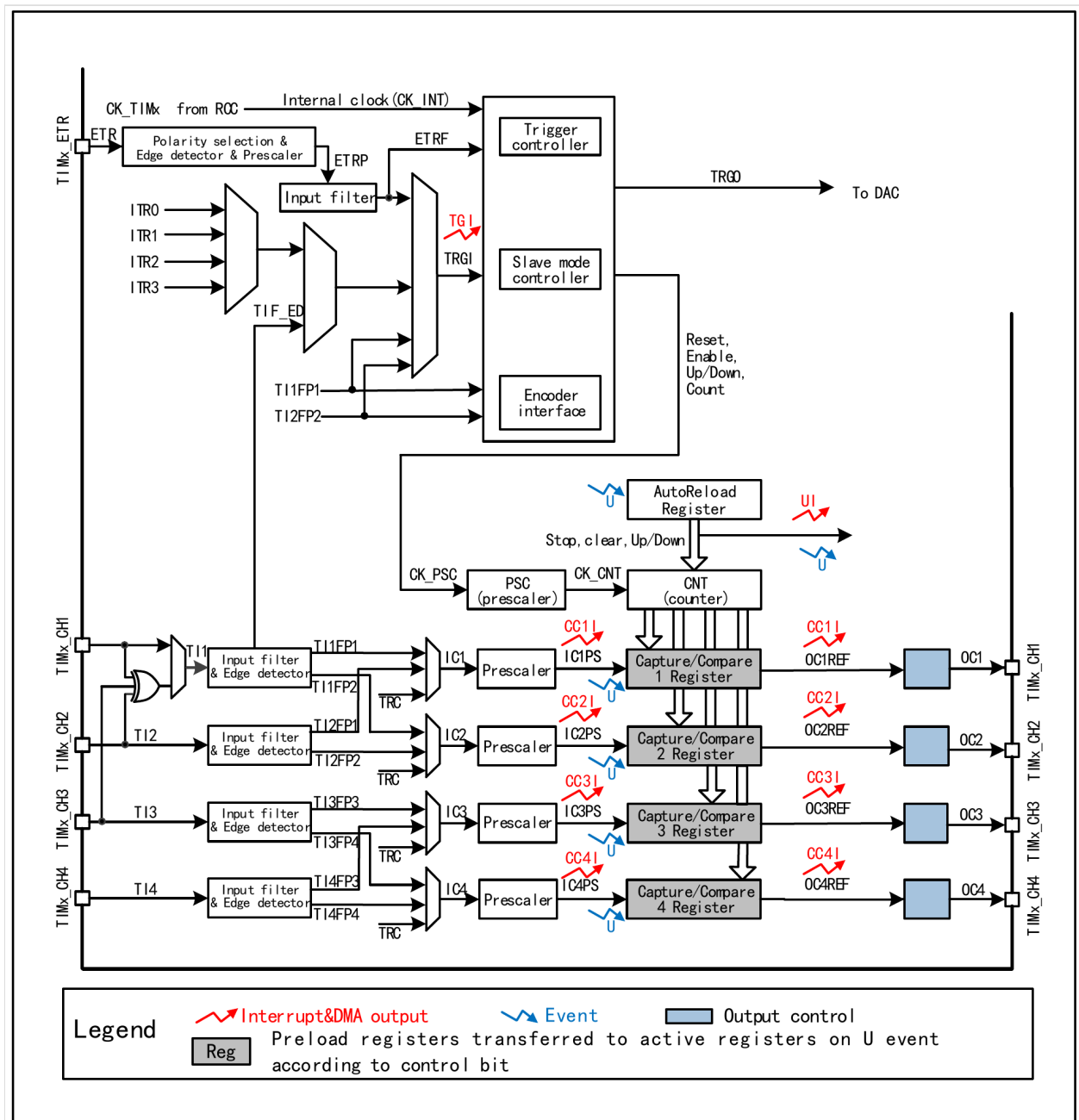


Figure 15-1: Block diagram of the general purpose timer

As shown in Figure 15-1, the structure of the general purpose timer can be roughly divided into three parts: the input clock section, the core counter section, and the compare/capture channel section.

The clock of the general purpose timer can come from the HB bus clock (CK_INT), from the external clock input pin (TIMx_ETRx), from other timers with clock output capability (ITRx), and from the input end of the compare/capture channels (TIMx_CHx). These input clock signals go through various configured filtering and division operations to become the CK_PSC clock, which is output to the core counter section. In addition, these complex clock sources can also be output as TRGO to other timers, ADCs, and other peripherals.

The core of the general purpose timer is a 16-bit counter (CNT). CK_PSC is divided by the prescaler (PSC) to become CK_CNT, which is finally fed to CNT. CNT supports up-counting mode, down-counting

mode, and up-down counting mode, and has an auto-reload register (ATRLR) that reloads the initial value to CNT after each counting cycle ends.

The general purpose timer has four sets of compare/capture channels. Each compare/capture channel can input pulses from its dedicated pin or output waveforms to the pin, i.e., the compare/capture channel supports both input and output modes. The input of each compare/capture register channel supports filtering, division, edge detection, and other operations, supports inter-channel mutual triggering, and can also provide the clock for the core counter CNT. Each compare/capture channel has a set of compare/capture registers (CHxCVR), which support comparing with the main counter (CNT) and outputting pulses.

15.2.2 Differences Between General Purpose Timer and Advanced Timer

Compared with the advanced timer, the general purpose timer lacks the following functions:

1. The general purpose timer lacks a repetition counter register that counts the counting period of the core counter.
2. The compare/capture channels of the general purpose timer lack dead-time generation and have no complementary output.
3. The general purpose timer has no break signal mechanism.
4. The default clock CK_INT of the general purpose timer comes from PB1, while the CK_INT of the advanced timer comes from PB2.

15.2.3 Clock Input

This section discusses the source of CK_PSC. Here is an excerpt of the clock source part from the overall block diagram of the general purpose timer.

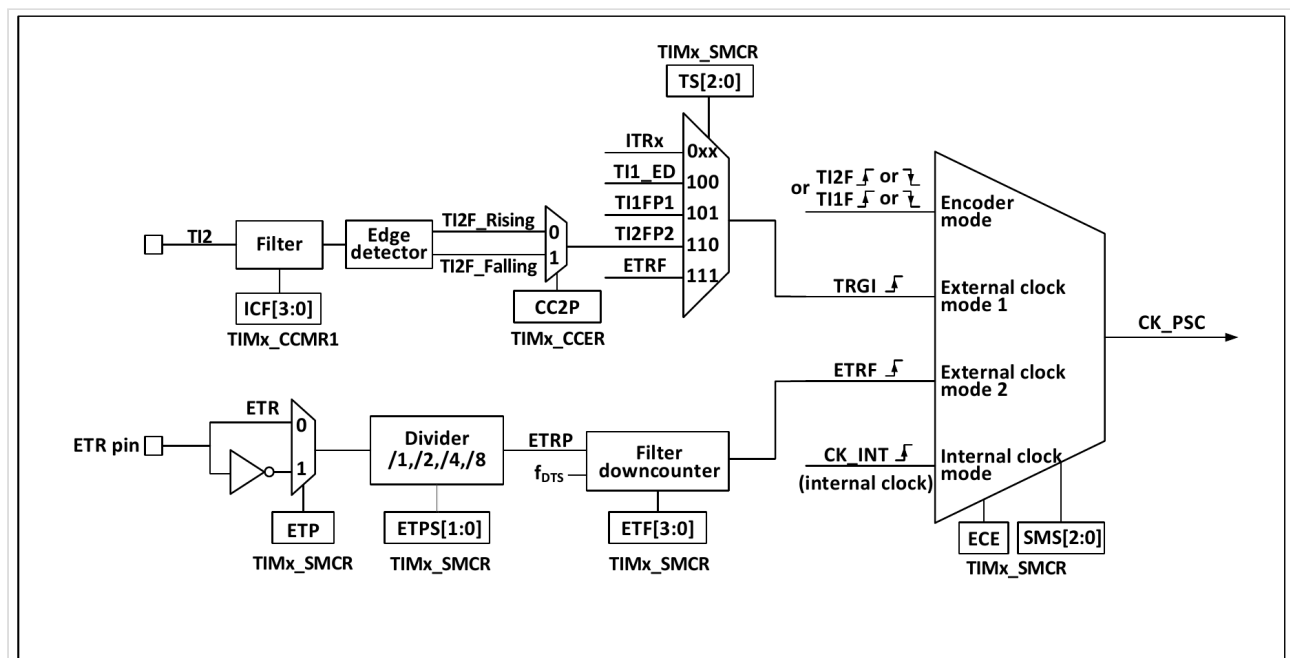


Figure 15-2: Block diagram of CK_PSC source for the general purpose timer

The available input clocks can be divided into 4 categories:

1. External clock pin (ETR) input path: ETR→ETRP→ETRF;
2. Internal PB clock input path: CK_INT;

3. Path from the compare/capture channel pins (TIMx_CHx): TIMx_CHx→TIX→TIXFPx, this path is also used in encoder mode;
4. Input from other internal timers: ITRx.

By determining the input pulse selection of SMS, which determines the source of CK_PSC, the actual operations can be divided into three categories:

1. Select internal clock source (CK_INT);
2. External clock source mode 1;
3. External clock source mode 2;
4. Encoder mode.

The 4 clock source sources mentioned above can all be selected through these 4 operations.

15.2.3.1 Internal Clock Source (CK_INT)

If the general purpose timer is started with the SMS field kept at 000b, then the internal clock source (CK_INT) is selected as the clock. In this case, CK_INT is CK_PSC.

15.2.3.2 External Clock Source Mode 1

If the SMS field is set to 111b, external clock source mode 1 is enabled. When external clock source 1 is enabled, TRGI is selected as the source of CK_PSC. It is worth noting that the user also needs to configure the TS field to select the source of TRGI. The TS field can select the following pulses as clock sources:

1. Internal triggers (ITRx, where x is 0, 1, 2, 3);
2. The signal from compare/capture channel 1 after the edge detector (TI1F_ED);
3. Compare/capture channel signals TI1FP1, TI2FP2;
4. The signal ETRF from the external clock pin input.

15.2.3.3 External Clock Source Mode 2

Using external trigger mode 2 allows counting on every rising or falling edge of the external clock pin input. When the ECE bit is set, external clock source mode 2 is used. When using external clock source mode 2, ETRF is selected as CK_PSC. The ETR pin goes through the optional inverter (ETP), divider (ETPS) to become ETRP, and then through the filter (ETF) to become ETRF.

When the ECE bit is set and SMS is set to 111b, it is equivalent to TS selecting ETRF as input.

15.2.3.4 Encoder Mode

Setting SMS to 001b, 010b, or 011b enables encoder mode. Enabling encoder mode allows selecting a specific level of TI1FP1 or TI2FP2 and using the other transition edge as the signal for output. This mode is used when an external encoder is connected. For specific functions, refer to Section 15.3.7.

15.2.4 Counter and Peripherals

CK_PSC is input to the prescaler (PSC) for division. PSC is 16-bit, and the actual division factor equals the value of R16_TIMx_PSC + 1. CK_PSC becomes CK_CNT after passing through PSC. Changing the value of R16_TIM1_PSC does not take effect immediately, but is updated to PSC after an update event. Update events include clearing the UG bit and reset.

15.2.5 Compare/Capture Channel

The compare/capture channel is the core of the timer's complex functions. Its core is the compare/capture register, supplemented by digital filtering of the peripheral input, division and inter-channel multiplexing, and the comparator and output control of the output section. The block diagram of the compare/capture channel is shown in Figure 15-3.

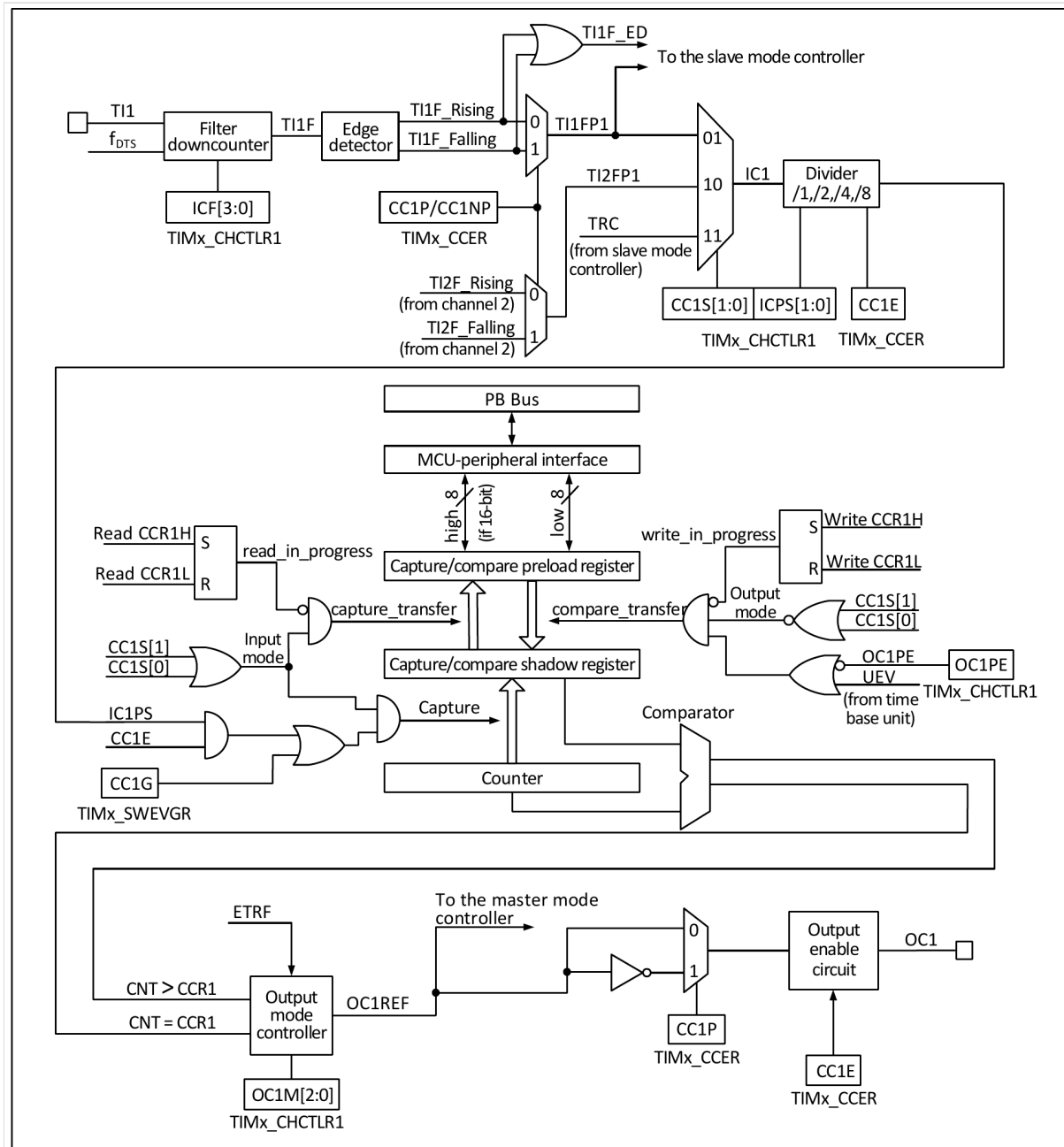


Figure 15-3: Block diagram of the compare/capture channel

After the signal enters from channel x pin, it can be selected as TIx (the source of $TI1$ can be more than just $CH1$, see the timer block diagram 15-1). Taking $TI1$ as an example: $TI1$ passes through the filter ($ICF[3:0]$) to generate $TI1F$, then through the edge detector to produce $TI1F_Rising$ and $TI1F_Falling$. These two signals pass through the selection ($CC1P$) to generate $TI1FP1$. $TI1FP1$ and $TI2FP1$ from channel 2 are sent together to $CC1S$ for selection to become $IC1$, which is then divided by $ICPS$ and sent to the compare/capture register.

The compare/capture register consists of a preload register and a shadow register. The read/write process only operates on the preload register. In capture mode, the capture occurs on the shadow register, and then is copied to the preload register; in compare mode, the contents of the preload register are copied to the shadow register, and then the contents of the shadow register are compared with the core counter (CNT).

15.3 Functions and Implementation

The complex functions of the general purpose timer are all implemented by operating the timer's compare/capture channels, clock input circuit, counter, and peripheral components. The timer's clock input can come from multiple clock sources, including the input of the compare/capture channel. The operations on the compare/capture channel and clock source selection directly determine its function. The compare/capture channel is bidirectional and can work in both input and output modes.

15.3.1 Input Capture Mode

Input capture mode is one of the basic functions of the timer. The principle of input capture mode is: when a determined edge is detected on the ICxPS signal, a capture event is generated, and the current value of the counter is latched into the compare/capture register (R16_TIMx_CHCTLRx). When a capture event occurs, CCxIF (in R16_TIMx_INTFR) is set. If interrupt or DMA is enabled, the corresponding interrupt or DMA is also generated. If CCxIF is already set when a capture event occurs, the CCxOF bit is set. CCxIF can be cleared by software or by hardware through reading the compare/capture register. CCxOF is cleared by software.

Using channel 1 as an example to illustrate the steps for using input capture mode:

1. Configure the CC1S field to select the source of the IC1 signal. For example, set it to 10b to select TI1FP1 as the source of IC1. The default setting cannot be used; the CC1S field defaults to configuring the compare/capture module as an output channel;
2. Configure the IC1F field. These bits set the sampling frequency and digital filter length of the TI1 input. The digital filter consists of an event counter, which produces an output transition after recording N events;
3. Configure the CC1P bit to set the polarity of TI1FP1. For example, keep the CC1P bit low to select rising edge transition;
4. Configure the IC1PS field to set the division factor between the IC1 signal becoming IC1PS. For example, keep IC1PS at 00b for no division;
5. Configure the CC1E bit to allow capturing the value of the core counter (CNT) into the compare/capture register;
6. Configure the CCxIE and CCxDE bits according to whether interrupts or DMA are needed.

When TI1 inputs a captured pulse, the value of the core counter (CNT) is recorded into the compare/capture register, CC1IF is set. If CC1IF was already set before, CC1OF is also set. If CC1IE is set, an interrupt is generated; if CC1DE is set, a DMA request is generated. An input capture event can be generated by software through writing the event generation register (R16_TIMx_SWEVGR).

15.3.2 Output Compare Mode

Output compare mode is one of the basic functions of the timer. The principle of output compare mode is that when the value of the core counter (CNT) matches the value of the compare/capture register, a specific change or waveform is output. The OCxM field (in R16_TIMx_CHCTLRx) and the CCxP

bit (in R16_TIMx_CCER) determine whether the output is a determined high/low level or a level toggle. When a compare match event occurs, the CCxIF bit is also set. If CCxIE is pre-set, an interrupt is generated; if CCxDE is pre-set, a DMA request is generated.

The steps to configure output compare mode are as follows:

1. Configure the clock source and auto-reload value of the core counter (CNT);
2. Set the desired compare count value into the compare/capture register (R16_TIMx_CHxCVR);
3. If an interrupt is needed, set the CCxIE bit;
4. Keep OCxPE at 0 to disable the preload register of the compare/capture register;
5. Set the output mode by configuring the OCxM field and CCxP bit;
6. Enable the output by setting the CCxE bit;
7. Set the CEN bit to start the timer.

15.3.3 Forced Output Mode

The output mode of the timer's compare/capture channel can be forced by software to output a determined level, without relying on the comparison between the shadow register of the compare/capture register and the core counter.

The specific method is to set OCxM to 100b, which forces OCxREF to low; or set OCxM to 101b, which forces OCxREF to high.

It should be noted that when OCxM is forced to 100b or 101b, the comparison between the internal main counter and the compare/capture register is still ongoing, the corresponding flag bits are still being set, and interrupt and DMA requests are still being generated.

15.3.4 PWM Input Mode

PWM input mode is used to measure the duty cycle and frequency of a PWM signal, and is a special case of input capture mode. Except for the following differences, the operation is the same as input capture mode: PWM occupies two compare/capture channels, and the input polarities of the two channels are set to opposite. One of the signals is set as the trigger input, and SMS is set to reset mode.

For example, to measure the period and frequency of a PWM wave input from TI1, the following operations are required:

1. Set TI1 (TI1FP1) as the input of the IC1 signal. Set CC1S to 01b;
2. Set TI1FP1 to rising edge active. Keep CC1P at 0;
3. Set TI1 (TI1FP2) as the input of the IC2 signal. Set CC2S to 10b;
4. Set TI1FP2 to falling edge active. Set CC2P to 1;
5. Select the clock source as TI1FP1. Set TS to 101b;
6. Set SMS to reset mode, i.e., 100b;
7. Enable input capture. Set CC1E and CC2E.

Note: Since only TI1FP1 and TI2FP2 are connected to the slave mode controller, PWM input mode can only use TIMx_CH1/TIMx_CH2.

15.3.5 PWM Output Mode

PWM output mode is one of the basic functions of the timer. The most common approach in PWM output mode is to use the auto-reload value to determine the PWM frequency and use the capture/compare register to determine the duty cycle. Set OCxM to 110b or 111b to use PWM mode 1 or mode 2, set the OCxPE bit to enable the preload register, and finally set the ARPE bit to enable auto-reload of the preload register. The value of the preload register is transferred to the shadow register only when an update event occurs, so before the core counter starts counting, the UG bit needs to be set to initialize all registers. In PWM mode, the core counter and the compare/capture register are continuously compared. According to the CMS bits, the timer can output edge-aligned or center-aligned PWM signals.

Edge-aligned

When using edge-aligned mode, the core counter counts up or down. In PWM mode 1, when the value of the core counter is greater than the compare/capture register, OCxREF goes high; when the value of the core counter is less than the compare/capture register (for example, when the core counter grows to the value of R16_TIMx_ATRLR and resets to all zeros), OCxREF goes low.

Center-aligned

When using center-aligned mode, the core counter operates in a mode of alternating up-counting and down-counting. OCxREF transitions up and down when the value of the core counter matches the compare/capture register. However, the timing of the compare flag being set differs among the three center-aligned modes. When using center-aligned mode, it is best to generate a software update flag (set the UG bit) before starting the core counter.

15.3.6 Single Pulse Mode

Single pulse mode can respond to a specific event and produce a pulse after a delay, with both the delay and pulse width being programmable. Setting the OPM bit causes the core counter to stop at the next update event (UEV) (counter rolls over to 0).

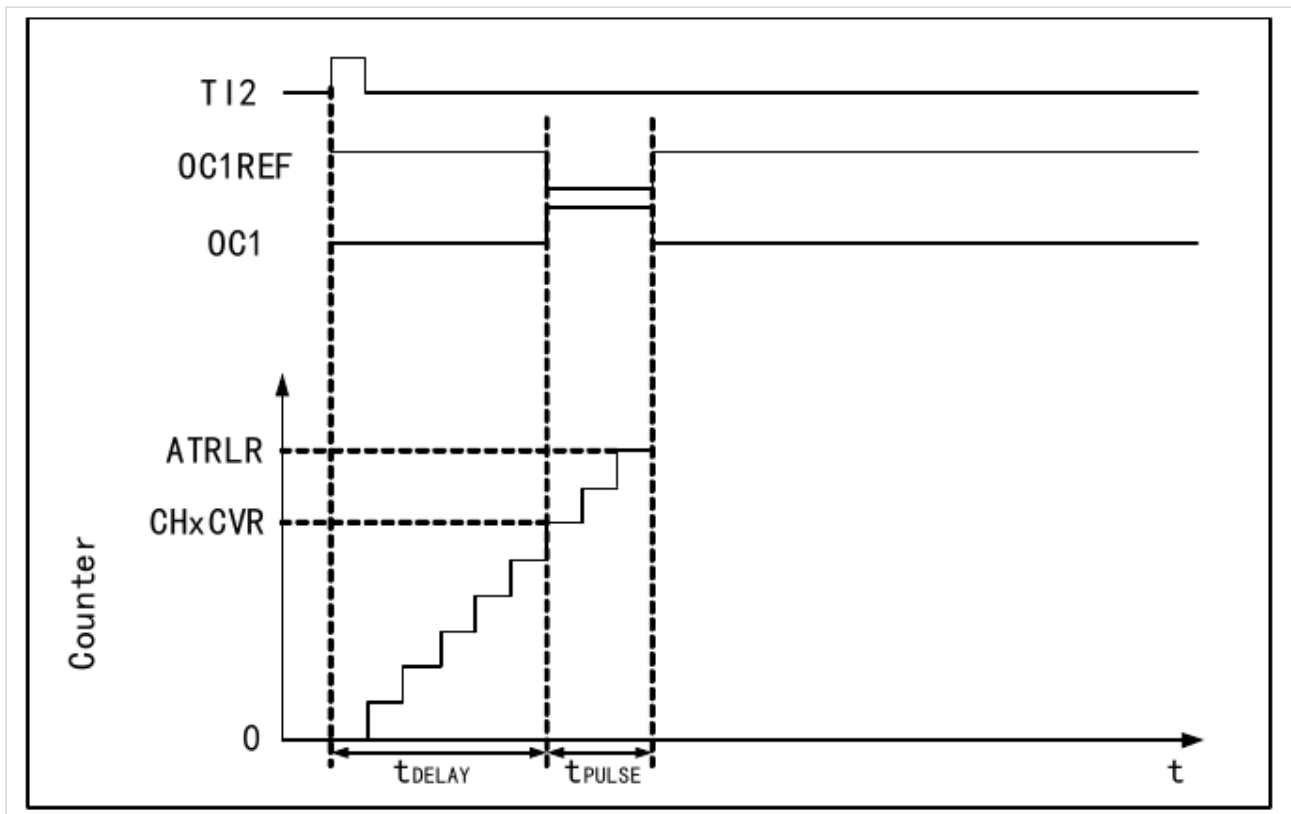


Figure 15-4: Event generation and pulse response

As shown in Figure 15-4, to detect a rising edge on the TI2 input pin, and after a delay T_{delay} , produce a positive pulse of length T_{pulse} on OC1:

1. Set TI2 as the trigger. Set CC2S to 01b to map TI2FP2 to TI2; set CC2P to 0b for TI2FP2 rising edge detection; set TS to 110b for TI2FP2 as the trigger source; set SMS to 110b so that TI2FP2 is used to start the counter;
2. T_{delay} is defined by the compare/capture register, and T_{pulse} is determined by the value of the auto-reload register and the compare/capture register.

15.3.7 Encoder Mode

Encoder mode is a typical application of the timer, which can be used to connect the dual-phase output of an encoder. The counting direction of the core counter is synchronized with the rotation direction of the encoder shaft. Each pulse output by the encoder causes the core counter to increment or decrement by one. The steps to use the encoder are: set the SMS field to 001b (count on TI2 edges only), 010b (count on TI1 edges only), or 011b (count on both TI1 and TI2 edges), connect the encoder to the input of compare/capture channels 1 and 2, and set a reload value for the counter, which can be set to a larger value. In encoder mode, the timer's internal compare/capture registers, prescaler, repetition counter register, etc., all work normally. The following table shows the relationship between the counting direction and the encoder signals.

Table 15-1: Relationship between counting direction and encoder signals in timer encoder mode

Active counting edge	Relative signal level	TI1FP1 signal edge		TI2FP2 signal edge	
		Rising	Falling	Rising	Falling
Count on TI1 edges only	High		Count up	No count	

Active counting edge	Relative signal level	TI1FP1 signal edge		TI2FP2 signal edge	
		Rising	Falling	Rising	Falling
		Count down			
	Low	Count up	Count down		
Count on TI2 edges only	High	No count		Count up	Count down
	Low			Count down	Count up
Count on both TI1 and TI2 edges	High	Count down	Count up	Count up	Count down
	Low	Count up	Count down	Count down	Count up

15.3.8 Timer Synchronization Mode

The timer can output clock pulses (TRGO) and can also receive inputs from other timers (ITRx). The sources of ITRx (TRGO of other timers) are different for different timers. The timer internal trigger connections are shown in Table 15-2.

Table 15-2: GPTM internal trigger connections

Slave timer	ITR0 (TS=000)	ITR1 (TS=001)	ITR2 (TS=010)	ITR3 (TS=011)
TIM2	TIM1	USB	TIM3	TIM4
TIM3	TIM1	TIM2	-	TIM4
TIM4	TIM1	TIM2	TIM3	-
TIM5	TIM2	TIM3	TIM4	-

15.3.9 Dual-Edge Capture Mode

Pulse measurement can be performed through the TIMx_AUX register.

For example: to capture the pulse width through channel 2, in the capture function configuration, select the source of the IC2 signal (CC2S bits of the TIMx_CHCTLR1 register set to 11b), and enable the dual-edge capture function of channel 2 (CAP_ED_CH2 bit of the TIMx_AUX register set to 1). The dual-edge capture configuration is now complete.

The captured dual-edge pulse width value can be read through the CCR2 bits of the TIMx_CH2CVR register.

15.3.10 Debug Mode

When the system enters debug mode, the timer can be controlled to continue running or stop according to the DBG module settings.

15.4 Register Description

Table 15-3: TIM2 related register list

Name	Address	Description	Reset Value
R16_TIM2_CTLR1	0x40000000	TIM2 control register 1	0x0000
R16_TIM2_CTLR2	0x40000004	TIM2 control register 2	0x0000
R16_TIM2_SMCFR	0x40000008	TIM2 slave mode control register	0x0000
R16_TIM2_DMAINTENR	0x4000000C	TIM2 DMA/interrupt enable register	0x0000
R16_TIM2_INTFR	0x40000010	TIM2 interrupt status register	0x0000
R16_TIM2_SWEVGR	0x40000014	TIM2 event generation register	0x0000
R16_TIM2_CHCTLR1	0x40000018	TIM2 compare/capture control register 1	0x0000
R16_TIM2_CHCTLR2	0x4000001C	TIM2 compare/capture control register 2	0x0000
R16_TIM2_CCER	0x40000020	TIM2 compare/capture enable register	0x0000
R16_TIM2_CNT	0x40000024	TIM2 counter	0x0000
R16_TIM2_PSC	0x40000028	TIM2 counter clock prescaler	0x0000
R16_TIM2_ATRLR	0x4000002C	TIM2 auto-reload register	0xFFFF
R16_TIM2_CH1CVR	0x40000034	TIM2 compare/capture register 1	0x0000
R16_TIM2_CH2CVR	0x40000038	TIM2 compare/capture register 2	0x0000
R16_TIM2_CH3CVR	0x4000003C	TIM2 compare/capture register 3	0x0000
R16_TIM2_CH4CVR	0x40000040	TIM2 compare/capture register 4	0x0000
R16_TIM2_DMACFR	0x40000048	TIM2 DMA control register	0x0000
R16_TIM2_DMAADR	0x4000004C	TIM2 DMA address register for continuous mode	0x0000
R16_TIM2_AUX	0x40000050	TIM2 dual-edge capture register	0x0000

Table 15-4: TIM3 related register list

Name	Address	Description	Reset Value
R16_TIM3_CTLR1	0x40000400	TIM3 control register 1	0x0000
R16_TIM3_CTLR2	0x40000404	TIM3 control register 2	0x0000
R16_TIM3_SMCFR	0x40000408	TIM3 slave mode control register	0x0000
R16_TIM3_DMAINTENR	0x4000040C	TIM3 DMA/interrupt enable register	0x0000
R16_TIM3_INTFR	0x40000410	TIM3 interrupt status register	0x0000
R16_TIM3_SWEVGR	0x40000414	TIM3 event generation register	0x0000
R16_TIM3_CHCTLR1	0x40000418	TIM3 compare/capture control register 1	0x0000
R16_TIM3_CHCTLR2	0x4000041C	TIM3 compare/capture control register 2	0x0000
R16_TIM3_CCER	0x40000420	TIM3 compare/capture enable register	0x0000
R16_TIM3_CNT	0x40000424	TIM3 counter	0x0000
R16_TIM3_PSC	0x40000428	TIM3 counter clock prescaler	0x0000
R16_TIM3_ATRLR	0x4000042C	TIM3 auto-reload register	0xFFFF
R16_TIM3_CH1CVR	0x40000434	TIM3 compare/capture register 1	0x0000

Name	Address	Description	Reset Value
R16_TIM3_CH2CVR	0x40000438	TIM3 compare/capture register 2	0x0000
R16_TIM3_CH3CVR	0x4000043C	TIM3 compare/capture register 3	0x0000
R16_TIM3_CH4CVR	0x40000440	TIM3 compare/capture register 4	0x0000
R16_TIM3_DMACHFR	0x40000448	TIM3 DMA control register	0x0000
R16_TIM3_DMAADR	0x4000044C	TIM3 DMA address register for continuous mode	0x0000
R16_TIM3_AUX	0x40000450	TIM3 dual-edge capture register	0x0000

Table 15-5: TIM4 related register list

Name	Address	Description	Reset Value
R16_TIM4_CTLR1	0x40000800	TIM4 control register 1	0x0000
R16_TIM4_CTLR2	0x40000804	TIM4 control register 2	0x0000
R16_TIM4_SMCFR	0x40000808	TIM4 slave mode control register	0x0000
R16_TIM4_DMAINTENR	0x4000080C	TIM4 DMA/interrupt enable register	0x0000
R16_TIM4_INTFR	0x40000810	TIM4 interrupt status register	0x0000
R16_TIM4_SWEVGR	0x40000814	TIM4 event generation register	0x0000
R16_TIM4_CHCTL1	0x40000818	TIM4 compare/capture control register 1	0x0000
R16_TIM4_CHCTL2	0x4000081C	TIM4 compare/capture control register 2	0x0000
R16_TIM4_CCER	0x40000820	TIM4 compare/capture enable register	0x0000
R16_TIM4_CNT	0x40000824	TIM4 counter	0x0000
R16_TIM4_PSC	0x40000828	TIM4 counter clock prescaler	0x0000
R16_TIM4_ATRLR	0x4000082C	TIM4 auto-reload register	0xFFFF
R16_TIM4_CH1CVR	0x40000834	TIM4 compare/capture register 1	0x0000
R16_TIM4_CH2CVR	0x40000838	TIM4 compare/capture register 2	0x0000
R16_TIM4_CH3CVR	0x4000083C	TIM4 compare/capture register 3	0x0000
R16_TIM4_CH4CVR	0x40000840	TIM4 compare/capture register 4	0x0000
R16_TIM4_DMACHFR	0x40000848	TIM4 DMA control register	0x0000
R16_TIM4_DMAADR	0x4000084C	TIM4 DMA address register for continuous mode	0x0000
R16_TIM4_AUX	0x40000850	TIM4 dual-edge capture register	0x0000

Table 15-6: TIM5 related register list

Name	Address	Description	Reset Value
R16_TIM5_CTLR1	0x40000C00	TIM5 control register 1	0x0000
R16_TIM5_CTLR2	0x40000C04	TIM5 control register 2	0x0000
R16_TIM5_SMCFR	0x40000C08	TIM5 slave mode control register	0x0000
R16_TIM5_DMAINTENR	0x40000C0C	TIM5 DMA/interrupt enable register	0x0000
R16_TIM5_INTFR	0x40000C10	TIM5 interrupt status register	0x0000
R16_TIM5_SWEVGR	0x40000C14	TIM5 event generation register	0x0000
R16_TIM5_CHCTL1	0x40000C18	TIM5 compare/capture control register 1	0x0000
R16_TIM5_CHCTL2	0x40000C1C	TIM5 compare/capture control register 2	0x0000

Name	Address	Description	Reset Value
R16_TIM5_CCER	0x40000C20	TIM5 compare/capture enable register	0x0000
R16_TIM5_CNT	0x40000C24	TIM5 counter	0x0000
R16_TIM5_PSC	0x40000C28	TIM5 counter clock prescaler	0x0000
R16_TIM5_ATRLR	0x40000C2C	TIM5 auto-reload register	0xFFFF
R16_TIM5_CH1CVR	0x40000C34	TIM5 compare/capture register 1	0x0000
R16_TIM5_CH2CVR	0x40000C38	TIM5 compare/capture register 2	0x0000
R16_TIM5_CH3CVR	0x40000C3C	TIM5 compare/capture register 3	0x0000
R16_TIM5_CH4CVR	0x40000C40	TIM5 compare/capture register 4	0x0000
R16_TIM5_DMACFGR	0x40000C48	TIM5 DMA control register	0x0000
R16_TIM5_DMAADR	0x40000C4C	TIM5 DMA address register for continuous mode	0x0000
R16_TIM5_AUX	0x40000C50	TIM5 dual-edge capture register	0x0000

15.4.1 Control Register 1 (TIMx_CTLR1) (x=2/3/4/5)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CKD[1:0]		ARPE	CMS[1:0]		DIR	OPM	URS	UDIS	CEN

Bits	Name	Access	Description	Reset
[15:10]	Reserved	RO	Reserved.	0
[9:8]	CKD[1:0]	RW	These 2 bits define the division ratio between the timer clock (CK_INT) frequency and the sampling clock used by the digital filter: 00: $T_{dts} = T_{ck_int}$; 01: $T_{dts} = 2 \times T_{ck_int}$; 10: $T_{dts} = 4 \times T_{ck_int}$; 11: Reserved.	00b
7	ARPE	RW	Auto-reload preload enable: 1: Auto-reload register (ATRLR) is enabled; 0: Auto-reload register (ATRLR) is disabled.	0
[6:5]	CMS[1:0]	RW	Center-aligned mode selection: 00: Edge-aligned mode. The counter counts up or down depending on the direction bit (DIR). 01: Center-aligned mode 1. The counter counts alternately up and down. The output compare interrupt flag of channels configured as output (CCxS=00 in CHCTLRx register) is set only when the counter is counting down. 10: Center-aligned mode 2. The counter counts alternately up and down. The output compare interrupt flag of channels configured as output (CCxS=00 in CHCTLRx register) is set only when the counter is counting up. 11: Center-aligned mode 3. The counter counts alternately up and down. The output compare interrupt flag of channels configured as output (CCxS=00 in CHCTLRx register) is set both when the counter is counting up and down.	00b

Bits	Name	Access	Description	Reset
			Note: It is not allowed to switch from edge-aligned mode to center-aligned mode while the counter is enabled (CEN=1).	
4	DIR	RW	Counter direction: 1: The counter counts down; 0: The counter counts up. Note: This bit is invalid when the counter is configured in center-aligned mode or encoder mode.	0
3	OPM	RW	Single pulse mode. 1: The counter stops at the next update event (CEN bit is cleared); 0: The counter does not stop at the next update event.	0
2	URS	RW	Update request source. Software uses this bit to select the source of UEV events. 1: If update interrupt or DMA request is enabled, only counter overflow/underflow generates update interrupt or DMA request; 0: If update interrupt or DMA request is enabled, any of the following events generates update interrupt or DMA request: - Counter overflow/underflow; - Setting the UG bit; - Update generated by the slave mode controller.	0
1	UDIS	RW	Update disable. Software uses this bit to enable/disable the generation of UEV events. 1: UEV disabled. No update events are generated, and the registers (ATRLR, PSC, CHCTLRx) keep their values. If the UG bit is set or a hardware reset is issued from the slave mode controller, the counter and prescaler are reinitialized. 0: UEV enabled. Update (UEV) events are generated by any of the following events: - Counter overflow/underflow; - Setting the UG bit; - Update generated by the slave mode controller. The buffered registers are loaded with their preload values.	0
0	CEN	RW	Counter enable. 1: Counter enabled; 0: Counter disabled. Note: External clock, gated mode, and encoder mode only work after the CEN bit is set by software. Trigger mode can automatically set the CEN bit through hardware.	0

15.4.2 Control Register 2 (TIMx_CTLR2) (x=2/3/4/5)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved								TI1S	MMS[2:0]			CCDS	CCUS	Reserved		CCPC

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
7	TI1S	RW	TI1 selection: 1: TIMx_CH1, TIMx_CH2, and TIMx_CH3 pins are XORed and connected to	0

Bits	Name	Access	Description	Reset
			the TI1 input; 0: TIMx_CH1 pin is directly connected to the TI1 input.	
[6:4]	MMS[2:0]	RW	Master mode selection. These 3 bits are used to select the synchronization information (TRGO) sent to the slave timer in master mode. The possible combinations are: 000: Reset – The UG bit is used as the trigger output (TRGO). If the reset is caused by a trigger input (slave mode controller in reset mode), there is a delay on the TRGO signal relative to the actual reset; 001: Enable – The counter enable signal CNT_EN is used as the trigger output (TRGO). Sometimes it is needed to start multiple timers at the same time or control enabling the slave timer for a period. The counter enable signal is produced by a logical OR of the CEN control bit and the trigger input signal in gated mode. When the counter enable signal is controlled by the trigger input, there is a delay on TRGO unless master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCFGR register); 010: Update event is selected as the trigger input (TRGO). For example, the clock of a master timer can be used as a prescaler for a slave timer; 011: Compare pulse. When a capture or a compare match occurs and the CC1IF flag is to be set (even if it is already high), the trigger output sends a positive pulse (TRGO); 100: OC1REF signal is used as the trigger output (TRGO); 101: OC2REF signal is used as the trigger output (TRGO); 110: OC3REF signal is used as the trigger output (TRGO); 111: OC4REF signal is used as the trigger output (TRGO).	000b
3	CCDS	RW	Compare/capture DMA selection: 1: DMA request of CHxCVR is sent when an update event occurs; 0: DMA request of CHxCVR is sent when a CHxCVR event occurs.	0
2	CCUS	RW	Compare/capture control update selection: 1: If CCPC is set, they can be updated by setting the COM bit or by a rising edge on TRGI; 0: If CCPC is set, they can only be updated by setting the COM bit. Note: This bit only applies to channels with complementary outputs.	0
1	Reserved	RO	Reserved.	0
0	CCPC	RW	Compare/capture preload control: 1: CCxE, CCxNE, and OCxM bits are preloaded. After setting this bit, they are only updated after the COM bit is set; 0: CCxE, CCxNE, and OCxM bits are not preloaded. Note: This bit only applies to channels with complementary outputs.	0

15.4.3 Slave Mode Control Register (TIMx_SMCFGR) (x=2/3/4/5)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETP	ECE	ETPS[1:0]		ETF[3:0]		MSM			TS[2:0]		Reserved			SMS[2:0]	

Bits	Name	Access	Description	Reset
15	ETP	RO	ETR trigger polarity selection. This bit selects whether to input ETR directly or the inverted ETR.	0

Bits	Name	Access	Description	Reset
			1: ETR is inverted, low level or falling edge active; 0: ETR, high level or rising edge active.	
14	ECE	RW	External clock mode 2 enable selection. 1: External clock mode 2 enabled; 0: External clock mode 2 disabled. Note 1: The slave mode can be used simultaneously with external clock mode 2: reset mode, gated mode, and trigger mode; however, TRGI cannot be connected to ETRF (TS bits cannot be 111b). Note 2: When both external clock mode 1 and external clock mode 2 are enabled, the external clock input is ETRF.	0
[13:12]	ETPS[1:0]	RW	External trigger signal (ETRP) prescaler. The frequency of this signal cannot exceed 1/4 of the TIMxCLK frequency. This field can be used to reduce the frequency. 00: Prescaler off; 01: ETRP frequency divided by 2; 10: ETRP frequency divided by 4; 11: ETRP frequency divided by 8.	00b
[11:8]	ETF[3:0]	RW	External trigger filter. In practice, the digital filter is an event counter that uses a certain sampling frequency and produces an output transition after recording N events. 0000: No filter, sampling at F_{dts} ; 0001: Sampling frequency $F_{sampling} = F_{ck_int'}$, $N=2$; 0010: Sampling frequency $F_{sampling} = F_{ck_int'}$, $N=4$; 0011: Sampling frequency $F_{sampling} = F_{ck_int'}$, $N=8$; 0100: Sampling frequency $F_{sampling} = F_{dts}/2$, $N=6$; 0101: Sampling frequency $F_{sampling} = F_{dts}/2$, $N=8$; 0110: Sampling frequency $F_{sampling} = F_{dts}/4$, $N=6$; 0111: Sampling frequency $F_{sampling} = F_{dts}/4$, $N=8$; 1000: Sampling frequency $F_{sampling} = F_{dts}/8$, $N=6$; 1001: Sampling frequency $F_{sampling} = F_{dts}/8$, $N=8$; 1010: Sampling frequency $F_{sampling} = F_{dts}/16$, $N=5$; 1011: Sampling frequency $F_{sampling} = F_{dts}/16$, $N=6$; 1100: Sampling frequency $F_{sampling} = F_{dts}/16$, $N=8$; 1101: Sampling frequency $F_{sampling} = F_{dts}/32$, $N=5$; 1110: Sampling frequency $F_{sampling} = F_{dts}/32$, $N=6$; 1111: Sampling frequency $F_{sampling} = F_{dts}/32$, $N=8$.	0000b
7	MSM	RW	Master/slave mode selection: 1: Events on the trigger input (TRGI) are delayed to allow perfect synchronization between the current timer (via TRGO) and its slave timer. This is very useful when several timers need to be synchronized to a single external event; 0: No effect.	0
[6:4]	TS[2:0]	RW	Trigger selection. These 3 bits select the trigger input source used to synchronize the counter. 000: Internal trigger 0 (ITR0); 100: TI1 edge detector (TI1F_ED); 001: Internal trigger 1 (ITR1);	000b

Bits	Name	Access	Description	Reset
			101: Filtered timer input 1 (TI1FP1); 010: Internal trigger 2 (ITR2); 110: Filtered timer input 2 (TI2FP2); 011: Internal trigger 3 (ITR3); 111: External trigger input (ETRF); The above can only be changed when SMS is 0.	
3	Reserved	RO	Reserved.	0
[2:0]	SMS[2:0]	RW	Input mode selection. Selects the clock and trigger mode of the core counter. 000: Driven by internal clock CK_INT; 001: Encoder mode 1. The core counter counts up/down on TI2FP2 edges depending on the level of TI1FP1; 010: Encoder mode 2. The core counter counts up/down on TI1FP1 edges depending on the level of TI2FP2; 011: Encoder mode 3. The core counter counts up/down on both TI1FP1 and TI2FP2 edges depending on the input level of the other signal; 100: Reset mode. The rising edge of the trigger input (TRGI) initializes the counter and generates a signal to update the registers; 101: Gated mode. The counter clock is enabled when the trigger input (TRGI) is high. When the trigger input goes low, the counter stops. Both the start and stop of the counter are controlled; 110: Trigger mode. The counter starts on the rising edge of the trigger input TRGI. Only the start of the counter is controlled; 111: External clock mode 1. The rising edge of the selected trigger input (TRGI) drives the counter.	000b

15.4.4 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=2/3/4/5)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	TDE	Reserved	CC4DE	CC3DE	CC2DE	CC1DE	UDE	Reserved	TIE	Reserved	CC4IE	CC3IE	CC2IE	CC1IE	UIE

Bits	Name	Access	Description	Reset
15	Reserved	RO	Reserved.	0
14	TDE	RW	Trigger DMA request enable. 1: Trigger DMA request enabled; 0: Trigger DMA request disabled.	0
13	Reserved	RW	Reserved.	0
12	CC4DE	RW	Compare/capture channel 4 DMA request enable. 1: DMA request of compare/capture channel 4 enabled; 0: DMA request of compare/capture channel 4 disabled.	0
11	CC3DE	RW	Compare/capture channel 3 DMA request enable. 1: DMA request of compare/capture channel 3 enabled; 0: DMA request of compare/capture channel 3 disabled.	0
10	CC2DE	RW	Compare/capture channel 2 DMA request enable. 1: DMA request of compare/capture channel 2 enabled; 0: DMA request of compare/capture channel 2 disabled.	0

Bits	Name	Access	Description	Reset
9	CC1DE	RW	Compare/capture channel 1 DMA request enable. 1: DMA request of compare/capture channel 1 enabled; 0: DMA request of compare/capture channel 1 disabled.	0
8	UDE	RW	Update DMA request enable. 1: Update DMA request enabled; 0: Update DMA request disabled.	0
7	Reserved	RO	Reserved.	0
6	TIE	RW	Trigger interrupt enable. 1: Trigger interrupt enabled; 0: Trigger interrupt disabled.	0
5	Reserved	RO	Reserved.	0
4	CC4IE	RW	Compare/capture channel 4 interrupt enable. 1: Compare/capture channel 4 interrupt enabled; 0: Compare/capture channel 4 interrupt disabled.	0
3	CC3IE	RW	Compare/capture channel 3 interrupt enable. 1: Compare/capture channel 3 interrupt enabled; 0: Compare/capture channel 3 interrupt disabled.	0
2	CC2IE	RW	Compare/capture channel 2 interrupt enable. 1: Compare/capture channel 2 interrupt enabled; 0: Compare/capture channel 2 interrupt disabled.	0
1	CC1IE	RW	Compare/capture channel 1 interrupt enable. 1: Compare/capture channel 1 interrupt enabled; 0: Compare/capture channel 1 interrupt disabled.	0
0	UIE	RW	Update interrupt enable. 1: Update interrupt enabled; 0: Update interrupt disabled.	0

15.4.5 Interrupt Status Register (TIMx_INTFR) (x=2/3/4/5)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CC4OF	CC3OF	CC2OF	CC1OF	Reserved	TIF	Reserved	CC4IF	CC3IF	CC2IF	CC1IF	UIF			

Bits	Name	Access	Description	Reset
[15:13]	Reserved	RO	Reserved.	0
12	CC4OF	RW0	Compare/capture channel 4 over-capture flag.	0
11	CC3OF	RW0	Compare/capture channel 3 over-capture flag.	0
10	CC2OF	RW0	Compare/capture channel 2 over-capture flag.	0
9	CC1OF	RW0	Compare/capture channel 1 over-capture flag, only used when the compare/capture channel is configured as input capture mode. This flag is set by hardware and cleared by software writing 0. 1: The counter value was captured to the capture/compare register while CC1IF was already set; 0: No over-capture occurred.	0
[8:7]	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
6	TIF	RW0	Trigger interrupt flag. This bit is set by hardware when a trigger event occurs and cleared by software. Trigger events include: when in modes other than gated mode, a valid edge is detected on the TRGI input; or in gated mode, any edge. 1: Trigger event generated; 0: No trigger event generated.	0
5	Reserved	RO	Reserved.	0
4	CC4IF	RW0	Compare/capture channel 4 interrupt flag.	0
3	CC3IF	RW0	Compare/capture channel 3 interrupt flag.	0
2	CC2IF	RW0	Compare/capture channel 2 interrupt flag.	0
1	CC1IF	RW0	Compare/capture channel 1 interrupt flag. If the compare/capture channel is configured as output mode, this bit is set by hardware when the counter value matches the compare value, except in center-aligned mode. This bit is cleared by software. 1: The core counter value matches the compare/capture register 1 value; 0: No match occurred. If compare/capture channel 1 is configured as input mode, this bit is set by hardware when a capture event occurs. It is cleared by software or by reading the compare/capture register. 1: The counter value has been captured to compare/capture register 1; 0: No input capture occurred.	0
0	UIF	RW0	Update interrupt flag. This bit is set by hardware when an update event is generated and cleared by software. 1: Update interrupt generated; 0: No update event generated. The following situations generate an update event: If UDIS=0, when the repetition counter overflows or underflows; If URS=0, UDIS=0, when the UG bit is set, or when the core counter is reinitialized by software; If URS=0, UDIS=0, when the counter CNT is reinitialized by a trigger event.	0

15.4.6 Event Generation Register (TIMx_SWEVGR) (x=2/3/4/5)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						TG	Reserved	CC4G	CC3G	CC2G	CC1G	UG			

Bits	Name	Access	Description	Reset
[15:7]	Reserved	RO	Reserved.	0
6	TG	WO	Trigger event generation. This bit is set by software and cleared by hardware. Used to generate a trigger event. 1: A trigger event is generated. TIF is set. If the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated; 0: No action.	0
5	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
4	CC4G	WO	Compare/capture event generation bit 4. Generates compare/capture event 4.	0
3	CC3G	WO	Compare/capture event generation bit 3. Generates compare/capture event 3.	0
2	CC2G	WO	Compare/capture event generation bit 2. Generates compare/capture event 2.	0
1	CC1G	WO	Compare/capture event generation bit 1. Generates compare/capture event 1. This bit is set by software and cleared by hardware. Used to generate a compare/capture event. 1: Generates a compare/capture event on compare/capture channel 1: If compare/capture channel 1 is configured as output: Sets the CC1IF bit. If the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated; If compare/capture channel 1 is configured as input: The current core counter value is captured into compare/capture register 1; the CC1IF bit is set. If the corresponding interrupt and DMA are enabled, the corresponding interrupt and DMA are generated. If CC1IF is already set, the CC1OF bit is set; 0: No action.	0
0	UG	WO	Update event generation. Generates an update event. This bit is set by software and automatically cleared by hardware. 1: Initializes the counter and generates an update event; 0: No action. Note: The prescaler counter is also cleared, but the prescaler division factor remains unchanged. If in center-aligned mode or up-counting mode, the core counter is cleared; if in down-counting mode, the core counter takes the value of the auto-reload register.	0

15.4.7 Compare/Capture Control Register 1 (TIMx_CHCTLR1) (x=2/3/4/5)

Offset address: 0x18

The channel can be used for input (capture mode) or output (compare mode). The direction of the channel is defined by the corresponding CCxS bits. The function of the other bits in this register differs between input and output modes. OCxx describes the channel function in output mode, and ICxx describes the channel function in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]		OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]		OC1PE	OC1FE	CC1S[1:0]			

Compare mode (pin direction is output):

Bits	Name	Access	Description	Reset
15	OC2CE	RW	Compare/capture channel 2 clear enable. 1: OC2REF is cleared to zero as soon as a high level is detected on the ETRF input; 0: OC2REF is not affected by the ETRF input.	0
[14:12]	OC2M[2:0]	RW	Compare/capture channel 2 mode configuration. These 3 bits define the action of the output reference signal OC2REF, which determines the values of OC2 and OC2N. OC2REF is active high, while the active level of	000b

Bits	Name	Access	Description	Reset
			OC2 and OC2N depends on the CC2P and CC2NP bits. 000: Frozen. The comparison between the compare/capture register value and the core counter has no effect on OC2REF; 001: Force active level. When the core counter matches the value of compare/capture register 2, OC2REF is forced high; 010: Force inactive level. When the core counter value matches compare/capture register 2, OC2REF is forced low; 011: Toggle. When the core counter matches the value of compare/capture register 2, OC2REF toggles; 100: Force inactive level. OC2REF is forced low; 101: Force active level. OC2REF is forced high; 110: PWM mode 1: In up-counting, channel 2 is active when the core counter is less than the compare/capture register value, otherwise inactive; in down-counting, channel 2 is inactive (OC2REF=0) when the core counter is greater than the compare/capture register value, otherwise active (OC2REF=1); 111: PWM mode 2: In up-counting, channel 2 is inactive when the core counter is less than the compare/capture register value, otherwise active; in down-counting, channel 2 is active (OC2REF=1) when the core counter is greater than the compare/capture register value, otherwise inactive (OC2REF=0); Note: Once the LOCK level is set to 3 and CC1S=00b, this bit cannot be modified. In PWM mode 1 or PWM mode 2, the OC2REF level changes only when the comparison result changes or when switching from frozen mode to PWM mode in output compare mode.	
11	OC2PE	RW	Compare/capture register 2 preload enable. 1: The preload function of the compare/capture register is enabled. Read/write operations only operate on the preload register. The preload value of compare/capture register 1 is loaded into the current shadow register when an update event occurs; 0: The preload function of compare/capture register 2 is disabled. Compare/capture register 2 can be written at any time, and the newly written value takes effect immediately. Note: Once the LOCK level is set to 3 and CC1S=00, this bit cannot be modified. Only in single pulse mode (OPM=1) can PWM mode be used without confirming the preload register, otherwise its behavior is undefined.	0
10	OC2FE	RW	Compare/capture channel 2 fast enable. This bit is used to speed up the response of the compare/capture channel output to trigger input events. 1: The effect of the active edge on the trigger is like a compare match occurred. Therefore, OC is set to the compare level regardless of the comparison result. The delay between the active edge of the trigger and the compare/capture channel 2 output is reduced to 3 clock cycles; 0: The compare/capture channel 2 operates normally based on the counter and compare/capture register 2 values, even if the trigger is enabled. When the trigger input has an active edge, the minimum delay to activate the compare/capture channel 2 output is 5 clock cycles. OC2FE only takes effect when the channel is configured in PWM1 or PWM2 mode.	0
[9:8]	CC2S[1:0]	RW		00b

Bits	Name	Access	Description	Reset
			Compare/capture channel 2 input selection. 00: Compare/capture channel 2 is configured as output; 01: Compare/capture channel 2 is configured as input, IC2 is mapped on TI2; 10: Compare/capture channel 2 is configured as input, IC2 is mapped on TI1; 11: Compare/capture channel 2 is configured as input, IC2 is mapped on TRC. This mode only works when the internal trigger input is selected (by TS bits). Note: Compare/capture channel 2 is writable only when the channel is off (CC2E is 0).	
7	OC1CE	RW	Compare/capture channel 1 clear enable.	0
[6:4]	OC1M[2:0]	RW	Compare/capture channel 1 mode configuration.	0
3	OC1PE	RW	Compare/capture register 1 preload enable.	0
2	OC1FE	RW	Compare/capture channel 1 fast enable.	0
[1:0]	CC1S[1:0]	RW	Compare/capture channel 1 input selection.	0

Capture mode (pin direction is input):

Bits	Name	Access	Description	Reset
[15:12]	IC2F[3:0]	RW	Input capture filter 2 configuration. These bits set the sampling frequency and digital filter length of the TI1 input. The digital filter consists of an event counter, which produces an output transition after recording N events. 0000: No filter, sampling at f_{DTS} ; 0001: Sampling frequency $F_{sampling} = F_{ck_int}$, N=2; 0010: Sampling frequency $F_{sampling} = F_{ck_int}$, N=4; 0011: Sampling frequency $F_{sampling} = F_{ck_int}$, N=8; 0100: Sampling frequency $F_{sampling} = F_{dts}/2$, N=6; 0101: Sampling frequency $F_{sampling} = F_{dts}/2$, N=8; 0110: Sampling frequency $F_{sampling} = F_{dts}/4$, N=6; 0111: Sampling frequency $F_{sampling} = F_{dts}/4$, N=8; 1000: Sampling frequency $F_{sampling} = F_{dts}/8$, N=6; 1001: Sampling frequency $F_{sampling} = F_{dts}/8$, N=8; 1010: Sampling frequency $F_{sampling} = F_{dts}/16$, N=5; 1011: Sampling frequency $F_{sampling} = F_{dts}/16$, N=6; 1100: Sampling frequency $F_{sampling} = F_{dts}/16$, N=8; 1101: Sampling frequency $F_{sampling} = F_{dts}/32$, N=5; 1110: Sampling frequency $F_{sampling} = F_{dts}/32$, N=6; 1111: Sampling frequency $F_{sampling} = F_{dts}/32$, N=8.	0000b
[11:10]	IC2PSC[1:0]	RW	Compare/capture channel 2 prescaler configuration. These 2 bits define the prescaler factor of compare/capture channel 2. Once CC1E=0, the prescaler is reset. 00: No prescaler, capture is triggered on every edge detected at the input; 01: Capture is triggered every 2 events;	00b

Bits	Name	Access	Description	Reset
			10: Capture is triggered every 4 events; 11: Capture is triggered every 8 events.	
[9:8]	CC2S[1:0]	RW	Compare/capture channel 2 input selection. These 2 bits define the channel direction (input/output) and the input pin selection. 00: Compare/capture channel 1 is configured as output; 01: Compare/capture channel 1 is configured as input, IC1 is mapped on TI1; 10: Compare/capture channel 1 is configured as input, IC1 is mapped on TI2; 11: Compare/capture channel 1 is configured as input, IC1 is mapped on TRC. This mode only works when the internal trigger input is selected (by TS bits). Note: CC1S is writable only when the channel is off (CC1E is 0).	00b
[7:4]	IC1F[3:0]	RW	Input capture filter 1 configuration.	0
[3:2]	IC1PSC[1:0]	RW	Compare/capture channel 1 prescaler configuration.	0
[1:0]	CC1S[1:0]	RW	Compare/capture channel 1 input selection.	0

15.4.8 Compare/Capture Control Register 2 (TIMx_CHCTLR2) (x=2/3/4/5)

Offset address: 0x1C

The channel can be used for input (capture mode) or output (compare mode). The direction of the channel is defined by the corresponding CCxS bits. The function of the other bits in this register differs between input and output modes. OCxx describes the channel function in output mode, and ICxx describes the channel function in input mode.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC4CE	OC4M[2:0]	OC4PE	OC4FE	CC4S[1:0]	OC3CE	OC3M[2:0]	OC3PE	OC3FE	CC3S[1:0]						

Compare mode (pin direction is output):

Bits	Name	Access	Description	Reset
15	OC4CE	RW	Compare/capture channel 4 clear enable.	0
[14:12]	OC4M[2:0]	RW	Compare/capture channel 4 mode configuration.	0
11	OC4PE	RW	Compare/capture register 4 preload enable.	0
10	OC4FE	RW	Compare/capture channel 4 fast enable.	0
[9:8]	CC4S[1:0]	RW	Compare/capture channel 4 input selection.	0
7	OC3CE	RW	Compare/capture channel 3 clear enable.	0
[6:4]	OC3M[2:0]	RW	Compare/capture channel 3 mode configuration.	0
3	OC3PE	RW	Compare/capture register 3 preload enable.	0
2	OC3FE	RW	Compare/capture channel 3 fast enable.	0
[1:0]	CC3S[1:0]	RW	Compare/capture channel 3 input selection.	0

Capture mode (pin direction is input):

Bits	Name	Access	Description	Reset
[15:12]	IC4F[3:0]	RW	Input capture filter 4 configuration.	0

Bits	Name	Access	Description	Reset
[11:10]	IC4PSC[1:0]	RW	Compare/capture channel 4 prescaler configuration.	0
[9:8]	CC4S[1:0]	RW	Compare/capture channel 4 input selection.	0
[7:4]	IC3F[3:0]	RW	Input capture filter 3 configuration.	0
[3:2]	IC3PSC[1:0]	RW	Compare/capture channel 3 prescaler configuration.	0
[1:0]	CC3S[1:0]	RW	Compare/capture channel 3 input selection.	0

15.4.9 Compare/Capture Enable Register (TIMx_CCER) (x=2/3/4/5)

Offset address: 0x20

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	CC4P	CC4E	Reserved	CC3P	CC3E	Reserved	CC2P	CC2E	Reserved	CC1P	CC1E				

Bits	Name	Access	Description	Reset
[15:14]	Reserved	RO	Reserved.	0
13	CC4P	RW	Compare/capture channel 4 output polarity.	0
12	CC4E	RW	Compare/capture channel 4 output enable.	0
[11:10]	Reserved	RO	Reserved.	0
9	CC3P	RW	Compare/capture channel 3 output polarity.	0
8	CC3E	RW	Compare/capture channel 3 output enable.	0
[7:6]	Reserved	RO	Reserved.	0
5	CC2P	RW	Compare/capture channel 2 output polarity.	0
4	CC2E	RW	Compare/capture channel 2 output enable.	0
[3:2]	Reserved	RO	Reserved.	0
1	CC1P	RW	Compare/capture channel 1 output polarity. CC1 channel configured as output: 1: OC1 active low; 0: OC1 active high. CC1 channel configured as input: This bit selects whether IC1 or the inverted IC1 signal is used as the trigger or capture signal. 1: Inverted: capture occurs on the falling edge of IC1; when used as an external trigger, IC1 is inverted; 0: Non-inverted: capture occurs on the rising edge of IC1; when used as an external trigger, IC1 is not inverted.	0
0	CC1E	RW	Compare/capture channel 1 output enable. CC1 channel configured as output: 1: On – OC1 signal is output to the corresponding output pin; 0: Off – OC1 output disabled. CC1 channel configured as input: This bit determines whether the counter value can be captured into the TIMx_CCR1 register. 1: Capture enabled; 0: Capture disabled.	0

15.4.10 General Purpose Timer Counter (TIMx_CNT) (x=2/3/4/5)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	CNT[15:0]	RW	Real-time value of the timer counter.	0											

15.4.11 Counter Clock Prescaler (TIMx_PSC) (x=2/3/4/5)

Offset address: 0x28

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	PSC[15:0]	RW	Division factor of the timer prescaler. The counter clock frequency equals the prescaler input frequency / (PSC + 1).	0											

15.4.12 Auto-Reload Register (TIMx_ATRLR) (x=2/3/4/5)

Offset address: 0x2C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	ARR[15:0]	RW	Counter threshold. For when ATRLR takes action and updates, please refer to Section 15.2.4. When ATRLR is empty, the counter stops.	0xFFFF											

15.4.13 Compare/Capture Register 1 (TIMx_CH1CVR) (x=2/3/4/5)

Offset address: 0x34

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	CCR1[15:0]	RW	Value of compare/capture register channel 1.	0											

15.4.14 Compare/Capture Register 2 (TIMx_CH2CVR) (x=2/3/4/5)

Offset address: 0x38

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR2[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	CCR2[15:0]	RW	Value of compare/capture register channel 2.	0											

15.4.15 Compare/Capture Register 3 (TIMx_CH3CVR) (x=2/3/4/5)

Offset address: 0x3C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR3[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	CCR3[15:0]	RW	Value of compare/capture register channel 3.	0											

15.4.16 Compare/Capture Register 4 (TIMx_CH4CVR) (x=2/3/4/5)

Offset address: 0x40

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR4[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	CCR4[15:0]	RW	Value of compare/capture register channel 4.	0											

15.4.17 DMA Control Register (TIMx_DMACFGR) (x=2/3/4/5)

Offset address: 0x48

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			DBL[4:0]					Reserved			DBA[4:0]				
Bits	Name	Access	Description											Reset	
[15:13]	Reserved	RO	Reserved.											0	
[12:8]	DBL[4:0]	RW	DMA burst transfer length. The actual value is this field's value + 1.											0	
[7:5]	Reserved	RO	Reserved.											0	
[4:0]	DBA[4:0]	RW	These bits define the address offset of the DMA in burst mode from control register 1.											0	

15.4.18 DMA Address Register for Continuous Mode (TIMx_DMAADR) (x=2/3/4/5)

Offset address: 0x4C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAB[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	DMAB[15:0]	RW	DMA address in continuous mode.	0											

15.4.19 Dual-Edge Capture Register (TIMx_AUX) (x=2/3/4/5)

Offset address: 0x50

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								CAP_ED_CH4				CAP_ED_CH3		CAP_ED_CH2	

Bits	Name	Access	Description	Reset
[15:3]	Reserved	R0	Reserved.	0
2	CAP_ED_CH4	RW	Dual-edge capture enable for channel 4: 1: Dual-edge capture enable for channel 4; 0: Dual-edge capture disable for channel 4.	0
1	CAP_ED_CH3	RW	Dual-edge capture enable for channel 3: 1: Dual-edge capture enable for channel 3; 0: Dual-edge capture disable for channel 3.	0
0	CAP_ED_CH2	RW	Dual-edge capture enable for channel 2: 1: Dual-edge capture enable for channel 2; 0: Dual-edge capture disable for channel 2.	0

Chapter 16: Basic Timer (BCTM)

The basic timer module contains two 16-bit auto-reload timers (TIM6 and TIM7), used for counting and generating interrupts or DMA requests on update events. TIM6 and TIM7 support a 16-bit programmable prescaler. They can provide a clock for the digital-to-analog converter (DAC) to trigger the DAC synchronization circuit. The basic timers are independent of each other and do not share any resources.

16.1 Main Features

The main features of the basic timer include:

- 16-bit auto-reload counter, supports up-counting mode
- 16-bit prescaler, with division factor dynamically adjustable from 1 to 65536
- Trigger DAC synchronization circuit
- Generate interrupt or DMA request on update event

16.2 Principle and Structure

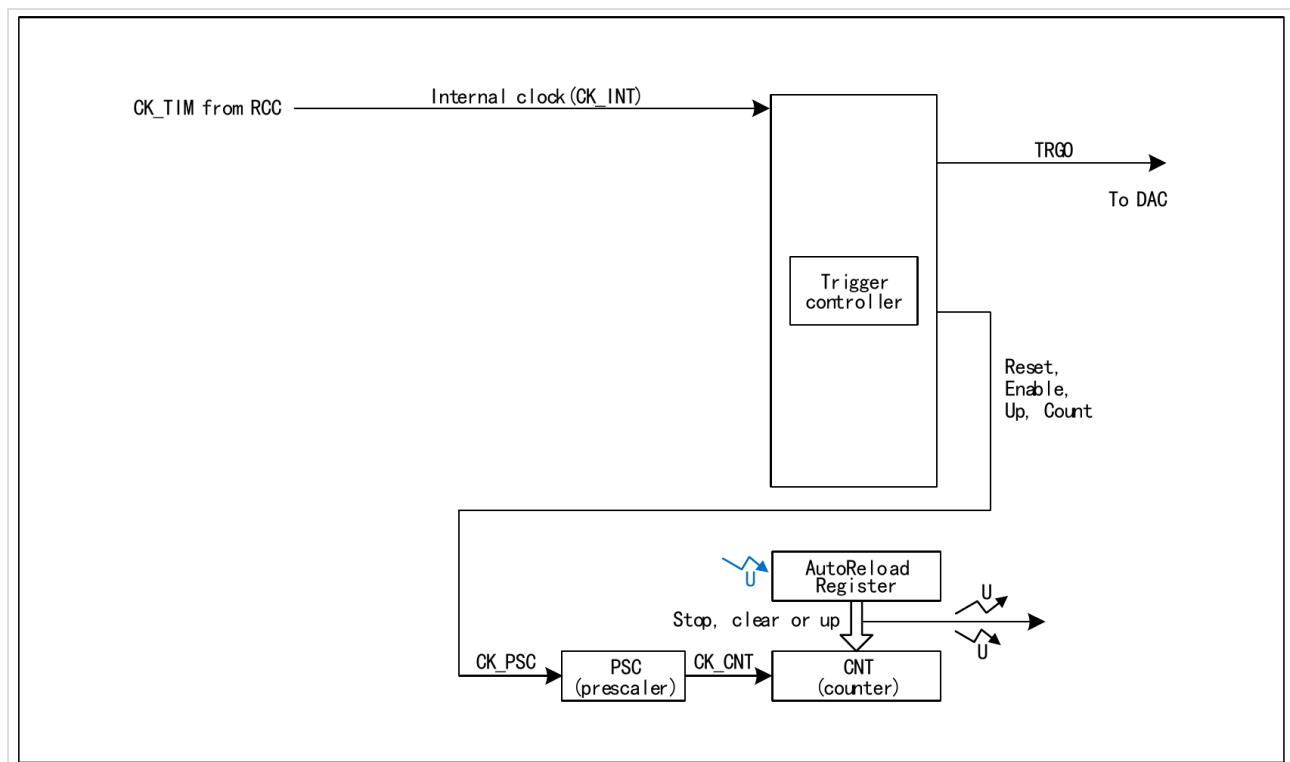


Figure 16-1: Basic timer block diagram

16.2.1 Overview

As shown in Figure 16-1, the structure of the basic timer can be roughly divided into two parts: the input clock section and the core counter section. The clock of the basic timer comes from the HB bus clock (CK_INT). These input clock signals undergo various configured filtering and division operations

to become the CK_PSC clock, which is output to the core counter section. In addition, these complex clock sources can also be output as TRGO to the DAC peripheral.

The core of the basic timer is a 16-bit counter (CNT). CK_PSC is divided by the prescaler (PSC) to become CK_CNT, which is finally fed to CNT. CNT supports up-counting mode and has an auto-reload register (ATRLR) that reloads the initialization value to CNT at the end of each counting cycle.

16.2.2 Differences Between Basic Timer and General-Purpose Timer

Compared with the general-purpose timer, the basic timer lacks the following functions:

1. The basic timer lacks down-counting mode and up-down counting mode.
2. The basic timer lacks four independent compare/capture channels.
3. The basic timer does not support external signal control of the timer.
4. The basic timer does not support incremental encoding, cascading and synchronization between timers.

16.2.3 Clock Input

The clock of the basic timer is provided by the internal clock CK_INT.

16.2.4 Counter and Peripherals

CK_PSC is input to the prescaler (PSC) for division. PSC is 16-bit, and the actual division factor is equal to the value of R16_TIMx_PSC + 1. CK_PSC becomes CK_CNT after passing through PSC. Changing the value of R16_TIM1_PSC does not take effect immediately, but is updated to PSC after an update event. Update events include UG bit clearing and reset.

16.3 Debug Mode

When the system enters debug mode, the timer can be controlled to continue running or stop according to the DBG module settings.

16.4 Register Description

Table 16-1: TIM6 related register list

Name	Address	Description	Reset Value
R16_TIM6_CTLR1	0x40001000	TIM6 control register 1	0x0000
R16_TIM6_CTLR2	0x40001004	TIM6 control register 2	0x0000
R16_TIM6_DMAINTENR	0x4000100C	TIM6 DMA/interrupt enable register	0x0000
R16_TIM6_INTFR	0x40001010	TIM6 interrupt status register	0x0000
R16_TIM6_SWEVGR	0x40001014	TIM6 event generation register	0x0000
R16_TIM6_CNT	0x40001024	TIM6 counter	0x0000
R16_TIM6_PSC	0x40001028	TIM6 counter clock prescaler	0x0000
R16_TIM6_ATRLR	0x4000102C	TIM6 auto-reload register	0xFFFF

Table 16-2: TIM7 related register list

Name	Address	Description	Reset Value
R16_TIM7_CTLR1	0x40001400	TIM7 control register 1	0x0000
R16_TIM7_CTLR2	0x40001404	TIM7 control register 2	0x0000
R16_TIM7_DMAINTENR	0x4000140C	TIM7 DMA/interrupt enable register	0x0000
R16_TIM7_INTFR	0x40001410	TIM7 interrupt status register	0x0000
R16_TIM7_CNT	0x40001424	TIM7 counter	0x0000
R16_TIM7_PSC	0x40001428	TIM7 counter clock prescaler	0x0000
R16_TIM7_ATRLR	0x4000142C	TIM7 auto-reload register	0xFFFF

16.4.1 Control Register 1 (TIMx_CTLR1) (x=6/7)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								ARPE	Reserved			OPM	URS	UDIS	CEN

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
7	ARPE	RW	Auto-reload preload enable bit: 1: Enable auto-reload register (ATRLR); 0: Disable auto-reload register (ATRLR).	0
[6:4]	Reserved	RO	Reserved.	0
3	OPM	RW	One-pulse mode. 1: The counter stops when the next update event occurs (clearing the CEN bit); 0: The counter does not stop when the next update event occurs.	0
2	URS	RW	Update request source. Software uses this bit to select the source of UEV events. 1: If update interrupt or DMA request is enabled, only counter overflow/underflow generates update interrupt or DMA request; 0: If update interrupt or DMA request is enabled, any of the following events generates update interrupt or DMA request. - Counter overflow/underflow - Setting the UG bit - Update generated by the slave mode controller.	0
1	UDIS	RW	Update disable. Software uses this bit to allow/disable the generation of UEV events. 1: Disable UEV. No update event is generated, and registers (ATRLR, PSC, CHCTLRx) keep their values. If the UG bit is set or the slave mode controller sends a hardware reset, the counter and prescaler are reinitialized. 0: Enable UEV. Update (UEV) events are generated by any of the following events: - Counter overflow/underflow - Setting the UG bit - Update generated by the slave mode controller. Buffered registers are loaded with their preload values.	0

Bits	Name	Access	Description	Reset
0	CEN	RW	Counter enable: 1: Enable counter; 0: Disable counter.	0

16.4.2 Control Register 2 (TIMx_CTLR2) (x=6/7)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										MMS[2:0]		Reserved			

Bits	Name	Access	Description	Reset
[15:7]	Reserved	RO	Reserved.	0
[6:4]	MMS[2:0]	RW	Master mode selection. These 3 bits are used to select the synchronization information (TRGO) sent to the slave timer in master mode. Possible combinations are: 000: Reset – The UG bit is used as the trigger output (TRGO). If the reset is generated by a trigger input (slave mode controller in reset mode), the signal on TRGO will have a delay relative to the actual reset; 001: Enable – The counter enable signal CNT_EN is used as the trigger output (TRGO). Sometimes multiple timers need to be started at the same time or the slave timer needs to be enabled for a period of time. The counter enable signal is generated by a logical OR of the CEN control bit and the trigger input signal in gated mode. When the counter enable signal is controlled by the trigger input, there will be a delay on TRGO, unless master/slave mode is selected (see the description of the MSM bit in the TIMx_SMCFG register); 010: Update event is selected as the trigger output (TRGO). For example, the clock of a master timer can be used as the prescaler of a slave timer.	0
[3:0]	Reserved	RO	Reserved.	0

16.4.3 DMA/Interrupt Enable Register (TIMx_DMAINTENR) (x=6/7)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								UDE		Reserved				UIE	

Bits	Name	Access	Description	Reset
[15:9]	Reserved	RO	Reserved.	0
8	UDE	RW	Update DMA request enable bit: 1: Enable update DMA request; 0: Disable update DMA request.	0
[7:1]	Reserved	RO	Reserved.	0
0	UIE	RW	Update interrupt enable bit: 1: Enable update interrupt; 0: Disable update interrupt.	0

16.4.4 Interrupt Status Register (R16_TIMx_INTFR) (x=6/7)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															UIF

Bits	Name	Access	Description	Reset
[15:1]	Reserved	RO	Reserved.	0
0	UIF	RW0	Update interrupt flag. This bit is set by hardware when an update event occurs and cleared by software. 1: Update interrupt generated; The following situations generate an update event: If UDIS=0, when the repetition counter value overflows or underflows; If URS=0, UDIS=0, when the UG bit is set, or when the core counter is reinitialized by software; 0: No update event generated.	0

16.4.5 Event Generation Register (TIMx_SWEVGR) (x=6/7)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															UG

Bits	Name	Access	Description	Reset
[15:1]	Reserved	RO	Reserved.	0
0	UG	WO	Update event generation bit. Generates an update event. This bit is set by software and automatically cleared by hardware. 1: Initialize the counter and generate an update event; 0: No action.	0

Note: The counter of the prescaler is also cleared, but the prescaler division factor remains unchanged. In up-counting mode, the core counter is cleared.

16.4.6 General-Purpose Timer Counter (TIMx_CNT) (x=6/7)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	CNT[15:0]	RW	Real-time value of the timer counter.	0

16.4.7 Counter Clock Prescaler (TIMx_PSC) (x=6/7)

Offset address: 0x28

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	PSC[15:0]	RW	Division factor of the timer prescaler. The counter clock frequency is equal to the prescaler input frequency / (PSC+1).	0

16.4.8 Auto-Reload Register (TIMx_ATRLR) (x=6/7)

Offset address: 0x2C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	ARR[15:0]	RW	The value of ATRLR[15:0] will be loaded into the counter. For when ATRLR takes effect and updates, refer to Section 16.2.4. When ATRLR is empty, the counter stops.	0xFFFF

Chapter 17: Universal Synchronous/ Asynchronous Receiver Transmitter (USART)

This module contains 10 universal synchronous/asynchronous receiver transmitters (USART1/2/3/4/5/6/7/8/9/10).

17.1 Main Features

- Full-duplex or half-duplex synchronous or asynchronous communication
- NRZ data format
- Fractional baud rate generator, up to 12.5 Mbps
- Programmable data length
- Configurable stop bits
- Supports LIN, IrDA encoder, smart card
- Supports DMA
- Multiple interrupt sources

17.2 Overview

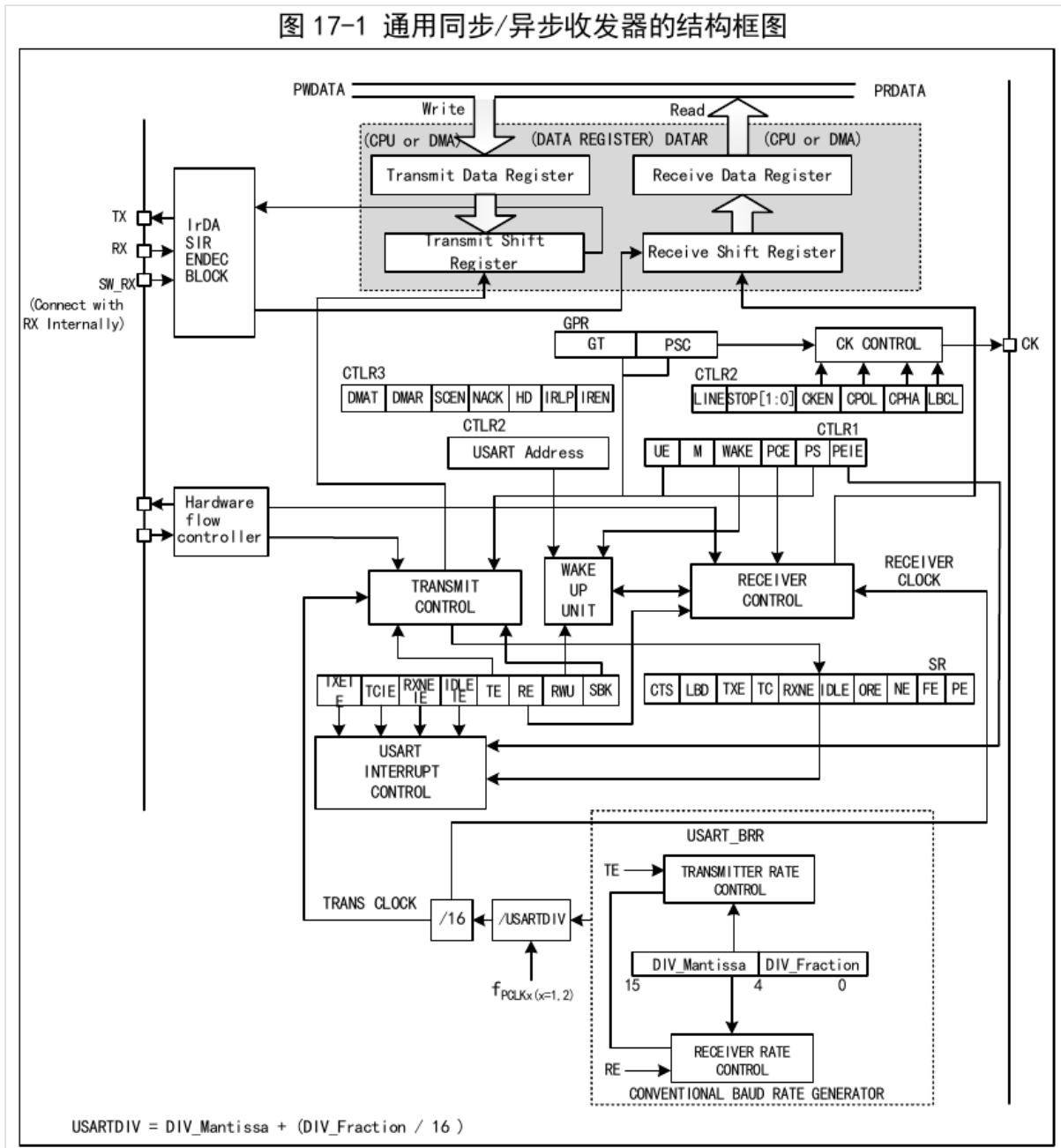


Figure 17-1: Structure block diagram of the universal synchronous/asynchronous receiver transmitter

When TE (transmit enable bit) is set, data in the transmit shift register is output on the TX pin, and the clock is output on the CK pin. During transmission, the least significant bit is shifted out first; each data frame begins with a low-level start bit, then the transmitter sends an 8-bit or 9-bit data word according to the setting of the M (word length) bit, followed by a configurable number of stop bits. If parity is enabled, the last bit of the data word is the parity bit. After TE is set, an idle frame is sent; the idle frame is 10 or 11 bits of high level, including the stop bit. A break frame is 10 or 11 bits of low level followed by a stop bit.

17.3 Baud Rate Generator

The baud rate of the transceiver = $F_{CLK} / (16 \times USARTDIV)$, where F_{CLK} is the clock of PBx, i.e. PCLK1 or PCLK2. The USART1 module uses PCLK2, and all others use PCLK1. The value of USARTDIV is determined by the DIV_M and DIV_F fields in USART_BRR. The calculation formula is:

$$USARTDIV = DIV_M + (DIV_F / 16)$$

Note: The baud rate generated by the baud rate generator may not exactly match the desired value; there may be a deviation. Besides choosing the closest value, another way to reduce the deviation is to increase the PBx clock. For example, when setting a baud rate of 9600 bps, the USARTDIV value is 78.125, and at 12 MHz a baud rate of 9600 bps can be obtained. However, if a baud rate of 115200 bps is needed, the calculated USARTDIV is 6.51, but the closest value that can be written to USART_BRR is 6.5, producing an actual baud rate of 115384 bps with an error of 0.16%.

When the serial waveform sent by the transmitter reaches the receiver, there is a certain error between the receiver's and the transmitter's baud rates. The error mainly comes from three aspects: the actual baud rates of the receiver and transmitter are inconsistent; the clocks of the receiver and transmitter have errors; and the waveform changes as it propagates over the line. The receiver of the peripheral module has a certain tolerance capacity. When the sum of the total deviations from the above three aspects is less than the module's tolerance limit, the total deviation does not affect communication. The module's tolerance limit is affected by whether fractional baud rate is used and the M bit (data field word length). Using fractional baud rate and a 9-bit data field length reduces the tolerance limit, but not below 3%.

17.4 Synchronous Mode

The synchronous mode allows the system to output a clock signal when using the USART module. When synchronous mode is enabled and data is being transmitted, the CK pin simultaneously outputs the clock.

Synchronous mode is enabled by setting the CLKEN bit in control register 2 (R16_USARTx_CTLR2). At the same time, LIN mode, smart card mode, IrDA mode, and half-duplex mode must be disabled, i.e. the SCEN, HDSEL, and IREN bits must be in the reset state. These three bits are in control register 3 (R16_USARTx_CTLR3).

The key point of synchronous mode is the control of clock output. The following should be noted:

- The USART module in synchronous mode only operates in master mode, i.e. the CK pin only outputs a clock and does not accept input;
- The clock signal is only output when data is transmitted on the TX pin;
- The LBCL bit determines whether a clock is output during the last data bit, the CPOL bit determines the clock polarity, and CPHA determines the clock phase. These three bits are in control register 2 (R16_USARTx_CTLR2) and must be set when TE and RE are not enabled. See Figure 17-2 for details.
- In synchronous mode, the receiver only samples when the clock is output. The slave device must maintain a certain signal setup time and hold time; see Figure 17-3 for details.

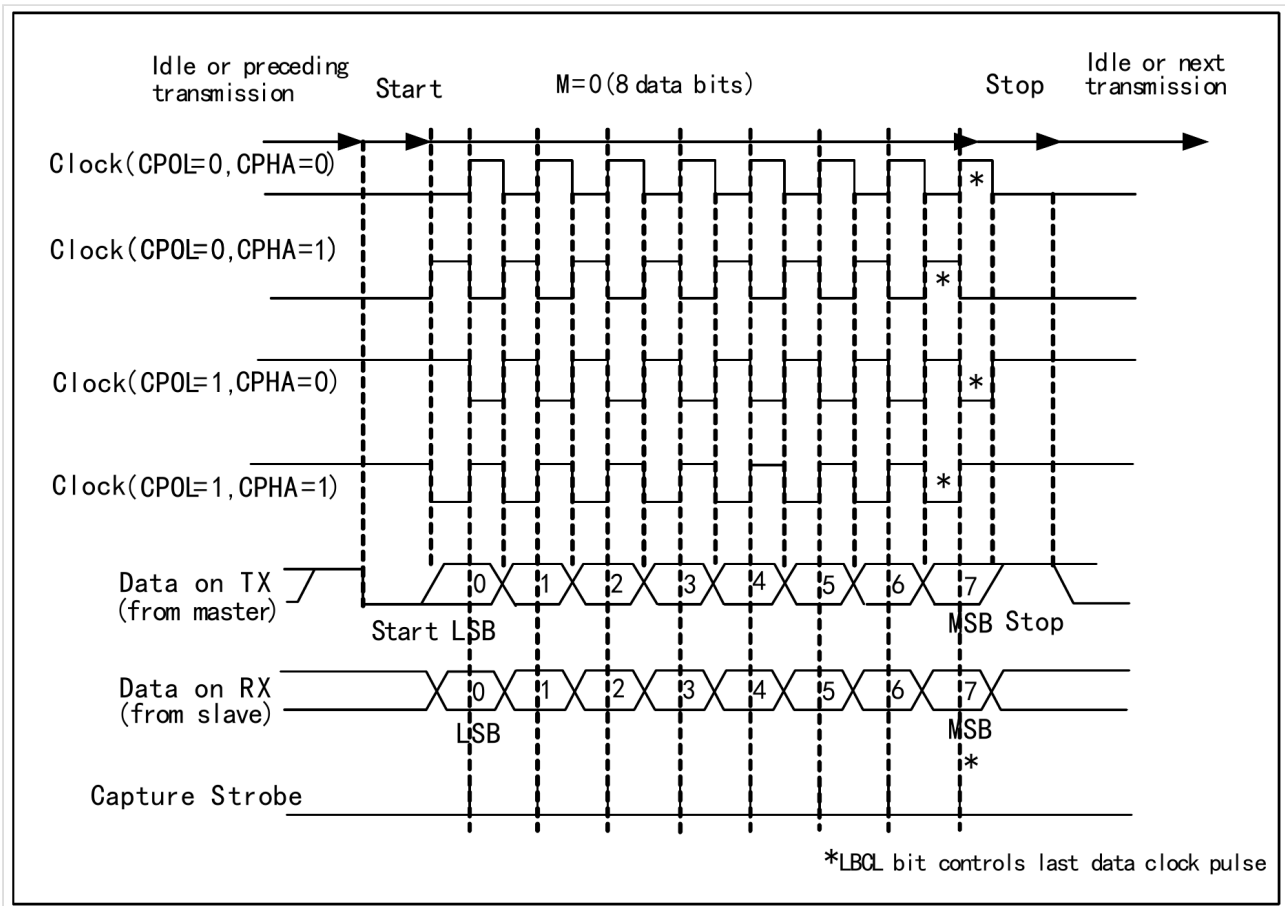


Figure 17-2: USART clock timing example (M=0)

图 17-2 USART 时钟时序示例 (M=0)

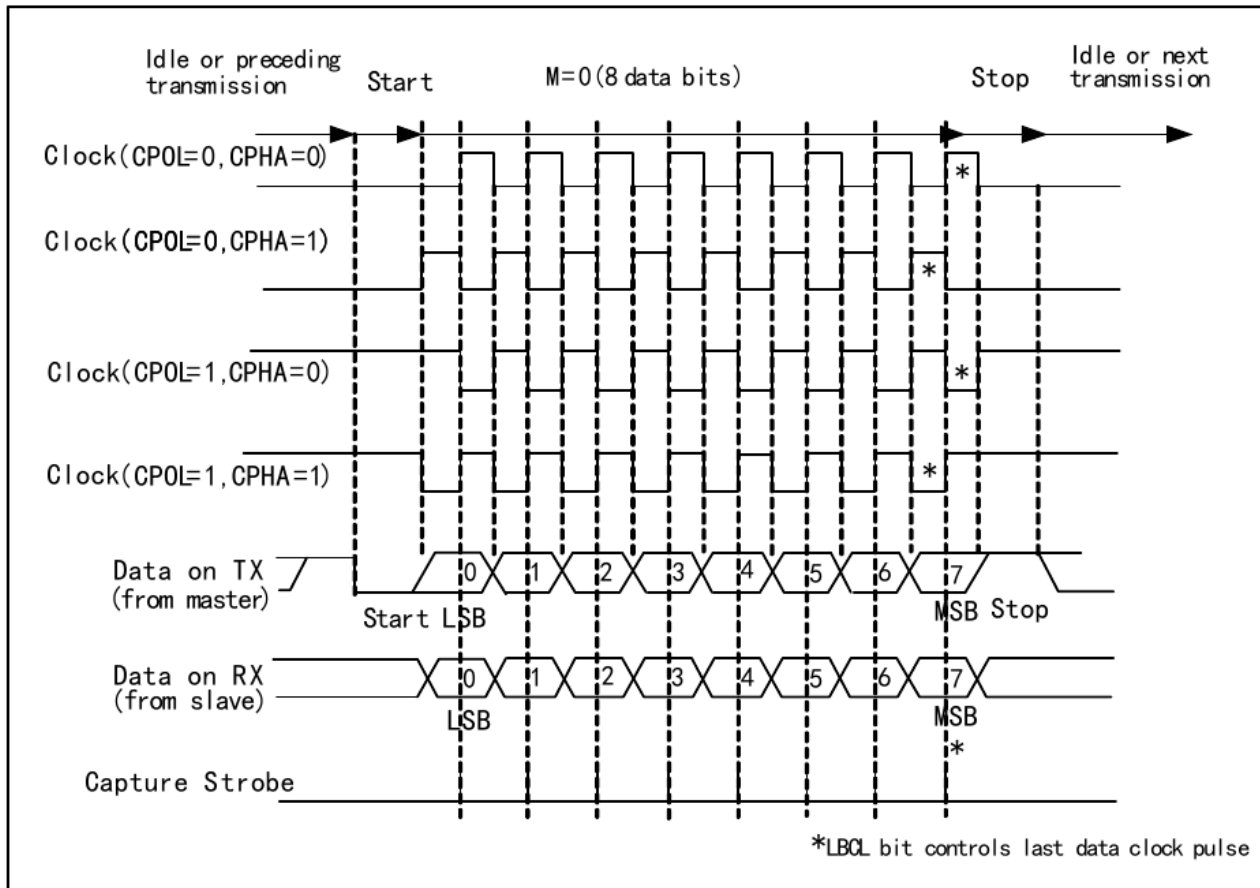


Figure 17-3: Data sample hold time

17.5 Single-Wire Half-Duplex Mode

Half-duplex mode supports using a single pin (only the TX pin) for both receiving and transmitting. The TX pin and RX pin are connected internally within the chip.

Half-duplex mode is enabled by setting the HDSEL bit in control register 3 (R16_USARTx_CTLR3). At the same time, LIN mode, smart card mode, IrDA mode, and synchronous mode must be disabled, i.e. the SCEN, CLKEN, and IREN bits must be in the reset state. These three bits are in control registers 2 and 3 (R16_USARTx_CTLR2 and R16_USARTx_CTLR3).

After configuring half-duplex mode, the TX I/O pin should be set to open-drain output-high mode. When TE is set, writing data to the data register will send it out. In particular, note that in half-duplex mode, bus conflicts may occur when multiple devices use a single bus for transmission and reception; the user must avoid this through software.

17.6 Smart Card

Smart card mode supports the ISO7816-3 protocol for accessing smart card controllers.

Smart card mode is enabled by setting the SCEN bit in control register 3 (R16_USARTx_CTLR3). At the same time, LIN mode, half-duplex mode, and IrDA mode must be disabled, i.e. the LINEN, HDSEL, and IREN bits must be in the reset state. However, CLKEN can be enabled to output a clock. These bits are in control registers 2 and 3 (R16_USARTx_CTLR2 and R16_USARTx_CTLR3).

To support smart card mode, the USART should be configured for 8 data bits plus 1 parity bit. The stop bits are recommended to be set to 1.5 bits for both transmission and reception. Smart card mode is a single-wire half-duplex protocol that uses the TX line for data communication and should be configured as open-drain output with a pull-up. When the receiver detects a parity error while receiving a frame, it sends a NACK signal during the stop bit period by actively pulling TX low for one cycle. When the transmitter detects the NACK signal, a frame error is generated, and the application can use this to retransmit. Figure 17-4 shows the waveform on the TX pin for both the correct case and the case with a parity error. The TC flag (transmission complete flag) of the USART can be delayed by GT (guard time) clock cycles, and the receiver will not interpret its own NACK signal as a start bit.

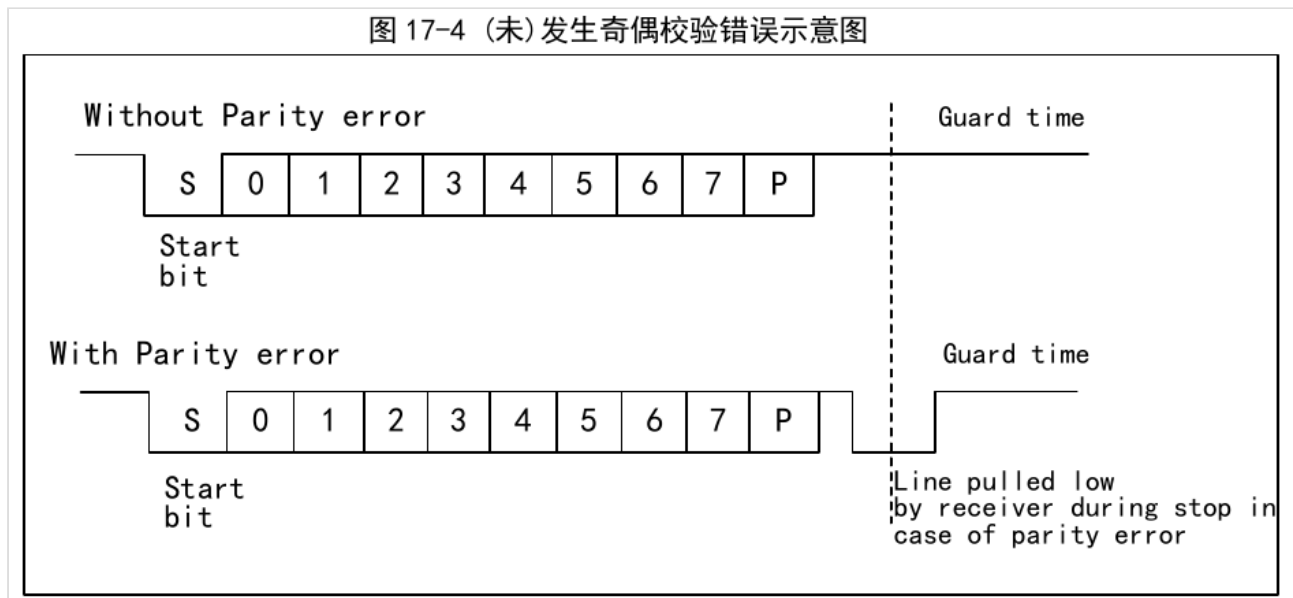


Figure 17-4: Schematic of (no) parity error occurrence

In smart card mode, when the CK pin is enabled, the output waveform is unrelated to communication; it only provides a clock to the smart card. Its value is the PB clock divided by a configurable 5-bit clock divider (the divider value is twice the PSC value, with a maximum division of 62).

17.7 IrDA

The USART module supports controlling an IrDA infrared transceiver for physical layer communication. To use IrDA, the LINEN, STOP, CLKEN, SCEN, and HDSEL bits must be cleared. The USART module and the SIR physical layer (infrared transceiver) use NRZ (non-return-to-zero) encoding, supporting rates up to 115200 bps.

IrDA is a half-duplex protocol. If the USART is sending data to the SIR physical layer, the IrDA decoder will ignore newly received infrared signals. If the USART is receiving data from the SIR, the SIR will not accept signals from the USART. The logic levels for USART-to-SIR and SIR-to-USART are different: in SIR receive logic, high level is 1 and low level is 0, but in SIR transmit logic, high level is 0 and low level is 1.

17.8 DMA

The USART module supports DMA functionality, which can be used for fast continuous transmission and reception. When DMA is enabled, the DMA writes data from the configured memory space to the

transmit buffer whenever TXE is set. When using DMA for reception, each time RXNE is set, the DMA transfers the data from the receive buffer to a specific memory space.

17.9 Interrupts

The USART module supports multiple interrupt sources, including transmit data register empty (TXE), CTS, transmission complete (TC), receive data ready (RXNE), data overrun (ORE), line idle (IDLE), parity error (PE), break flag (LBD), noise (NE), overrun in multi-buffer communication (ORT), and frame error (FE), etc.

Table 17-1: Relationship between interrupts and corresponding enable bits

Interrupt source	Enable bit
Transmit data register empty (TXE)	TXEIE
CTS enable (CTS)	CTSIE
Transmission complete (TC)	TCIE
Receive data ready (RXNE)	RXNEIE
Data overrun (ORE)	
Line idle (IDLE)	IDLEIE
Parity error (PE)	PEIE
Break flag (LBD)	LBDIE
Noise (NE)	EIE
Overrun in multi-buffer communication (ORT)	
Frame error in multi-buffer communication (FE)	

17.10 Register Description

Table 17-2: USART1 related register list

Name	Address	Description	Reset Value
R32_USART1_STATR	0x40013800	USART1 status register	0x000000C0
R32_USART1_DATAR	0x40013804	USART1 data register	0x000000XX
R32_USART1_BRR	0x40013808	USART1 baud rate register	0x00000000
R32_USART1_CTLR1	0x4001380C	USART1 control register 1	0x00000000
R32_USART1_CTLR2	0x40013810	USART1 control register 2	0x00000000
R32_USART1_CTLR3	0x40013814	USART1 control register 3	0x00000000
R32_USART1_GPR	0x40013818	USART1 guard time and prescaler register	0x00000000
R32_USART1_CTRL4	0x4001381C	USART1 control register 4	0x00000000

Table 17-3: USART2 related register list

Name	Address	Description	Reset Value
R32_USART2_STATR	0x40004400	USART2 status register	0x000000C0
R32_USART2_DATAR	0x40004404	USART2 data register	0x000000XX
R32_USART2_BRR	0x40004408	USART2 baud rate register	0x00000000

R32_USART2_CTLR1	0x4000440C	USART2 control register 1	0x00000000
R32_USART2_CTLR2	0x40004410	USART2 control register 2	0x00000000
R32_USART2_CTLR3	0x40004414	USART2 control register 3	0x00000000
R32_USART2_GPR	0x40004418	USART2 guard time and prescaler register	0x00000000
R32_USART2_CTRL4	0x4000441C	USART2 control register 4	0x00000000

Table 17-4: USART3 related register list

Name	Address	Description	Reset Value
R32_USART3_STATR	0x40004800	USART3 status register	0x000000C0
R32_USART3_DATAR	0x40004804	USART3 data register	0x000000XX
R32_USART3_BRR	0x40004808	USART3 baud rate register	0x00000000
R32_USART3_CTLR1	0x4000480C	USART3 control register 1	0x00000000
R32_USART3_CTLR2	0x40004810	USART3 control register 2	0x00000000
R32_USART3_CTLR3	0x40004814	USART3 control register 3	0x00000000
R32_USART3_GPR	0x40004818	USART3 guard time and prescaler register	0x00000000
R32_USART3_CTRL4	0x4000481C	USART3 control register 4	0x00000000

Table 17-5: USART4 related register list

Name	Address	Description	Reset Value
R32_USART4_STATR	0x40004C00	USART4 status register	0x000000C0
R32_USART4_DATAR	0x40004C04	USART4 data register	0x000000XX
R32_USART4_BRR	0x40004C08	USART4 baud rate register	0x00000000
R32_USART4_CTLR1	0x40004C0C	USART4 control register 1	0x00000000
R32_USART4_CTLR2	0x40004C10	USART4 control register 2	0x00000000
R32_USART4_CTLR3	0x40004C14	USART4 control register 3	0x00000000
R32_USART4_GPR	0x40004C18	USART4 guard time and prescaler register	0x00000000
R32_USART4_CTRL4	0x40004C1C	USART4 control register 4	0x00000000

Table 17-6: USART5 related register list

Name	Address	Description	Reset Value
R32_USART5_STATR	0x40005000	USART5 status register	0x000000C0
R32_USART5_DATAR	0x40005004	USART5 data register	0x00000000
R32_USART5_BRR	0x40005008	USART5 baud rate register	0x00000000
R32_USART5_CTLR1	0x4000500C	USART5 control register 1	0x00000000
R32_USART5_CTLR2	0x40005010	USART5 control register 2	0x00000000
R32_USART5_CTLR3	0x40005014	USART5 control register 3	0x00000000
R32_USART5_GPR	0x40005018	USART5 guard time and prescaler register	0x00000000
R32_USART5_CTRL4	0x4000501C	USART5 control register 4	0x00000000

Table 17-7: USART6 related register list

Name	Address	Description	Reset Value
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R32_USART6_STATR	0x40001800	USART6 status register	0x000000C0
R32_USART6_DATAR	0x40001804	USART6 data register	0x00000000
R32_USART6_BRR	0x40001808	USART6 baud rate register	0x00000000
R32_USART6_CTLR1	0x4000180C	USART6 control register 1	0x00000000
R32_USART6_CTLR2	0x40001810	USART6 control register 2	0x00000000
R32_USART6_CTLR3	0x40001814	USART6 control register 3	0x00000000
R32_USART6_GPR	0x40001818	USART6 guard time and prescaler register	0x00000000
R32_USART6_CTRL4	0x4000181C	USART6 control register 4	0x00000000

Table 17-8: USART7 related register list

Name	Address	Description	Reset Value
R32_USART7_STATR	0x40001C00	USART7 status register	0x000000C0
R32_USART7_DATAR	0x40001C04	USART7 data register	0x00000000
R32_USART7_BRR	0x40001C08	USART7 baud rate register	0x00000000
R32_USART7_CTLR1	0x40001C0C	USART7 control register 1	0x00000000
R32_USART7_CTLR2	0x40001C10	USART7 control register 2	0x00000000
R32_USART7_CTLR3	0x40001C14	USART7 control register 3	0x00000000
R32_USART7_GPR	0x40001C18	USART7 guard time and prescaler register	0x00000000
R32_USART7_CTRL4	0x40001C1C	USART7 control register 4	0x00000000

Table 17-9: USART8 related register list

Name	Address	Description	Reset Value
R32_USART8_STATR	0x40002000	USART8 status register	0x000000C0
R32_USART8_DATAR	0x40002004	USART8 data register	0x00000000
R32_USART8_BRR	0x40002008	USART8 baud rate register	0x00000000
R32_USART8_CTLR1	0x4000200C	USART8 control register 1	0x00000000
R32_USART8_CTLR2	0x40002010	USART8 control register 2	0x00000000
R32_USART8_CTLR3	0x40002014	USART8 control register 3	0x00000000
R32_USART8_GPR	0x40002018	USART8 guard time and prescaler register	0x00000000
R32_USART8_CTRL4	0x4000201C	USART8 control register 4	0x00000000

Table 17-10: USART9 related register list

Name	Address	Description	Reset Value
R32_USART9_STATR	0x40007800	USART9 status register	0x000000C0
R32_USART9_DATAR	0x40007804	USART9 data register	0x00000000
R32_USART9_BRR	0x40007808	USART9 baud rate register	0x00000000
R32_USART9_CTLR1	0x4000780C	USART9 control register 1	0x00000000
R32_USART9_CTLR2	0x40007810	USART9 control register 2	0x00000000
R32_USART9_CTLR3	0x40007814	USART9 control register 3	0x00000000
R32_USART9_GPR	0x40007818	USART9 guard time and prescaler register	0x00000000
R32_USART9_CTRL4	0x4000781C	USART9 control register 4	0x00000000

Table 17-11: USART10 related register list

Name	Address	Description	Reset Value
R32_USART10_STATR	0x40007C00	USART10 status register	0x000000C0
R32_USART10_DATAR	0x40007C04	USART10 data register	0x00000000
R32_USART10_BRR	0x40007C08	USART10 baud rate register	0x00000000
R32_USART10_CTLR1	0x40007C0C	USART10 control register 1	0x00000000
R32_USART10_CTLR2	0x40007C10	USART10 control register 2	0x00000000
R32_USART10_CTLR3	0x40007C14	USART10 control register 3	0x00000000
R32_USART10_GPR	0x40007C18	USART10 guard time and prescaler register	0x00000000
R32_USART10_CTRL4	0x40007C1C	USART10 control register 4	0x00000000

17.10.1 USART Status Register (USARTx_STATR) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		MS_ERR		RX_BUSY		CTS	LBD	TXE	TC	RXNE	IDLE	ORE	NE	FE	PE

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
11	MS_ERR	RO	MARK or SPACE parity error flag. In receive mode, if a MARK or SPACE parity error occurs, hardware sets this bit. Reading this bit followed by reading the data register resets this bit. If MS_ERRIE was previously set, setting this bit generates a corresponding interrupt.	0
10	RX_BUSY	RO	Receive status indication bit: 1: This bit is set when operating in receive state; 0: Not in receive state.	0
9	CTS	RW0	CTS status change flag. If the CTSE bit is set, this bit is set by hardware when the nCTS output state changes. Cleared by software. If the CTSIE bit is already set, an interrupt is generated. 1: A change is detected on the nCTS status line; 0: No change on the nCTS status line.	0
8	LBD	RW0	LIN break detection flag. When a LIN break is detected, this bit is set by hardware. Cleared by software. If LBDIE is already set, an interrupt is generated. 1: LIN break detected; 0: No LIN break detected.	0
7	TXE	RO	Transmit data register empty flag. When the data in the TDR register is transferred to the shift register by hardware, this bit is set by hardware. If TXEIE is already set, an interrupt is generated. Writing to the data register resets this bit. 1: Data has been transferred to the shift register; 0: Data has not been transferred to the shift register.	1
6	TC	RW0	Transmission complete flag. When a frame containing data has been sent and TXE is set, hardware sets this bit. If TCIE is set, a corresponding	1

			interrupt is generated. Reading this bit followed by writing to the data register clears this bit. It can also be cleared by writing 0 directly. 1: Transmission complete; 0: Transmission not yet complete.	
5	RXNE	RW0	Read data register not empty flag. When data from the shift register is transferred to the data register, this bit is set by hardware. If RXNEIE is already set, a corresponding interrupt is generated. Reading the data register clears this bit. It can also be cleared by writing 0 directly. 1: Data received and can be read; 0: Data not yet received.	0
4	IDLE	RO	Bus idle flag. When the bus is idle, this bit is set by hardware. If IDLEIE is already set, a corresponding interrupt is generated. Reading the status register followed by reading the data register clears this bit. 1: Bus is idle; 0: No bus idle detected. Note: This bit will not be set again until RXNE is set.	0
3	ORE	RO	Overrun error flag. When the receive shift register has data to transfer to the data register but the receive field of the data register still has unread data, this bit is set. If RXNEIE is set, a corresponding interrupt is generated. 1: Overrun error occurred; 0: No overrun error. Note: When an overrun error occurs, the value in the data register is not lost, but the value in the shift register is overwritten. If the EIE bit is set, in multi-buffer communication mode, setting the ORE flag generates an interrupt.	0
2	NE	RO	Noise error flag. When a noise error is detected, this bit is set by hardware. Reading the status register followed by reading the data register resets this bit. 1: Noise detected; 0: No noise detected. Note: This bit does not generate an interrupt. If the EIE bit is set, in multi-buffer communication mode, setting the NE flag generates an interrupt.	0
1	FE	RO	Frame error flag. When a synchronization error, excessive noise, or a break character is detected, this bit is set by hardware. Reading this bit followed by reading the data register resets this bit. 1: Frame error detected; 0: No frame error detected. Note: This bit does not generate an interrupt. If the EIE bit is set, in multi-buffer communication mode, setting the FE flag generates an interrupt.	0
0	PE	RO	Parity error flag. In receive mode, if a parity error occurs, hardware sets this bit. Reading this bit followed by reading the data register resets this bit. Before clearing this bit, software must wait for the RXNE flag to be set. If PEIE was previously set, setting this bit generates a corresponding interrupt. 1: Parity error occurred; 0: No parity error.	0

17.10.2 USART Data Register (USARTx_DATAR) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DR[8:0]							

Bits	Name	Access	Description	Reset
[31:9]	Reserved	RO	Reserved.	0
[8:0]	DR[8:0]	RW	Data register. This register is actually composed of two registers: the receive data register (RDR) and the transmit data register (TDR). Read and write operations on DR are effectively reading the RDR and writing the TDR, respectively.	X

17.10.3 USART Baud Rate Register (USARTx_BRR) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV_Mantissa[11:0]												DIV_Fraction[3:0]			

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:4]	DIV_Mantissa[11:0]	RW	These 12 bits define the integer part of the divider factor.	0
[3:0]	DIV_Fraction[3:0]	RW	These 4 bits define the fractional part of the divider factor.	0

17.10.4 USART Control Register 1 (USARTx_CTLR1) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
M_EXT[1:0]		UE	M	WAKE	PCE	PS	PEIE	TXEIE	TCIE	RXNEIE	IDLEIE	TE	RE	RWU	SBK

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:14]	M_EXT[1:0]	RW	Word length extension bits: 00: Invalid, the M bit determines the data length. 01: 7 data bits; 10: 6 data bits; 11: 5 data bits.	0
13	UE	RW	USART enable bit. When this bit is cleared, after the current byte transmission completes, the USART divider and outputs stop working.	0
12	M	RW	Word length bit. 1: 9 data bits; 0: 8 data bits.	0
11	WAKE	RW		0

			Wakeup bit. This bit determines the method for waking up the USART: 1: Address mark; 0: Bus idle.	
10	PCE	RW	Parity control enable. 1: Parity control enabled; 0: Parity control disabled. For the receiver, it performs parity checking on data; for the transmitter, it inserts the parity bit. Once this bit is set, parity control only takes effect after the current byte transmission completes.	0
9	PS	RW	Parity selection: 1: Odd parity; 0: Even parity. After setting this bit, parity control only takes effect after the current byte transmission completes.	0
8	PEIE	RW	Parity error interrupt enable bit. 1: Parity error interrupt enabled; 0: Parity error interrupt disabled.	0
7	TXEIE	RW	Transmit buffer empty interrupt enable: 1: Transmit buffer empty interrupt enabled; 0: Transmit buffer empty interrupt disabled.	0
6	TCIE	RW	Transmission complete interrupt enable: 1: Transmission complete interrupt enabled; 0: Transmission complete interrupt disabled.	0
5	RXNEIE	RW	Receive buffer not empty interrupt enable: 1: Receive buffer not empty interrupt enabled; 0: Receive buffer not empty interrupt disabled.	0
4	IDLEIE	RW	Bus idle interrupt enable: 1: Bus idle interrupt enabled; 0: Bus idle interrupt disabled.	0
3	TE	RW	Transmit enable. 1: Transmitter enabled; 0: Transmitter disabled.	0
2	RE	RW	Receive enable. 1: Receiver enabled, the receiver begins detecting start bits on the RX pin; 0: Receiver disabled.	0
1	RWU	RW	Receive wakeup. This bit determines whether to place the USART in mute mode: 1: Receiver in mute mode; 0: Receiver in normal operating mode. Note 1: Before setting the RWU bit, the USART must first receive a data byte; otherwise, in mute mode, it cannot be woken up by bus idle. Note 2: When configured for address mark wakeup, the RWU bit cannot be modified by software while RXNE is set.	0
0	SBK	RW	Send break character control bit. Set this bit to send a break character. It is reset by hardware during the stop bit of the break frame. 1: Send; 0: Do not send.	0

17.10.5 USART Control Register 2 (USARTx_CTLR2) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	LINEN	STOP	CLKEN	CPOL	CPHA	LBCL	Reserved	LBDIE	LBDL	Reserved	ADD[3:0]				

Bits	Name	Access	Description	Reset
[31:15]	Reserved	RO	Reserved.	0
14	LINEN	RW	LIN mode enable bit. Setting this bit enables LIN mode. In LIN mode, the SBK bit can be used to send LIN break characters and to detect LIN break characters.	0
[13:12]	STOP	RW	Stop bit setting field. These two bits set the stop bits. 00: 1 stop bit; 01: 0.5 stop bits; 10: 2 stop bits; 11: 1.5 stop bits.	00b
11	CLKEN	RW	Clock enable, enables the CK pin. 1: Enabled; 0: Disabled.	0
10	CPOL	RW	Clock polarity setting bit. In synchronous mode, this bit selects the polarity of the clock output on the SCLK pin, combined with CPHA to produce the required clock/data sampling relationship. 1: CK pin stays high when bus is idle; 0: CK pin stays low when bus is idle. Note: This bit cannot be modified after the transmitter is enabled.	0
9	CPHA	RW	Clock phase setting bit. In synchronous mode, this bit selects the phase of the clock output on the SCLK pin, combined with CPOL to produce the required clock/data sampling relationship. 1: Data capture on the second edge of the clock; 0: Data capture on the first edge of the clock. Note: This bit cannot be modified after the transmitter is enabled.	0
8	LBCL	RW	Last clock pulse control bit. In synchronous mode, this bit controls whether the clock pulse corresponding to the last transmitted data byte is output on the CK pin. 1: The clock pulse of the last data bit is not output on CK; 0: The clock pulse of the last data bit is output on CK. Note: This bit cannot be modified after the transmitter is enabled.	0
7	Reserved	RW	Reserved.	0
6	LBDIE	RW	LIN break detection interrupt enable. Setting this bit enables the interrupt caused by LBD.	0
5	LBDL	RW	LIN break detection length. This bit selects whether break detection is 11-bit or 10-bit. 1: 11-bit break detection; 0: 10-bit break detection.	0
4	Reserved	RW	Reserved.	0

[3:0]	ADD[3:0]	RW	Address field, used to set the USART node address of this device. Used in mute mode during multi-processor communication to wake up a specific USART device using an address mark.	0
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17.10.6 USART Control Register 3 (USARTx_CTLR3) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					CTSIE	CTSE	RTSE	DMAT	DMAR	SCEN	NACK	HDSEL	IRLP	IREN	EIE

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
10	CTSIE	RW	CTSIE interrupt enable bit: 1: An interrupt is generated when CTS is set; 0: No interrupt is generated when CTS is set.	0
9	CTSE	RW	CTS enable bit: 1: CTS flow control enabled; 0: CTS flow control disabled.	0
8	RTSE	RW	RTS enable bit: 1: RTS flow control enabled; 0: RTS flow control disabled.	0
7	DMAT	RW	DMA transmit enable bit: 1: DMA is used during transmission; 0: DMA is not used during transmission.	0
6	DMAR	RW	DMA receive enable bit: 1: DMA is used during reception; 0: DMA is not used during reception.	0
5	SCEN	RW	Smart card mode enable bit: 1: Smart card mode enabled; 0: Smart card mode disabled.	0
4	NACK	RW	Smart card NACK enable bit: 1: NACK is sent when a parity error occurs; 0: NACK is not sent when a parity error occurs.	0
3	HDSEL	RW	Half-duplex mode selection bit: 1: Half-duplex mode enabled; 0: Half-duplex mode disabled.	0
2	IRLP	RW	Infrared low-power selection bit: 1: Low-power mode enabled when IrDA is selected; 0: Low-power mode disabled when IrDA is selected.	0
1	IREN	RW	Infrared enable bit: 1: IrDA mode enabled; 0: IrDA mode disabled.	0
0	EIE	RW	Error interrupt enable bit: 1: Error interrupt enabled; 0: Error interrupt disabled.	0

		When this bit is set, if DMAR is set and FE, ORE, or NE is set, an interrupt is generated.	
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17.10.7 USART Guard Time and Prescaler Register (USARTx_GPR) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GT[7:0]								PSC[7:0]							

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:8]	GT[7:0]	RW	Guard time value field. This field specifies the guard time in units of baud rate clock. In smart card mode, the transmission complete flag is only set after the guard time has elapsed.	0
[7:0]	PSC[7:0]	RW	Prescaler value field. In IrDA low-power mode, the source clock is divided by this value (all 8 bits valid); a value of 0 means reserved. In IrDA normal mode, this bit can only be set to 1. In smart card mode, the source clock is divided by twice this value (lower 5 bits valid) to provide the clock to the smart card; a value of 0 means reserved.	0

17.10.8 USART Control Register 4 (USARTx_CTLR4) (x=1/2/3/4/5/6/7/8/9/10)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								CHECK_SEL[1:0]				MS_ERRIE		Reserved	

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
[3:2]	CHECK_SEL[1:0]	RW	Check function selection bits: 0x: MARK and SPACE check disabled. 10: MARK check enabled. 11: SPACE check enabled.	0
1	MS_ERRIE	RW	SPACE or MARK check error interrupt enable bit: 1: SPACE or MARK check error interrupt enabled; 0: SPACE or MARK check error interrupt disabled.	0
0	Reserved	RO	Reserved.	0

Chapter 18: Two-Wire Communication Bus (I2C)

The Inter-Integrated Circuit bus (I2C) is widely used for communication between microcontrollers and sensors or other off-chip modules. It natively supports multi-master and multi-slave modes, and can communicate at two speeds — 100 kHz (standard) and 400 kHz (fast) — using only two lines (SDA and SCL). The I2C bus is also compatible with the SMBus protocol, supporting not only I2C timing but also arbitration, timing, and DMA, and provides a CRC checksum function.

18.1 Main Features

- Supports master mode and slave mode
- Supports 7-bit or 10-bit addressing
- Slave device supports dual 7-bit address
- Supports two speed modes: 100 kHz and 400 kHz
- Multiple status modes, multiple error flags
- Supports clock stretching function
- 2 interrupt vectors
- Supports DMA
- Supports PEC
- SMBus compatible

18.2 Overview

I2C is a half-duplex bus; it can only operate in one of the following four modes at a time: master transmitter mode, master receiver mode, slave transmitter mode, and slave receiver mode. The I2C module operates in slave mode by default. After generating a START condition, it automatically switches to master mode; when arbitration is lost or a STOP condition is generated, it switches back to slave mode. The I2C module supports multi-master functionality.

In master mode, the I2C module actively sends out data and addresses. Both data and addresses are transmitted in 8-bit units, MSB first and LSB last. Following a START event, one byte (in 7-bit addressing mode) or two bytes (in 10-bit addressing mode) of address is sent. After every 8 bits of data or address transmitted by the master, the slave must reply with an acknowledge (ACK) by pulling the SDA line low, as shown in Figure 18-1.

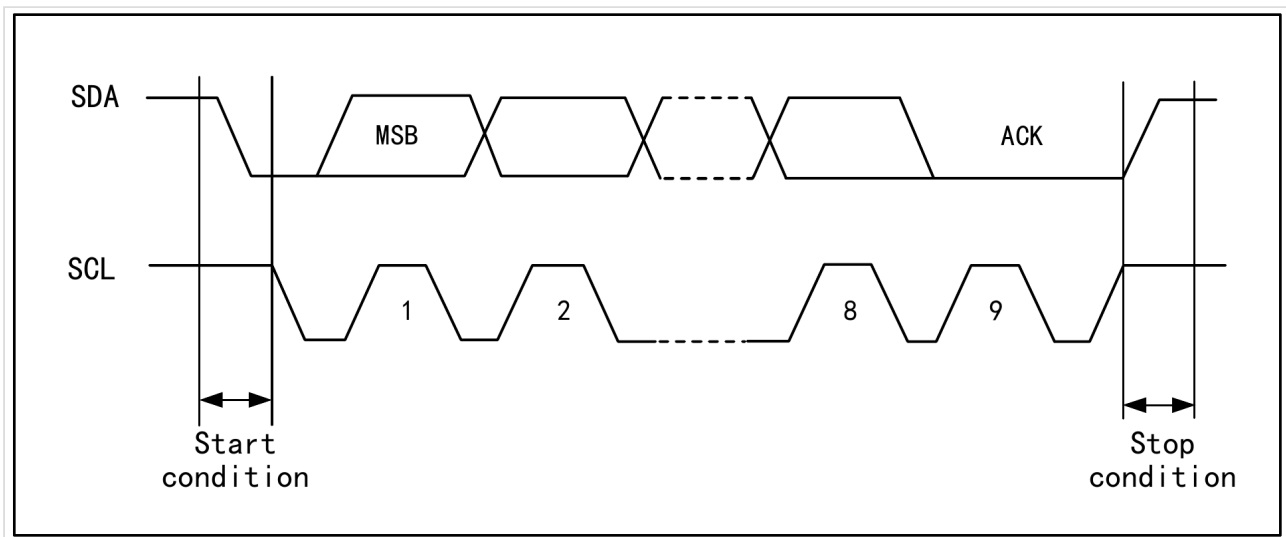


Figure 18-1: I2C Timing Diagram

To operate correctly, the I2C must be provided with a proper input clock. In standard mode, the minimum input clock is 2 MHz; in fast mode, the minimum input clock is 4 MHz.

Figure 18-2 shows the functional block diagram of the I2C module.

图 18-2 I2C 功能框图

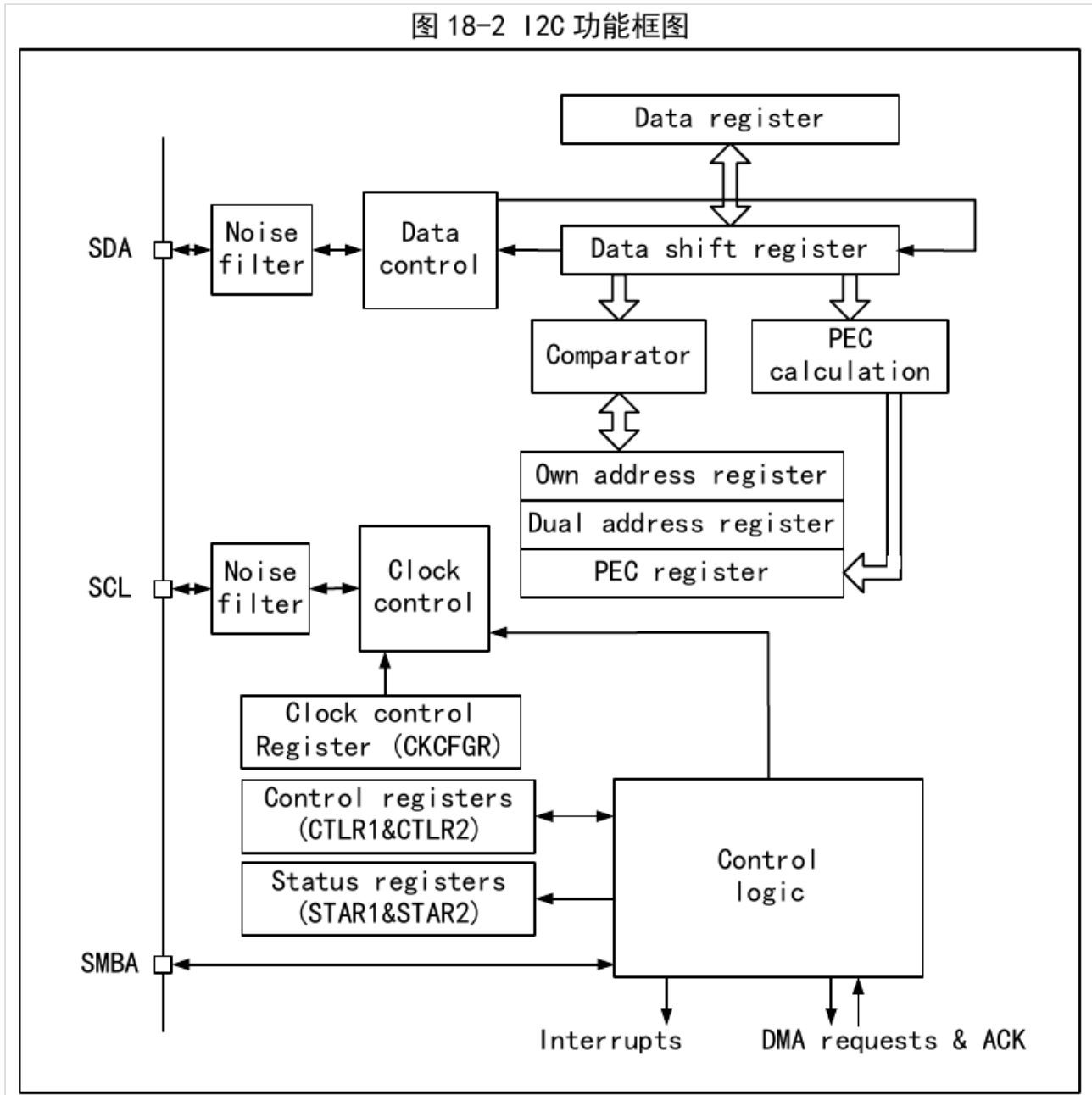


Figure 18-2: I2C Functional Block Diagram

18.3 Master Mode

In master mode, the I2C module initiates data transfer and outputs the clock signal. Data transfer begins with a START event and ends with a STOP event. The steps for master-mode communication are as follows:

- Set the correct clock in Control Register 2 (R16_I2C_CTLR2) and the Clock Control Register (R16_I2C_CKCFGR); set an appropriate rise time in the Rise Time Register (R16_I2C_RTR).
- Set the PE bit in Control Register 1 (R16_I2C_CTLR1) to enable the peripheral.
- Set the START bit in Control Register 1 (R16_I2C_CTLR1) to generate a START event. After setting the START bit, the I2C module automatically switches to master mode, the MSL bit is set, and a START event is generated. After the START event is generated, the SB bit is set; if the ITEVTEN bit (in R16_I2C_CTLR2) is set, an interrupt is generated. At this point, read Status Register 1

(R16_I2C_STAR1), then write the slave address to the data register; the SB bit is cleared automatically.

- If 10-bit addressing mode is used, write the data register to send the header sequence (the header sequence is 11110xx0b, where xx represents the upper 2 bits of the 10-bit address).
- After sending the header sequence, the ADD10 bit in the status register is set. If the ITEVTEN bit is set, an interrupt is generated. At this point, read R16_I2C_STAR1, then write the second address byte to the data register to clear the ADD10 bit.
- Then write the data register to send the second address byte. After sending the second address byte, the ADDR bit in the status register is set. If the ITEVTEN bit is set, an interrupt is generated. At this point, read R16_I2C_STAR1 and then read R16_I2C_STAR2 to clear the ADDR bit.
- If 7-bit addressing mode is used, write the data register to send the address byte. After sending the address byte, the ADDR bit in the status register is set. If the ITEVTEN bit is set, an interrupt is generated. At this point, read R16_I2C_STAR1 and then read R16_I2C_STAR2 to clear the ADDR bit.

In 7-bit addressing mode, the first byte transmitted is the address byte: the upper 7 bits represent the target slave device address, and the 8th bit determines the direction of the subsequent message — 0 means the master writes data to the slave, and 1 means the master reads information from the slave.

In 10-bit addressing mode, as shown in Figure 18-3, during the address phase the first byte is 11110xx0 (xx is the upper 2 bits of the 10-bit address), and the second byte is the lower 8 bits of the 10-bit address. If the master subsequently enters transmitter mode, it continues sending data; if it prepares to enter receiver mode, it must re-issue a START condition followed by a byte 11110xx1, and then enter master receiver mode.

图 18-3 10 位地址时主机收发数据示意图

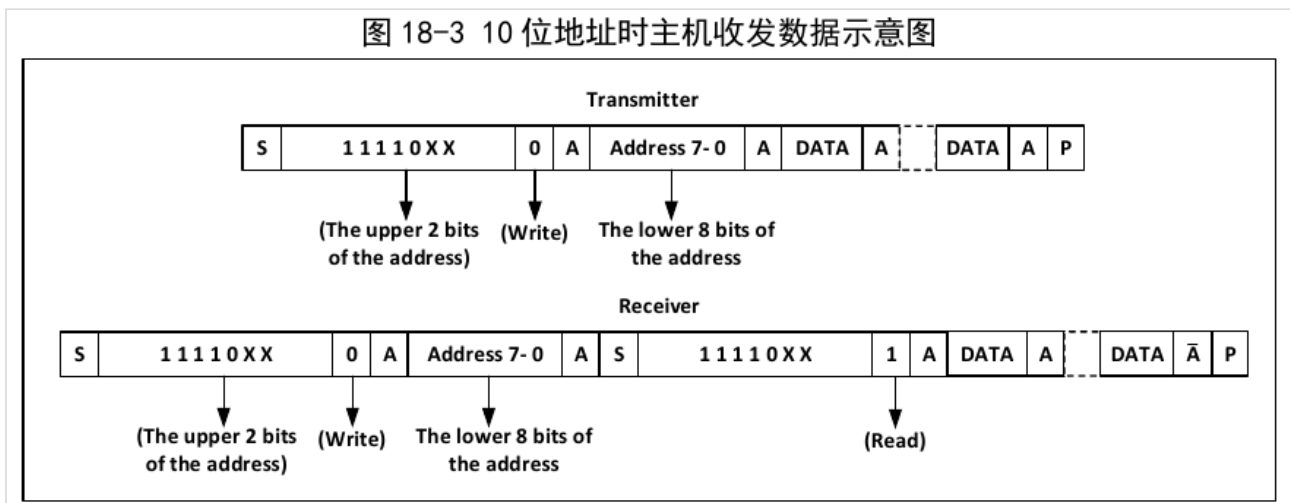


Figure 18-3: Master Transmit/Receive Data Diagram for 10-bit Addressing

In transmitter mode, the master's internal shift register sends data from the data register onto the SDA line. When the master receives an ACK, the TxE bit in Status Register 1 (R16_I2C_STAR1) is set. If ITEVTEN and ITBUFEN are set, an interrupt is also generated. Writing data to the data register clears the TxE bit. If the TxE bit is set and no new data has been written to the data register before the last data transmission, the BTF bit is set; SCL is held low until it is cleared. Reading R16_I2C_STAR1 and then writing data to the data register clears the BTF bit.

In receiver mode, the I2C module receives data from the SDA line and writes it into the data register through the shift register. After each byte, if the ACK bit is set, the I2C module sends an acknowledge (low level), and the RxNE bit is set. If ITEVTEN and ITBUFEN are set, an interrupt is also generated. If

RxNE is set and the previous data has not been read before new data is received, the BTF bit is set; SCL is held low until BTF is cleared. Reading R16_I2C_STAR1 and then reading the data register clears the BTF bit.

When the master finishes transmitting data, it actively generates a STOP event by setting the STOP bit. In receiver mode, the master must NAK the acknowledge position of the last data byte. Note that after generating a NAK, the I2C module switches to slave mode.

18.4 Slave Mode

In slave mode, the I2C module can recognize its own address and the general call address. Software can enable or disable recognition of the general call address. Once a START event is detected, the I2C module compares the SDA data (via the shift register) with its own address (bit count depends on ENDUAL and ADDMODE) or the general call address (when ENGCG is set). If there is no match, it ignores the data until a new START event is generated. If the header sequence matches, an ACK is generated and the module waits for the second address byte; if the second address byte also matches, or in 7-bit addressing the full address matches, then:

- First, an ACK is generated; the ADDR bit is set. If the ITEVTEN bit is set, a corresponding interrupt is generated.
- If dual address mode is used (ENDUAL bit set), the DUALF bit must also be read to determine which address the master is addressing.

The slave mode defaults to receiver mode. When the last bit of the received header sequence is 1, or the last bit of a 7-bit address is 1 (depending on whether a header sequence or a normal 7-bit address was received first), the I2C module enters transmitter mode; the TRA bit indicates whether the current mode is receiver or transmitter.

In transmitter mode, after the ADDR bit is cleared, the I2C module sends bytes from the data register to the SDA line through the shift register. After receiving an ACK, the TxE bit is set. If ITEVTEN and ITBUFEN are set, an interrupt is generated. If TxE is set but no new data is written to the data register before the next data transmission completes, the BTF bit is set. SCL is held low until BTF is cleared. Reading Status Register 1 (R16_I2C_STAR1) and then writing data to the data register clears the BTF bit.

In receiver mode, after ADDR is cleared, the I2C module stores SDA data into the data register through the shift register. After each received byte, the I2C module sets an ACK bit and sets the RxNE bit. If ITEVTEN and ITBUFEN are set, an interrupt is generated. If RxNE is set and the old data has not been read before new data is received, BTF is set. SCL is held low until BTF is cleared. Reading Status Register 1 (R16_I2C_STAR1) and reading the data in the data register clears the BTF bit.

When the I2C module detects a STOP event, it sets the STOPF bit. If ITEVFEN is set, an interrupt is generated. The user must read the status register (R16_I2C_STAR1) and then write to the control register (e.g., reset the control word SWRST) to clear it.

18.5 Errors

18.5.1 Bus Error (BERR)

During transmission of an address or data, when the I2C module detects an external START or STOP event, a bus error is generated. When a bus error occurs, the BERR bit is set; if ITERREN is set, an

interrupt is also generated. In slave mode, the data is discarded and the hardware releases the bus. If it was a START signal, the hardware treats it as a repeated START and begins waiting for an address or STOP signal; if it was a STOP signal, it operates as a normal stop condition in advance. In master mode, the hardware does not release the bus and the current transfer is not affected; the user code decides whether to abort the transfer.

18.5.2 Acknowledge Failure (AF)

When the I2C module detects that no acknowledge follows a byte, an acknowledge error is generated. When an acknowledge error occurs: the AF bit is set; if ITERREN is set, an interrupt is also generated. Upon an AF error, if the I2C module is in slave mode, the hardware must release the bus; if in master mode, the software must generate a STOP event.

18.5.3 Arbitration Lost (ARLO)

When the I2C module detects arbitration loss, an arbitration lost error is generated. When an arbitration lost error occurs: the ARLO bit is set; if ITERREN is set, an interrupt is also generated. The I2C module switches to slave mode and no longer responds to transfers addressed to its own slave address unless a master initiates a new START event; the hardware releases the bus.

18.5.4 Overrun/Underrun Error (OVR)

Overrun error: In slave mode, if clock stretching is disabled and the I2C module is receiving data, if a new byte of data has been received but the previously received data has not yet been read, an overrun error occurs. When an overrun error occurs, the last received byte is discarded and the sender should retransmit the last byte sent.

Underrun error: In slave mode, if clock stretching is disabled and the I2C module is transmitting data, if new data has not been written to the data register before the clock of the next byte arrives, an underrun error occurs. When an underrun error occurs, the previous data in the data register is sent twice; the receiver should discard the duplicate data. To avoid underrun errors, the I2C module should write data to the data register before the first rising edge of the next byte.

18.6 Clock Stretching

If clock stretching is disabled, the possibility of overrun/underrun errors exists. However, if clock stretching is enabled:

- In transmitter mode, if TxE is set and BTF is set, SCL is held low until the user reads the status register and writes the data to be transmitted to the data register.
- In receiver mode, if RxNE is set and BTF is set, SCL is held low after receiving data until the user reads the status register and reads the data register.

Thus, enabling clock stretching avoids overrun/underrun errors.

18.7 SMBus

SMBus is also a two-wire interface, typically used between system and power management. SMBus and I2C share many similarities; for example, SMBus uses the same 7-bit addressing mode as I2C. The following are common features of SMBus and I2C:

1. Master-slave communication mode; the master provides the clock and multi-master/multi-slave is supported;
2. Two-wire communication structure, where SMBus can optionally have an alert line;
3. Both support 7-bit address format.

SMBus and I2C also differ:

1. I2C supports speeds up to 400 kHz, while SMBus supports up to 100 kHz and has a minimum speed limit of 10 kHz;
2. When the SMBus clock is low for more than 35 ms, a timeout is reported, but I2C has no such limit;
3. SMBus has fixed logic levels, whereas I2C does not — it depends on VDD;
4. SMBus has bus protocols, whereas I2C does not.

SMBus also includes device identification, address resolution protocol, unique device identifiers, SMBus alert, and various bus protocols. Refer to the SMBus specification version 2.0 for details. When using SMBus, simply set the SMBUS bit in the control register, and configure the SMBTYPE and ENAARP bits as needed.

18.8 Interrupts

Each I2C module has two interrupt vectors: the event interrupt and the error interrupt. Both interrupt types support the interrupt sources shown in Figure 18-4.

图 18-4 I2C 中断请求

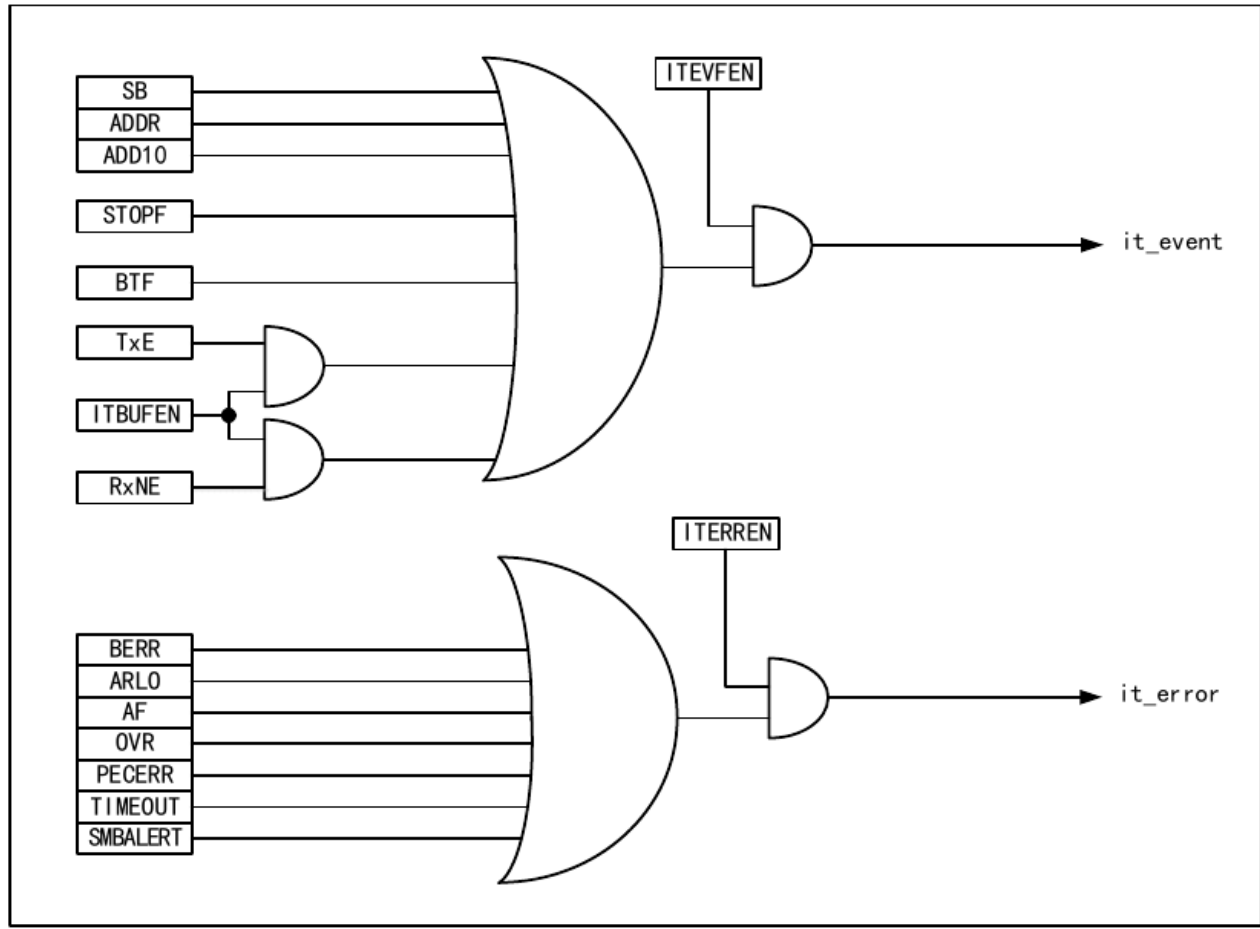


Figure 18-4: I2C Interrupt Requests

18.9 DMA

DMA can be used for bulk data transmission and reception. When using DMA, the ITBUFEN bit in the control register must not be set.

DMA Transmission

DMA mode is activated by setting the DMAEN bit in the CTLR2 register. Whenever the TxNE bit is set, data is loaded from the configured memory into the I2C data register by the DMA. The following settings are required to allocate a channel for the I2C:

1. Set the I2C_DATAR register address in the DMAy_PADDRx register, and set the memory address in the DMAy_MADDRx register, so that after each TxNE event, data is transferred from memory to the I2C_DATAR register.
2. Set the desired number of bytes to transfer in the DMAy_CNTRx register. This value is decremented after each TxNE event.
3. Configure the channel priority using the PL[0:1] bits in the DMAy_CFGRx register.
4. Set the DIR bit in the DMAy_CFGRx register, and optionally configure an interrupt request to be issued when half or all of the transfer is complete, according to application requirements.
5. Activate the channel by setting the EN bit in the DMAy_CFGRx register.

When the number of data transfers configured in the DMA controller has been completed, the DMA controller sends an end-of-transfer EOT/EOT_1 signal to the I2C interface. If interrupts are enabled, a DMA interrupt is generated.

DMA Reception

After setting the DMAEN bit in the CTLR2 register, DMA receive mode can be used. When using DMA reception, the DMA transfers data from the data register to a preset memory area. The following steps are required to allocate a channel for the I2C:

1. Set the I2C_DATAR register address in the DMAy_PADDRx register, and set the memory address in the DMAy_MADDRx register, so that after each RxNE event, data is written from the I2C_DATAR register to memory.
2. Set the desired number of bytes to transfer in the DMAy_CNTRx register. This value is decremented after each RxNE event.
3. Configure the channel priority using the PL[0:1] bits in the DMAy_CFGRx register.
4. Clear the DIR bit in the DMAy_CFGRx register, and optionally configure an interrupt request to be issued when half or all of the data transfer is complete, according to application requirements.
5. Activate the channel by setting the EN bit in the DMAy_CFGRx register.

When the number of data transfers configured in the DMA controller has been completed, the DMA controller sends an end-of-transfer EOT/EOT_1 signal to the I2C interface. If interrupts are enabled, a DMA interrupt is generated.

18.10 Packet Error Checking

Packet Error Checking (PEC) adds a CRC8 checksum step to improve transmission reliability. The following polynomial is used to calculate each serial data bit:

$$C = X^8 + X^2 + X + 1$$

PEC calculation is activated by the ENPEC bit in the control register and is computed over all message bytes, including the address and read/write bit. In transmission, when PEC is enabled, a one-byte CRC8 result is appended after the last data byte. In receive mode, the last byte is treated as the CRC8 checksum result; if it does not match the internally computed result, a NAK is returned. For a master receiver, a NAK is returned regardless of whether the checksum is correct or not.

18.11 Debug Mode

After the system enters debug mode, the DBG_I2C_SMBUS_TIMEOUT bit of the DEBUG module can be used to determine whether the I2C SMBus timeout control continues to operate or stops.

18.12 Register Description

Table 18-1: I2C Register List

Name	Address	Description	Reset Value
R16_I2C_CTLR1	0x40005400	I2C Control Register 1	0x0000

Name	Address	Description	Reset Value
R16_I2C_CTLR2	0x40005404	I2C Control Register 2	0x0000
R16_I2C_OADDR1	0x40005408	I2C Address Register 1	0x0000
R16_I2C_OADDR2	0x4000540C	I2C Address Register 2	0x0000
R16_I2C_DATAR	0x40005410	I2C Data Register	0x0000
R16_I2C_STAR1	0x40005414	I2C Status Register 1	0x0000
R16_I2C_STAR2	0x40005418	I2C Status Register 2	0x0000
R16_I2C_CKCFGR	0x4000541C	I2C Clock Register	0x0000
R16_I2C_RTR	0x40005420	I2C Rise Time Register	0x0002

18.12.1 I2C Control Register 1 (I2C_CTLR1)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWRST	Reserved	ALERT	PEC	POS	ACK	STOP	START	NOSTRETCH	ENGCG	ENPEC	ENARP	SMBTYPE	Reserved	SMBUS	PE

Bits	Name	Access	Description	Reset
15	SWRST	RW	Software reset. Setting this bit by user code resets the I2C peripheral. Before resetting, ensure the I2C bus pins are released and the bus is idle. Note: This bit can be used to reset the I2C module when no STOP condition is detected on the bus but the BUSY bit is 1.	0
14	Reserved	RO	Reserved.	0
13	ALERT	RW	SMBus alert bit. User code can set or clear this bit; when PE is set, this bit can be cleared by hardware. 1: Drives the SMBusALERT pin low, and the response address header should follow the ACK signal; 0: Releases the SMBusALERT pin high, and the response address header should follow the NACK signal.	0
12	PEC	RW	Packet error checking enable bit. Setting this bit enables PEC. User code can set or clear this bit; it is cleared by hardware after PEC is transferred, a START or STOP is generated, or PE is cleared. 1: With PEC; 0: Without PEC. Note: PEC is invalidated upon arbitration loss.	0
11	POS	RW	ACK and PEC position control. This bit can be set or cleared by user code, and can be cleared by hardware when PE is cleared. 1: The ACK bit controls the ACK/NAK of the next byte received in the shift register; the PEC indicates that the next byte in the shift register is PEC. 0: The ACK bit controls the ACK/NAK of the byte currently being received in the shift register; the PEC bit indicates that the current byte in the shift register is PEC. Note: The usage of the POS bit in 2-byte data reception is as follows: It must be configured before reception. To NACK the 2nd byte, the ACK bit must be cleared immediately after clearing the ADDR bit; to	0

Bits	Name	Access	Description	Reset
			detect the PEC of the second byte, the PEC bit must be set after configuring the POS bit following the ADDR event.	
10	ACK	RW	Acknowledge enable bit. Can be set or cleared by user code; can be cleared by hardware when PE is set. 1: Returns an acknowledge after receiving a byte; 0: No acknowledge.	0
9	STOP	RW	STOP event generation bit. Can be set or cleared by user code, cleared by hardware when a STOP event is detected, or set by hardware when a timeout error is detected. Master mode: 1: Generates a STOP event after the current byte transfer or current START condition; 0: No STOP event generated. Slave mode: 1: Releases SCL and SDA lines after the current byte transfer; 0: No STOP event generated.	0
8	START	RW	START event generation bit. Can be set or cleared by user code, cleared by hardware when a START condition is issued or PE is cleared. Master mode: 1: Repeatedly generates a START event; 0: No START event generated. Slave mode: 1: Generates a START event when the bus is idle; 0: No START event generated.	0
7	NOSTRETCH	RW	Clock stretching disable bit. Used to disable clock stretching in slave mode when the ADDR or BTF flag is set, until cleared by software. 1: Disable clock stretching; 0: Enable clock stretching.	0
6	ENGCG	RW	General call enable bit. Setting this bit enables general call, acknowledging the general call address 00h.	0
5	ENPEC	RW	PEC enable bit. Setting this bit enables PEC calculation.	0
4	ENARP	RW	ARP enable bit. Setting this bit enables ARP. If SMBTYPE=0, the default address of the SMBus device is used; if SMBTYPE=1, the SMBus master address is used.	0
3	SMBTYPE	RW	SMBus device type. Set to 1 for SMBus master, set to 0 for SMBus slave.	0
2	Reserved	RO	Reserved.	0
1	SMBUS	RW	SMBus mode select. Set to 1 to use SMBus mode; set to 0 to use I2C mode.	0
0	PE	RW	I2C peripheral enable bit. 1: Enable I2C module; 0: Disable I2C module.	0

18.12.2 I2C Control Register 2 (I2C_CTLR2)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	LAST	DMAEN	ITBUFEN	ITEVTEN	ITERREN	Reserved	Reserved	FREQ[5]	FREQ[4]	FREQ[3]	FREQ[2]	FREQ[1]	FREQ[0]

Bits	Name	Access	Description	Reset
[15:13]	Reserved	RO	Reserved.	0
12	LAST	RW	DMA last transfer bit. 1: The next DMA EOT is the last transfer; 0: The next DMA EOT is not the last transfer. Note: This bit is used in master receiver mode to generate a NAK on the last received data byte.	0

Bits	Name	Access	Description	Reset
11	DMAEN	RW	DMA request enable bit. Setting this bit allows DMA requests when TxE or RxNE is set.	0
10	ITBUFEN	RW	Buffer interrupt enable bit. 1: Generates an event interrupt when TxE or RxNE is set; 0: No interrupt when TxE or RxNE is set.	0
9	ITEVTEN	RW	Event interrupt enable bit. Setting this bit enables event interrupts. An interrupt is generated under the following conditions: SB=1 (master mode); ADDR=1 (master/slave mode); ADD10=1 (master mode); STOPF=1 (slave mode); BTF=1 with no TxE or RxNE event; if ITBUFEN=1, TxE event is 1; if ITBUFEN=1, RxNE event is 1.	0
8	ITERREN	RW	Error interrupt enable bit. Setting this bit enables error interrupts. An interrupt is generated under the following conditions: BERR=1; ARLO=1; AF=1; OVR=1; PECERR=1; TIMEOUT=1; SMBAlert=1.	0
[7:6]	Reserved	RO	Reserved.	0
[5:0]	FREQ[5:0]	RW	I2C module clock frequency field. The correct clock frequency must be entered to generate proper timing. The allowed range is 4–60 MHz. Must be set between 000100b and 111100b, in MHz.	0

18.12.3 I2C Address Register 1 (I2C_OADDR1)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDMODE	Reserved	Reserved	Reserved	Reserved	Reserved	ADD[9]	ADD[8]	ADD[7]	ADD[6]	ADD[5]	ADD[4]	ADD[3]	ADD[2]	ADD[1]	ADD0

Bits	Name	Access	Description	Reset
15	ADDMODE	RW	Addressing mode. 1: 10-bit slave address (does not respond to 7-bit address); 0: 7-bit slave address (does not respond to 10-bit address).	0
[14:10]	Reserved	RO	Reserved.	0
[9:8]	ADD[9:8]	RW	Interface address; bits 9–8 when using 10-bit addressing, ignored when using 7-bit addressing.	0
[7:1]	ADD[7:1]	RW	Interface address, bits 7–1.	0
0	ADD0	RW	Interface address; bit 0 when using 10-bit addressing, ignored when using 7-bit addressing.	0

18.12.4 I2C Address Register 2 (I2C_OADDR2)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	ADD2[7]	ADD2[6]	ADD2[5]	ADD2[4]	ADD2[3]	ADD2[2]	ADD2[1]	ENDUAL

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
[7:1]	ADD2[7:1]	RW	Interface address; bits 7–1 of the address in dual address mode.	0
0	ENDUAL	RW	Dual address mode enable bit. Setting this bit allows ADD2 to also be recognized.	0

18.12.5 I2C Data Register (I2C_DATAR)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	DR[7]	DR[6]	DR[5]	DR[4]	DR[3]	DR[2]	DR[1]	DR[0]

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
[7:0]	DR[7:0]	RW	Data register. This field holds received data or data to be transmitted on the bus.	0

18.12.6 I2C Status Register 1 (I2C_STAR1)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SMBALERT	TIMEOUT	Reserved	PECERR	OVR	AF	ARLO	BERR	TxE	RxNE	Reserved	STOPF	ADD10	BTF	ADDR	SB

Bits	Name	Access	Description	Reset
15	SMBALERT	RW0	SMBus alert bit. Can be reset by user writing 0, or by hardware when PE goes low. In SMBus master mode: 1: SMBus alert generated on the pin; 0: No SMBus alert. In SMBus slave mode: 1: SMBAlert response address header received until SMBAlert goes low; 0: No SMBAlert response address header received.	0
14	TIMEOUT	RW0	Timeout or Tlow error flag. Can be reset by user writing 0, or by hardware when PE goes low. 1: SCL has been low for 25 ms, or master low-level accumulated clock stretch time exceeds 10 ms, or slave low-level accumulated time exceeds 25 ms; 0: No timeout error. Note: In slave mode when this bit is set, the slave resets communication and the hardware releases the bus; in master mode when this bit is set, the hardware generates a stop condition.	0
13	Reserved	RO	Reserved.	0
12	PECERR	RW0	PEC error flag during reception. Can be reset by user writing 0, or by hardware when PE goes low. 1: PEC error; returns NAK after receiving PEC; 0: No PEC error.	0
11	OVR	RW0	Overrun/underrun flag. 1: Overrun/underrun event occurred: when NOSTRETCH=1, in receive mode a new byte is received but the content in the data register has not been read, so the newly received byte is lost; in transmit mode, no new data is written to the data register and the same byte is sent twice; 0: No overrun/underrun event.	0
10	AF	RW0	Acknowledge failure flag. Can be reset by user writing 0, or by hardware when PE goes low. 1: Acknowledge error; 0: Acknowledge normal.	0
9	ARLO	RW0	Arbitration lost flag. Can be reset by user writing 0, or by hardware when PE goes low. 1: Arbitration lost detected, the module lost control of the bus; 0: Arbitration normal.	0
8	BERR	RW0	Bus error flag. Can be reset by user writing 0, or by hardware when PE goes low. 1: START or STOP condition error; 0: Normal.	0

Bits	Name	Access	Description	Reset
7	TxE	RO	Data register empty flag. Cleared by writing data to the data register, or automatically by hardware after a START or STOP is generated, or when PE is 0. 1: Transmit data register empty; 0: Data register not empty.	0
6	RxNE	RO	Data register not empty flag. Cleared by read/write operation on the data register, or by hardware when PE is 0. 1: Data register not empty during reception; 0: Normal.	0
5	Reserved	RO	Reserved.	0
4	STOPF	RO	STOP event flag. Cleared by a write operation to Control Register 1 after the user reads Status Register 1, or by hardware when PE is 0. 1: The slave detected a STOP event on the bus after an acknowledge; 0: No STOP event detected.	0
3	ADD10	RO	10-bit address header sent flag. Cleared by a write operation to Control Register 1 after the user reads Status Register 1, or by hardware when PE is 0. 1: In 10-bit addressing mode, the master has sent the first address byte; 0: None.	0
2	BTF	RO	Byte transfer finished flag. Cleared by a read/write operation on the data register after the user reads Status Register 1; during transfer, cleared by hardware after a START or STOP is generated, or when PE is 0. 1: Byte transfer finished. When NOSTRETCH=0: in transmit mode, when new data is sent and the data register has not been written with new data; in receive mode, when a new byte is received but the data register has not been read; 0: None.	0
1	ADDR	RW0	Address sent/matched flag. Cleared by a read operation on Status Register 2 after the user reads Status Register 1, or by hardware when PE is 0. Master mode: 1: Address sent complete — in 10-bit addressing mode, set after receiving the ACK of the second address byte; in 7-bit addressing mode, set after receiving the ACK of the address; 0: Address sending not complete. Slave mode: 1: Received address matched; 0: Address did not match or no address received.	0
0	SB	RO	Start bit sent flag. Cleared by a write operation to the data register after reading Status Register 1, or by hardware when PE is 0. 1: Start bit sent; 0: Start bit not sent.	0

18.12.7 I2C Status Register 2 (I2C_STAR2)

Offset address: 0x18

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PEC[7]	PEC[6]	PEC[5]	PEC[4]	PEC[3]	PEC[2]	PEC[1]	PEC[0]	DUALF	SMBHOST	SMBDEFAULT	GENCALL	Reserved	TRA	BUSY	MSL

Bits	Name	Access	Description	Reset
[15:8]	PEC[7:0]	RO	Packet error checking field. When PEC is enabled (ENPEC set), this field holds the PEC value.	0
7	DUALF	RO	Dual flag (matched). Cleared by hardware when a STOP or START is generated, or when PE=0. 1: The received address matches the content of OAR2; 0: The received address matches the content of OAR1.	0
6	SMBHOST	RO		0

Bits	Name	Access	Description	Reset
			SMBus host header flag. Cleared by hardware when a STOP or START is generated, or when PE=0. 1: When SMBTYPE=1 and ENARP=1, the SMBus host address was received; 0: SMBus host address not received.	
5	SMBDEFAULT	RO	SMBus device default address flag. Cleared by hardware when a STOP or START is generated, or when PE=0. 1: When ENARP=1, the SMBus device default address was received; 0: Address not received.	0
4	GENCALL	RO	General call address flag. Cleared by hardware when a STOP or START is generated, or when PE=0. 1: When ENGC=1, the general call address was received; 0: General call address not received.	0
3	Reserved	RO	Reserved.	0
2	TRA	RO	Transmitter/receiver flag. Cleared by hardware when a STOP event is detected (STOPF=1), a repeated START condition, bus arbitration is lost (ARLO=1), or PE=0. 1: Data transmitted; 0: Data received. This bit is determined by the R/W bit of the address byte.	0
1	BUSY	RO	Bus busy flag. Cleared when a STOP condition is detected. This information is still updated when the interface is disabled (PE=0). 1: Bus busy: SDA or SCL is low; 0: Bus idle, no communication.	0
0	MSL	RO	Master/slave mode indicator. Set by hardware when the interface is in master mode (SB=1); cleared by hardware when a STOP is detected on the bus, arbitration is lost, or PE=0.	0

18.12.8 I2C Clock Register (I2C_CKCFGR)

Offset address: 0x1C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F/S	DUTY	Reserved	Reserved	CCR[11]	CCR[10]	CCR[9]	CCR[8]	CCR[7]	CCR[6]	CCR[5]	CCR[4]	CCR[3]	CCR[2]	CCR[1]	CCR[0]

Bits	Name	Access	Description	Reset
15	F/S	RW	Master mode select bit. 1: Fast mode; 0: Standard mode.	0
14	DUTY	RW	Duty cycle in fast mode: 1: Tlow/Thigh = 16/9; 0: Tlow/Thigh = 2.	0
[13:12]	Reserved	RO	Reserved.	0
[11:0]	CCR[11:0]	RW	Clock divider coefficient field; determines the SCL clock frequency waveform.	0

18.12.9 I2C Rise Time Register (I2C_RTR)

Offset address: 0x20

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	TRISE[5]	TRISE[4]	TRISE[3]	TRISE[2]	TRISE[1]	TRISE[0]

Bits	Name	Access	Description	Reset
[15:6]	Reserved	RO	Reserved.	0
[5:0]	TRISE[5:0]	RW	Maximum rise time field. Sets the SCL rise time in master mode. The maximum rise time equals (TRISE – 1) clock cycles. This bit can only be	000010b

Bits	Name	Access	Description	Reset
			set when PE is cleared. For example, if the I2C module input clock period is 125 ns and TRISE is 9, the maximum rise time is $(9 - 1) \times 125$ ns = 1000 ns.	

Chapter 19: I3C Bus (I3C)

The I3C bus is a two-wire serial single-ended multi-drop bus designed to improve upon the traditional I2C bus. The I3C interface handles communication between this device and other devices connected on the I3C bus. As a master device, it enhances the functionality of the I2C interface while maintaining a degree of backward compatibility.

19.1 Main Features

- Supports master device
- Supports MIPI I3C specification v1.1
- Supports DMA function
- In-Band Interrupt (IBI) function
- Hot-Join (HJ) function
- Built-in error detection and recovery
- Supports dynamic address assignment, direct and broadcast Common Command Code (CCC) and private read/write transfers

19.2 Overview

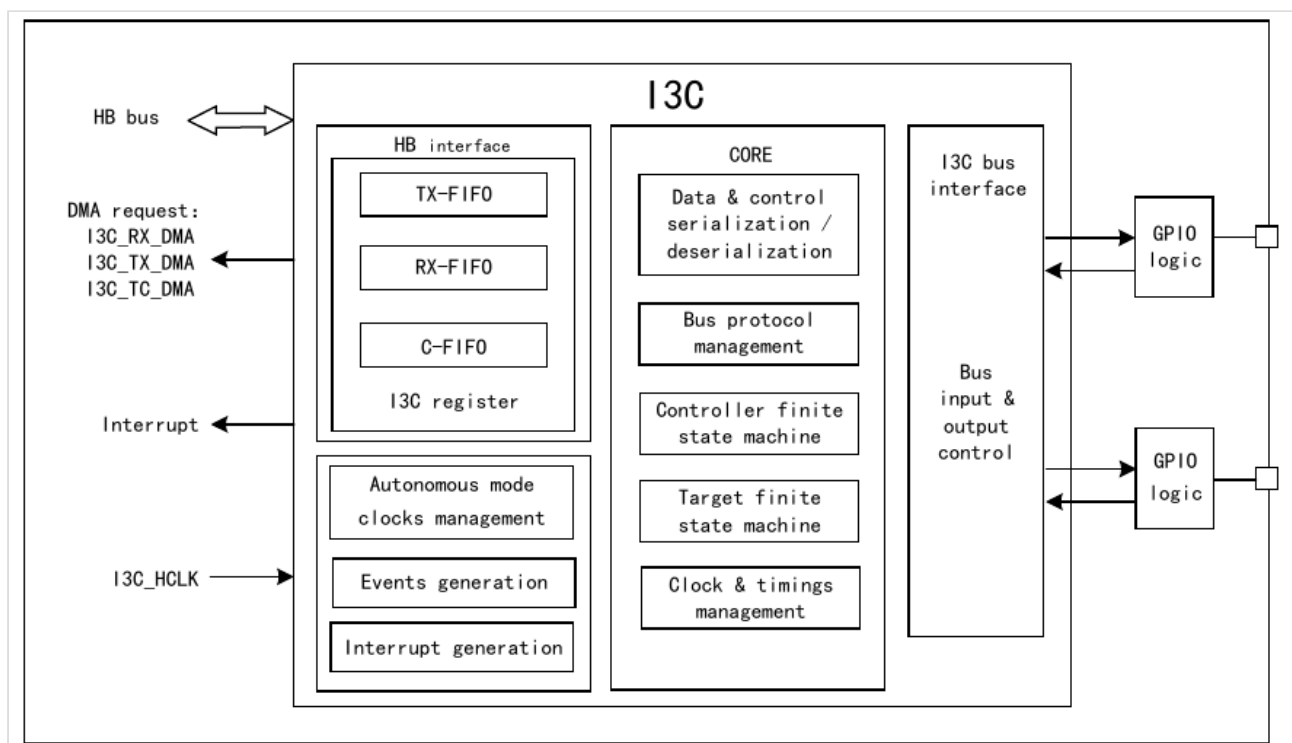


Figure 19-1: I3C block diagram

19.3 Functional Description

19.3.1 I3C Peripheral States

The I3C peripheral operates only as an I3C master. Under any circumstances, the peripheral is in one of the following states:

- **Disabled state**

Operating conditions: After the I3C peripheral is reset (the I3CRST bit in the RCC module is set to 1 to perform I3C reset), the peripheral is in the disabled state.

Transition from disabled state to idle state: When software sets the EN bit in the R32_I3C_CFGR register to 1, the peripheral switches from the disabled state to the idle state after completing the verification of internal configuration parameters.

- **Idle state**

In the idle state, software can perform partial updates to the I3C peripheral configuration.

Transition from idle state to active state:

1) Software-controlled trigger:

In master mode, software triggers by writing to set the TSFSET bit in the R32_I3C_CFGR register to 1 when initiating frame transfer;

2) Hardware event trigger:

The master triggers after receiving a start request from a slave during bus arbitration and after the maximum TCAS time.

- **Active state**

In the active state, the peripheral performs transfers on the bus.

In the active state, when the requested transfer is completed, the software is notified through events in the R32_I3C_EVR register, and the corresponding interrupts are enabled through the R32_I3C_INTENR register. After the peripheral switches from the active state to the idle state, communication can still continue on the bus, and partial reconfiguration can be performed.

As a master: In the active state, the events/flags generated after the requested transfer is completed can be frame complete (FCF), IBI/master-role/hot-join request complete (IBIF/CRF/HJF), or transfer error (ERRF).

19.3.2 I3C Master States and Programming Sequences

In this section, we provide the complete programming sequence for the I3C peripheral as a master, covering state transitions, main subtasks, and conditions.

Master initialization

When the master is in the disabled state (EN bit in R32_I3C_CFGR register is 0), software must follow these steps for initialization:

- Configure the R32_I3C_CFGR register, set CRINIT to 1 to mark the master role
- Clear the HST_SIE_RST bit in the R32_I3C_RESET register
- Configure I3C bus timing:
 1. Configure the R32_I3C_TIMINGR0 register to set the SCL clock high-level period and low-level period for legacy I2C and I3C open-drain/push-pull phases
 2. Configure the R32_I3C_TIMINGR1 register to set the SDA hold time for push-pull phase (tHD_PP), bus idle condition time (I3C tCAS, legacy I2C tBUF), I3C repeated start bit timing (tCASr, tCBSr), I3C stop timing (tCBP), maximum SCL clock low-level stall time (for ENTDA)

CCC (tSTALLDAA), legacy I2C ACK/NACK, I3C transfer address phase, write data transfer parity bit, legacy I2C write transfer ACK/NACK data phase, I3C read transfer transition bit or legacy I2C write transfer ACK/NACK phase (tSTALL)). The peripheral used as master can adjust the SCL clock low-level stall time and tNEWCRLock according to its own needs to accommodate the master role switching process (after GETACCCR CCC command)

3. Configure the R32_I3C_TIMINGR2 register to set the SCL clock low-level stall time (each phase can be individually enabled or disabled, thus adjusting the SCL clock low-level stall time to meet the requirements of the addressed I3C slave or legacy I2C slave during SDA switching)
 - Configure the DA[6:0] bits in the R32_I3C_DEVR0 register to set its own dynamic address
 - Configure the R32_I3C_DEVRx registers to set the related management for any slave x (x = 1/2/3/4)
 - Configure the R32_I3C_CFGR register to set the execution mode for frame transfer or target request transfer
 - Configure the R32_I3C_INTENR register to set interrupt generation or polling mode for any event
 - After initialization is complete, set the EN bit to 1 to enable the I3C peripheral

Initiating master-initiated frame transfer

When the master is in the enabled state (EN bit in R32_I3C_CFGR register is 1), software can initiate a frame transfer using any of the following configuration methods:

1. Software trigger: By performing a write operation to set the TSFSET bit in the R32_I3C_CFGR register to 1;
 - This operation causes the hardware to set the CFNFF flag in the R32_I3C_EVR register to request writing the first control word to the R32_I3C_CTLR register.
2. No trigger: By software directly writing the first control word to the R32_I3C_CTLR register.

Regardless of the initiation method, the I3C peripheral then switches to the active state. When the control word is not the last message of the I3C frame (i.e., the MEND bit in the R32_I3C_CTLR register is not set), and no transfer error has occurred (i.e., the ERRF bit in the R32_I3C_EVR register is set to 1), the hardware will continuously request the next control word and continue executing the frame transfer:

1. If the C-FIFO is in non-DMA mode (CDMAEN bit in R32_I3C_CFGR register is 0), software will write the next control word after the CFNFF flag in the R32_I3C_EVR register is set or the corresponding interrupt is triggered (CFNFIE bit in R32_I3C_INTENR register is set to 1);
2. If the C-FIFO is in DMA mode (CDMAEN bit in R32_I3C_CFGR register is 1), the corresponding I3C DMA request will be automatically handled by the assigned DMA channel, which will write the next control word.

Initiating and receiving slave-initiated transfers

When the master is in the enabled state (EN bit in R32_I3C_CFGR register is 1), the master can initiate transfers, and slaves can also initiate transfers by issuing a start request (driving SDA low), provided that the master has allowed hot-join requests, IBI requests, or master-role requests through the R32_I3C_DEVR0 register.

Executing master-initiated frame transfer

On the bus, the master continuously executes frame transfers until the last message transfer is completed (FCF bit in R32_I3C_EVR register is 1), or a transfer error occurs (ERRF bit in R32_I3C_EVR

register is 1), and the corresponding interrupt is triggered (if enabled). This process relies on the R32_I3C_CTLR and R32_I3C_TDW/TDR registers (explicitly written by software or automatically pushed by the assigned DMA channel) and R32_I3C_RDW/RDR (explicitly read by software or by the assigned DMA channel). After the transfer is completed, the I3C master returns to the idle state.

Updating master transfer configuration

After returning to the idle state, software can update the I3C peripheral configuration before the next transfer:

- Configure the R32_I3C_TIMINGR2 register to modify the SCL clock stall;
- Configure the R32_I3C_INTENR register to modify the interrupt/polling mode strategy;
- Configure the R32_I3C_CFGR register TXTHRES, RXTHRES, TMODE, TXDMAEN, CDMAEN, RXDMAEN, EXITPTRN, RSTPTRN, NOARBH bits;
- Modify/prepare the control words, status words, and read/write data for the next frame transfer to be executed through software and/or DMA.

Master receiving (broadcast CCC, direct read/write CCC, or private read/write) messages

When the slave is in the idle state (EN bit in R32_I3C_CFGR register is 1), it indicates that it is ready to receive communication messages from the master on the I3C bus, and is ready to switch to the active state.

In the active state, the slave first receives the broadcast ENTDAACCCC (possibly after receiving the optional broadcast ENEC/DISEC CCC), and is then assigned a dynamic address. The DAUPF event in the R32_I3C_EVR register is set to 1, and the corresponding interrupt is generated (if enabled), after which the slave returns to the idle state.

After that, the slave in the idle state is ready to receive any other broadcast CCC messages, direct read/write CCC, or private read/write messages from the master.

19.3.3 I3C Bus Transfers and Programming

19.3.3.1 I3C Broadcast/Direct CCC Transfer (except ENTDAACCCC and RSTACT) (as Master)

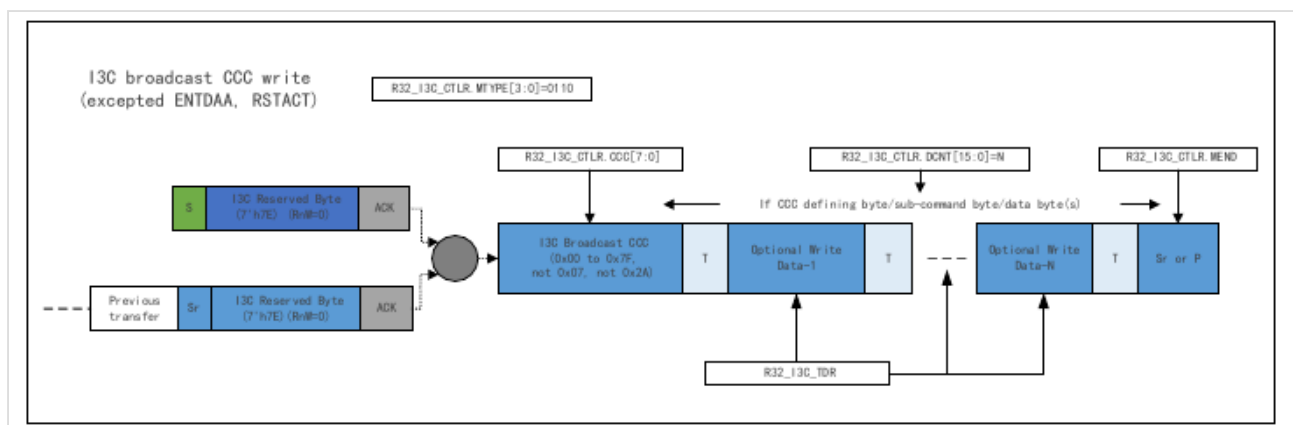


Figure 19-2: I3C CCC message

When the peripheral acts as a master, it supports I3C broadcast CCC write (except ENTDAACCCC and RSTACT), I3C direct CCC write (except RSTACT, first part and $n \times$ second part), and I3C direct CCC read (except RSTACT, first part and $n \times$ second part). The control word configuration uses MTYPE[3:0]=0110 for the first part (broadcast/direct CCC) and MTYPE[3:0]=0011 for the second part (direct addressing of individual slaves).

19.3.3.2 I3C Broadcast ENTDAACCC Transfer (as Master)

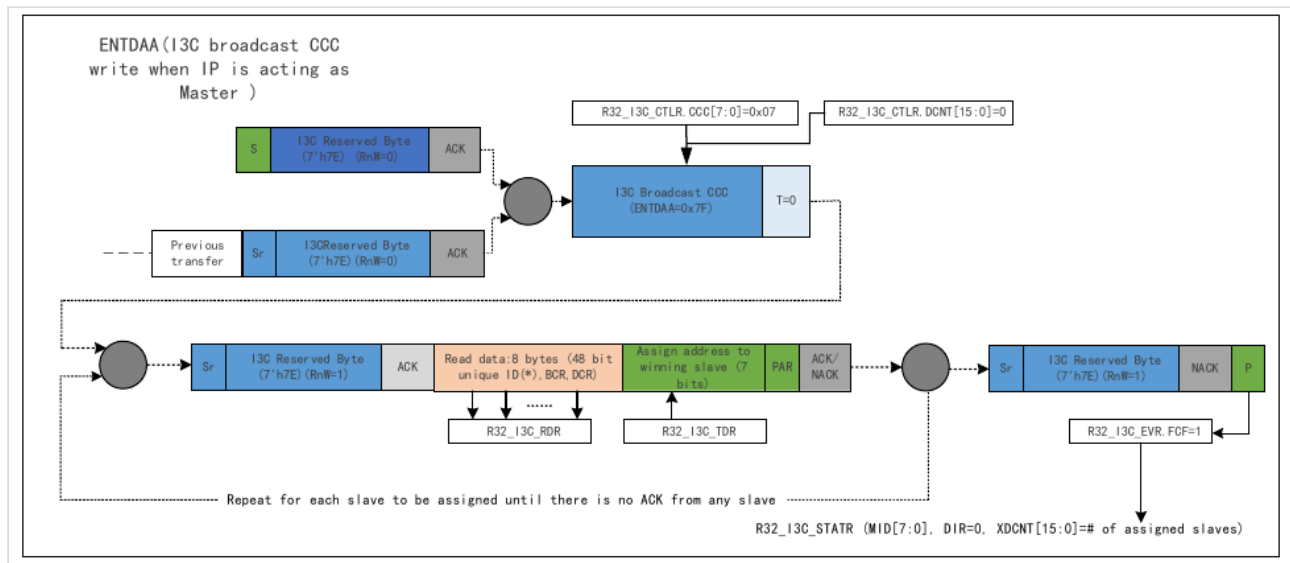


Figure 19-3: I3C broadcast ENTDAACCC transfer (as master)

The ENTDAAC broadcast CCC is used for dynamic address assignment. The master issues the broadcast ENTDAAC CCC (0x7F) with DCNT=0. The process is repeated for each slave to be assigned until there is no ACK from any slave. During each iteration, the master reads 8 bytes (48-bit unique ID, BCR, DCR) from the winning slave, then assigns a dynamic address. The 48-bit unique/provisioned ID is subject to arbitration; the slave that wins arbitration continues to provide its BCR and DCR.

19.3.3.3 I3C Broadcast/Direct RSTACT CCC Transfer (as Master)

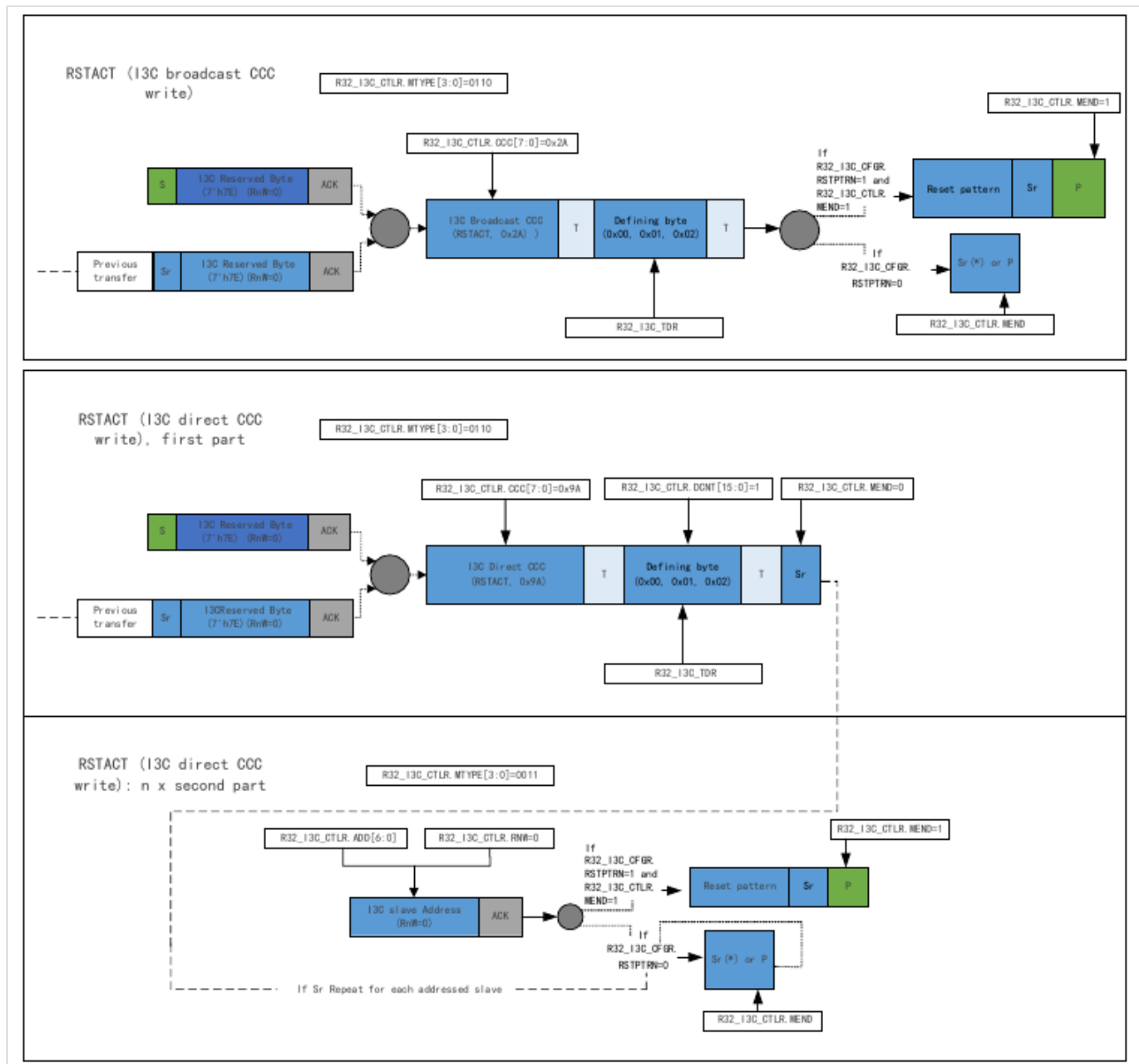


Figure 19-4: I3C broadcast/direct read/write RSTACT CCC (as master)

The RSTACT CCC is used to reset the activity state of target devices. It can be issued as a broadcast CCC write (0x2A) or a direct CCC write/read (0x9A), with a defining byte (0x00, 0x01, or 0x02). When $R32_I3C_CFGR.RSTPTRN=1$ and $R32_I3C_CTLR.MEND=1$, a reset pattern is inserted before the stop condition. The direct version consists of a first part (addressing the CCC) and an n× second part (addressing each individual slave).

19.3.3.4 I3C Private Read/Write Transfer (as Master)

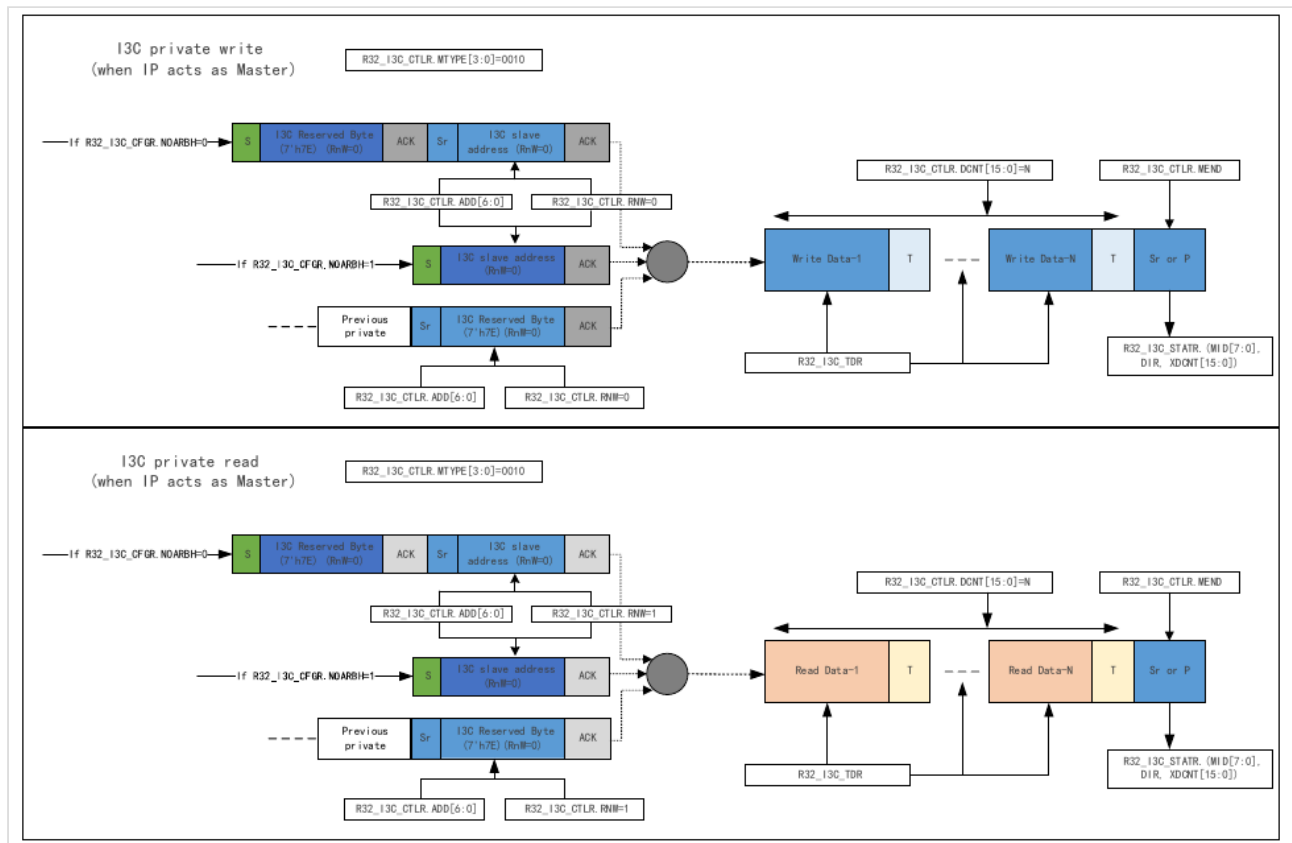


Figure 19-5: I3C private read/write message (as master)

When the peripheral acts as a master, I3C private write and private read transfers use $MTYPE[3:0]=0010$. If $R32_I3C_CFGR.NOARBH=0$, an arbitrable header (7'h7E, RnW=0) is issued after the start condition, followed by a repeated start and the I3C slave address. If $NOARBH=1$, the slave address is issued directly after the start condition without the arbitrable header. The control word includes $ADD[6:0]$, RNW , $DCNT[15:0]$, and $MEND$.

19.3.3.5 Legacy I2C Read/Write Transfer (as Master)

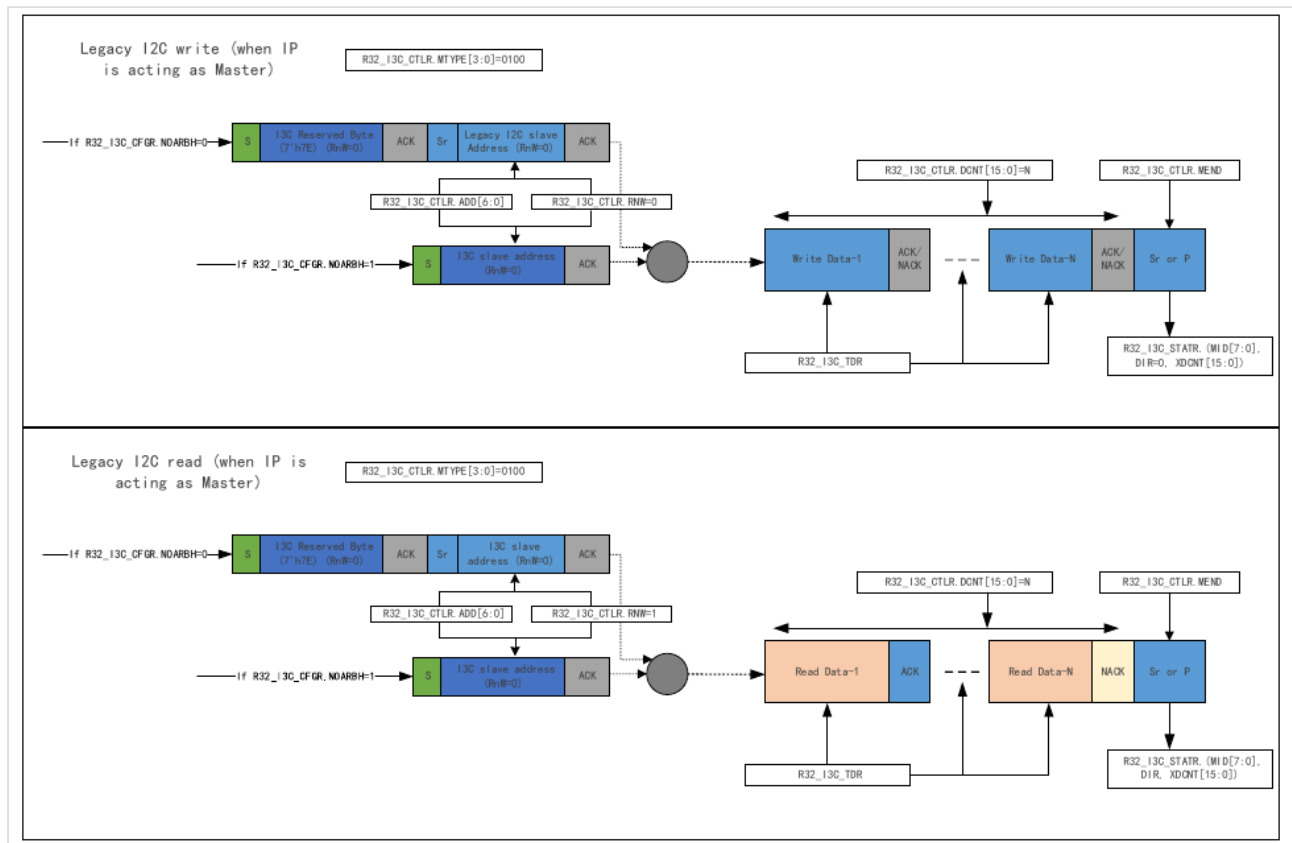


Figure 19-6: Legacy I2C read/write transfer (as master)

Legacy I2C write and read transfers use $MTYPE[3:0]=0100$. Similar to I3C private transfers, if $R32_I3C_CFGR.NOARBH=0$, an arbitrable header is inserted after the start condition. The legacy I2C transfer uses standard ACK/NACK handshake on the open-drain bus, without transition bits (T). Write data uses ACK/NACK from the slave, and read data uses ACK/NACK from the master (ACK to continue, NACK to end).

19.3.3.6 I3C IBI Transfer (as Master)

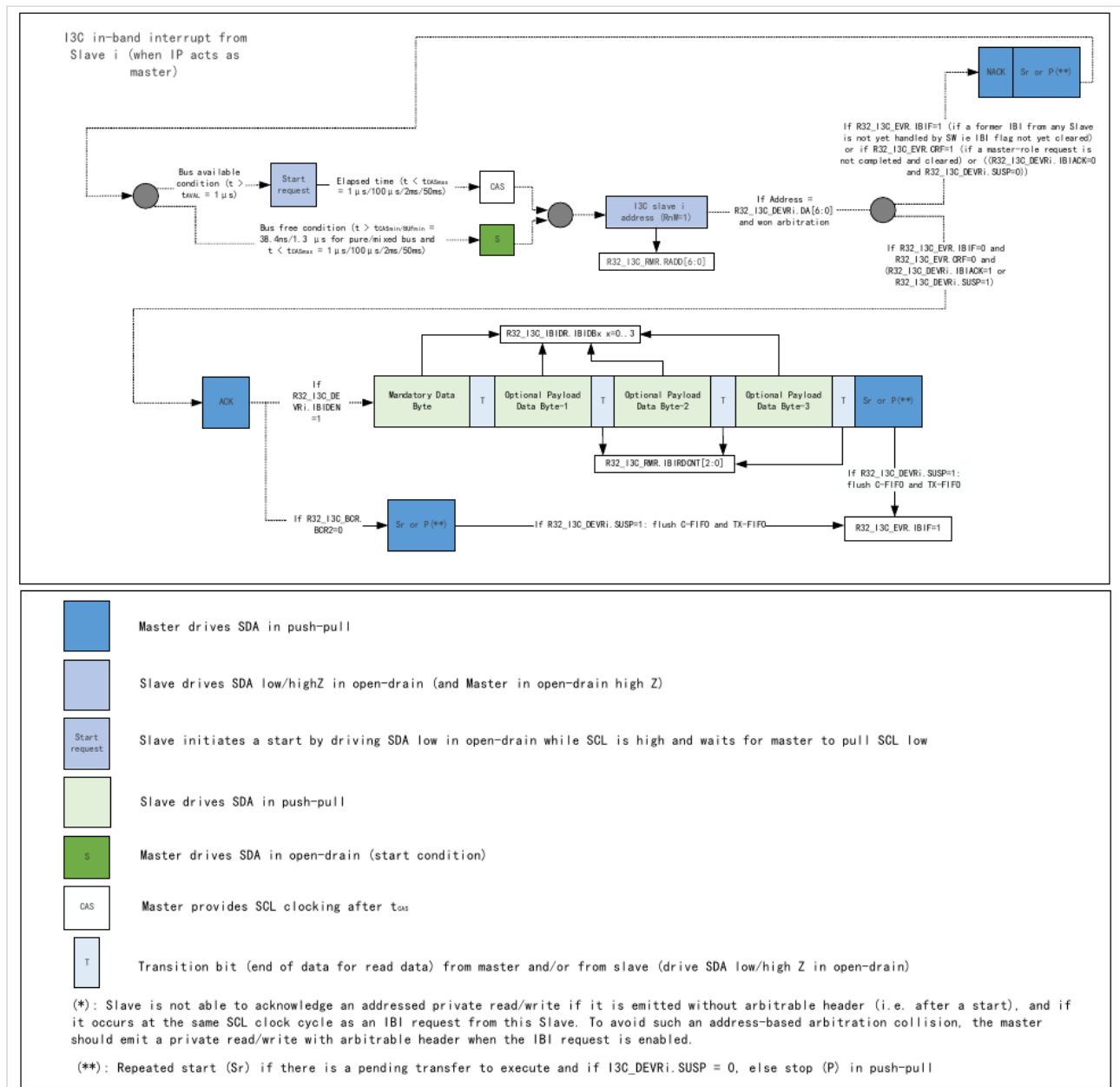


Figure 19-7: I3C IBI transfer (as master)

When the peripheral acts as a master, slaves can initiate in-band interrupt (IBI) requests by driving SDA low in open-drain while SCL is high. The master provides SCL clocking after tCAS. The IBI request is acknowledged or not acknowledged based on the R32_I3C_DEV*Ri*.IBIACK and R32_I3C_DEV*Ri*.SUSP configuration. If acknowledged, the slave address is read, and if IBIDEN=1, a mandatory data byte (MDB) follows, along with optional payload data bytes. The IBI payload data is received in the R32_I3C_IBIDR register.

Note: When the peripheral acts as a master, the R32_I3C_IBIDR register is used to receive IBI data payloads. Therefore, IBI requests from slaves must not exceed 4 bytes of data payload. If more information needs to be exchanged in the context of this in-band interrupt, the master software must issue a private read control word.

19.3.3.7 I3C Hot-Join Request Transfer (as Master)

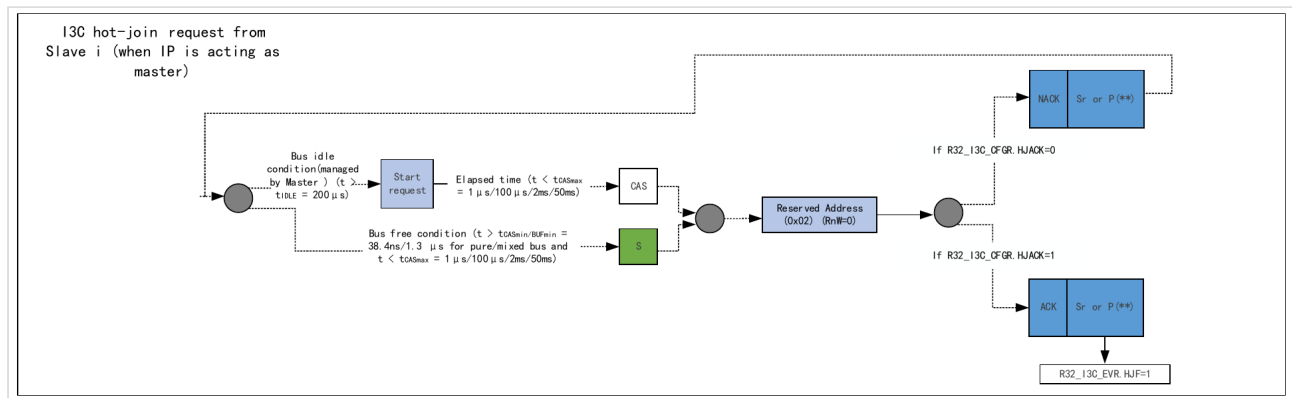


Figure 19-8: I3C hot-join request transfer (as master)

When the peripheral acts as a master, slaves can initiate hot-join requests by driving SDA low in open-drain while SCL is high when the bus is idle ($t > tIDLE = 200\mu s$). The master provides SCL clocking after $tCAS$. If $R32_I3C_CFGR.HJACK=0$, the hot-join request is NACKED. If $HJACK=1$, the reserved address (0x02, RnW=0) is acknowledged, and the HJF flag in $R32_I3C_EVR$ is set to 1. A repeated start (Sr) is issued if there is a pending transfer to execute, otherwise a stop (P) is issued.

19.3.4 I3C FIFO Management

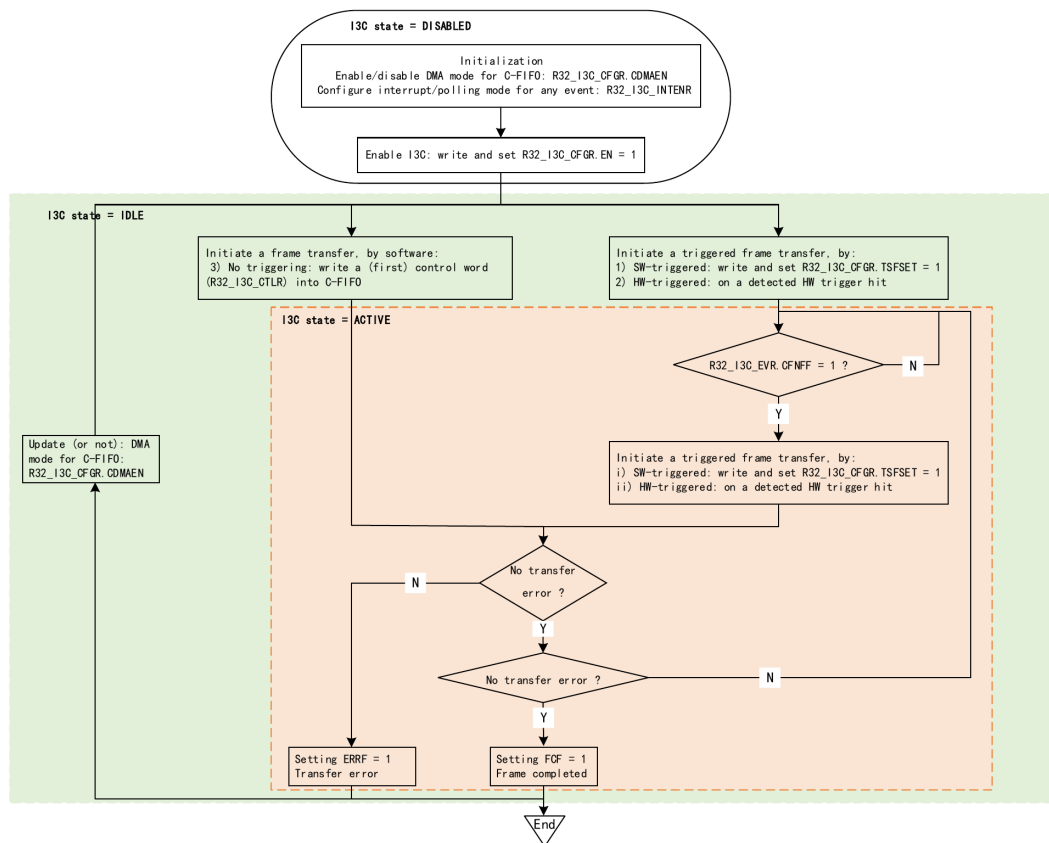
Table 19-1: I3C FIFO implementation

FIFO	Content	Unit	Size	Used as master (basic principle)
C-FIFO	32-bit control word	Word	2	Master (a frame can be based on multiple control words; this is not the case when used as a slave)
TX-FIFO	Data to be transmitted	Byte	8	Master
RX-FIFO	Data to be received	Byte	8	Master

19.3.4.1 C-FIFO Management (as Master)

When used as a master, the C-FIFO can be used during broadcast CCC, direct read/write CCC, private read/write, legacy I2C read/write, and software-initiated error recovery.

图 19-9 C-FIFO 管理（作为主设备）



首先，软件必须通过 R32_I3C_CFGR 寄存器 CDMAEN 来初始化 C-FIFO 管理，具体步骤如下：

- 软件直接按控制字进行写入（当 CDMAEN=0 时）：
 - 通过轮询模式（设置 R32_I3C_INTENR 寄存器中的 CFNFIE=0）：在向 R32_I3C_CTLR 寄存器显式写入之前，等待硬件请求下一个控制字（R32_I3C_INTENR 寄存器中的 CFNFF=1）
 - 通过使能的中断通知（当 CFNFIE=1 时）：
- 由分配的 DMA 通道写入（当 CDMAEN=1 时），以使能相应的 I3C 外设 DMA 请求：
 - 根据模块级别的配置，DMA 会自动将控制字从其存储器源缓冲区推入/写入 R32_I3C_CTLR 寄存器，

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Figure 19-9: C-FIFO management (as master)

Figure 19-9 illustrates how the C-FIFO is managed to queue control words on the I3C bus when the I3C peripheral is used as a master.

First, software must initialize C-FIFO management through the CDMAEN bit in the R32_I3C_CFGR register, with the following steps:

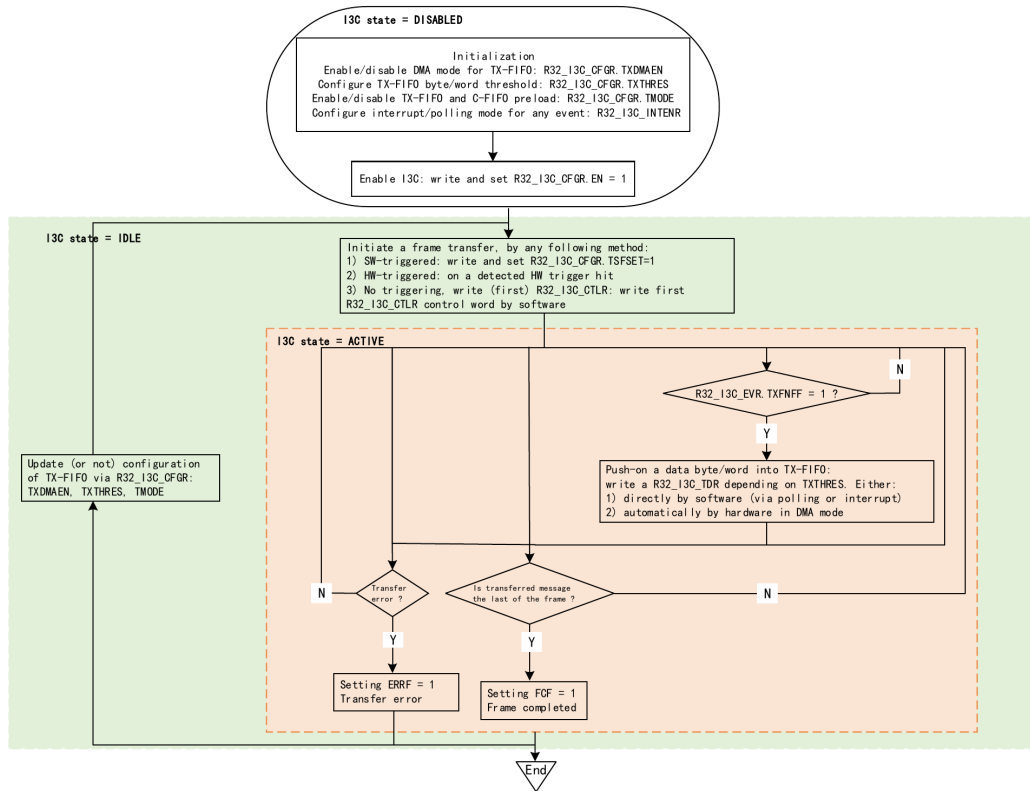
- Software directly writes control words (when CDMAEN=0):
 - Through polling mode (set CFNFIE=0 in R32_I3C_INTENR register): Before explicitly writing to the R32_I3C_CTLR register, wait for the hardware to request the next control word (CFNFF=1 in R32_I3C_INTENR register)
 - Through enabled interrupt notification (when CFNFIE=1)

- Written by the assigned DMA channel (when CDMAEN=1), to enable the corresponding I3C peripheral DMA request:
 - According to the module-level configuration, the DMA automatically pushes/writes control words from its memory source buffer into the R32_I3C_CTLR register until the frame is completed (a stop bit is issued on the I3C bus after the last message of the frame), except when a transfer error occurs.
- In any case, whenever the C-FIFO is empty and a repeated start bit for a new control word must be issued, a C-FIFO underflow is reported (ERRF=1 in R32_I3C_EVR register and COVR=1 in R32_I3C_STATER register). If the ERRIE bit in the R32_I3C_INTENR register is enabled, an interrupt is generated.
- The DMA mode configuration for C-FIFO management can be modified when the I3C peripheral is not in the active state.
- When used as a master, if a transfer error occurs (ERRF=1 in R32_I3C_EVR register), the hardware will automatically flush the C-FIFO.

19.3.4.2 TX-FIFO Management (as Master)

When used as a master, the TX-FIFO can be used during broadcast or direct CCC (including ENTDA and RSTACT), if there are defining bytes/sub-command bytes or data bytes, private write, and legacy I2C write.

图 19-10 TX-FIFO 管理（作为主设备）



首先，必须通过 R32_I3C_CFGR 寄存器中的以下位域初始化 TX-FIFO 管理：

- TXDMAEN：使能/禁止 TX-FIFO 的 DMA 模式
- TXTHRES：将数据字节或字推入 TX-FIFO
- TMODE：使能/禁止 TX-FIFO 和 C-FIFO 预装载

然后，根据 R32_I3C_CFGR 寄存器 TXDMAEN 位，有两种方法可以向 TX-FIFO 写入数据：

- 软件直接按字节/字写入（当 TXDMAEN=0 时）：
 - 通过轮询模式（R32_I3C_INTENR 寄存器 TXFNFIE=0）：在显式写入 R32_I3C_TDR 或 R32_I3C_TDWR 寄存器之前，等待硬件请求下一个数据字节/字（R32_I3C_INTENR 寄存器 TXFNF = 1），

Figure 19-10: TX-FIFO management (as master)

Figure 19-10 illustrates how the TX-FIFO is managed to queue data bytes or words to be transmitted on the I3C bus when the I3C peripheral is used as a master.

First, TX-FIFO management must be initialized through the following bit fields in the R32_I3C_CFGR register:

- TXDMAEN: Enable/disable DMA mode for TX-FIFO
- TXTHRES: Push data bytes or words into TX-FIFO
- TMODE: Enable/disable TX-FIFO and C-FIFO preload

Then, depending on the TXDMAEN bit in the R32_I3C_CFGR register, there are two methods to write data to the TX-FIFO:

- Software directly writes by byte/word (when TXDMAEN=0):
 - Through polling mode (TXFNFI=0 in R32_I3C_INTENR register): Before explicitly writing to the R32_I3C_TDR or R32_I3C_TDWR register, wait for the hardware to request the next data byte/word (TXFNFI=1 in R32_I3C_INTENR register), depending on the TXTHRES bit in the R32_I3C_CFGR register
 - Through enabled interrupt notification (when TXFNFI=1)
- Written by the assigned DMA channel (when TXDMAEN=1), to enable the corresponding I3C peripheral DMA request:
 - According to the DMA module-level configuration, the DMA automatically pushes/writes data bytes/words from its memory source buffer into the R32_I3C_TDR or R32_I3C_TDWR register (depending on the TXTHRES bit in the R32_I3C_CFGR register), until the frame is completed (a stop bit is issued on the I3C bus after the last message of the frame), except when a transfer error occurs.

An I3C message begins with a start bit or repeated start bit, and ends with a stop bit or repeated start bit. At the message level, the last data byte/word to be transmitted is indicated by TXLASTF=1 in the R32_I3C_EVR register. When an I3C frame contains multiple messages (separated by repeated start bits), software can use this event to update the corresponding pointer to the buffer used to store the bytes/words of the next message.

When frame completion is reported (FCF=1 in R32_I3C_EVR register, and the corresponding interrupt is enabled), the TX-FIFO is empty. If the TX-FIFO is empty and a data byte must be transmitted on the I3C bus, a TX-FIFO underflow is reported (ERRF=1 in R32_I3C_EVR register and DOVR=1 in R32_I3C_STATERR register). If the corresponding interrupt is enabled (ERRIE=1 in R32_I3C_INTENR register), an interrupt is generated. The TX-FIFO management configuration can be modified when the I3C peripheral is not in the active state.

When used as a master, if a transfer error occurs (ERRF=1), the hardware will automatically flush the TX-FIFO.

Without C-FIFO/TX-FIFO preload

The C-FIFO size is 2 words, and the TX-FIFO size is 8 bytes. Without C-FIFO/TX-FIFO preload enabled (TMODE=0 in R32_I3C_CFGR register), once the C-FIFO receives the first control word, the I3C peripheral will immediately issue a start bit on the I3C bus. The peripheral then decodes the R32_I3C_CTLR register and writes the next data byte or word as needed. Once the hardware requires the next control word to be sent on the I3C bus (e.g., a repeated start bit needs to be issued on the I3C bus or the C-FIFO needs available space), the peripheral immediately requests the control word to be written to the C-FIFO, until the last message (i.e., MEND=1 in R32_I3C_CTLR register). Similarly, once the hardware requires another data byte or word to be sent on the I3C bus (e.g., a repeated start bit needs to be issued on the I3C bus, the TX-FIFO is not full, or a data byte or word must be sent during this I3C message), the data byte/word must be immediately written to the TX-FIFO.

C-FIFO and TX-FIFO preload

When C-FIFO/TX-FIFO preload is configured (TMODE=1 in R32_I3C_CFGR register), before issuing the start bit on the bus, the I3C peripheral waits as much as possible for the C-FIFO and TX-FIFO to be loaded, as follows:

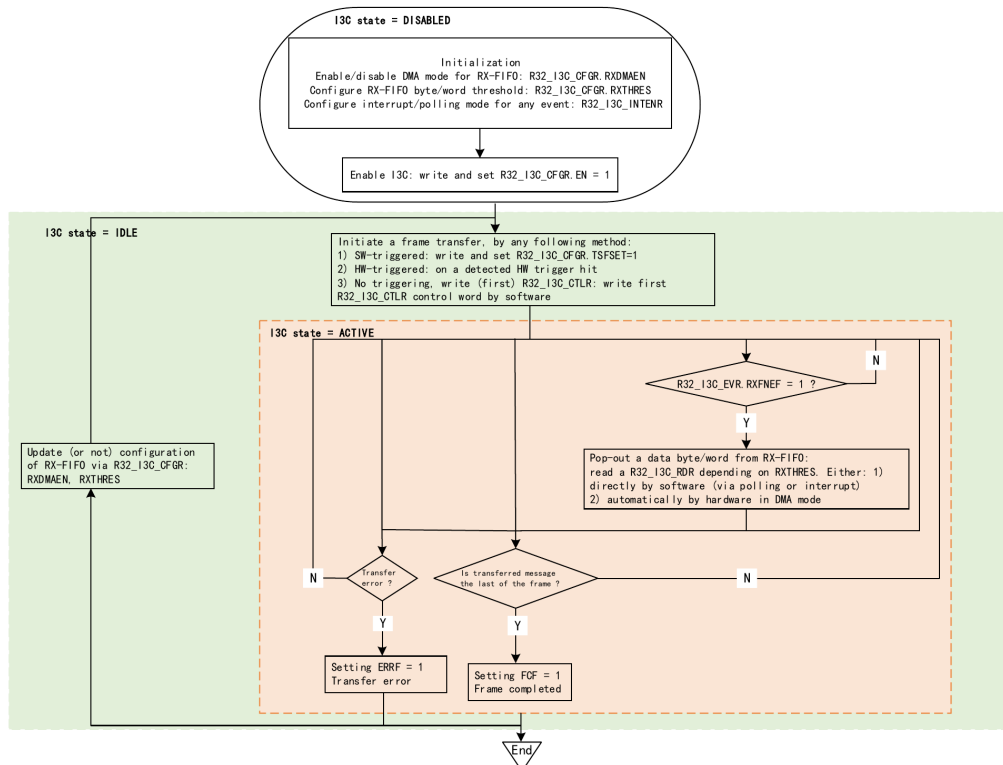
- Wait for the first control word to be written to the C-FIFO

- Wait for data bytes/words to be written to the TX-FIFO (if any), as defined by the first control word (if RNW=0 and DCNT[15:0]=0 in R32_I3C_CTLR register), up to the TX-FIFO size
- If the TX-FIFO is not full and the first control word is not the last message of the frame (MEND=0 in R32_I3C_CTLR register):
 - Wait for the second control word to be written to the C-FIFO, at which point the C-FIFO is full.
 - If the TX-FIFO is not full, wait for data bytes/words to be written to the TX-FIFO (if any), as defined by the second control word (RNW=0 and DCNT[15:0] in R32_I3C_CTLR register), up to the TX-FIFO size.
- Then, whenever the hardware requires the next control word to be sent on the I3C bus (if a repeated start bit must be issued on the I3C bus), the control word is immediately requested to be written to the C-FIFO, until the last message (MEND=1 in R32_I3C_CTLR register).
- Similarly, whenever the hardware requires the next data byte/word to be sent on the I3C bus (if a repeated start bit must be issued on the I3C bus, the TX-FIFO is not full, or a data byte/word will be sent during this I3C message), the data byte/word must be immediately written to the TX-FIFO.

19.3.4.3 RX-FIFO Management (as Master)

When used as a master, the RX-FIFO can be used during broadcast ENTDAACCC, direct CCC read (including direct RSTACT CCC read), private read, and legacy I2C read.

图 19-11 RX-FIFO 管理 (作为主设备)



首先，软件必须通过 R32_I3C_CFGR 寄存器的以下位域初始化 RX-FIFO 管理：

- RXDMAEN：使能/禁止 RX-FIFO 的 DMA 模式
- RXTHRES：从 RX-FIFO 弹出数据字节或字

然后，根据 RXDMAEN 位读取 RX-FIFO：

- 软件直接按字节/字读取 (RXDMAEN=0)：
 - 通过轮询模式 (R32_I3C_INTENR 寄存器中的 RXFNEIE=0)：在显式读取 R32_I3C_RDR 或 R32_I3C_RDWR 寄存器之前等待硬件请求下一个数据字节/字 (R32_I3C_INTENR 寄存器中的 RXFNEF=1)，具体取决于 R32_I3C_CFGFR 寄存器中的 RXTHRES 位
 - 通过使能的中断通知 (如果 R32_I3C_INTENR 寄存器 RXFNEIE=1)
- 由分配的 DMA 通道读取 (如果 RXDMAEN=1)，以使能相应的 I3C 外设 DMA 请求：
 - 根据 DMA 模块级别的配置，DMA 会自动从 R32_I3C_RDR 或 R32_I3C_RDWR 寄存器弹出/读取数据字节/字 (具体取决于 RXTHRES 位)，并将它们写入其存储器从设备缓冲区，直到帧完成 (在帧的最后一条消息后，I3C 总线上会发出停止位)，但发生传输错误时除外。

Figure 19-11: RX-FIFO management (as master)

Figure 19-11 illustrates how the RX-FIFO is managed to queue and pop data bytes or words received on the I3C bus when the I3C peripheral is used as a master.

First, software must initialize RX-FIFO management through the following bit fields in the R32_I3C_CFGFR register:

- RXDMAEN: Enable/disable DMA mode for RX-FIFO
- RXTHRES: Pop data bytes or words from the RX-FIFO

Then, the RX-FIFO is read depending on the RXDMAEN bit:

- Software directly reads by byte/word (RXDMAEN=0):
 - Through polling mode (RXFNEIE=0 in R32_I3C_INTENR register): Before explicitly reading the R32_I3C_RDR or R32_I3C_RDWR register, wait for the hardware to request the next

- data byte/word (RXFNEF=1 in R32_I3C_INTENR register), depending on the RXTHRES bit in the R32_I3C_CFGR register
 - Through enabled interrupt notification (if RXFNEIE=1 in R32_I3C_INTENR register)
- Read by the assigned DMA channel (if RXDMAEN=1), to enable the corresponding I3C peripheral DMA request:
 - According to the DMA module-level configuration, the DMA automatically pops/reads data bytes/words from the R32_I3C_RDR or R32_I3C_RDWR register (depending on the RXTHRES bit), and writes them to its memory slave buffer, until the frame is completed (a stop bit is issued on the I3C bus after the last message of the frame), except when a transfer error occurs.

An I3C message begins with a start bit or repeated start bit, and ends with a stop bit or repeated start bit. At the message level, the last data byte/word received from the I3C bus is indicated by IRXLASTF=1 in the I3C_EVR register. When an I3C frame contains multiple messages (separated by repeated start bits), software can use this event to update the corresponding pointer to the buffer used to store the data bytes/words of the next message.

If the RX-FIFO is full and a data byte is received on the I3C bus, an RX-FIFO overflow is reported (ERRF=1 in R32_I3C_EVR register and DOVR=1 in R32_I3C_STATR register). If the corresponding interrupt is enabled (ERRIE=1 in R32_I3C_INTENR register), an interrupt is generated. The RX-FIFO management configuration can be modified when the I3C peripheral is not in the active state.

Slave early termination of read transfer

With the S-FIFO disabled (SMODE=0 in R32_I3C_CFGR register), software is notified of early termination of a read transfer through RXTGTENDF=1 in the R32_I3C_EVR register and the corresponding interrupt (if enabled). Software can then read the status register R32_I3C_STATR to check information related to the last message and obtain the number of data bytes received in the prematurely terminated read transfer (XDCNT[15:0] in R32_I3C_STATR register).

In any case, if the S-FIFO is enabled (SMODE=1 in R32_I3C_CFGR register), software or DMA (depending on SDMAEN in R32_I3C_CFGR register) must read the status register R32_I3C_STATR once for each message. The number of valid data bytes received in the prematurely terminated read message is reported by XDCNT[15:0] in R32_I3C_STATR register (followed by ABT=1).

19.3.5 I3C Master Error Management

Table 19-2: Master error management

Error type	Description	Error detection	Master operation	Error reporting
CE0	Illegal formatted CCC (e.g., slave returns fewer data bytes during direct CCC read transfer)	Detected at the end of legacy I2C read transfer when incorrect ACK is detected	Hardware holds SCL running for nine clock cycles, can exchange another byte, then issues NACK again followed by a stop bit	ERRF=1, PERR=1 and CODERR[3:0]=0000b
CE1	Watchdog error	Unable to generate repeated start bit or stop bit at the end of I3C SDR read transfer	SCL is held running. Software can stop SCL by writing a control word with MTYPE[3:0]=0000 (header message) to the R32_I3C_CTLR register, then	ERRF=1, PERR=1 and CODERR[3:0]=0001b

Error type	Description	Error detection	Master operation	Error reporting
			must usually wait at least 150µs before issuing another message. After CE1 error is detected, unable to generate start bit	
CE2	No response to broadcast address (1111110b)	During message transmission (except upgrade fault or reset timing message), header (1111110b+RnW=0) is detected as NACK	Hardware issues HDR exit timing followed by a stop bit	ERRF=1, PERR=1 and CODERR[3:0]=0010b
CE3	Master role switch failure	After the new master drives SDA low (initiated by test header or start request from slave) and after the delay time defined in its active state has ended, SCL has not been driven low	Hardware issues start bit + 1111110b+RnW=0, followed by ACK/NACK from slave, and finally a stop bit	ERRF=1, PERR=1 and CODERR[3:0]=0011b
-	Incorrect 7-bit slave address with parity bit returned in GETACCCR CCC	Detected incorrect dynamic address and/or parity bit	Hardware cancels GETACCCR CCC by issuing repeated start bit + 1111110b+RnW=0, followed by ACK/NACK from slave, and finally a stop bit	ERRF=1 and DERR=1
-	Addressed slave is NACKed for the first time in direct CCC read transfer	NACK detected	Hardware performs a single retry by issuing repeated start bit + 7-bit same slave address + RnW=1	-
-	Addressed slave is NACKed for the second time in direct CCC write transfer, I3C private read/write transfer, legacy I2C transfer, or direct CCC read transfer	NACK detected	Hardware issues a stop bit	ERRF=1 and ANACK=1
-	Assigned/sent dynamic address with parity bit is NACKed for the first time in ENTDAACCCC	NACK detected	Hardware performs a single retry assignment loop by issuing repeated start bit + 1111110b+RnW=1, followed by ACK from the (highest priority) slave and 8-byte read data, then the assigned address + parity bit	-
-	Assigned/sent dynamic address	NACK detected	Hardware issues a stop bit	ERRF=1 and DNACK=1

Error type	Description	Error detection	Master operation	Error reporting
	with parity bit is NACKed for the second time in ENTDAACCC			
-	Write data is NACKed in legacy I2C write transfer	NACK detected	Hardware issues a stop bit	ERRF=1 and DNACK=1
-	Relative to I3C bus timing, control words, status words, transmit data, or read data are not written/read in time	SCL stall timeout	Hardware issues a stop bit	ERRF=1 and (COVR=1 or DOVR=1)

19.3.6 I3C Low-Power Mode

When the I3C is in sleep mode under low-power mode, it is not affected. I3C interrupts can wake the device from sleep mode. When in low-power mode, changes on the SCL and SDA buses will wake the system, exiting low-power mode.

19.3.7 I3C Interrupts

Table 19-3: I3C interrupt requests

Interrupt event	Interrupt partition	Interrupt enable control	Event flag	Event clear control
Request control word	Request	CFNFIE	CFNFF	-
Request data to be transmitted		TXFNIE	TXFNFF	-
Request data to be received		RXFNEIE	RXFNEF	-
Master: frame transfer complete		FCIE	FCF	CFCF
IBI request received		IBIIE	IBIF	CIBIF
Hot-join request received		HJIE	HJF	CHJF
Error occurred	Error	ERRIE	ERRF	CERRF

19.4 Register Description

Table 19-4: I3C related register list

Name	Address	Description	Reset Value
R32_I3C_CTLR	0x40024C00	I3C control register	0x00000000
R32_I3C_CFGR	0x40024C04	I3C configuration register	0x00000040
R32_I3C_RDR	0x40024C10	I3C receive data byte register	0x00000000

Name	Address	Description	Reset Value
R32_I3C_TDR	0x40024C18	I3C transmit data byte register	0x00000000
R32_I3C_IBIDR	0x40024C20	I3C IBI payload data register	0x00000000
R32_I3C_RESET	0x40024C2C	I3C reset register	0x03000000
R32_I3C_STATR	0x40024C30	I3C status register	0x00000000
R32_I3C_STATER	0x40024C34	I3C status error register	0x00000000
R32_I3C_RMR	0x40024C40	I3C receive message register	0x00000000
R32_I3C_EVR	0x40024C50	I3C event register	0x00000003
R32_I3C_INTENR	0x40024C54	I3C interrupt enable register	0x00000000
R32_I3C_CEV	0x40024C58	I3C clear event register	0x00000000
R32_I3C_DEVR0	0x40024C60	I3C device register 0	0x00000000
R32_I3C_DEVR1	0x40024C64	I3C device register 1	0x00000000
R32_I3C_DEVR2	0x40024C68	I3C device register 2	0x00000000
R32_I3C_DEVR3	0x40024C6C	I3C device register 3	0x00000000
R32_I3C_DEVR4	0x40024C70	I3C device register 4	0x00000000
R32_I3C_TIMINGR0	0x40024CA0	I3C timing register 0	0x00000000
R32_I3C_TIMINGR1	0x40024CA4	I3C timing register 1	0x00000000
R32_I3C_TIMINGR2	0x40024CA8	I3C timing register 2	0x00000000
R32_I3C_GETMSR	0x40024CD0	I3C get maximum data speed register	0x00000000

19.4.1 I3C Control Register (R32_I3C_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEND	MTYPE[3:0]				Reserved				ADD[6:0]						RNW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCNT[15:0]															

Bits	Name	Access	Description	Reset
31	MEND	WO	End of message type/last message of frame: 1: This message from the master ends with a stop bit and is the last message of the frame; 0: This message from the master is followed by a repeated start bit before another message can be issued. Note: This bit cannot be 0 when MTYPE[3:0]=0100b.	0
[30:27]	MTYPE[3:0]	WO	Message type. When I3C is used as master: 0000: SCL clock forced stop until the next control word is executed; 0001: Header message; 0010: Private message; 0011: Direct message (second part of I3C SDR direct CCC command); 0100: Legacy I2C message; Others: Reserved.	0

Bits	Name	Access	Description	Reset
[26:24]	Reserved	RO	Reserved.	0
[23:17]	ADD[6:0]	WO	7-bit I3C dynamic/I2C static slave address. Note: When I3C is used as master, this bit field is used if MTYPE[3:0] is 0010b, 0011b, or 0100b.	0
16	RNW	WO	Read/not-write message. 1: Read message; 0: Write message. Note: When I3C is used as master, this bit field is used if MTYPE[3:0] is 0010b, 0011b, or 0100b.	0
[15:0]	DCNT[15:0]	WO	Data count to be transferred during read/write messages, in bytes: 0x0000: No data to transfer; 0x0001: 1 byte; 0x0002: 2 bytes; ... 0xFFFF: 64KB - 1 byte. When I3C is used as master, this bit field is used to set the number of data bytes exchanged on the bus if MTYPE[3:0] is 0010b, 0011b, or 0100b. For private or legacy I2C read/write messages, this bit field must be non-empty.	0

19.4.2 I3C Control Register [Multiplexed] (R32_I3C_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEND	MTYPE[3:0]				Reserved				CCC[7:0]						
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DCNT[15:0]															

Bits	Name	Access	Description	Reset
31	MEND	WO	End of message type/last message of frame: 1: This message from the master ends with a stop bit and is the last message of the frame; 0: This message from the master is followed by a repeated start bit before another message can be issued. Note: This bit cannot be 0 when MTYPE[3:0]=0100b.	0
[30:27]	MTYPE[3:0]	WO	Message type. When I3C is used as master: 0110: Broadcast/direct CCC command; Bits [23:16] (CCC[7:0]) are the 8-bit CCC code issued; Bits [15:0] (DCNT[15:0]) are the number of CCC defining bytes, CCC sub-command bytes, or CCC data bytes. If bit[23] = CCC[7] = 1: The message is the first part of a direct CCC command for I3C SDR direct CCC command; the direct CCC command (first part) message is: – {S/S + 0b111_1110 + RnW = 0/Sr+*} + (direct CCC + T) + (8-bit data + T)* + Sr; – After S (start bit), the arbitrable header (0b111_1110 + RnW = 0) is inserted or not depending on the NOARBH bit in the R32_I3C_CFGR	0

Bits	Name	Access	Description	Reset
			register. – Sr+*: After Sr (repeated start bit), the hardware automatically inserts (0b111_1110 + R/W). If bit[23] = CCC[7] = 0: The message is an I3C SDR broadcast CCC command; the broadcast CCC command message transmitted is: – {S/S + 0b111_1110 + RnW = 0/Sr + *} + (broadcast CCC + T) + (8-bit data + T)* + Sr/P – After S (start bit), the arbitrable header (0b111_1110 + RnW = 0) is inserted or not depending on the NOARBH bit. – Sr+*: After Sr (repeated start bit), the hardware automatically inserts (0b111_1110 + R/W). Other values: Reserved.	
[26:24]	Reserved	RO	Reserved.	0
[23:16]	CCC[7:0]	WO	8-bit CCC code: If bit[23] = CCC[7] = 1, the message is the first part of an I3C SDR direct CCC command. If bit[23] = CCC[7] = 0, the message is an I3C SDR broadcast CCC command (including ENTDAAs).	0
[15:0]	DCNT[15:0]	WO	Count of CCC command-related data to be transmitted as CCC defining bytes, CCC sub-command bytes, or CCC data bytes (in bytes): Linear encoding up to 64KB - 1. 0x0000: No data to transfer; Note: Value forced when issuing ENTDAAs broadcast CCC. 0x0001: 1 byte; Note: Value forced when issuing RSTACT direct/broadcast CCC. 0x0002: 2 bytes; ... 0xFFFF: 64KB - 1 byte.	0

19.4.3 I3C Configuration Register (R32_I3C_CFGR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved	TSFSET	Reserved				CFLUSH				CDMAEN	TMODE	Reserved			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	TXTHRES	TXFLUSH	TXDMAEN	Reserved	RXTHRES	RXFLUSH	RXDMAEN	HJACK	PULLUP_EN	Reserved	EXITPTRN	RSTPTRN	NOARBH	CRIN	

Bits	Name	Access	Description	Reset
31	Reserved	RO	Reserved.	0
30	TSFSET	WO	Frame transfer setting: 1: Hardware sets the CFNFF bit to initiate frame transfer; 0: No operation. If this bit is not set to 1, frame transfer can be initiated by software by directly writing the first control word register (R32_I3C_CTLR) when the C-FIFO is empty (CFEF bit in R32_I3C_EVR register is 1). Then, if the first written control word is not marked as end of message (MEND bit in R32_I3C_CTLR register is 0), the hardware will set CFNFF to 1.	0
[29:22]	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
21	CFLUSH	WO	C-FIFO flush: 1: Flush the C-FIFO; 0: No operation.	0
20	CDMAEN	RW	C-FIFO DMA request enable: 1: Enable DMA mode for C-FIFO; 0: Disable DMA mode for C-FIFO.	0
19	TMODE	RW	Frame issued on the I3C bus to manage C-FIFO and TX-FIFO: 1: Preload C-FIFO and TX-FIFO before initiating frame transfer. 0: Do not preload C-FIFO and TX-FIFO before initiating frame transfer. Frame transfer starts immediately when the first control word appears in the C-FIFO.	0
[18:15]	Reserved	RO	Reserved.	0
14	TXTHRES	RW	TX-FIFO threshold: 1: 1 word/4-byte threshold; 0: 1-byte threshold.	0
13	TXFLUSH	WO	TX-FIFO flush: 1: Flush the TX-FIFO; 0: No operation.	0
12	TXDMAEN	RW	TX-FIFO DMA request enable: 1: Enable DMA mode for TX-FIFO; 0: Disable DMA mode for TX-FIFO.	0
11	Reserved	RO	Reserved.	0
10	RXTHRES	RW	RX-FIFO threshold: 1: 1 word/4-byte threshold; 0: 1-byte threshold.	0
9	RXFLUSH	WO	RX-FIFO flush: 1: Flush the RX-FIFO; 0: No operation.	0
8	RXDMAEN	RW	RX-FIFO DMA request enable: 1: Enable DMA mode for RX-FIFO; 0: Disable DMA mode for RX-FIFO.	0
7	HJACK	RW	Hot-join request acknowledge: 1: Hot-join request is acknowledged; 0: Hot-join request is not acknowledged.	0
6	PULLUP_EN	RW	SDA pull-up enable: 1: Enable SDA pull-up; 0: In low-power mode, the pull-up can be turned off to reduce power consumption.	1
5	Reserved	RO	Reserved.	0
4	EXITPTRN	RW	HDR exit timing enable: 1: Send HDR exit timing after issuing the message header (MTYPE[3:0] bit in R32_I3C_CTLR register is 0001b). This is used for master error detection and escalation handling when the slave does not respond to a private message or direct read CCC. 0: Do not send HDR exit timing after issuing the message header (MTYPE[3:0] bit in R32_I3C_CTLR register is 0001b). This is used to send	0

Bits	Name	Access	Description	Reset
			a header to test bus ownership after a master role switch if there are concerns (the new master has not enabled its master role by accessing the previous master within the delay time defined in the active state).	
3	RSTPTRN	RW	HDR reset timing enable: 1: Insert HDR reset timing before the stop bit of any issued frame (including RSTACT CCC command); 0: Issue a standard stop bit at the end of the frame.	0
2	NOARBH	RW	No arbitrable header after start bit: 1: No arbitrable header - For legacy I2C messages or I3C SDR private read/write messages, the slave address is issued directly after the start bit. - This is a more efficient option (when issuing 0x7E arbitrable header is useless), but can only be used when the master is certain that the addressed slave device cannot simultaneously issue an IBI or master-role request (to ensure that there is no erroneous parsing and potential conflict between the address issued by the master in open-drain mode and the same address that the slave device could issue after the start bit for IBI or MR). 0: Issue an arbitrable header (0b111_1110 + RnW = 0) after the start bit, before the legacy I2C message or I3C SDR private read/write message (default).	0
1	CRINIT	RW	Mark master/slave role: 1: Master role; 0: Slave role.	0
0	EN	RW	I3C enable: 1: Enable I3C; 0: Disable I3C.	0

19.4.4 I3C Receive Data Byte Register (R32_I3C_RDR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								RDB0[7:0]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	RDB0[7:0]	RO	8-bit received data on the I3C bus.	0

19.4.5 I3C Transmit Data Byte Register (R32_I3C_TDR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									TDB0[7:0]						

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	TDB0[7:0]	RO	8-bit transmitted data on the I3C bus.	0

19.4.6 I3C IBI Payload Data Register (R32_I3C_IBIDR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IBIDB3[7:0]									IBIDB2[7:0]						

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIDB1[7:0]									IBIDB0[7:0]						

Bits	Name	Access	Description	Reset
[31:24]	IBIDB3[7:0]	RW	8-bit IBI payload data (latest byte on the I3C bus).	0
[23:16]	IBIDB2[7:0]	RW	8-bit IBI payload data (byte following IBIDB1[7:0] on the I3C bus).	0
[15:8]	IBIDB1[7:0]	RW	8-bit IBI payload data (byte following IBIDB0[7:0] on the I3C bus).	0
[7:0]	IBIDB0[7:0]	RW	8-bit IBI payload data (earliest byte on the I3C bus, MDB[7:0] mandatory data byte).	0

19.4.7 I3C Reset Register (R32_I3C_RESET)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23		22	21	20	19	18	17	16
Reserved									HST_SIE_RST	Reserved						

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved															

Bits	Name	Access	Description	Reset
[31:25]	Reserved	RO	Reserved.	0x1
24	HST_SIE_RST	RW	Master reset register: 1: The master protocol layer is in reset state; 0: The master protocol layer is in operating state.	1
[23:0]	Reserved	RO	Reserved.	0

19.4.8 I3C Status Register (R32_I3C_STATR)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MID[7:0]									Reserved			DIR	ABT	Reserved	

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XDCNT[15:0]															

Bits	Name	Access	Description	Reset
[31:24]	MID[7:0]	RO	Message identifier/counter for a given frame: When I3C is used as master, this bit field identifies the control word message (R32_I3C_CTLR) referenced by the R32_I3C_STATR status register. The first message of a frame is identified with MID[7:0] = 0. When a new message control word (R32_I3C_CTLR) on the I3C bus is completed, this bit field is incremented (by hardware). This bit field is reset each time a new frame starts.	0
[23:19]	Reserved	RO	Reserved.	0
18	DIR	RO	Message direction: 1: Read; 0: Write. Note: ENTDAACCC is considered a write command.	0
17	ABT	RO	Slave early termination of private read message: 1: Slave completed early; 0: Slave did not complete early.	0
16	Reserved	RO	Reserved.	0
[15:0]	XDCNT[15:0]	RO	Data counter: Condition: During dynamic address assignment process (ENTDAACCC) - When I3C is used as master: Number of slaves detected on the bus; Condition: For other transfers during a message - Regardless of whether I3C is used as master: Number of data bytes read from the I3C bus or sent on the I3C bus during the message.	0

19.4.9 I3C Status Error Register (R32_I3C_STATER)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DERR		DNACK		ANACK		COVR		DOVR		Reserved	
												PERR		CODERR[3:0]	

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
10	DERR	RO	Data error: 1: When the received slave address and/or parity bit does not match, the master detects a data error during the master role switching process (GETACCCR CCC, formerly known as GETACCMST). The active master retains the master role. 0: No error detected.	0
9	DNACK	RO	Data not acknowledged: 1: The master detects that the slave did not acknowledge a data byte during: i) Legacy I2C write transfer; ii) Second attempt to send dynamic address during ENTDAAC process. 0: No error detected.	0

Bits	Name	Access	Description	Reset
8	ANACK	RO	Address not acknowledged: 1: The master detects that the slave did not acknowledge the static/ dynamic address during: i) Legacy I2C read/write transfer; ii) Direct CCC write transfer; iii) Second attempt of direct CCC read transfer; iv) Private read/write transfer; 0: No error detected.	0
7	COVR	RO	C-FIFO underflow: 1: The master detects: i) C-FIFO underflow: The control FIFO is empty and a repeated start bit must be issued; 0: No error detected.	0
6	DOVR	RO	RX-FIFO overflow or TX-FIFO underflow: i) TX-FIFO underflow: The TX-FIFO is empty and a write data byte must be sent; ii) RX-FIFO overflow: The RX-FIFO is full and a new data byte is received; 0: No error detected.	0
5	Reserved	RO	Reserved.	0
4	PERR	RO	Protocol error: 1: Protocol error detected; 0: No error detected.	0
[3:0]	CODERR[3:0]	RO	0000: CE0 error (transaction after sending CCC): The master detects an illegally formatted CCC; 0001: CE1 error (watchdog error): The master detects that the data sent on the bus does not match expectations; 0010: CE2 error (no response to broadcast address): The master detects that the broadcast address (0b111_1110) is not acknowledged; 0011: CE3 error (master role switch failure): The master detects that the new master did not drive the bus after the master role switch; Other values: Reserved.	0

19.4.10 I3C Receive Message Register (R32_I3C_RMR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								RADD[6:0]						Reserved	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RCODE[7:0]								Reserved				IBIRDCNT[2:0]			

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
[23:17]	RADD[6:0]	RO	Received slave address: When I3C is configured as master, this bit field records the dynamic	0

Bits	Name	Access	Description	Reset
			address received from the slave during an acknowledged IBI or master-role request.	
16	Reserved	RO	Reserved.	0
[15:8]	RCODE[7:0]	RO	Received CCC code.	0
[7:3]	Reserved	RO	Reserved.	0
[2:0]	IBIRDCNT[2:0]	RO	IBI received payload data count: When I3C is configured as master, this bit field records the number of valid data bytes received in the R32_I3C_IBIDR register. Note: For chips with a lot number second-to-last digit greater than 0, it becomes 0x7 after clearing the IBI request flag.	0x7

19.4.11 I3C Event Register (R32_I3C_EVR)

Offset address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												HJF	Reserved		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIF	Reserved	ERRF	RXTGTENDF	FCF	Reserved	RXLASTF	TXLASTF	RXFNEF	TXFNFF	Reserved	CFNFF	TXFEF	CFEF		

Bits	Name	Access	Description	Reset
[31:20]	Reserved	RO	Reserved.	0
19	HJF	RO	Hot-join flag: When I3C is used as master, this flag is set to 1 by hardware to indicate that a hot-join request has been received. This flag is cleared by software writing 1 to the CHJF bit in the R32_I3C_CEV register.	0
[18:16]	Reserved	RO	Reserved.	0
15	IBIF	RO	IBI flag: When I3C is used as master, this flag is set to 1 by hardware to indicate that an IBI request has been received. This flag is cleared by software writing 1 to the CIBIF bit in the R32_I3C_CEV register.	0
[14:12]	Reserved	RO	Reserved.	0
11	ERRF	RO	Error flag: This flag is set to 1 by hardware to indicate that an error has occurred. Software must then read R32_I3C_STATR to obtain the error type. This flag is cleared by software writing 1 to the CERRF bit in the R32_I3C_CEV register.	0
10	RXTGTENDF	RO	Slave-initiated read end flag: When I3C is used as master, this flag indicates that the slave has terminated the read transfer early. Software must then read the status register R32_I3C_STATR to check information related to the last message and obtain the number of data bytes received in the prematurely terminated read transfer (XDCNT in R32_I3C_STATR register).	0

Bits	Name	Access	Description	Reset
			This flag is cleared by software writing 1 to the CRXTGTENDF bit in the R32_I3C_CEV register.	
9	FCF	RO	Frame complete flag: When I3C is used as master, this flag is set to 1 by hardware to indicate that the frame on the I3C bus has completed (normally), e.g., when a stop bit is issued. This flag is cleared by software writing 1 to the CFCF bit in the R32_I3C_CEV register.	0
8	Reserved	RO	Reserved.	0
7	RXLASTF	RO	Last read data byte/word flag: When I3C is used as master, this flag is set to 1 by hardware to indicate the last data byte/word of the message that must be read from the RX-FIFO (depending on the RXTHRES bit in the I3C_CFGR register). When the last data byte/word of the message is read, this flag is cleared by hardware.	0
6	TXLASTF	RO	Last written data byte/word flag: This flag is set to 1 by hardware to indicate the last data byte/word of the message that must be written to the TX-FIFO (depending on the TXTHRES bit in the R32_I3C_CFGR register). When the last data byte/word of the message is written, this flag is cleared by hardware.	0
5	RXFNEF	RO	RX-FIFO not empty flag: This flag is set/cleared by hardware to indicate that data bytes must/must not be read from the RX-FIFO. Note: Before reading a data byte from the RX-FIFO (reading R32_I3C_RDR, depending on RXTHRES), software must wait for RXFNEF = 1 (through polling or through an enabled interrupt).	0
4	TXFNFF	RO	TX-FIFO not full flag: This flag is set/cleared by hardware to indicate that data bytes must/must not be written to the TX-FIFO. Note: Before writing to the TX-FIFO (writing R32_I3C_TDR, depending on TXTHRES), software must wait for TXFNFF = 1 (through polling or through an enabled interrupt).	0
3	Reserved	RO	Reserved.	0
2	CFNFF	RO	C-FIFO not full flag: When I3C is used as master, this flag is set to 1 by hardware to indicate that a control word must be written to the C-FIFO. This flag is cleared by hardware to indicate that no control word should be written to the C-FIFO. Note: Before writing to the C-FIFO (writing R32_I3C_CTLR), software must wait for CFNFF = 1 (through polling or through an enabled interrupt).	0
1	TXFEF	RO	TX-FIFO empty flag: This flag is set to 1 by hardware to indicate that the TX-FIFO is empty; This flag is cleared by hardware to indicate that the TX-FIFO is not empty.	1
0	CFEF	RO	C-FIFO empty flag: When used as master: This flag is set to 1 by hardware to indicate that the C-FIFO is empty;	1

Bits	Name	Access	Description	Reset
			This flag is cleared by hardware to indicate that the C-FIFO is not empty. When used as slave: This flag is set to 1 by hardware to indicate that the R32_I3C_CTLR register does not contain a control word (no IBI/CR/HJ request); This flag is cleared by hardware to indicate that the R32_I3C_CTLR register contains a control word (pending IBI/CR/HJ request). Note: If C-FIFO and TX-FIFO preload is configured (TMODE bit in R32_I3C_CFGR register is 1), software must wait for TXFEF = 1 and CFEF = 1 before starting a new frame transfer.	

19.4.12 I3C Interrupt Enable Register (R32_I3C_INTENR)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												HJIE	Reserved		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IBIIE	Reserved		ERRIE	RXTGTENDIE		FCIE	Reserved		RXFNEIE	TXFNEIE	Reserved		CFNFIE	Reserved	

Bits	Name	Access	Description	Reset
[31:20]	Reserved	RO	Reserved.	0
19	HJIE	RW	Hot-join interrupt enable: 1: Enable; 0: Disable.	0
[18:17]	Reserved	RO	Reserved.	0
15	IBIIE	RW	IBI request interrupt enable: 1: Enable; 0: Disable.	0
[14:12]	Reserved	RO	Reserved.	0
11	ERRIE	RW	Error interrupt enable: 1: Enable; 0: Disable.	0
10	RXTGTENDIE	RW	Slave-initiated read end interrupt enable: 1: Enable; 0: Disable.	0
9	FCIE	RW	Frame complete interrupt enable: 1: Enable; 0: Disable.	0
[8:6]	Reserved	RO	Reserved.	0
5	RXFNEIE	RW	RX-FIFO not empty interrupt enable: 1: Enable; 0: Disable.	0
4	TXFNEIE	RW		0

Bits	Name	Access	Description	Reset
			TX-FIFO not full interrupt enable: 1: Enable; 0: Disable.	
3	Reserved	RO	Reserved.	0
2	CFNFIE	RW	C-FIFO not full interrupt enable: 1: Enable; 0: Disable.	0
[1:0]	Reserved	RO	Reserved.	0

19.4.13 I3C Clear Event Register (R32_I3C_CEVr)

Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												CHJF	Reserved		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CIBIF	Reserved			CERRF	CRXTGTENDF			CFCF	Reserved						

Bits	Name	Access	Description	Reset
[31:20]	Reserved	RO	Reserved.	0
19	CHJF	WO	Clear hot-join flag: 1: Clear HJF; 0: No effect.	0
[18:16]	Reserved	RO	Reserved.	0
15	CIBIF	WO	Clear IBI request flag: 1: Clear IBIF; 0: No effect.	0
[14:12]	Reserved	RO	Reserved.	0
11	CERRF	WO	Clear error flag: 1: Clear ERRF; 0: No effect.	0
10	CRXTGTENDF	WO	Clear slave-initiated read end flag: 1: Clear RXTGTENDF; 0: No effect.	0
9	CFCF	WO	Clear frame complete flag: 1: Clear FCF; 0: No effect.	0
[8:0]	Reserved	RO	Reserved.	0

19.4.14 I3C Device Register 0 (R32_I3C_DEVR0)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DA[6:0]				DAVAL			

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:1]	DA[6:0]	RW	7-bit dynamic address: When I3C is used as master, this bit field can be written by software to define its own dynamic address.	0
0	DAVAL	RW	Dynamic address valid: When I3C is used as master, this bit can be written by software to validate its own dynamic address.	0

19.4.15 I3C Device Register x (R32_I3C_DEVRx) (x=1/2/3/4)

Offset address: 0x64, 0x68, 0x6C, 0x70

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DIS	Reserved										SUSP	IBIDEN	Reserved		IBIACK

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DA[6:0]				Reserved			

Bits	Name	Access	Description	Reset
31	DIS	RO	DA[6:0] write disable: 1: Disable/lock write operations to DA[7:0] and IBIDEN; 0: Allow write operations to DA[7:0] and IBIDEN in the R32_I3C_DEVRx register.	0
[30:20]	Reserved	RO	Reserved.	0
19	SUSP	RW	Suspend/stop I3C transfer upon receiving IBI: 1: After receiving an IBI request from slave x, stop I3C transfer and flush C-FIFO and TX-FIFO; 0: After acknowledging and completing the IBI request from slave x, do not flush C-FIFO and TX-FIFO, and issue a repeated start bit or stop bit depending on whether there is a next control word.	0
18	IBIDEN	RW	IBI data enable: 1: The acknowledged IBI from slave x is followed by a mandatory data byte MDB[7:0]; 0: The acknowledged IBI from slave x has no data byte.	0
17	Reserved	RO	Reserved.	0
16	IBIACK	RW	IBI request acknowledge: 1: The IBI request from slave x must be ACKed (via the 7-bit dynamic address DA[6:0]) - The DIS bit field is set to 1 by hardware to protect DA[6:0] from software modification, while the hardware can internally store the current DA[6:0] into the core clock domain. - After ACK, the master records the IBI payload data (if any) according to I3C_DEVRx.IBIDEN. - Software is notified through the IBI flag (IBIF=1) and/or the corresponding interrupt (if enabled); - Regardless of the IBIACK configuration of this device or other devices,	0

Bits	Name	Access	Description	Reset
			all other IBI requests will be NACKed until the IBI request flag (IBIF) and master-role request flag (CRF) are both cleared. 0: The IBI request from slave x must be NACKed. After NACK, the message continues transfer as initially programmed (the slave has received NACK and can issue another IBI request later).	
[15:8]	Reserved	RO	Reserved.	0
[7:1]	DA[6:0]	RW	I3C dynamic address assigned to slave x: When I3C is used as master, this bit field must be written by software to store the 7-bit dynamic address sent by the master through the broadcast ENTDA or direct SETNEWDA CCC acknowledged by slave x.	0
0	Reserved	RO	Reserved.	0

19.4.16 I3C Timing Register 0 (R32_I3C_TIMINGR0)

Offset address: 0xA0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SCLH_I2C[7:0]								SCLL_OD[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SCLH_I3C[7:0]								SCLL_PP[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	SCLH_I2C[7:0]	RW	SCL high-level duration for legacy I2C messages, in core clock cycles: $t_{SCLH_I2C} = (SCLH_I2C + 1) \times t_{HCLK}$	0
[23:16]	SCLL_OD[7:0]	RW	SCL low-level duration for the open-drain phase, for legacy I2C messages and I3C open-drain phase (address phase after start bit, ACK phase during master-initiated messages, and T bit phase during direct/private IBI payload), in core clock cycles: $t_{SCLL_OD} = (SCLL_OD + 1) \times t_{HCLK}$	0
[15:8]	SCLH_I3C[7:0]	RW	SCL high-level duration for I3C messages (in push-pull and open-drain phases), in core clock cycles: $t_{SCLH_I3C} = (SCLH_I3C + 1) \times t_{HCLK}$	0
[7:0]	SCLL_PP[7:0]	RW	SCL low-level duration for the I3C push-pull phase, in core clock cycles: $t_{SCLL_PP} = (SCLL_PP + 1) \times t_{HCLK}$	0

19.4.17 I3C Timing Register 1 (R32_I3C_TIMINGR1)

Offset address: 0xA4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				SDA_HD		Reserved				FREE[6:0]					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						ASNCR[1:0]			AVAL[7:0]						

Bits	Name	Access	Description	Reset
[31:29]	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
28	SDA_HD	RW	SDA hold time, in core clock cycles. $\text{SDA hold time} = (\text{SDA_HD} + 0.5) \times \text{tHCLK}$	0
[27:23]	Reserved	RO	Reserved.	0
[22:16]	FREE[6:0]	RW	Core clock cycles used to set some MIPI timings (e.g., bus idle condition time).	0
[15:10]	Reserved	RO	Reserved.	0
[9:8]	ASNCR[1:0]	RW	Active state of the new master: The time that must be waited before being accessed as a new slave, see AVAL[7:0].	0
[7:0]	AVAL[7:0]	RW	Core clock cycles used to set the 1 μ s time unit: The hardware then uses this time unit to build some internal timers, corresponding to the following MIPI I3C timings: When I3C is used as master, the time it stalls the clock must not exceed the maximum stall time (stalling SCL clock low), as follows: 1. For the first bit of the address assigned during dynamic address assignment: The clock stall time must not exceed the MIPI timing $\text{tSTALLDAA} = 15\text{ms}$. This timing is defined as: $\text{tSTALLDAAmax} = (\text{AVAL}[7:0] + 1) \times 15000 \times \text{tHCLK}$ 2. For the ACK/NACK phase of I3C/I2C transfers, the parity bit of write data transfers, and the transition bit of I3C read transfers: The clock stall time must not exceed the MIPI timing $\text{tSTALL} = 100\mu\text{s}$. This timing is defined as: $\text{tSTALLmax} = (\text{AVAL}[7:0] + 1) \times 100 \times \text{tHCLK}$	0

19.4.18 I3C Timing Register 2 (R32_I3C_TIMINGR2)

Offset address: 0xA8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STALL[7:0]						Reserved			STALLA		STALLC		STALLD		STALLT

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:8]	STALL[7:0]	RW	Master clock stall time, in core clock cycles: $\text{tSCLL_STALL} = \text{STALL} \times \text{tHCLK}$	0
[7:4]	Reserved	RO	Reserved.	0
3	STALLA	RW	Master clock stall enable for ACK phase: 1: Enable stall; 0: Disable stall.	0
2	STALLC	RW	Master clock stall enable for CCC PAR phase: 1: Enable stall; 0: Disable stall.	0
1	STALLD	RW		0

Bits	Name	Access	Description	Reset
			Master clock stall enable for data PAR phase: 1: Enable stall; 0: Disable stall.	
0	STALLT	RW	Master clock stall enable for data T bit phase (and ACK/NACK phase of data bytes in legacy I2C read messages): 1: Enable stall; 0: Disable stall.	0

19.4.19 I3C Get Maximum Data Speed Register (R32_I3C_GETMSDR)

Offset address: 0xD0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							TSCO	Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								Reserved							

Bits	Name	Access	Description	Reset
[31:25]	Reserved	RO	Reserved.	0
24	TSCO	RW	Clock-to-data turnaround time (tSCO): 1: tSCO > 12ns; 0: tSCO ≤ 12ns.	0
[23:0]	Reserved	RO	Reserved.	0

Chapter 20: Serial Peripheral Interface (SPI/I2S)

SPI supports data exchange in three-wire synchronous serial mode. With the chip select line, it supports hardware switching between master and slave modes, and supports communication over a single data line.

I2S is also a three-wire synchronous serial interface communication protocol. It supports four audio standards, including the Philips I2S standard, MSB-aligned standard, LSB-aligned standard, and PCM standard.

20.1 Main Features

20.1.1 SPI Features

- Supports full-duplex synchronous serial mode
- Supports single-wire half-duplex mode
- Supports master mode and slave mode, multi-slave mode
- Supports 8-bit or 16-bit data structure
- Maximum clock frequency supports up to half of Fpclk
- Data order supports MSB-first or LSB-first
- Supports hardware or software controlled NSS pin
- Transmit and receive support hardware CRC verification
- Transmit and receive buffers support DMA transfer
- Supports configurable clock phase and polarity

20.1.2 I2S Features

- Supports simplex communication
- Supports master mode and slave mode
- Supports 16-bit, 24-bit, and 32-bit data formats
- Audio sampling frequency range: 8 kHz – 562.2 kHz
- Supports programmable clock polarity
- Supports common I2S protocols: Philips standard, MSB-aligned standard, LSB-aligned standard, and PCM standard
- Transmit and receive buffers support DMA transfer
- Supports master clock output to external audio devices

20.2 SPI Functional Description

20.2.1 Overview

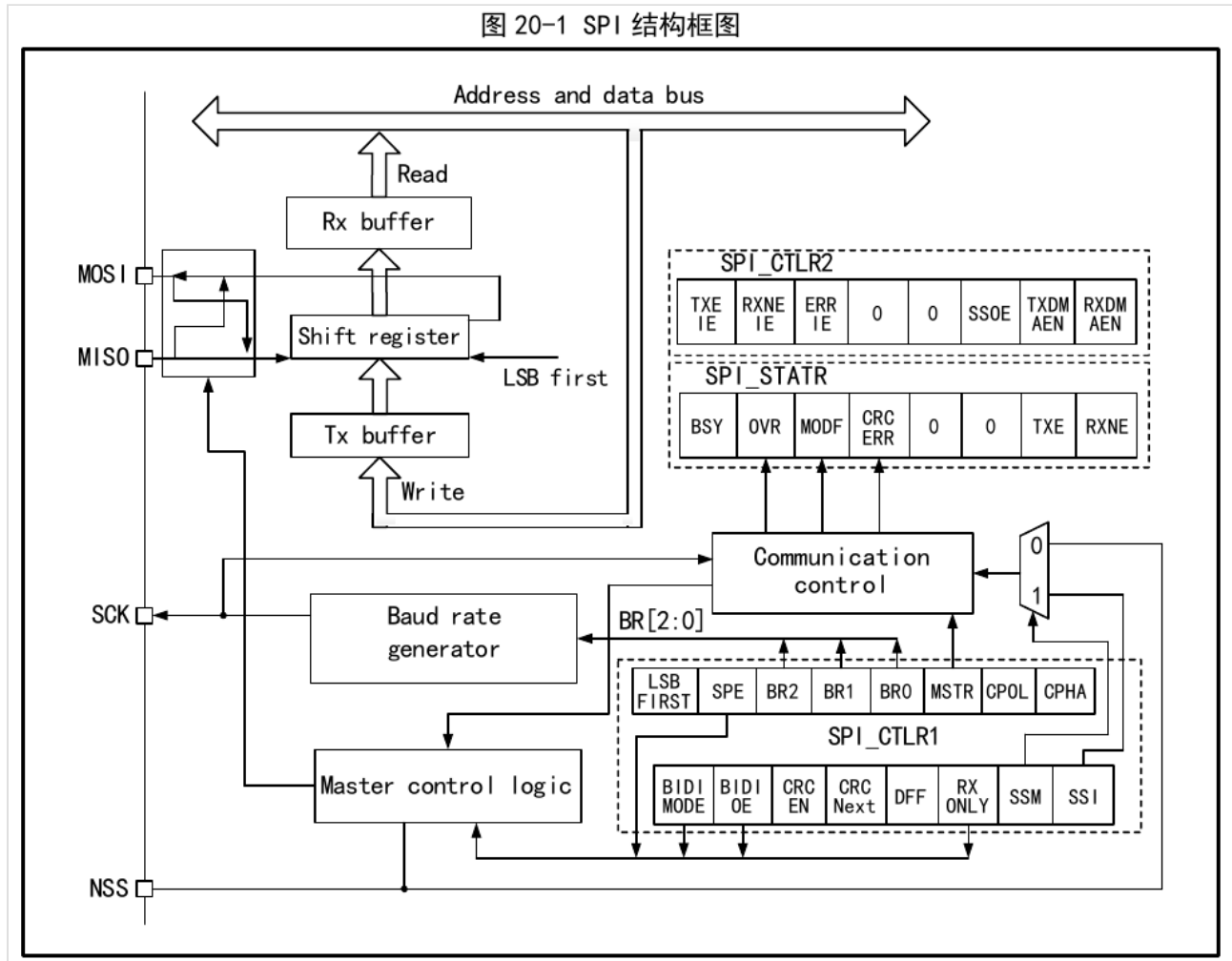


Figure 20-1: SPI block diagram

As can be seen from Figure 20-1, the four pins mainly related to SPI are MISO, MOSI, SCK, and NSS. Among them, the MISO pin is the data input pin when the SPI module operates in master mode, and the data output pin when operating in slave mode. The MOSI pin is the data output pin when operating in master mode, and the data input pin when operating in slave mode. SCK is the clock pin; the clock signal is always output by the master, and the slave receives the clock signal and synchronizes data transmission/reception. The NSS pin is the chip select pin, with the following usage:

1. NSS controlled by software: In this case SSM is set, and the internal NSS signal level (high or low) is determined by SSI. This is generally used in SPI master mode.
2. NSS controlled by hardware: When NSS output is enabled (SSOE set), the SPI master actively pulls the NSS pin low when transmitting outward. If the NSS pin cannot be successfully pulled low, it indicates that another master device is still communicating on the bus, and a hardware error is generated. When SSOE is not set, it can be used in multi-master mode; if it is pulled low, the device is forced into slave mode and the MSTR bit is automatically cleared.

The operating mode of SPI can be configured through CPHA and CPOL. CPHA set means the module samples data on the second clock edge and data is latched; CPHA not set means the SPI module

samples data on the first clock edge and data is latched. CPOL indicates whether the clock stays high or low when there is no data. See Figure 20-2 below for details.

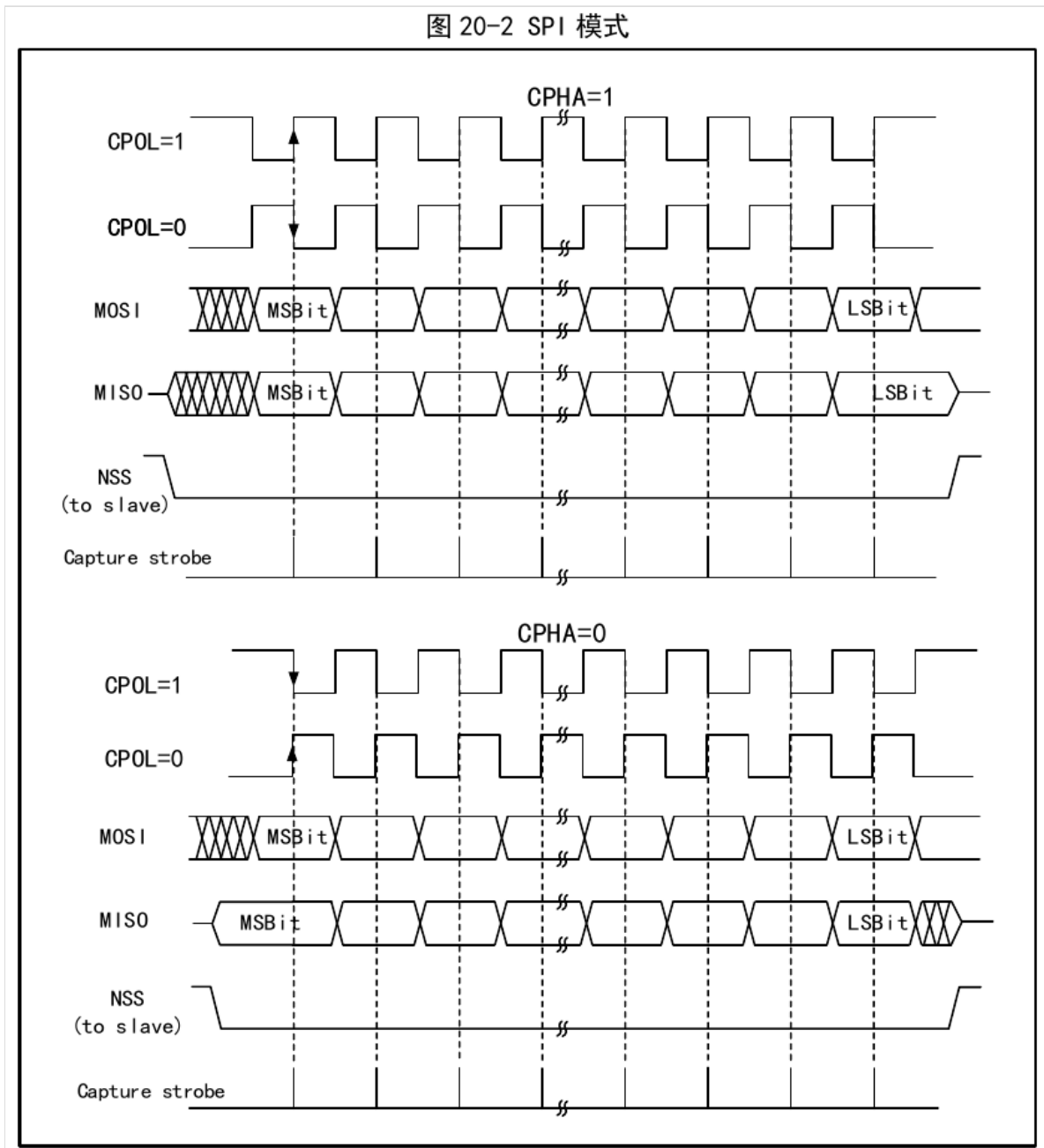


Figure 20-2: SPI modes

The master and device need to be set to the same SPI mode. Before configuring the SPI mode, the SPE bit must be cleared. The DFF bit determines whether a single SPI data length is 8-bit or 16-bit. LSBFIRST controls whether a single data word is sent MSB-first or LSB-first.

20.2.2 Master Mode

When the SPI module operates in master mode, SCK generates the serial clock. The steps to configure master mode are as follows:

- Configure the BR[2:0] field of the control register to determine the clock frequency;
- Configure CPOL and CPHA bits to determine the SPI mode;

- Configure DFF to determine the data word length;
- Configure LSBFIRST to determine the frame format;
- Configure the NSS pin, e.g., set the SSOE bit to let hardware control NSS. Alternatively, set the SSM bit and set SSI high;
- Set the MSTR bit and SPE bit; ensure NSS is already high at this point.

To send data, simply write the data to be transmitted to the data register. SPI transfers data from the transmit buffer to the shift register in parallel, then shifts data out from the shift register according to the LSBFIRST setting. When data has been moved to the shift register, the TXE flag is set; if TXEIE has been set, an interrupt is generated. When the TXE flag is set, data needs to be written to the data register to maintain a complete data stream.

When receiving data, upon the last sampling clock edge of the data word, data is transferred in parallel from the shift register to the receive buffer, and the RXNE bit is set. If RXNEIE was previously set, an interrupt is also generated. At this point the data register should be read as soon as possible to retrieve the data.

20.2.3 Slave Mode

When the SPI module operates in slave mode, SCK is used to receive the clock from the master, and the local baud rate setting has no effect. The steps to configure slave mode are as follows:

- Configure the DFF bit to set the data bit length;
- Configure CPOL and CPHA bits to match the master mode;
- Configure LSBFIRST to match the master data frame format;
- In hardware management mode, the NSS pin must be kept low. If NSS is set to software management (SSM set), keep SSI not set;
- Clear the MSTR bit, set the SPE bit to enable the SPI module.

During transmission, when the first slave sampling clock edge appears on SCK, the slave begins transmitting. The transmission process moves data from the transmit buffer to the transmit shift register; once the data in the transmit buffer has been moved to the shift register, the TXE flag is set. If TXEIE was previously set, an interrupt is generated.

During reception, after the last sampling clock edge, the RXNE bit is set, and the byte received by the shift register is transferred to the receive buffer. Reading the data register retrieves the data from the receive buffer. If RXNEIE was set before RXNE is set, an interrupt is generated.

20.2.4 Simplex Mode

The SPI interface can operate in half-duplex mode, where the master device uses the MOSI pin and the slave device uses the MISO pin for communication. To use half-duplex communication, BIDIMODE must be set, and BIDIOE controls the transfer direction.

In normal full-duplex mode, setting the RXONLY bit puts the SPI module into receive-only simplex mode. After RXONLY is set, one data pin is released; the released pin differs between master and slave modes. Alternatively, the received data can be ignored and the SPI can be set to transmit-only mode.

20.2.5 CRC

The SPI module uses CRC verification to ensure the reliability of full-duplex communication. Separate CRC calculators are used for data transmission and reception. The polynomial used for CRC calculation

is determined by the polynomial register. Different calculation methods are used for 8-bit data width and 16-bit data width respectively.

Setting the CRCEN bit enables CRC verification and simultaneously resets the CRC calculators. After transmitting the last data byte, setting the CRCNEXT bit causes the calculation result of the TXCRCR calculator to be transmitted after the current byte is sent. If the value of the last received shift register does not match the locally calculated RXCRCR value, the CRCERR bit is set. To use CRC verification, the polynomial calculator must be configured and the CRCEN bit set when configuring the SPI operating mode. The CRCNEXT bit should be set on the last word or half-word to send the CRC and verify the received CRC. Note: the CRC calculation polynomial must be the same for both the transmitter and receiver.

20.2.6 DMA

The SPI module supports using DMA to accelerate data communication speed. DMA can be used to fill data into the transmit buffer or to promptly retrieve data from the receive buffer. DMA uses RXNE and TXE as signals to promptly retrieve or send data. DMA can also operate in simplex or CRC-verified mode.

20.2.7 Errors

Master mode fault error

When the SPI operates in NSS pin hardware management mode and an external operation pulls the NSS pin low; or in NSS pin software management mode, the SSI bit is cleared; or the SPE bit is cleared, causing the SPI to be disabled; or the MSTR bit is cleared, the SPI enters slave mode. If the ERRIE bit has been set, an interrupt is generated. To clear the MODF bit: first perform a read or write operation on R16_SPI1_STATR, then write R16_SPI1_CTLR1.

Overrun error

If the master sends data while the slave's receive buffer still contains unread data, an overrun error occurs and the OVR bit is set. If ERRIE is set, an interrupt is generated. An overrun error should cause the current transmission to restart. Reading the data register followed by reading the status register clears this bit.

CRC error

When the received CRC verification word does not match the RXCRCR value, a CRC verification error occurs and the CRCERR bit is set.

20.2.8 Interrupts

The SPI module supports five interrupt sources. The transmit buffer empty and receive buffer not empty events set the TXE and RXNE flags respectively, generating interrupts when the TXEIE and RXNEIE bits are set. In addition, the three errors mentioned above — MODF, OVR, and CRCERR — also generate interrupts when the ERRIE bit is enabled.

20.3 I2S Functional Description

20.3.1 I2S Overview

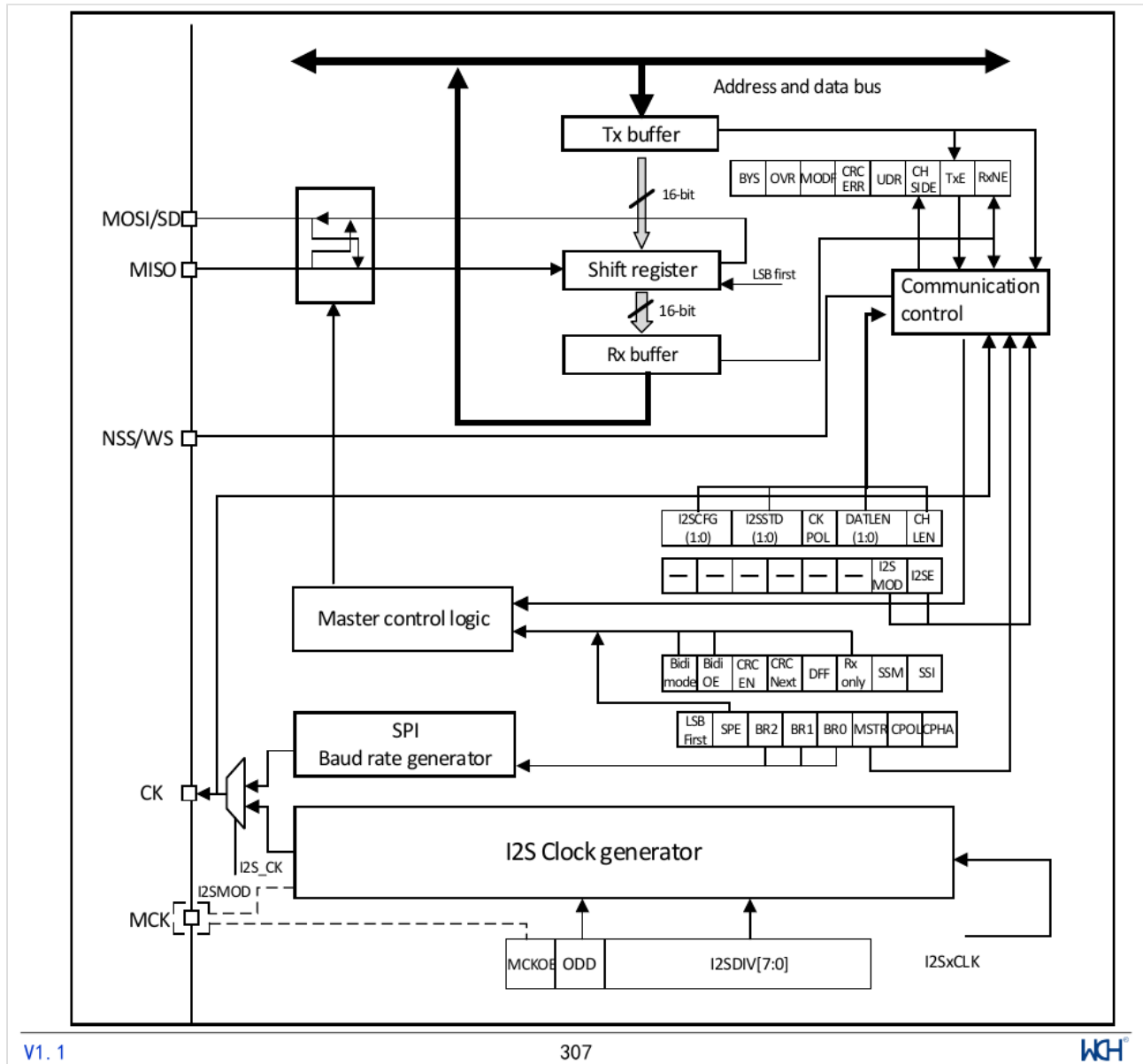


Figure 20-3: I2S block diagram

By setting the I2SMOD bit of the I2SCFGR register, the I2S function is enabled. At this point, the SPI module can be used as an I2S audio interface. I2S shares three pins with SPI:

- SD: Serial data (mapped to the MOSI pin), used to transmit and receive time-division multiplexed data for 2 channels;
- WS: Word select (mapped to the NSS pin), output as a data control signal in master mode, input in slave mode;
- CK: Serial clock (mapped to the SCK pin), output as a clock signal in master mode, input in slave mode.

When some external audio devices require a master clock, an additional pin can output the clock:

- **MCK:** Master clock (independently mapped). When I2S is configured as master mode and the MCKOE bit of the I2SPR register is 1, it is used as a pin to output an additional clock signal. The output clock signal frequency is preset to $256 \times F_s$, where F_s is the sampling frequency of the audio signal.

When configured as master mode, I2S uses its own clock generator to produce the clock signal for communication. This clock generator is also the clock source for the master clock output. In I2S mode there are 2 additional registers: one is the I2SPR register related to clock generator configuration, and the other is the I2S general configuration register I2SCFGR (which can set parameters such as audio standard, slave/master mode, data format, data packet frame, clock polarity, etc.). In I2S mode, the CTLR1 register and all CRCR registers are not used. Likewise, in I2S mode the SSOE bit of the CTLR2 register, and the MODF and CRCERR bits of the STATR register are not used. I2S uses the same DATAR register as SPI for 16-bit-wide mode data transfer.

20.3.2 Supported Audio Protocols

The three-wire bus supports time-division multiplexing of audio data on 2 channels: left channel and right channel, but only one 16-bit register is used for transmission or reception. Therefore, software must write the corresponding data to the data register according to the channel currently being transmitted. Similarly, when reading register data, the CHSIDE bit of the STATR register is checked to determine which channel the received data belongs to. The left channel is always transmitted before the right channel (the CHSIDE bit has no meaning under the PCM protocol). There are four available data and packet frame combinations. Data can be sent in the following four data formats:

- 16-bit data packed into a 16-bit frame
- 16-bit data packed into a 32-bit frame
- 24-bit data packed into a 32-bit frame
- 32-bit data packed into a 32-bit frame

When extending 16-bit data to a 32-bit frame, the first 16 bits (MSB) are meaningful data and the last 16 bits (LSB) are forced to 0. This operation requires no software intervention and no additional DMA request (only one read or write operation is needed). The 24-bit and 32-bit data frames require the CPU to perform 2 read or write operations on the DATAR register; when using DMA, 2 DMA transfers are needed. For 24-bit data, after extension to 32 bits, the lowest 8 bits are set to 0 by hardware. For all data formats and communication standards, the most significant bit (MSB) is always sent first. The I2S interface supports four audio standards, selectable by setting the I2SSTD[1:0] bits and the PCMSYNC bit of the I2SCFGR register.

20.3.2.1 I2S Philips Standard

In this standard, the WS pin is used to indicate which channel the data being transmitted belongs to. The WS signal becomes active 1 clock cycle before the first data bit (MSB) is transmitted. The transmitter changes data on the falling edge of the clock signal (CK), and the receiver reads data on the rising edge. The WS signal also changes on the falling edge of the clock signal.

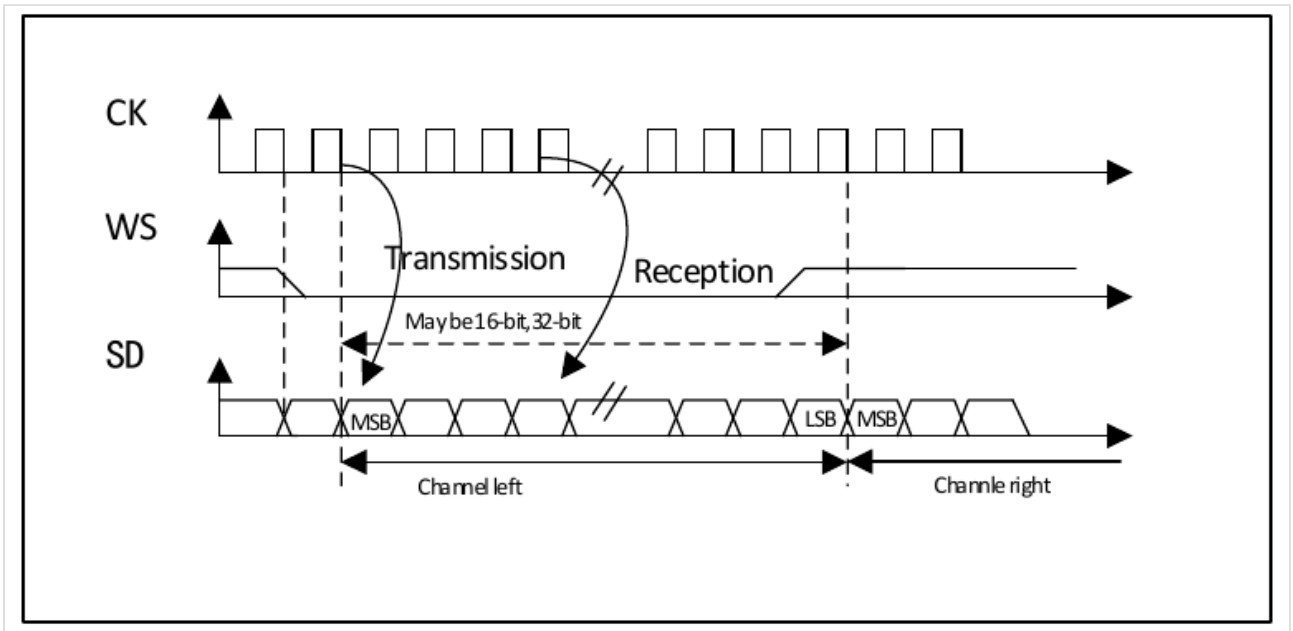
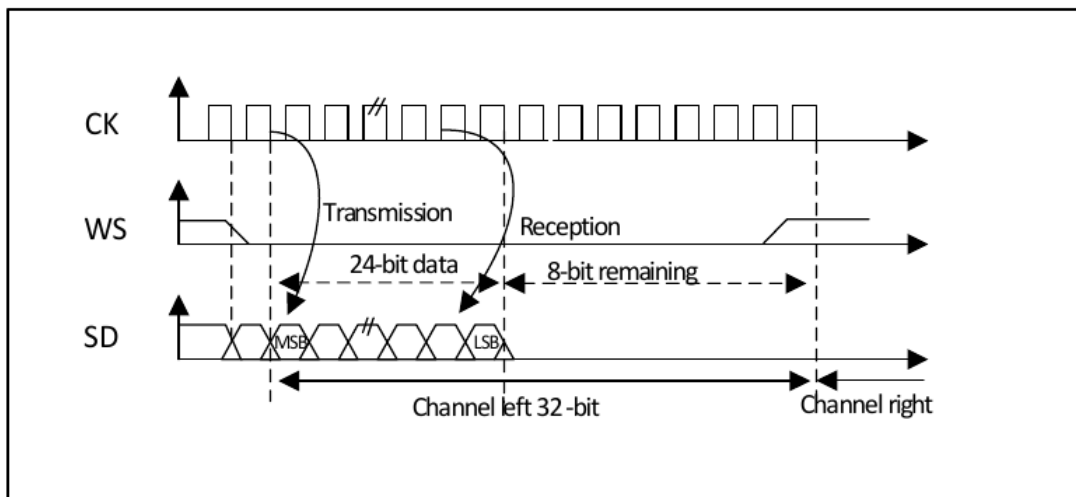


Figure 20-4: Philips protocol waveform (16/32-bit full precision, CPOL=0)

图 20-5 飞利浦协议波形（24 位帧，CPOL=0）



此模式需要对寄存器 SPI_DATAR 进行 2 次读或写操作。在发送模式下：如果需要发送 0x8EAA33

Figure 20-5: Philips protocol waveform (24-bit frame, CPOL=0)

This mode requires 2 read or write operations to the SPI_DATAR register. In transmit mode, to send 0x8EAA33 (24-bit):

- First write to Data register: 0x8EAA
- Second write to Data register: 0x33XX — Only the 8 MSB are sent to complete the 24 bits. The 8 LSB have no meaning and could be anything.

In receive mode, to receive 0x8EAA33:

- First read from Data register: 0x8EAA
- Second read from Data register: 0x3300 — Only the 8 MSB are valid; the 8 LSB will always be 00.

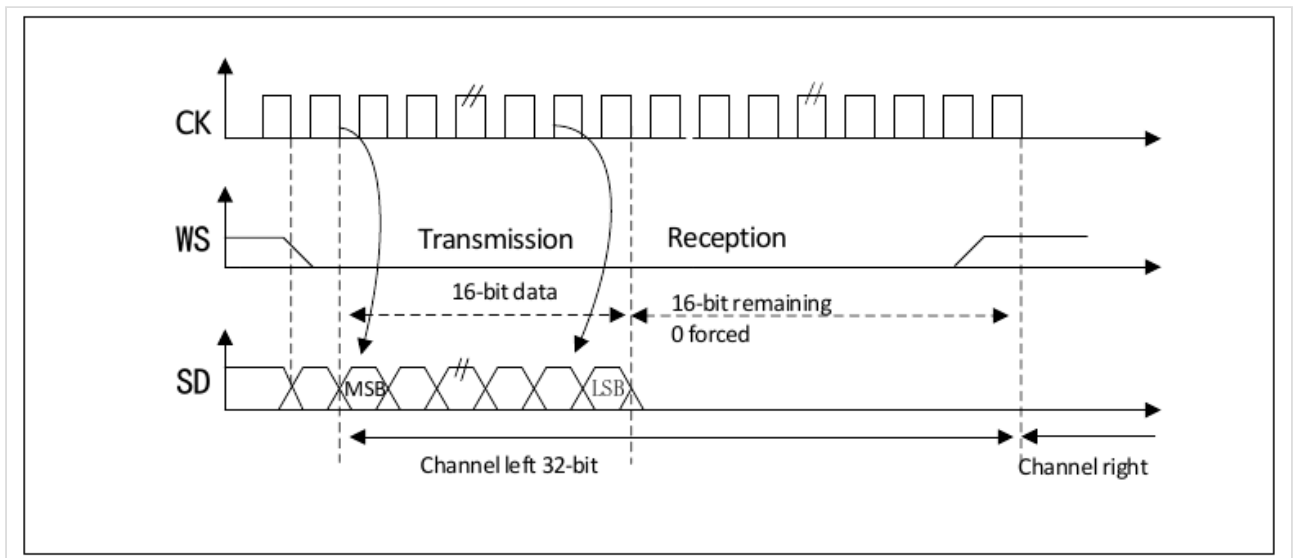


Figure 20-6: Philips protocol waveform (16-bit extended to 32-bit packet frame, CPOL=0)

During the I2S configuration phase, if 16-bit data is selected to be extended to a 32-bit channel frame, only one access to the DATAR register is needed. The lower 16 bits used for extension to 32 bits are set to 0x0000 by hardware. If the data to be transmitted or received is 0x76A3 (extended to 32 bits: 0x76A30000), only one operation on DATAR is needed. When transmitting, the MSB must be written to the DATAR register; the TXE flag being 1 indicates that new data can be written, and if the corresponding interrupt is enabled, an interrupt can be generated. Transmission is completed by hardware; even before the trailing 16 bits of 0x0000 have been sent out, TXE is set and the corresponding interrupt is generated. When receiving, each time the upper 16-bit half-word (MSB) is received, the RXNE flag is set to 1, and if the corresponding interrupt is enabled, an interrupt can be generated. This gives more time between 2 reads and writes, preventing underflow or overflow conditions.

20.3.2.2 MSB-Aligned Standard

In this standard, the WS signal is generated simultaneously with the first data bit, i.e., the most significant bit (MSB).

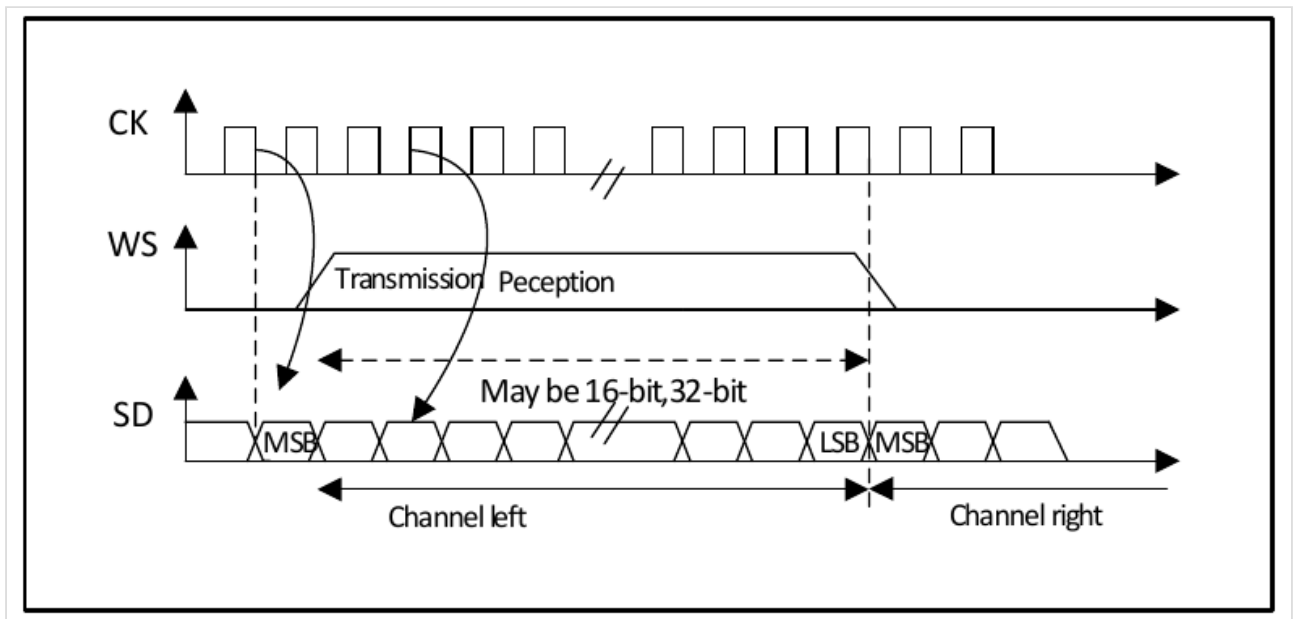


Figure 20-7: MSB-aligned 16-bit or 32-bit full precision (CPOL=0)

The transmitter changes data on the falling edge of the clock signal; the receiver reads data on the rising edge.

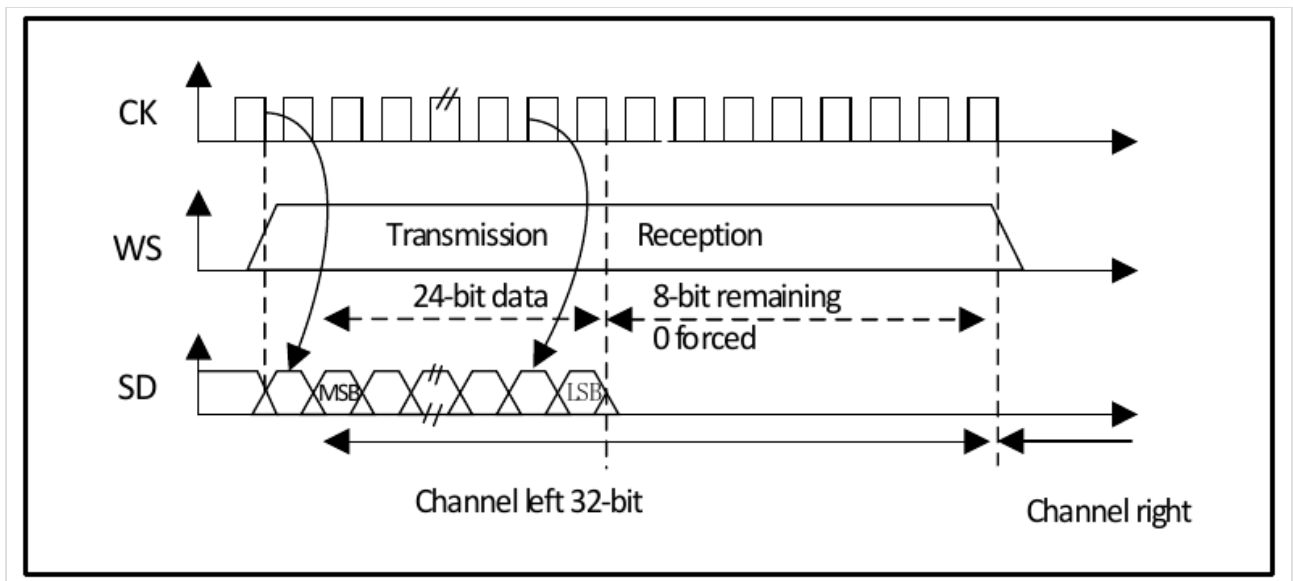


Figure 20-8: MSB-aligned 24-bit data, CPOL=0

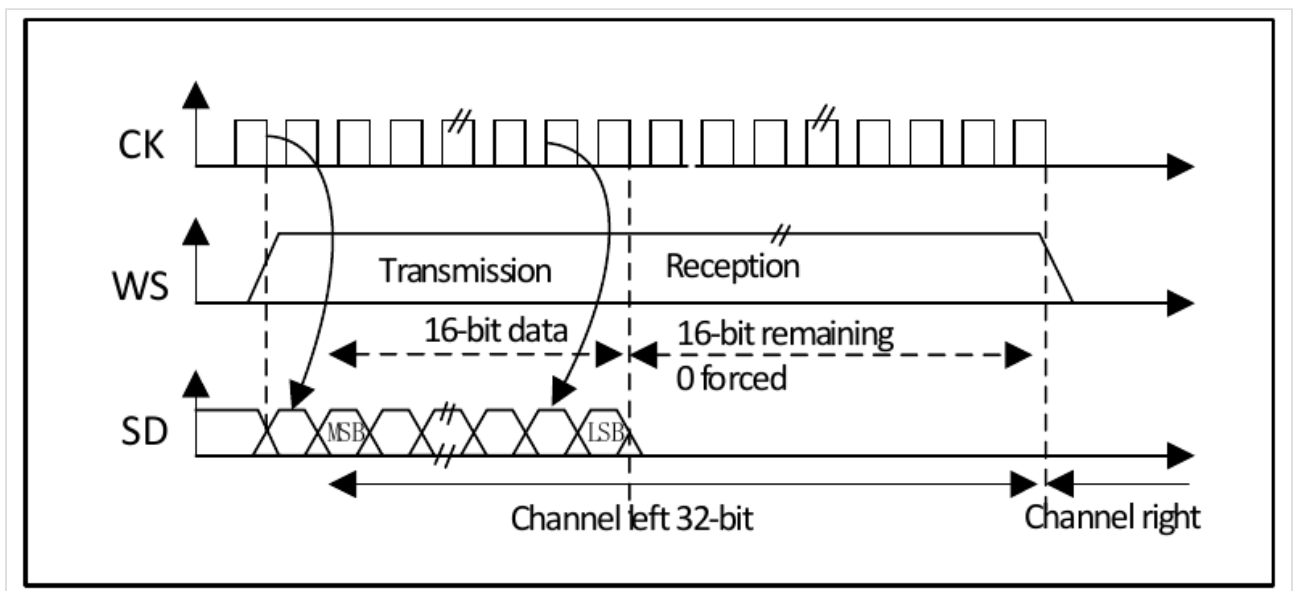


Figure 20-9: MSB-aligned 16-bit data extended to 32-bit packet frame, CPOL=0

20.3.2.3 LSB-Aligned Standard

This standard is similar to the MSB-aligned standard (no difference in 16-bit or 32-bit full-precision frame format).

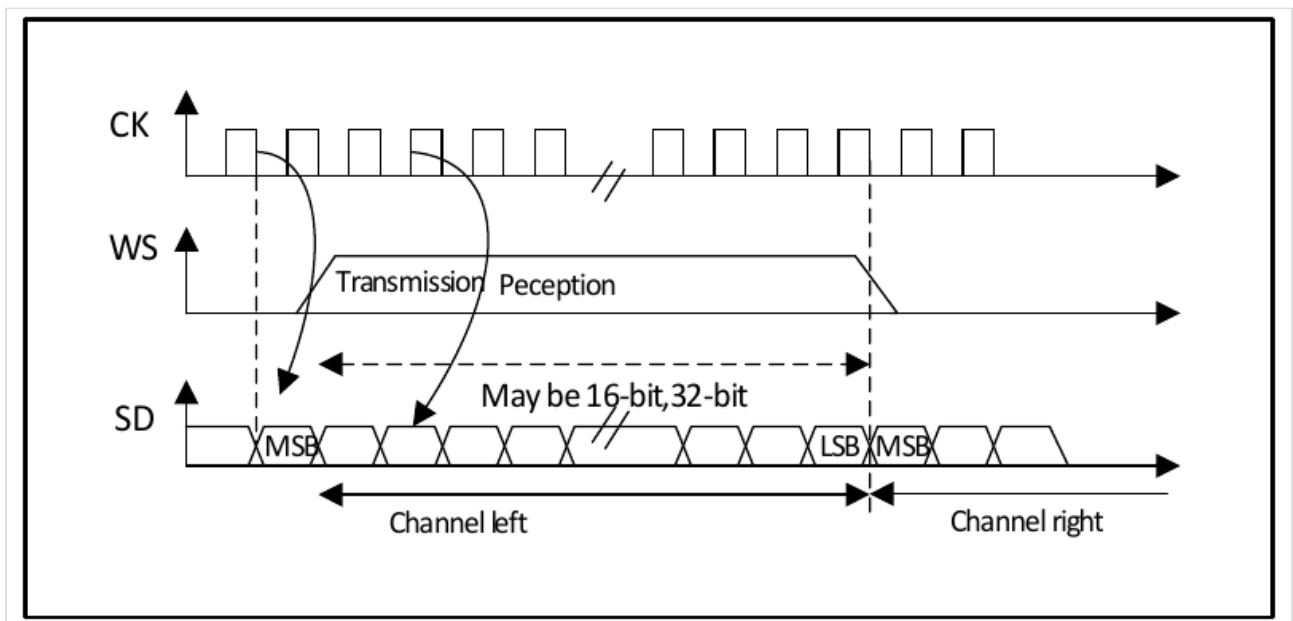


Figure 20-10: LSB-aligned 16 or 32-bit full precision, CPOL=0

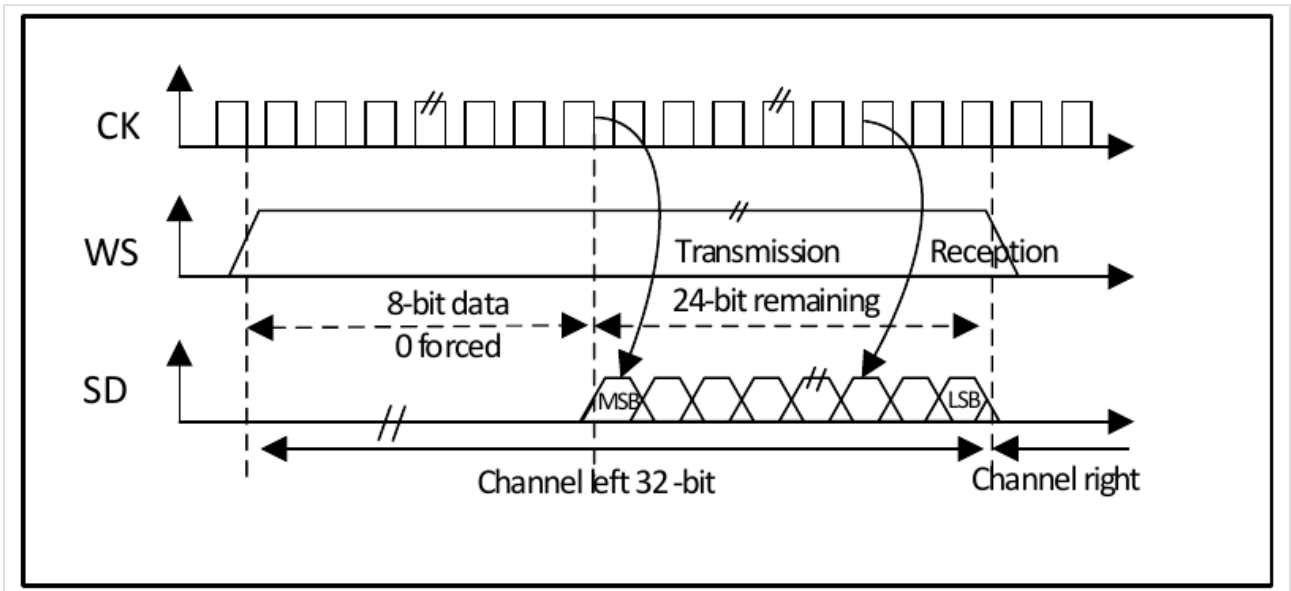


Figure 20-11: LSB-aligned 24-bit data, CPOL=0

In transmit mode, to send data 0x3478AE, 2 write operations to the DATAR register are needed via software or DMA:

- First time TXE='1' when the data register is written: 0xXX34 — Only the lower 8 bits in the halfword have meaning; the higher 8 bits are forced to 0x00.
- Second time TXE='1' when the data register is written: 0x78AE.

In receive mode, to receive data 0x3478AE, 1 read operation of the DATAR register is needed for each of 2 consecutive RXNE events:

- First time RXNE='1' when the data register is read: 0xXX34 — Only the lower 8 bits in the halfword have meaning; the higher 8 bits are forced to 0x00.
- Second time RXNE='1' when the data register is read: 0x78AE.

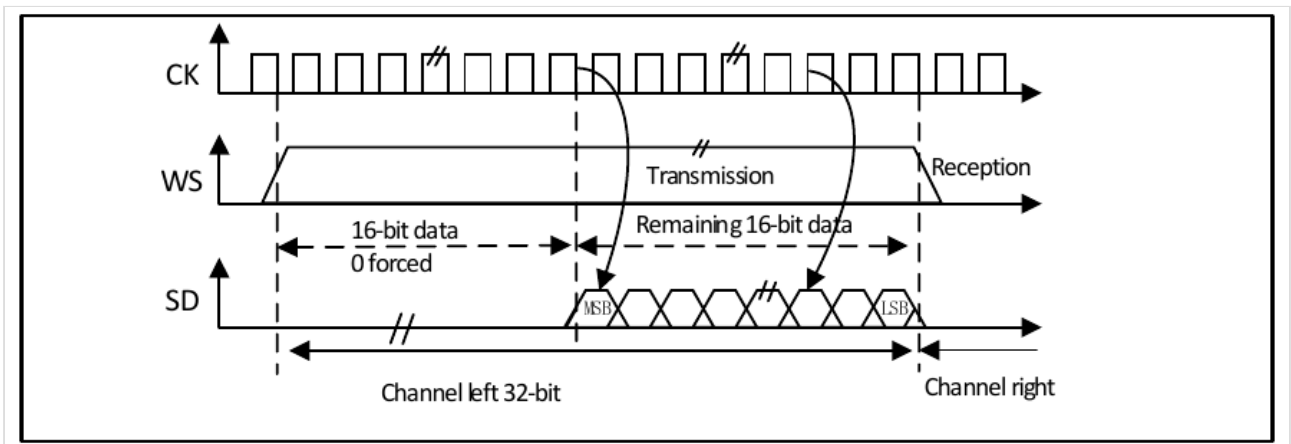


Figure 20-12: LSB-aligned 16-bit data extended to 32-bit packet frame, CPOL=0

During the I2S configuration phase, if 16-bit data is selected to be extended to a 32-bit channel frame, only one access to the DATAR register is needed. At this point, the upper half-word (16-bit MSB) after extension to 32 bits is set to 0x0000 by hardware. If the data to be transmitted or received is 0x76A3 (extended to 32 bits: 0x000076A3), only one operation on DATAR is needed. When transmitting, if TXE is 1, the user needs to write the data to be transmitted (i.e., 0x76A3). The 0x0000 portion used for

extension to 32 bits is sent out by hardware first; once valid data starts being sent from the SD pin, the next TXE event occurs. When receiving, once valid data is received (rather than the 0x0000 portion), the RXNE event occurs. This gives more time between 2 reads and writes, preventing underflow or overflow conditions.

20.3.2.4 PCM Standard

In the PCM standard, there is no channel selection information. The PCM standard has 2 available frame structures: short frame or long frame, selectable by setting the PCMSYNC bit of the I2SCFGR register.

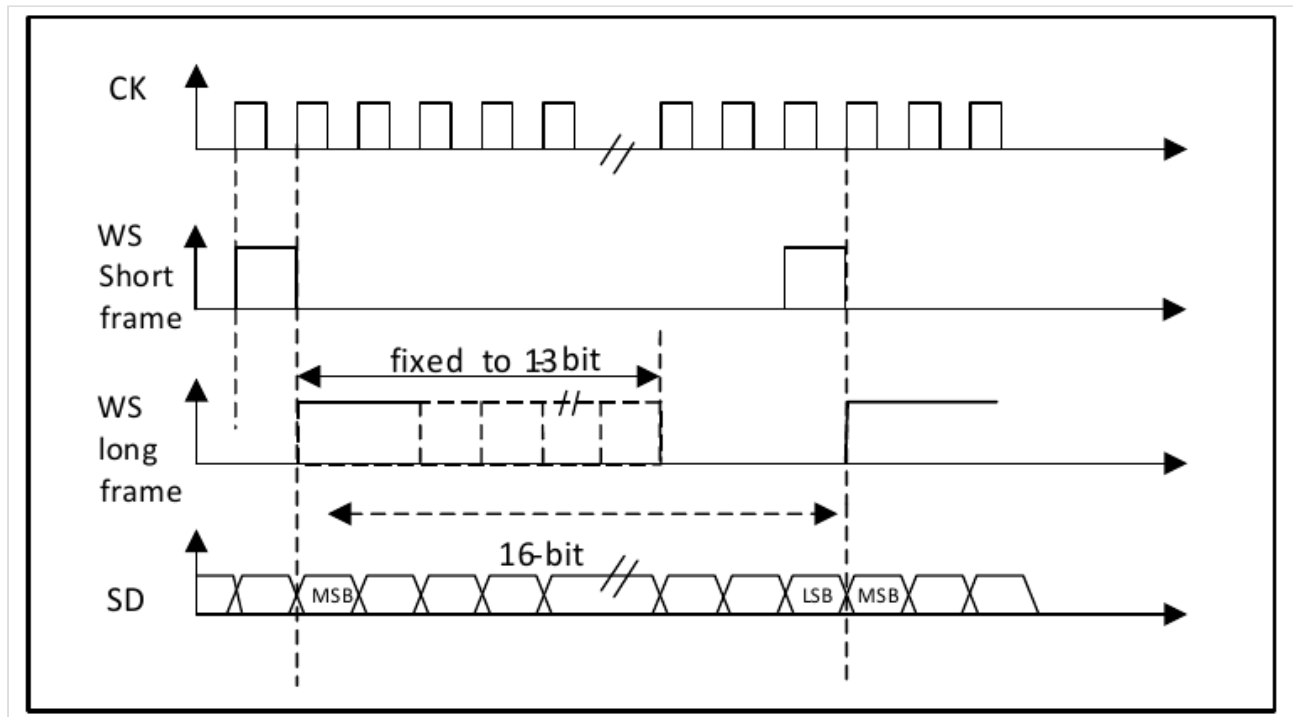


Figure 20-13: PCM standard waveform (16-bit)

For the long frame, in master mode, the WS signal used for synchronization is active for a fixed duration of 13 bits. For the short frame, the WS signal used for synchronization is only 1 bit long.

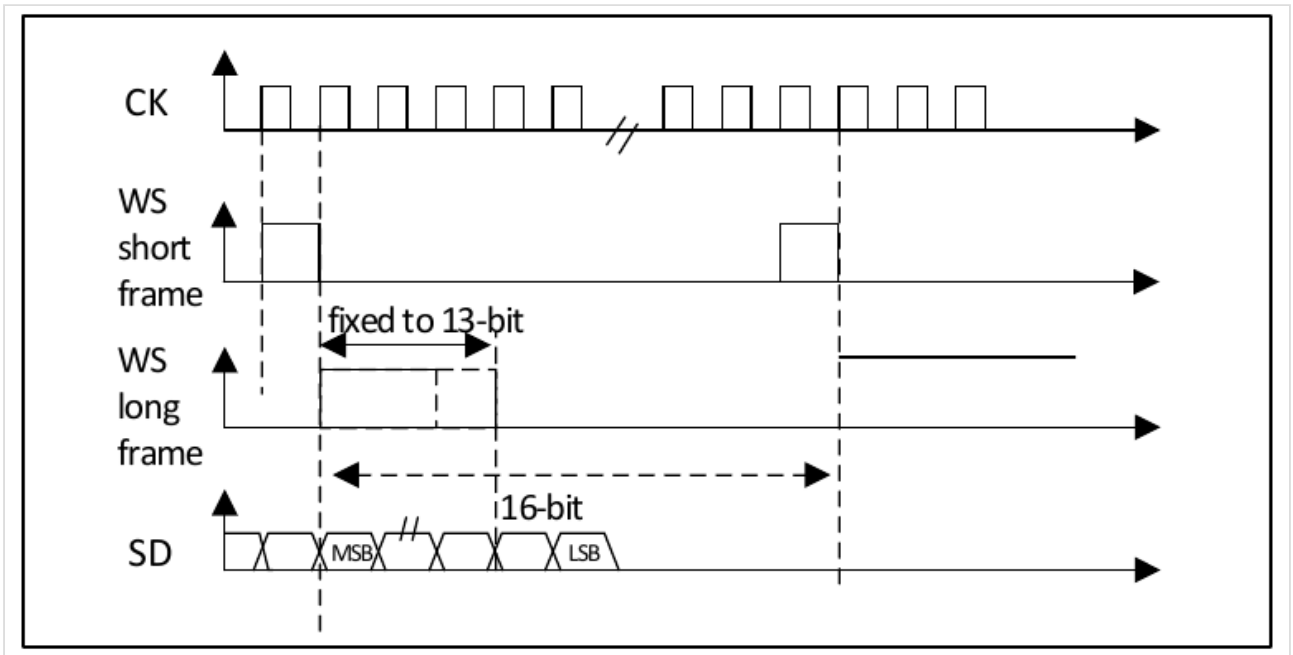


Figure 20-14: PCM standard waveform (16-bit)

Regardless of the mode (master or slave) or the synchronization method (short frame or long frame), the time difference between 2 consecutive data frames and between 2 synchronization signals (even in slave mode) must be determined by setting the DATLEN and CHLEN bits of the I2SCFGR register.

20.3.3 Clock Generator

The I2S bit rate determines the data stream on the I2S data line and the I2S clock signal frequency. I2S bit rate = number of bits per channel × number of channels × audio sampling frequency.

For a signal with left and right channels and 16-bit audio, the I2S bit rate is calculated as follows:

$$\text{I2S bit rate} = 16 \times 2 \times F_s$$

If the packet length is 32 bits, the I2S bit rate is calculated as follows:

$$\text{I2S bit rate} = 32 \times 2 \times F_s$$

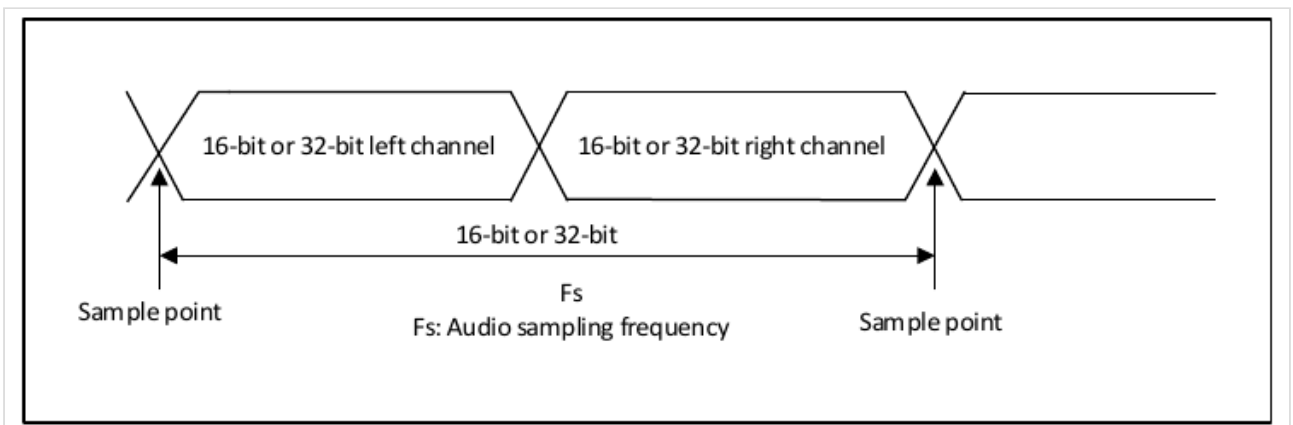


Figure 20-15: Audio sampling frequency definition

In master mode, to obtain the required audio frequency, the linear divider must be set correctly.

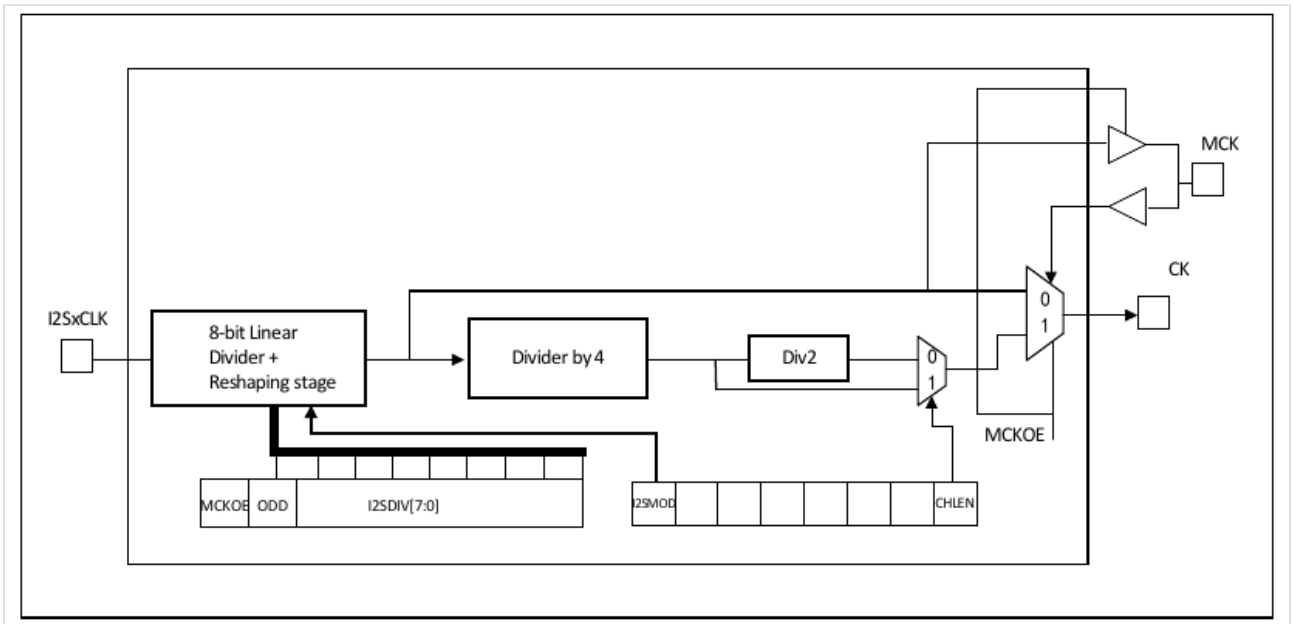


Figure 20-16: I2S clock generator structure

In the figure, the clock source of I2SxCLK is the system clock (i.e., the HSI, HSE, or PLL that drives the HB clock). I2SxCLK can come from SYSCLK, ETH_PLL, or USB_PLL clock, selectable via the I2S2SRC and I2S3SRC bits of the RCC_CFGR2 register. The audio sampling frequency can be 96 kHz, 48 kHz, 44.1 kHz, 32 kHz, 22.05 kHz, 16 kHz, 11.025 kHz, or 8 kHz (or any value within this range). To obtain the required frequency, set the linear divider according to the following formulas:

When the master clock needs to be generated (MCKOE bit of the SPI_I2SPR register is 1):

For 16-bit channel frame length: $F_s = I2SxCLK / [(16 \times 2) \times ((2 \times I2SDIV) + ODD) \times 8]$

For 32-bit channel frame length: $F_s = I2SxCLK / [(32 \times 2) \times ((2 \times I2SDIV) + ODD) \times 4]$

When the master clock is disabled (MCKOE bit is '0'):

For 16-bit channel frame length: $F_s = I2SxCLK / [(16 \times 2) \times ((2 \times I2SDIV) + ODD)]$

For 32-bit channel frame length: $F_s = I2SxCLK / [(32 \times 2) \times ((2 \times I2SDIV) + ODD)]$

20.3.4 I2S Master Mode

To set I2S to operate in master mode, the serial clock is output from the CK pin, and the word select signal is generated from the WS pin. The MCKOE bit of the I2SPR register can be set to choose whether or not to output the master clock (MCK).

20.3.4.1 Configuration Procedure

- Set the I2SDIV[7:0] of the I2SPR register to define the serial clock baud rate matching the audio sampling frequency. Also define the ODD bit of the I2SPR register.
- Set the CKPOL bit to define the idle-state level of the communication clock. If a master clock MCK needs to be provided to an external DAC/ADC audio device, the MCKOE bit of the I2SPR register must be set.
- Set the I2SMOD bit of the I2SCFGR register to 1 to activate I2S functionality. Set I2SSTD[1:0] and the PCMSYNC bit to select the I2S standard to be used. Set CHLEN to select the number of data bits per channel. Also set the I2SCFG[1:0] of the SPI_I2SCFGR register to select I2S master mode and direction (transmitter or receiver).

- If needed, enable the required interrupt and DMA functions by setting the CR2 register.
- The I2SE bit of the I2SCFGR register must be set to 1.
- The WS and CK pins need to be configured as output mode. If the MCKOE bit of the SPI_I2SPR register is 1, the MCK pin must also be configured as output mode.

20.3.4.2 Transmit Procedure

The transmit procedure begins when 1 half-word (16-bit) of data is written to the transmit buffer. Assume the first data written to the transmit buffer corresponds to the left channel. When data is moved from the transmit buffer to the shift register, the TXE flag is set, and the data corresponding to the right channel must be written to the transmit buffer. The CHSIDE flag indicates which channel the data currently being transmitted corresponds to. The CHSIDE flag value is updated when TXE is 1, so it is meaningful when TXE is 1. A complete data frame is considered complete only after both the left channel and then the right channel data have been transmitted. It is not permissible to transmit only a partial data frame, e.g., only left channel data.

When the first data bit is sent, the half-word data is transferred in parallel to the 16-bit shift register, and then the subsequent bits are sent out from the MOSI/SD pin in MSB-first order. Each time data is moved from the transmit buffer to the shift register, the TXE flag is set to 1; if the TXEIE bit of the CR2 register is 1, an interrupt is generated.

To ensure continuous audio data transmission, it is recommended to write the next data to be transmitted to the DATAR register before the current transmission completes. When disabling the I2S function, it is recommended to wait for TXE=1 and BSY=0, then clear the I2SE bit to '0'.

20.3.4.3 Receive Procedure

The configuration steps for the receive procedure are the same as for the transmit procedure (see "Transmit Procedure" above), except for step 3: the master receive mode must be selected by configuring I2SCFG[1:0]. Regardless of the data and channel length, audio data is always received in 16-bit packets. That is, each time the receive buffer is filled, the RXNE flag is set to 1; if the RXNEIE bit of the CR2 register is 1, an interrupt is generated. Depending on the configured data and channel length, receiving left or right channel data may require 1 or 2 transfers of data to the receive buffer. A read operation on the DATAR register clears the RXNE flag. CHSIDE is updated after each reception. Its value depends on the WS signal generated by the I2S unit. If the previously received data has not been read and new data is received, an overrun occurs, the OVR flag is set to 1, and if the ERRIE bit of the CR2 register is 1, an interrupt is generated indicating an error. To disable the I2S function, special operations are required to ensure the I2S module can properly complete the transmission cycle without starting a new data transfer. The procedure depends on the data configuration, channel length, and audio protocol mode:

- 16-bit data extended to 32-bit channel length (DATLEN=00 and CHLEN=1), using LSB-aligned mode (I2SSTD=10):
 1. Wait for the second-to-last (n-1) RXNE=1;
 2. Wait for 17 I2S clock cycles (using software delay);
 3. Disable I2S (I2SE=0).
- 16-bit data extended to 32-bit channel length (DATLEN=00 and CHLEN=1), using MSB-aligned, I2S, or PCM mode (I2SSTD=00, I2SSTD=01, or I2SSTD=11 respectively):
 1. Wait for the last RXNE=1;

2. Wait for 1 I2S clock cycle (using software delay);
 3. Disable I2S (I2SE=0).
- All other combinations of DATLEN and CHLEN, with any audio mode selected by I2SSTD, disable I2S as follows:
 1. Wait for the second-to-last (n-1) RXNE=1;
 2. Wait for 1 I2S clock cycle (using software delay);
 3. Disable I2S (I2SE=0).

Note: During transmission, the BSY flag is always low.

20.3.5 I2S Slave Mode

In slave mode, I2S can be set to transmit and receive modes. The slave mode configuration basically follows the same procedure as master mode configuration. In slave mode, the I2S interface does not need to provide a clock. The clock signal and WS signal are provided by the external master I2S device, connected to the corresponding pins. Therefore, the user does not need to configure the clock.

Configuration steps are as follows:

- Set the I2SMOD bit of the I2SCFGR register to activate I2S functionality; set I2SSTD[1:0] to select the I2S standard to be used; set DATLEN[1:0] to select the number of data bits; set CHLEN to select the number of data bits per channel. Set I2SCFG[1:0] of the I2SCFGR register to select the I2S slave mode data direction (transmitter or receiver).
- As needed, set the CR2 register to enable the required interrupt and DMA functions.
- The I2SE bit of the I2SCFGR register must be set to 1.

20.3.5.1 Transmit Procedure

When the external master device sends a clock signal and the NSS_WS signal requests data transfer, the transmit procedure begins. The slave must be enabled and data written to the I2S data register before the external master can begin communication. For I2S MSB-aligned and LSB-aligned modes, the first data item written to the data register corresponds to the left channel data. When communication starts, data is transferred from the transmit buffer to the shift register, and the TXE flag is set to 1; at this point, the data item corresponding to the right channel must be written to the I2S data register. The CHSIDE flag indicates which channel the data currently being transmitted corresponds to. Compared to the master mode transmit procedure, in slave mode CHSIDE depends on the WS signal from the external master I2S. This means the slave I2S must have the first data to be transmitted ready before receiving the clock signal generated by the master. A WS signal of 1 indicates that the left channel is sent first.

20.3.5.2 Receive Procedure

The configuration steps are the same as the transmit procedure, except for step 1. The master receive mode must be selected by configuring I2SCFG[1:0]. Regardless of the data and channel length, audio data is always received in 16-bit packets, i.e., each time the receive buffer is filled, the RXNE flag is set to 1; if the RXNEIE bit of the I2S_CTLR2 register is 1, an interrupt is generated. Depending on the data and channel length settings, receiving left or right channel data may require 1 or 2 transfers of data to the receive buffer. CHSIDE is updated after each data reception (about to be read from DATAR), corresponding to the WS signal generated by the I2S unit. Reading the SPI_DATAR register clears the

RXNE bit. When new data is received before the previously received data has been read, an overrun occurs and the OVR flag is set to 1; if the ERRIE bit of the I2S_CTLR2 register is 1, an interrupt is generated indicating an error. To disable the I2S function, the I2SE bit must be cleared to 0 upon receiving the last RXNE=1.

20.3.6 Status Flags

There are 3 status flags for the user to monitor the status of the I2S bus.

20.3.6.1 Status Flag (BSY)

The BSY flag is set and cleared by hardware (writing to this bit has no effect). This flag indicates the status of the I2S communication layer. When this bit is 1, it indicates I2S communication is in progress, with one exception: in master receive mode (I2SCFG=11), the BSY flag is always low during reception. Before software disables the SPI module, the BSY flag can be used to detect whether transmission has ended, avoiding corruption of the last transmission. The following procedure must be strictly followed. When transmission starts, the BSY flag is set to 1 unless the I2S module is in master receive mode. When transmission ends or when the I2S module is disabled, this flag is cleared. In continuous communication, in master transmit mode, the BSY flag remains high throughout the entire transmission; in slave mode, the BSY flag goes low for 1 I2S clock cycle between each data item transmission.

20.3.6.2 Transmit Buffer Flag (TXE)

This flag being 1 indicates the transmit buffer is empty and new data to be transmitted can be written to the transmit buffer. When the transmit buffer already contains data, the flag is cleared to 0. When I2S is disabled (I2SE bit is 0), this flag is also 0.

20.3.6.3 Receive Buffer Not Empty Flag (RXNE)

This flag being set to 1 indicates that valid received data is in the receive buffer. When the DATAR register is read, this bit is cleared to 0.

20.3.6.4 Channel Flag (CHSIDE)

In transmit mode, this flag is updated when TXE is high, indicating the channel of the data being sent from the SD pin. If an underflow error occurs in slave transmit mode, the flag value is invalid; the I2S must be disabled and re-enabled before resuming communication. In receive mode, this flag is updated when data is received in the DATAR register, indicating the channel of the received data. Note: if an error occurs, this flag is meaningless; the I2S must be disabled and re-enabled. Under the PCM standard, this flag has no meaning in either short-frame or long-frame format. If the OVR or UDR flag of the STATR register is 1, and the ERRIE bit of the CR2 register is 1, an interrupt is generated. (After the interrupt source has been cleared) the interrupt flag can be cleared by reading the STATR register.

20.3.7 Error Flags

20.3.7.1 Underflow Flag (UDR)

In slave transmit mode, if new data has not been written to the DATAR register when the first clock edge of data transfer arrives, this flag is set to 1. This flag is valid only after the I2SMOD bit of the

I2SCFGR register is set to 1. If the ERRIE bit of the CR2 register is 1, an interrupt is generated. This flag is cleared by performing a read operation on the STATR register.

20.3.7.2 Overrun Flag (OVR)

If new data is received before the previously received data has been read, an overrun occurs and this flag is set to 1. If the ERRIE bit of the CTLR2 register is 1, an interrupt is generated indicating an error. In this case, the content of the receive buffer is not updated with the new data from the transmitting device. A read operation on the DATAR register returns the last correctly received data. All other 16-bit data sent by the transmitter after the overrun occurs are lost. This flag is cleared by first reading the DATAR register and then reading the STATR register.

20.3.8 I2S Interrupts

I2S has 4 interrupt sources. The transmit buffer empty and receive buffer not empty events set TXE and RXNE respectively, generating interrupts when TXEIE and RXNEIE are set. If new data is received before the previously received data has been read, an overrun occurs; if ERRIE is set, an overrun interrupt is generated. In slave transmit mode, if new data has not been written to the DATAR register when the first clock edge of data transfer arrives, if ERRIE is set, an underflow interrupt is generated.

20.3.9 DMA Function

The DMA operation in I2S mode is identical to that in SPI mode, except that the CRC function is not available. This is because there is no data transfer protection system in I2S mode.

20.4 Register Description

Table 20-1: SPI1 related register list

Name	Address	Description	Reset Value
R16_SPI1_CTLR1	0x40013000	SPI1 control register 1	0x0000
R16_SPI1_CTLR2	0x40013004	SPI1 control register 2	0x0000
R16_SPI1_STATR	0x40013008	SPI1 status register	0x0002
R16_SPI1_DATAR	0x4001300C	SPI1 data register	0x0000
R16_SPI1_CRCR	0x40013010	SPI1 polynomial register	0x0007
R16_SPI1_RCRRCR	0x40013014	SPI1 receive CRC register	0x0000
R16_SPI1_TCRRCR	0x40013018	SPI1 transmit CRC register	0x0000
R16_SPI1_I2S_CFGR	0x4001301C	SPI1_I2S configuration register	0x0000
R16_SPI1_HSCR	0x40013024	SPI1 high-speed control register	0x0000

Table 20-2: SPI2 related register list

Name	Address	Description	Reset Value
R16_SPI2_CTLR1	0x40003800	SPI2 control register 1	0x0000
R16_SPI2_CTLR2	0x40003804	SPI2 control register 2	0x0000
R16_SPI2_STATR	0x40003808	SPI2 status register	0x0002
R16_SPI2_DATAR	0x4000380C	SPI2 data register	0x0000

Name	Address	Description	Reset Value
R16_SPI2_CRCR	0x40003810	SPI2 polynomial register	0x0007
R16_SPI2_RCR	0x40003814	SPI2 receive CRC register	0x0000
R16_SPI2_TCR	0x40003818	SPI2 transmit CRC register	0x0000
R16_SPI2_I2S_CFGR	0x4000381C	SPI2_I2S configuration register	0x0000
R16_SPI2_I2SPR	0x40003820	SPI2_I2S prescaler register	0x0000
R16_SPI2_HSCR	0x40003824	SPI2 high-speed control register	0x0000

Table 20-3: SPI3 related register list

Name	Address	Description	Reset Value
R16_SPI3_CTLR1	0x40003C00	SPI3 control register 1	0x0000
R16_SPI3_CTLR2	0x40003C04	SPI3 control register 2	0x0000
R16_SPI3_STATR	0x40003C08	SPI3 status register	0x0002
R16_SPI3_DATAR	0x40003C0C	SPI3 data register	0x0000
R16_SPI3_CRCR	0x40003C10	SPI3 polynomial register	0x0007
R16_SPI3_RCR	0x40003C14	SPI3 receive CRC register	0x0000
R16_SPI3_TCR	0x40003C18	SPI3 transmit CRC register	0x0000
R16_SPI3_I2S_CFGR	0x40003C1C	SPI3_I2S configuration register	0x0000
R16_SPI3_I2SPR	0x40003C20	SPI3_I2S prescaler register	0x0000
R16_SPI3_HSCR	0x40003C24	SPI3 high-speed control register	0x0000

20.4.1 SPI Control Register 1 (SPIx_CTLR1) (x=1/2/3)

Offset address: 0x00

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BIDIMODE	BIDIOE	CRCEN	CRCNEXT	DFF	RXONLY	SSM	SSI	LSBFIRST	SPE	BR[2:0]	MSTR	CPOL	CPHA		

Bits	Name	Access	Description	Reset
15	BIDIMODE	RW	Unidirectional data mode enable bit. 1: Select single-wire bidirectional mode; 0: Select two-wire bidirectional mode.	0
14	BIDIOE	RW	Single-wire output enable bit, used together with BIDIMODE. 1: Enable output, transmit only; 0: Disable output, receive only.	0
13	CRCEN	RW	Hardware CRC verification enable bit. This bit can only be written when SPE is 0. This bit can only be used in full-duplex mode. 1: Enable CRC calculation; 0: Disable CRC calculation.	0
12	CRCNEXT	RW	Transmit the CRC register value after the next data transfer. This bit should be set immediately after writing the last data to the data register. 1: Transmit CRC verification result; 0: Continue transmitting data register data.	0
11	DFF	RW		0

Bits	Name	Access	Description	Reset
			Data frame length bit. This bit can only be written when SPE is 0. 1: Use 16-bit data length for transmit/receive; 0: Use 8-bit data length for transmit/receive.	
10	RXONLY	RW	Receive-only bit in two-wire mode, used together with BIDIMODE. Setting this bit allows the device to only receive without transmitting. 1: Receive only, simplex mode; 0: Full-duplex mode.	0
9	SSM	RW	Chip select pin management bit. This bit determines whether the NSS pin level is controlled by hardware or software. 1: Software controls NSS pin; 0: Hardware controls NSS pin.	0
8	SSI	RW	Chip select pin control bit. When SSM is set, this bit determines the NSS pin level. 1: NSS is high; 0: NSS is low.	0
7	LSBFIRST	RW	Frame format control bit. This bit cannot be modified during communication. 1: Send LSB first; 0: Send MSB first.	0
6	SPE	RW	SPI enable bit. 1: Enable SPI; 0: Disable SPI.	0
[5:3]	BR[2:0]	RW	Baud rate setting field. This field cannot be modified during communication. When HSRXEN bit is 0, SCK frequency is: 000: FPCLK/2; 001: FPCLK/4; 010: FPCLK/8; 011: FPCLK/16; 100: FPCLK/32; 101: FPCLK/64; 110: FPCLK/128; 111: FPCLK/256. When HSRXEN bit is 1, SCK frequency is: 000: FPCLK/2; 001: FPCLK/3; 010: FPCLK/4; 011: FPCLK/5; 100: FPCLK/6; 101: FPCLK/7; 110: FPCLK/8; 111: FPCLK/9.	000b
2	MSTR	RW	Master/slave selection bit. This bit cannot be modified during communication. 1: Configure as master device; 0: Configure as slave device.	0
1	CPOL	RW	Clock polarity selection bit. This bit cannot be modified during communication. 1: SCK stays high in idle state; 0: SCK stays low in idle state.	0
0	CPHA	RW	Clock phase setting bit. This bit cannot be modified during communication. 1: Data sampling starts from the second clock edge; 0: Data sampling starts from the first clock edge.	0

20.4.2 SPI Control Register 2 (SPIx_CTLR2) (x=1/2/3)

Offset address: 0x04

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					TXEIE	RXNEIE	ERRIE	Reserved		SSOE	TXDMAEN	RXDMAEN			

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
7	TXEIE	RW	Transmit buffer empty interrupt enable bit: 1: Enable interrupt when TXE is set; 0: Disable interrupt when TXE is set.	0
6	RXNEIE	RW	Receive buffer not empty interrupt enable bit: 1: Enable interrupt when RXNE is set; 0: Disable interrupt when RXNE is set.	0
5	ERRIE	RW	Error interrupt enable bit: 1: Enable interrupt on error (CRCERR, OVR, MODF); 0: Disable interrupt on error (CRCERR, OVR, MODF).	0
[4:3]	Reserved	RO	Reserved.	0
2	SSOE	RW	SS output enable. Disabling SS output allows operation in multi-master mode. 1: Enable SS output; 0: Disable SS output in master mode.	0
1	TXDMAEN	RW	Transmit buffer DMA enable bit. 1: Enable transmit buffer DMA; 0: Disable transmit buffer DMA.	0
0	RXDMAEN	RW	Receive buffer DMA enable bit. 1: Enable receive buffer DMA; 0: Disable receive buffer DMA.	0

20.4.3 SPI Status Register (SPIx_STATR) (x=1/2/3)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					BSY	OVR	MODF	CRCERR	UDR	CHSIDE	TXE	RXNE			

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
7	BSY	RO	Busy flag bit. This bit is set or cleared by hardware. 1: SPI is communicating, or transmit buffer is not empty; 0: SPI is not communicating.	0
6	OVR	RO	Overrun flag bit. This bit is set by hardware, cleared by software. 1: Overrun error occurred; 0: No overrun error.	0
5	MODF	RO	Mode fault flag bit. This bit is set by hardware, cleared by software. 1: Mode fault occurred; 0: No mode fault.	0

Bits	Name	Access	Description	Reset
4	CRCERR	RW0	CRC error flag bit. This bit is set by hardware, cleared by software. 1: Received CRC value does not match RCRCR value; 0: Received CRC value matches RCRCR value.	0
3	UDR	RO	Underflow flag bit. This bit is set by hardware, cleared by software. 1: Underflow occurred; 0: No underflow.	0
2	CHSIDE	RO	Channel. This bit is set by hardware, cleared by software. 1: Right channel to be transmitted or received; 0: Left channel to be transmitted or received.	0
1	TXE	RO	Transmit buffer empty flag bit: 1: Transmit buffer is empty; 0: Transmit buffer is not empty.	1
0	RXNE	RO	Receive buffer not empty flag bit: 1: Receive buffer is not empty; 0: Receive buffer is empty. Note: Reading DATAR automatically clears this bit.	0

20.4.4 SPI Data Register (SPIx_DATAR) (x=1/2/3)

Offset address: 0x0C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DR[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	DR[15:0]	RW	Data register. The data register is used to store received data or pre-store data to be transmitted. Therefore, reads and writes of the data register actually correspond to different areas: reads correspond to the receive buffer, writes correspond to the transmit buffer. Data reception and transmission can be 8-bit or 16-bit; the number of data bits must be determined before transmission. When using 8-bit data transfer, only the lower 8 bits of the data register are used, and the upper 8 bits are forced to 0 upon reception. Using 16-bit data structure causes all 16 bits of the data register to be used.	0

20.4.5 SPI Polynomial Register (SPIx_CRCR) (x=1/2/3)

Offset address: 0x10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CRCPOLY[15:0]															

Bits	Name	Access	Description	Reset
[15:0]	CRCPOLY[15:0]	RW	CRC polynomial. This field defines the polynomial used for CRC calculation.	0x0007

20.4.6 SPI Receive CRC Register (SPIx_RCRCR) (x=1/2/3)

Offset address: 0x14

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXCRC[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	RXCRC[15:0]	RO	Receive CRC value. Stores the calculated CRC verification result of received bytes. Setting CRCEN resets this register. The calculation method uses the polynomial defined by CRCPOLY. In 8-bit mode, only the lower 8 bits participate in the calculation; in 16-bit mode, all 16 bits participate. This register should be read when BSY is 0.	0											

20.4.7 SPI Transmit CRC Register (SPIx_TCR CR) (x=1/2/3)

Offset address: 0x18

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TXCRC[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	TXCRC[15:0]	RO	Transmit CRC value. Stores the calculated CRC verification result of transmitted bytes. Setting CRCEN resets this register. The calculation method uses the polynomial defined by CRCPOLY. In 8-bit mode, only the lower 8 bits participate in the calculation; in 16-bit mode, all 16 bits participate. This register should be read when BSY is 0.	0											

20.4.8 SPI_I2S Configuration Register (SPI_I2S_CFGR) (x=1/2/3)

Offset address: 0x1C

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved		I2SMOD		I2SE	I2SCFG[1:0]		PCMSYNC		Reserved		I2SSTD[1:0]		CKPOL	DATLEN[1:0]		CHLEN

Bits	Name	Access	Description	Reset
[15:12]	Reserved	RO	Reserved.	0
11	I2SMOD	RW	I2S mode selection. This bit can only be set when SPI or I2S is disabled. 1: Select I2S mode; 0: Select SPI mode.	0
10	I2SE	RW	I2S enable. Not used in SPI mode. 1: I2S enabled; 0: I2S disabled.	0
[9:8]	I2SCFG[1:0]	RW	I2S mode selection. This bit can only be set when I2S is disabled: 00: Slave transmit; 01: Slave receive; 10: Master transmit; 11: Master receive.	00b
7	PCMSYNC	RW	PCM frame synchronization. This bit is meaningful only when I2SSTD = 11 (PCM standard). 1: Long frame synchronization; 0: Short frame synchronization.	0
6	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
[5:4]	I2SSTD[1:0]	RW	I2S standard selection. This bit can only be set when I2S is disabled. 00: I2S Philips standard; 01: MSB-aligned standard (left-aligned); 10: LSB-aligned standard (right-aligned); 11: PCM standard.	00b
3	CKPOL	RW	Steady-state clock polarity. For correct operation, this bit can only be set when I2S is disabled. 1: I2S clock steady state is high; 0: I2S clock steady state is low.	0
[2:1]	DATLEN[1:0]	RW	Data length to be transmitted. For correct operation, this bit can only be set when I2S is disabled. 00: 16-bit data length; 01: 24-bit data length; 10: 32-bit data length; 11: Not allowed.	00b
0	CHLEN	RW	Channel length. Writing this bit is meaningful only when DATLEN = 00; otherwise the channel length is fixed to 32 bits by hardware. 1: 32-bit wide; 0: 16-bit wide.	0

20.4.9 SPI_I2S Prescaler Register (SPIx_I2SPR) (x=2/3)

Offset address: 0x20

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						MCKOE	ODD	I2SDIV[7:0]							

Bits	Name	Access	Description	Reset
[15:10]	Reserved	RO	Reserved.	0
9	MCKOE	RW	Master clock output enable. For correct operation, this bit can only be set when I2S is disabled. Used only in I2S master mode. 1: Master clock output enabled; 0: Master clock output disabled.	0
8	ODD	RW	Odd factor prescaler. For correct operation, this bit can only be set when I2S is disabled. Used only in I2S master mode. 1: Actual division factor = (I2SDIV × 2) + 1; 0: Actual division factor = I2SDIV × 2.	0
[7:0]	I2SDIV[7:0]	RW	I2S linear prescaler. For correct operation, this bit can only be set when I2S is disabled. Used only in I2S master mode. Setting I2SDIV[7:0] = 0 or I2SDIV[7:0] = 1 is prohibited. See section 20.3.3.	0

20.4.10 SPI High-Speed Control Register (SPIx_HSCR) (x=1/2/3)

Offset address: 0x24

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						I2S_WS_SYNC		HSRXEN2		Reserved		HSRXEN			

Bits	Name	Access	Description	Reset
[15:4]	Reserved	RO	Reserved.	0
3	I2S_WS_SYNC	RW	Enhanced I2S slave synchronization capability: 1: Enabled; 0: Disabled.	0
2	HSRXEN2	RW	SPI high-speed read mode 2: This mode is only used when HSRXEN is enabled and SPI PCLK is $\geq$ 120 MHz; below 120 MHz, using this bit is not recommended. Can achieve higher master read speed than enabling HSRXEN alone. 1: Enable high-speed read mode 2; 0: Disable high-speed read mode 2.	0
1	Reserved	RO	Reserved.	0
0	HSRXEN	RW	High-speed read mode enable bit: 1: Enable high-speed read mode; 0: Disable high-speed read mode. This mode is only valid when the clock is divided by 2 (i.e., BR = 000 in the CTLR1 register).	0

Chapter 21: Operational Amplifier (OPA)

The Operational Amplifier module (OPA) contains one independently configurable operational amplifier. Both the input and output of the operational amplifier are connected to I/O ports with selectable pins. The operational amplifier (OPA or PGA) supports gain selection.

21.1 Main Features

- Selectable input pins
- Selectable output pins
- Supports PGA gain selection

21.2 Functional Description

In the R32_OPA_CTLR register: setting EN1 enables the corresponding OPA1; configuring MODE1 selects the output channel of OPA1; configuring PSEL1 selects the positive input pin of OPA1; configuring NSEL1 selects the negative input channel of OPA1, or the gain when used as a PGA; setting FB_EN1 enables the PGA mode of OPA1; setting PGADIF1 enables the differential input PGA mode; setting HS1 enables the high-speed mode of OPA1.

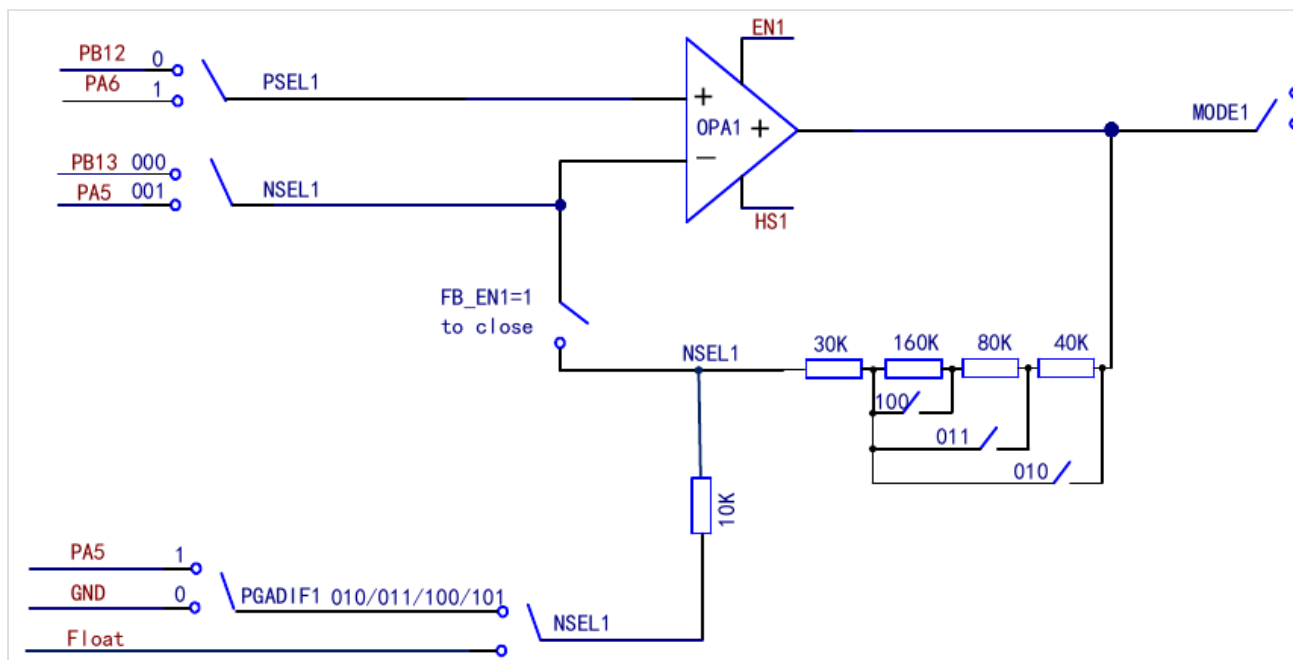


Figure 21-1: OPA block diagram

21.3 Register Description

Table 21-1: OPA related register list

Name	Address	Description	Reset Value
R32_OPA_CTLR	0x40023804	OPA configuration register	0x00000000

21.3.1 OPA Configuration Register (OPA_CTLR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								HS1	FB_EN1	PGADIF1	PSEL1	NSEL1[2:0]		MODE1	EN1

Bits	Name	Access	Description	Reset
[31:9]	Reserved	R0	Reserved.	0
8	HS1	RW	OPA1 high-speed mode enable: 1: Enable; 0: Disable.	0
7	FB_EN1	RW	OPA1 PGA mode feedback enable: 1: Enable; 0: Disable.	0
6	PGADIF1	RW	OPA1 works as a PGA with NSEL, and the N terminal is connected to OPA1_CHN1 (PA5): 1: Enable differential input PGA mode; 0: Disable differential input PGA mode.	0
5	PSEL1	RW	OPA1 positive terminal channel selection: 1: OPA1_CHP1 (PA6); 0: OPA1_CHP0 (PB12).	0
[4:2]	NSEL1[2:0]	RW	OPA1 negative terminal channel selection and PGA gain selection: 000: OPA1_CHN0 (PB13); 001: OPA1_CHN1 (PA5); 010: PGA mode, no negative input channel, internal gain is 4; 011: PGA mode, no negative input channel, internal gain is 8; 100: PGA mode, no negative input channel, internal gain is 16; 101: PGA mode, no negative input channel, internal gain is 32; Others: Shutdown.	0
1	MODE1	RW	OPA1 output channel selection: 1: Output signal through PA7; 0: Output signal through PB1.	0
0	EN1	RW	OPA1 enable: 1: Enable; 0: Disable.	0

Chapter 22: USB High-Speed Host/Device Controller (USBHS)

The chip integrates two sets of USB2.0 high-speed controllers with dual roles of host controller and device controller, and a built-in 480Mbps USB-PHY physical layer transceiver. When acting as a host controller, it can support low-speed, full-speed, and high-speed USB devices. When acting as a device controller, it can be flexibly configured to low-speed, full-speed, or high-speed mode to suit various applications.

22.1 USB Function Description

- Supports USB 2.1, USB 2.0, USB 1.1, and USB 1.0 protocol specifications
- Supports USB Host function and USB Device function
- Supports control transfer, bulk transfer, interrupt transfer, and real-time/isochronous transfer
- Provides bus reset, suspend, wake-up, and resume functions
- Host supports high-speed USB HUB
- All non-0 endpoints support up to 1024-byte data packets, with built-in FIFO, supporting interrupt and DMA
- Supports USART serial or I2C pin mapping, also usable for two-wire debug

22.2 Register Description

USB-related registers are divided into two parts:

- USB high-speed device control registers
- USB high-speed host control registers

The peripheral base address of the USBHS1 module is: 0x40024000;

The peripheral base address of the USBHS2 module is: 0x40023400.

22.2.1 Device Register Description

Table 22-1: USBHS1 Device-Related Register List

Name	Address	Description	Reset Value
R8_USB_CTRL	0x40024000	USBHS control register	0x07
R8_USB_BASE_MODE	0x40024001	USBHS mode control register	0x00
R8_USB_INT_EN	0x40024002	USBHS interrupt enable register	0x00
R8_USB_DEV_AD	0x40024003	USBHS device address register	0x00
R8_USB_WAKE_CTRL	0x40024004	USBHS remote wakeup register	0x00
R8_USB_TEST_MODE	0x40024005	USBHS test mode register	0x00
R16_USB_LPM_DATA	0x40024006	USBHS power management register	0x8000

Name	Address	Description	Reset Value
R8_USB_INT_FG	0x40024008	USBHS interrupt flag register	0x00
R8_USB_INT_ST	0x40024009	USBHS interrupt status register	0x00
R8_USB_MIS_ST	0x4002400A	USBHS miscellaneous status register	0x00
R16_USB_FRAME_NO	0x4002400C	USBHS frame number register	0x0000
R16_USB_BUS	0x4002400E	USBHS bus status register	0x0000
R16_UEP_TX_EN	0x40024010	USBHS endpoint transmit enable register	0x0000
R16_UEP_RX_EN	0x40024012	USBHS endpoint receive enable register	0x0000
R16_UEP_T_TOG_AUTO	0x40024014	USBHS endpoint transmit auto-toggle enable register	0x0000
R16_UEP_R_TOG_AUTO	0x40024016	USBHS endpoint receive auto-toggle enable register	0x0000
R8_UEP_T_BURST	0x40024018	USBHS endpoint transmit burst register	0x00
R8_UEP_T_BURST_MODE	0x40024019	USBHS endpoint transmit burst mode register	0x00
R8_UEP_R_BURST	0x4002401A	USBHS endpoint receive burst register	0x00
R8_UEP_R_RES_MODE	0x4002401B	USBHS endpoint receive response mode register	0x00
R32_UEP_AF_MODE	0x4002401C	USBHS endpoint multiplex register	0x00000000
R32_UEP0_DMA	0x40024020	Endpoint 0 buffer start address register	0x000XXXXX
R32_UEP1_RX_DMA	0x40024024	Endpoint 1 receive buffer start address register	0x000XXXXX
R32_UEP2_RX_DMA	0x40024028	Endpoint 2 receive buffer start address register	0x000XXXXX
R32_UEP3_RX_DMA	0x4002402C	Endpoint 3 receive buffer start address register	0x000XXXXX
R32_UEP4_RX_DMA	0x40024030	Endpoint 4 receive buffer start address register	0x000XXXXX
R32_UEP5_RX_DMA	0x40024034	Endpoint 5 receive buffer start address register	0x000XXXXX
R32_UEP6_RX_DMA	0x40024038	Endpoint 6 receive buffer start address register	0x000XXXXX
R32_UEP7_RX_DMA	0x4002403C	Endpoint 7 receive buffer start address register	0x000XXXXX
R32_UEP1_TX_DMA	0x40024040	Endpoint 1 transmit buffer start address register	0x000XXXXX
R32_UEP2_TX_DMA	0x40024044	Endpoint 2 transmit buffer start address register	0x000XXXXX
R32_UEP3_TX_DMA	0x40024048	Endpoint 3 transmit buffer start address register	0x000XXXXX
R32_UEP4_TX_DMA	0x4002404C	Endpoint 4 transmit buffer start address register	0x000XXXXX
R32_UEP5_TX_DMA	0x40024050	Endpoint 5 transmit buffer start address register	0x000XXXXX
R32_UEP6_TX_DMA	0x40024054	Endpoint 6 transmit buffer start address register	0x000XXXXX
R32_UEP7_TX_DMA	0x40024058	Endpoint 7 transmit buffer start address register	0x000XXXXX
R32_UEP0_MAX_LEN	0x4002405C	Endpoint 0 max length packet register	0x000000XX
R32_UEP1_MAX_LEN	0x40024060	Endpoint 1 max length packet register	0x00000XXX
R32_UEP2_MAX_LEN	0x40024064	Endpoint 2 max length packet register	0x00000XXX
R32_UEP3_MAX_LEN	0x40024068	Endpoint 3 max length packet register	0x00000XXX
R32_UEP4_MAX_LEN	0x4002406C	Endpoint 4 max length packet register	0x00000XXX
R32_UEP5_MAX_LEN	0x40024070	Endpoint 5 max length packet register	0x00000XXX
R32_UEP6_MAX_LEN	0x40024074	Endpoint 6 max length packet register	0x00000XXX
R32_UEP7_MAX_LEN	0x40024078	Endpoint 7 max length packet register	0x00000XXX

Name	Address	Description	Reset Value
R16_UEP0_RX_LEN	0x4002407C	Endpoint 0 receive length register	0x00XX
R16_UEP1_RX_LEN	0x40024080	Endpoint 1 single receive length register	0XXXXX
R16_UEP1_R_SIZE	0x40024082	Endpoint 1 total receive data length register	0XXXXX
R16_UEP2_RX_LEN	0x40024084	Endpoint 2 single receive length register	0XXXXX
R16_UEP2_R_SIZE	0x40024086	Endpoint 2 total receive data length register	0XXXXX
R16_UEP3_RX_LEN	0x40024088	Endpoint 3 single receive length register	0XXXXX
R16_UEP3_R_SIZE	0x4002408A	Endpoint 3 total receive data length register	0XXXXX
R16_UEP4_RX_LEN	0x4002408C	Endpoint 4 single receive length register	0XXXXX
R16_UEP4_R_SIZE	0x4002408E	Endpoint 4 total receive data length register	0XXXXX
R16_UEP5_RX_LEN	0x40024090	Endpoint 5 single receive length register	0XXXXX
R16_UEP5_R_SIZE	0x40024092	Endpoint 5 total receive data length register	0XXXXX
R16_UEP6_RX_LEN	0x40024094	Endpoint 6 single receive length register	0XXXXX
R16_UEP6_R_SIZE	0x40024096	Endpoint 6 total receive data length register	0XXXXX
R16_UEP7_RX_LEN	0x40024098	Endpoint 7 single receive length register	0XXXXX
R16_UEP7_R_SIZE	0x4002409A	Endpoint 7 total receive data length register	0XXXXX
R16_UEP0_T_LEN	0x4002409C	Endpoint 0 transmit length register	0x00XX
R8_UEP0_TX_CTRL	0x4002409E	Endpoint 0 transmit control register	0x00
R8_UEP0_RX_CTRL	0x4002409F	Endpoint 0 receive control register	0x00
R16_UEP1_T_LEN	0x400240A0	Endpoint 1 transmit length register	0x0000
R8_UEP1_TX_CTRL	0x400240A2	Endpoint 1 transmit control register	0x00
R8_UEP1_RX_CTRL	0x400240A3	Endpoint 1 receive control register	0x00
R16_UEP2_T_LEN	0x400240A4	Endpoint 2 transmit length register	0x0000
R8_UEP2_TX_CTRL	0x400240A6	Endpoint 2 transmit control register	0x00
R8_UEP2_RX_CTRL	0x400240A7	Endpoint 2 receive control register	0x00
R16_UEP3_T_LEN	0x400240A8	Endpoint 3 transmit length register	0x0000
R8_UEP3_TX_CTRL	0x400240AA	Endpoint 3 transmit control register	0x00
R8_UEP3_RX_CTRL	0x400240AB	Endpoint 3 receive control register	0x00
R16_UEP4_T_LEN	0x400240AC	Endpoint 4 transmit length register	0x0000
R8_UEP4_TX_CTRL	0x400240AE	Endpoint 4 transmit control register	0x00
R8_UEP4_RX_CTRL	0x400240AF	Endpoint 4 receive control register	0x00
R16_UEP5_T_LEN	0x400240B0	Endpoint 5 transmit length register	0x0000
R8_UEP5_TX_CTRL	0x400240B2	Endpoint 5 transmit control register	0x00
R8_UEP5_RX_CTRL	0x400240B3	Endpoint 5 receive control register	0x00
R16_UEP6_T_LEN	0x400240B4	Endpoint 6 transmit length register	0x0000
R8_UEP6_TX_CTRL	0x400240B6	Endpoint 6 transmit control register	0x00
R8_UEP6_RX_CTRL	0x400240B7	Endpoint 6 receive control register	0x00
R16_UEP7_T_LEN	0x400240B8	Endpoint 7 transmit length register	0x0000

Name	Address	Description	Reset Value
R8_UEP7_TX_CTRL	0x400240BA	Endpoint 7 transmit control register	0x00
R8_UEP7_RX_CTRL	0x400240BB	Endpoint 7 receive control register	0x00
R16_UEP_T_ISO	0x400240BC	USBHS endpoint transmit isochronous mode enable register	0x0000
R16_UEP_R_ISO	0x400240BE	USBHS endpoint receive isochronous mode enable register	0x0000

Table 22-2: USBHS2 Device-Related Register List

Name	Address	Description	Reset Value
R8_USB_CTRL	0x40023400	USBHS control register	0x07
R8_USB_BASE_MODE	0x40023401	USBHS mode control register	0x00
R8_USB_INT_EN	0x40023402	USBHS interrupt enable register	0x00
R8_USB_DEV_AD	0x40023403	USBHS device address register	0x00
R8_USB_WAKE_CTRL	0x40023404	USBHS remote wakeup register	0x00
R8_USB_TEST_MODE	0x40023405	USBHS test mode register	0x00
R16_USB_LPM_DATA	0x40023406	USBHS power management register	0x8000
R8_USB_INT_FG	0x40023408	USBHS interrupt flag register	0x00
R8_USB_INT_ST	0x40023409	USBHS interrupt status register	0x00
R8_USB_MIS_ST	0x4002340A	USBHS miscellaneous status register	0x00
R16_USB_FRAME_NO	0x4002340C	USBHS frame number register	0x0000
R16_USB_BUS	0x4002340E	USBHS bus status register	0x0000
R16_UEP_TX_EN	0x40023410	USBHS endpoint transmit enable register	0x0000
R16_UEP_RX_EN	0x40023412	USBHS endpoint receive enable register	0x0000
R16_UEP_T_TOG_AUTO	0x40023414	USBHS endpoint transmit auto-toggle enable register	0x0000
R16_UEP_R_TOG_AUTO	0x40023416	USBHS endpoint receive auto-toggle enable register	0x0000
R8_UEP_T_BURST	0x40023418	USBHS endpoint transmit burst register	0x00
R8_UEP_T_BURST_MODE	0x40023419	USBHS endpoint transmit burst mode register	0x00
R8_UEP_R_BURST	0x4002341A	USBHS endpoint receive burst register	0x00
R8_UEP_R_RES_MODE	0x4002341B	USBHS endpoint receive response mode register	0x00
R32_UEP_AF_MODE	0x4002341C	USBHS endpoint multiplex register	0x00000000
R32_UEP0_DMA	0x40023420	Endpoint 0 buffer start address register	0x000XXXXX
R32_UEP1_RX_DMA	0x40023424	Endpoint 1 receive buffer start address register	0x000XXXXX
R32_UEP2_RX_DMA	0x40023428	Endpoint 2 receive buffer start address register	0x000XXXXX
R32_UEP3_RX_DMA	0x4002342C	Endpoint 3 receive buffer start address register	0x000XXXXX
R32_UEP4_RX_DMA	0x40023430	Endpoint 4 receive buffer start address register	0x000XXXXX
R32_UEP5_RX_DMA	0x40023434	Endpoint 5 receive buffer start address register	0x000XXXXX
R32_UEP6_RX_DMA	0x40023438	Endpoint 6 receive buffer start address register	0x000XXXXX

Name	Address	Description	Reset Value
R32_UEP7_RX_DMA	0x4002343C	Endpoint 7 receive buffer start address register	0x000XXXXX
R32_UEP1_TX_DMA	0x40023440	Endpoint 1 transmit buffer start address register	0x000XXXXX
R32_UEP2_TX_DMA	0x40023444	Endpoint 2 transmit buffer start address register	0x000XXXXX
R32_UEP3_TX_DMA	0x40023448	Endpoint 3 transmit buffer start address register	0x000XXXXX
R32_UEP4_TX_DMA	0x4002344C	Endpoint 4 transmit buffer start address register	0x000XXXXX
R32_UEP5_TX_DMA	0x40023450	Endpoint 5 transmit buffer start address register	0x000XXXXX
R32_UEP6_TX_DMA	0x40023454	Endpoint 6 transmit buffer start address register	0x000XXXXX
R32_UEP7_TX_DMA	0x40023458	Endpoint 7 transmit buffer start address register	0x000XXXXX
R32_UEP0_MAX_LEN	0x4002345C	Endpoint 0 max length packet register	0x000000XX
R32_UEP1_MAX_LEN	0x40023460	Endpoint 1 max length packet register	0x00000XXX
R32_UEP2_MAX_LEN	0x40023464	Endpoint 2 max length packet register	0x00000XXX
R32_UEP3_MAX_LEN	0x40023468	Endpoint 3 max length packet register	0x00000XXX
R32_UEP4_MAX_LEN	0x4002346C	Endpoint 4 max length packet register	0x00000XXX
R32_UEP5_MAX_LEN	0x40023470	Endpoint 5 max length packet register	0x00000XXX
R32_UEP6_MAX_LEN	0x40023474	Endpoint 6 max length packet register	0x00000XXX
R32_UEP7_MAX_LEN	0x40023478	Endpoint 7 max length packet register	0x00000XXX
R16_UEP0_RX_LEN	0x4002347C	Endpoint 0 receive length register	0x00XX
R16_UEP1_RX_LEN	0x40023480	Endpoint 1 single receive length register	0XXXXX
R16_UEP1_R_SIZE	0x40023482	Endpoint 1 total receive data length register	0XXXXX
R16_UEP2_RX_LEN	0x40023484	Endpoint 2 single receive length register	0XXXXX
R16_UEP2_R_SIZE	0x40023486	Endpoint 2 total receive data length register	0XXXXX
R16_UEP3_RX_LEN	0x40023488	Endpoint 3 single receive length register	0XXXXX
R16_UEP3_R_SIZE	0x4002348A	Endpoint 3 total receive data length register	0XXXXX
R16_UEP4_RX_LEN	0x4002348C	Endpoint 4 single receive length register	0XXXXX
R16_UEP4_R_SIZE	0x4002348E	Endpoint 4 total receive data length register	0XXXXX
R16_UEP5_RX_LEN	0x40023490	Endpoint 5 single receive length register	0XXXXX
R16_UEP5_R_SIZE	0x40023492	Endpoint 5 total receive data length register	0XXXXX
R16_UEP6_RX_LEN	0x40023494	Endpoint 6 single receive length register	0XXXXX
R16_UEP6_R_SIZE	0x40023496	Endpoint 6 total receive data length register	0XXXXX
R16_UEP7_RX_LEN	0x40023498	Endpoint 7 single receive length register	0XXXXX
R16_UEP7_R_SIZE	0x4002349A	Endpoint 7 total receive data length register	0XXXXX
R16_UEP0_T_LEN	0x4002349C	Endpoint 0 transmit length register	0x00XX
R8_UEP0_TX_CTRL	0x4002349E	Endpoint 0 transmit control register	0x00
R8_UEP0_RX_CTRL	0x4002349F	Endpoint 0 receive control register	0x00
R16_UEP1_T_LEN	0x400234A0	Endpoint 1 transmit length register	0x0000
R8_UEP1_TX_CTRL	0x400234A2	Endpoint 1 transmit control register	0x00
R8_UEP1_RX_CTRL	0x400234A3	Endpoint 1 receive control register	0x00

Name	Address	Description	Reset Value
R16_UEP2_T_LEN	0x400234A4	Endpoint 2 transmit length register	0x0000
R8_UEP2_TX_CTRL	0x400234A6	Endpoint 2 transmit control register	0x00
R8_UEP2_RX_CTRL	0x400234A7	Endpoint 2 receive control register	0x00
R16_UEP3_T_LEN	0x400234A8	Endpoint 3 transmit length register	0x0000
R8_UEP3_TX_CTRL	0x400234AA	Endpoint 3 transmit control register	0x00
R8_UEP3_RX_CTRL	0x400234AB	Endpoint 3 receive control register	0x00
R16_UEP4_T_LEN	0x400234AC	Endpoint 4 transmit length register	0x0000
R8_UEP4_TX_CTRL	0x400234AE	Endpoint 4 transmit control register	0x00
R8_UEP4_RX_CTRL	0x400234AF	Endpoint 4 receive control register	0x00
R16_UEP5_T_LEN	0x400234B0	Endpoint 5 transmit length register	0x0000
R8_UEP5_TX_CTRL	0x400234B2	Endpoint 5 transmit control register	0x00
R8_UEP5_RX_CTRL	0x400234B3	Endpoint 5 receive control register	0x00
R16_UEP6_T_LEN	0x400234B4	Endpoint 6 transmit length register	0x0000
R8_UEP6_TX_CTRL	0x400234B6	Endpoint 6 transmit control register	0x00
R8_UEP6_RX_CTRL	0x400234B7	Endpoint 6 receive control register	0x00
R16_UEP7_T_LEN	0x400234B8	Endpoint 7 transmit length register	0x0000
R8_UEP7_TX_CTRL	0x400234BA	Endpoint 7 transmit control register	0x00
R8_UEP7_RX_CTRL	0x400234BB	Endpoint 7 receive control register	0x00
R16_UEP_T_ISO	0x400234BC	USBHS endpoint transmit isochronous mode enable register	0x0000
R16_UEP_R_ISO	0x400234BE	USBHS endpoint receive isochronous mode enable register	0x0000

22.2.1.1 USBHS Control Register (R8_USB_CTRL)

Offset address: 0x00

Bits	Name	Access	Description	Reset
7	RB_UD_LPM_EN	RW	LPM enable: 1: Enable; 0: Disable.	0
6	Reserved	RO	Reserved.	0
5	RB_UD_DEV_EN	RW	USB device enable: 1: Enable; 0: Disable.	0
4	RB_UD_DMA_EN	RW	DMA transfer enable: 1: Enable; 0: Disable.	0
3	RB_UD_PHY_SUSPENDM	RW	USB PHY suspend: 1: Normal operation; 0: Suspend.	0

Bits	Name	Access	Description	Reset
2	RB_UD_CLR_ALL	RW	Clear all interrupt flags: 1: Clear USB interrupt flags and FIFO, requires software to clear to 0; 0: Do not clear.	1
1	RB_UD_RST_SIE	RW	USB protocol processor reset: 1: Force reset USB protocol processor (SIE), including endpoint-related registers, requires software to clear to 0; 0: No reset.	1
0	RB_UD_RST_LINK	RW	LINK layer reset: 1: USB Link layer reset; 0: No reset.	1

22.2.1.2 USBHS Mode Control Register (R8_USB_BASE_MODE)

Offset address: 0x01

Bits	Name	Access	Description	Reset
[7:2]	Reserved	RO	Reserved.	0x0
[1:0]	RB_UD_SPEED_TYPE[1:0]	RW	Device expected speed mode: 00: Full-speed device; 01: High-speed device; 10: Low-speed device; 11: Reserved.	0

22.2.1.3 USBHS Interrupt Enable Register (R8_USB_INT_EN)

Offset address: 0x02

Bits	Name	Access	Description	Reset
7	RB_UDIE_FIFO_OVER	RW	FIFO overflow interrupt: 1: Enable interrupt; 0: Disable interrupt.	0
6	RB_UDIE_LINK_RDY	RW	USB connect interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
5	RB_UDIE_SOF_ACT	RW	Receive SOF packet interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
4	RB_UDIE_TRANSFER	RW	USB transfer complete interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
3	RB_UDIE_LPM_ACT	RW	LPM transfer complete interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
2	RB_UDIE_BUS_SLEEP	RW	USB bus sleep interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0

Bits	Name	Access	Description	Reset
1	RB_UDIE_SUSPEND	RW	USB bus suspend interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0
0	RB_UDIE_BUS_RST	RW	USB bus reset interrupt enable: 1: Enable interrupt; 0: Disable interrupt.	0

22.2.1.4 USBHS Device Address Register (R8_USB_DEV_AD)

Offset address: 0x03

Bits	Name	Access	Description	Reset
7	Reserved	RO	Reserved.	0
[6:0]	RB_UD_DEV_ADDR[6:0]	RW	USB device address.	0

22.2.1.5 USBHS Remote Wakeup Register (R8_USB_WAKE_CTRL)

Offset address: 0x04

Bits	Name	Access	Description	Reset
[7:1]	Reserved	RO	Reserved.	0
0	RB_UD_REMOTE_WKUP	RW1	Remote wakeup, hardware auto-clears.	0

22.2.1.6 USBHS Test Mode Register (R8_USB_TEST_MODE)

Offset address: 0x05

Bits	Name	Access	Description	Reset
7	RB_UD_TEST_EN	RW	Test mode enable.	0
[6:4]	Reserved	RO	Reserved.	0
3	RB_UD_TEST_SE0NAK	RW	Test mode, output SE0.	0
2	RB_UD_TEST_PKT	RW	Test mode, output one packet. DATA0, endpoint 4 data address and length, TOG is DATA0.	0
1	RB_UD_TEST_K	RW	Test mode, output K.	0
0	RB_UD_TEST_J	RW	Test mode, output J.	0

22.2.1.7 USBHS Power Management Register (R16_USB_LPM_DATA)

Offset address: 0x06

Bits	Name	Access	Description	Reset
15	RB_UD_LPM_BUSY	RW	Power management busy.	1
[14:11]	Reserved	RO	Reserved.	0
[10:0]	RB_UD_LPM_DATA[10:0]	RO	Power management data.	0

22.2.1.8 USBHS Interrupt Flag Register (R8_USB_INT_FG)

Offset address: 0x08

Bits	Name	Access	Description	Reset
7	RB_UDIF_FIFO_OV	RW1Z	FIFO overflow interrupt flag, write 1 to clear. 1: FIFO overflow triggered; 0: No event.	0
6	RB_UDIF_LINK_RDY	RW1Z	USB connect interrupt flag, write 1 to clear. 1: USB connect event triggered; 0: No event.	0
5	RB_UDIF_RX_SOF	RW1Z	Receive SOF packet interrupt flag, write 1 to clear. 1: Receive SOF packet event triggered; 0: No event.	0
4	RB_UDIF_RTX_ACT	RO	USB transfer complete interrupt flag: Receive is cleared by ENDP_R_DONE; Transmit is cleared by ENDP_T_DONE.	0
3	RB_UDIF_LPM_ACT	RW1Z	LPM transfer complete interrupt flag, write 1 to clear. 1: LPM transfer complete event triggered; 0: No event.	0
2	RB_UDIF_BUS_SLEEP	RW1Z	USB bus sleep interrupt flag, write 1 to clear. 1: USB bus sleep event triggered; 0: No event.	0
1	RB_UDIF_SUSPEND	RW1Z	USB bus suspend interrupt flag, write 1 to clear. 1: USB suspend event triggered; 0: No event.	0
0	RB_UDIF_BUS_RST	RW1Z	USB bus reset interrupt flag, write 1 to clear. 1: USB bus reset event triggered; 0: No event.	0

22.2.1.9 USBHS Interrupt Status Register (R8_USB_INT_ST)

Offset address: 0x09

Bits	Name	Access	Description	Reset
[7:5]	Reserved	RO	Reserved.	0
4	RB_UDIS_EP_DIR	RO	Endpoint data transfer direction: 1: Endpoint IN data; 0: Endpoint OUT/SETUP data.	0
3	Reserved	RO	Reserved.	0
[2:0]	RB_UDIS_EP_ID_MASK[2:0]	RO	Endpoint number where data transfer occurred.	0

22.2.1.10 USBHS Miscellaneous Status Register (R8_USB_MIS_ST)

Offset address: 0x0A

Bits	Name	Access	Description	Reset
7	RB_UDMS_HS_MOD	RO	Whether the host is high-speed.	0

Bits	Name	Access	Description	Reset
[6:5]	Reserved	RO	Reserved.	0
4	RB_UDMS_SUSP_REQ	RO	USB suspend request.	0
3	RB_UDMS_SIE_FREE	RO	USB idle status.	0
2	RB_UDMS_SLEEP	RO	USB sleep status.	0
1	RB_UDMS_SUSPEND	RO	USB suspend status: 1: USB bus is in suspend state, no USB activity for a period of time; 0: USB bus is in non-suspend state.	0
0	RB_UDMS_READY	RO	USB connection status.	0

22.2.1.11 USBHS Frame Register (R16_USB_FRAME_NO)

Offset address: 0x0C

Bits	Name	Access	Description	Reset
[15:13]	RB_UD_MFRAME_NO[2:0]	RO	Received micro-frame number.	0
[12:11]	Reserved	RO	Reserved.	0
[10:0]	RB_UD_FRAME_NO[10:0]	RO	Received frame number.	0

22.2.1.12 USBHS Bus Status Register (R16_USB_BUS)

Offset address: 0x0E

Bits	Name	Access	Description	Reset
[15:4]	Reserved	RO	Reserved.	0
3	RB_USB_DM_ST	RO	UDM status.	0
2	RB_USB_DP_ST	RO	UDP status.	0
1	Reserved	RO	Reserved.	0
0	RB_USB_WAKEUP	RO	USB wakeup (active high).	0

22.2.1.13 USBHS Endpoint Transmit Enable Register (R16_UEP_TX_EN)

Offset address: 0x10

Bits	Name	Access	Description	Reset
[15:0]	RB_UEP_TX_EN[15:0]	RW	Endpoints 0~15 transmit enable: 1: Enable; 0: Disable.	0

22.2.1.14 USBHS Endpoint Receive Enable Register (R16_UEP_RX_EN)

Offset address: 0x12

Bits	Name	Access	Description	Reset
[15:0]	RB_UEP_RX_EN[15:0]	RW		0

Bits	Name	Access	Description	Reset
			Endpoints 0~15 receive enable: 1: Enable; 0: Disable.	

22.2.1.15 USBHS Endpoint Transmit Auto-Toggle Enable Register (R16_UEP_T_TOG_AUTO)

Offset address: 0x14

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
[7:1]	RB_UEP_T_TOG_AUTO[7:1]	RW	Endpoints 1~7 sync toggle bit auto-toggle enable: 1: Auto-toggle after successful data transmission; 0: Manual toggle control.	0
0	Reserved	RO	Reserved.	0

22.2.1.16 USBHS Endpoint Receive Auto-Toggle Enable Register (R16_UEP_R_TOG_AUTO)

Offset address: 0x16

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
[7:1]	RB_UEP_R_TOG_AUTO[7:1]	RW	Endpoints 1~7 sync toggle bit auto-toggle enable: 1: Auto-toggle after successful data reception; 0: Manual toggle control.	0
0	Reserved	RO	Reserved.	0

22.2.1.17 USBHS Endpoint Transmit Burst Register (R8_UEP_T_BURST)

Offset address: 0x18

Bits	Name	Access	Description	Reset
[7:0]	RB_UEP_T_BURST_EN[7:0]	RW	Endpoints 0~7 burst transmit enable: 1: Enable; 0: Disable.	0

22.2.1.18 USBHS Endpoint Transmit Mode Register (R8_UEP_T_BURST_MODE)

Offset address: 0x19

Bits	Name	Access	Description	Reset
[7:0]	RB_UEP_T_BURST_MODE[7:0]	RW	Endpoints 0~7 burst transmit mode: 1: Do not send 0-length data in burst mode; 0: Send 0-length data in burst mode.	0

22.2.1.19 USBHS Endpoint Receive Burst Register (R8_UEP_R_BURST)

Offset address: 0x1A

Bits	Name	Access	Description	Reset
[7:0]	RB_UEP_R_BURST_EN[7:0]	RW	Endpoints 0~7 burst receive enable: 1: Enable; 0: Disable.	0

22.2.1.20 USBHS Endpoint Response Mode Register (R8_UEP_R_RES_MODE)

Offset address: 0x1B

Bits	Name	Access	Description	Reset
[7:0]	RB_UEP_R_RES_MODE[7:0]	RW	Endpoints 0~7 receive response mode: 1: NYET; 0: ACK.	0

22.2.1.21 USBHS Endpoint Multiplex Register (R32_UEP_AF_MODE)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							UEP_T_AF[6:0]					Reserved			

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:1]	RB_UEP_T_AF[6:0]	RW	Endpoints 1~7 multiplex enable: 1: Multiplex to 9-15; 0: Multiplex to 1-7.	0
0	Reserved	RO	Reserved.	0

22.2.1.22 Endpoint 0 Buffer Start Address Register (R32_UEP0_DMA)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UEP0_BUF_DMA[15:0]															

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
[16:0]	UEP0_BUF_DMA[16:0]	RW	Endpoint 0 buffer start address.	X

22.2.1.23 Endpoint n Receive Buffer Start Address Register (R32_UEPn_RX_DMA) (n=1~7)

Offset address: $0x24 + 4*(n-1)$

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
[16:0]	UEPn_RX_DMA[16:0]	RW	Endpoint n receive buffer start address. The address must be 4-byte aligned.	X

22.2.1.24 Endpoint n Transmit Buffer Start Address Register (R32_UEPn_TX_DMA) (n=1~7)

Offset address: $0x40 + 4*(n-1)$

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
[16:0]	UEPn_TX_DMA[16:0]	RW	Endpoint n transmit buffer start address. The address must be 4-byte aligned.	X

22.2.1.25 Endpoint 0 Max Length Packet Register (R32_UEP0_MAX_LEN)

Offset address: $0x5C$

Bits	Name	Access	Description	Reset
[31:7]	Reserved	RO	Reserved.	0
[6:0]	UEP0_MAX_LEN[6:0]	RW	Maximum limit length that endpoint 0 DMA can offset.	X

22.2.1.26 Endpoint n Max Length Packet Register (R32_UEPn_MAX_LEN) (n=1~7)

Offset address: $0x5C + 4*n$

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
[10:0]	UEPn_MAX_LEN[10:0]	RW	Maximum limit length that endpoint n DMA can offset.	X

22.2.1.27 Endpoint 0 Receive Length Register (R16_UEP0_RX_LEN)

Offset address: $0x7C$

Bits	Name	Access	Description	Reset
[15:7]	Reserved	RO	Reserved.	0
[6:0]	UEP0_RX_LEN[6:0]	RW	Endpoint 0 received data length.	X

22.2.1.28 Endpoint n Single Receive Length Register (R16_UEPn_RX_LEN) (n=1~7)

Offset address: $0x80 + 4*(n-1)$

Bits	Name	Access	Description	Reset
[15:0]	UEPn_RX_LEN[15:0]	RW	Endpoint n single receive data length.	X

22.2.1.29 Endpoint n Total Receive Data Length Register (R16_UEPn_R_SIZE) (n=1~7)

Offset address: 0x82 + 4*(n-1)

Bits	Name	Access	Description	Reset
[15:0]	UEPn_R_SIZE[15:0]	RW	Endpoint n total received data length.	X

22.2.1.30 Endpoint 0 Transmit Length Register (R16_UEP0_T_LEN)

Offset address: 0x9C

Bits	Name	Access	Description	Reset
[15:7]	Reserved	RO	Reserved.	0
[6:0]	UEP0_T_LEN[6:0]	RW	Endpoint 0 transmit data length.	X

22.2.1.31 Endpoint 0 Transmit Control Register (R8_UEP0_TX_CTRL)

Offset address: 0x9E

Bits	Name	Access	Description	Reset
7	RB_UEP_T_DONE	RW0	Endpoint 0 transmit complete flag, write 0 to clear.	0
6	RB_UEP_T_NAK_ACT	RW0	Endpoint 0 transmit NAK complete flag, write 0 to clear.	0
[5:3]	Reserved	RO	Reserved.	0
2	RB_UEP_T_TOG_MASK	RW	Endpoint 0 transmit expected sync toggle bit: 1: DATA1; 0: DATA0.	0
[1:0]	RB_UEP_T_RES_MASK[1:0]	RW	Endpoint 0 transmit response control: 00: Respond NAK or busy; 01: Respond STALL or error; 10: Respond ACK; 11: Reserved.	0

22.2.1.32 Endpoint 0 Receive Control Register (R8_UEP0_RX_CTRL)

Offset address: 0x9F

Bits	Name	Access	Description	Reset
7	RB_UEP_R_DONE	RW0	Endpoint 0 receive complete flag, write 0 to clear.	0
6	RB_UEP_R_NAK_ACT	RW0	Endpoint 0 receive NAK complete flag, write 0 to clear.	0
5	RB_UEP_R_NAK_TOG	RO	Endpoint 0 receive returned NAK, data packet type: 1: DATA1; 0: DATA0.	0
4	RB_UEP_R_TOG_MATCH	RO		0

Bits	Name	Access	Description	Reset
			Received sync toggle bit match status with expected sync toggle bit: 1: Synced; 0: Not synced.	
3	RB_UEP_R_SETUP_IS	RO	Whether endpoint 0 received a SETUP transaction.	0
2	RB_UEP_R_TOG_MASK	RW	Endpoint 0 receive expected sync toggle bit: 1: DATA1; 0: DATA0.	0
[1:0]	RB_UEP_R_RES_MASK[1:0]	RW	Endpoint 0 receive response control: 00: Respond NAK or busy; 01: Respond STALL or error; 10: Respond ACK; 11: Reserved.	0

22.2.1.33 Endpoint n Transmit Length Register (R16_UEPn_T_LEN) (n=1~7)

Offset address: 0xA0 + 4*(n-1)

Bits	Name	Access	Description	Reset
[15:0]	UEPn_T_LEN[15:0]	RW	Endpoint n transmit data length.	0

22.2.1.34 Endpoint n Transmit Control Register (R8_UEPn_TX_CTRL) (n=1~7)

Offset address: 0xA2 + 4*(n-1)

Bits	Name	Access	Description	Reset
7	RB_UEP_T_DONE	RW0	Endpoint n transmit complete flag, write 0 to clear.	0
6	RB_UEP_T_NAK_ACT	RW0	Endpoint n transmit NAK complete flag, write 0 to clear.	0
[5:4]	Reserved	RO	Reserved.	0
[3:2]	RB_UEP_T_TOG_MASK[1:0]	RW	Endpoint n transmit expected sync toggle bit: 00: DATA0; 01: DATA1; 10: DATA2; 11: MDATA. Note: This bit is configurable in manual mode, not configurable in auto mode.	0
[1:0]	RB_UEP_T_RES_MASK[1:0]	RW	Endpoint n transmit response control: 00: Respond NAK or busy; 01: Respond STALL or error; 10: Respond ACK; 11: Reserved.	0

22.2.1.35 Endpoint n Receive Control Register (R8_UEPn_RX_CTRL) (n=1~7)

Offset address: 0xA3 + 4*(n-1)

Bits	Name	Access	Description	Reset
7	RB_UEP_R_DONE	RW0	Endpoint n receive complete flag, write 0 to clear.	0
6	RB_UEP_R_NAK_ACT	RW0	Endpoint n receive NAK complete flag, write 0 to clear.	0
5	RB_UEP_R_NAK_TOG	RO	Endpoint n receive returned NAK, data packet type: 1: DATA1; 0: DATA0.	0
4	RB_UEP_R_TOG_MATCH	RO	Endpoint n received sync toggle bit match status with expected sync toggle bit: 1: Synced; 0: Not synced.	0
[3:2]	RB_UEP_R_TOG_MASK[1:0]	RW	Endpoint n receive expected sync toggle bit: 00: DATA0; 01: DATA1; 10: DATA2; 11: MDATA. Note: This bit is configurable in manual mode, not configurable in auto mode.	0
[1:0]	RB_UEP_R_RES_MASK[1:0]	RW	Endpoint n receive response control: 00: Respond NAK or busy; 01: Respond STALL or error; 10: Respond ACK; 11: Respond ACK (interrupt OUT).	0

22.2.1.36 USB Endpoint Transmit Isochronous Mode Enable Register (R16_UEP_T_ISO)

Offset address: 0xBC

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
[7:1]	RB_UEPn_T_ISO_EN[6:0]	RW	Upload endpoint (IN) isochronous mode enable.	0
0	Reserved	RO	Reserved.	0

22.2.1.37 USB Endpoint Receive Isochronous Mode Enable Register (R16_UEP_R_ISO)

Offset address: 0xBE

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
[7:1]	RB_UEPn_R_ISO_EN[6:0]	RW	Download endpoint (OUT) isochronous mode enable.	0
0	Reserved	RO	Reserved.	0

22.2.2 Host Register Description

Table 22-3: USBHS1 Host-Related Register List

Name	Address	Description	Reset Value
R8_UH_CFG	0x40023500	USBHS host configuration register	0x07
R8_UH_INT_EN	0x40023502	USBHS host interrupt enable register	0x00
R8_UH_DEV_AD	0x40023503	USBHS host device address register	0x00
R32_UH_CONTROL	0x40023504	USBHS host control register	0x00XXXXXX
R8_UH_INT_FLAG	0x40023508	USBHS host interrupt flag register	0x00
R8_UH_INT_ST	0x40023509	USBHS host interrupt status register	0xFF
R8_UH_MIS_ST	0x4002350A	USBHS host miscellaneous status register	0xFF
R32_UH_LPM_DATA	0x4002350C	USBHS host power management data register	0x00000XXX
R32_UH_SPLIT_DATA	0x40023510	USBHS host SPLIT data register	0x000XXXXX
R32_UH_FRAME	0x40023514	USBHS host frame register	0x00000000
R32_UH_TX_LEN	0x40023518	USBHS host transmit length register	0x00000XXX
R32_UH_RX_LEN	0x4002351C	USBHS host receive length register	0x00000XXX
R32_UH_RX_MAX_LEN	0x40023520	USBHS host receive max length register	0x00000XXX
R32_UH_RX_DMA	0x40023524	DMA receive address register	0x000XXXXX
R32_UH_TX_DMA	0x40023528	DMA transmit address register	0x000XXXXX
R32_UH_PORT_CTRL	0x4002352C	USBHS host port control register	0x0000X000
R8_UH_PORT_CFG	0x40023530	USBHS host port configuration register	0x00
R8_UH_PORT_INT_EN	0x40023532	USBHS host port interrupt enable register	0x00
R8_UH_PORT_TEST_CT	0x40023533	USBHS host port test mode register	0x00
R16_UH_PORT_ST	0x40023534	USBHS host port status register	0x0010
R8_UH_PORT_CHG	0x40023536	USBHS host port status change register	0x00
R32_UH_BC_CTRL	0x4002353C	USBHS host BC charging control register	0x00000000

Table 22-4: USBHS2 Host-Related Register List

Name	Address	Description	Reset Value
R8_UH_CFG	0x40024100	USBHS host configuration register	0x07
R8_UH_INT_EN	0x40024102	USBHS host interrupt enable register	0x00
R8_UH_DEV_AD	0x40024103	USBHS host device address register	0x00
R32_UH_CONTROL	0x40024104	USBHS host control register	0x00XXXXXX
R8_UH_INT_FLAG	0x40024108	USBHS host interrupt flag register	0x00
R8_UH_INT_ST	0x40024109	USBHS host interrupt status register	0xFF
R8_UH_MIS_ST	0x4002410A	USBHS host miscellaneous status register	0xFF
R32_UH_LPM_DATA	0x4002410C	USBHS host power management data register	0x00000XXX
R32_UH_SPLIT_DATA	0x40024110	USBHS host SPLIT data register	0x000XXXXX
R32_UH_FRAME	0x40024114	USBHS host frame register	0x00000000
R32_UH_TX_LEN	0x40024118	USBHS host transmit length register	0x00000XXX
R32_UH_RX_LEN	0x4002411C	USBHS host receive length register	0x00000XXX
R32_UH_RX_MAX_LEN	0x40024120	USBHS host receive max length register	0x00000XXX

Name	Address	Description	Reset Value
R32_UH_RX_DMA	0x40024124	DMA receive address register	0x000XXXXX
R32_UH_TX_DMA	0x40024128	DMA transmit address register	0x000XXXXX
R32_UH_PORT_CTRL	0x4002412C	USBHS host port control register	0x0000X000
R8_UH_PORT_CFG	0x40024130	USBHS host port configuration register	0x00
R8_UH_PORT_INT_EN	0x40024132	USBHS host port interrupt enable register	0x00
R8_UH_PORT_TEST_CT	0x40024133	USBHS host port test mode register	0x00
R16_UH_PORT_ST	0x40024134	USBHS host port status register	0x0010
R8_UH_PORT_CHG	0x40024136	USBHS host port status change register	0x00
R32_UH_BC_CTRL	0x4002413C	USBHS host BC charging control register	0x00000000

22.2.2.1 USBHS Host Configuration Register (R8_UH_CFG)

Offset address: 0x00

Bits	Name	Access	Description	Reset
7	RB_UH_LPM_EN	RW	LPM enable.	0
6	RB_UH_FORCE_FS	RW	Force use of USB_FS.	0
5	RB_UH_SOF_EN	RW	Enable SOF packet transmission.	0
4	RB_UH_DMA_EN	RW	Enable DMA transmission.	0
3	RB_UH_PHY_SUSPENDM	RW	PHY suspend.	0
2	RB_UH_CLR_ALL	RW	Clear interrupt flags and FIFO, requires software to clear to 0.	1
1	RB_UH_RST_SIE	RW	USB protocol processor reset, requires software to clear to 0.	1
0	RB_UH_RST_LINK	RW	USB connection control module reset.	1

22.2.2.2 USBHS Host Interrupt Enable Register (R8_UH_INT_EN)

Offset address: 0x02

Bits	Name	Access	Description	Reset
7	RB_UHIE_FIFO_OVER	RW	FIFO overflow interrupt enable.	0
6	RB_UHIE_TX_HALT	RW	Transmit halt interrupt enable.	0
5	RB_UHIE_SOF_ACT	RW	SOF packet transmission interrupt enable.	0
4	RB_UHIE_TRANSFER	RW	USB completion or transfer complete interrupt enable.	0
3	RB_UHIE_RESUME_ACT	RW	Bus resume interrupt enable.	0
2	RB_UHIE_WKUP_ACT	RW	Wakeup interrupt enable.	0
[1:0]	Reserved	RO	Reserved.	0

22.2.2.3 USBHS Host Device Address Register (R8_UH_DEV_AD)

Offset address: 0x03

Bits	Name	Access	Description	Reset
[7:0]	RB_UH_DEV_ADDR[7:0]	RW	Address of the USB device currently being operated.	0

22.2.2.4 USBHS Host Control Register (R32_UH_CONTROL)

Offset address: 0x04

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
23	RB_UH_RX_NO_RES	RW	IN-DATA no response, used for isochronous transfer or high-speed SPLIT packets.	X
22	RB_UH_TX_NO_RES	RW	OUT/SETUP-DATA no expected response, used for isochronous transfer or high-speed SPLIT packets.	X
21	RB_UH_RX_NO_DATA	RW	No data packet expected after IN token packet, used for high-speed SPLIT packets.	X
20	RB_UH_TX_NO_DATA	RW	No data packet after OUT/SETUP token packet, used for high-speed SPLIT packets.	X
19	RB_UH_PRE_PID_EN	RW	Low-speed preamble PRE PID enable bit. When the port operates in full-speed and needs to send a low-speed packet (PRE), this bit must be enabled. 1: Enable, used to communicate with low-speed USB devices through an external full-speed HUB; 0: Disable low-speed preamble packet.	X
18	RB_UH_SPLIT_VALID	RW	Send SPLIT packet valid.	X
17	RB_UH_LPM_VALID	RW	Send LPM packet valid.	X
16	RB_UH_HOST_ACTION	RW	HOST transaction execute enable, this bit is automatically cleared after the transaction completes.	0
[15:11]	Reserved	RO	Reserved.	0
10	RB_UH_BUF_MODE	RW	Data buffer control bit. 1: Use R32_UH_RX_DMA for transmit, R32_UH_TX_DMA for receive; 0: Use R32_UH_TX_DMA for transmit, R32_UH_RX_DMA for receive.	X
[9:8]	RB_UH_T_TOG_MASK[1:0]	RW	Data PID to send: 00: PID_DATA0; 01: PID_DATA1; 10: PID_DATA2; 11: PID_MDATA.	X
[7:4]	RB_UH_T_ENDP_MASK[3:0]	RW	Host transmit transaction token packet endpoint number.	X
[3:0]	RB_UH_T_TOKEN_MASK[3:0]	RW	Host transmit transaction token packet PID.	X

22.2.2.5 USBHS Host Interrupt Flag Register (R8_UH_INT_FLAG)

Offset address: 0x08

Bits	Name	Access	Description	Reset
7	RB_UHIF_FIFO_OVER	RW1Z	FIFO overflow interrupt flag.	0
6	RB_UHIF_TX_HALT	RW1Z	Transmit halt interrupt flag. In host mode, used for data overflow during downstream read, or when transmission is not completed at the EOF2 time point, abort transmission, write 1 to clear.	0
5	RB_UHIF_SOF_ACT	RW1Z	SOF packet transmission complete interrupt flag, write 1 to clear.	0
4	RB_UHIF_TRANSFER	RW1Z	USB transaction transfer complete interrupt flag, write 1 to clear.	0
3	RB_UHIF_RESUME_ACT_IF	RW1Z	Bus resume interrupt flag, write 1 to clear.	0
2	RB_UHIF_WKUP_ACT	RW1Z	Wakeup interrupt flag, write 1 to clear.	0
[1:0]	Reserved	RO	Reserved.	0

22.2.2.6 USBHS Host Interrupt Status Register (R8_UH_INT_ST)

Offset address: 0x09

Bits	Name	Access	Description	Reset
[7:5]	Reserved	RO	Reserved.	X
4	RB_UHIS_PORT_RX_RESUME	RW	This bit being 1 indicates the port received a wakeup signal.	X
[3:0]	RB_UH_R_TOKEN_MASK[3:0]	RO	Received PID.	X

22.2.2.7 USBHS Host Miscellaneous Status Register (R8_UH_MIS_ST)

Offset address: 0x0A

Bits	Name	Access	Description	Reset
7	RB_UHMS_BUS_SE0	RO	USB bus is SE0.	X
6	RB_UHMS_BUS_J	RO	USB bus is J.	X
[5:4]	RB_UHMS_LINESTATE[1:0]	RO	PHY Linestate signal.	X
3	RB_UHMS_USB_WAKEUP	RO	USB bus wakeup status: 1: Bus wakeup in progress; 0: Bus wakeup not in progress.	X
2	RB_UHMS_SOF_ACT	RO	USB bus SOF packet transmission status bit: 1: SOF packet being sent; 0: Transmission complete or idle.	X
1	RB_UHMS_SOF_PRE	RO	USB bus SOF packet pending status: 1: About to send SOF packet; 0: No SOF packet to send.	X
0	RB_UHMS_SOF_FREE	RO	Port enable status: 1: Port enabled; 0: Port not enabled.	X

22.2.2.8 USBHS Host Power Management Data Register (R32_UH_LPM_DATA)

Offset address: 0x0C

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
[10:0]	RB_UH_LPM_DATA[10:0]	RW	Link power management packet data content.	X

22.2.2.9 USBHS Host SPLIT Register (R32_UH_SPLIT_DATA)

Offset address: 0x10

Bits	Name	Access	Description	Reset
[31:19]	Reserved	RO	Reserved.	0
[18:0]	RB_UH_SPLIT_DATA[18:0]	RW	Host endpoint transmit SPLIT packet data content. From high bit to low bit: ET, E, S, Port, SC, Hub Addr.	X

22.2.2.10 USBHS Host Frame Register (R32_UH_FRAME)

Offset address: 0x14

Bits	Name	Access	Description	Reset
[31:26]	Reserved	RO	Reserved.	0
25	RB_UH_SOF_CNT_CLR	RW	SOF count clear.	0
24	RB_UH_SOF_CNT_EN	RW	SOF count enable.	0
[23:19]	Reserved	RO	Reserved.	0
[18:16]	RB_UH_MFRAME_NO[2:0]	RO	Micro-frame number of the SOF packet about to be sent.	0
[15:11]	Reserved	RO	Reserved.	0
[10:0]	RB_UH_FRAME_NO[10:0]	RW	Frame number of the SOF packet about to be sent.	0

22.2.2.11 USBHS Host Transmit Length Register (R32_UH_TX_LEN)

Offset address: 0x18

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
[10:0]	RB_UH_TX_LEN[10:0]	RW	Transmit data length.	X

22.2.2.12 USBHS Host Receive Length Register (R32_UH_RX_LEN)

Offset address: 0x1C

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
[10:0]	RB_UH_RX_LEN[10:0]	RW	Receive data length.	X

22.2.2.13 USBHS Host Receive Max Length Register (R32_UH_RX_MAX_LEN)

Offset address: 0x20

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
[10:0]	RB_UH_RX_MAX_LEN[10:0]	RW	Receive max length.	X

22.2.2.14 DMA Receive Address Register (R32_UH_RX_DMA)

Offset address: 0x24

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
[16:0]	R32_UH_RX_DMA[16:0]	RW	Receive address. The lower two bits are invalid, must be 4-byte aligned.	X

22.2.2.15 DMA Transmit Address Register (R32_UH_TX_DMA)

Offset address: 0x28

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
[16:0]	R16_UH_TX_DMA[16:0]	RW	Transmit address.	X

22.2.2.16 USBHS Host Port Control Register (R32_UH_PORT_CTRL)

Offset address: 0x2C

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
16	RB_UH_BUS_RST_LONG	RW	Bus reset time selection: 1: Long reset; 0: Normal reset.	0
[15:12]	RB_UH_PORT_SLEEP_BESL[3:0]	RW	Wakeup time control: 0000: 125us; 1000: 3ms; 0001: 150us; 1001: 4ms; 0010: 200us; 1010: 5ms; 0011: 300us; 1011: 6ms; 0100: 400us; 1100: 7ms; 0101: 500us; 1101: 8ms; 0110: 1ms; 1110: 9ms; 0111: 2ms; 1111: 10ms.	X
[11:9]	Reserved	RO	Reserved.	0
8	RB_UH_CLR_PORT_SLEEP	WO	PORT exits SLEEP state (LPM).	0
[7:6]	Reserved	RO	Reserved.	0
5	RB_UH_CLR_PORT_CONNECT	WO	Make the port enter port state.	0

Bits	Name	Access	Description	Reset
4	RB_UH_CLR_PORT_EN	WO	PORT exits enabled state and enters DISABLED state.	0
3	RB_UH_SET_PORT_SLEEP	WO	PORT enters SLEEP state (LPM).	0
2	RB_UH_CLR_PORT_SUSP	WO	PORT exits suspend state.	0
1	RB_UH_SET_PORT_SUSP	WO	PORT enters suspend state.	0
0	RB_UH_SET_PORT_RESET	WO	Port sends reset.	0

22.2.2.17 USBHS Host Port Configuration Register (R8_UH_PORT_CFG)

Offset address: 0x30

Bits	Name	Access	Description	Reset
7	RB_UH_PD_EN	RW	15K resistor pull-down enable in host mode.	0
[6:1]	Reserved	RO	Reserved.	0
0	RB_UH_HOST_EN	RW	USB port mode selection: 1: Port is host mode; 0: Port is device mode.	0

22.2.2.18 USBHS Host Port Interrupt Enable Register (R8_UH_PORT_INT_EN)

Offset address: 0x32

Bits	Name	Access	Description	Reset
[7:6]	Reserved	RO	Reserved.	0
5	RB_UHIE_PORT_SLP	RW	Port sleep state change interrupt enable.	0
4	RB_UHIE_PORT_RESET	RW	Port reset state change interrupt enable.	0
3	Reserved	RO	Reserved.	0
2	RB_UHIE_PORT_SUSP	RW	Port suspend state change interrupt enable.	0
1	RB_UHIE_PORT_EN	RW	Port enable state change interrupt enable.	0
0	RB_UHIE_PORT_CONNECT	RW	Port connect state change interrupt enable.	0

22.2.2.19 USBHS Host Port Test Mode Register (R8_UH_PORT_TEST_CT)

Offset address: 0x33

Bits	Name	Access	Description	Reset
[7:3]	Reserved	RO	Reserved.	0
2	RB_UH_TEST_FORCE_EN	RW	Test mode enable.	0
1	RB_UH_TEST_K	RW	Test output K.	0
0	RB_UH_TEST_J	RW	Test output J.	0

22.2.2.20 USBHS Host Port Status Register (R16_UH_PORT_ST)

Offset address: 0x34

Bits	Name	Access	Description	Reset
[15:12]	Reserved	RO	Reserved.	0
11	RB_UHIS_PORT_TEST	RO	Whether the port is in test mode.	0
10	RB_UHIS_PORT_HS	RO	Whether the port connection speed is high-speed.	0
9	RB_UHIS_PORT_LS	RO	Whether the port connection speed is low-speed.	0
[8:6]	Reserved	RO	Reserved.	0
5	RB_UHIS_PORT_SLP	RO	Port sleep.	0
4	RB_UHIS_PORT_RST	RO	Port reset status.	1
3	Reserved	RO	Reserved.	0
2	RB_UHIS_PORT_SUSP	RO	Port suspend status.	0
1	RB_UHIS_PORT_EN	RO	Port enable.	0
0	RB_UHIS_PORT_CONNECT	RO	Port connection status.	0

22.2.2.21 USBHS Host Port Status Change Register (R8_UH_PORT_CHG)

Offset address: 0x36

Bits	Name	Access	Description	Reset
[7:6]	Reserved	RO	Reserved.	0
5	RB_UHIF_PORT_SLP	RW1Z	Port sleep state change, write 1 to clear.	0
4	RB_UHIF_PORT_RESET	RW1Z	Port reset state change, write 1 to clear.	0
3	Reserved	RO	Reserved.	0
2	RB_UHIF_PORT_SUSP	RW1Z	Port suspend state change, write 1 to clear.	0
1	RB_UHIF_PORT_EN	RW1Z	Port enable state change, write 1 to clear.	0
0	RB_UHIF_PORT_CONNECT	RW1Z	Port connection state change, write 1 to clear.	0

22.2.2.22 USBHS Host BC Charging Control Register (R32_UH_BC_CTRL)

Offset address: 0x3C

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
10	RB_UDM_VSRC_ACT	RO	In auto mode, this bit indicates UDM outputs VBC_SRC; otherwise it is controlled by RB_UDM_BC_CMPE. Note: This bit is 1 when UDP is higher than VBC_SRC but UDP is not at high level, otherwise it is 0.	0
9	RB_UDM_BC_VSRC	RW	UDM pin BC protocol source voltage enable: 1: UDM pin outputs BC protocol source voltage VBC_SRC; 0: Disable output.	0
8	RB_UDP_BC_VSRC	RW	UDP pin BC protocol source voltage enable: 1: UDP pin outputs BC protocol source voltage VBC_SRC; 0: Disable output.	0

Bits	Name	Access	Description	Reset
7	Reserved	RO	Reserved.	0
6	RB_BC_AUTO_MODE	RW	Auto mode enable: 1: Enable; 0: Disable.	0
5	RB_UDM_BC_CMPE	RW	UDM pin BC protocol comparator enable: 1: Enable; 0: Disable.	0
4	RB_UDP_BC_CMPE	RW	UDP pin BC protocol comparator enable: 1: Enable; 0: Disable.	0
[3:2]	Reserved	RO	Reserved.	0
1	RB_UDM_BC_CMPO	RO	UDM pin BC protocol comparator status: 1: UDM pin voltage is higher than BC protocol reference value VBC_REF; 0: UDM pin voltage is lower than BC protocol reference value VBC_REF.	0
0	RB_UDP_BC_CMPO	RO	UDP pin BC protocol comparator status: 1: UDP pin voltage is higher than BC protocol reference value VBC_REF; 0: UDP pin voltage is lower than BC protocol reference value VBC_REF.	0

Chapter 23: Digital Video Port (DVP)

The Digital Video Port (DVP) is used to connect to camera modules to acquire image data streams. It provides an 8/10/12-bit parallel interface for communication and supports a maximum pixel clock input frequency of 150 MHz. It supports image data organized in raw line and frame formats, such as YUV, RGB, etc., and also supports compressed image data such as JPEG format. It can receive high-speed parallel data streams output by external 8-bit, 10-bit, and 12-bit camera modules. During reception, synchronization is primarily achieved through VSYNC and HSYNC signals. Image cropping functionality is supported.

23.1 Main Features

- Configurable 8/10/12-bit data width mode
- Supports YUV and RGB format data
- Supports JPEG compressed format data
- Built-in FIFO, supports DMA transfer
- Supports double-buffer reception
- Supports cropping function
- Supports continuous mode and snapshot mode

23.2 Functional Description

23.2.1 Connecting to the Sensor

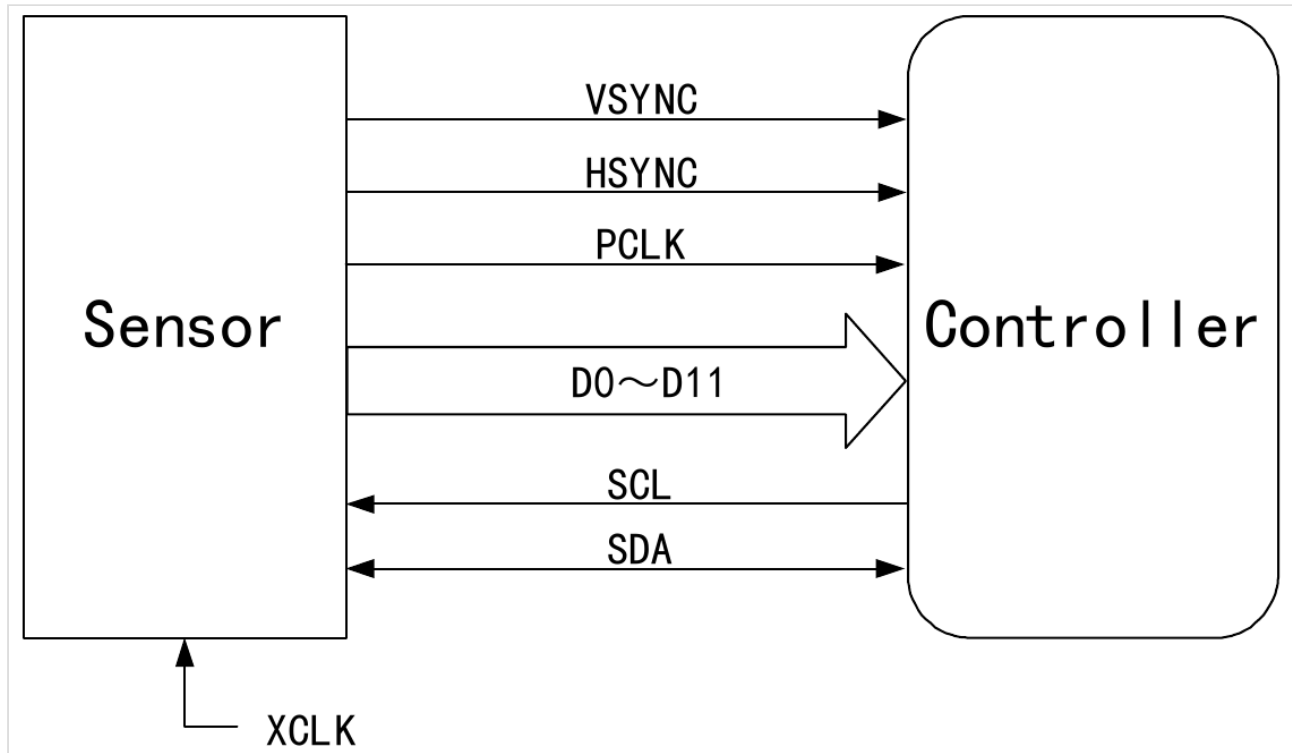


Figure 23-1: DVP interface connection

- **PCLK (Pixel clk)**: Pixel clock; each clock corresponds to one pixel of data (uncompressed data). The maximum supported HCLK clock frequency output by the external DVP interface sensor is 150 MHz.
- **HSYNC (horizontal synchronization)**: Line synchronization signal.
- **VSYNC (vertical synchronization)**: Frame synchronization signal.
- **DATA**: Pixel data or compressed data; bit width supports 8/10/12 bits.
- **XCLK**: Reference clock for the sensor; can be provided by the microcontroller or externally, typically using a crystal oscillator.
- **I2C interface**: Used to configure the sensor's registers; can be implemented by bit-banging I2C timing through ordinary GPIO pins, or by using the hardware I2C interface.

Table 23-1: DVP pins

Name	Signal Type	Description
PCLK	Input	Pixel clock input
DATA[11:0]	Input	Pixel data input
VSYNC	Input	Frame synchronization signal input
HSYNC	Input	Line synchronization signal input

23.2.2 Working Timing

The digital signal data stream output by a commonly used DVP interface sensor module has a certain relationship with the image size. The following is an example using one image data.

As shown in Figure 23-2, the sensor internally has a complete image size of 3576×1145. After internal scaling processing, the image data size finally output from the interface is 1920×1080, with an image refresh rate of 25.

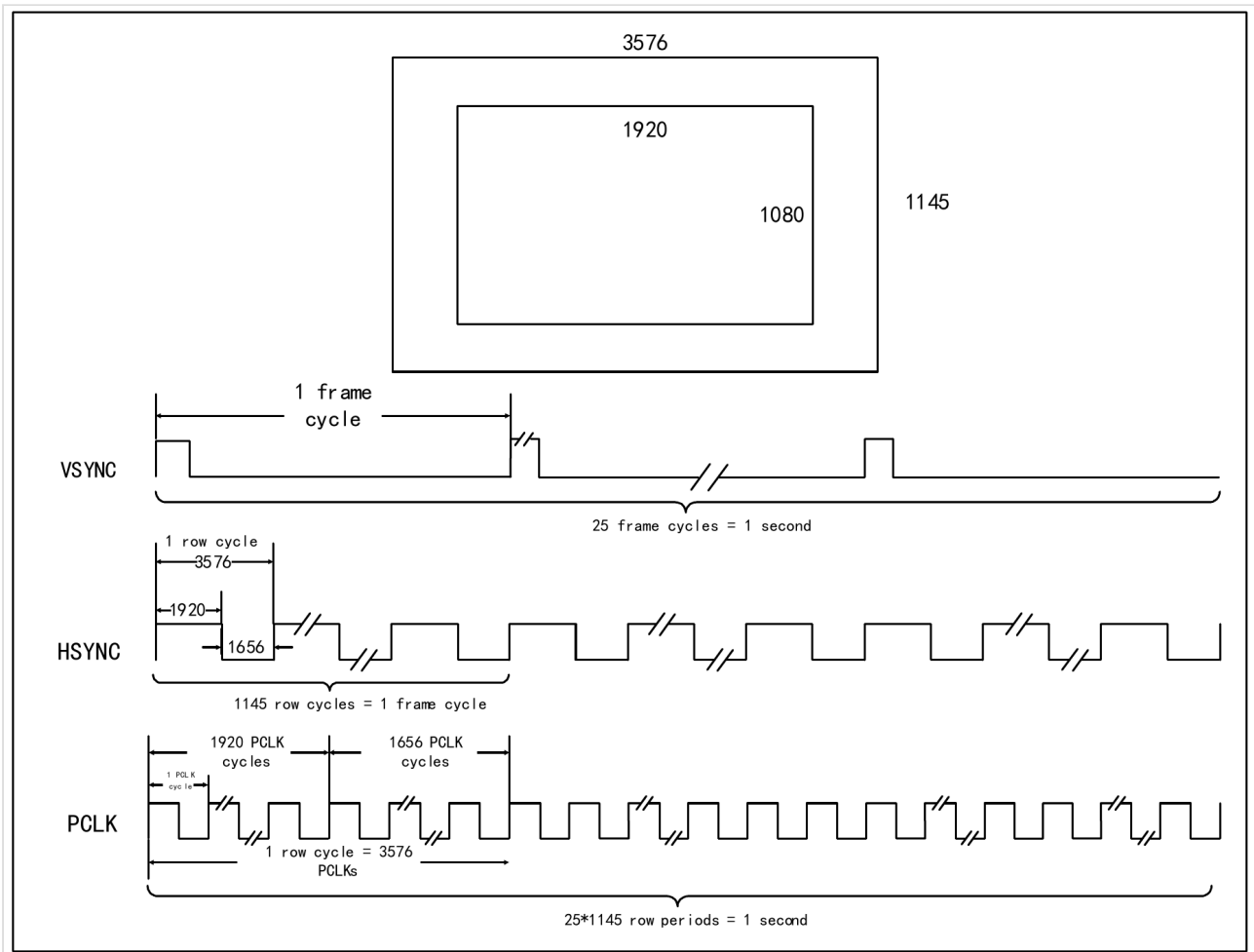


Figure 23-2: Timing description

HCLK is the time for one pixel transfer, so HSYNC is 3576 times HCLK. Among the 3576 pixels, only 1920 pixels are valid; during the remaining 1656 pixel periods, the sensor does not transmit data. VSYNC is the frame synchronization signal, so the VSYNC period is 3576×1145 times HCLK. Similarly, only during the 1920×1080 valid pixel periods does the sensor transmit data. If the sensor transmits JPEG compressed data, the HSYNC signal may not be needed.

The relationship between DVP interface signals and image data is primarily based on the datasheet of the selected sensor.

23.2.3 RGB/YUV/JPEG Compressed Data Format Description

- **RGB**

Three primary colors: red, green, blue.

- **YUV**

Luminance signal Y, chrominance signals U and V, Pb and Pr, Cb and Cr.

- **JPEG (Joint Photographic Experts Group)**

Lossy compression, but the lost portions are those not easily perceived by human vision. It exploits the characteristic that the human eye is insensitive to high-frequency information in

computer color. It removes visually redundant information (spatial redundancy) and removes inherently redundant information in the data (structural redundancy).

23.3 Digital Video Port Application

23.3.1 DVP Configuration Description

1. In DVP data reception, each frame of data is stored alternately in BUF0 and BUF1, starting from BUF0. For RGB and YUV data streams, the hardware resets and selects BUF0 to start each time the frame signal transitions from inactive level to active level. When one row of data is filled, it switches to BUF1, achieving alternate storage. For JPEG compressed data, the hardware sets the switching threshold between BUF0 and BUF1 based on the configured DMA reception length.
2. When the data bus width is 10 bits or 12 bits, each time a data is received, the system automatically performs unsigned extension to 16 bits before storing.
3. The R16_DVP_ROW_NUM and R16_DVP_COL_NUM registers must match the actual image size output by the sensor.
4. In video stream RGB mode, R16_DVP_COL_NUM represents the number of valid HCLK periods in one row of data, and R16_DVP_ROW_NUM represents the number of rows contained in one frame of image data. In image JPEG mode, R16_DVP_COL_NUM is used to configure the DMA length; in this mode, the R16_DVP_ROW_NUM register has no effect.

23.3.2 DVP Application Description

When using the digital video port to receive image data, the DVP-related control registers must be correctly configured to match the mode of the image sensor. The specific operation steps are as follows:

1. Clear the RB_DVP_ALL_CLR and RB_DVP_RCV_CLR fields through the R8_DVP_CR1 register.
2. Configure the image mode, data bit width, HCLK polarity, HSYNC polarity, and VSYNC polarity through the R8_DVP_CR0 register to match the SENSOR output.
3. Based on the valid image pixels output by the configured image sensor, configure the R16_DVP_ROW_NUM and R16_DVP_COL_NUM registers to match the SENSOR output. In image JPEG mode, only the R16_DVP_COL_NUM register needs to be configured.
4. Configure the DMA receive address through the R32_DVP_DMA_BUF0/1 registers.
5. If using snapshot mode, configure the RB_DVP_CM field through the R8_DVP_CR1 register to enable snapshot mode.
6. If using cropping mode, configure the RB_DVP_CROP and RB_DVP_FCRC fields through the R8_DVP_CR1 register to enable the cropping function and control the frame capture rate. At the same time, configure R16_DVP_HOFFCNT, R16_DVP_VST, R16_DVP_CAPCNT, and R16_DVP_VLINE to set the cropped image dimensions.
7. As needed, enable the corresponding interrupts through the R8_DVP_IER register, and configure the interrupt priority and enable the DVP interrupt through the interrupt controller (PFIC).
8. Enable DMA through the R8_DVP_CR1 register, and enable the DVP interface through the R8_DVP_CR0 register.
9. Wait for the relevant reception interrupts to be generated and process the received data in a timely manner.

23.3.3 Capture Mode Description

The DVP interface supports two capture modes: snapshot (single frame) mode and continuous mode.

23.3.3.1 Snapshot Mode

In this mode, only a single frame is captured (RB_DVP_CM set to 1 in the R8_DVP_CR1 register). After the DVP interface is enabled, the system waits for the frame start to be detected, then begins image data sampling. After receiving a complete frame of data, the DVP interface is turned off (the RB_DVP_ENABLE field in the R8_DVP_CR0 register is cleared to 0). The single-frame capture waveform in snapshot mode is shown in Figure 23-3.

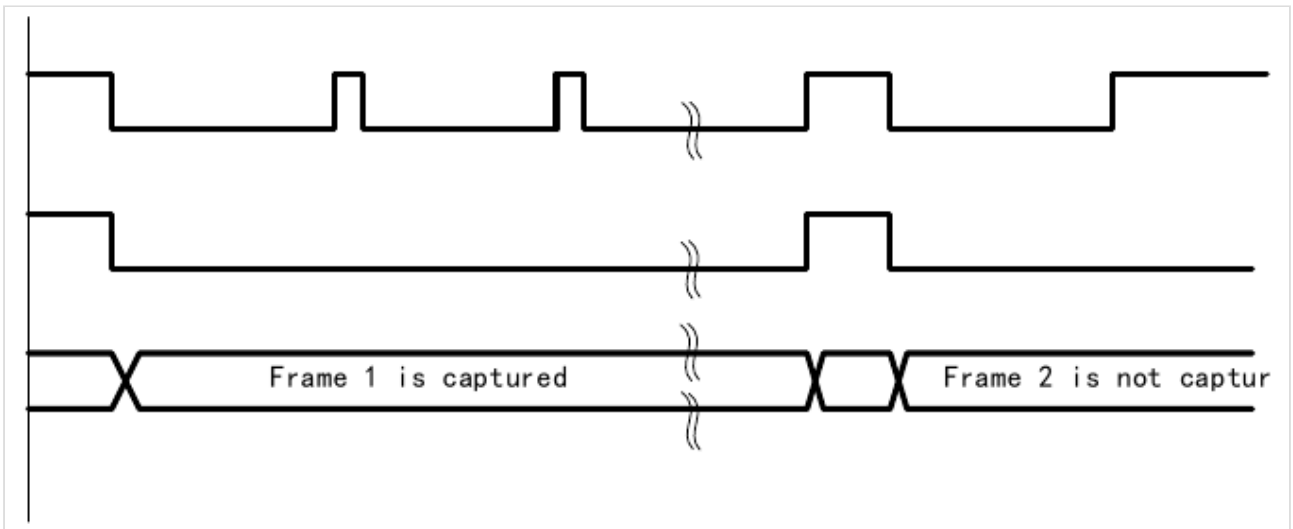


Figure 23-3: Single-frame capture waveform in snapshot mode

23.3.3.2 Continuous Mode

In this mode (RB_DVP_CM cleared to 0 in the R8_DVP_CR1 register), after the DVP interface is enabled, each frame of data is continuously sampled until the RB_DVP_ENABLE field in the R8_DVP_CR0 register is cleared to 0. The frame capture waveform in continuous mode is shown in Figure 23-4.

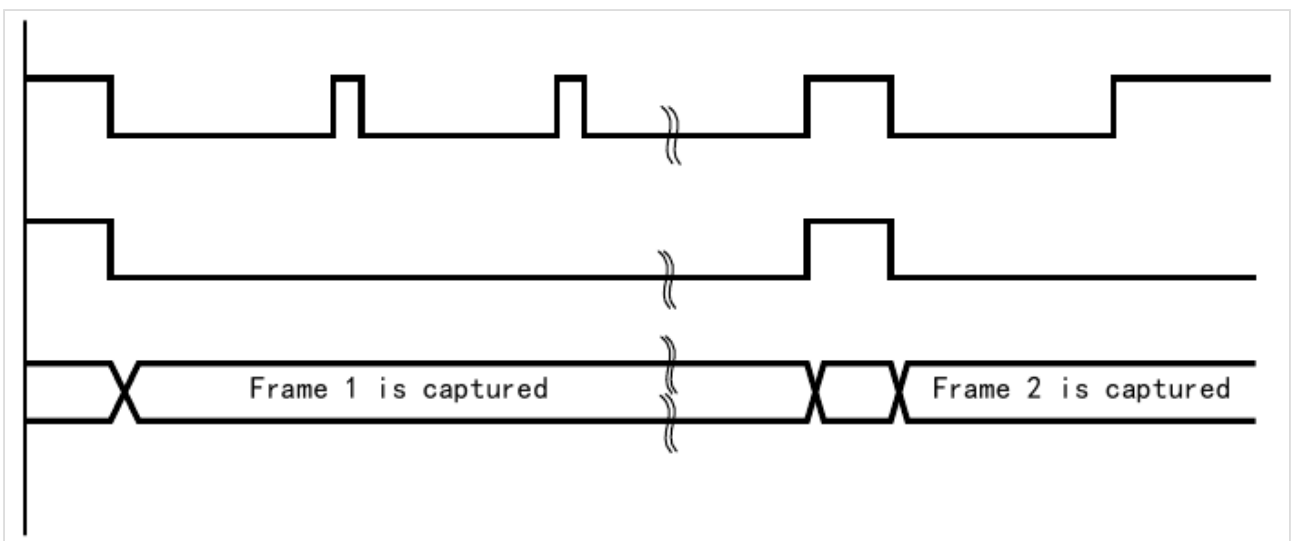


Figure 23-4: Frame capture waveform in continuous mode

23.3.4 Cropping Function Description

The DVP can use the cropping function to extract a rectangular window from the received image. The rectangular window's starting coordinates (the rectangle's top-left X coordinate R16_DVP_HOFFCNT, Y coordinate R16_DVP_VST) and window size (R16_DVP_CAPCNT for horizontal size, R16_DVP_VLINE for vertical size) are configurable. The coordinates and size of the cropping window are shown in Figure 23-5. The cropping window data capture waveform is shown in Figure 23-6.

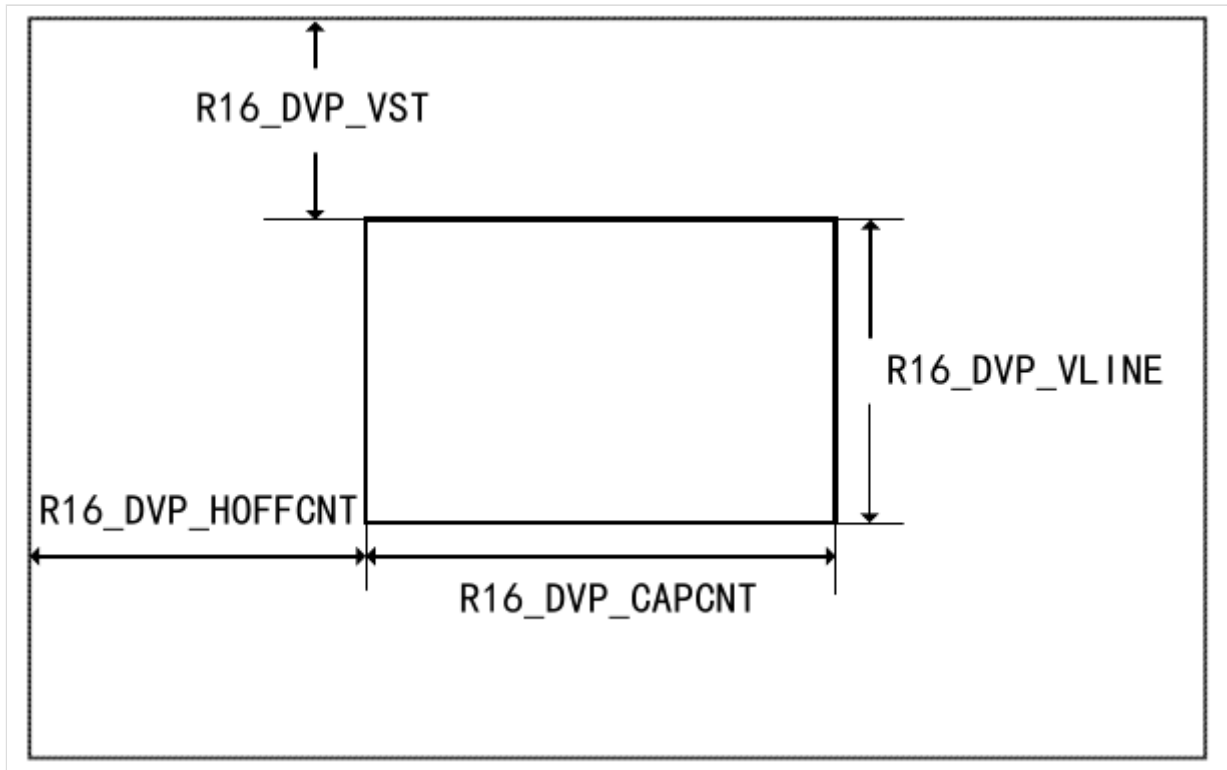


Figure 23-5: Coordinates and size of the cropping window

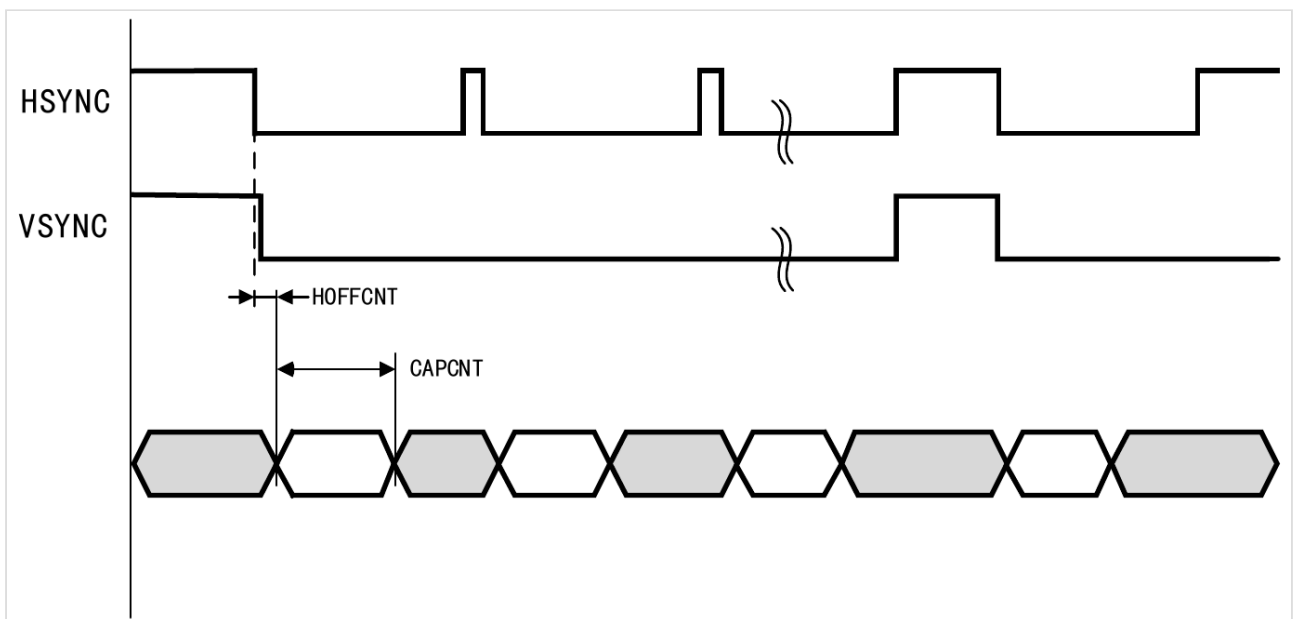


Figure 23-6: Cropping window data capture waveform

23.4 Register Description

Table 23-2: DVP-related register list

Name	Address	Description	Reset Value
R8_DVP_CR0	0x50050000	DVP control register 0	0x00
R8_DVP_CR1	0x50050001	DVP control register 1	0x06
R8_DVP_IER	0x50050002	DVP interrupt enable register	0x00
R16_DVP_ROW_NUM	0x50050004	Image row count configuration register	0x0000
R16_DVP_COL_NUM	0x50050006	Image column count configuration register	0x0000
R32_DVP_DMA_BUF0	0x50050008	DVP DMA address 0 register	0x00000000
R32_DVP_DMA_BUF1	0x5005000C	DVP DMA address 1 register	0x00000000
R8_DVP_IFR	0x50050010	DVP interrupt flag register	0x00
R8_DVP_STATUS	0x50050011	DVP receive FIFO status register	0x00
R16_DVP_ROW_CNT	0x50050014	DVP row counter register	0x0000
R16_DVP_HOFFCNT	0x50050018	Window start horizontal displacement register	0x0000
R16_DVP_VST	0x5005001A	Window start row register	0x0000
R16_DVP_CAPCNT	0x5005001C	Capture count register	0x0000
R16_DVP_VLINE	0x5005001E	Vertical line count register	0x0000
R32_DVP_DR	0x50050020	Data register	0x00000000

23.4.1 DVP Configuration Register 0 (R8_DVP_CR0)

Offset address: 0x00

Bits	Name	Access	Description	Reset
7	Reserved	RO	Reserved.	0
6	RB_DVP_JPEG	RW	JPEG mode enable: 1: JPEG compressed format; 0: Raw data format.	0
[5:4]	RB_DVP_MSK_DAT_MOD[1:0]	RW	DVP data bit width configuration: 00: 8-bit mode; 01: 10-bit mode; 1x: 12-bit mode.	00b
3	RB_DVP_P_POLAR	RW	HCLK polarity configuration: 1: Sample data on HCLK falling edge; 0: Sample data on HCLK rising edge.	0
2	RB_DVP_H_POLAR	RW	HSYNC polarity configuration: 1: HSYNC low level data valid; 0: HSYNC high level data valid.	0
1	RB_DVP_V_POLAR	RW	VSYSNC polarity configuration: 1: VSYSNC high level data valid; 0: VSYSNC low level data valid.	0
0	RB_DVP_ENABLE	RW		0

Bits	Name	Access	Description	Reset
			DVP function enable: 1: Enable DVP; 0: Disable DVP.	

23.4.2 DVP Configuration Register 1 (R8_DVP_CR1)

Offset address: 0x01

Bits	Name	Access	Description	Reset
[7:6]	RB_DVP_FCRC[1:0]	RW	DVP frame capture rate control: 00: Capture all frames; 01: Capture every other frame; 10: Capture every fourth frame; 11: Reserved.	00b
5	RB_DVP_CROP	RW	Cropping function control: 1: Only capture data within the window specified by the crop registers; 0: Capture the complete image.	0
4	RB_DVP_CM	RW	Capture mode: 1: Snapshot mode; 0: Continuous mode.	0
3	RB_DVP_BUF_TOG	RWT	Buffer address flag bit. Hardware-controlled toggle; software writes 1 to toggle this bit, writing 0 has no effect. 1: Data stored in receive address 1; 0: Data stored in receive address 0.	0
2	RB_DVP_RCV_CLR	RW	Receive logic reset control: 1: Reset receive logic circuit; 0: Cancel reset operation.	1
1	RB_DVP_ALL_CLR	RW	Flag and FIFO clear control, software writes 1 or 0: 1: Reset flags and FIFO; 0: Cancel reset operation.	1
0	RB_DVP_DMA_ENABLE	RW	DMA enable control bit: 1: Enable DMA; 0: Disable DMA.	0

23.4.3 DVP Interrupt Enable Register (R8_DVP_IER)

Offset address: 0x02

Bits	Name	Access	Description	Reset
[7:5]	Reserved	RO	Reserved.	0
4	RB_DVP_IE_STP_FRM	RW	Frame end interrupt enable. (Interrupt is generated when VSYNC transitions from active level to inactive level) 1: Enable frame end interrupt; 0: Disable frame end interrupt.	0
3	RB_DVP_IE_FIFO_OV	RW		0

Bits	Name	Access	Description	Reset
			Receive FIFO overflow interrupt enable: 1: Enable FIFO overflow interrupt; 0: Disable FIFO overflow interrupt.	
2	RB_DVP_IE_FRM_DONE	RW	Frame reception complete interrupt enable. (Interrupt is generated when the counter reaches the ROW/COL_NUM configured value, indicating the last data has been written to RAM) 1: Enable frame reception complete interrupt; 0: Disable frame reception complete interrupt.	0
1	RB_DVP_IE_ROW_DONE	RW	Row end interrupt enable. (Interrupt is generated when the counter reaches the COL_NUM configured value) 1: Enable row end interrupt; 0: Disable row end interrupt.	0
0	RB_DVP_IE_STR_FRM	RW	New frame start interrupt enable. (Interrupt is generated when VSYNC transitions from inactive level to active level, indicating a new frame is starting and data is about to arrive) 1: Enable new frame start interrupt; 0: Disable new frame start interrupt.	0

23.4.4 DVP Image Valid Row Count Configuration Register (R16_DVP_ROW_NUM)

Offset address: 0x04

Bits	Name	Access	Description	Reset
[15:0]	RB_DVP_ROW_NUM[15:0]	RW	In RGB mode, indicates the number of rows contained in one frame of image data. In JPEG mode, this register has no practical meaning.	0

23.4.5 DVP Image Valid Column Configuration Register (R16_DVP_COL_NUM)

Offset address: 0x06

Bits	Name	Access	Description	Reset
[15:0]	RB_DVP_COL_NUM[15:0]	RW	In RGB mode, indicates the number of HCLK periods contained in one row of data. In JPEG mode, used to configure the DMA reception length.	0

23.4.6 DVP DMA Receive Address 0 Register (R32_DVP_DMA_BUF0)

Offset address: 0x08

Bits	Name	Access	Description	Reset
[31:0]	RB_DVP_DMA_BUF0[31:0]	RW	DMA receive address 0.	0

23.4.7 DVP DMA Receive Address 1 Register (R32_DVP_DMA_BUF1)

Offset address: 0x0C

Bits	Name	Access	Description	Reset
[31:0]	RB_DVP_DMA_BUF1[31:0]	RW	DMA receive address 1.	0

23.4.8 DVP Interrupt Flag Register (R8_DVP_IFR)

Offset address: 0x10

Bits	Name	Access	Description	Reset
[7:5]	Reserved	RO	Reserved.	0
4	RB_DVP_IF_STP_FRM	RW1Z	Frame end interrupt flag, active high, write 1 to clear.	0
3	RB_DVP_IF_FIFO_OV	RW1Z	Receive FIFO overflow interrupt flag, active high, write 1 to clear.	0
2	RB_DVP_IF_FRM_DONE	RW1Z	Frame reception complete interrupt flag, active high, write 1 to clear.	0
1	RB_DVP_IF_ROW_DONE	RW1Z	Row end interrupt flag, active high, write 1 to clear.	0
0	RB_DVP_IF_STR_FRM	RW1Z	Frame start interrupt flag, active high, write 1 to clear.	0

23.4.9 DVP Receive FIFO Status Register (R8_DVP_STATUS)

Offset address: 0x11

Bits	Name	Access	Description	Reset
7	Reserved	RO	Reserved.	0
[6:4]	RB_DVP_FIFO_CNT[2:0]	RO	FIFO counter.	0
3	Reserved	RO	Reserved.	0
2	RB_DVP_FIFO_OV	RO	FIFO overflow status: 1: FIFO overflowed; 0: FIFO not overflowed.	0
1	RB_DVP_FIFO_FULL	RO	FIFO full status: 1: Buffer full; 0: FIFO not full.	0
0	RB_DVP_FIFO_RDY	RO	FIFO ready status: 1: FIFO has data; 0: FIFO has no data.	0

23.4.10 DVP Received Image Row Count Register (R16_DVP_ROW_CNT)

Offset address: 0x14

Bits	Name	Access	Description	Reset
[15:0]	RB_DVP_ROW_CNT[15:0]	RO	During actual reception, the number of rows contained in one frame of image data. This register is updated at frame end. In JPEG format, the value of this register is meaningless.	0

23.4.11 DVP Window Start Horizontal Displacement Register (R16_DVP_HOFFCNT)

Offset address: 0x18

Bits	Name	Access	Description	Reset
[15:0]	RB_DVP_HOFFCNT[15:0]	RW	Within the window rows, the number of HCLK periods to skip before capturing data in each row. The lower 14 bits are valid.	0

23.4.12 DVP Window Start Row Register (R16_DVP_VST)

Offset address: 0x1A

Bits	Name	Access	Description	Reset
[15:0]	RB_DVP_VST[15:0]	RW	The row number at which image capture begins; data before this row is not captured. The lower 13 bits are valid.	0

23.4.13 DVP Capture Count Register (R16_DVP_CAPCNT)

Offset address: 0x1C

Bits	Name	Access	Description	Reset
[15:0]	RB_DVP_CAPCNT[15:0]	RW	The number of HCLK periods to capture within the cropping window. The lower 14 bits are valid.	0

23.4.14 DVP Vertical Line Count Register (R16_DVP_VLINE)

Offset address: 0x1E

Bits	Name	Access	Description	Reset
[15:0]	RB_DVP_VLINE[15:0]	RW	The number of rows to capture within the cropping window. The lower 14 bits are valid.	0

23.4.15 DVP Data Register (R32_DVP_DR)

Offset address: 0x20

Bits	Name	Access	Description	Reset
[31:0]	RB_DVP_DR[31:0]	RO	The DVP interface triggers a DMA request only after receiving 4 bytes of data each time. The 4-byte-deep FIFO allows sufficient time for DMA transfer, effectively preventing DMA overflow.	0

Chapter 24: SDIO Interface (SDIO)

The term "SDIO" in this chapter refers to a communication interface designed on the microcontroller for operating external memory cards such as SD cards or other devices; it is a peripheral of the microcontroller. The microcontroller's SDIO is directly connected to the HB bus, clocked directly by HCLK, enabling high communication speeds. The microcontroller's SDIO acts as the SDIO host, and the controlled devices are collectively referred to as SDIO devices. In applications, SDIO is generally used to read and write SD cards, TF cards, or eMMC cards, or to control other devices that use SDIO as the communication interface, such as WiFi/4G modules.

24.1 Main Features

24.1.1 Features

- Supports SD cards, SDIO cards, and MMC cards
- Supports 1-bit, 4-bit, and 8-bit bus
- SDIO supports selectable clock frequency, up to 80 MHz in 4-wire mode, up to 50 MHz in 8-wire mode
- Compatible with MMC specification 4.5 (forward compatible)
- Compatible with SD card specification 3.0, SDIO card specification 2.0
- Not compatible with SPI

24.1.2 Overview

The microcontroller's SDIO supports communication with memory cards such as SD cards or MMC cards. It should be noted that SDIO only provides a set of clock, data, and command control timing sequences required for a single command transfer as defined by the SD card and MMC card specifications. The sequencing and combination of commands between each other must be determined by the user through programming. In addition, for various memory cards, SDIO can only implement read and write functions. The file system functions provided for files need to be implemented by the user through programming to build the file system.

SDIO is different from the QSPI interface in that it has no chip-select pin, and has an additional CMD pin. The CMD can be considered a special data line dedicated to transmitting commands and responses. SDIO has three selectable data line widths: 1-bit, 4-bit, and 8-bit. The SDIO clock is selectable during configuration. During actual data transmission, the clock supports up to 80 MHz in 4-wire mode and up to 50 MHz in 8-wire mode. According to the protocol, when the clock received by the SDIO device exceeds a certain threshold, the peak waveform of the clock line, data line, and command line needs to be reduced to save the rise and fall time of the waveform.

Unlike SD cards, SDIO cards often refer to non-storage devices such as WiFi/Bluetooth modules and 4G modules that use the SDIO interface. Unless otherwise specified, the content described in this chapter definitely applies to SD cards, and content that only applies to SDIO cards or MMC cards will be specifically noted. In this chapter, SD cards are treated as the primary potential operation target, followed by SDIO cards, and finally MMC cards.

24.2 Interface and Clock

The SDIO receives control from the CPU through the HB bus. The SDIO registers contain a FIFO interface through which the CPU or DMA reads or writes data to obtain or send data. The SDIO is clocked directly by HCLK and has an interrupt entry that supports multiple interrupt sources. The pins directly controlled by the SDIO are SDIO_CK, SDIO_CMD, and SDIO_D[7:0] — a total of ten pins — which connect to the SDIO device. The SDIO is a host-driven communication interface; all transfers must be initiated by the microcontroller.

24.2.1 Peripheral Structure

The structure of the SDIO is shown in Figure 24-1.

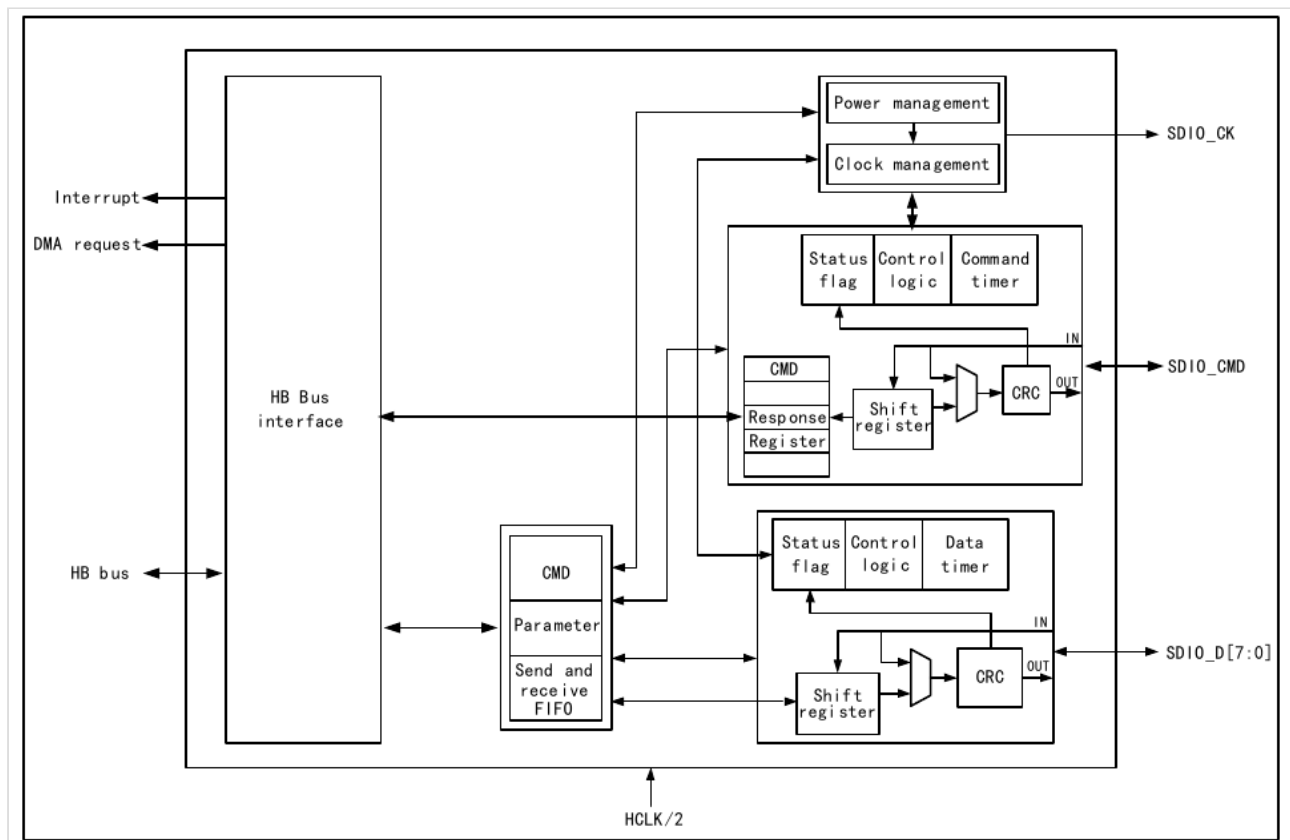


Figure 24-1: SDIO structure diagram

The SDIO is a peripheral through which the microcontroller directly operates SDIO devices as the host. It roughly consists of five modules: the HB interface, clock control section, CMD line control section, data line control section, and control register section. The SDIO is a half-duplex peripheral. The CPU writes the commands and data to be sent into the control registers. The command line and data line control modules are responsible for pushing the commands or data onto the IO lines with CRC added. The SDIO data flow is handled by the general-purpose DMA, which moves data from RAM to the SDIO FIFO. The SDIO FIFO has a size of 32 words (32-bit each).

24.2.2 Pins and Configuration

The pins and their modes that need to be configured for the SDIO are shown in Table 24-1.

Table 24-1: SDIO pin configuration

SDIO alternate function	Pin mode to be configured	Pin speed to be configured
SDIO_D[0:3]	Push-pull alternate function output	50 MHz
SDIO_CK	Push-pull alternate function output	50 MHz
SDIO_CMD	Push-pull alternate function output	50 MHz
SDIO_D4	Push-pull alternate function output	50 MHz
SDIO_D5	Push-pull alternate function output	50 MHz
SDIO_D6	Push-pull alternate function output	50 MHz
SDIO_D7	Push-pull alternate function output	50 MHz

Note: In slave mode, the SDIO_CK pin should be configured as floating input.

24.2.3 Clock

The clock pin of the SDIO is SDIO_CK. Its output clock is derived by dividing HCLK. The division factor can be configured to any integer value between 2 and 261. SDIO devices generally only support a single-bus mode with a maximum clock of 400 kHz during initialization. After initialization, the host generally initiates a switch to low-voltage operation while raising the clock to the maximum clock acceptable to both the microcontroller and the SDIO device. The supported clock speeds and switching procedures differ among different versions and speed classes of SD cards; the user needs to understand this independently.

24.2.4 Command State Machine

The command workflow of the SDIO follows the state machine shown below.

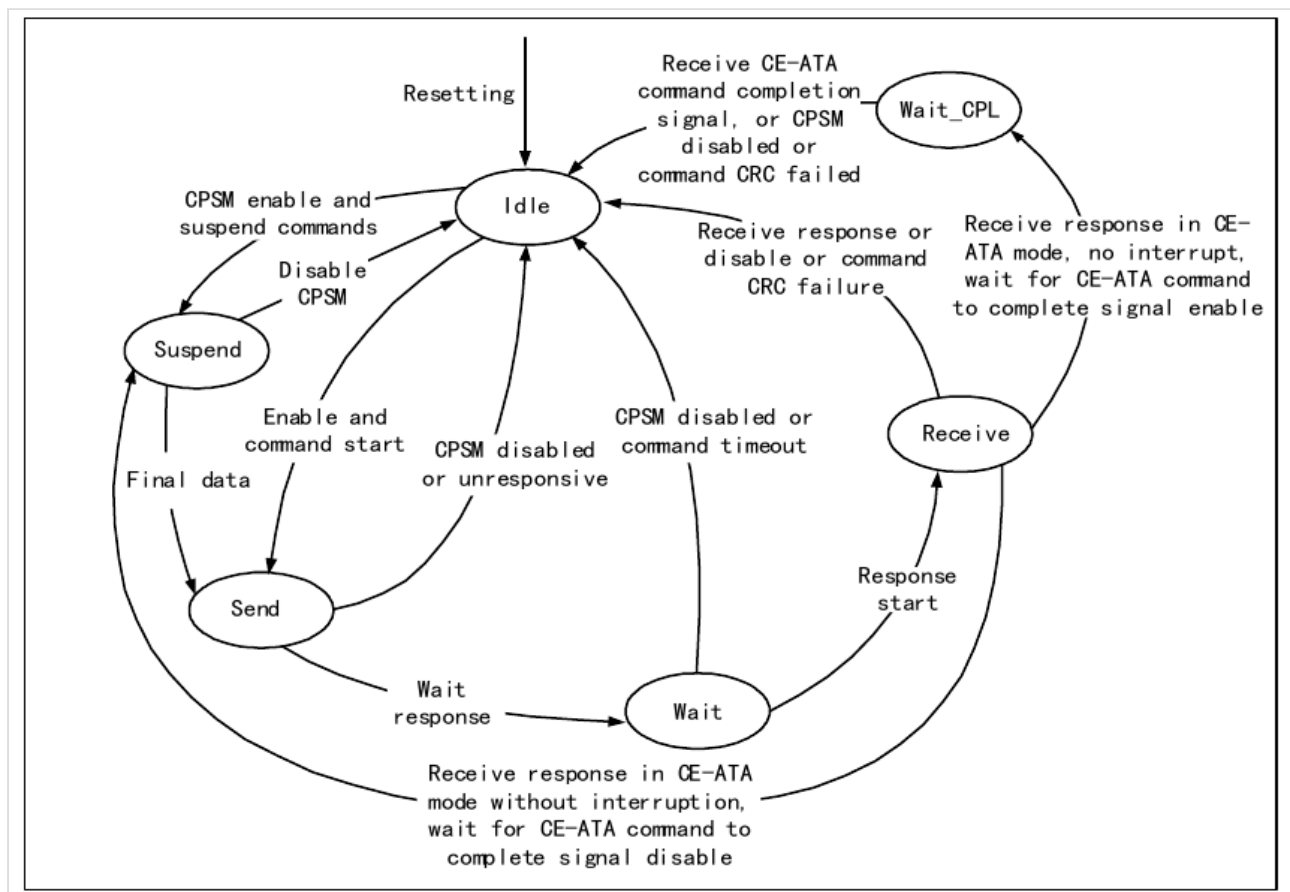


Figure 24-2: Command state machine

24.2.5 Data State Machine

The data workflow of the SDIO follows the state machine shown below.

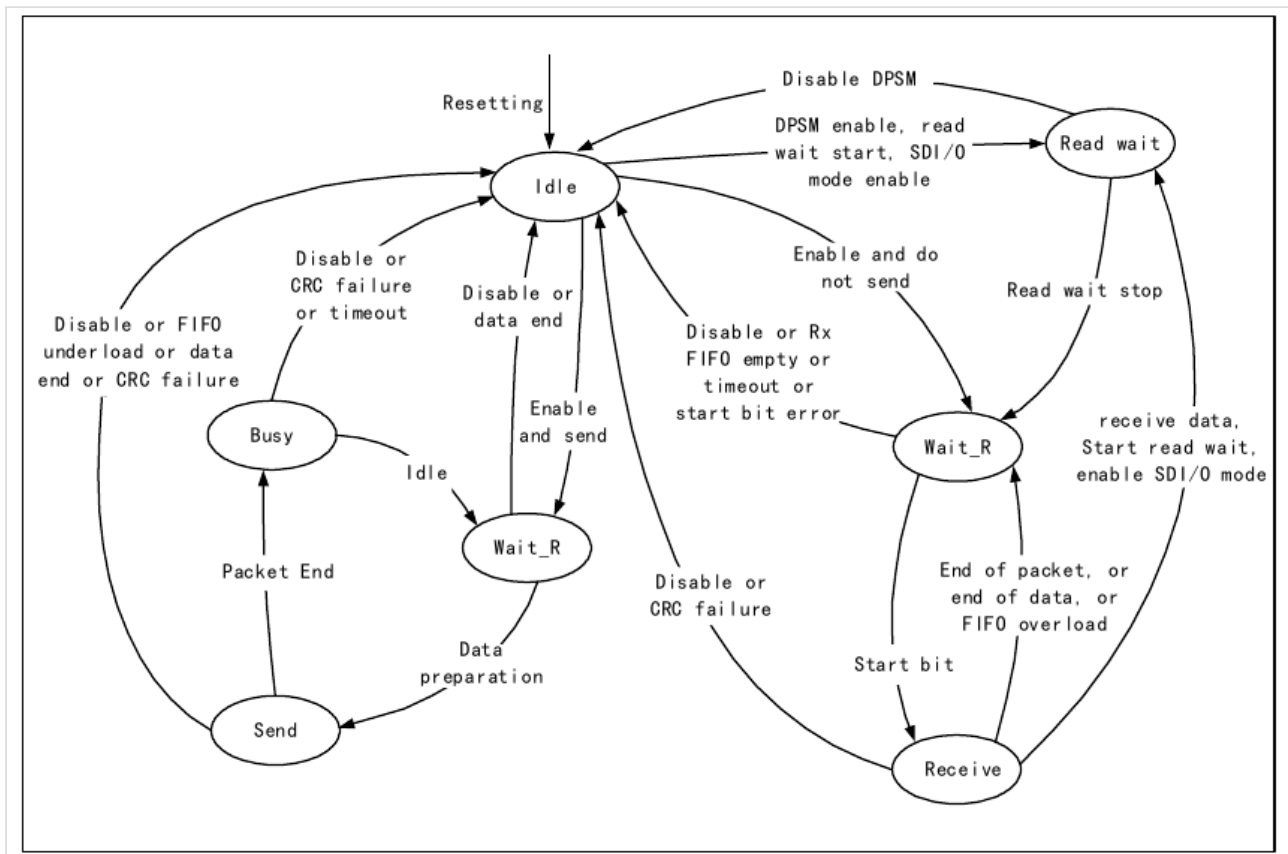


Figure 24-3: Data state machine

24.3 SDIO Protocol Overview

Communication on the SDIO uses the transfer as the minimum unit. Each transfer always starts with the host sending a command on the CMD line. After some commands are sent, the SDIO device will also send a segment of data on the CMD line to reply to the host, called a "response." Sometimes it is also accompanied by data transfer, which takes place on the D line. The formats of commands and responses are fixed, and the bit definitions of each field are determined by the specific command or response. Responses and data transfers must be sent or stopped within a specified time after the command or response ends; otherwise, a timeout error will occur.

The purpose of this section is to give users a preliminary understanding of some necessary specification details for using SDIO in a concise manner. It does not guarantee completeness or timely updates. The microcontroller's SDIO only guarantees the hardware operation basis for implementing the SD 2.0, SDIO 2.0, and MMC 4.5 specifications. For functions defined by higher-version specifications, such as double-edge sampling, support is not necessarily provided. Users should program SDIO interactions based on the SD specification, SDIO specification, and MMC specification during actual development.

24.3.1 Bus Timing

SD card transfers are always initiated by the host sending a CMD. The SD card may not reply with a response, may reply with a short response or a long response, and some responses are followed by data transfers. In SDIO card or MMC card communication, there may also be cases where the SDIO device actively reports an interrupt. The timing diagrams are shown in the figure set below.

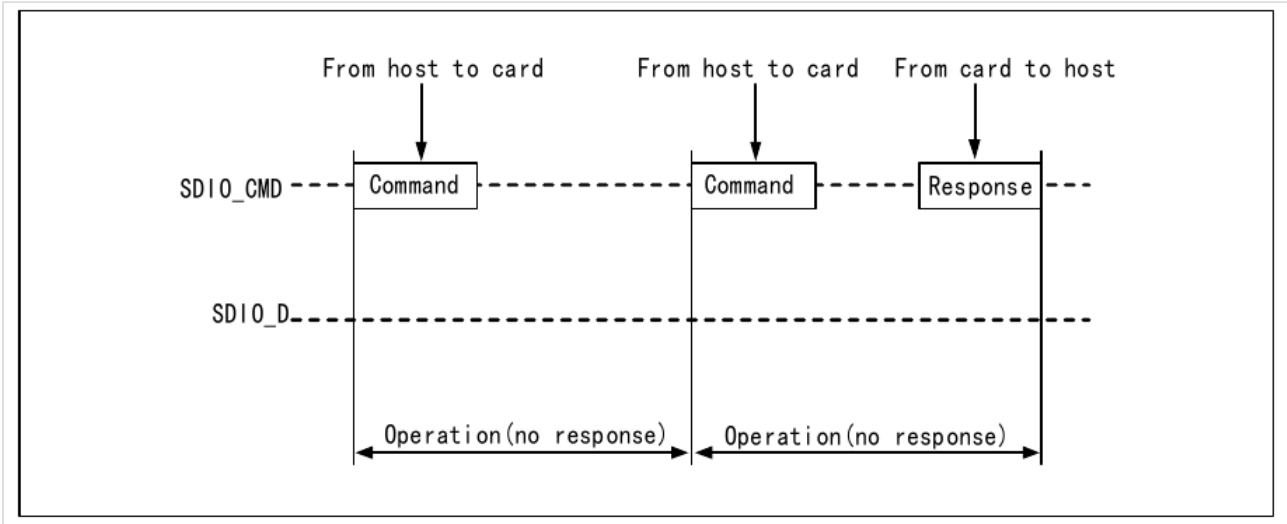


Figure 24-4: SDIO no-response timing and no-data timing (SD card example)

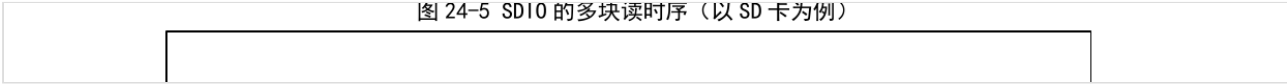


Figure 24-5: SDIO multi-block read timing (SD card example)

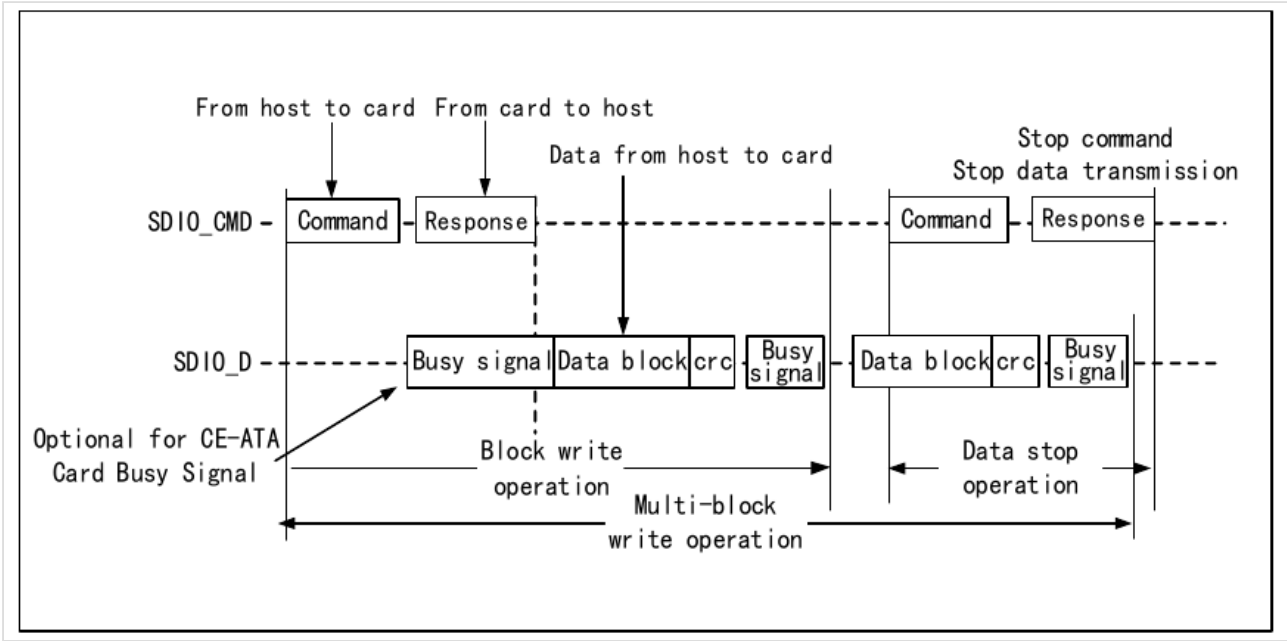


Figure 24-6: SDIO multi-block write timing (SD card example)

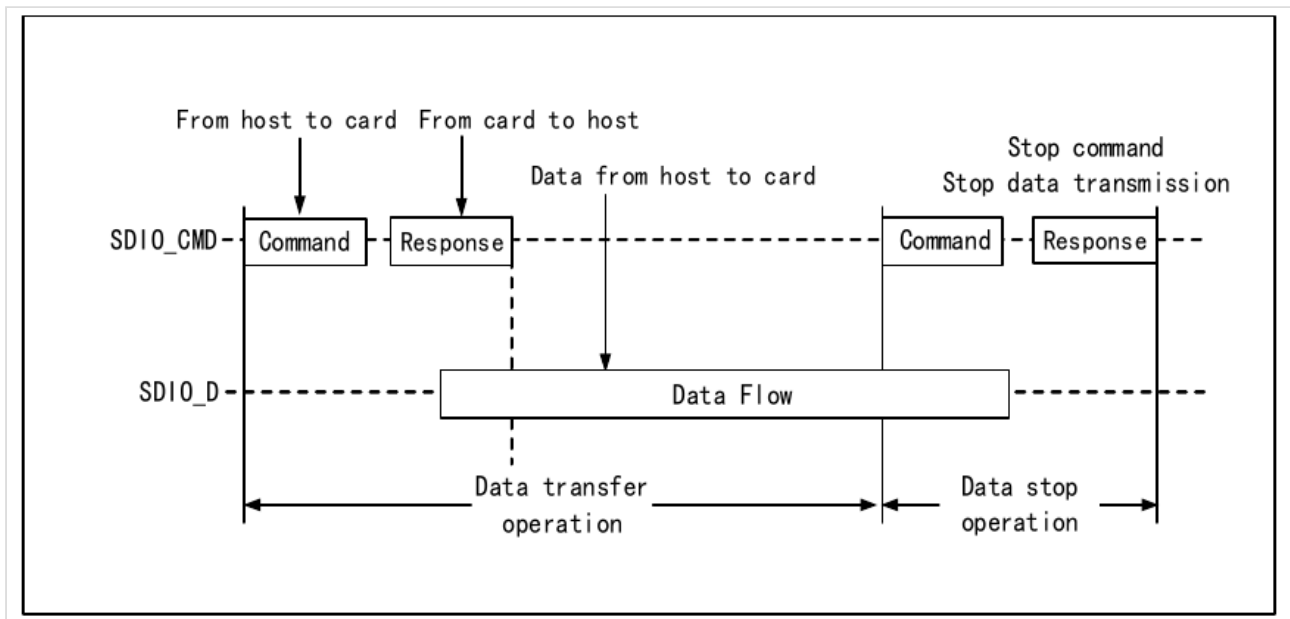


Figure 24-7: SDIO data stream read timing (SDIO card example)

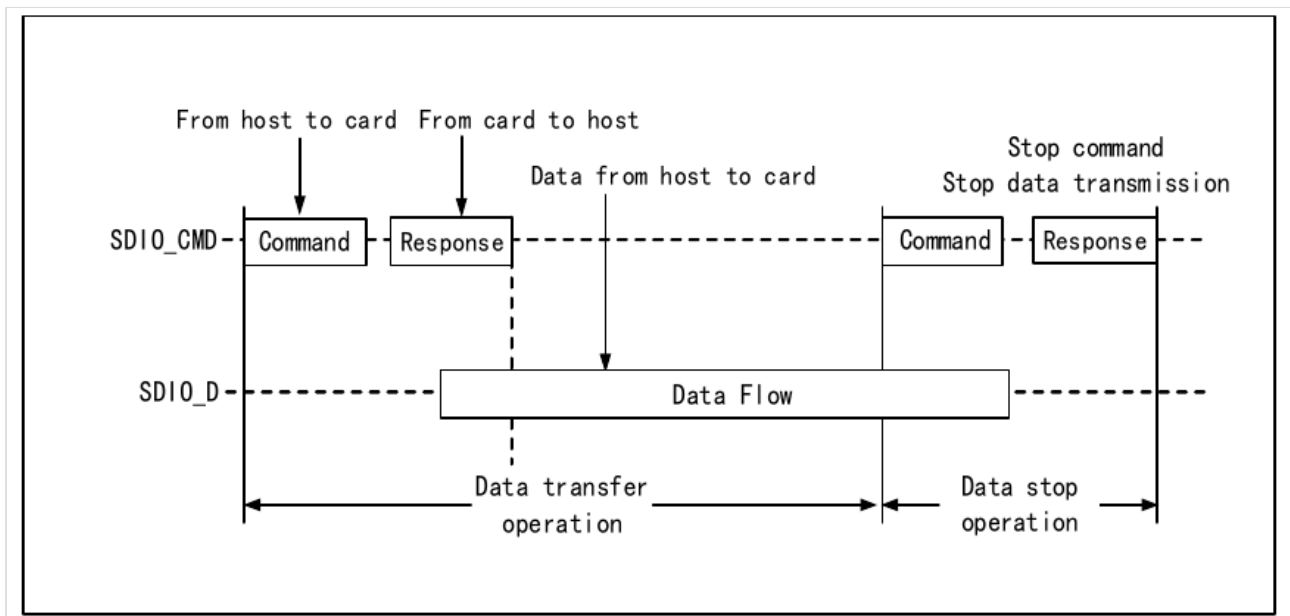


Figure 24-8: SDIO data stream write timing (SDIO card example)

24.3.2 Commands

Most SDIO transfers begin with a command (CMD). The SDIO host informs the device of its intention through commands. The command format is shown in the table below.

Table 24-2: Command format

Bit/Field name	Start bit	Transmitter bit	Command index	Command parameter	CRC7	End bit
Bit position/width	47	46	[45:40]/6 bits	[39:8]/32 bits	[7:1]/7 bits	0
Value	0b	1b	X	X	X	1b

There are roughly four types of commands:

1. Broadcast command (bc): sent to all cards on the bus, with no response returned;
2. Broadcast command with response (bcr): sent to all cards on the bus, with a response returned;
3. Addressed (point-to-point) command (ac): sent to a specific card, but with no data transfer;
4. Addressed (point-to-point) command with data transfer (adtc): sent to a specific card, with accompanying data transfer.

The following are some commonly used commands.

24.3.2.1 Basic Commands

Basic commands are some relatively fundamental functions supported by SD cards.

Table 24-3: SD card basic commands

Command index	Type	Parameter	Response type	Abbreviation	Description
CMD0	bc	[31:0] stuffing bits, no meaning	None	GO_IDLE_STATE	Reset all cards to IDLE state.
CMD2	bcr	[31:0] stuffing bits, no meaning	R2	ALL_SEND_CID	All cards reply with CID.
CMD3	bcr	[31:0] stuffing bits, no meaning	R6	SEND_RELATIVE_ADDR	Reply with a new RCA.
CMD7	ac	[31:16] RCA; [15:0] stuffing bits, no meaning	R1b	SELECT/DESELECT_CARD	Select or deselect a card.
CMD8	bcr	[31:12] reserved; [11:8] supply voltage; [7:0] check pattern	R7	SEND_IF_COND	Send SD card interface condition.
CMD9	ac	[31:16] RCA; [15:0] stuffing bits, no meaning	R2	SEND_CSD	Request CSD.
CMD10	ac	[31:16] RCA; [15:0] stuffing bits, no meaning	R2	SEND_CID	Request CID.
CMD11	ac	[31:0] stuffing bits, no meaning	R1	VOLATGE_SWITCH	Switch to 1.8 V voltage level.
CMD12	ac	[31:0] stuffing bits, no meaning	R1b	STOP_TRANSMISSION	Force the card to stop transmission.
CMD13	ac	[31:16] RCA; [15] send task status register; [14:0] stuffing bits	R1	SEND_STATUS/ SEND_TASK_STATUS	Send status or task status register.
CMD15	ac	[31:16] RCA; [15:0] stuffing bits, no meaning	None	GO_INACTIVE_STATE	Request the card to enter INACTIVE mode.

24.3.2.2 Erase Commands

SD cards also have a FLASH structure and require erasing before writing. However, SD cards have integrated erase logic; if a write command is executed and it is found that erasing has not been done, the erase operation is automatically added. In many cases, especially before large-scale writes, proactively performing an erase helps improve efficiency.

Table 24-4: SD card erase commands

Command index	Type	Parameter	Response type	Abbreviation	Description
CMD32	ac	[31:0] start erase address ^[2]	R1	ERASE_WR_BLK_START	Set the first erase address
CMD33	ac	[31:0] end erase address ^[2]	R1	ERASE_WR_BLK_END	Set the last erase address
CMD38	ac	[31:0] erase mode	R1b	ERASE	Parameter 0 means normal erase

Note 1: The erase commands here are the SD card erase commands defined by the SD protocol specification. The erase commands for SDIO cards and MMC cards differ from these.

Note 2: For commonly used SDHC/SDXC (2 GB to 2 TB) class cards, the erase address for writing must be a block address, i.e., 512-byte aligned.

24.3.2.3 Block Transfer Read Commands

Table 24-5: SD card block read commands

Command index	Type	Parameter	Response type	Abbreviation	Description
CMD16	ac	[31:0] block length	R1	SET_BLOCKLEN	Set block length, 512
CMD17	adtc	[31:0] single block address ^[1]	R1	READ_SINGLE_BLOCK	Set the start address for a single read
CMD18	adtc	[31:0] multiple block address ^[1]	R1	READ_MULTIPLE_BLOCK	Continuously transfer data blocks from the card to the host until interrupted by a STOP_TRANSMISSION command
CMD19	adtc	[31:0] reserved bits	R1	SEND_TUNING_BLOCK	Send a 64-byte sequence indicating mode change
CMD20	ac	[31:28] speed control bits; [27:0] see command description	R1b	SPEED_CLASS_CONTROL	Speed class control command: [27:0] speed class; [27:0] UHS speed class; [27:0] video speed class
CMD22	ac	[31:6] reserved bits; [5:0] extended address	R1	ADDRESS_EXTENSION	Only used by SDUC

Command index	Type	Parameter	Response type	Abbreviation	Description
CMD23	ac	[31:0] block counter	R1	SET_BLOCK_COUNT	Block counter

Note 1: For commonly used SDHC/SDXC (2 GB to 2 TB) class cards, the erase address for writing must be a block address, i.e., 512-byte aligned.

24.3.2.4 Block Transfer Write Commands

Table 24-6: SD card block write commands

Command index	Type	Parameter	Response type	Abbreviation	Description
CMD16	ac	[31:0] block length	R1	SET_BLOCKLEN	Set block length, 512
CMD24	adtc	[31:0] block address ^[1]	R1	WRITE_BLOCK	Set the start address for a single block write
CMD25	adtc	[31:0] multiple block address ^[1]	R1	WRITE_MULTIPLE_BLOCK	Set the start address for a multiple block write
CMD27	adtc	[31:0] stuffing bits	R1	PROGRAM_CSD	Program the programmable fields of the CSD

Note 1: For commonly used SDHC/SDXC (2 GB to 2 TB) class cards, the erase address for writing must be a block address, i.e., 512-byte aligned.

24.3.3 Responses

Responses, as the necessary replies from the SDIO device to the host, are also transmitted on the CMD line and must be replied within a specified time. The response is transmitted with the most significant bit first and the least significant bit last. The response length and the definition of each bit and field are determined by the response type, but all responses begin with a start bit fixed at 0, followed by a transmission direction bit fixed at 0^[1]. All responses end with a stop bit fixed at 1^[1]. There are roughly 7 types of responses. SD cards support R1/R1b/R2/R3/R6/R7, and SDIO cards also support R4/R5. The formats are described below.

24.3.3.1 R1 Response

A normal response, 48 bits total, with CRC7 check, and the card status field is 32 bits. The format is shown in the table below.

Table 24-7: R1 format

Bit/Field name	Start bit	Transmitter bit	Response index	Card status	CRC7	End bit
Bit position/width	47	46	[45:40]/6 bits	[39:8]/32 bits	[7:1]/7 bits	0
Value	0b	0b	Follows CMD index	X	X	1b

24.3.3.2 R1b Response

The format of R1b is the same as R1, but a busy signal can be added after the response, i.e., clamping the data line D2 low. After the host receives the busy signal (detecting SDIO_D2 low), it needs to handle it accordingly.

24.3.3.3 R2 Response

A response to certain specific commands, 136 bits total, with CRC7 check included in the card status field, and the card status field is 128 bits. The card status field stores the value of the CID register or CSD register. The CID register is generally the reply to CMD2/CMD10, and the CSD register is generally the reply to CMD9. The specific meaning of the CID/CSD registers is described in section 28.4.2 Device Registers. Note that R2 only replies with the [127:1] segment of the CID/CSD register. The reserved bit CID/CSD[0] fixed at 1^[2] is occupied by the end bit, which is also fixed at 1. The format is shown in the table below.

Table 24-8: R2 format

Bit/Field name	Start bit	Transmitter bit	Command index	Card status	End bit
Bit position/width	135	134	[133:128]/6 bits	[127:1]/127 bits	0
Value	0b	0b	111111b	CID/CSD	1b

24.3.3.4 R3 Response

A dedicated response that replies with the OCR register, 48 bits total, with no CRC7 check. The OCR register is generally the reply to ACMD41. The format is shown in the table below.

Table 24-9: R3 format

Bit/Field name	Start bit	Transmitter bit	Command index	Card status	Reserved	End bit
Bit position/width	47	46	[45:40]/6 bits	[39:8]/32 bits	[7:1]/7 bits	0
Value	0b	0b	111111b	OCR	1111111b	1b

24.3.3.5 R4 Response

A dedicated response to CMD5 that replies with the OCR and related registers, 48 bits total, with CRC7 check. R4 is used in SDIO cards. The format is shown in the table below.

Table 24-10: R4 format

Bit/Field name	Bit position	Width (bits)	Value
Start bit	47	1	0b
Transmitter bit	46	1	0b
Reserved	[45:40]	6	111111b
Card ready	39	1	X
Number of IO functions	[38:36]	3	X
Current register	[35]	1	X
Stuff bits	[34:33]	2	00b

Bit/Field name	Bit position	Width (bits)	Value
S18A	32	1	X
IO OCR	[31:8]	24	OCR
CRC check field	[7:1]	7	1111111b
End bit	0	1	1b

Note: The R4 format for MMC cards differs from that of SDIO cards.

24.3.3.6 R5 Response

A dedicated response to CMD5, 48 bits total, with CRC7 check. R5 is used in SDIO cards. The format is shown in the table below.

Table 24-11: R5 format

Bit/Field name	Start bit	Transmitter bit	Command index	Stuff bits	Response format	Read/write data	CRC7	End bit
Bit position	47	46	[45:40]	[39:24]	[23:16]	[15:8]	[7:1]	0
Width	1	1	6	16	8	8	7	1
Value	0b	0b	110100b	0000h	X	X	X	1b

Note: The R5 format for MMC cards differs from that of SDIO cards.

24.3.3.7 R6 Response

A dedicated response that replies with the RCA, 48 bits total, with CRC7 check. The format is shown in the table below.

Table 24-12: R6 format

Bit/Field name	Start bit	Transmitter bit	Command index	Card RCA	Card status bits	CRC7	End bit
Bit position	47	46	[45:40]	[39:24]	[23:8]	[7:1]	0
Width	1	1	6	16	16	7	1
Value	0b	0b	000011b	X	X	X	1b

24.3.3.8 R7 Response

A dedicated response to CMD8, indicating the supported voltage information, 48 bits total, with CRC7 check. The format is shown in the table below.

Table 24-13: R7 format

Bit/Field name	Start bit	Transmitter bit	Command index	Reserved	PCIe 1V2 support	PCIe response	Accepted voltage	Check pattern	CRC7	End bit
Bit position	47	46	[45:40]	[39:22]	21	20	[19:16]	[15:8]	[7:1]	0
Width	1	1	6	18	1	1	4	8	7	1

Bit/ Field name	Start bit	Transmitter bit	Command index	Reserved	PCIe 1V2 support	PCIe response	Accepted voltage	Check pattern	CRC7	End bit
Value	0b	0b	001000b	00000h	X	X	X	X	X	1b

24.3.4 Data Transfer

Data transfer takes place on the data line SDIO_D, with three widths: 1/4/8 bits. During data transfer, a start bit of one clock period generally precedes the data, and each byte is transmitted with the most significant bit first and the least significant bit last. For SD cards, which only support block transfer mode, each block of data transfer is followed by a CRC check.

By enabling the RANDOM_LEN_EN bit, configuring the byte length represented by DBLOCKSIZE2, configuring the total transfer data length represented by R32_SDIO_DLEN, enabling the DTEN bit, and writing data to R32_SDIO_FIFO, block transfer of any byte length with CRC check can be accomplished.

MMC cards also support data stream transfer mode, which does not include CRC. The format of data transfer is shown in the figure below.

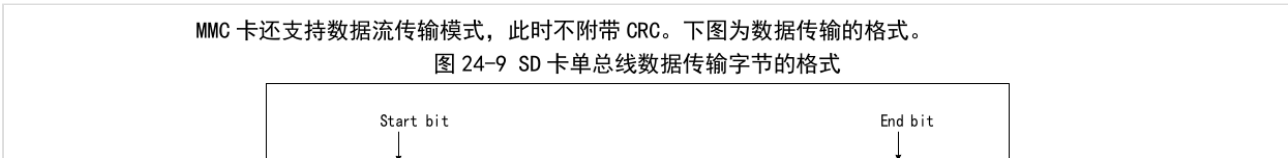


Figure 24-9: SD card single-bus data transfer byte format

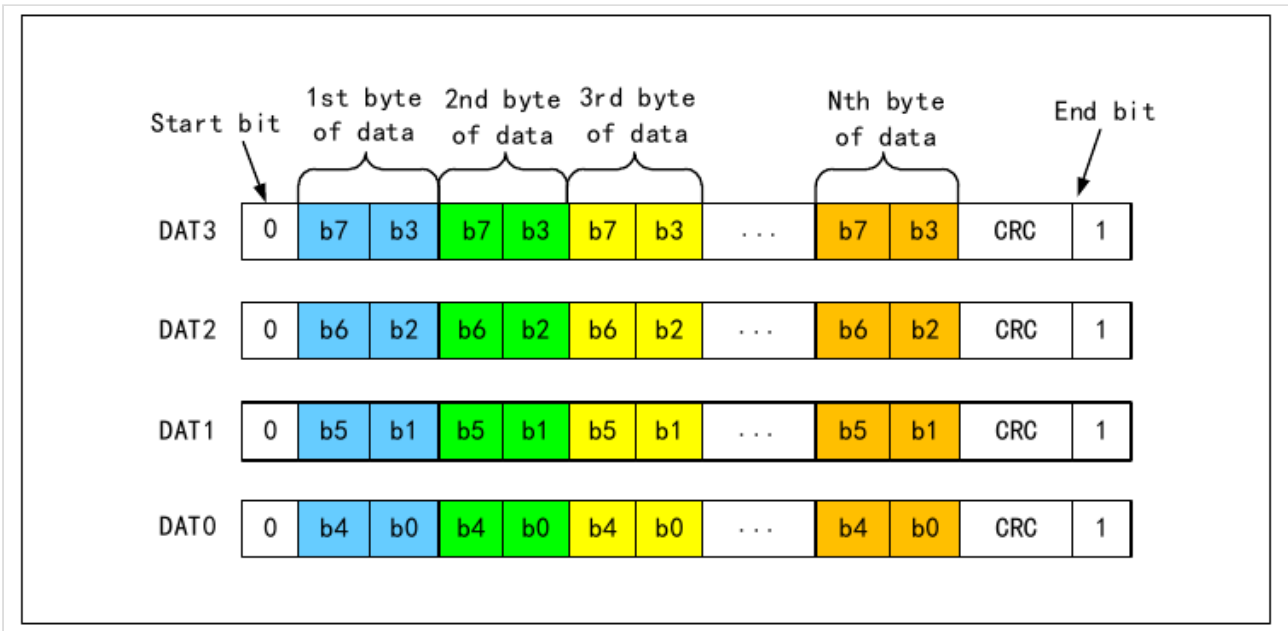


Figure 24-10: SD card 4-bus data transfer byte format

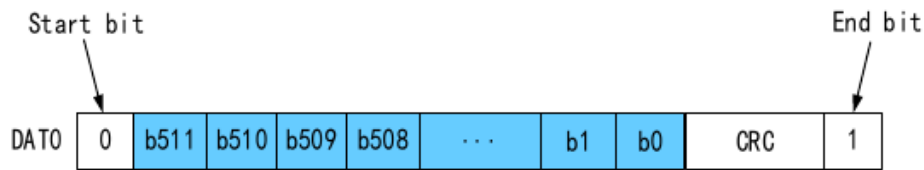


Figure 24-11: SD card single-bus data transfer format for a 512-bit word

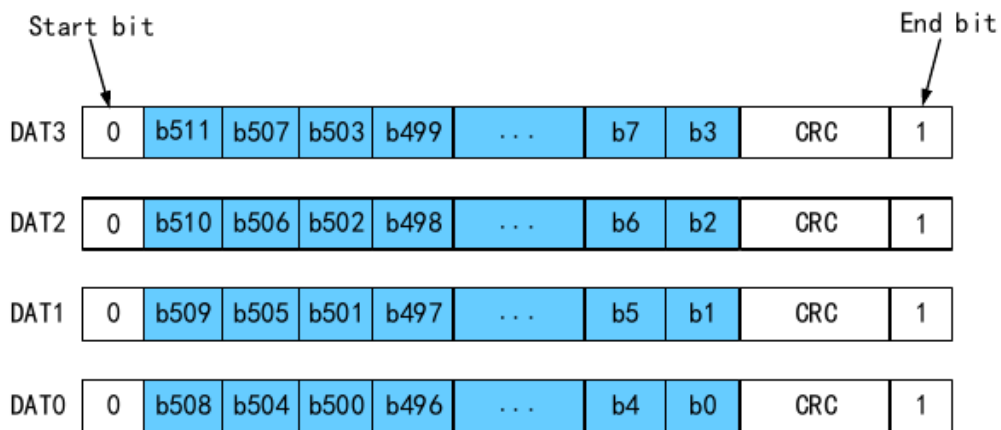


Figure 24-12: SD card 4-bus data transfer format for a 512-bit word

24.3.5 Slave Mode

By enabling the SLV_MODE bit, the controller enters a state waiting for host commands. The received command parameters are placed in RESP1, the received command index is placed in RESP2, and an R1-type response is automatically sent to the host, with the response parameter being the value of CMDARG.

For data transmission and reception in slave mode, refer to the master mode. However, the slave can use the SLV_FORCE_ERR bit to force a data block CRC error, thereby indicating an undesired read or write.

Note: The slave only supports R1-type responses. The command indices and parameter meanings used for master-slave communication are all software-defined.

24.4 Application

24.4.1 Device Initialization and Device Registers

24.4.1.1 OCR Register

The Operation Conditions Register stores information about the supply voltage accepted by the SD card and related status bits. The relevant bit definitions are shown in the table below.

Table 24-14: OCR register bit definitions

Bit position	Bit definition	Description
[0:3]	Reserved	
4	Reserved	
5	Reserved	
6	Reserved	
7	Reserved for low voltage range	
8	Reserved	
9	Reserved	
10	Reserved	
11	Reserved	
12	Reserved	
13	Reserved	Supported VDD33 voltage range, in volts
14	Reserved	
15	2.7-2.8	
16	2.8-2.9	
17	2.9-3.0	
18	3.0-3.1	
19	3.1-3.2	
20	3.2-3.3	
21	3.3-3.4	
22	3.4-3.5	
23	3.5-3.6	
24	Accept switching to 1.8 V	
[25:26]	Reserved	
27	Over 2 TB support status bit	
28	Reserved	
29	UHS-II card status bit	

Bit position	Bit definition	Description
		When this bit is set, it indicates that this card supports the UHS-II interface
30	Card capacity status (CCS)	When this bit is set, it indicates that the card capacity is greater than 2 GB
31	Card power-up status bit (busy)	This bit is set after the card completes power-up. After power-up is complete, the other bits become meaningful

24.4.1.2 CID Register

The CID stores some identification information.

Table 24-15: CID register field definitions

Bit/Field name	Abbreviation	Width	Bit position
Manufacturer ID	MID	8	[127:120]
Application ID	OID	16	[119:104]
Product name	PNM	40	[103:64]
Product revision	PRV	8	[63:56]
Product serial number	PSN	32	[55:24]
Reserved	None	4	[23:20]
Manufacturing date	MDT	12	[19:8]
CRC7	CRC	7	[7:1]
Fixed bit, fixed at 1	None	1	[0]

24.4.1.3 CSD Register

The CSD register stores the characteristic data of the SD card. Taking the most commonly used second-version CSD for SDHC and SDXC cards as an example, the field definitions are shown in the table below.

Table 24-16: CSD register field meanings

Name	Abbreviation	Width	Value	Read/Write	Bit position
CSD version	CSD_STRUCTURE	2	01b	RO	[127:126]
Reserved	None	6	00_0000b	RO	[125:120]
Read access time	TAAC	8	0Eh	RO	[119:112]
Read access time in clock cycles	NSAC	8	00h	RO	[111:104]
Maximum data transfer speed	TRAN_SPEED	8	32h/5Ah/0Bh/2Bh	RO	[103:96]
Card command classes	CCC	12	X1X1101101X1b	RO	[95:84]
Maximum read data block length	READ_BL_LEN	4	9	RO	[83:80]

Name	Abbreviation	Width	Value	Read/Write	Bit position
Partial blocks read allowed	READ_BLK_PARTIAL	1	0	RO	[79]
Write block misalignment	WRITE_BLK_MISALIGN	1	0	RO	[78]
Read block misalignment	READ_BLK_MISALIGN	1	0	RO	[77]
DSR implemented	DSR_IMP	1	X	RO	[76]
Reserved	None	6	00_0000b	RO	[75:70]
Device size	C_SIZE	22	XXXXXXh	RO	[69:48]
Reserved	None	1	0	RO	[47]
Single block erase enabled	ERASE_BLK_EN	1	1	RO	[46]
Erase sector size	SECTOR_SIZE	7	7Fh	RO	[45:39]
Write protect group size	WP_GRP_SIZE	7	0000000b	RO	[38:32]
Write protect group enable	WP_GRP_ENABLE	1	0	RO	[31]
Reserved	None	2	00b	RO	[30:29]
Write speed factor	R2W_FACTOR	3	010b	RO	[28:26]
Maximum write data block length	WRITE_BLK_LEN	4	9	RO	[25:22]
Partial blocks write allowed	WRITE_BLK_PARTIAL	1	0	RO	[21]
Reserved	None	5	00000b	RO	[20:16]
File format group	FILE_FORMAT_GRP	1	0	RO	[15]
Copy flag	TMP_WRITE_PROTECT	1	X	RW OTP	[14]
Permanent write protect	PERM_WRITE_PROTECT	1	X	RW OTP	[13]
Temporary write protect	TMP_WRITE_PROTECT	1	X	RW	[12]
File format	FILE_FOMAT	2	00b	RO	[11:10]
Reserved	None	2	00b	RO	[9:8]
CRC	CRC	7	0000000b	RW	[7:1]
Unused, must be 1	None	1	1b	RO	[0]

24.4.1.4 RCA Register

The Relative Card Address register stores the card address, is 16 bits, and the default value is 0.

24.4.2 Voltage Switching

In the later stages of SD card initialization, an interface level switch is required to switch the IO level of the SD card's clock line, data line, and command line to the 1.8 V level. For devices with insufficient slew rate, using a lower level standard helps improve frequency. However, the SD power supply voltage does not necessarily change; only in newer versions of the protocol do SD cards with low-voltage power supply appear.

The voltage switching steps are shown in the figure below.

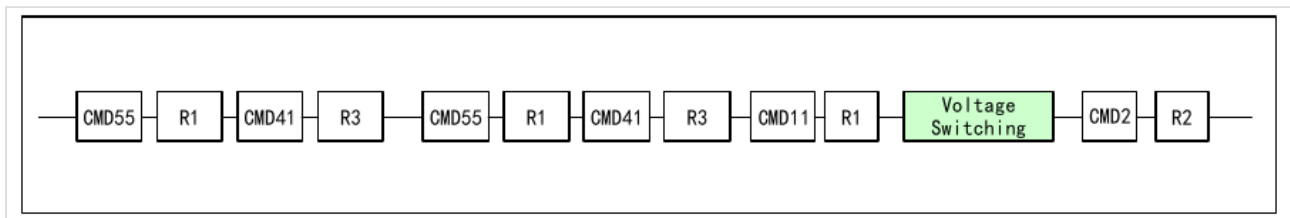


Figure 24-13: Voltage switching sequence

24.4.3 Clock Switching

During initialization, the SD card clock is only 400 kHz. After the voltage switch is complete, the clock can be raised to a higher level, for example, the first speed class of UHS-I mode for SDHC cards, where the bus clock can reach 80 MHz. Given the IO output capability of the microcontroller, the clock should be limited to within 50 MHz.

24.5 Interrupts

24.5.1 SDIO Interrupt

The SDIO supports multiple interrupt sources. As shown in the interrupt enable register (R32_SDIO_IER), there are 24 conditions that can trigger interrupts. The user can enable them as needed.

24.5.2 SDIO Device Interrupt

It should be noted that not only can the SDIO peripheral report interrupts to the CPU, but external SDIO cards and MMC cards can also report interrupts to the SDIO peripheral. In 4-bit bus mode, the interrupt line is D1; in 8-bit bus mode, the interrupt line is D7, both active low. If the SDIO detects D1 or D7 low in the idle state, it should read the SDIO device's status register or interrupt flag register to respond to the interrupt in a timely manner. The CPU can determine whether the SDIO host has received an interrupt through the SDIOIT bit of the R32_SDIO_STA register.

24.6 Register Description

Table 24-17: SDIO related register list

Name	Address	Description	Reset Value
R32_SDIO_POWER	0x40018000	Power register	0x00000000
R32_SDIO_CLKCR	0x40018004	Clock register	0x00000000
R32_SDIO_ARG	0x40018008	Command argument register	0x00000000
R32_SDIO_CMD	0x4001800C	Command register	0x00000000
R32_SDIO_RESPCMD	0x40018010	Response register	0x00000000
R128_SDIO_RESPX	0x40018014	Response parameter register	0x00000000
R32_SDIO_DTIMER	0x40018024	Data timer register	0x00000000
R32_SDIO_DLEN	0x40018028	Transfer length register	0x00000000
R32_SDIO_DCTLR	0x4001802C	Data control register	0x00000000

Name	Address	Description	Reset Value
R32_SDIO_DCOUNT	0x40018030	Transfer count register	0x00000000
R32_SDIO_STA	0x40018034	Status register	0x00000000
R32_SDIO_ICR	0x40018038	Interrupt clear register	0x00000000
R32_SDIO_MASK	0x4001803C	Interrupt enable register	0x00000000
R32_SDIO_FIFOCNT	0x40018048	FIFO counter	0x00000000
R32_SDIO_DCTRL2	0x40018060	Data control register 2	0x00000000
R32_SDIO_FIFO	0x40018080	FIFO register	0x00000000

24.6.1 Power Register (R32_SDIO_POWER)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													PWRCTRL[1:0]		

Bits	Name	Access	Description	Reset
[31:2]	Reserved	RO	Reserved.	0
[1:0]	PWRCTRL[1:0]	RW	Power control bits: 00: Power off, clock stopped; 01: Reserved; 10: Reserved power-up state; 11: Power-up state, card clock enabled.	0

24.6.2 Clock Register (R32_SDIO_CLKCR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		HWFC_EN	NEGEDGE	WIDBUS[1:0]		BYPASS	PWRSV	CLKEN	CLKDIV[7:0]						

Bits	Name	Access	Description	Reset
[31:15]	Reserved	RO	Reserved.	0
14	HWFC_EN	RW	Hardware flow control enable bit. When this bit is set, the TXFIFOE and RXFIFO signals take effect. 1: Enable hardware flow control; 0: Disable hardware flow control.	0
13	NEGEDGE	RW	SDIO_CLK phase selection bit: 1: Generate SDIO_CLK on the falling edge of HCLK; 0: Generate SDIO_CLK on the rising edge of HCLK.	0
[12:11]	WIDBUS[1:0]	RW	Bus width configuration field: 00: 1-bit bus mode, uses SDIO_D0;	0

Bits	Name	Access	Description	Reset
			01: 4-bit bus mode, uses SDIO_D[3:0]; 10: 8-bit bus mode, uses SDIO_D[7:0]; 11: Not used.	
10	BYPASS	RW	Clock bypass enable bit: 1: SDIO_CK is directly connected to HCLK; 0: SDIO_CK is derived through the divider. Note: When using this bit, the HWFC_EN bit must be enabled.	0
9	PWRSAPV	RW	Clock state configuration when idle. When this bit is set, SDIO_CK output is turned off when the bus is idle to save power. 1: SDIO_CK output only when needed; 0: SDIO_CK always outputs.	0
8	CLKEN	RW	Clock enable bit: 1: SDIO_CK output enabled; 0: SDIO_CK output disabled.	0
[7:0]	CLKDIV[7:0]	RW	Clock division factor field. This field represents the relationship between SDIO_CK and HCLK. $SDIO_CK = HCLK / (CLKDIV + 2)$. Note that during the initialization phase, SDIO_CK should be below 400 kHz.	0

Note: (1) The clock configuration register is used to control parameters related to SDIO_CK. Note that this register must not be changed within 7 HCLK cycles during read/write data operations.

(2) In slave mode, there is no need to configure the clock register.

24.6.3 Command Argument Register (R32_SDIO_ARG)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CMDARG[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMDARG[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	CMDARG	RW	Master mode: Command argument field. This field stores the parameters of the command, which are sent as part of the command onto the CMD line. Slave mode: 32-bit response parameter register for slave reply.	0

24.6.4 Command Register (R32_SDIO_CMD)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	ATACMD	NIEN	ENCMDCompl	SDIOSuspend	CPSMEN	WAITPEND	WAITINT	WAITRESP[1:0]	CMDINDEX[5:0]						

Bits	Name	Access	Description	Reset
[31:15]	Reserved	RO	Reserved.	0
14	ATACMD	RW	CE-ATA command. If this bit is set, CPSM will go to CMD61.	0
13	NIEN	RW	CE-ATA interrupt not enable bit. If this bit is set, CE-ATA will not generate an interrupt.	0
12	ENCMDCompl	RW	Enable CMD completion signal enable bit. If this bit is set, command completion will generate a signal.	0
11	SDIOSuspend	RW	Suspend command send bit. If this bit is set, a suspend signal will be sent. Only applicable to SDIO cards.	0
10	CPSMEN	RW	CPSM (Command Path State Machine) enable bit. If this bit is set, CPSM is enabled.	0
9	WAITPEND	RW	Command wait control bit. If this bit is set, CPSM waits for data transfer to complete before sending the command.	0
8	WAITINT	RW	Command wait interrupt control bit. If this bit is set, CPSM disables timeout control and waits for an interrupt to be generated.	0
[7:6]	WAITRESP[1:0]	RW	Response type field. This field indicates the type of response expected by CPSM. 00: No response, wait for CMDSENT flag; 01: Short response, wait for CMDREND or CCRCFAIL flag; 10: No response, wait for CMDSENT flag; 11: Long response, wait for CMDREND or CCRCFAIL flag.	0
[5:0]	CMDINDEX[5:0]	RW	Command index field. This field indicates the specific command value.	0

24.6.5 Response Register (R32_SDIO_RESPCMD)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										RESPCMD[5:0]					

Bits	Name	Access	Description	Reset
[31:6]	Reserved	RO	Reserved.	0
[5:0]	RESPCMD[5:0]	RO	This field records the index value of the received response.	0

24.6.6 Response Parameter Register (R128_SDIO_RESPX)

Bits	Name	Access	Description	Reset
[127:0]	CARDSTATUSx	RO	Master mode: When the response is a long response, all 128 bits represent the card status; when the response is a short response, the lower 32 bits represent the card status. The SDIO peripheral receives the most significant bit of the card status first and stores it starting from the least significant bit of R128_SDIO_RESPX.	0

24.6.6.1 Response Parameter Register High 32 Bits (R128_SDIO_RESP1[127:96])

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS1[127:112]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS1[111:96]															
Bits	Name	Access	Description	Reset											
[31:0]	CARDSTATUS1	RO	Master mode: Card status [127:96]; Slave mode: Received command parameters.	0											

24.6.6.2 Response Parameter Register Second-High 32 Bits (R128_SDIO_RESP2[95:64])

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS2[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS2[79:64]															
Bits	Name	Access	Description	Reset											
[31:0]	CARDSTATUS2	RO	Master mode: Card status [95:64]; Slave mode: Received command index.	0											

24.6.6.3 Response Parameter Register Second-Low 32 Bits (R128_SDIO_RESP3[63:32])

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS3[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS3[47:32]															
Bits	Name	Access	Description	Reset											
[31:0]	CARDSTATUS3	RO	Master mode: Card status [63:32].	0											

24.6.6.4 Response Parameter Register Low 32 Bits (R128_SDIO_RESP4[31:0])

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CARDSTATUS4[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CARDSTATUS4[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	CARDSTATUS4	RO	Master mode: Card status [31:0].	0											

24.6.7 Data Timer Register (R32_SDIO_DTIMER)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATETIME[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATETIME[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	DATETIME	RW	Data timeout duration. In units of SDIO_CK cycles.	0											

24.6.8 Transfer Length Register (R32_SDIO_DLEN)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							DATALENGTH[24:16]								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATALENGTH[15:0]															

Bits	Name	Access	Description	Reset
[31:25]	Reserved	RO	Reserved.	0
[24:0]	DATALENGTH[24:0]	RW	Transfer data length field. The value of this field is loaded into the transfer counter when the transfer is started. For block transfer, the value of this field must be an integer multiple of the block size. The block size is defined by the SDIO device and is stored in R32_SDIO_DCTLR[7:4]; a common value is 512 B, etc.	0

24.6.9 Data Control Register (R32_SDIO_DCTRL)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	SDIOEN	RWMOD	RWSTOP	RWSTART	DBLOCKSIZE[3:0]		DMAEN		DTMODE	DTDIR	DTEN				

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
11	SDIOEN	RW	SDIO enable bit. When this bit is set, DPSM can perform some operations specific to SDIO cards.	0
10	RWMOD	RW	Read wait mode: 1: Use SDIO_D2 to control read wait; 0: Stop SDIO_CK to control read wait.	0
9	RWSTOP	RW	Read wait stop bit. If the RWSTART bit is set, the read wait will be stopped.	0

Bits	Name	Access	Description	Reset
8	RWSTART	RW	Read wait start bit. Setting this bit will execute a read wait operation.	0
[7:4]	DBLOCKSIZE[3:0]	RW	Data block length field. This field stores the length of the data block. The block transfer length must be defined before block transfer. The value that can be written to this field is between 0 and 1110b, representing a block transfer length of 2^{BLKLEN} , i.e., between 0 and 16384 bytes.	0
3	DMAEN	RW	DMA enable bit: 1: Enable DMA; 0: Disable DMA.	0
2	DTMODE	RW	Transfer mode setting bit. Setting this bit sets the transfer mode. 1: Stream transfer; 0: Block transfer.	0
1	DTDIR	RW	Transfer direction setting bit. Setting this bit sets the transfer direction. 1: Card to controller; 0: Controller to card.	0
0	DTEN	RW	Transfer enable bit. Setting this bit starts data transfer. The specific process is: after setting this bit, DPSM enters the Wait_S or Wait_R flow (depending on the transfer direction).	0

24.6.10 Transfer Count Register (R32_SDIO_DCOUNT)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								DATACOUNT[24:16]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATACOUNT[15:0]															

Bits	Name	Access	Description	Reset
[31:25]	Reserved	RO	Reserved.	0
[24:0]	DATACOUNT[24:0]	RO	Transfer data counter field. When the transfer is started, the value of the transfer length register is loaded into this counter and decremented during the transfer process.	0

24.6.11 Status Register (R32_SDIO_STA)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CEATAEND	SDIOIT	RXDAVL	TXDAVL	RXFIFOE	TXFIFOE	RXFIFOE	TXFIFOE
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXFIFOH	TXFIFOH	RXACT	TXACT	CMDACT	DBCKEND	STBITERR	DATAEND	CMDSENT	CMDREND	RXOVERR	TXUNDERR	DTIMEOUT	CTIMEOUT	DOVERR	DOVERR

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
23	CEATAEND	RO	When this bit is set, CMD61 received CE-ATA completion signal.	0
22	SDIOIT	RO	When this bit is set, SDIO received a device interrupt.	0
21	RXDAVL	RO	When this bit is set, receive FIFO data available.	0
20	TXDAVL	RO	When this bit is set, data available in transmit FIFO.	0
19	RXFIFOE	RO	When this bit is set, receive FIFO empty.	0
18	TXFIFOE	RO	When this bit is set, transmit FIFO empty.	0
17	RXFIFOE	RO	When this bit is set, receive FIFO full.	0
16	TXFIFOE	RO	When this bit is set, transmit FIFO full.	0
15	RXFIFOHF	RO	When this bit is set, receive FIFO half full.	0
14	TXFIFOHE	RO	When this bit is set, transmit FIFO half empty.	0
13	RXACT	RO	When this bit is set, data is being received.	0
12	TXACT	RO	When this bit is set, data is being transmitted.	0
11	CMDACT	RO	When this bit is set, command is being transferred.	0
10	DBCKEND	RO	When this bit is set, a data block has been sent or received and CRC check passed.	0
9	STBITERR	RO	When this bit is set, in wide bus mode, no start signal was detected on any data line.	0
8	DATAEND	RO	When this bit is set, data transfer ended (transfer counter is zero).	0
7	CMDSENT	RO	When this bit is set, the command has been sent out.	0
6	CMDREND	RO	When this bit is set, a response has been received and CRC check succeeded.	0
5	RXOVERR	RO	When this bit is set, receive FIFO overrun.	0
4	TXUNDERR	RO	When this bit is set, transmit FIFO underrun.	0
3	DTIMEOUT	RO	When this bit is set, data timeout.	0
2	CTIMEOUT	RO	When this bit is set, command timeout exceeded 64 SDIO_CLK cycles.	0
1	DCRCFAIL	RO	When this bit is set, a data block has been sent or received but CRC check failed.	0
0	CCRCFAIL	RO	When this bit is set, a response has been received but CRC check failed.	0

24.6.12 Interrupt Clear Register (R32_SDIO_ICR)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21				20	19	18	17	16					
Reserved									CEATAENDC			SDIOITC			Reserved								
15	14	13	12	11	10	9		8		7		6		5		4		3		2		1	
Reserved			DBCKENDC		STBITERRC		DATAENDC		CMDSENTC		CMDREND		RXOVERRC		TXUNDERRC		DTIMEOUTC		CTIMEOUTC		DCRC		

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
23	CEATAENDC	RW	Setting this bit clears the CEATAEND bit in the status register.	0

Bits	Name	Access	Description	Reset
22	SDIOITC	RW	Setting this bit clears the SDIOIT bit in the status register.	0
[21:11]	Reserved	RW	Reserved.	0
10	DBCKENDC	RW	Setting this bit clears the DBCKEND bit in the status register.	0
9	STBITERRC	RW	Setting this bit clears the STBITERR bit in the status register.	0
8	DATAENDC	RW	Setting this bit clears the DATAEND bit in the status register.	0
7	CMDSENTC	RW	Setting this bit clears the CMDSENT bit in the status register.	0
6	CMDREND	RW	Setting this bit clears the CMDREND bit in the status register.	0
5	RXOVERRC	RW	Setting this bit clears the RXOVERR bit in the status register.	0
4	TXUNDERRC	RW	Setting this bit clears the TXUNDERR bit in the status register.	0
3	DTIMEOUTC	RW	Setting this bit clears the DTIMEOUT bit in the status register.	0
2	CTIMEOUTC	RW	Setting this bit clears the CTIMEOUT bit in the status register.	0
1	DCRCFAILC	RW	Setting this bit clears the DCRCFAIL bit in the status register.	0
0	CCRCFAILC	RW	Setting this bit clears the CCRCFAIL bit in the status register.	0

24.6.13 Interrupt Enable Register (R32_SDIO_MASK)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16		
										CEATAENDIE	SDIOITIE	RXDAVLIE	TXDAVLIE	RXFIFOEIE	TXFIFOEIE	RXFIFOEIE	TXFIFOEIE
15	14	13	12	11	10	9	8	7	6	5	4	3					
RXFIFOHFIE	TXFIFOHEIE	RXACTIE	TXACTIE	CMDACTIE	DBCKENDIE	STBITERRIE	DATAENDIE	CMDSENTIE	CMDRENDIE	RXOVERRIE	TXUNDERRIE	DTIMEOUTIE					

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
23	CEATAENDIE	RW	When this bit is set, an interrupt will be generated when the CEATAEND bit is set in the status register.	0
22	SDIOITIE	RW	When this bit is set, an interrupt will be generated when the SDIOIT bit is set in the status register.	0
21	RXDAVLIE	RW	When this bit is set, an interrupt will be generated when the RXDAVL bit is set in the status register.	0
20	TXDAVLIE	RW	When this bit is set, an interrupt will be generated when the TXDAVL bit is set in the status register.	0
19	RXFIFOEIE	RW	When this bit is set, an interrupt will be generated when the RXFIFOE bit is set in the status register.	0
18	TXFIFOEIE	RW	When this bit is set, an interrupt will be generated when the TXFIFOE bit is set in the status register.	0
17	RXFIFOEIE	RW	When this bit is set, an interrupt will be generated when the RXFIFOE bit is set in the status register.	0
16	TXFIFOEIE	RW	When this bit is set, an interrupt will be generated when the TXFIFOE bit is set in the status register.	0
15	RXFIFOHFIE	RW		0

Bits	Name	Access	Description	Reset
			When this bit is set, an interrupt will be generated when the RXFIFOHF bit is set in the status register.	
14	TXFIFOHEIE	RW	When this bit is set, an interrupt will be generated when the TXFIFOHE bit is set in the status register.	0
13	RXACTIE	RW	When this bit is set, an interrupt will be generated when the RXACT bit is set in the status register.	0
12	TXACTIE	RW	When this bit is set, an interrupt will be generated when the TXACT bit is set in the status register.	0
11	CMDACTIE	RW	When this bit is set, an interrupt will be generated when the CMDACT bit is set in the status register.	0
10	DBCKENDIE	RW	When this bit is set, an interrupt will be generated when the DBCKEND bit is set in the status register.	0
9	STBITERRIE	RW	When this bit is set, an interrupt will be generated when the STBITERR bit is set in the status register.	0
8	DATAENDIE	RW	When this bit is set, an interrupt will be generated when the DATAEND bit is set in the status register.	0
7	CMDSENTIE	RW	When this bit is set, an interrupt will be generated when the CMDSENT bit is set in the status register.	0
6	CMDRENDIE	RW	When this bit is set, an interrupt will be generated when the CMDREND bit is set in the status register.	0
5	RXOVERRIE	RW	When this bit is set, an interrupt will be generated when the RXOVERR bit is set in the status register.	0
4	TXUNDERRIE	RW	When this bit is set, an interrupt will be generated when the TXUNDERR bit is set in the status register.	0
3	DTIMEOUTIE	RW	When this bit is set, an interrupt will be generated when the DTIMEOUT bit is set in the status register.	0
2	CTIMEOUTIE	RW	When this bit is set, an interrupt will be generated when the CTIMEOUT bit is set in the status register.	0
1	DCRCFAILIE	RW	When this bit is set, an interrupt will be generated when the DCRCFAIL bit is set in the status register.	0
0	CCRCFAILIE	RW	When this bit is set, an interrupt will be generated when the CCRCFAIL bit is set in the status register.	0

24.6.14 FIFO Count Register (R32_SDIO_FIFOCNT)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FIFOCOUNT[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FIFOCOUNT[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	FIFOCOUNT[31:0]	RO	The number of words (32-bit) that the FIFO contains that have not yet been written to the FIFO or have not yet been read from the	0

Bits	Name	Access	Description	Reset
			FIFO. When setting R32_SDIO_DCTLR, if DPSM is idle, the FIFO counter will load the transfer length value from R32_SDIO_TLEN. If this value is not divisible by 4, the last 1 to 3 bytes will be treated as one word.	

24.6.15 Data Control Register 2 (R32_SDIO_DCTRL2)

Offset address: 0x60

31	30	29	28	27	26	25			24			23			22	21	20	19	18	17	16	
Reserved						SLV_CK_PHASE		SLV_FORCE_ERR			SLV_MODE			Reserved						RANDOM_LEN_EN		
15	14		13		12		11		10		9		8	7	6	5	4	3	2	1	0	
Reserved							DBLOCKSIZE2															

Bits	Name	Access	Description	Reset
[31:27]	Reserved	RO	Reserved.	0
26	SLV_CK_PHASE	RW	When SLV_MODE is 1, phase selection bit for DATA output in slave mode: 1: Output DATA on the falling edge of internal SDCK; 0: Output DATA on the rising edge of internal SDCK. When SLV_MODE is 0: 1: Writing 0 to DTEN bit does not clear the FIFO; 0: Writing 0 to DTEN bit clears the FIFO.	0
25	SLV_FORCE_ERR	RW	Software-forced data block CRC error in slave mode: 1: Software forces the data block CRC to an error value; 0: Use the hardware-correct CRC value.	0
24	SLV_MODE	RW	Slave mode enable bit: 1: The controller CPSM and DPSM operate in slave mode; 0: The controller operates in master mode.	0
[23:17]	Reserved	RO	Reserved.	0
16	RANDOM_LEN_EN	RW	Data block arbitrary byte length enable bit: 1: The data block can be any length from 0 to 2048, with the length being DBLOCKSIZE2; 0: The data block can only be 2 ^{DBLOCKSIZE} bytes.	0
[15:12]	Reserved	RO	Reserved.	0
[11:0]	DBLOCKSIZE2	RW	Data block length field for arbitrary byte length mode.	0

24.6.16 FIFO Register (R32_SDIO_FIFO)

Offset address: 0x80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FIFODATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FIFODATA[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	FIFODATA[31:0]	RW	FIFO data field. This field is the FIFO data. Reading or writing this field reads out received data or writes data to be sent. The SDIO FIFO has a total of 32 words (one word is 32 bits).	0

Chapter 25: Controller Area Network (CAN)

The Controller Area Network is a high-performance communication protocol for serial data communication. The CAN controller provides a complete implementation of the CAN protocol, supporting CAN protocol 2.0A and 2.0B. The CAN controller can be used to build powerful local networks to achieve safe distributed real-time control, handling a large amount of data messages with a small CPU load, and is widely used in industrial and automotive fields.

25.1 Main Features

- Compatible with CAN specification 2.0A and 2.0B
- Programmable transmission rate, up to 1 Mbit/s
- Supports time-triggered communication function to avoid low-priority message blocking
- Supports three transmit mailboxes; transmit message priority can be determined by the message identifier or the order of transmit requests, and can record the timestamp of the transmit message SOF moment
- Supports two receive FIFOs with three-level mailbox depth, 14 configurable message filter banks; each filter bank can be configured in 32-bit or 16-bit mode, mask or identifier list mode, minimizing software intervention in message filtering; flexible FIFO overflow handling, and can record the timestamp of the received message SOF moment
- Occupies 4 interrupt vectors; each interrupt source can be independently configured

25.2 CAN Controller Operating Modes

The CAN controller can switch between three operating modes — initialization mode, sleep mode, and normal mode — by operating the SLEEP or INRQ bits in the CAN1_CTLR register.

25.2.1 Initialization Mode

After reset, the CAN defaults to sleep mode to reduce power consumption; message reception and transmission are disabled, the internal pull-up resistor of the TX pin is enabled, and the TX pin outputs a recessive bit. Setting the INRQ bit in the CAN1_CTLR register to 1 requests the CAN controller to enter initialization mode; when the INAK bit in the CAN1_STATR register is automatically set to 1, initialization mode is successfully entered. Similarly, clearing the INRQ bit in the CAN1_CTLR register to 0 requests the CAN controller to exit initialization mode; when the INAK bit in the CAN1_STATR register is automatically cleared to 0, initialization mode is successfully exited.

Initialization of the filter banks can be performed in non-initialization mode, but the FINIT bit in the CAN1_FCTLR register must be set to 1, at which point reception of messages is disabled.

25.2.2 Sleep Mode

Setting the SLEEP bit in the CAN1_CTLR register to 1 requests the CAN controller to enter sleep mode; when the SNAK bit in the CAN1_STATR register is automatically set to 1, the CAN has successfully entered sleep mode. At this time, the CAN controller clock stops, but the mailbox registers remain accessible.

To enter initialization mode from sleep mode, the SLEEP bit in CAN1_CTLR must be cleared to 0 and the INRQ bit set to 1; when the INAK bit in the CAN1_STATR register is automatically set to 1, the switch to initialization mode is complete.

To enter normal mode from sleep mode, the SLEEP bit in CAN1_CTLR must be cleared to 0; when the SNAK bit in the CAN1_STATR register is automatically cleared to 0, normal mode is entered.

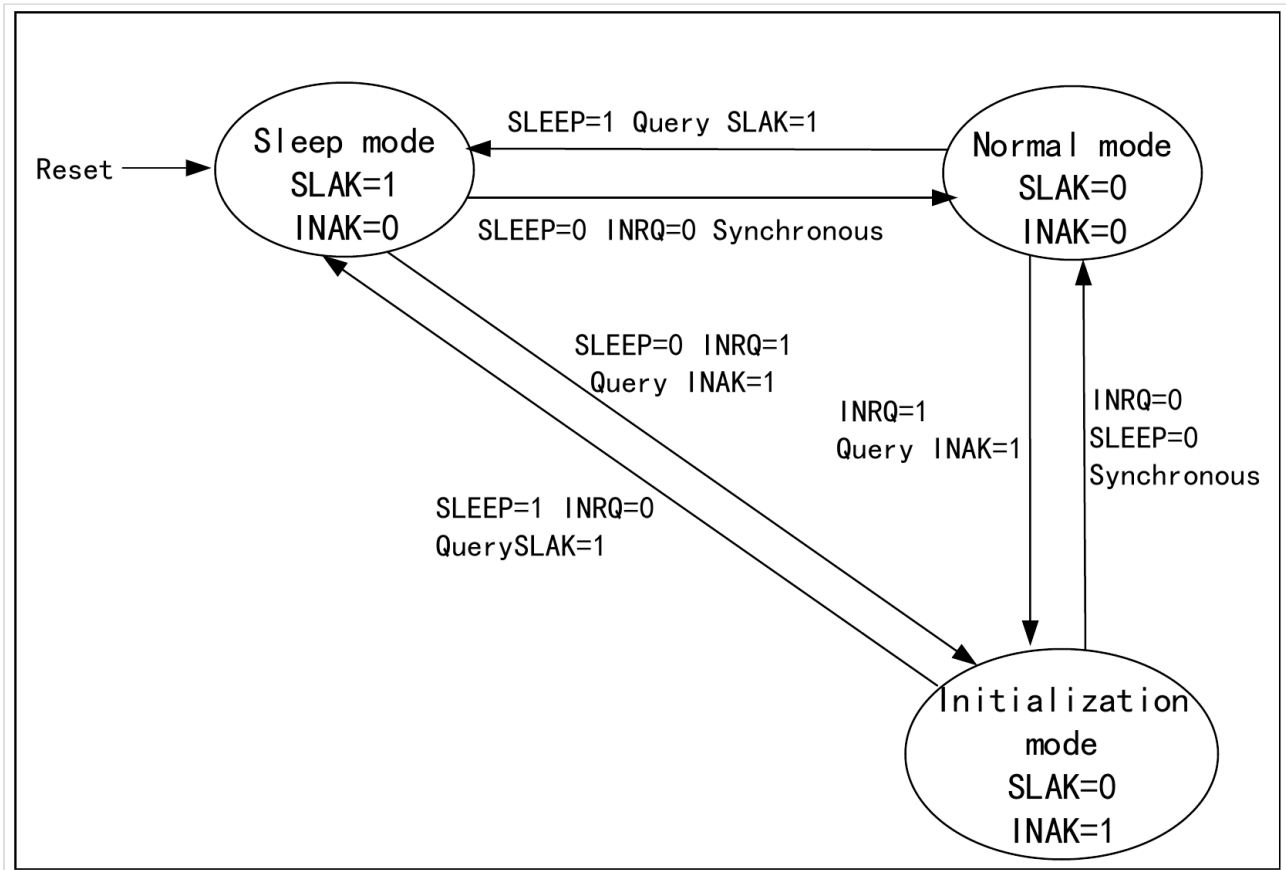


Figure 25-1: CAN operating mode switching

25.3 CAN Controller Test Modes

In initialization mode, by operating the SILM and LBKM bits in the CAN1_BTMR register, a test mode can be selected. Then by clearing the INRQ bit in the CAN1_CTLR register, initialization mode is exited and the test mode is entered. The test modes are divided into three types: silent mode, loopback mode, and silent loopback mode.

25.3.1 Silent Mode

Setting the SILM bit in the CAN1_BTMR register to 1 selects silent mode. In this mode, the CAN controller can receive but cannot transmit messages externally; it always presents a recessive bit externally, avoiding influence on the bus, but messages can be received by the local node's controller. Silent mode is typically used for CAN bus state analysis.

25.3.2 Loopback Mode

Setting the LBKM bit in the CAN1_BTMR register to 1 selects loopback mode. In this mode, the CAN controller can transmit messages externally but cannot receive external messages; however,

transmitted messages can be received by the local node's controller, and the receive filtering mechanism is effective. Loopback mode is typically used for CAN controller transceiver testing.

25.3.3 Silent Loopback Mode

Setting both the SILM and LBKM bits in the CAN1_BTMR register to 1 selects silent loopback mode. This mode is typically used for closed self-testing of the CAN controller. In this mode, there is no effect on the CAN bus; the RX pin is disconnected from the bus, and the TX pin is set to a recessive bit.

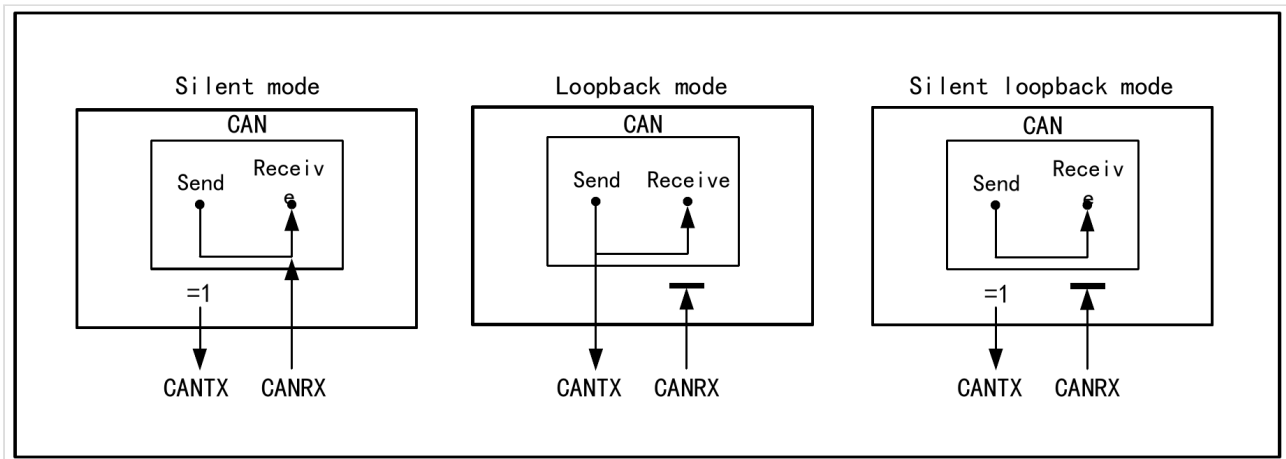


Figure 25-2: Three test modes of the CAN bus

25.4 CAN Controller Operating State When MCU is in Debug Mode

When the MCU enters debug mode, the core is in a halted state, but configuration bits in the debug module can be used to determine whether the CAN controller continues to operate normally or stops.

25.5 CAN Controller Functional Description

25.5.1 Transmit Processing Flow

The transmit processing flow is as follows: if there is an idle mailbox among the three transmit mailboxes, the application layer software has write access only to the idle mailbox registers. By operating the registers CAN1_TXMIyR, CAN1_TXMDTyR, CAN1_TXMDLyR, and CAN1_TXMDHyR, the message identifier, message length, timestamp, and message data can be set. After the data is ready, set the TXRQ bit in the CAN1_TXMIyR register to 1 to request transmission; the mailbox enters the pending state and is prioritized in the queue. Once it becomes the highest-priority mailbox, it transitions to the scheduled transmit state, waiting for the CAN bus to become idle. When the CAN bus is idle, the scheduled transmit mailbox message immediately enters the transmit state. After the message transmission is complete, the mailbox becomes idle again, and the RQCP and TXOK bits in the CAN1_TSTATR register are set to 1 to indicate successful transmission. If arbitration fails during transmission, the ALST bit in the CAN1_TSTATR register is set to 1; if a transmission error occurs, the TERR bit is set to 1.

25.5.2 Transmit Priority

The transmit priority can be determined by the identifier or the order of transmit requests. When the TXFP bit in the CAN1_CTLR register is set to 1, transmission follows the order of transmit requests; transmit-by-request-order is mainly used for segmented transmission. When the TXFP bit is cleared to 0, the transmit order is determined by identifier priority — the smaller the identifier, the higher the priority; for identical identifiers, the lower-numbered mailbox has higher priority.

25.5.3 Transmit Abort Handling

Setting the ABRQ bit in the CAN1_TSTATR register to 1 can abort a transmit request. When the mailbox is in the pending or scheduled transmit state, the transmit request is directly aborted; when the mailbox is in the transmitting state, the abort request may succeed (transmission stopped) or fail (transmission completed). The result can be queried via the TXOK bit in the CAN1_TSTATR register.

25.5.4 Time-Triggered Mode

In traditional CAN communication, when the bus is busy, low-priority messages can be blocked for a long time, and may even fail to meet their deadline requirements. To solve this bottleneck, time-triggered mode protocols have been introduced. Such protocols have a certain scale of application in industry. The time-triggered mode function is designed to support the application of such protocols.

In time-triggered mode, there are two modes available for selection. Using this mode requires disabling the automatic retransmission function. The default mode and enhanced mode are selected by configuring the MODE bit in the CAN1_TTCTLR register. Setting the TTCM and NART bits in the CAN1_CTLR register to 1 enables time-triggered mode and disables automatic retransmission. The MODE bit in the CAN1_TTCTLR register defaults to 0, meaning it operates in default mode; the internal timer is activated to generate timestamps for transmit and receive mailboxes. The timer accumulates in CAN bit time, and the internal timer is sampled at the sampling point of the start-of-frame bit of receive and transmit frames to produce a timestamp. To use enhanced mode, set the MODE bit in the CAN1_TTCTLR register to 1 to enable enhanced mode. In this mode, there must be three or more nodes in the entire CAN network; one node transmits a time reference, and other nodes, after receiving the reference node's timestamp, write 1 to the TIMRST bit in the CAN1_TTCTLR register to reset the internal counter and synchronize the internal counter. In this way, all CAN nodes except the one transmitting the time reference achieve time synchronization. Afterward, the data to be transmitted is written to the transmit mailbox, and the time-triggered count value (TIMCNT in the CAN1_TTCNT register) and the internal counter terminal value (TIMCMV in the CAN1_TTCTLR register) are configured for each node. The time-triggered count value and internal counter terminal value are determined by the number of CAN nodes, the CAN communication rate, and the number of bits per frame. After configuration is complete, each node waits for the internal counter to count to the time-triggered count value, which then triggers the transmit action.

25.5.5 Receive Processing Flow

Reception of CAN bus messages is completed by the controller hardware without MCU intervention, reducing the MCU processing load. Received messages are stored, according to the settings of the CAN1_FAFIFOR register, into two FIFOs each with 3-level mailbox depth. The application layer can only read valid received messages through the receive FIFO mailbox.

Initially, the receive FIFO is empty, and the FMR[1:0] value of the receive FIFO register CAN1_RFIFOx is binary 00b. After receiving a valid received message, it transitions to pending-1 state, and the controller automatically sets the FMR[1:0] value of the receive FIFO register CAN1_RFIFOx to binary 01b. If the mailbox data registers CAN1_RXMDLyR and CAN1_RXMDHyR are read at this time, setting the RFOM bit in the receive FIFO register CAN1_RFIFOx to 1 releases the mailbox, and the receive FIFO state returns to empty. If the mailbox is not released in the pending-1 state, the next valid received message causes the receive FIFO state to switch to pending-2 state, at which point the FMR[1:0] value of the receive FIFO register CAN1_RFIFOx is automatically set to binary 10b. If the mailbox data registers are read and the mailbox is released, the state returns to pending-1. If the mailbox is not released in the pending-2 state, the receive FIFO enters pending-3 state. Similarly, reading the message and releasing the mailbox in the pending-3 state returns the state to pending-2. If the mailbox is not released in the pending-3 state, receiving the next valid message will inevitably result in message loss.

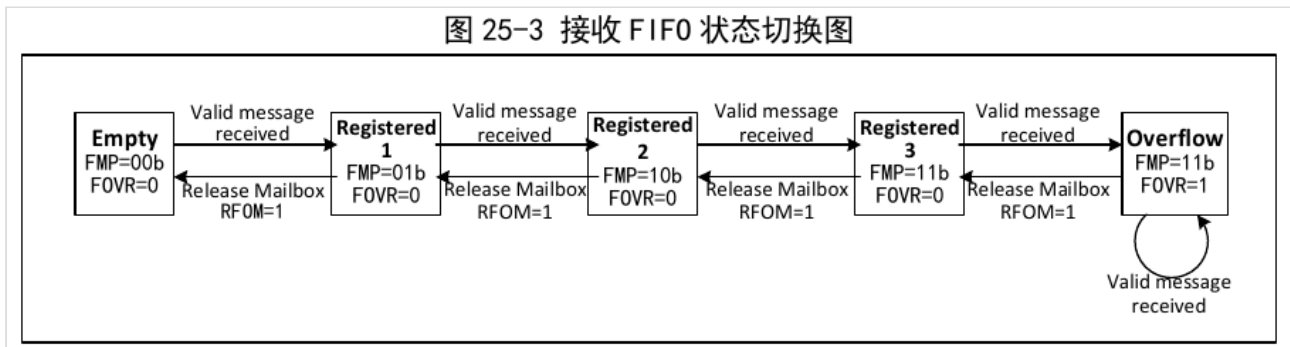


Figure 25-3: Receive FIFO state transition diagram

The message loss situation described above means the receive FIFO is full and message overflow causes message loss. The FOVR bit in the receive FIFO register CAN1_RFIFOx is automatically set to 1 by hardware for overflow query. Setting the RFLM bit in the CAN1_CTLR register to 1 enables the receive FIFO locking function, and the discarded message is the newly received message. Clearing the RFLM bit in the CAN1_CTLR register to 0 disables the receive FIFO locking function; among the three original messages in the receive FIFO, the last received message will be overwritten by the new message.

When the relevant bits in the CAN1_INTENR register are set, an interrupt can be generated when the receive FIFO state transitions, enabling more efficient processing of received messages. See Section 25.6 CAN Interrupts for details.

25.5.6 Received Message Identifier Filtering

The module has up to 14 filter banks. By configuring the filter banks, each CAN node can receive messages that match the filter rules; messages that do not match the filter rules are discarded by hardware without software intervention.

Each filter bank consists of two 32-bit registers, CAN1_FxR0 and CAN1_FxR1. The bit width of each filter bank can be independently configured as one 32-bit filter or two 16-bit filters by setting the respective bits in the CAN1_FSCFGR register. Each filter bank can be configured in mask or identifier list mode by setting the respective bits in the CAN1_FMCFG register. Each filter bank can be enabled or disabled by setting the respective bits in the CAN1_FWR register. Setting the respective bits in the CAN1_FAFIFOR register selects which receive FIFO stores the messages that pass the filter.

As shown in Table 25-1 below, in mask mode, the two registers are the identifier register and the mask register, which must be used together. Each bit of the identifier register indicates whether the corresponding bit is expected to be dominant or recessive, and each bit of the mask register indicates whether the corresponding bit must match the expected value of the identifier register bit.

Table 25-1: 32-bit mask mode

	CAN1_FxR1[31:24]	CAN1_FxR1[23:16]	CAN1_FxR1[15:8]	CAN1_FxR1[7:0]
Identifier register				
Mask register	CAN1_FxR2[31:24]	CAN1_FxR2[23:16]	CAN1_FxR2[15:8]	CAN1_FxR2[7:0]
Mapping	STID[10:3]	STID[2:0] EXID[17:13]	EXID[12:5]	EXID[4:0] IDE RTR 0

In identifier list mode, both registers are used as identifier registers. The received message identifier must match one of the registers to pass the filtering.

Table 25-2: 32-bit identifier list mode

	CAN1_FxR1[31:24]	CAN1_FxR1[23:16]	CAN1_FxR1[15:8]	CAN1_FxR1[7:0]
Identifier register				
Mask register	CAN1_FxR2[31:24]	CAN1_FxR2[23:16]	CAN1_FxR2[15:8]	CAN1_FxR2[7:0]
Mapping	STID[10:3]	STID[2:0] EXID[17:13]	EXID[12:5]	EXID[4:0] IDE RTR 0

In 16-bit mode, the register bank is split into four registers. In mask mode, each filter group's mask mode can have 2 filters, each containing one 16-bit identifier register and one 16-bit mask register. In identifier list mode, all four registers are used as identifier registers.

Table 25-3: 16-bit mask mode

	CAN1_FxR1[15:8]	CAN1_FxR1[7:0]
Identifier register n		
Mask register n	CAN1_FxR1[31:24]	CAN1_FxR1[23:16]
Identifier register n+1	CAN1_FxR2[15:8]	CAN1_FxR2[7:0]
Mask register n+1	CAN1_FxR2[31:24]	CAN1_FxR2[23:16]
Mapping	STID[10:3]	STID[2:0] RTR IDE EXID[17:15]

Table 25-4: 16-bit identifier list mode

	CAN1_FxR1[15:8]	CAN1_FxR1[7:0]
Identifier register n		
Mask register n	CAN1_FxR1[31:24]	CAN1_FxR1[23:16]
Identifier register n+1	CAN1_FxR2[15:8]	CAN1_FxR2[7:0]
Mask register n+1	CAN1_FxR2[31:24]	CAN1_FxR2[23:16]
Mapping	STID[10:3]	STID[2:0] RTR IDE EXID[17:15]

When a message enters the FIFO mailbox, it is read and stored by the application. Typically, the application distinguishes message data by the message identifier. The CAN controller provides a filter number for messages filtered through different filter banks in the receive FIFO. The number is stored in the FMI[7:0] field of the CAN1_RXMDTyR register, and the numbering does not consider whether the filter bank is enabled. The numbering rules are detailed in the example in Figure 25-4.

When a message passes through multiple filters, the filter number stored in the receive mailbox is determined by the filter priority rules, which are as follows:

- All 32-bit filters have higher priority than 16-bit filters
- For filters of the same width, identifier list filters have higher priority than mask mode filters
- For filters with the same width and mode, the lower-numbered filter has higher priority

As shown in Figure 25-5: when receiving a message, the identifier is first matched against 32-bit identifier list mode filters; if no match, it is matched against 32-bit mask mode filters; if no match, it continues to be matched against 16-bit identifier list mode filters; if no match, it is finally matched against 16-bit mask mode filters. If there is still no match, the message is discarded. If a match is found, the message is stored in the receive FIFO mailbox, and the identifier number is stored in the FMI field of the CAN1_RXMDTyR register.

图 25-4 过滤器编号的示例

Filter group number	FIFO0	Filter number	Filter group number	FIFO1	Filter number
0	32-bit shield mode	0	1	16-bit list mode	0 1 2 3
2	16-bit list mode	1 2 3 4	4	16-bit shield mode	4 5
3	32-bit list mode	5 6	6	32-bit list mode	6 7
5	Disabled 16-bit shield Mode	7 8	9	32-bit shield mode	8
7	32-bit shield mode	9	11	Disabled 16-bit shield Mode	9 10
8	32-bit list mode	10 11	12	32-bit list mode	11 12
10	16-bit shield mode	12 13	13	32-bit list mode	13 14

Figure 25-4: Example of filter numbering

图 25-5 过滤器过滤示例

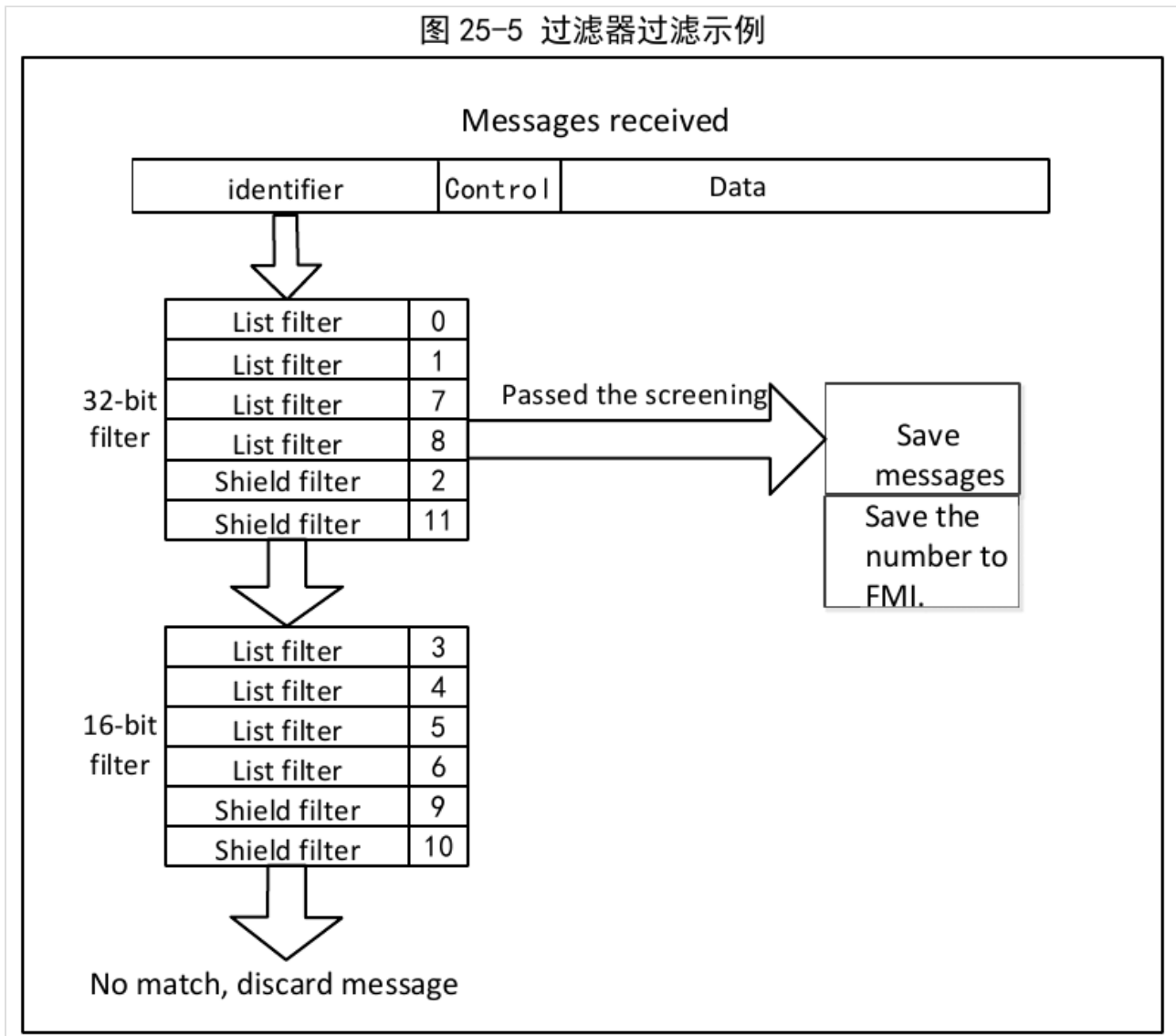


Figure 25-5: Filter filtering example

25.5.7 Error Handling

The CAN controller relies on the error status register CAN1_ERRSR for bus error management. The TEC and REC fields in the error status register CAN1_ERRSR represent the transmit and receive error count values, respectively. They increase with more transceiver errors and decrease with successful transceiver operations; their values can be used to judge the stability of the CAN bus.

When the TEC and REC in the error status register CAN1_ERRSR are less than 128, the current CAN node is in the error-active state, can normally participate in bus communication, and issues an active error flag when an error is detected.

When the TEC and REC in the error status register CAN1_ERRSR are greater than 127, the current CAN node is in the error-passive state and is not allowed to issue an active error flag when an error is detected; it can only issue a passive error flag.

When the TEC in the error status register CAN1_ERRSR is greater than 255, the current CAN node enters the bus-off state.

When the bus detects 128 occurrences of 11 consecutive recessive bits, it recovers to the error-active state. This recovery method is affected by the ABOM bit in the main control register CAN1_CTLR. If

ABOM is set to 1, the hardware automatically exits the bus-off state. If ABOM is 0, software must operate the INRQ bit to enter initialization mode and then exit initialization mode to exit the bus-off state.

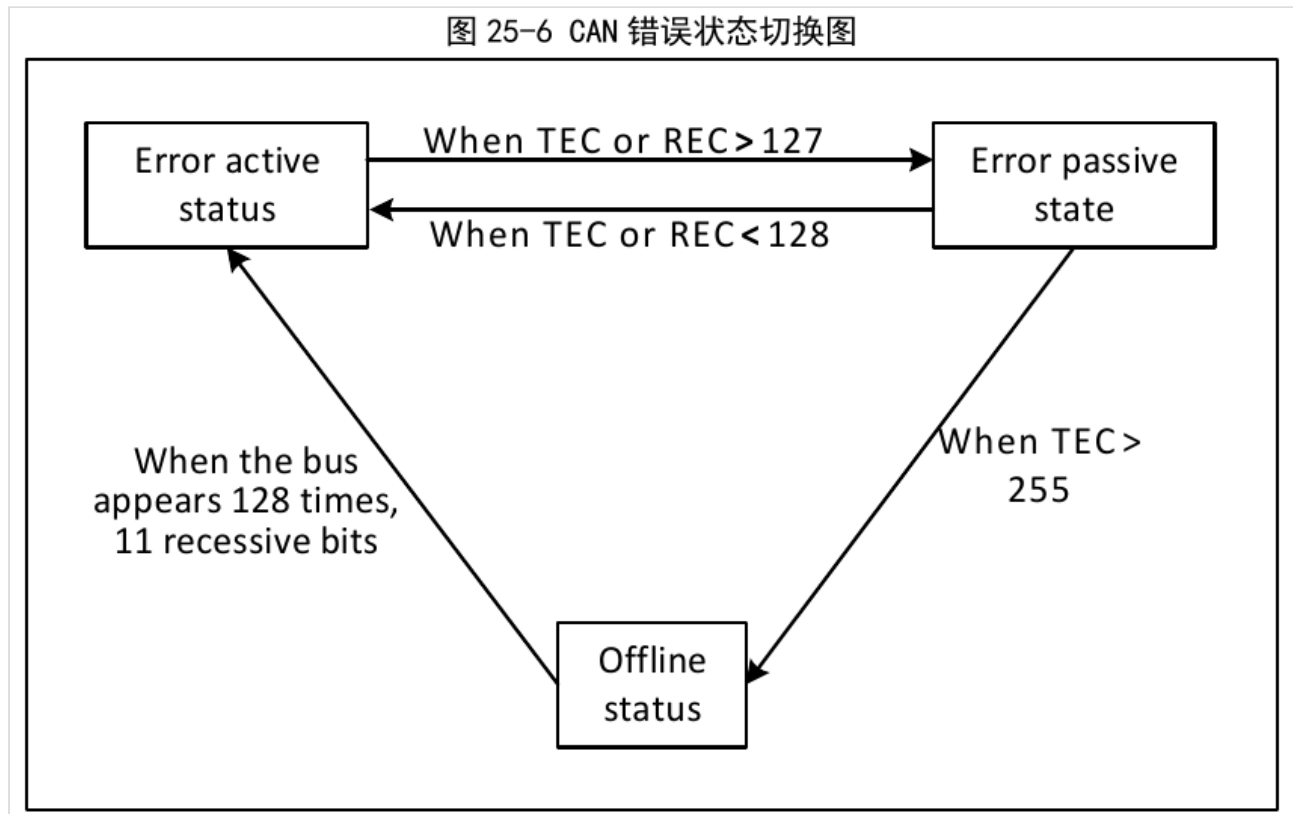


Figure 25-6: CAN error state transition diagram

25.5.8 Bit Timing

According to the CAN bus standard, each bit time is divided into four segments: the synchronization segment, the propagation time segment, the phase buffer segment 1, and the phase buffer segment 2. These segments are composed of the minimum time unit T_q . The CAN controller monitors CAN bus changes through sampling and synchronizes via the edge of the frame start bit.

The CAN controller regroups the above four segments into three segments:

- **Synchronization segment (SS):** This is the synchronization segment in the CAN standard, fixed at 1 minimum time unit. Under normal conditions, the expected bit transition occurs within this time segment.
- **Time segment 1 (BS1):** Includes the propagation time segment and phase buffer segment 1 in the CAN standard. It can be set to include 1 to 16 minimum time units and can be automatically extended to compensate for positive phase drift caused by frequency accuracy errors of different nodes on the CAN bus. The end of this time segment is the sampling point position.
- **Time segment 2 (BS2):** This is the phase buffer segment 2 in the CAN standard. It can be set to 1 to 8 minimum time units and can be automatically shortened to compensate for negative phase drift caused by frequency accuracy errors of different nodes on the CAN bus.

The resynchronization jump width (SJW) is the upper limit of the number of minimum time units that can be extended or shortened per bit, and the range can be set from 1 to 4 minimum time units.

All the above parameters can be configured in the CAN bus timing register CAN1_BTMR.

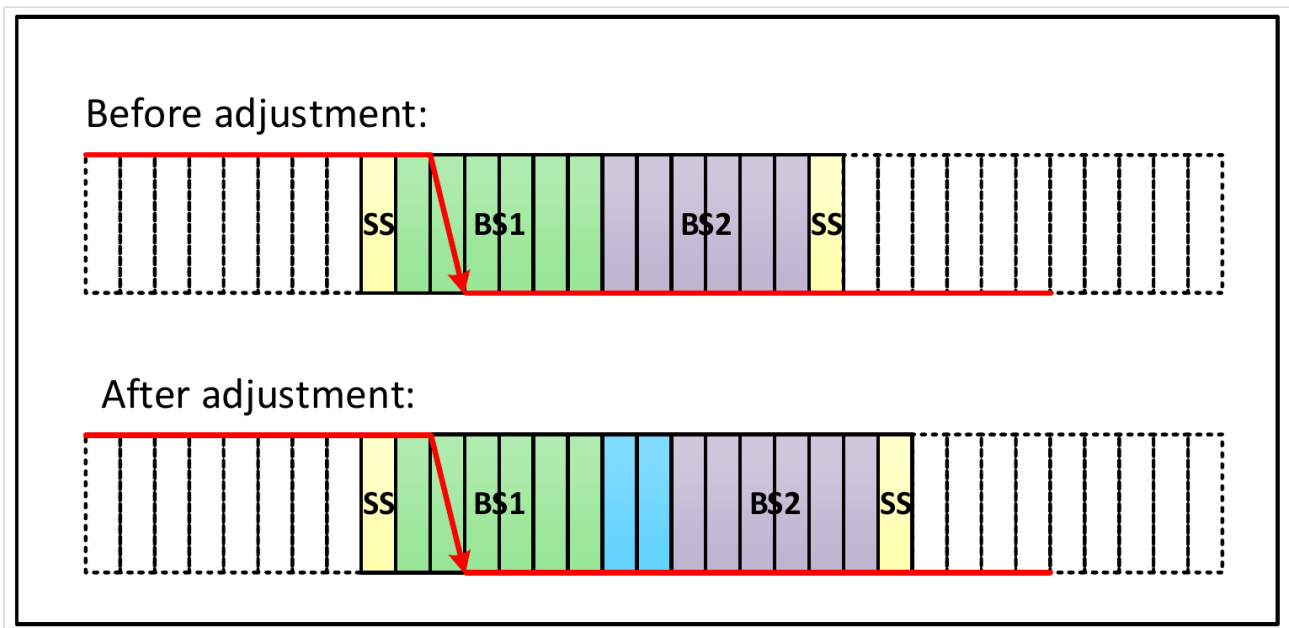


Figure 25-7: Transition occurring within BS1

As shown in Figure 25-7, with $SJW = 2$, when the bus level transition is detected in time segment 1, the length of time segment 1 needs to be extended by a maximum of SJW , thereby delaying the sampling point position.

图 25-8 跳变出现在 BS2 中

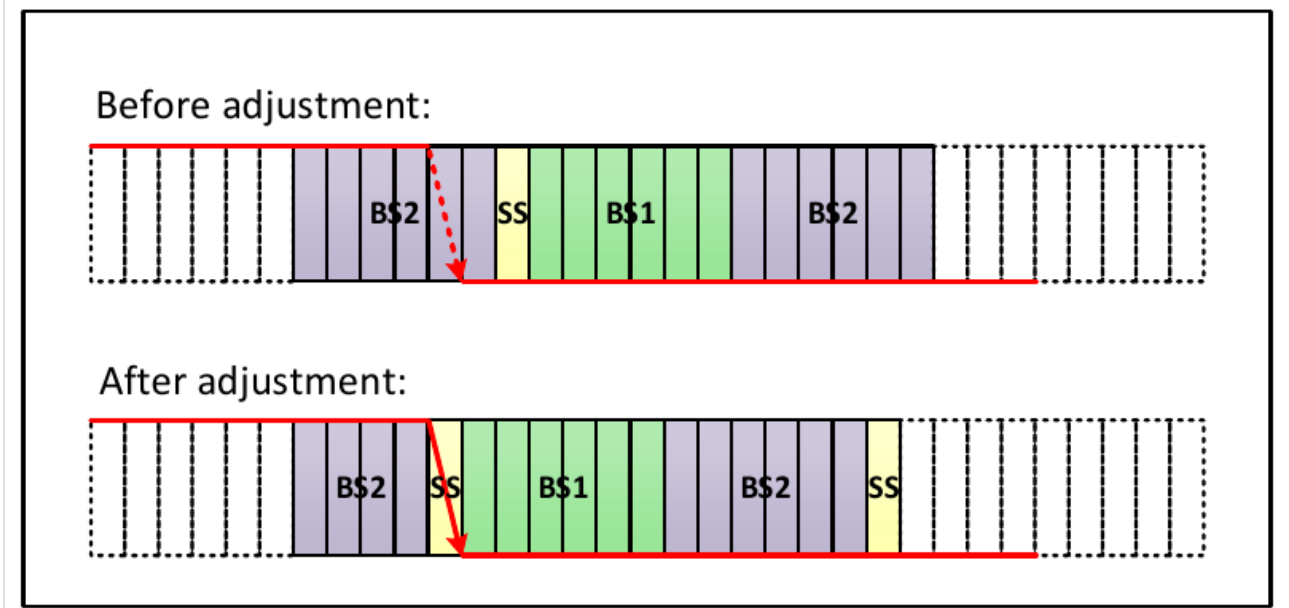


Figure 25-8: Transition occurring within BS2

As shown in Figure 25-8, with $SJW = 2$, when the bus level transition is detected in time segment 2, the length of time segment 2 needs to be shortened by a maximum of SJW , thereby advancing the sampling point position.

25.6 CAN Interrupts

The CAN controller has four interrupt vectors: transmit interrupt, FIFO_0 interrupt, FIFO_1 interrupt, and error and status change interrupt.

Setting the CAN interrupt enable register CAN1_INTENR can enable or disable each interrupt source.

The transmit interrupt is generated by the transmit mailbox becoming empty event. After the interrupt is generated, query the RQCP0, RQCP1, and RQCP2 bits in the CAN1_TSTATR register to determine which mailbox empty event generated the interrupt.

The FIFO0 interrupt is generated by receive new message, receive mailbox full, and overflow events. After the interrupt is generated, query the FMP0, FULL0, and FOVER0 bits in the CAN1_RFIFO0 register to determine which mailbox event generated the interrupt.

The FIFO1 interrupt is generated by receive new message, receive mailbox full, and overflow events. After the interrupt is generated, query the FMP1, FULL1, and FOVER1 bits in the CAN1_RFIFO1 register to determine which mailbox event generated the interrupt.

The error and status change interrupt is generated by error, wakeup, and sleep events.

图 25-9 CAN 中断逻辑图

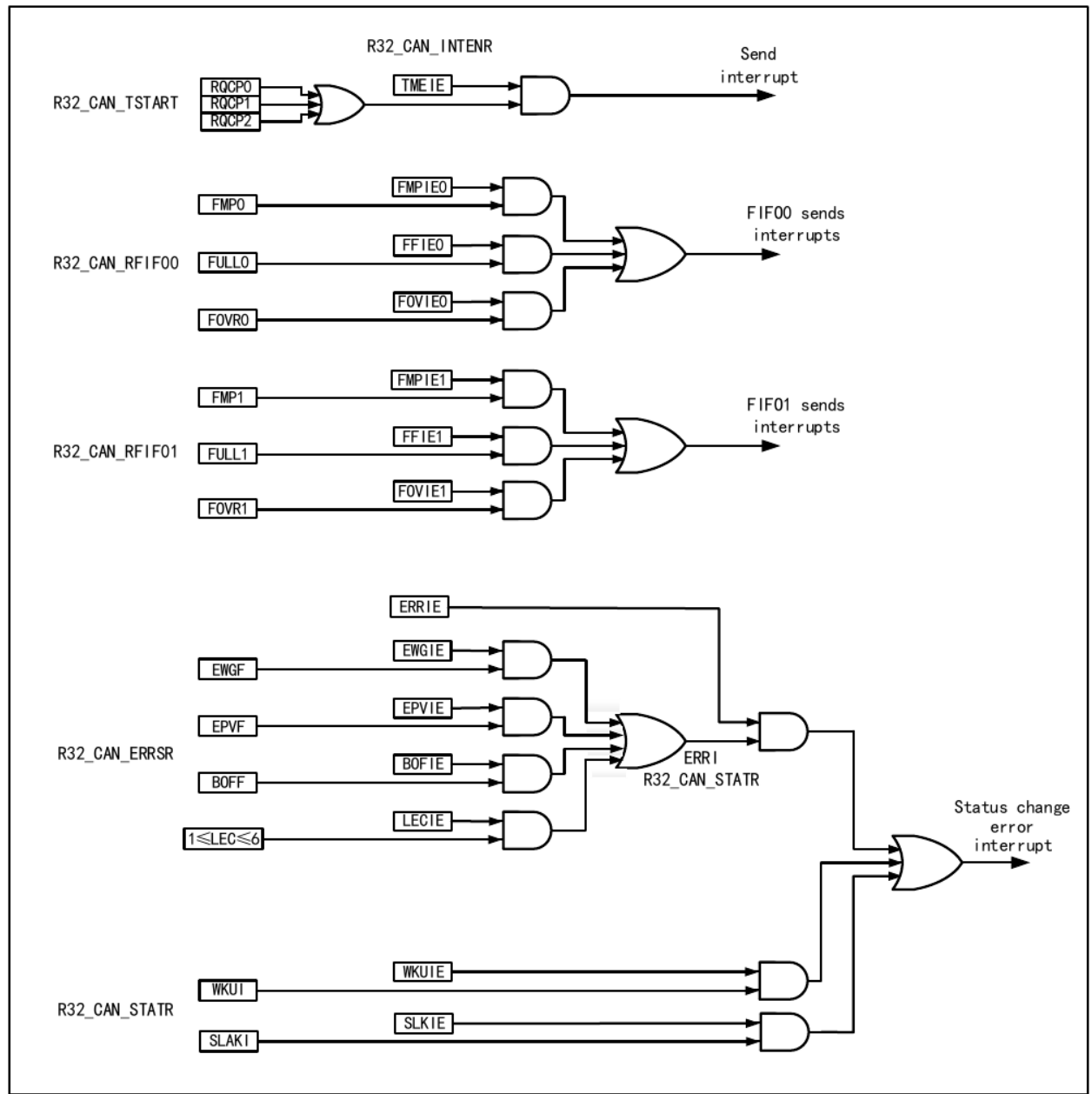


Figure 25-9: CAN interrupt logic diagram

25.8 Register Description

The registers related to the CAN controller must be operated in 32-bit word mode, and the CAN peripheral clock must be enabled during operation. To avoid the current node's influence on the entire CAN bus, application software can only modify the bit timing register CAN1_BTMR in initialization mode.

Table 25-5: CAN related register list

Name	Address	Description	Reset Value
R32_CAN1_CTLR	0x40006400	CAN1 main control register	0x00010002
R32_CAN1_STATR	0x40006404	CAN1 main status register	0x00000x02
R32_CAN1_TSTATR	0x40006408	CAN1 transmit status register	0x1C000000

Name	Address	Description	Reset Value
R32_CAN1_RFIFO0	0x4000640C	CAN1 receive FIFO0 control and status register	0x00000000
R32_CAN1_RFIFO1	0x40006410	CAN1 receive FIFO1 control and status register	0x00000000
R32_CAN1_INTENR	0x40006414	CAN1 interrupt enable register	0x00000000
R32_CAN1_ERRSR	0x40006418	CAN1 error status register	0x00000000
R32_CAN1_BTMR	0x4000641C	CAN1 bit timing register	0x01230000
R32_CAN1_TTCTLR	0x40006420	CAN1 time-triggered control register	0x0000FFFF
R32_CAN1_TTCNT	0x40006424	CAN1 time-triggered count value register	0x00000000
R32_CAN1_TERR_CNT	0x40006428	CAN1 bus-off recovery error counter	0x00000000

Table 25-6: CAN mailbox related register list

Name	Address	Description	Reset Value
R32_CAN1_TXMI0R	0x40006580	CAN1 transmit mailbox 0 identifier register	X
R32_CAN1_TXMDT0R	0x40006584	CAN1 transmit mailbox 0 data length and timestamp register	X
R32_CAN1_TXMDL0R	0x40006588	CAN1 transmit mailbox 0 low byte data register	X
R32_CAN1_TXMDH0R	0x4000658C	CAN1 transmit mailbox 0 high byte data register	X
R32_CAN1_TXMI1R	0x40006590	CAN1 transmit mailbox 1 identifier register	X
R32_CAN1_TXMDT1R	0x40006594	CAN1 transmit mailbox 1 data length and timestamp register	X
R32_CAN1_TXMDL1R	0x40006598	CAN1 transmit mailbox 1 low byte data register	X
R32_CAN1_TXMDH1R	0x4000659C	CAN1 transmit mailbox 1 high byte data register	X
R32_CAN1_TXMI2R	0x400065A0	CAN1 transmit mailbox 2 identifier register	X
R32_CAN1_TXMDT2R	0x400065A4	CAN1 transmit mailbox 2 data length and timestamp register	X
R32_CAN1_TXMDL2R	0x400065A8	CAN1 transmit mailbox 2 low byte data register	X
R32_CAN1_TXMDH2R	0x400065AC	CAN1 transmit mailbox 2 high byte data register	X
R32_CAN1_RXMI0R	0x400065B0	CAN1 receive FIFO0 mailbox identifier register	X
R32_CAN1_RXMDT0R	0x400065B4	CAN1 receive FIFO0 mailbox data length and timestamp register	X
R32_CAN1_RXMDL0R	0x400065B8	CAN1 receive FIFO0 mailbox low byte data register	X
R32_CAN1_RXMDH0R	0x400065BC	CAN1 receive FIFO0 mailbox high byte data register	X
R32_CAN1_RXMI1R	0x400065C0	CAN1 receive FIFO1 mailbox identifier register	X
R32_CAN1_RXMDT1R	0x400065C4	CAN1 receive FIFO1 mailbox data length and timestamp register	X
R32_CAN1_RXMDL1R	0x400065C8	CAN1 receive FIFO1 mailbox low byte data register	X
R32_CAN1_RXMDH1R	0x400065CC	CAN1 receive FIFO1 mailbox high byte data register	X

Table 25-7: CAN filter related register list

Name	Address	Description	Reset Value
R32_CAN1_FCTLR	0x40006600	CAN1 filter master control register	0x2A1C0E01
R32_CAN1_FMCFGR	0x40006604	CAN1 filter mode register	0x00000000
R32_CAN1_FSCFGR	0x4000660C	CAN1 filter bit width register	0x00000000
R32_CAN1_FAFIFOR	0x40006614	CAN1 filter FIFO association register	0x00000000
R32_CAN1_FWR	0x4000661C	CAN1 filter activation register	0x00000000
R32_CAN1_F0R1	0x40006640	CAN1 filter bank 0 register 1	X
R32_CAN1_F0R2	0x40006644	CAN1 filter bank 0 register 2	X
R32_CAN1_F1R1	0x40006648	CAN1 filter bank 1 register 1	X
R32_CAN1_F1R2	0x4000664C	CAN1 filter bank 1 register 2	X
R32_CAN1_F2R1	0x40006650	CAN1 filter bank 2 register 1	X
R32_CAN1_F2R2	0x40006654	CAN1 filter bank 2 register 2	X
R32_CAN1_F3R1	0x40006658	CAN1 filter bank 3 register 1	X
R32_CAN1_F3R2	0x4000665C	CAN1 filter bank 3 register 2	X
R32_CAN1_F4R1	0x40006660	CAN1 filter bank 4 register 1	X
R32_CAN1_F4R2	0x40006664	CAN1 filter bank 4 register 2	X
R32_CAN1_F5R1	0x40006668	CAN1 filter bank 5 register 1	X
R32_CAN1_F5R2	0x4000666C	CAN1 filter bank 5 register 2	X
R32_CAN1_F6R1	0x40006670	CAN1 filter bank 6 register 1	X
R32_CAN1_F6R2	0x40006674	CAN1 filter bank 6 register 2	X
R32_CAN1_F7R1	0x40006678	CAN1 filter bank 7 register 1	X
R32_CAN1_F7R2	0x4000667C	CAN1 filter bank 7 register 2	X
R32_CAN1_F8R1	0x40006680	CAN1 filter bank 8 register 1	X
R32_CAN1_F8R2	0x40006684	CAN1 filter bank 8 register 2	X
R32_CAN1_F9R1	0x40006688	CAN1 filter bank 9 register 1	X
R32_CAN1_F9R2	0x4000668C	CAN1 filter bank 9 register 2	X
R32_CAN1_F10R1	0x40006690	CAN1 filter bank 10 register 1	X
R32_CAN1_F10R2	0x40006694	CAN1 filter bank 10 register 2	X
R32_CAN1_F11R1	0x40006698	CAN1 filter bank 11 register 1	X
R32_CAN1_F11R2	0x4000669C	CAN1 filter bank 11 register 2	X
R32_CAN1_F12R1	0x400066A0	CAN1 filter bank 12 register 1	X
R32_CAN1_F12R2	0x400066A4	CAN1 filter bank 12 register 2	X
R32_CAN1_F13R1	0x400066A8	CAN1 filter bank 13 register 1	X
R32_CAN1_F13R2	0x400066AC	CAN1 filter bank 13 register 2	X

25.8.1 CAN1 Main Control Register (CAN1_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															DBF
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						TTCM		ABOM	AWUM	NART	RFLM	TXFP	SLEEP	INRQ	

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
16	DBF	RW	Whether debugging disables CAN bus operation. 1: During debugging, CAN transmission and reception are disabled, but the control and read/write operations of the receive FIFO work normally; 0: During debugging, the CAN controller operates normally.	1
15	RST	RW1	CAN controller software reset request. Writing 0 has no effect. 1: Resets the CAN controller; after reset, the controller enters sleep mode, and then hardware automatically clears this bit; 0: CAN controller normal state.	0
[14:8]	Reserved	RO	Reserved	0
7	TTCM	RW	Time-triggered mode enable: 1: Enable time-triggered mode; 0: Disable time-triggered mode. Time-triggered mode is mainly used in conjunction with the TTCAN protocol.	0
6	ABOM	RW	Bus-off automatic recovery control: 1: When hardware detects 128 occurrences of 11 consecutive recessive bits, it automatically exits the bus-off state; 0: Software must set the INRQ bit in the CAN1_CTLR register to 1 and then clear it to 0; after detecting 128 occurrences of 11 consecutive recessive bits, the bus-off state is exited.	0
5	AWUM	RW	CAN controller automatic wakeup enable: 1: When a message is detected, hardware automatically wakes up; the SLEEP and SLAK bits in the CAN1_STATR register are automatically cleared; 0: Software must clear the SLEEP bit in the CAN1_CTLR register to wake up the CAN controller.	0
4	NART	RW	Message automatic retransmission disable: 1: The message is transmitted only once, regardless of whether transmission succeeds; 0: The CAN controller keeps retransmitting until transmission succeeds.	0
3	RFLM	RW	Receive FIFO message lock mode enable: 1: When the receive FIFO overflows and received mailbox messages have not been read (mailbox not released), newly received messages are discarded; 0: When the receive FIFO overflows and received mailbox messages have not been read (mailbox not released), newly received messages overwrite the original messages.	0
2	TXFP	RW	Transmit mailbox priority selection: 1: Priority determined by the order of transmit requests; 0: Priority determined by the message identifier.	0
1	SLEEP	RW		1

Bits	Name	Access	Description	Reset
			Sleep mode request bit: 1: Set to 1 to request the CAN controller to enter sleep mode; the controller enters sleep mode after the current activity completes; if AWUM is set to 1, the controller clears the SLEEP bit upon receiving a message; 0: After software clears it, the controller exits sleep mode.	
0	INRQ	RW	Initialization mode request bit: 1: Set to 1 to request the CAN controller to enter initialization mode; the controller enters initialization mode after the current activity completes, and hardware sets the INAK bit in the CAN1_STATR register; 0: Set to 0 to request the CAN controller to exit initialization mode and enter normal mode; hardware clears the INAK bit in the CAN1_STATR register.	0

25.8.2 CAN1 Main Status Register (CAN1_STATR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Reserved																
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved				RX	SAMP	RXM	TXM	Reserved				SLAKI	WKUI	ERRI	SLAK	INAK

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
11	RX	RO	Current actual logic level of the CAN controller RX receive pin.	1
10	SAMP	RO	Logic level of the last received bit on the CAN controller RX receive pin.	1
9	RXM	RO	Receive mode query bit. 1: The CAN controller is currently in receive mode; 0: The CAN controller is not in receive mode.	0
8	TXM	RO	Transmit mode query bit. 1: The CAN controller is currently in transmit mode; 0: The CAN controller is not in transmit mode.	0
[7:5]	Reserved	RO	Reserved.	0
4	SLAKI	RW1	When sleep interrupt is enabled (i.e., SLKIE bit in CAN1_INTENR is set to 1), this is the interrupt generation flag bit. Writing 1 clears it; writing 0 has no effect. 1: An interrupt is generated when entering sleep mode; hardware sets this bit; 0: Hardware clears this bit when exiting sleep mode; software can also clear it.	0
3	WKUI	RW1	Wakeup interrupt flag bit. When the WKUIE bit in the CAN1_INTENR register is set to 1, if the CAN controller is in sleep mode and a SOF bit is detected, hardware sets this bit. Software writing 1 clears it; writing 0 has no effect.	0
2	ERRI	RW1		0

Bits	Name	Access	Description	Reset
			Error interrupt status flag bit. When the ERRIE bit in the CAN1_INTENR register is set to 1, an error and status change interrupt is generated. Software writing 1 clears this bit; writing 0 has no effect.	
1	SLAK	RO	Sleep mode indication bit. 1: The CAN controller is in sleep mode; 0: The CAN controller is not in sleep mode.	1
0	INAK	RO	Initialization mode indication bit. 1: The CAN controller is in initialization mode; 0: The CAN controller is operating in non-initialization mode.	0

25.8.3 CAN1 Transmit Status Register (CAN1_TSTATR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
LOW2	LOW1	LOW0	TME2	TME1	TME0	CODE[1:0]	ABRQ2	Reserved	TERR2	ALST2	TXOK2	RQCP2			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABRQ1	Reserved	TERR1	ALST1	TXOK1	RQCP1	ABRQ0	Reserved	TERR0	ALST0	TXOK0	RQCP0				

Bits	Name	Access	Description	Reset
31	LOW2	RO	Lowest priority flag bit for transmit mailbox 2. 1: Transmit mailbox 2 has the lowest priority; 0: Transmit mailbox 2 does not have the lowest priority.	0
30	LOW1	RO	Lowest priority flag bit for transmit mailbox 1. 1: Transmit mailbox 1 has the lowest priority; 0: Transmit mailbox 1 does not have the lowest priority.	0
29	LOW0	RO	Lowest priority flag bit for transmit mailbox 0. 1: Transmit mailbox 0 has the lowest priority; 0: Transmit mailbox 0 does not have the lowest priority.	0
28	TME2	RO	Empty flag bit for transmit mailbox 2. 1: Transmit mailbox 2 has no message waiting to be sent; 0: Transmit mailbox 2 has a message waiting to be sent.	1
27	TME1	RO	Empty flag bit for transmit mailbox 1. 1: Transmit mailbox 1 has no message waiting to be sent; 0: Transmit mailbox 1 has a message waiting to be sent.	1
26	TME0	RO	Empty flag bit for transmit mailbox 0. 1: Transmit mailbox 0 has no message waiting to be sent; 0: Transmit mailbox 0 has a message waiting to be sent.	1
[25:24]	CODE[1:0]	RO	Mailbox number. When more than one mailbox is empty, it indicates the next empty mailbox number; when all mailboxes are empty, it indicates the lowest-priority mailbox number.	0
23	ABRQ2	RW1	Transmit abort request for transmit mailbox 2. Software setting to 1 can abort the transmit request of mailbox 2; hardware clears it when the transmit message is cleared. If mailbox 2 is empty, software setting to 1 has no effect.	0
[22:20]	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
19	TERR2	RW1	Transmit failure flag bit for transmit mailbox 2. When transmit mailbox 2 fails to transmit, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	0
18	ALST2	RW1	Arbitration failure flag bit for transmit mailbox 2. When transmit mailbox 2 fails to transmit due to low arbitration priority, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	0
17	TXOK2	RW1	Transmit success flag bit for transmit mailbox 2. 1: Last transmission succeeded; 0: Last transmission failed. Software writing 1 clears it; writing 0 has no effect.	0
16	RQCP2	RW1	Request complete flag bit for transmit mailbox 2. When the transmit or abort request of transmit mailbox 2 completes, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	0
15	ABRQ1	RW1	Transmit abort request for transmit mailbox 1. Software setting to 1 can abort the transmit request of mailbox 1; hardware clears it when the transmit message is cleared. Writing 0 has no effect.	0
[14:12]	Reserved	RO	Reserved.	0
11	TERR1	RW1	Transmit failure flag bit for transmit mailbox 1. When transmit mailbox 1 fails to transmit, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	0
10	ALST1	RW1	Arbitration failure flag bit for transmit mailbox 1. When transmit mailbox 1 fails to transmit due to low arbitration priority, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	0
9	TXOK1	RW1	Transmit success flag bit for transmit mailbox 1. 1: Last transmission succeeded; 0: Last transmission failed. Software writing 1 clears it; writing 0 has no effect.	0
8	RQCP1	RW1	Request complete flag bit for transmit mailbox 1. When the transmit or abort request of transmit mailbox 1 completes, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	0
7	ABRQ0	RW1	Transmit abort request for transmit mailbox 0. Software setting to 1 can abort the transmit request of mailbox 0; hardware clears it when the transmit message is cleared. Writing 0 has no effect.	0
[6:4]	Reserved	RO	Reserved.	0
3	TERR0	RW1	Transmit failure flag bit for transmit mailbox 0. When transmit mailbox 0 fails to transmit, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	0
2	ALST0	RW1	Arbitration failure flag bit for transmit mailbox 0. When transmit mailbox 0 fails to transmit due to low arbitration priority, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	0
1	TXOK0	RW1	Transmit success flag bit for transmit mailbox 0. 1: Last transmission succeeded; 0: Last transmission failed. Software writing 1 clears it; writing 0 has no effect.	0
0	RQCP0	RW1		0

Bits	Name	Access	Description	Reset
			Request complete flag bit for transmit mailbox 0. When the transmit or abort request of transmit mailbox 0 completes, this bit is automatically set. Software writing 1 clears it; writing 0 has no effect.	

25.8.4 CAN1 Receive FIFO 0 Status Register (CAN1_RFIFO0)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						RFOM0		FOVR0		FULL0		Reserved		FMP0[1:0]	

Bits	Name	Access	Description	Reset
[31:6]	Reserved	RO	Reserved.	0
5	RFOM0	RW1	Software setting this bit to 1 releases the current mailbox message of receive FIFO_0. It is automatically cleared after release; writing 0 has no effect.	0
4	FOVR0	RW1	Receive FIFO_0 overflow flag bit. When there are three messages in FIFO_0 and a new message is received, hardware sets this bit. Software must write 1 to clear it; writing 0 has no effect.	0
3	FULL0	RW1	Receive FIFO_0 full flag bit. When there are three messages in FIFO_0, hardware sets this bit. Software must write 1 to clear it; writing 0 has no effect.	0
2	Reserved	RO	Reserved.	0
[1:0]	FMP0[1:0]	RO	Number of messages in receive FIFO_0.	0

25.8.5 CAN1 Receive FIFO 1 Status Register (CAN1_RFIFO1)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						RFOM1		FOVR1		FULL1		Reserved		FMP1[1:0]	

Bits	Name	Access	Description	Reset
[31:6]	Reserved	RO	Reserved.	0
5	RFOM1	RW1	Software setting this bit to 1 releases the current mailbox message of receive FIFO_1. It is automatically cleared after release; writing 0 has no effect.	0
4	FOVR1	RW1	Receive FIFO_1 overflow flag bit. When there are three messages in FIFO_1 and a new message is received, hardware sets this bit. Software must write 1 to clear it; writing 0 has no effect.	0
3	FULL1	RW1		0

Bits	Name	Access	Description	Reset
			Receive FIFO_1 full flag bit. When there are three messages in FIFO_1, hardware sets this bit. Software must write 1 to clear it; writing 0 has no effect.	
2	Reserved	RF	Reserved.	0
[1:0]	FMP1[1:0]	RO	Number of messages in receive FIFO_1.	0

25.8.6 CAN1 Interrupt Enable Register (CAN1_INTENR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17		16	
Reserved															SLKIE	WKUIE	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
ERRIE	Reserved	LECIE	BOFIE	EPVIE	EWGIE	Reserved	FOVIE1	FFIE1	FMPIE1	FOVIE0	FFIE0	FMPIE0	TMEIE				

Bits	Name	Access	Description	Reset
[31:18]	Reserved	RO	Reserved.	0
17	SLKIE	RW	Sleep interrupt enable bit. 1: An interrupt is generated when entering sleep state; 0: No interrupt is generated when entering sleep state.	0
16	WKUIE	RW	Wakeup interrupt enable bit. 1: An interrupt is generated when the CAN controller is woken up; 0: No interrupt is generated when the CAN controller is woken up.	0
15	ERRIE	RW	Error interrupt enable bit; CAN error interrupt global enable bit. 1: An interrupt is generated when the CAN controller produces an error; 0: No interrupt is generated when the CAN controller produces an error.	0
[14:12]	Reserved	RF	Reserved.	0
11	LECIE	RW	Last error code interrupt enable bit. 1: When an error is detected, hardware updates LEC[2:0], sets ERRI to 1, and triggers an error interrupt; 0: When an error is detected, hardware updates LEC[2:0] but does not update ERRI, and does not trigger an error interrupt.	0
10	BOFIE	RW	Bus-off interrupt enable bit. 1: When entering bus-off state, ERRI is set to 1 and an error interrupt is triggered; 0: When entering bus-off state, ERRI is not updated and no error interrupt is triggered.	0
9	EPVIE	RW	Error passive interrupt enable bit. 1: When entering error passive state, ERRI is set to 1 and an error interrupt is triggered; 0: When entering error passive state, ERRI is not updated and no error interrupt is triggered.	0
8	EWGIE	RW	Error warning interrupt enable bit. 1: When the error count reaches the warning threshold, ERRI is set to 1 and an error interrupt is triggered; 0: When the error count reaches the warning threshold, ERRI is not updated and no error interrupt is triggered.	0

Bits	Name	Access	Description	Reset
7	Reserved	RF	Reserved.	0
6	FOVIE1	RW	Receive FIFO_1 overflow interrupt enable bit. 1: When FIFO_1 overflows, a FIFO_1 interrupt is triggered; 0: When FIFO_1 overflows, no FIFO_1 interrupt is triggered.	0
5	FFIE1	RW	Receive FIFO_1 full interrupt enable bit. 1: When FIFO_1 is full, a FIFO_1 interrupt is triggered; 0: When FIFO_1 is full, no FIFO_1 interrupt is triggered.	0
4	FMPIE1	RW	Receive FIFO_1 message pending interrupt enable bit. 1: When FIFO_1 updates FMP and it is non-zero, a FIFO_1 interrupt is triggered; 0: When FIFO_1 updates FMP and it is non-zero, no FIFO_1 interrupt is triggered.	0
3	FOVIE0	RW	Receive FIFO_0 overflow interrupt enable bit. 1: When FIFO_0 overflows, a FIFO_0 interrupt is triggered; 0: When FIFO_0 overflows, no FIFO_0 interrupt is triggered.	0
2	FFIE0	RW	Receive FIFO_0 full interrupt enable bit. 1: When FIFO_0 is full, a FIFO_0 interrupt is triggered; 0: When FIFO_0 is full, no FIFO_0 interrupt is triggered.	0
1	FMPIE0	RW	Receive FIFO_0 message pending interrupt enable bit. 1: When FIFO_0 updates FMP and it is non-zero, a FIFO_0 interrupt is triggered; 0: When FIFO_0 updates FMP and it is non-zero, no FIFO_0 interrupt is triggered.	0
0	TMEIE	RW	Transmit mailbox empty interrupt. 1: An interrupt is generated when the transmit mailbox is empty; 0: No interrupt is generated when the transmit mailbox is empty.	0

25.8.7 CAN1 Error Status Register (CAN1_ERRSR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REC[7:0]								TEC[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						LEC[2:0]		Reserved		BOFF		EPVF		EWGF	

Bits	Name	Access	Description	Reset
[31:24]	REC[7:0]	RO	Receive error counter. When a CAN receive error occurs, this counter is incremented by 1 or 8 depending on the error condition; after a successful receive, the counter is decremented by 1 or set to 120 (when the error count is greater than 127). When the counter value exceeds 127, the CAN enters the error-passive state.	0
[23:16]	TEC[7:0]	RO	Transmit error counter. When a CAN transmit error occurs, this counter is incremented by 1 or 8 depending on the error condition; after a successful transmit, the counter is decremented by 1 or set to 120 (when the error count is	0

Bits	Name	Access	Description	Reset
			greater than 127). When the counter value exceeds 127, the CAN enters the error-passive state.	
[15:7]	Reserved	RO	Reserved.	0
[6:4]	LEC[2:0]	RW	Last error code. When a CAN bus transmission error is detected, the controller sets this according to the error condition; when messages are correctly received and transmitted, it is set to 000b. 000: No error; 001: Bit stuffing error; 010: Form format error; 011: ACK acknowledgment error; 100: Recessive bit error; 101: Dominant bit error; 110: CRC error; 111: Set by software. Typically, when application software reads an error, it sets the code to 111b to detect the code update.	0
3	Reserved	RO	Reserved.	0
2	BOFF	RO	Bus-off status flag bit. When the CAN controller enters the bus-off state, hardware automatically sets this bit; when exiting the bus-off state, hardware automatically clears it.	0
1	EPVF	RO	Error passive flag bit. When the transmit/receive error counter reaches the error-passive threshold, i.e., greater than 127, hardware sets this bit.	0
0	EWGF	RO	Error warning flag bit. When the transmit/receive error counter reaches the warning threshold, i.e., greater than or equal to 96, hardware sets this bit.	0

25.8.8 CAN1 Bit Timing Register (CAN1_BTMR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
SILM	LBKM	Reserved		SJW[3:0]				TS2[3:0]				TS1[3:0]				
15	14	13	12	11	10		9	8	7	6	5	4	3	2	1	0
BTR_TS1_T				Reserved				BRP[9:0]								

Bits	Name	Access	Description	Reset
31	SILM	RW	Silent mode setting bit. 1: Enter silent mode; 0: Exit silent mode.	0
30	LBKM	RW	Loopback mode setting bit. 1: Enter loopback mode; 0: Exit loopback mode.	0
[29:28]	Reserved	RO	Reserved.	0
[27:24]	SJW[3:0]	RW	Defines the resynchronization jump width setting value. The upper limit of the number of minimum time units that can be extended or	0001b

Bits	Name	Access	Description	Reset
			shortened per bit during resynchronization. The actual value is (SJW[1:0]+1), with a range of 1 to 4 minimum time units.	
[23:20]	TS2[3:0]	RW	Time segment 2 setting value. Defines how many minimum time units time segment 2 occupies; the actual value is (TS2[1:0]+1).	0010b
[19:16]	TS1[3:0]	RW	Time segment 1 setting value. Defines how many minimum time units time segment 1 occupies; the actual value is (TS1[1:0]+1).	0011b
[15:12]	BTR_TS1_T	RW	For traditional CAN, when CLAS_LONG_TS1=0, TS1 is TS[3:0] (4-bit); when CLAS_LONG_TS1=1, TS1 is TS[1:0]+BTR_TS1_T[15:12] (6-bit).	0
[11:10]	Reserved	RO	Reserved.	0
[9:0]	BRP[9:0]	RW	Minimum time unit length setting value: $T_q = (BRP[9:0]+1) \times t_{PCLK}$	0

25.8.9 CAN1 Time-Triggered Control Register (CAN1_TTCTLR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17		16
Reserved														MODE		TIMRST
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
TIMCMV[15:0]																

Bits	Name	Access	Description	Reset
[31:18]	Reserved	RO	Reserved.	0
17	MODE	RW	Time-triggered mode selection bit. 1: Enhanced mode; 0: Default mode.	0
16	TIMRST	WZ	Internal counter reset control bit. Writing 1 resets the internal counter; hardware automatically clears it.	0
[15:0]	TIMCMV[15:0]	RW	Internal counter terminal value.	ffffh

25.8.10 CAN1 Time-Triggered Count Value Register (CAN1_TTCNT)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMCNT[15:0]															

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	TIMCNT[15:0]	RW	Time-triggered count value.	0

25.8.11 CAN1 Bus-Off Recovery Error Counter (CAN1_TERR_CNT)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved									TX_ERR_CNT						
Bits	Name	Access	Description												Reset
[31:9]	Reserved	RO	Reserved.												0
[8:0]	TX_ERR_CNT	RW	Current bus-off recovery error count value. Modifying this count value allows immediate recovery from bus-off.												0

25.8.12 CAN1 Transmit Mailbox Identifier Register (CAN1_TXMIyR) (y=0/1/2)

Offset address: 0x180, 0x190, 0x1A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STID[10:0]/EXID[28:18]											EXID[17:13]				
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXID[12:0]											IDE	RTR	TXRQ		

Bits	Name	Access	Description	Reset
[31:21]	STID[10:0]/EXID[28:18]	RW	Standard identifier or the upper 11 bits of the extended identifier.	x
[20:3]	EXID[17:0]	RW	Lower 18 bits of the extended identifier.	x
2	IDE	RW	Identifier selection flag bit. 1: Extended identifier selected; 0: Standard identifier selected.	x
1	RTR	RW	Remote frame (also called remote control frame) selection flag bit. 1: Current frame is a remote frame; 0: Current frame is a data frame.	x
0	TXRQ	RW	Data transmit request flag bit. When software sets it to 1, it requests to transmit the data in the mailbox; when transmission is complete and the mailbox is empty, hardware clears it.	0

25.8.13 CAN1 Transmit Mailbox Data Length and Timestamp Register (CAN1_TXMDTyR) (y=0/1/2)

Offset address: 0x184, 0x194, 0x1A4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TIME[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							TGT		Reserved				DLC[3:0]		
Bits	Name	Access	Description											Reset	
[31:16]	TIME[15:0]	RW	16-bit timer value for the transmit message SOF moment.											x	
[15:9]	Reserved	RO	Reserved.											0	

Bits	Name	Access	Description	Reset
8	TGT	RW	Message timestamp transmit selection flag bit. This bit is valid when TTCM is set to 1 and the message length is 8. 1: Transmit timestamp, the value is the instant value of TIME[15:0], replacing the last two bytes of the 8-byte message; 0: Do not transmit timestamp.	x
[7:4]	Reserved	RO	Reserved.	0
[3:0]	DLC[3:0]	RW	Data length of the data frame or requested data length of the remote frame. The data length can be set in the range of 0 to 64.	0

25.8.14 CAN1 Transmit Mailbox Low Byte Data Register (CAN1_TXMDLyR) (y=0/1/2)

Offset address: 0x188, 0x198, 0x1A8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA3[7:0]								DATA2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA1[7:0]								DATA0[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	DATA3[7:0]	RW	Content of transmit data byte 3.	x
[23:16]	DATA2[7:0]	RW	Content of transmit data byte 2.	x
[15:8]	DATA1[7:0]	RW	Content of transmit data byte 1.	x
[7:0]	DATA0[7:0]	RW	Content of transmit data byte 0.	x

25.8.15 CAN1 Transmit Mailbox High Byte Data Register (CAN1_TXMDHyR) (y=0/1/2)

Offset address: 0x18C, 0x19C, 0x1AC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA7[7:0]								DATA6[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA5[7:0]								DATA4[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	DATA7[7:0]	RW	Content of transmit data byte 7.	x
[23:16]	DATA6[7:0]	RW	Content of transmit data byte 6.	x
[15:8]	DATA5[7:0]	RW	Content of transmit data byte 5.	x
[7:0]	DATA4[7:0]	RW	Content of transmit data byte 4.	x

25.8.16 CAN1 Receive Mailbox Identifier Register (CAN1_RXMIyR) (y=0/1)

Offset address: 0x1B0, 0x1C0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STID[10:0]/EXID[28:18]												EXID[17:13]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXID[12:0]												IDE	RTR	Reserved	

Bits	Name	Access	Description	Reset
[31:21]	STID[10:0]/EXIDH[28:18]	RO	Standard identifier or the upper 11 bits of the extended identifier.	x
[20:3]	EXIDL[17:0]	RO	Lower 18 bits of the extended identifier.	x
2	IDE	RO	Identifier selection flag bit. 1: Extended identifier selected; 0: Standard identifier selected.	x
1	RTR	RO	Remote frame (also called remote control frame) selection flag bit. 1: Current frame is a remote frame; 0: Current frame is a data frame.	x
0	Reserved	RO	Reserved.	0

25.8.17 CAN1 Receive Mailbox Data Length and Timestamp Register (CAN1_RXMDTyR) (y=0/1)

Offset address: 0x1B4, 0x1C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
TIME[15:0]																
15	14	13	12	11	10	9	8	7	6		5	4	3	2	1	0
FMI[7:0]								Reserved		RES	ESI	BRS	DLC[3:0]			

Bits	Name	Access	Description	Reset
[31:16]	TIME[15:0]	RO	16-bit timer value for the received message SOF moment.	0
[15:8]	FMI[7:0]	RO	Filter number matched by the message.	x
7	Reserved	RO	Reserved.	0
6	RES	RO	RES bit of the currently received frame.	0
5	ESI	RO	ESI bit of the currently received frame.	0
4	BRS	RO	BRS bit of the currently received frame.	0
[3:0]	DLC[3:0]	RO	Received message data length. DLC=0-8: same for standard CAN and CANFD, indicating data frame length 0 to 8; remote frame is 0.	x

25.8.18 CAN1 Receive Mailbox Low Byte Data Register (CAN1_RXMDLyR) (y=0/1)

Offset address: 0x1B8, 0x1C8

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA3[7:0]								DATA2[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA1[7:0]								DATA0[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	DATA3[7:0]	RO	Data byte 3 of the received message.	x
[23:16]	DATA2[7:0]	RO	Data byte 2 of the received message.	x
[15:8]	DATA1[7:0]	RO	Data byte 1 of the received message.	x
[7:0]	DATA0[7:0]	RO	Data byte 0 of the received message.	x

25.8.19 CAN1 Receive Mailbox High Byte Data Register (CAN1_RXMDHyR) (y=0/1)

Offset address: 0x1BC, 0x1CC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DATA7[7:0]								DATA6[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA5[7:0]								DATA4[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	DATA7[7:0]	RO	Data byte 7 of the received message.	x
[23:16]	DATA6[7:0]	RO	Data byte 6 of the received message.	x
[15:8]	DATA5[7:0]	RO	Data byte 5 of the received message.	x
[7:0]	DATA4[7:0]	RO	Data byte 4 of the received message.	x

25.8.20 CAN1 Filter Master Control Register (CAN1_FCTLR)

Offset address: 0x200

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														FINIT	

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0x2A1C
[15:1]	Reserved	RO	Reserved.	0
0	FINIT	RW	Filter initialization mode enable flag bit. 1: Filter banks are in initialization mode; 0: Filter banks are in normal mode.	1

25.8.21 CAN1 Filter Mode Register (CAN1_FMCFR)

Offset address: 0x204

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FBM13	FBM12	FBM11	FBM10	FBM9	FBM8	FBM7	FBM6	FBM5	FBM4	FBM3	FBM2	FBM1	FBM0	

Bits	Name	Access	Description	Reset
[31:14]	Reserved	RO	Reserved.	0
[13:0]	FBMx	RW	Filter bank x operating mode control bit; can only be written when FINIT is 1. 1: Filter bank x registers are in identifier list mode; 0: Filter bank x registers are in mask mode.	0

25.8.22 CAN1 Filter Bit Width Register (CAN1_FSCFGR)

Offset address: 0x20C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FSC13	FSC12	FSC11	FSC10	FSC9	FSC8	FSC7	FSC6	FSC5	FSC4	FSC3	FSC2	FSC1	FSC0	

Bits	Name	Access	Description	Reset
[31:14]	Reserved	RO	Reserved.	0
[13:0]	FSCx	RW	Filter bank x bit width control bit; can only be written when FINIT is 1. 1: Filter bank x registers are a single 32-bit; 0: Filter bank x registers are 2 × 16-bit.	0

25.8.23 CAN1 Filter FIFO Association Register (CAN1_FAFIFOR)

Offset address: 0x214

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FFA13	FFA12	FFA11	FFA10	FFA9	FFA8	FFA7	FFA6	FFA5	FFA4	FFA3	FFA2	FFA1	FFA0	

Bits	Name	Access	Description	Reset
[31:14]	Reserved	RO	Reserved.	0
[13:0]	FFAx	RW	Filter bank x associated FIFO control bit; can only be written when FINIT is 1. 1: Filter bank x is associated with FIFO_1; 0: Filter bank x is associated with FIFO_0.	0

25.8.24 CAN1 Filter Activation Register (CAN1_FWR)

Offset address: 0x21C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	FACT13	FACT12	FACT11	FACT10	FACT9	FACT8	FACT7	FACT6	FACT5	FACT4	FACT3	FACT2	FACT1	FACT0	

Bits	Name	Access	Description	Reset
[31:14]	Reserved	RO	Reserved.	0
[13:0]	FACTx	RW	Filter bank x activation control bit. 1: Filter bank x is activated; 0: Filter bank x is disabled.	0

25.8.25 CAN1 Filter Bank Filter Registers (CAN1_FiRx) (i=0-13, x=1/2)

Offset address: 0x240-0x31C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FB31	FB30	FB29	FB28	FB27	FB26	FB25	FB24	FB23	FB22	FB21	FB20	FB19	FB18	FB17	FB16

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FB15	FB14	FB13	FB12	FB11	FB10	FB9	FB8	FB7	FB6	FB5	FB4	FB3	FB2	FB1	FB0

Bits	Name	Access	Description	Reset
[31:0]	FB	RW	Flag bits of the filter bank registers; can only be written when FINIT is 1. Identifier mode: 1: The corresponding bit expects a recessive bit level; 0: The corresponding bit expects a dominant bit level. Mask mode: 1: Must match the corresponding identifier register bit; 0: Does not need to match the corresponding identifier register bit.	0

Chapter 26: Flexible Static Memory Controller (FSMC)

The Flexible Static Memory Controller (FSMC) supports multiple static memory types and a rich set of memory operation methods. It can be used to expand different types of large-capacity static memories according to system application requirements.

26.1 Main Features

- Supports operation of SRAM, ROM, NOR FLASH, and PSRAM
- Supports operation of NAND FLASH with built-in hardware ECC, capable of detecting up to 8 KByte of data
- Supports operation of synchronous devices, such as PSRAM
- Supports 8-bit or 16-bit data bus width
- Timing signals are software-programmable

26.2 Functional Description

26.2.1 Module Structure

The FSMC mainly consists of the HB bus interface, NOR memory controller, NAND memory controller, and external device interface.

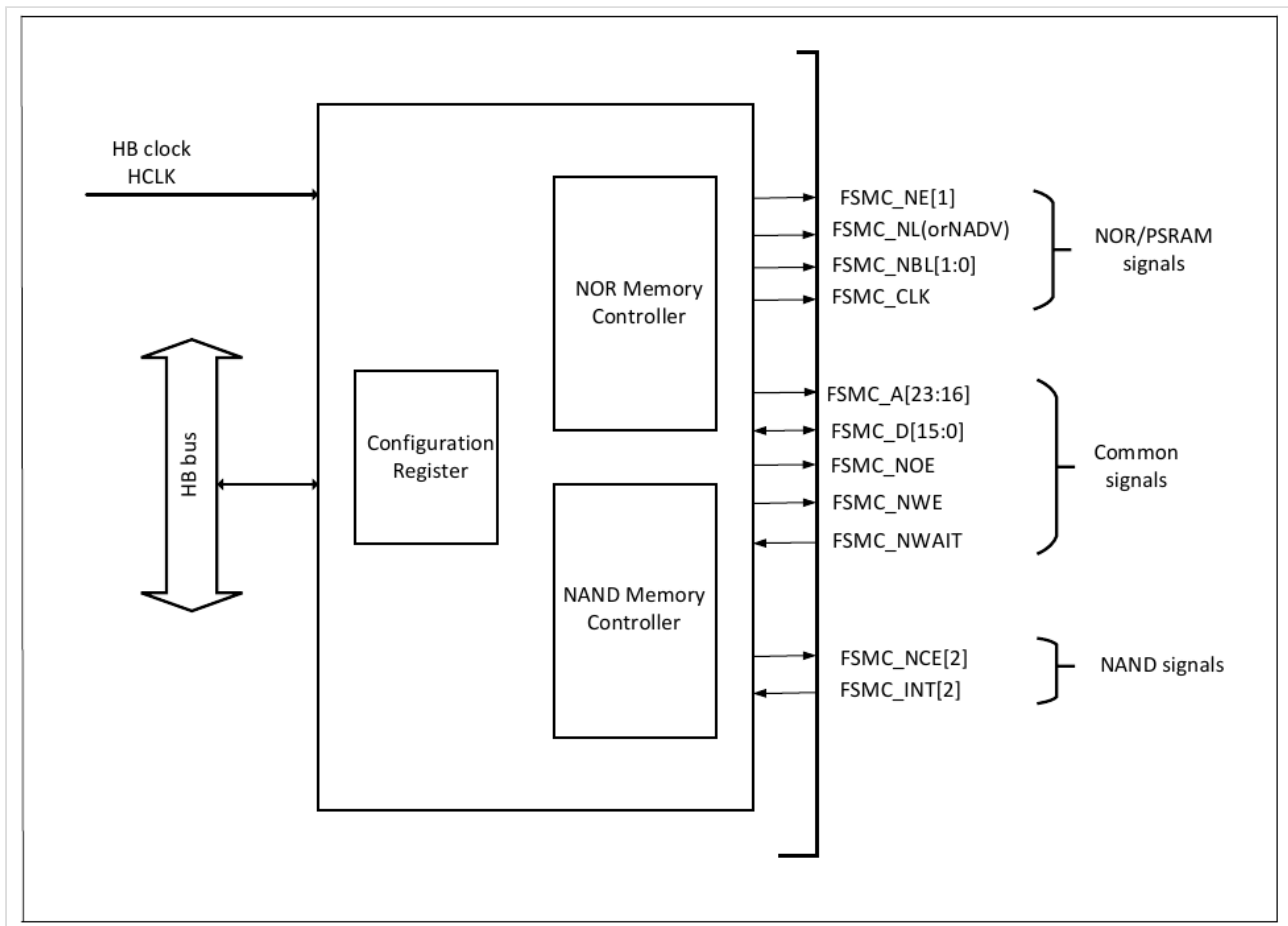


Figure 26-1: FSMC module block diagram

26.2.2 External Device Address Mapping

The FSMC divides memory blocks into two fixed-size 16 MByte memory blocks according to the address lines, as shown in Figure 26-2.

图 26-2 FSMC 存储块

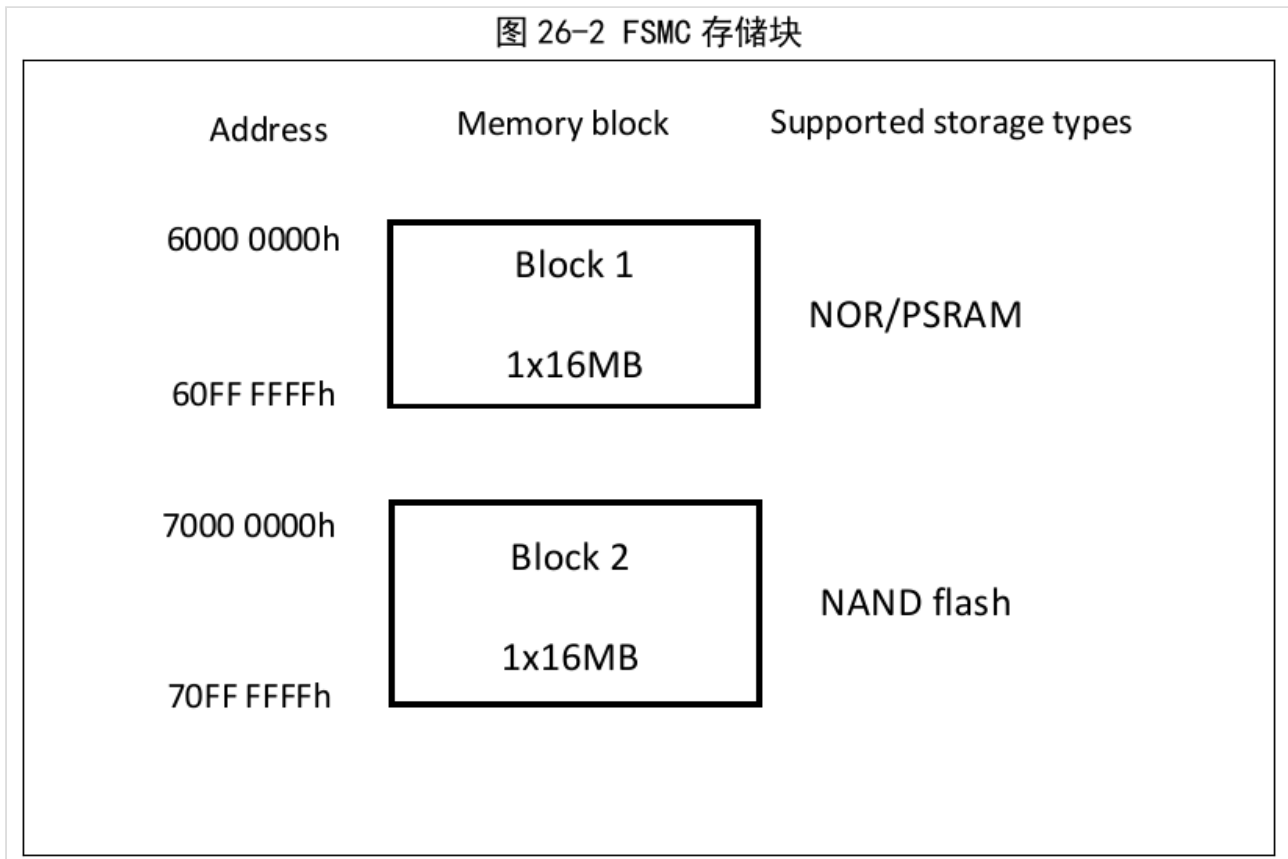


Figure 26-2: FSMC memory blocks

Figure 26-2: FSMC memory block mapping

Address	Memory block	Supported storage types
6000 0000h – 60FF FFFFh	Block 1 (1×16MB)	NOR/PSRAM
7000 0000h – 70FF FFFFh	Block 2 (1×16MB)	NAND flash

26.2.2.1 NOR/PSRAM Address Mapping

Table 26-1: External memory address

Data width	Address lines connected to memory	Maximum accessible memory space (bit)
8-bit	HADDR[23:0] correspondingly connected to FSMC_A[23:0]	16MB×8=128Mbit
16-bit	HADDR[23:1] correspondingly connected to FSMC_A[22:0], HADDR[0] not connected	16MB/2×16=128Mbit

Note: 100-pin chips support FSMC and only support address/data multiplexed mode.

NOR FLASH and PSRAM support unaligned access. For asynchronous mode, each operation requires an exact address; for synchronous mode, the address signal is issued only once, after which data is transferred in bulk sequentially via CLK. For NOR FLASH that supports unaligned bulk access, the unaligned access mode of the memory can be set to the same mode as HB. If this cannot be set, disable the unaligned access mode and split the unaligned access request into two consecutive access operations.

26.2.2.2 NAND Address Mapping

NAND FLASH mapping and timing registers are shown in Table 26-2.

Table 26-2: Memory mapping and timing registers

Start address	End address	FSMC memory block	Memory space	Timing register
0x78000000	0x78FFFFFF	Block 2 — NAND FLASH	Attribute	FSMC_PATT2 (0x6C)
0x70000000	0x70FFFFFF		Common	FSMC_PMEM2 (0x68)

The common and attribute spaces can be divided within the low 256 KB into the address area (second 128 KB region), command area (second 64 KB region), and data area (first 64 KB region), as shown in Table 26-3.

The specific process for software to access NAND FLASH by operating these three regions:

- Send read/write and other operation commands to NAND FLASH: software can send commands by operating any address in the command area;
- Send the address to be operated to NAND FLASH: software can send commands by operating any address in the address area;
- Read/write data from/to NAND FLASH: software can write or read data by operating any address in the data area.

Table 26-3: NAND memory block selection

Region name	HADDR[17:16]	Address range
Address area	1X	0x020000–0x03FFFF
Command area	01	0x010000–0x01FFFF
Data area	00	0x000000–0x00FFFF

26.2.3 NOR/PSRAM Controller

The FSMC supports 8-bit, 16-bit, and 32-bit asynchronous operations for SRAM and ROM, asynchronous mode and burst mode operations for PSRAM, and asynchronous mode and burst mode operations for NOR FLASH. All controller output signals change on the rising edge of the internal clock HCLK. For synchronous write mode (PSRAM), the output data changes on the falling edge of the memory clock (CLK). Refer to the synchronous and asynchronous transfer timing diagrams for details. The memory read/write parameters are software-configurable, as shown in Table 26-4.

Table 26-4: Software-controllable NOR/PSRAM read/write parameters

Parameter	Read/Write mode	Parameter value range
Address setup time	Asynchronous	$1 \leq T \leq 16$ (HB HCLK)
Address hold time	Asynchronous	$1 \leq T \leq 16$ (HB HCLK)
Data setup time	Asynchronous	$2 \leq T \leq 256$ (HB HCLK)
Bus turnaround time	Asynchronous or synchronous read	$1 \leq T \leq 16$ (HB HCLK)
Clock division factor	Synchronous	$2 \leq T \leq 16$ (HB HCLK)
Data generation time	Synchronous	$2 \leq T \leq 17$ (Memory CLK)

26.2.3.1 External Memory Multiplexed Interface Signals

NOR FLASH and PSRAM interfaces are shown in Table 26-5 and Table 26-6.

Table 26-5: Multiplexed NOR FLASH interface

FSMC pin	Direction	Description
CLK	Output	Clock line (only for synchronous burst mode)
A[23:16]	Output	Address lines
AD[15:0]	Input/Output	16-bit address/data lines (multiplexed)
NE[1]	Output	Chip select line
NOE	Output	Output enable
NWE	Output	Write enable
NL (NADV)	Output	Latch enable
NWAIT	Input	Wait signal line (only for NOR FLASH)

Table 26-6: Multiplexed PSRAM interface

FSMC pin	Direction	Description
CLK	Output	Clock line (only for synchronous burst mode)
A[23:16]	Output	Address lines
AD[15:0]	Input/Output	16-bit address/data lines (multiplexed)
NE[1]	Output	Chip select line
NOE	Output	Output enable
NWE	Output	Write enable
NL (NADV)	Output	Latch enable
NWAIT	Input	Wait signal line
NBL[1]	Output	High byte enable (NUB)
NBL[0]	Output	Low byte enable (NLB)

Note: Signals with the "N" prefix indicate active-low signals.

26.2.3.2 Supported Memories and Operation Modes

Supported memories and operation modes are shown in Table 26-7.

Table 26-7: Supported memories and operation modes

Memory	Mode	HB data width	Memory width	Description
NOR FLASH	Async read	8	16	
	Async read	16	16	
	Async write	16	16	
	Async read	32	16	Split into 2 FSMC accesses
	Async write	32	16	Split into 2 FSMC accesses
	Sync read	16	16	

	Sync read	32	16	Split into 2 FSMC accesses
PSRAM	Async read	8	16	Uses byte signals NBL[1:0]
	Async write	8	16	Uses byte signals NBL[1:0]
	Async read	16	16	
	Async write	16	16	
	Async read	32	16	Split into 2 FSMC accesses
	Async write	32	16	Split into 2 FSMC accesses
	Sync read	16	16	
	Sync read	32	16	Split into 2 FSMC accesses
	Sync write	8	16	Uses byte signals NBL[1:0]
	Sync write	16/32	16	
SRAM and ROM	Async read	8/16/32	8/16	Uses byte signals NBL[1:0]
	Async write	8/16/32	8/16	Uses byte signals NBL[1:0]

26.2.3.3 NOR/PSRAM Asynchronous Transfer Address/Data Multiplexed Timing Diagrams

For asynchronous static memory (NOR FLASH and PSRAM) operations, note the following:

- When extended mode is enabled, modes A, B, C, and D can be mixed for read and write operations to the memory.

图 26-3 模式 1 读操作（复用）

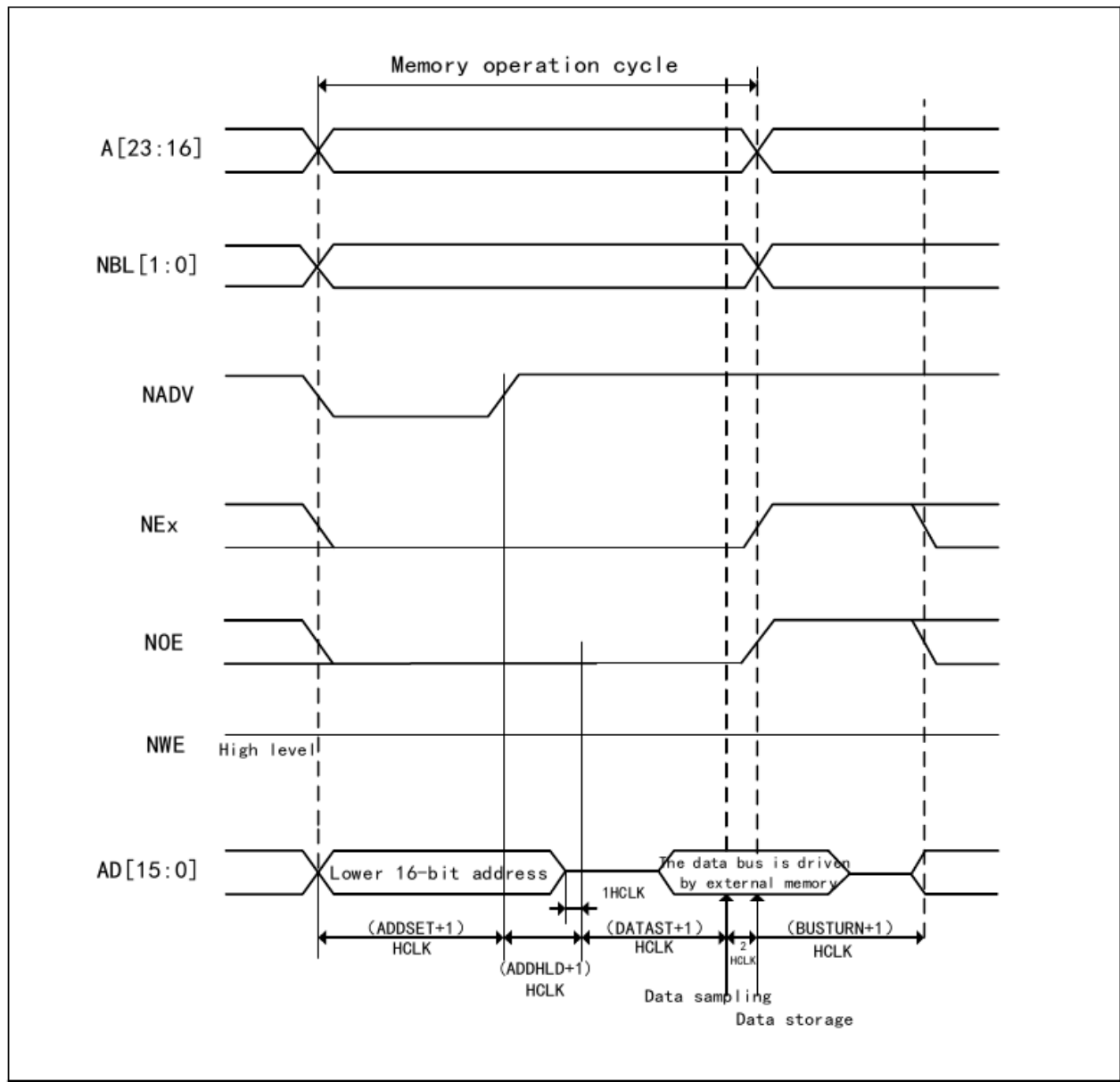


Figure 26-3: Mode 1 read operation (multiplexed)

图 26-3 模式 1 读操作（复用）

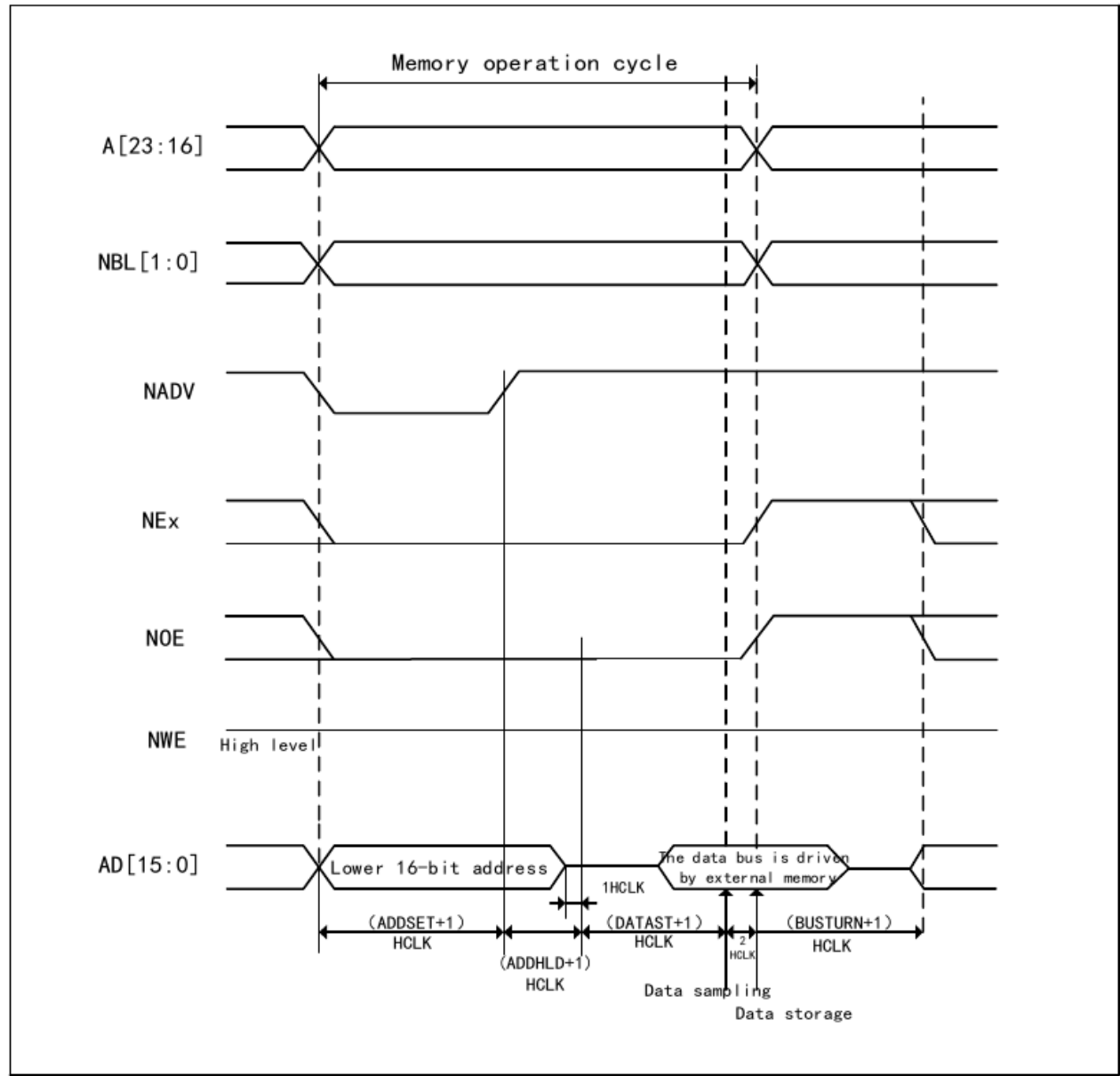


Figure 26-4: Mode 1 write operation (multiplexed)

Note: The last HCLK cycle of the write operation guarantees the address and data hold time after the NWE rising edge. Because this HCLK cycle exists, DATAST must be greater than 0 (DATAST > 0).

Table 26-8: Mode 1 FSMC_BCR1 bit fields

Bitx	Name	Configuration value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCAWAIT	1 if the memory supports this feature, otherwise 0
14	EXTMOD	0x0
13	WAITEN	0x0
12	WREN	Set as needed

11	WAITCFG	This bit has no effect
10	Reserved	0x0
9	WAITPOL	Meaningful when bit15 is 1
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	This bit has no effect
[5:4]	MWID	Set as needed
[3:2]	MTYP	Set as needed, excluding 0x2 (NOR FLASH)
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-9: Mode 1 FSMC_BTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	This bit has no effect
[27:24]	DATLAT	This bit has no effect
[23:20]	CLKDIV	This bit has no effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Write operation is (DATAST+1 HCLK), read operation is (DATAST+3 HCLK). Minimum value is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. (ADDSET+1 HCLK).

图 26-5 模式 A 读操作（复用）

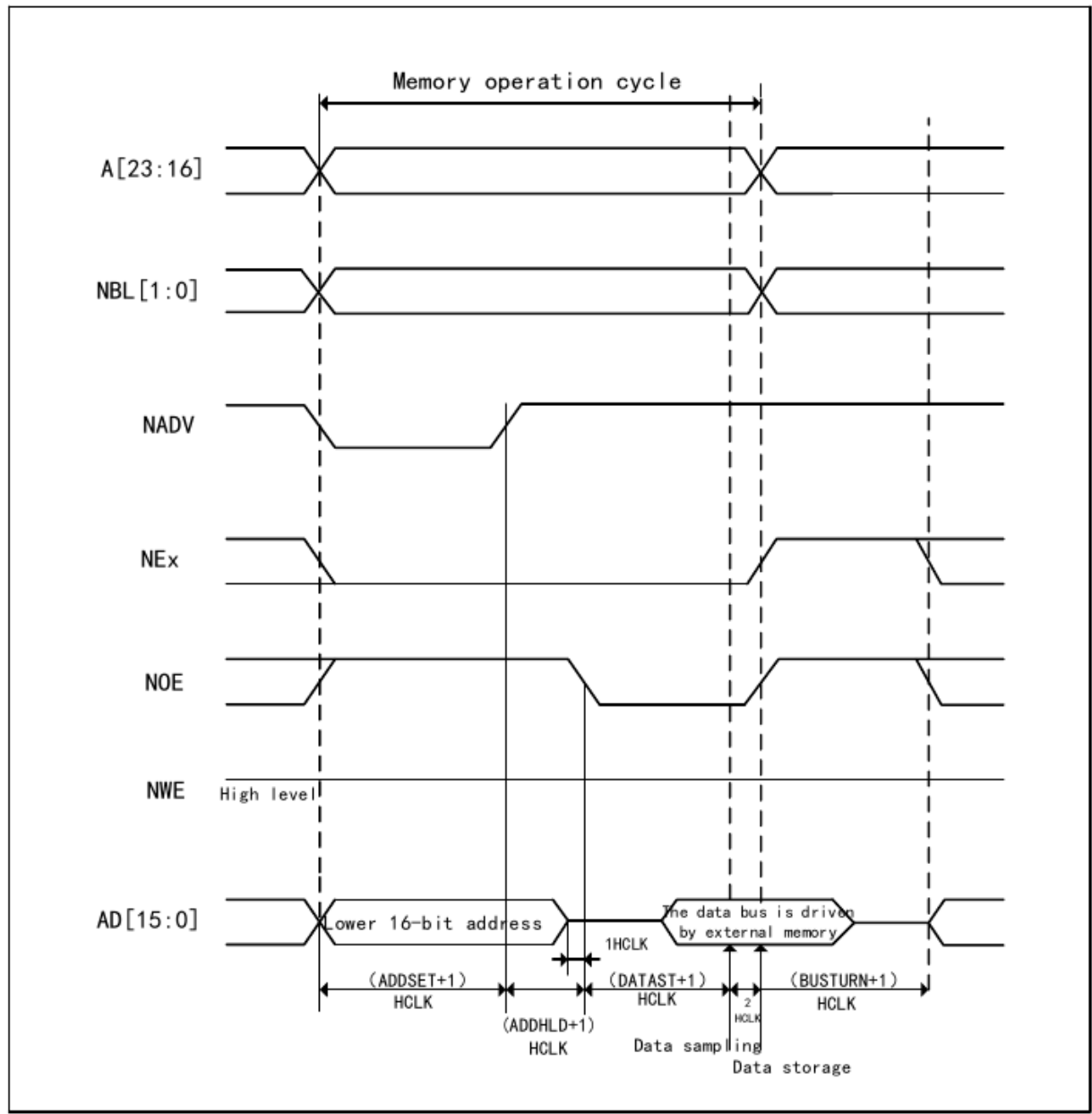


Figure 26-5: Mode A read operation (multiplexed)

图 26-5 模式 A 读操作（复用）

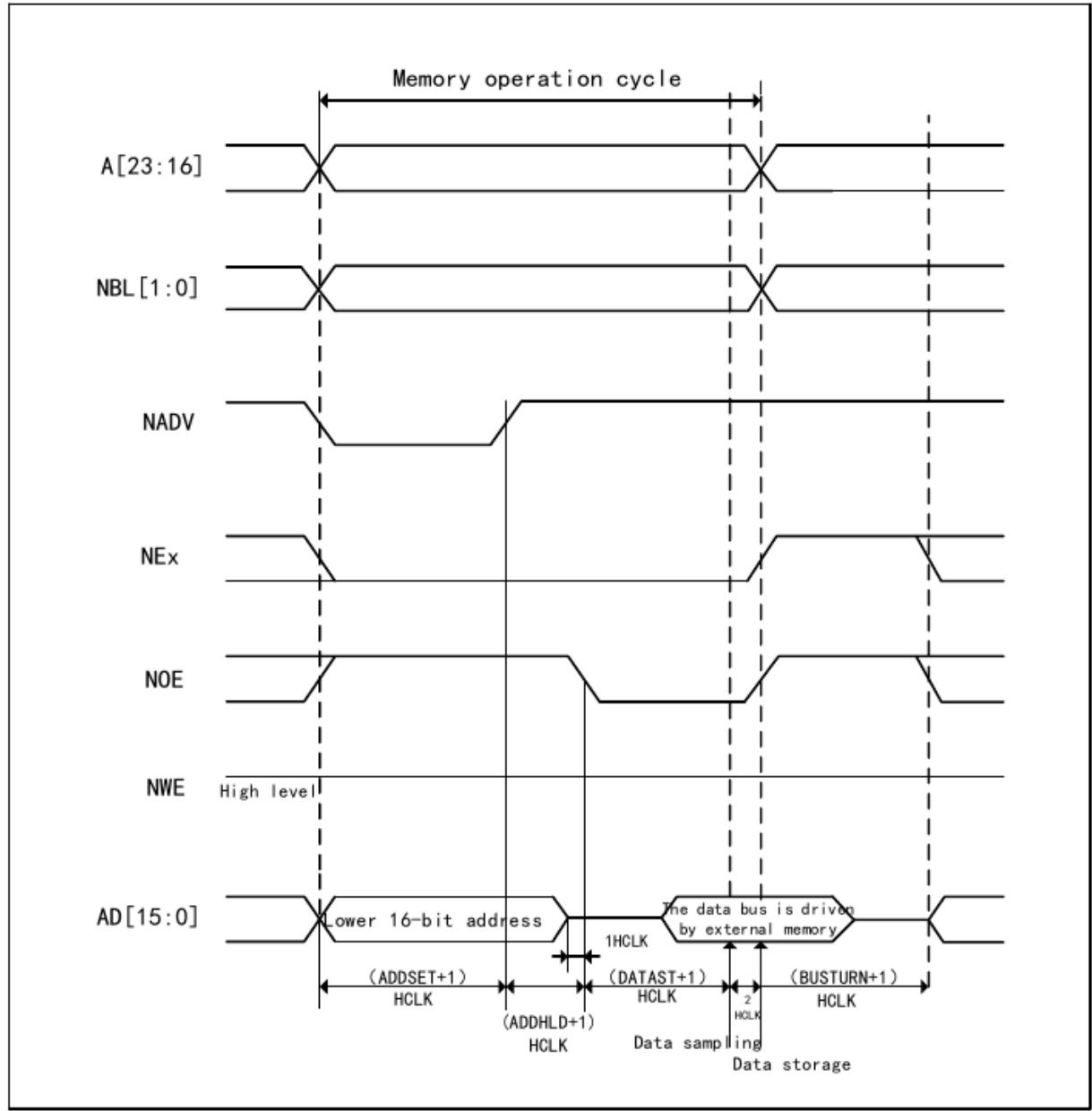


Figure 26-6: Mode A write operation (multiplexed)

Table 26-10: Mode A FSMC_BCR1 bit fields

Bitx	Name	Configuration value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCAWAIT	1 if the memory supports this feature, otherwise 0
14	EXTMOD	0x1
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	This bit has no effect

10	Reserved	0x0
9	WAITPOL	Meaningful when bit15 is 1
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	This bit has no effect
[5:4]	MWID	Set as needed
[3:2]	MTYP	Set as needed, excluding 0x2 (NOR FLASH)
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-11: Mode A FSMC_BTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	This bit has no effect
[23:20]	CLKDIV	This bit has no effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Read operation is (DATAST+3 HCLK). Minimum value is 1.
[7:4]	ADDHLD	This bit has no effect
[3:0]	ADDSET	Address setup time. Read operation is (ADDSET+1 HCLK).

Table 26-12: Mode A FSMC_BWTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	This bit has no effect
[23:20]	CLKDIV	This bit has no effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Write operation is (DATAST+1 HCLK). Minimum value is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. Write operation is (ADDSET+1 HCLK).

图 26-7 模式 B 读操作（复用）

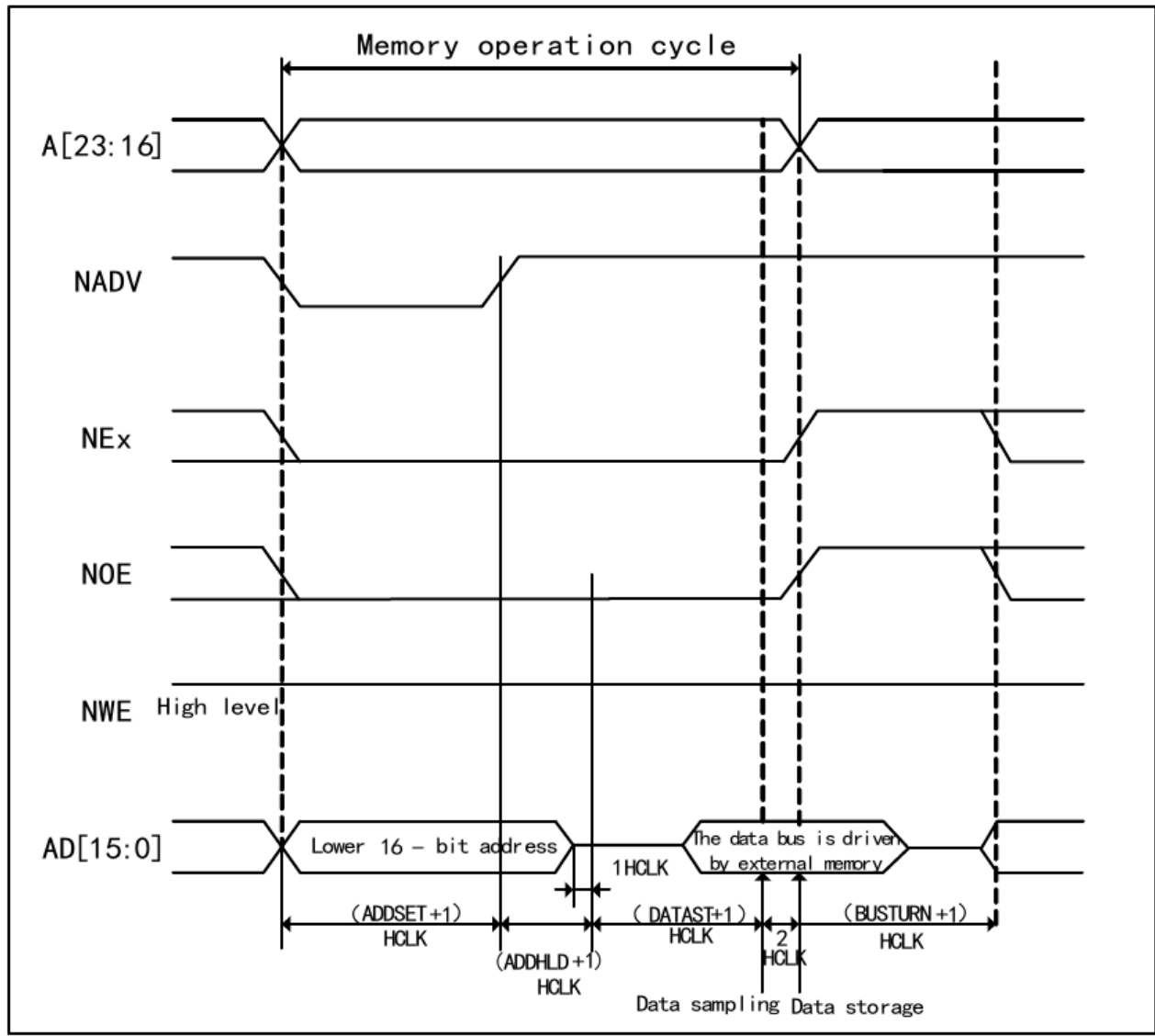


Figure 26-7: Mode B read operation (multiplexed)

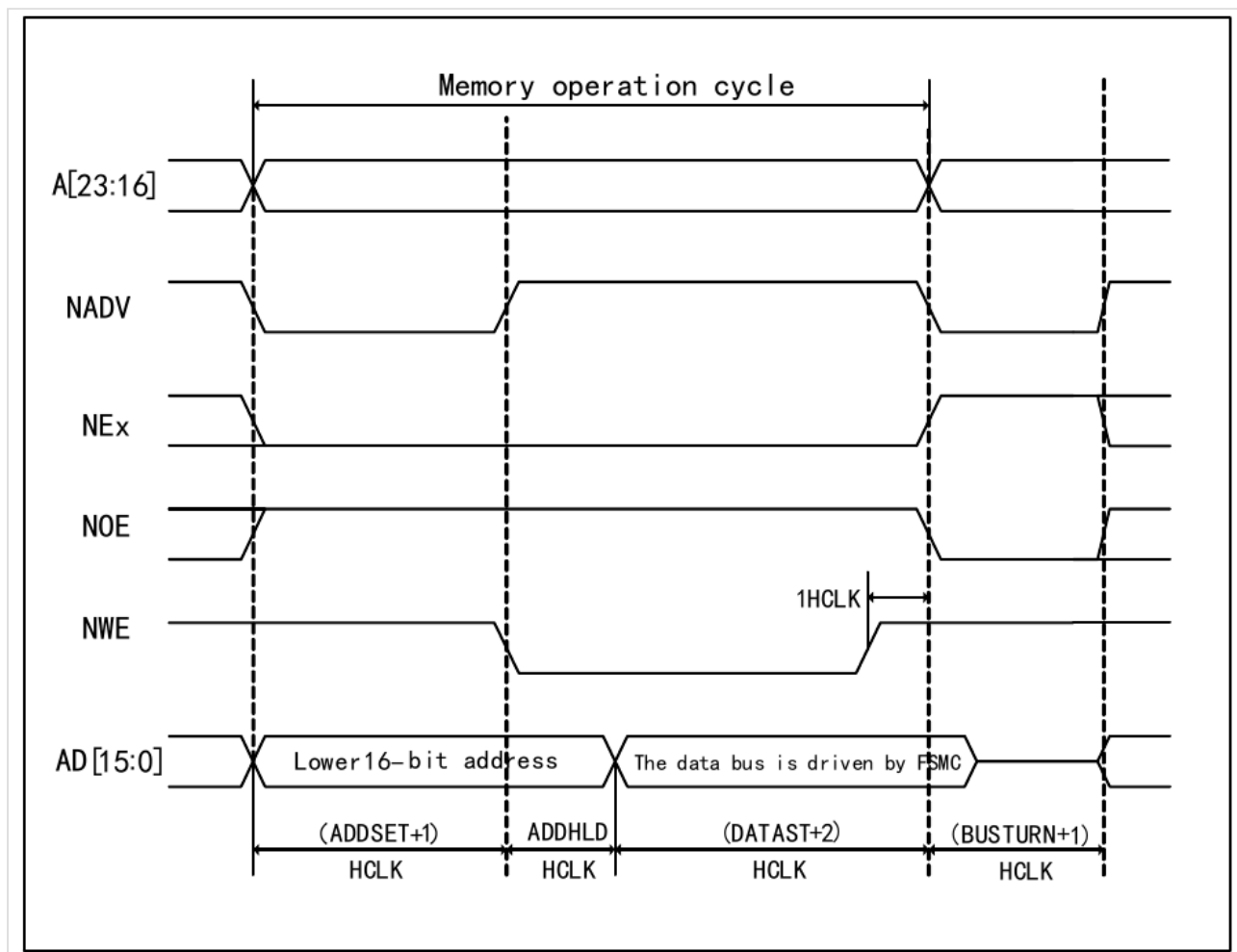


Figure 26-8: Mode B write operation (multiplexed)

Table 26-13: Mode B FSMC_BCR1 bit fields

Bitx	Name	Configuration value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCAWAIT	1 if the memory supports this feature, otherwise 0
14	EXTMOD	0x1
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	This bit has no effect
10	Reserved	0x0
9	WAITPOL	Meaningful when bit15 is 1.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
[5:4]	MWID	Set as needed
[3:2]	MTYP	0x2 (NOR FLASH)

1	MUXEN	0x1
0	MBKEN	0x1

Table 26-14: Mode B FSMC_BTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x1
[27:24]	DATLAT	This bit has no effect
[23:20]	CLKDIV	This bit has no effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Read operation is (DATAST+3 HCLK). Minimum value is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. Read operation is (ADDSET+1 HCLK).

Table 26-15: Mode B FSMC_BWTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x1
[27:24]	DATLAT	This bit has no effect
[23:20]	CLKDIV	This bit has no effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Write operation is (DATAST+1 HCLK). Minimum value is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. Write operation is (ADDSET+1 HCLK).

图 26-9 模式 C 读操作（复用）

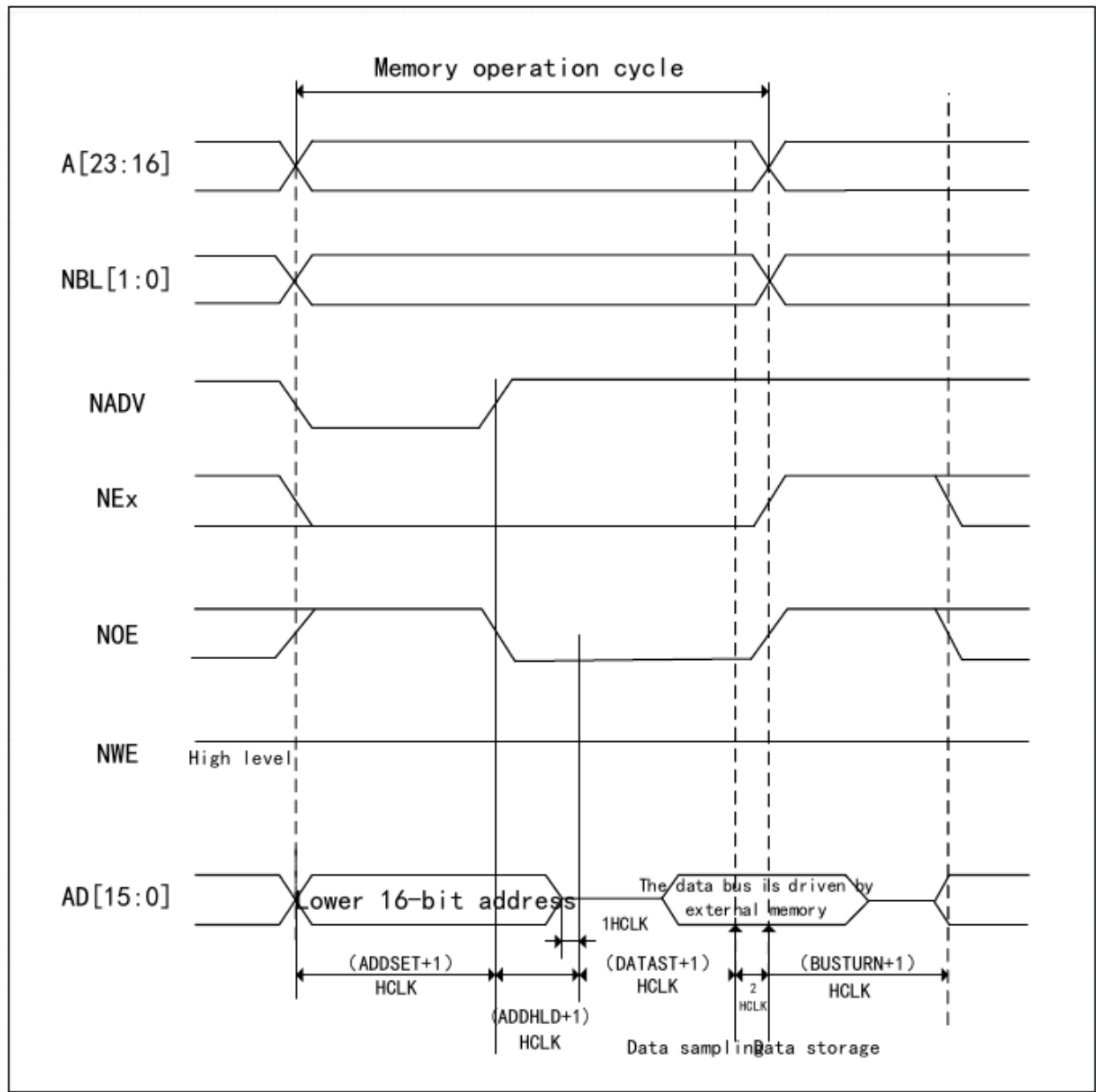


Figure 26-9: Mode C read operation (multiplexed)

图 26-9 模式 C 读操作（复用）

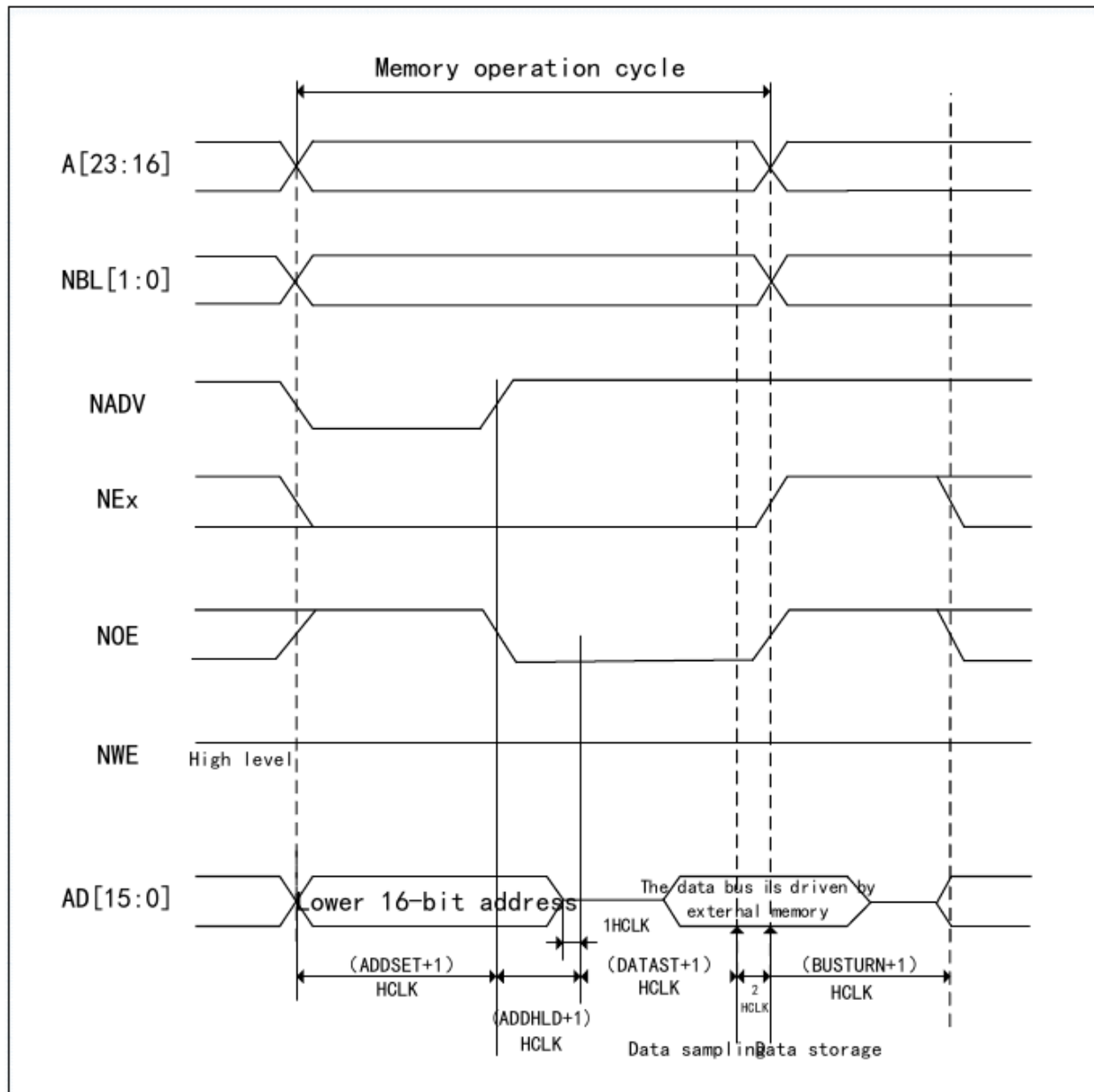


Figure 26-10: Mode C write operation (multiplexed)

Table 26-16: Mode C FSMC_BCR1 bit fields

Bitx	Name	Configuration value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCAWAIT	1 if the memory supports this feature, otherwise 0
14	EXTMOD	0x1
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	This bit has no effect

10	Reserved	0x0
9	WAITPOL	Meaningful when bit15 is 1.
8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	0x1
[5:4]	MWID	Set as needed
[3:2]	MTYP	0x2 (NOR FLASH)
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-17: Mode C FSMC_BTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x2
[27:24]	DATLAT	This bit has no effect
[23:20]	CLKDIV	This bit has no effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Read operation is (DATAST+3 HCLK). Minimum value is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. Read operation is (ADDSET+1 HCLK).

Table 26-18: Mode C FSMC_BWTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x2
[27:24]	DATLAT	This bit has no effect
[23:20]	CLKDIV	This bit has no effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Write operation is (DATAST+1 HCLK). Minimum value is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK)
[3:0]	ADDSET	Address setup time. Write operation is (ADDSET+1 HCLK).

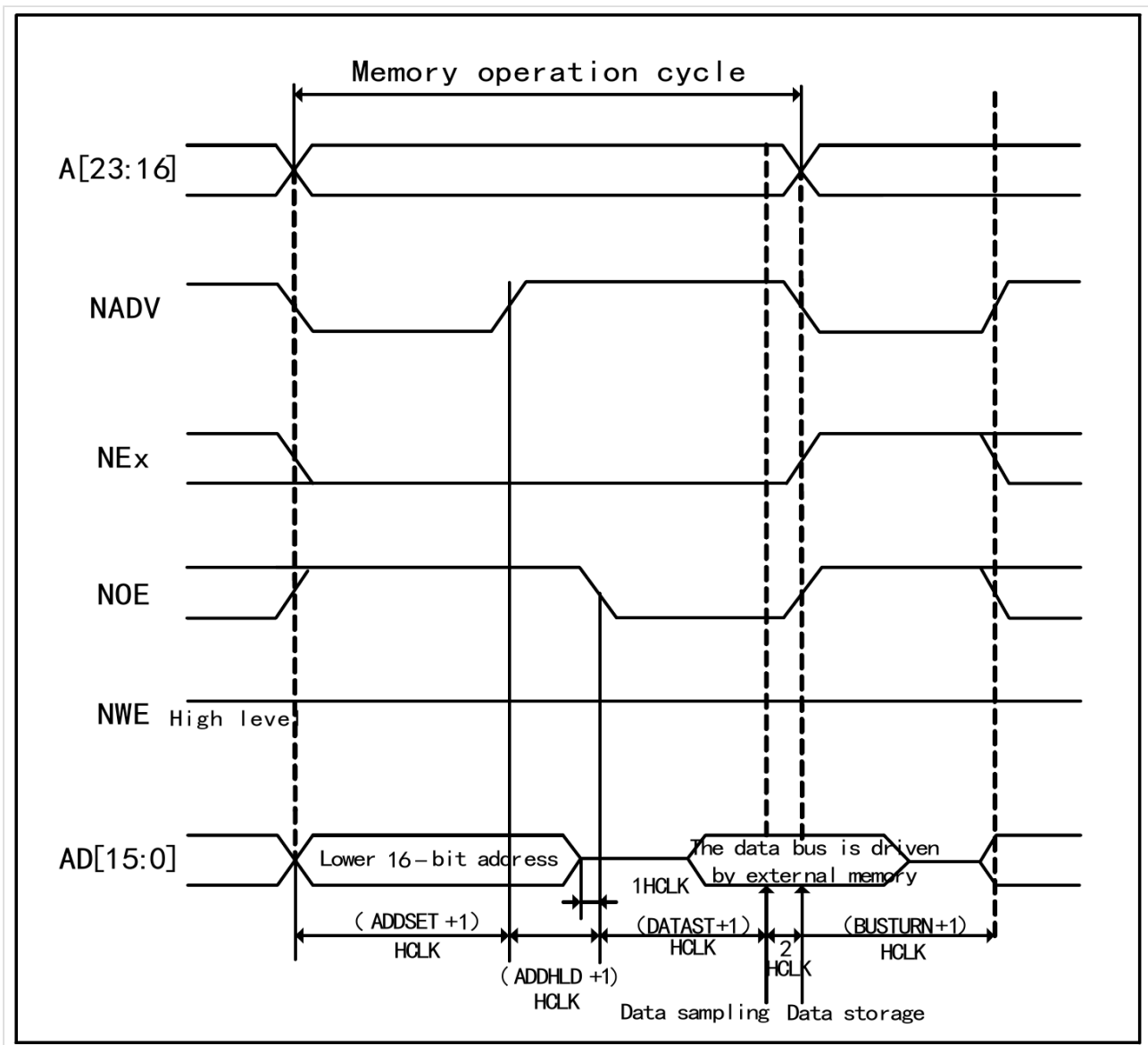


Figure 26-11: Mode D read operation (multiplexed)

图 26-11 模式 D 读操作（复用）

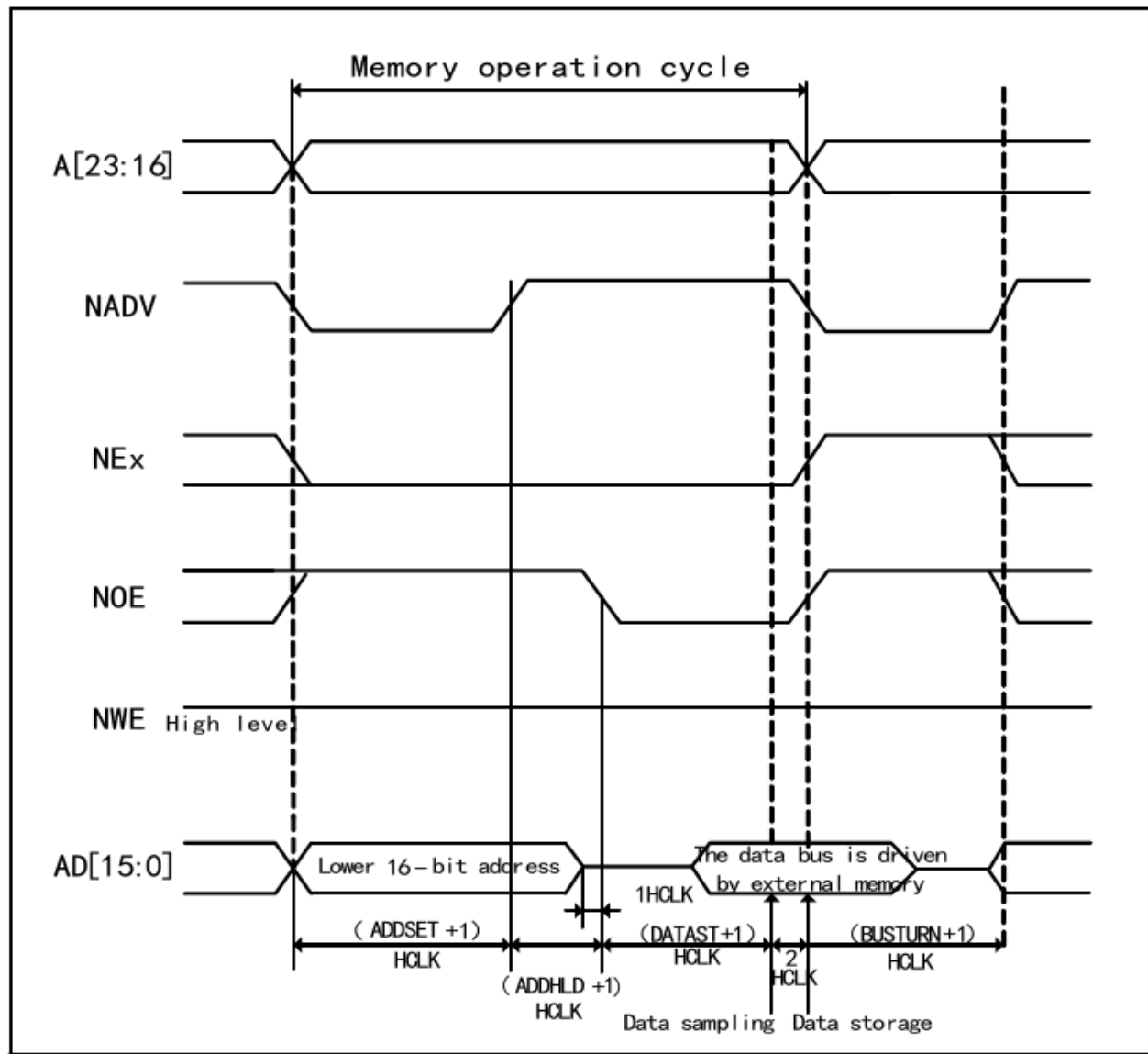


Figure 26-12: Mode D write operation (multiplexed)

Table 26-19: Mode D FSMC_BCR1 bit fields

Bitx	Name	Configuration value
[31:20]	Reserved	0x000
19	CBURSTRW	0x0
[18:16]	Reserved	0x0
15	ASYNCAWAIT	1 if the memory supports this feature, otherwise 0
14	EXTMOD	0x1
13	WAITEN	0x0
12	WREN	Set as needed
11	WAITCFG	This bit has no effect
10	Reserved	0x0
9	WAITPOL	Meaningful when bit15 is 1.

8	BURSTEN	0x0
7	Reserved	0x1
6	FACCEN	Set as needed
[5:4]	MWID	Set as needed
[3:2]	MTYP	Set as needed
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-20: Mode D FSMC_BTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x3
[27:24]	DATLAT	This bit has no effect
[23:20]	CLKDIV	This bit has no effect
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Read operation is (DATAST+3 HCLK). Minimum value is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK).
[3:0]	ADDSET	Address setup time. Read operation is (ADDSET+1 HCLK).

Table 26-21: Mode D FSMC_BWTR1 bit fields

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x3
[27:24]	DATLAT	0x0
[23:20]	CLKDIV	0x0
[19:16]	BUSTURN	0x0
[15:8]	DATAST	Data setup time. Write operation is (DATAST+1 HCLK). Minimum value is 1.
[7:4]	ADDHLD	Address hold time. (ADDHLD+1 HCLK).
[3:0]	ADDSET	Address setup time. Write operation is (ADDSET+1 HCLK).

26.2.3.4 NOR/PSRAM Synchronous Transfer Address/Data Multiplexed Timing Diagrams

Note the data delay and NOR FLASH latency: the DATLAT value must be consistent with the definition in the NOR FLASH configuration register. The clock cycles when the NADV signal is low are not counted in the delay parameter. The FSMC DATLAT parameter can be DATLAT+2 or DATLAT+3. For special memories that generate an NWAIT signal during the data hold time, DATLAT must be set to the minimum value. For other memories that do not generate an NWAIT signal during the data hold time, the data hold time of the FSMC and the memory must be set consistently.

For single bulk transfers, note that the memory is configured in synchronous burst mode. If 16-bit data is transferred, the FSMC performs one burst transfer of length 1. If 32-bit data is transferred, the FSMC splits it into two 16-bit transfers and performs one burst transfer of length 2.

During NOR FLASH synchronous burst mode access, after the hold time (DATLAT+1 CLK), if the NWAIT signal is detected as low, the FSMC must insert wait cycles before the NWAIT goes high. When the NWAIT becomes high, the FSMC considers the data valid. During the wait state insertion controlled by the NWAIT signal, the controller continuously sends clock pulses to the memory, maintains the chip select signal and output enable signal, and ignores invalid data signals. In burst transfer mode, the NWAIT signal of the NOR FLASH can select two timing configurations by configuring the WAITCFG bit.

图 26-13 同步总线复用读操作（复用）

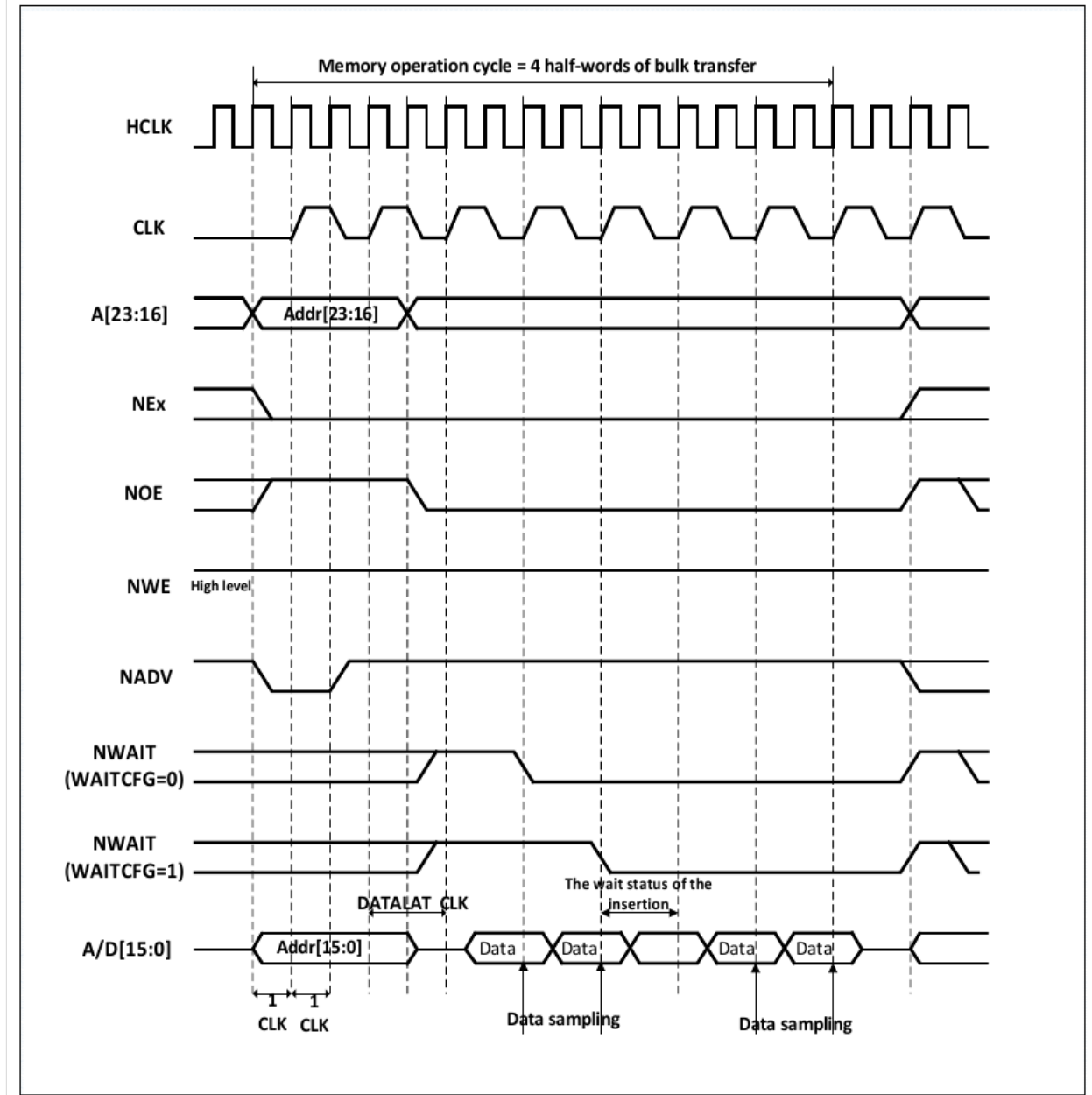


Figure 26-13: Synchronous bus multiplexed read operation (multiplexed)

Table 26-22: FSMC_BCR1 bit fields (synchronous read mode)

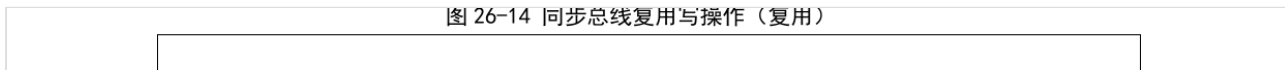
Bitx	Name	Configuration value
[31:20]	Reserved	0x000
19	CBURSTRW	This bit has no effect
[18:16]	Reserved	0x0

15	ASYNCWAIT	0x0
14	EXTMOD	0x0
13	WAITEN	0x0
12	WREN	This bit has no effect
11	WAITCFG	Set as needed
10	Reserved	0x0
9	WAITPOL	Set as needed
8	BURSTEN	0x1
7	Reserved	0x1
6	FACCEN	Set as needed
[5:4]	MWID	Set as needed
[3:2]	MTYP	0x1 or 0x2
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-23: FSMC_BTR1 bit fields (synchronous read mode)

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	Data hold time
[23:20]	CLKDIV	0x0 – Reserved; 0x1 - CLK=2 HCLK
[19:16]	BUSTURN	This bit has no effect
[15:8]	DATAST	This bit has no effect
[7:4]	ADDHLD	This bit has no effect
[3:0]	ADDSET	This bit has no effect

图 26-14 同步总线复用与操作（复用）

**Figure 26-14: Synchronous bus multiplexed write operation (multiplexed)****Table 26-24: FSMC_BCR1 bit fields (synchronous write mode)**

Bitx	Name	Configuration value
[31:20]	Reserved	0x000
19	CBURSTRW	0x1
[18:16]	Reserved	0x0
15	ASYNCWAIT	0x0
14	EXTMOD	0x0
13	WAITEN	1 if the memory supports this feature, otherwise 0
12	WREN	0x1
11	WAITCFG	0x0

10	Reserved	0x0
9	WAITPOL	Set as needed
8	BURSTEN	This bit has no effect
7	Reserved	0x1
6	FACCEN	Set as needed
[5:4]	MWID	Set as needed
[3:2]	MTYP	0x1
1	MUXEN	0x1
0	MBKEN	0x1

Table 26-25: FSMC_BTR1 bit fields (synchronous write mode)

Bitx	Name	Configuration value
[31:30]	Reserved	0x0
[29:28]	ACCMOD	0x0
[27:24]	DATLAT	Data hold time
[23:20]	CLKDIV	0x0 – Reserved; 0x1 - CLK=2 HCLK
[19:16]	BUSTURN	This bit has no effect
[15:8]	DATAST	This bit has no effect
[7:4]	ADDHLD	This bit has no effect
[3:0]	ADDSET	This bit has no effect

26.2.4 NAND FLASH Controller

The FSMC supports 8-bit and 16-bit NAND FLASH operations. The FSMC can generate multiple address cycles as needed, and theoretically there is no limit to the NAND FLASH capacity that the FSMC can address. The NAND FLASH read/write parameters are software-configurable, as shown in Table 26-26.

Table 26-26: Software-controllable NAND read/write parameters

Parameter	Function	Read/ Write	Parameter value range
Memory setup time	Address setup time before issuing command	Read/ Write	$1 \leq T \leq 255$ (HB HCLK)
Memory wait time	Minimum duration of command	Read/ Write	$1 \leq T \leq 255$ (HB HCLK)
Memory hold time	Address hold time after command ends; also data hold time during write	Read/ Write	$1 \leq T \leq 255$ (HB HCLK)
Memory data bus high-z time	Duration the data bus remains high-impedance after write operation starts	Write	$0 \leq T \leq 255$ (HB HCLK)

26.2.4.1 External Memory Interface Signals

The 8-bit and 16-bit NAND FLASH interfaces are shown in Table 26-27 and Table 26-28.

Table 26-27: 8-bit NAND FLASH

FSMC pin	Direction	Description
A[17]	Output	Address latch enable signal (ALE)
A[16]	Output	Command latch enable signal (CLE)
D[7:0]	Input/Output	8-bit bidirectional address/data multiplexed bus
NCE[2]	Output	Chip select line
NOE	Output	Output enable
NWE	Output	Write enable
NWAIT	Input	Wait signal line

Table 26-28: 16-bit NAND FLASH

FSMC pin	Direction	Description
A[17]	Output	Address latch enable signal (ALE)
A[16]	Output	Command latch enable signal (CLE)
D[15:0]	Input/Output	16-bit bidirectional address/data multiplexed bus
NCE[2]	Output	Chip select line
NOE	Output	Output enable
NWE	Output	Write enable
NWAIT	Input	Wait signal line

26.2.4.2 Supported Memories and Operation Modes

Supported memories and operation modes are shown in Table 26-29.

Table 26-29: Supported memories and operation modes

Memory	Mode	HB data width	Memory width	Description
8-bit NAND	Async read	8	8	
	Async write	8	8	
	Async read	16	8	Split into 2 FSMC accesses
	Async write	16	8	Split into 2 FSMC accesses
	Async read	32	8	Split into 4 FSMC accesses
	Async write	32	8	Split into 4 FSMC accesses
16-bit NAND	Async read	8	16	
	Async read	16	16	
	Async write	16	16	
	Async read	32	16	Split into 2 FSMC accesses
	Async write	32	16	Split into 2 FSMC accesses

26.2.4.3 NAND FLASH Timing Diagram (Including Pre-wait Function)

NAND FLASH operation timing control involves the FSMC_PMEM2 and FSMC_PATT2 timing registers, each containing 4 parameters. MEMHOLD, MEMWAIT, and ATTSET correspond to the number of HCLK cycles for the three stages of NAND FLASH operation. The MEMHIZ parameter corresponds to when

the FSMC starts driving the data bus. The NAND FLASH controller common memory space access timing is shown in Figure 26-15.

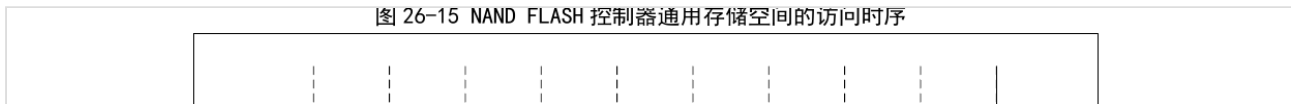


Figure 26-15: NAND FLASH controller common memory space access timing

26.2.4.4 NAND FLASH Operation Flow

The NAND FLASH command latch enable signal (CLE) and address latch enable signal (ALE) are driven by FSMC address lines A16 and A17, respectively. Therefore, when sending commands or addresses to NAND FLASH, a write operation must be performed to a specific address in the memory space. The NAND FLASH read operation process is as follows:

1. According to the characteristics of the NAND FLASH to be operated, configure the PWID, PTYP, PWAITEN, and PBKEN of the FSMC_PCR2, FSMC_PMEM2, and FSMC_PATT2 registers.
2. Write the command byte to the common memory space. During the period when NWE is low, CLE outputs a high level. At this time, the command byte is recognized by the NAND FLASH as a command and latched. The subsequent page read operation does not need to repeat sending this command.
3. Then write 4 bytes to the memory space as the start address for the read operation. During the period when NWE is low, ALE outputs a high level. At this time, the 4 bytes are recognized by the NAND FLASH as the start address for the read operation. When operating the attribute memory space, the FSMC can generate different timing to implement the pre-wait function of some NAND FLASH devices.
4. Before starting a new operation, the FSMC controller must wait for the NAND FLASH R/NB signal to go high. During the wait period, the FSMC controller must keep the NCE signal low.
5. The FSMC controller can read out the memory page byte by byte from the NAND FLASH by operating the common memory space.

26.2.4.5 NAND FLASH Pre-wait Function

Certain special NAND FLASH devices require the R/NB signal to go low after the last address byte is input, as shown in Figure 26-16.

In the figure: (1) CPU writes command byte 0x00 to 0x70010000; (2) CPU writes NAND FLASH address A7-A0 to 0x70020000; (3) CPU writes NAND FLASH address A15-A8 to 0x70020000; (4) CPU writes NAND FLASH address A23-A16 to 0x70020000; (5) CPU writes NAND FLASH address A31-A24 to 0x70020000.

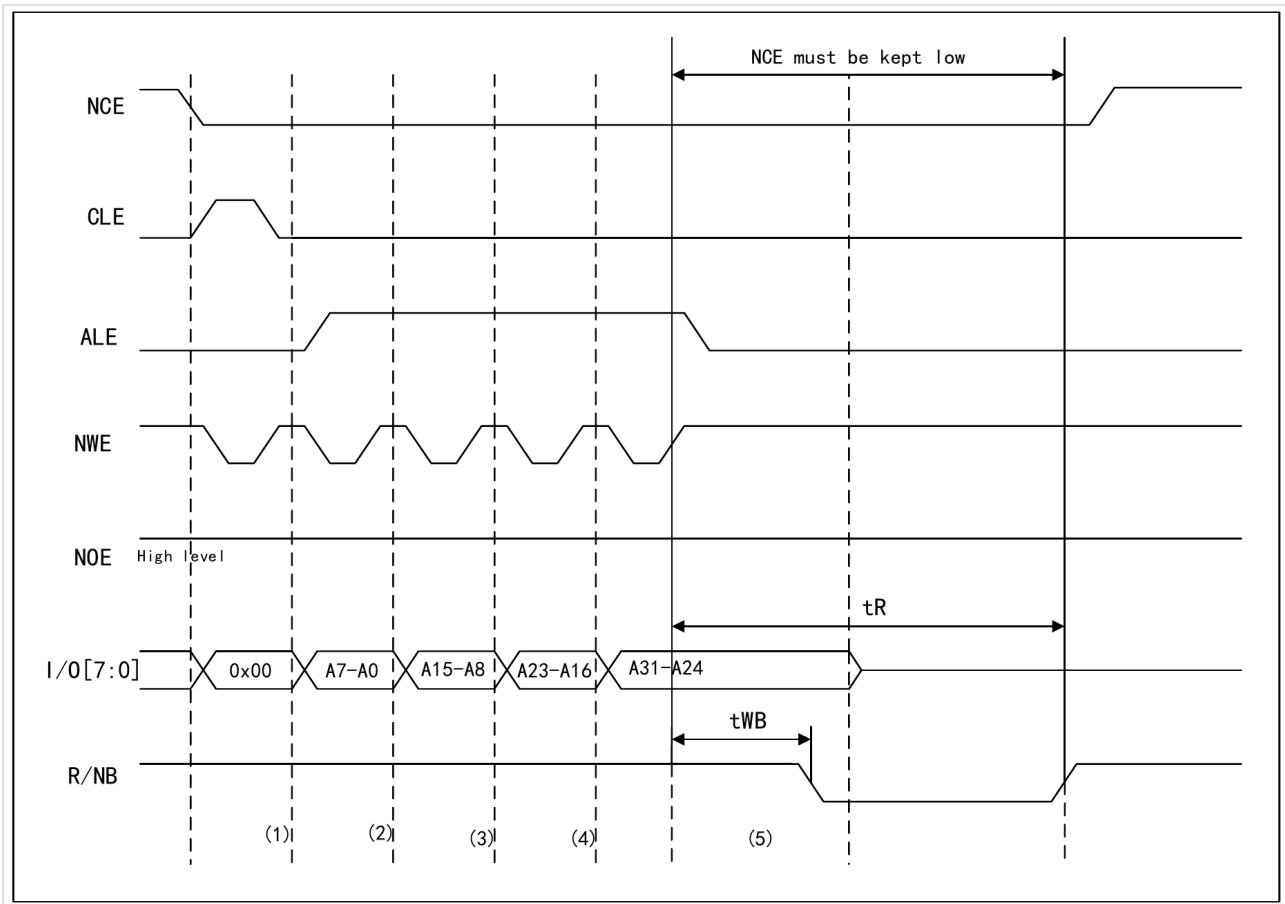


Figure 26-16: Operating special NAND FLASH

When using this function, configure the MEMHOLD field of the FSMC_PMEM2 register to guarantee the tWB timing. After that, for all read/write operations to the NAND FLASH, the FSMC controller inserts (MEMHOLD+1) HCLK hold delay between the rising edge of the NWE signal and the next operation. To solve this issue, use the attribute space to configure the ATTHOLD value to meet the tWB timing, while keeping MEMHOLD at the minimum value. In this case, only when writing the last byte of the NAND FLASH address does the FSMC controller need to write to the attribute memory space; all other NAND FLASH read/write operations use the common memory space.

26.2.4.6 NAND FLASH ECC Function

The FSMC NAND FLASH controller includes a hardware error correction code (ECC) calculation module, which can effectively reduce the software workload on the CPU when processing error correction codes. The ECC module supports correcting 1-bit errors and detecting 2-bit errors per 256, 512, 1024, 2048, 4096, or 8192 bytes during NAND FLASH read/write operations. The valid ECC result bits corresponding to page sizes are shown in Table 26-30. After the ECC calculation circuit is enabled, the ECC module monitors the NAND FLASH data bus and read/write signals (NCE and NWE). Notes for using ECC:

- When accessing NAND FLASH, data appearing on the D[15:0] bus is latched and used for ECC calculation.
- After the specified number of bytes has been written to or read from NAND FLASH, the software reads the ECC value from the FSMC_ECCR2 register. To recalculate ECC, first clear ECCEN to 0, then write 1 to re-enable ECC calculation.

Table 26-30: Page size corresponding to valid ECC result bits

ECCPS[2:0]	Page size (bytes)	ECC valid bits
000	256	ECC[21:0]
001	512	ECC[23:0]
010	1024	ECC[25:0]
011	2048	ECC[27:0]
100	4096	ECC[29:0]
101	8192	ECC[31:0]

26.3 Register Description

Table 26-31: FSMC related register list

Name	Address	Description	Reset Value
R32_FSMC_BCR1	0xA0000000	SRAM/NOR-Flash chip select control register 1	0x000030DX
R32_FSMC_BTR1	0xA0000004	SRAM/NOR-Flash chip select timing register 1	0x0FFFFFFF
R32_FSMC_PCR2	0xA0000060	NAND-Flash control register 2	0x00000018
R32_FSMC_SR2	0xA0000064	FIFO status and interrupt register 2	0x00000040
R32_FSMC_PMEM2	0xA0000068	Common memory space timing register 2	0xFCFCFCFC
R32_FSMC_PATT2	0xA000006C	Attribute memory space timing register 2	0xFCFCFCFC
R32_FSMC_ECCR2	0xA0000074	ECC result register 2	0x00000000
R32_FSMC_BWTR1	0xA0000104	SRAM/NOR-Flash write timing register 1	0x0FFFFFFF

26.3.1 SRAM/NOR-Flash Chip Select Control Register 1 (FSMC_BCR1)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												CBURSTRW	Reserved		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ASYNCWAIT	EXTMOD	WAITEN	WREN	WAITCFG	Reserved	WAITPOL	BURSTEN	Reserved	FACCEN	MWID[1:0]	MTYP[1:0]	MUXEN	MBKEN		

Bits	Name	Access	Description	Reset
[31:20]	Reserved	RO	Reserved.	0
19	CBURSTRW	RW	Synchronous write enable: 1: write operation in synchronous mode; 0: write operation in asynchronous mode.	0
[18:16]	Reserved	RO	Reserved.	0
15	ASYNCWAIT	RW	Wait signal during asynchronous transfer: 1: enable wait signal during asynchronous transfer; 0: disable wait signal during asynchronous transfer.	0
14	EXTMOD	RW	Extended mode enable: 1: allow write to use FSMC_BWTR register; 0: do not allow write to use FSMC_BWTR register.	0
13	WAITEN	RW		1

			Wait enable. When the flash memory is in burst transfer mode, this bit controls whether wait signals are inserted based on the NWAIT signal. 1: use NWAIT signal; 0: disable NWAIT signal.	
12	WREN	RW	Write operation enable: 1: allow FSMC to write to memory; 0: disable FSMC from writing to memory.	1
11	WAITCFG	RW	Wait timing configuration: 1: NWAIT signal is active during the wait state; 0: NWAIT signal is active one data cycle before the wait state.	0
10	Reserved	RO	Reserved.	0
9	WAITPOL	RW	Wait signal polarity: 1: NWAIT wait signal is active high; 0: NWAIT wait signal is active low.	0
8	BURSTEN	RW	Synchronous read enable: 1: enable synchronous read mode, perform read access in synchronous mode; 0: disable synchronous read mode, perform read access in asynchronous mode.	0
7	Reserved	RO	Reserved.	1
6	FACCEN	RW	NOR FLASH access enable: 1: allow NOR FLASH access; 0: disable NOR FLASH access.	1
[5:4]	MWID[1:0]	RW	Data bus width: 00: 8-bit; 01: 16-bit; 10: Reserved; 11: Reserved.	01b
[3:2]	MTYP[1:0]	RW	Memory type: 00: SRAM, ROM; 01: PSRAM; 10: NOR FLASH; 11: Reserved.	xxb
1	MUXEN	RW	Address/data multiplex enable: 1: address/data multiplexed; 0: address/data not multiplexed.	x
0	MBKEN	RW	Memory block enable: 1: enable the corresponding memory block; 0: disable the corresponding memory block.	x

26.3.2 SRAM/NOR-Flash Chip Select Timing Register 1 (FSMC_BTR1)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		ACCMOD[1:0]		DATLAT[3:0]				CLKDIV[3:0]				BUSTURN[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATAST[7:0]								ADDHLD[3:0]				ADDSET[3:0]			

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
[29:28]	ACCMOD[1:0]	RW	Asynchronous access mode: 00: Mode A; 01: Mode B; 10: Mode C; 11: Mode D.	00b
[27:24]	DATLAT[3:0]	RW	Data hold time. Used for synchronous burst mode (only for NOR FLASH). 0000: first data hold time is 2 CLK clocks; ...; 1110: first data hold time is 16 CLK clocks; 1111: first data hold time is 17 CLK clocks.	1111b
[23:20]	CLKDIV[3:0]	RW	Clock division ratio (CLK signal). This parameter has no effect when accessing asynchronous NOR FLASH, SRAM, and ROM. 0000: Reserved; 0001: 1 CLK cycle = 2 HCLK cycles; ...; 1110: 1 CLK cycle = 15 HCLK cycles; 1111: 1 CLK cycle = 16 HCLK cycles.	1111b
[19:16]	BUSTURN[3:0]	RW		1111b

			Bus turnaround time. (Only for NOR FLASH) 0000: bus turnaround time = 1 HCLK cycle; ...; 1110: bus turnaround time = 15 HCLK cycles; 1111: bus turnaround time = 16 HCLK cycles.	
[15:8]	DATAST[7:0]	RW	Data hold time. 00000000: data hold time = 1 HCLK cycle; 00000001: data hold time = 2 HCLK cycles; ...; 11111110: data hold time = 255 HCLK cycles; 11111111: data hold time = 256 HCLK cycles. For the data hold time of each memory type and access method, refer to the corresponding diagrams (Figures 26-3 to 26-12). For example: Mode 1 and Mode A, read operation with DATAST=1: data hold time = DATAST+3 = 4 HCLK cycles.	0xFF
[7:4]	ADDHLD[3:0]	RW	Address hold time. (Only for asynchronous operations) 0000: address hold time = 1 HCLK cycle; ...; 1110: address hold time = 15 HCLK cycles; 1111: address hold time = 16 HCLK cycles.	1111b
[3:0]	ADDSET[3:0]	RW	Address setup time. (Only for asynchronous operations) 0000: address setup time = 1 HCLK cycle; ...; 1110: address setup time = 15 HCLK cycles; 1111: address setup time = 16 HCLK cycles. For the data hold time of each memory type and access method, refer to the corresponding diagrams (Figures 26-3 to 26-12). For example: Mode 1 and Mode A, read operation with DATAST=1: data hold time = DATAST+3 = 4 HCLK cycles.	1111b

26.3.3 NAND-Flash Control Register 2 (FSMC_PCR2)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												ECCPS[2:0]			TAR[3]
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TAR[2:0]			TCLR[3:0]			Reserved			ECCEN	PWID[1:0]		PTYP	PBKEN	PWAITEN	Reserved

Bits	Name	Access	Description	Reset
[31:20]	Reserved	RO	Reserved.	0
[19:17]	ECCPS[2:0]	RW	ECC page size: 000: 256 Byte; 001: 512 Byte; 010: 1024 Byte; 011: 2048 Byte; 100: 4096 Byte; 101: 8192 Byte; 111: Reserved.	000b
[16:13]	TAR[3:0]	RW	ALE to RE delay: 0000: 1 HCLK cycle; ...; 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	0000b
[12:9]	TCLR[3:0]	RW	CLE to RE delay: 0000: 1 HCLK cycle; ...; 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	0000b
[8:7]	Reserved	RO	Reserved.	0
6	ECCEN	RW	ECC enable: 1: enable ECC; 0: disable ECC.	0
[5:4]	PWID[1:0]	RW	Data bus width: 00: 8-bit; 01: 16-bit; 10: Reserved; 11: Reserved.	01b
3	PTYP	RW	Memory type: 1: NAND flash; 0: Reserved.	1
2	PBKEN	RW	NAND memory enable: 1: enable the corresponding memory block; 0: disable the corresponding memory block.	0
1	PWAITEN	RW	Wait function enable: 1: enable; 0: disable.	0
0	Reserved	RO	Reserved.	0

26.3.4 FIFO Status and Interrupt Register 2 (FSMC_SR2)

Offset address: 0x64

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							FEMPT	Reserved							

Bits	Name	Access	Description	Reset
[31:7]	Reserved	RO	Reserved.	0
6	FEMPT	RO	FIFO empty flag. 1: FIFO empty; 0: FIFO not empty.	1
[5:0]	Reserved	RO	Reserved.	0

26.3.5 Common Memory Space Timing Register 2 (FSMC_PMEM2)

Offset address: 0x68

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MEMHIZx[7:0]								MEMHOLDx[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MEMWAITx[7:0]								MEMSETx[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	MEMHIZx[7:0]	RW	Common space data bus high-impedance time: 00000000: NAND FLASH 0 HCLK cycles; ...; 11111110: NAND FLASH 254 HCLK cycles; 11111111: NAND FLASH 255 HCLK cycles.	0xFC
[23:16]	MEMHOLDx[7:0]	RW	Common space hold time: 00000000: Reserved; 00000001: 1 HCLK cycle; ...; 11111110: 254 HCLK cycles; 11111111: 255 HCLK cycles.	0xFC
[15:8]	MEMWAITx[7:0]	RW	Common space wait time. (Must add the wait cycles introduced by the NWAIT signal going low) 00000000: Reserved; 00000001: 2 HCLK cycles; ...; 11111110: 255 HCLK cycles; 11111111: 256 HCLK cycles.	0xFC
[7:0]	MEMSETx[7:0]	RW	Common space setup time: 00000000: NAND FLASH 2 HCLK cycles; ...; 11111110: NAND FLASH 256 HCLK cycles; 11111111: NAND FLASH 257 HCLK cycles.	0xFC

26.3.6 Attribute Memory Space Timing Register 2 (FSMC_PATT2)

Offset address: 0x6C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ATTHIZx[7:0]								ATTHOLDx[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ATTWAITx[7:0]								ATTSETx[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	ATTHIZx[7:0]	RW		0xFC

			Attribute space data bus high-impedance time: 00000000: 0 HCLK cycles; ...; 11111110: 254 HCLK cycles; 11111111: 255 HCLK cycles.	
[23:16]	ATTHOLDx[7:0]	RW	Attribute space hold time: 00000000: Reserved; 00000001: 1 HCLK cycle; ...; 11111110: 254 HCLK cycles; 11111111: 255 HCLK cycles.	0xFC
[15:8]	ATTWAITx[7:0]	RW	Attribute space wait time. (Must add the wait cycles introduced by the NWAIT signal going low) 00000000: 1 HCLK cycle; 00000001: 2 HCLK cycles; ...; 11111110: 255 HCLK cycles; 11111111: 256 HCLK cycles.	0xFC
[7:0]	ATTSETx[7:0]	RW	Attribute space setup time: 00000000: 1 HCLK cycle; ...; 11111110: 255 HCLK cycles; 11111111: 256 HCLK cycles.	0xFC

26.3.7 ECC Result Register 2 (FSMC_ECCR2)

Offset address: 0x74

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ECC[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ECC[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	ECC[31:0]	RO	ECC calculation result.	0

26.3.8 SRAM/NOR-Flash Write Timing Register 1 (FSMC_BWTR1)

Offset address: 0x104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved		ACCMOD[1:0]				Reserved						BUSTYRN[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATAST[7:0]								ADDHLD[3:0]				ADDSET[3:0]			

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
[29:28]	ACCMOD[1:0]	RW	Access mode. (This bit is valid when EXTMOD is 1) 00: Access mode A; 01: Access mode B; 10: Access mode C; 11: Access mode D.	00b
[27:20]	Reserved	RO	Reserved.	0
[19:16]	BUSTYRN[3:0]	RW	Bus turnaround time. 0000: 1 HCLK cycle; 0001: 2 HCLK cycles; ...; 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	1111b
[15:8]	DATAST[7:0]	RW	Data hold time: 00000000: Reserved; 00000001: 2 HCLK cycles; ...; 11111110: 255 HCLK cycles; 11111111: 256 HCLK cycles.	0xFF
[7:4]	ADDHLD[3:0]	RW	Address hold time: 0000: Reserved; 0001: 2 HCLK cycles; ...; 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	1111b
[3:0]	ADDSET[3:0]	RW	Address setup time: 0000: 1 HCLK cycle; 0001: 2 HCLK cycles; ...; 1110: 15 HCLK cycles; 1111: 16 HCLK cycles.	1111b

Chapter 27: LCD-TFT Display Controller (LTDC)

The LCD-TFT (Thin-Film Transistor Liquid Crystal Display) display controller (LTDC) mainly provides parallel digital RGB signals along with horizontal synchronization, vertical synchronization, pixel clock, and data enable signals, which can be used as output signals to various LCD and TFT panel interfaces.

27.1 Main Features

- Provides 1 display layer, including a dedicated 8×32-bit FIFO
- Supports color lookup table (CLUT), up to 256 colors per layer
- Supports programmable timing for different display panels
- Supports programmable background color
- Supports programmable polarity for HSYNC, VSYNC, and data enable signals
- Display layer supports up to 8 input color formats
- Supports programmable window position and size
- Supports thin-film transistor (TFT) color displays
- Supports up to 3 programmable interrupt timings

27.2 Overview

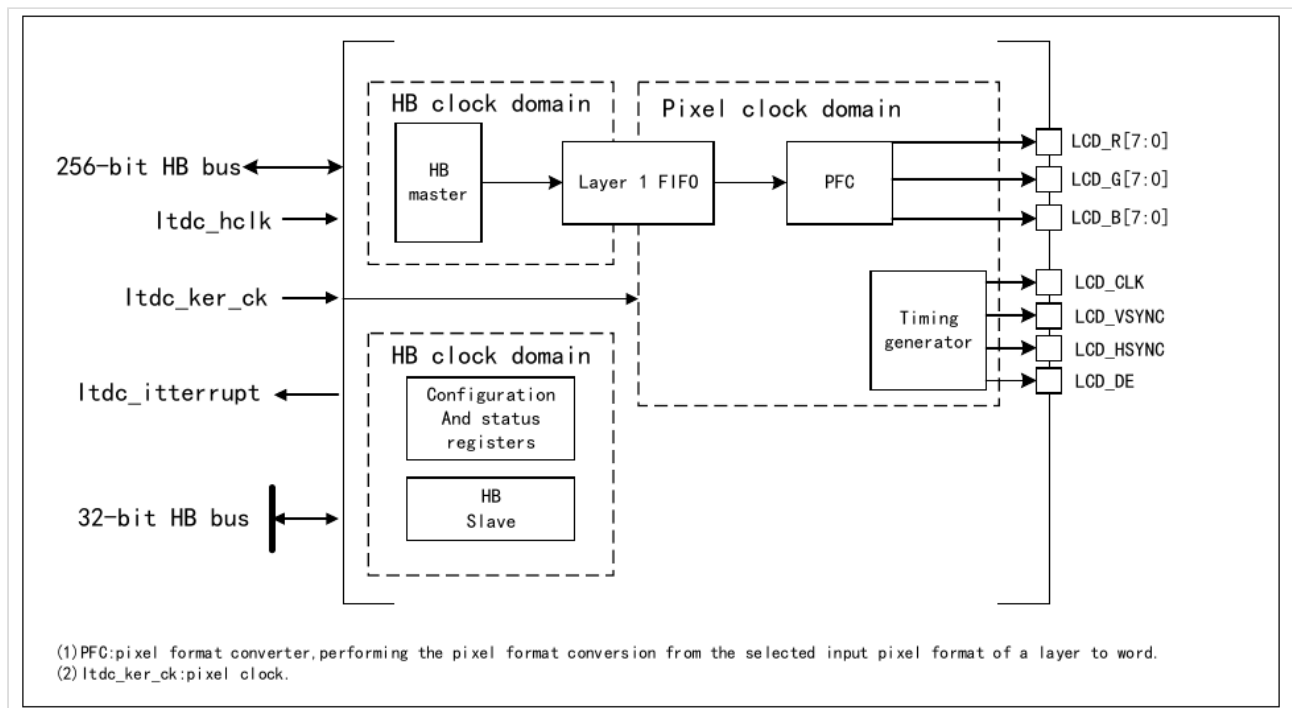


Figure 27-1: LTDC Block Diagram

- (1) PFC: Pixel Format Converter, performing the pixel format conversion from the selected input pixel format of a layer to word.
 (2) Itdc_ker_ck: pixel clock.

27.3 Functional Description

27.3.1 LTDC Synchronous Timing

The LTDC interface is a synchronous data interface, including pixel clock, pixel data, and horizontal and vertical synchronization signals. Figure 27-2 shows the configurable timing parameters generated by the LTDC synchronous timing generator module.

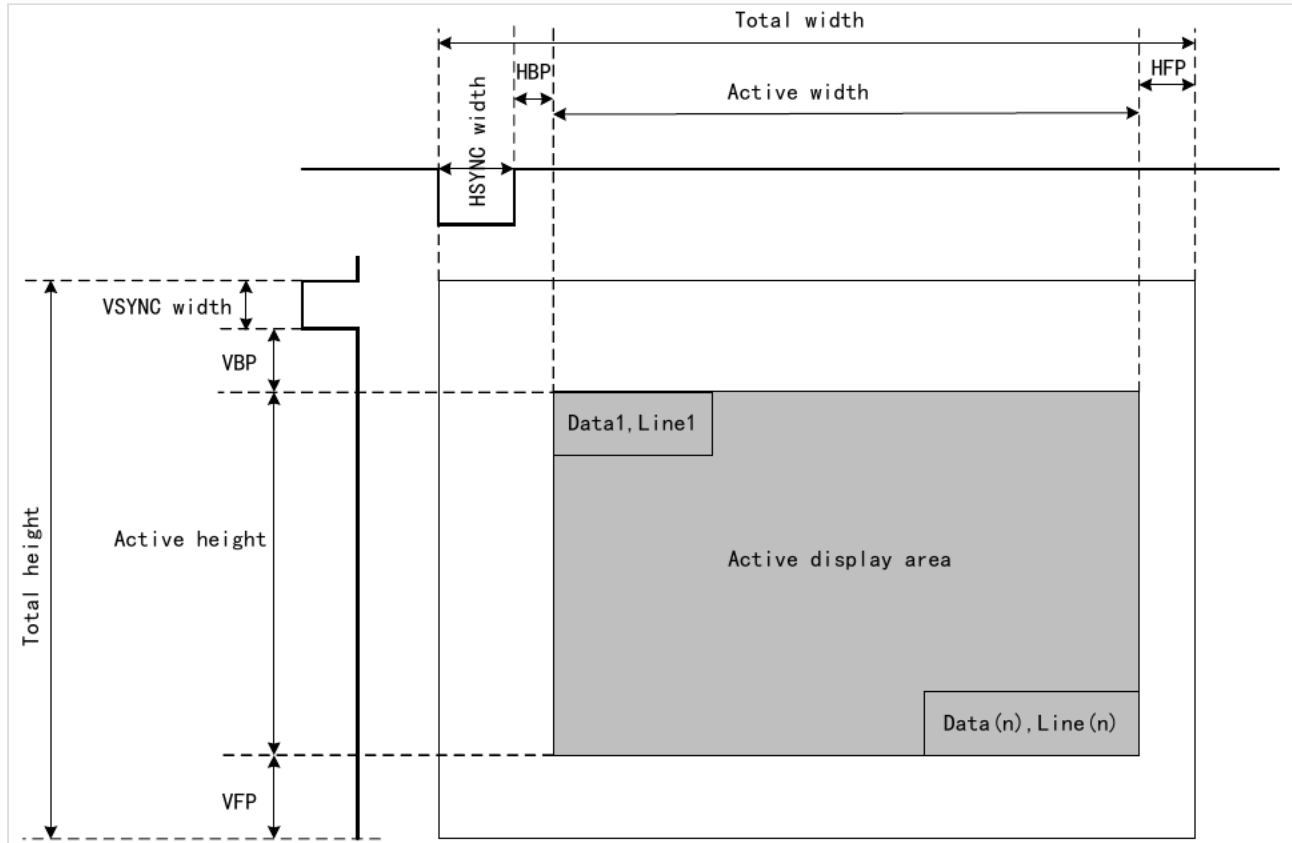


Figure 27-2: LTDC Synchronous Timing

Note: (1) HSYNC and VSYNC widths are the horizontal and vertical synchronization widths, respectively; HBP and HFP are the horizontal back porch and horizontal front porch periods, respectively; VBP and VFP are the vertical back porch and vertical front porch periods, respectively.

(2) When LTDC is enabled, the generated timing uses the X/Y=0/0 position as the first horizontal sync pixel in the vertical synchronization region, followed by the back porch, active data display area, and front porch.

(3) When LTDC is disabled, the timing generator module resets to X=total width-1, Y=total height-1, and retains the previous pixel during the vertical synchronization phase and before FIFO refresh. Therefore, only blanking data is continuously output.

The synchronous timing configuration example is as follows:

1. LCD-TFT timing (should be extracted from the panel datasheet);
2. Horizontal and vertical synchronization width: 0xA pixels, 0x2 lines;
3. Horizontal and vertical back porch: 0x14 pixels, 0x2 lines;
4. Active width and active height: 0x140 pixels, 0xF0 lines (320×240);
5. Horizontal front porch: 0xA pixels;
6. Vertical front porch: 0x4 lines;

The values programmed in the LTDC timing registers will be:

1. R32_LTDC_SSCR register: will be programmed to 0x00090001. (HSW[11:0] = 0x9 and VSH[10:0] = 0x1);
2. R32_LTDC_BPCR register: will be programmed to 0x001D0003. (AHBP[11:0] = 0x1D (0xA + 0x13), AVBP[10:0] = 0x3 (0x2 + 0x1));
3. R32_LTDC_AWCR register: will be programmed to 0x015D00F3. (AAW[11:0] = 0x15D (0xA + 0x14 + 0x13F), AAH[10:0] = 0xF3 (0x2 + 0x2 + 0xEF));
4. R32_LTDC_TWCR register: will be programmed to 0x00000167. (TOTALW[11:0] = 0x167 (0xA + 0x14 + 0x140 + 0x9));
5. R32_LTDC_THCR register: will be programmed to 0x000000F7. (TOTALH[10:0] = 0xF7 (0x2 + 0x2 + 0xF0 + 3)).

27.3.2 Pixel Format

The programmable pixel format is used for the data stored in the layer's frame buffer. Up to 8 input pixel formats can be configured for each layer via the R32_LTDC_PFCR register.

The ARGB8888 format requires 8 bits per channel (Alpha, Red, Green, and Blue). However, some channels in the ARGB1555 and ARGB4444 formats have fewer than 8 bits. The pixel processing unit converts them to ARGB8888 by copying the most significant bits to fill the least significant bits. When processing RGB888 and RGB565 formats, the pixel processing unit assumes Alpha=255, and if a channel has fewer than 8 bits, it also copies the most significant bits to fill the least significant bits.

27.3.3 Color Lookup Table (CLUT)

The CLUT can be enabled for each layer at runtime via the CLUTEN bit in the R32_LTDC_LCR register. The CLUT only applies to indexed colors when using L8, AL44, and AL88 input pixel formats.

First, the CLUT must be loaded with R, G, and B values to replace the original R, G, and B values of the corresponding pixels (indexed colors). Each color (RGB value) has its own corresponding address within the CLUT. The R, G, and B values and their respective addresses are programmed through the R32_LTDC_LCLUTWR register.

When using L8 and AL88 input pixel formats, the CLUT must be loaded with 256 colors. The address of each color is configured in the CLUTADD bits of the R32_LTDC_LCLUTWR register.

When using the AL44 input pixel format, the CLUT must only be loaded with 16 colors. The address of each color must be filled by repeating the 4-bit L channel to 8 bits, as follows:

- L0 (index color 0), at address 0x00;
- L1, at address 0x11;
- L2, at address 0x22;
- ...
- L15, at address 0xFF.

27.3.4 Color Frame Buffer

27.3.4.1 Color Frame Buffer Address

Each layer's color frame buffer has a starting address, which is configured through the R32_LTDC_LCFBAR register. When a layer is enabled, data is fetched from the color frame buffer.

27.3.4.2 Color Frame Buffer Length

Each layer has a configurable total line length (in bytes) and number of lines for the color frame buffer, which can be configured through the R32_LTDC_LCFBLR and R32_LTDC_LCFBLNR registers, respectively.

The line length and number of lines settings prevent data beyond the end of the frame buffer from being prefetched into the layer's corresponding FIFO.

If set lower than the required bytes, a FIFO underrun interrupt will be generated (if previously enabled).

If set higher than the actual required bytes, the useless data read from the FIFO will be discarded and not displayed.

27.3.4.3 Color Frame Buffer Pitch

Each layer's color frame buffer has a configurable pitch, which is the distance (in bytes) from the start of one line to the start of the next line. It is configured through the R32_LTDC_LCFBLR register.

27.3.5 Color Keying

The color key (RGB) can be configured for pixel replacement. At runtime, pixels matching the color key value can be replaced.

When color keying is enabled, the current pixel (after format conversion, before CLUT and blending respectively) is compared with the color key. If the current pixel matches the programmed RGB value, the RGB channels of that pixel are set to the default color.

The color key is configured through the LTDC_LCKCR register. The programmed value depends on the pixel format, as it is compared with the current pixel after pixel format conversion to RGB888.

For example, using medium yellow (50% red + 50% green) as the replacement color key:

- In RGB565, medium yellow is 0x8400. Set LTDC_LCKCR to 0x848200.
- In ARGB8888, medium yellow is 0x808000. Set LTDC_LCKCR to 0x808000.
- In all CLUT-based color modes (L8, AL88, AL44), set one palette entry to medium yellow 0x808000, and set LTDC_LCKCR to 0x808000.

27.3.6 Reload of Shadow Registers

Some configuration registers are shadowed. Shadow register values can be reloaded into the active registers either immediately upon a write to the active register, or at the start of the vertical blanking period after the configuration phase in the R32_LTDC_SRCR register. If immediate reload is selected, the reload should only be activated after all new registers have been written.

Shadow registers should not be modified again before the reload is complete. Reading a shadow register returns the actual valid value. Newly written values can only be read after the reload occurs.

If correspondingly enabled in the R32_LTDC_IER register, a register reload interrupt can be generated.

All Layer 1 and Layer 2 registers are shadow registers, except the R32_LTDC_LCLUTWR register.

27.3.7 Interrupts

The LTDC provides 3 maskable interrupts, which are logically ORed into 1 interrupt vector.

Interrupt sources can be individually enabled or disabled through the R32_LTDC_IER register; the corresponding interrupt flags can be read through the R32_LTDC_ISR register; the corresponding interrupt flag bits can be cleared by setting the R32_LTDC_ICR register to 1.

Interrupts are generated when the following events occur:

- Line interrupt: generated when the programmed line is reached. The position of the line interrupt is programmed in the R32_LTDC_LIPCR register;
- Register reload interrupt: generated when a shadow register reload is performed during the vertical blanking period;
- FIFO underrun interrupt: generated when a pixel is requested from an empty layer FIFO.

27.4 Register Description

Table 27-1: LTDC Register List

Name	Address	Description	Reset Value
R32_LTDC_SSCR	0x40024800	LTDC Synchronization Size Configuration Register	0x00000000
R32_LTDC_BPCR	0x40024804	LTDC Back Porch Configuration Register	0x00000000
R32_LTDC_AWCR	0x40024808	LTDC Active Width Configuration Register	0x00000000
R32_LTDC_TWCR	0x4002480C	LTDC Total Width Configuration Register	0x00000000
R32_LTDC_GCR	0x40024810	LTDC Global Control Register	0x00002220
R32_LTDC_SRCR	0x40024814	LTDC Shadow Reload Configuration Register	0x00000000
R32_LTDC_BCCR	0x40024818	LTDC Background Color Configuration Register	0x00000000
R32_LTDC_IER	0x4002481C	LTDC Interrupt Enable Register	0x00000000
R32_LTDC_ISR	0x40024820	LTDC Interrupt Status Register	0x00000000
R32_LTDC_ICR	0x40024824	LTDC Interrupt Clear Register	0x00000000
R32_LTDC_LIPCR	0x40024828	LTDC Line Interrupt Position Configuration Register	0x00000000
R32_LTDC_CPSR	0x4002482C	LTDC Current Position Status Register	0x00000000
R32_LTDC_CDSR	0x40024830	LTDC Current Display Status Register	0x0000000F
R32_LTDC_LCR	0x40024834	LTDC Layer Control Register	0x00000000
R32_LTDC_LWHPCR	0x40024838	LTDC Layer Window Horizontal Position Configuration Register	0x00000000
R32_LTDC_LWVPCR	0x4002483C	LTDC Layer Window Vertical Position Configuration Register	0x00000000
R32_LTDC_LCKCR	0x40024840	LTDC Layer Color Key Configuration Register	0x00000000
R32_LTDC_LPFCR	0x40024844	LTDC Layer Pixel Format Configuration Register	0x00000000
R32_LTDC_LDCCR	0x4002484C	LTDC Layer Default Color Configuration Register	0x00000000
R32_LTDC_LCFBAR	0x40024854	LTDC Layer Color Frame Buffer Address Register	0x00000000
R32_LTDC_LCFBLR	0x40024858	LTDC Layer Color Frame Buffer Length Register	0x00000000
R32_LTDC_LCFBLNR	0x4002485C	LTDC Layer Color Frame Buffer Line Number Register	0x00000000
R32_LTDC_LCLUTWR	0x40024860	LTDC Layer CLUT Write Register	0x00000000

27.4.1 LTDC Synchronization Size Configuration Register (R32_LTDC_SSCR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													HSW[11:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					VSH[10:0]										

Bits	Name	Access	Description	Reset
[31:28]	Reserved	RO	Reserved.	0
[27:16]	HSW[11:0]	RW	Horizontal synchronization width (in pixel clock cycles): This field defines the number of horizontal sync pixels minus 1.	0
[15:11]	Reserved	RO	Reserved.	0
[10:0]	VSH[10:0]	RW	Vertical synchronization height (in horizontal scan lines): This field defines the vertical sync height minus 1. It represents the number of horizontal sync lines.	0

27.4.2 LTDC Back Porch Configuration Register (R32_LTDC_BPCR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													AHBP[11:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					AVBP[10:0]										

Bits	Name	Access	Description	Reset
[31:28]	Reserved	RO	Reserved.	0
[27:16]	AHBP[11:0]	RW	Accumulated horizontal back porch (in pixel clock cycles): This field defines the accumulated horizontal back porch width (horizontal sync pixels plus horizontal back porch pixels minus 1). The horizontal back porch is the interval between the horizontal sync signal becoming inactive and the start of the active display of the next scan line.	0
[15:11]	Reserved	RO	Reserved.	0
[10:0]	AVBP[10:0]	RW	Accumulated vertical back porch (in horizontal scan lines): This field defines the accumulated vertical back porch width (vertical sync lines plus vertical back porch lines minus 1). The vertical back porch is the number of horizontal scan lines from frame start to the start of the first active scan line of the next frame.	0

27.4.3 LTDC Active Width Configuration Register (R32_LTDC_AWCR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													AAW[11:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					AAH[10:0]										

Bits	Name	Access	Description	Reset
[31:28]	Reserved	RO	Reserved.	0
[27:16]	AAW[11:0]	RW	Accumulated active width (in pixel clock cycles): This field defines the accumulated active width (horizontal sync pixels plus horizontal back porch pixels plus active pixels minus 1). The active width is the number of pixels in the active display area of a panel scan line.	0
[15:11]	Reserved	RO	Reserved.	0
[10:0]	AAH[10:0]	RW	Accumulated active height (in horizontal scan lines): This field defines the accumulated height (vertical sync lines plus vertical back porch lines plus active height lines minus 1). The active height is the number of active lines in the panel.	0

27.4.4 LTDC Total Width Configuration Register (R32_LTDC_TWCR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													TOTALW[11:0]		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					TOTALH[10:0]										

Bits	Name	Access	Description	Reset
[31:28]	Reserved	RO	Reserved.	0
[27:16]	TOTALW[11:0]	RW	Total width (in pixel clock cycles): This field defines the accumulated total width (horizontal sync pixels plus horizontal back porch pixels plus active width plus horizontal front porch pixels minus 1).	0
[15:11]	Reserved	RO	Reserved.	0
[10:0]	TOTALH[10:0]	RW	Total height (in horizontal scan lines): This field defines the accumulated height (vertical sync lines plus vertical back porch lines plus active height plus vertical front porch lines minus 1).	0

27.4.5 LTDC Global Control Register (R32_LTDC_GCR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HSPOL	VSPOL	DEPOL	PCPOL	Reserved											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved														LTDCCEN	

Bits	Name	Access	Description	Reset
31	HSPOL	RW	Horizontal synchronization polarity: 1: Horizontal sync polarity active high; 0: Horizontal sync polarity active low.	0
30	VSPOL	RW	Vertical synchronization polarity: 1: Vertical sync active high; 0: Vertical sync active low.	0
29	DEPOL	RW	Not data enable polarity: 1: Not data enable polarity active high; 0: Not data enable polarity active low.	0

Bits	Name	Access	Description	Reset
28	PCPOL	RW	Pixel clock polarity: 1: Pixel clock active high; 0: Pixel clock polarity active low.	0
[27:1]	Reserved	RO	Reserved.	0x444
0	LTDCEEN	RW	LCD-TFT controller enable bit: 1: Enable; 0: Disable.	0

27.4.6 LTDC Shadow Reload Configuration Register (R32_LTDC_SRCR)

Offset address: 0x14

31~16: Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved													VBR		IMR

Bits	Name	Access	Description	Reset
[31:2]	Reserved	RO	Reserved.	0
1	VBR	RW	Vertical blanking reload: 1: Shadow registers are reloaded during the vertical blanking period (at the start of the first line after the active display area); 0: No effect. Note: This bit is set to 1 by software and only cleared by hardware after the reload is complete.	0
0	IMR	RW	Immediate reload: 1: Shadow registers are reloaded immediately; 0: No effect. Note: This bit is set to 1 by software and only cleared by hardware after the reload is complete.	0

Note: When shadow registers are read back, the old value is returned until the reload is complete.

27.4.7 LTDC Background Color Configuration Register (R32_LTDC_BCCR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								BCRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				BCGREEN[7:0]				BCBLUE[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
[23:16]	BCRED[7:0]	RW	Background red value.	0
[15:8]	BCGREEN[7:0]	RW	Background green value.	0
[7:0]	BCBLUE[7:0]	RW	Background blue value.	0

27.4.8 LTDC Interrupt Enable Register (R32_LTDC_IER)

Offset address: 0x1C

31~16: Reserved																
15	14	13	12	11	10	9	8	7	6	5	4	3	2		1	0
Reserved												RRIE	DMA_DISABLE		FUIE	LIE

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
3	RRIE	RW	Register reload interrupt enable: 1: Enable; 0: Disable.	0
2	DMA_DISABLE	RW	DMA disable function bit: 1: DMA disabled; when this bit is set to 1, the current fetch data address is forcibly switched to the reloaded CFBADD, and the internal FIFO is cleared. It is recommended to use this bit to switch the fetch address during line blanking or field blanking (configure the address, reload immediately, then pull this bit low to switch to the target address); 0: DMA enabled. Note: Only applicable to chips where the fifth-to-last digit of the lot number is greater than 0.	0
1	FUIE	RW	FIFO underrun interrupt enable: 1: Enable; 0: Disable.	0
0	LIE	RW	Line interrupt enable: 1: Enable; 0: Disable.	0

27.4.9 LTDC Interrupt Status Register (R32_LTDC_ISR)

Offset address: 0x20

31~16: Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											RRIF	Reserved	FUIF	LIF	

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
3	RRIF	RO	Register reload interrupt flag: 1: Register reload interrupt generated when a vertical blanking reload occurs (and when the first line after the active area is reached); 0: No register reload interrupt generated.	0
2	Reserved	RO	Reserved.	0
1	FUIF	RO	FIFO underrun interrupt flag: 1: FIFO underrun interrupt generated when one of the layer FIFOs is empty and pixel data is read from the FIFO; 0: No FIFO underrun interrupt generated.	0
0	LIF	RO	Line interrupt flag: 1: Line interrupt generated when the programmed line is reached; 0: No line interrupt generated.	0

27.4.10 LTDC Interrupt Clear Register (R32_LTDC_ICR)

Offset address: 0x24

31~16: Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											CRRIF	Reserved	CFUIF	CLIF	

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
3	CRRIF	RW1Z	Register reload interrupt clear flag, write 1 to clear, write 0 has no effect.	0
2	Reserved	RO	Reserved.	0
1	CFUIF	RW1Z	FIFO underrun interrupt clear flag, write 1 to clear, write 0 has no effect.	0

Bits	Name	Access	Description	Reset
0	CLIF	RW1Z	Line interrupt clear flag, write 1 to clear, write 0 has no effect.	0

27.4.11 LTDC Line Interrupt Position Configuration Register (R32_LTDC_LIPCR)

Offset address: 0x28

31~16: Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				LIPOS[10:0]											

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
[10:0]	LIPOS[10:0]	RW	Line interrupt position.	0

27.4.12 LTDC Current Position Status Register (R32_LTDC_CPSR)

Offset address: 0x2C

31~16: CXPOS[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYPOS[15:0]															

Bits	Name	Access	Description	Reset
[31:16]	CXPOS[15:0]	RW	Current X position.	0
[15:0]	CYPOS[15:0]	RW	Current Y position.	0

27.4.13 LTDC Current Display Status Register (R32_LTDC_CDSR)

Offset address: 0x30

31~16: Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										HSYNCS		VSYNCS		HDES	VDES

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
3	HSYNCS	RO	Horizontal sync display status: 1: Active high; 0: Active low.	1
2	VSYNCS	RO	Vertical sync display status: 1: Active high; 0: Active low.	1
1	HDES	RO	Horizontal data enable display status: 1: Active high; 0: Active low.	1
0	VDES	RO	Vertical data enable display status: 1: Active high; 0: Active low.	1

27.4.14 LTDC Layer Control Register (R32_LTDC_LCR)

Offset address: 0x34

31~16: Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Reserved	CLUTEN	Reserved	Reserved	COLKEN	Reserved	LEN
----------	--------	----------	----------	--------	----------	-----

Bits	Name	Access	Description	Reset
[31:5]	Reserved	RO	Reserved.	0
4	CLUTEN	RW	Color lookup table enable: 1: Enable; 0: Disable. Note: CLUT is only meaningful for L8, AL44, and AL88 pixel formats.	0
[3:2]	Reserved	RO	Reserved.	0
1	COLKEN	RW	Color keying enable: 1: Enable; 0: Disable.	0
0	LEN	RW	Layer enable: 1: Enable; 0: Disable.	0

27.4.15 LTDC Layer Window Horizontal Position Configuration Register (R32_LTDC_LWHPCR)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WHSPPOS[11:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WHSTPOS[11:0]											

Bits	Name	Access	Description	Reset
[31:28]	Reserved	RO	Reserved.	0
[27:16]	WHSPPOS[11:0]	RW	Window horizontal stop position: WHSPPOS[11:0] >= AHBP[11:0] + 1.	0
[15:12]	Reserved	RO	Reserved.	0
[11:0]	WHSTPOS[11:0]	RW	Window horizontal start position: WHSTPOS[11:0] <= AAW[11:0].	0

For example, if the R32_LTDC_BPCR register is configured as 0x000E0005 (AHBP[11:0] = 0xE), and the R32_LTDC_AWCR register is configured as 0x028E01E5 (AAW[11:0] = 0x28E). To configure the horizontal position of a 630×460 window (horizontal start offset of 5 pixels in the active data area):

1. First pixel of the layer window: WHSTPOS[11:0] should be programmed to 0x14 (0xE + 1 + 0x5);
2. Last pixel of the layer window: WHSPPOS[11:0] should be programmed to 0x28A.

27.4.16 LTDC Layer Window Vertical Position Configuration Register (R32_LTDC_LWVPCR)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				WVSPPOS[10:0]											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				WVSTPOS[10:0]											

Bits	Name	Access	Description	Reset
[31:27]	Reserved	RO	Reserved.	0
[26:16]	WVSPPOS[10:0]	RW	Window vertical stop position: WVSPPOS[10:0] >= AVBP[10:0] + 1.	0

Bits	Name	Access	Description	Reset
[15:11]	Reserved	RO	Reserved.	0
[10:0]	WVSTPOS[10:0]	RW	Window vertical start position: WVSTPOS[10:0] <= AAH[10:0].	0

For example, if the R32_LTDC_BPCR register is configured as 0x000E0005 (AVBP[10:0] = 0x5), and the R32_LTDC_AWCR register is configured as 0x028E01E5 (AAH[10:0] = 0x1E5). To configure the vertical position of a 630×460 window (vertical start offset of 8 lines in the active data area):

1. First line of the layer window: WVSTPOS[10:0] should be programmed to 0xE (0x5 + 1 + 0x8);
2. Last line of the layer window: WVSPPOS[10:0] should be programmed to 0x1DA.

27.4.17 LTDC Layer Color Key Configuration Register (R32_LTDC_LCKCR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								CKRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				CKGREEN[7:0]				CKBLUE[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
[23:16]	CKRED[7:0]	RW	Color key red value.	0
[15:8]	CKGREEN[7:0]	RO	Color key green value.	0
[7:0]	CKBLUE[7:0]	RW	Color key blue value.	0

27.4.18 LTDC Layer Pixel Format Configuration Register (R32_LTDC_LPFCR)

Offset address: 0x44

31~16: Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved												PF[2:0]			

Bits	Name	Access	Description	Reset
[31:3]	Reserved	RO	Reserved.	0
[2:0]	PF[2:0]	RW	Pixel format: 000: ARGB8888; 001: RGB888; 010: RGB565; 011: ARGB1555; 100: ARGB4444; 101: L8 (8-bit Luminance); 110: AL44 (4-bit Alpha, 4-bit Luminance); 111: AL88 (8-bit Alpha, 8-bit Luminance).	0

27.4.19 LTDC Layer Default Color Configuration Register (R32_LTDC_LDCCR)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								DCRED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DCGREEN[7:0]				DCBLUE[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
[23:16]	DCRED[7:0]	RW	Default color red.	0
[15:8]	DCGREEN[7:0]	RW	Default color green.	0
[7:0]	DCBLUE[7:0]	RW	Default color blue.	0

27.4.20 LTDC Layer Color Frame Buffer Address Register (R32_LTDC_LCFBAR)

Offset address: 0x54

31~16: CFBADD[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFBADD[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	CFBADD[31:0]	RW	Color frame buffer start address.	0

27.4.21 LTDC Layer Color Frame Buffer Length Register (R32_LTDC_LCFBLR)

Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			CFBP[12:0]												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CFBLL[12:0]												

Bits	Name	Access	Description	Reset
[31:29]	Reserved	RO	Reserved.	0
[28:16]	CFBP[12:0]	RW	Color frame buffer pitch (in bytes): This field defines the increment from the start of a pixel line to the start of the next line.	0
[15:13]	Reserved	RO	Reserved.	0
[12:0]	CFBLL[12:0]	RW	Color frame buffer line length: Defines the length of one line of pixels (in bytes) + 31. The line length is calculated as: active width × bytes per pixel + 31.	0

27.4.22 LTDC Layer Color Frame Buffer Line Number Register (R32_LTDC_LCFBLNR)

Offset address: 0x5C

31~16: Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				CFBLNBR[10:0]											

Bits	Name	Access	Description	Reset
[31:12]	Reserved	RO	Reserved.	0
[10:0]	CFBLNBR[10:0]	RW		0

Bits	Name	Access	Description	Reset
			Frame buffer line number: This field defines the number of lines in the buffer, which corresponds to the active height.	

Note: The line number and line length settings define the amount of data fetched from each frame for each layer. If configured lower than the required bytes, a FIFO underrun interrupt will be generated (if enabled). On the other hand, the start address and pitch settings define the correct start position of each line in memory.

27.4.23 LTDC Layer CLUT Write Register (R32_LTDC_LCLUTWR)

Offset address: 0x60

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
CLUTADD[7:0]								RED[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GREEN[7:0]								BLUE[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	CLUTADD[7:0]	WO	Configures the CLUT address for each RGB value.	0
[23:16]	RED[7:0]	WO	Red value.	0
[15:8]	GREEN[7:0]	WO	Green value.	0
[7:0]	BLUE[7:0]	WO	Blue value.	0

Note: (1) The CLUT write register should only be configured during blanking periods or when the layer is disabled.

(2) The CLUT can be enabled and disabled through the R32_LTDC_LCR register.

(3) The CLUT is only meaningful for L8, AL44, and AL88 pixel formats.

Chapter 28: Addressable RGB (ARGB)

The chip integrates a set of ARGB (Addressable RGB) modules, supporting configurable high/low level timing and period for 0-code and 1-code output, and supporting DMA.

The ARGB module can implement single-data-line RGB light strip timing control, supporting single-wire RGB daisy-chaining. Compared with traditional RGB, it can support driving more different models of ARGB lamp control ICs, producing more lighting effects.

28.1 Main Features

- Supports configurable high/low level timing and period for 0-code and 1-code output
- Supports DMA

28.2 Register Description

Table 28-1: ARGB Related Register List

Name	Address	Description	Reset Value
R32_ARGB_CTLR	0x40025000	ARGB control register	0x00000000
R32_ARGB_DAT_ARR	0x40025004	ARGB 0/1 code period configuration register	0x000003FF
R32_ARGB_RST_ARR	0x40025008	ARGB RESET code width configuration	0x0001FFFF
R32_ARGB_CCRL	0x4002500C	ARGB 0-code high level T0H width configuration register	0x00000000
R32_ARGB_CCRH	0x40025010	ARGB 1-code high level T1H width configuration register	0x00000000
R32_ARGB_DAT_CYC	0x40025014	ARGB data refresh cycle length configuration register	0x00000000
R32_ARGB_DATAR	0x40025018	ARGB data register	0x00000000
R32_ARGB_STATR	0x4002501C	ARGB status register	0x00000020
R32_ARGB_CNTR	0x40025020	ARGB counter register	0x00000000

28.2.1 ARGB Control Register (R32_ARGB_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		TXEIE	HALFIE	BYTEIE	TCIE	RSTIE	FRHIE	Reserved		LSB		MODE	DMAEN		TE

Bits	Name	Access	Description	Reset
[31:14]	Reserved	RO	Reserved.	0
13	TXEIE	RW	Transmit buffer empty interrupt enable: 1: Enable; 0: Disable.	0

Bits	Name	Access	Description	Reset
12	HALFIE	RW	Data refresh cycle half-transmitted interrupt enable: 1: Enable; 0: Disable.	0
11	BYTEIE	RW	Byte transmit complete interrupt enable: 1: Enable; 0: Disable.	0
10	TCIE	RW	Transmit complete interrupt enable: 1: Enable; 0: Disable.	0
9	RSTIE	RW	RESET code transmit complete interrupt enable: 1: Enable; 0: Disable.	0
8	FRHIE	RW	Data refresh cycle end interrupt enable: 1: Enable; 0: Disable.	0
[7:4]	Reserved	RO	Reserved.	0
3	LSB	RW	Data transmit mode enable bit: 1: Low-bit data transmitted first; 0: High-bit data transmitted first.	0
2	MODE	RW	Transmit mode configuration bit: 1: Send data code first, then send RESET code; 0: Send RESET code first, then send data code.	0
1	DMAEN	RW	DMA enable: 1: Enable; 0: Disable.	0
0	TE	RW	Transmit enable: 1: Enable; 0: Disable.	0

28.2.2 ARGB 0/1 Code Period Configuration Register (R32_ARGB_DAT_ARR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved							DAT_ARR[9:0]								

Bits	Name	Access	Description	Reset
[31:10]	Reserved	RO	Reserved.	0
[9:0]	DAT_ARR[9:0]	RW	0/1 code period, unit: system clock cycles.	0x3FF

28.2.3 ARGB RESET Code Width Configuration Register (R32_ARGB_RST_ARR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved, RST_ARR[16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RST_ARR[15:0]															
Bits	Name	Access	Description	Reset											
[31:17]	Reserved	RO	Reserved.	0											
[16:0]	RST_ARR[16:0]	RW	Reset code period, unit: system clock cycles.	0x1FFFF											

28.2.4 ARGB 0-Code High Level T0H Width Configuration Register (R32_ARGB_CCRL)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								CCRL[9:0]							
Bits	Name	Access	Description											Reset	
[31:10]	Reserved	RO	Reserved.											0	
[9:0]	CCRL[9:0]	RW	0-code high level T0H width, unit: system clock cycles.											0	

28.2.5 ARGB 1-Code High Level T1H Width Configuration Register (R32_ARGB_CCRH)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								CCRH[9:0]							
Bits	Name	Access	Description											Reset	
[31:10]	Reserved	RO	Reserved.											0	
[9:0]	CCRH[9:0]	RW	1-code high level T1H width, unit: system clock cycles.											0	

28.2.6 ARGB Data Refresh Cycle Length Configuration Register (R32_ARGB_DAT_CYC)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DAT_CYC[15:0]															

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	DAT_CYC[15:0]	RW	ARGB data refresh cycle length, actual value is this field's value + 1, unit: bytes.	0

28.2.7 ARGB Data Register (R32_ARGB_DATAR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DATA[7:0]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	DATA[7:0]	RW	ARGB transmit data.	0

28.2.8 ARGB Status Register (R32_ARGB_STATR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BYTE_CNT[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			BIT_CNT[2:0]			STATE[1:0]		TXEIF	HALFIF	BYTEIF	TCIF	RSTIF	FRHIF		

Bits	Name	Access	Description	Reset
[31:16]	BYTE_CNT[15:0]	RO	Transmit byte counter.	0
[15:11]	Reserved	RO	Reserved.	0
[10:8]	BIT_CNT[2:0]	RO	Transmit bit counter.	0
[7:6]	STATE[1:0]	RO	State machine indication signal: 00: Idle; 01: Data transmit phase; 10: RESET code transmit phase; 11: Reserved.	0
5	TXEIF	RO	Transmit buffer empty interrupt flag, writing DATAR clears this flag.	1
4	HALFIF	RW1Z	Data refresh cycle half-transmitted interrupt flag, write 1 to clear.	0
3	BYTEIF	RW1Z	Byte transmit complete interrupt flag, write 1 to clear.	0
2	TCIF	RW1Z	Transmit complete interrupt flag, write 1 to clear.	0
1	RSTIF	RW1Z	RESET code transmit complete interrupt flag, write 1 to clear.	0
0	FRHIF	RW1Z	Data refresh cycle transmit complete interrupt flag, write 1 to clear.	0

28.2.9 ARGB Counter Register (R32_ARGB_CNTR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved, CNT[16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
[16:0]	CNT[16:0]	RO	ARGB base counter.	0

Chapter 29: Ethernet Controller (ETH)

The product provides an Ethernet controller (MAC) compliant with the IEEE 802.3-2002 standard, acting as the data link layer, and also integrates a 10 Mbps and 100 Mbps Ethernet PHY physical layer transceiver, supporting speed auto-negotiation, enabling Ethernet communication on a single chip. In application, combined with a TCP/IP protocol stack interface, network products can be developed.

29.1 Functional Description

29.1.1 MAC Features

- Supports 10M/100M Ethernet
- Supports full-duplex and half-duplex
- Supports automatic insertion of preamble and SFD
- Supports automatic insertion and verification of CRC
- Supports automatic calculation and verification of IP/ICMP/TCP/UDP protocol packet checksums
- Supports transmit frame length control
- Supports transmit inter-frame gap adjustment
- Supports VLAN frames
- Supports perfect address filtering and HASH filtering for receive frame addresses
- Supports transmit frame address filtering
- Supports multicast/broadcast frame receive control
- Supports promiscuous mode
- Supports magic frame and custom wake-up frame to wake up the microcontroller
- Supports a dedicated Ethernet wake-up interrupt entry
- Supports data loopback at the link layer

29.1.2 DMA Features

- Dedicated MAC DMA, minimizing CPU operations
- Supports byte-aligned access to RAM
- Manages transmit/receive buffers using descriptors
- Part of the transmit/receive status is fed back in the descriptors
- Descriptors and buffers not currently in use can be dynamically modified
- Supports manual stop or start
- Supports chained or ring-structured descriptor linking
- Supports multiple interrupt sources including transmit/receive completion interrupts

29.1.3 MMC Module Features

- Supports manual reset, stop, or freeze
- Supports multiple interrupts
- Multiple counters supporting various transmit and receive counting modes

29.1.4 PTP Module Features

- Supports IEEE 1588 protocol
- Supports automatic saving of the current time on transmit and receive
- Built-in 32-bit seconds timer and 32-bit signed sub-seconds timer
- Supports both fine and coarse adjustment for time correction
- Supports one interrupt source

29.1.5 Internal 10BASE-T/100BASE-TX Physical Layer Features

- Managed via the SMI interface
- Supports 10BASE-T and 100BASE-TX with auto-negotiation
- Supports power-down mode
- Supports Auto-MDIX to swap TX/RX, automatically identifying positive/negative signal lines
- Supports full-duplex and half-duplex operation
- Supports UTP CAT5E, CAT6 twisted pair cables, supporting up to 120 m transmission distance
- Supports WOL (Wake-on-LAN)
- Supports interrupt functionality

29.1.6 Frame Filtering

Using the MAC frame filter, perfect filtering or HASH filtering can be performed on the destination MAC address and source MAC address of received frames, and frames that do not pass filtering can be configured to be discarded, received, or marked in the receive descriptor. For details on this function, refer to the description of the MAC frame filter register (R32_ETH_MACFFR). The MAC has four built-in MAC address registers, of which MAC address register 0 is used by default to store the device's own MAC address, and the remaining three MAC address registers can be used for perfect filtering, comparing against the source or destination address of received frames. When R32_ETH_MACFFR:RA is set, all received frames will be received, but the corresponding status will be marked in the status field of the first word of the descriptor described later. If R32_ETH_MACFFR:PM is set, a similar effect occurs, but the status will not be marked. When R32_ETH_MACFFR:DAIF/SAIF is set, the result is inverted: frames that originally passed filtering will be discarded or marked, while frames that did not pass filtering will be transferred to the buffer in RAM.

29.1.6.1 Unicast Filtering

The MAC uses the HPF bit and the HU bit to determine whether unicast frames are HASH-filtered or perfect-address-filtered.

29.1.6.2 Multicast Filtering

The MAC uses the HPF bit and the HM bit to determine whether unicast frames are HASH-filtered or perfect-address-filtered. When the PAM bit is set, all multicast packets can pass through the filter.

29.1.6.3 Broadcast Filtering

By setting the BFD bit, the MAC can block all broadcast packets.

29.1.6.4 Source Address Filtering Selection

By setting the AE bit in the MAC address register, that MAC address register is enabled, and setting the SA bit determines whether the MAC address register is used as a source address sample or a destination address sample for comparison.

29.1.6.5 Summary

The filtering settings for the destination address and source address are shown in Table 29-1 and Table 29-2, respectively.

Table 29-1: Acceptance of received frame destination MAC address based on R32_ETH_MACFFR bit settings

Frame type	PM	HPF	HU	DAIF	HM	PAM	Effect
Broadcast	1	-	-	-	-	-	Pass
Broadcast	0	-	-	-	-	-	Not passed (BFD set)
Unicast	1	-	-	-	-	-	All frames pass
Unicast	0	-	0	0	-	-	Pass when perfect filter matches
Unicast	0	-	0	1	-	-	Not passed when perfect filter matches
Unicast	0	0	1	0	-	-	Pass when HASH filter matches
Unicast	0	0	1	1	-	-	Not passed when HASH filter matches
Unicast	0	1	1	0	-	-	Pass when perfect filter or HASH filter matches
Unicast	0	1	1	1	-	-	Not passed when perfect filter or HASH filter matches
Multicast	1	-	-	-	-	-	Pass
Multicast	-	-	-	-	-	1	Pass
Multicast	0	-	-	0	0	0	Pass when perfect filter matches
Multicast	0	0	-	0	1	0	Pass when HASH filter matches
Multicast	0	1	-	0	1	0	Pass when perfect filter or HASH filter matches
Multicast	0	-	-	1	0	0	Not passed when perfect filter matches
Multicast	0	0	-	1	1	0	Not passed when HASH filter matches
Multicast	0	1	-	1	1	0	Not passed when perfect filter or HASH filter matches

Table 29-2: Acceptance of received frame source MAC address based on R32_ETH_MACFFR bit settings

Frame type	RA	SAIF	SAF	Effect
Unicast	1	-	-	All frames pass
Unicast	0	0	0	Pass when perfect filter matches; mark but do not discard frames that do not pass
Unicast	0	1	0	Not passed when perfect filter matches; mark but do not discard frames that do not pass
Unicast	0	0	1	Pass when perfect filter matches; discard frames that do not pass
Unicast	0	1	1	Not passed when perfect filter matches; discard frames that do not pass

Note: "-" indicates the bit setting is don't-care.

29.1.7 MMC

The Management Counters (MMC) module primarily counts various indicators of frame receive/transmit status and MAC operation status. It can be configured to generate interrupts. In general, the "good" frame receive counter (MMCRGUFCR) can be used to obtain the number of intact frames currently received, the "good" frame transmit counter (MMCTGFCR) can be used to obtain the number of successfully transmitted frames, and the receive CRC error frame counter (MMCRFCECR) can be used to check whether there are frames received with CRC errors.

Users need to understand a concept: what constitutes a "good" frame. Under normal conditions, as long as a frame starts transmission, its frame length meets the requirement (with auto-padding enabled), and automatic CRC calculation is enabled, it is a "good" frame. For a received frame, as long as its CRC is correct, the frame length is within the Ethernet frame length range, the frame length is consistent with the value in the length field (if the length/type field indicates length), or there is no misalignment, it is considered a "good" received frame.

29.1.8 PMT

29.1.8.1 Overview

The PMT (Power Management) module's function is mainly to wake up the microcontroller from low-power mode via Ethernet. The Gigabit Ethernet controller supports two types of frames to wake up the system: magic frames and (remote) wake-up frames. When the Ethernet receives these two types of frames, since the microcontroller is in low-power mode, a receive interrupt is not necessarily generated even if the frame is valid. However, if the frame passes the MAC's magic frame and wake-up frame recognition, a PMT interrupt will be generated. The PMT interrupt is independent of the Ethernet interrupt; by querying the PMT control and status register, the type of Ethernet frame that caused the interrupt can be determined.

29.1.8.2 Magic Frame

The magic packet (also known as magic packet) defined by AMD is a commonly used Ethernet frame to wake up a microcontroller. It has a fixed and special frame format: through frame filtering, the target network card can receive it; it can be encapsulated as a broadcast frame, containing 6 consecutive bytes of all-high (0xFF), followed by 16 repetitions of the target network card's MAC address. This combination can exist anywhere in the frame, so the magic frame can be encapsulated as a MAC frame, IP packet, or even a UDP packet. The following is the payload format of a magic frame.

```
xx xx xx xx xx xx xx xx (preceding data is unrestricted, may even be absent) — ff ff ff ff ff ff
84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02
84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02
84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02 84 c2 e4 01 02 02
84 c2 e4 01 02 02 xx xx xx xx xx xx (some network cards require a password appended at the end of
the magic frame).
```

29.1.8.3 Wake-up Frame

Since the magic frame format has limitations, the (remote) wake-up frame format can be user-defined. Any Ethernet frame that passes through the remote wake-up frame filter register group (ETH_MACRWUFFR) is considered a wake-up frame. Users define wake-up frames that meet their needs by configuring the remote wake-up frame filter register group. Note that the remote wake-up frame filter register group has only one address entry; users can write data into it through 8 consecutive write operations, and read back the previously written data through 8 consecutive read operations.

The structure of the remote wake-up frame filter register group is shown in the table below:

Table 29-3: Structure of the remote wake-up frame filter register group

Filter register 0	Byte mask of filter register 0							
Filter register 1	Byte mask of filter register 1							
Filter register 2	Byte mask of filter register 2							
Filter register 3	Byte mask of filter register 3							
Filter register 4	Reserved	Reserved	Reserved	Reserved	Filter 3 command	Filter 2 command	Filter 1 command	Filter 0 command
Filter register 5	Filter 3 offset	Filter 2 offset	Filter 1 offset	Filter 0 offset				
Filter register 6	Filter 1 CRC-16	Filter 0 CRC-16						
Filter register 7	Filter 3 CRC-16	Filter 2 CRC-16						

As can be seen from the table above, the four domains — byte mask, command, offset, and filter — work together to determine whether a frame is a remote wake-up frame. In practice, four different frames satisfying the requirements can be configured.

The most significant bit of the 32-bit byte mask must be 0. Bits [30:0] correspond to the first 31 bytes of data as defined by the offset domain. If a bit is set to 1, the corresponding byte participates in the CRC-16 calculation, with a maximum of 31 bytes participating in the calculation.

The most significant bit of the 4-bit command domain indicates the frame type to which it applies: 1 means valid only for multicast addresses, 0 means valid only for unicast addresses. Bits 2 and 1 of the command domain are reserved. Bit 0 is the enable bit; when set high, this group of filters is enabled.

The offset domain indicates how many bytes from the beginning of the frame header to start calculating the CRC-16 value. The minimum value is 12. If the offset domain is 12, CRC-16 calculation starts from the 13th byte (the parameter model uses CRC-16-IBM with default parameters).

The filter stores the user's expected CRC result value. The MAC compares the CRC-16 value it calculates with the value in this domain. If they match, the frame is identified as a (remote) wake-up frame. If the wake-up frame interrupt and PMT interrupt are enabled, a PMT interrupt will also be generated.

Additionally, according to the bit definitions of the PMT control and status register, if the GU bit is set, unicast frames that pass frame filtering will also be considered wake-up frames.

29.1.9 IEEE1588 PTP

29.1.9.1 PTP Principle and Implementation

The IEEE 1588 standard defines a precise protocol for obtaining time, with the goal of achieving time synchronization within 10 microseconds of error. Although the NTP protocol could already achieve time synchronization at the 200-microsecond level, this level was insufficient to meet the time synchronization requirements in industrial automation. Therefore, the Network Precision Clock Synchronization Committee drafted the PTP protocol, which was approved by the IEEE Standards Committee at the end of 2002 as the IEEE 1588 standard.

Implementing the PTP protocol requires both the master and slave to accurately record the times at which the MAC receives and transmits Ethernet frames, which requires both the master and slave to have their own high-precision time counters. Then, master and slave devices supporting PTP perform a time synchronization procedure through which the slave can determine and correct its time difference with the master. The figure below shows the master-slave time synchronization procedure.

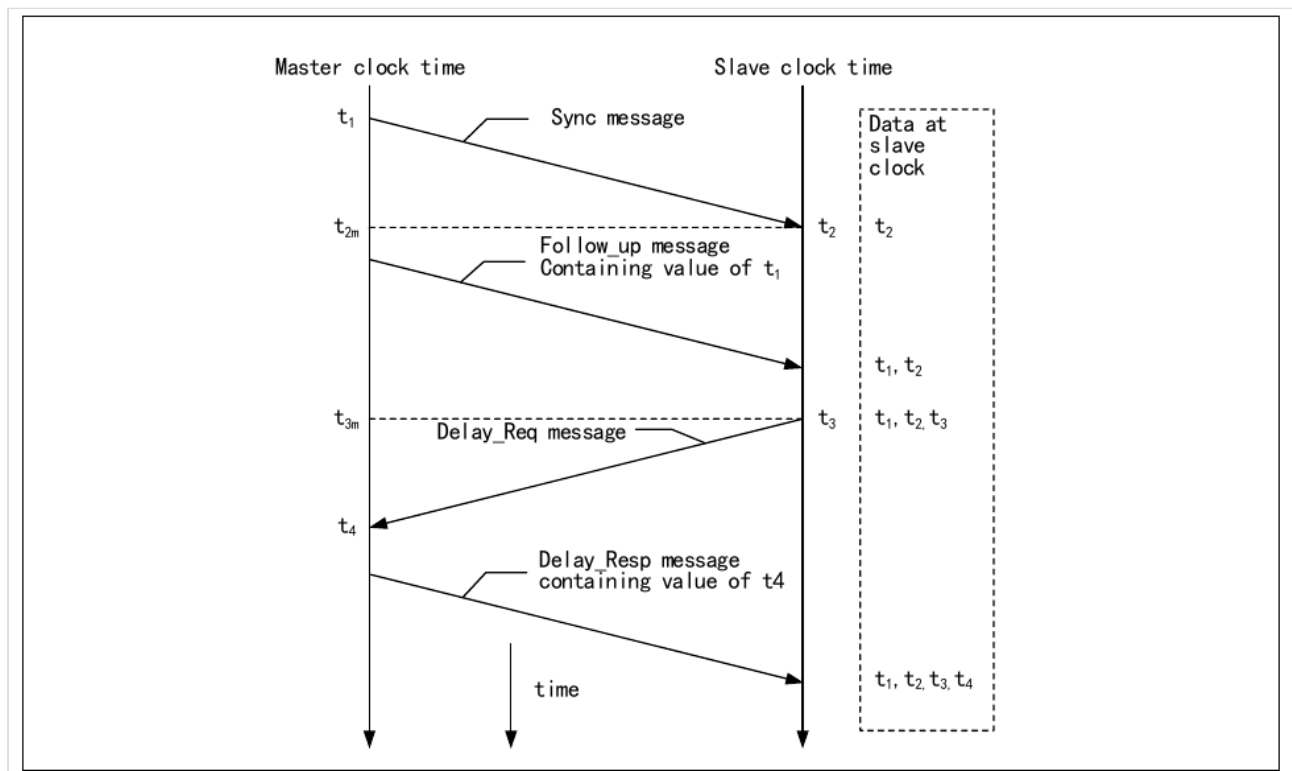


Figure 29-1: IEEE1588 PTP protocol synchronization message timing diagram

The master sends a Sync message to the slave; the slave receives this message and records the local time t_2 when the Sync message is received.

The master sends a Follow_up message to the slave, containing the master time t_1 when the master sent the Sync message.

The slave sends a Delay_Req message to the master; the slave records the send time t_3 .

The master sends a Delay_Resp message to the slave, containing the receive time t_4 of the Delay_Req message.

In actual use, the master sends a Sync message once every two seconds, and each Sync message can be considered the Follow_up message of the previous Sync message, both carrying the send time of the previous Sync message. Through this process, the slave can determine the network delay time T_{delay} from the master network to the slave network, and then calculate the time offset between the master time and the slave time.

$$T_{delay} = \frac{(t_2 - t_1) + (t_4 - t_3)}{2}$$

Any time sent by the master minus the offset gives the master's time.

The PTP synchronization procedure is generally implemented using the UDP protocol, though users can also implement it with a custom protocol. PTP is highly dependent on the delay stability of the internal network.

29.1.9.2 Local Time Update and Correction

To implement PTP on the master, the device needs a local high-precision time counter, at least accurate to the nanosecond level. The PTP module of the microcontroller's Gigabit Ethernet controller has a 32-bit seconds counter and a 31-bit sub-seconds counter. When the sub-seconds counter overflows, the seconds counter increments, so the local time resolution can reach approximately 0.46 nanoseconds.

Local time can be updated in two ways: coarse adjustment and fine adjustment. In coarse adjustment, the update timing is determined externally. When a time update is needed, the time update bit (PTPTSCR:TSSTU) in the timestamp control register is set, and the microcontroller's PTP module will subtract or add the value of the timestamp update register (TSHUR, TSLUR) to/from the seconds counter and sub-seconds counter. Coarse adjustment is a simpler and more convenient time update mechanism, but with poorer precision.

Fine adjustment is the more commonly used time update method. Its update procedure is as follows:

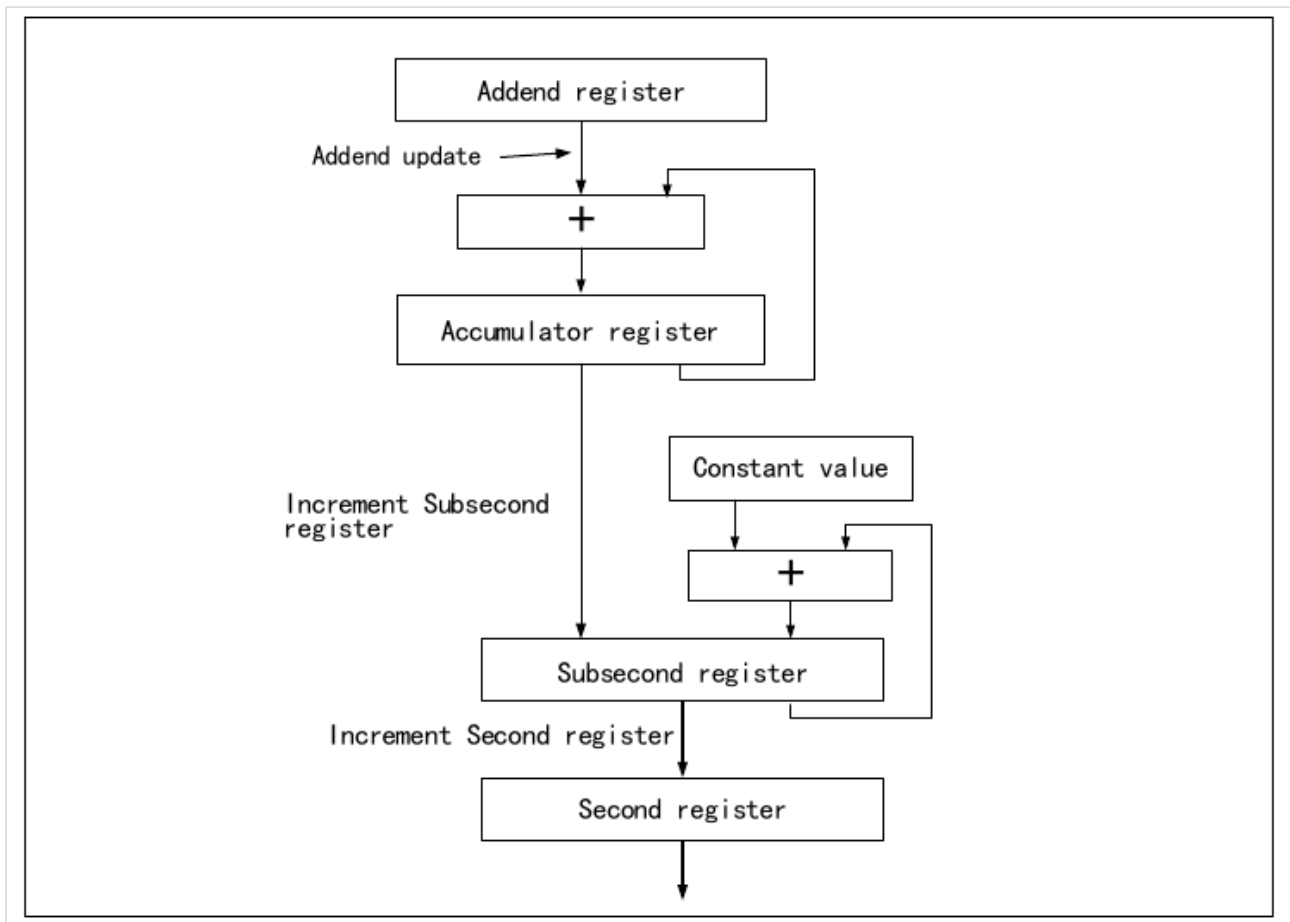


Figure 29-2: Procedure for updating time using fine adjustment

Using fine adjustment mode requires a clear understanding of the system main frequency and the fine adjustment procedure. Unlike coarse adjustment, the fine adjustment update event timing is triggered by an overflow of the accumulator (32-bit, not listed in the register list; shown as "Accumulator register" in the figure). The accumulator increments by the value of the addend register (PTPTSAR, shown as "Addend register" in the figure) on every system main clock cycle. Once it overflows, a time update event is generated: the value of the sub-second increment register (PTPSSIR, shown as "Constant value" in the figure) is added to the sub-seconds counter (Subsecond register), completing the time update. When the sub-seconds counter overflows, the seconds counter increments. In practice, the time for one sub-seconds counter increment is $1/(2^{31}) = 0.46566128730...$ nanoseconds. The user must ensure that the time for the accumulator to overflow equals the value of the sub-second increment register multiplied by the time for one sub-seconds counter increment.

Local time correction is relatively simple: by setting the time correction bit (PTPTSCR:TSSTI) in the timestamp control register, the seconds counter and sub-seconds counter values will be replaced by the values of the timestamp update counter (TSHUR, TSLUR).

29.1.10 DMA Operation

29.1.10.1 Overview

The advantage of the Ethernet transceiver is its high speed and large throughput. To maintain this advantage, CPU intervention during the Ethernet frame receive/transmit process must be minimized. This is where the Ethernet transceiver dedicated DMA is used.

The DMA used by Ethernet is 32-bit. It is managed by the CPU through two types of data structures: traditional control and status registers, and receive/transmit descriptors. Since the DMA is 32-bit wide, the base address of the receive/transmit descriptor queue in RAM must be 4-byte aligned. The receive/transmit buffers have no alignment requirement.

29.1.10.2 DMA Descriptor

Users typically operate traditional peripherals by writing control bits to registers and reading status registers to obtain the peripheral's status and return information. These registers exist independently of the core, SRAM, and non-volatile memory; they are physically present and can be called "hardware registers." Traditional communication peripherals, such as USART or SPI, have a data register to temporarily store transmitted/received data, and a set of DMA to uniformly save all received data to a specific address space.

However, the Ethernet transceiver, with its extremely high data transfer speed, enormous data throughput, and unique data frame organization, operates differently from traditional communication transceivers. The Ethernet transceiver receives large amounts of data as densely as possible. It must store the received data stream into separate memory spaces in units of frames, autonomously completing receive and transmit operations so that the CPU can read and process this data with minimal operations, freeing the corresponding memory space, and ensuring that the DMA controller and CPU do not conflict over memory access rights. The individual size of the buffer opened in memory to temporarily store Ethernet frames is generally set to the maximum packet length of an Ethernet frame, which IEEE 802.3 specifies as 1518 bytes (including the source/destination hardware address, length/type field, and CRC32 checksum field, totaling 18 bytes). The number of Ethernet frame transmit/receive buffers is determined by the user based on the actual interaction frequency and microcontroller memory resources.

Ethernet frame buffers are organized in a queue-like manner. In the receive direction, Ethernet frames are enqueued as the DMA writes them into memory via the MAC, and dequeued as the CPU reads them. In the transmit direction, Ethernet frames are enqueued as the CPU writes them into memory, and dequeued as the DMA reads them and pushes them into the transmit FIFO. The queue depth is the number of buffers. Unlike ordinary communication peripherals, the start address and number of Ethernet frame buffers are not fixed in a specific register, but are managed by a special type of data structure. This data structure is stored in memory; a single such data structure unit manages one buffer, storing the buffer's start address, length, the settings the DMA controller needs when invoking this buffer, the transmit status written back by the DMA upon transmit completion, and the address of the next such data structure. This data structure serves the role of traditional communication peripheral control/status registers, but is physically located in memory, so it can be called a "software register," formally known as a "DMA descriptor."

DMA descriptors are divided into transmit and receive types, each with a fixed format. The storage structure of DMA descriptors comes in two forms: one is the linked-list form, where the fourth word of each descriptor holds the address of the next descriptor, and the DMA controller reads the next descriptor directly from TDes3/RDes3. The other is the ring structure, where all descriptors must be tightly packed, and the DMA controller takes the next descriptor from the position immediately following the current descriptor. When the DMA controller detects the TER/RER bit set in TDes4/RDes4, it takes the next descriptor from the beginning of the descriptor list. The descriptor array address must be 4-byte aligned, and a single descriptor is 16 bytes in size.

Figure 29-3 describes a buffer allocation scheme for the ring structure and the chained structure, respectively, for reference when users allocate space.

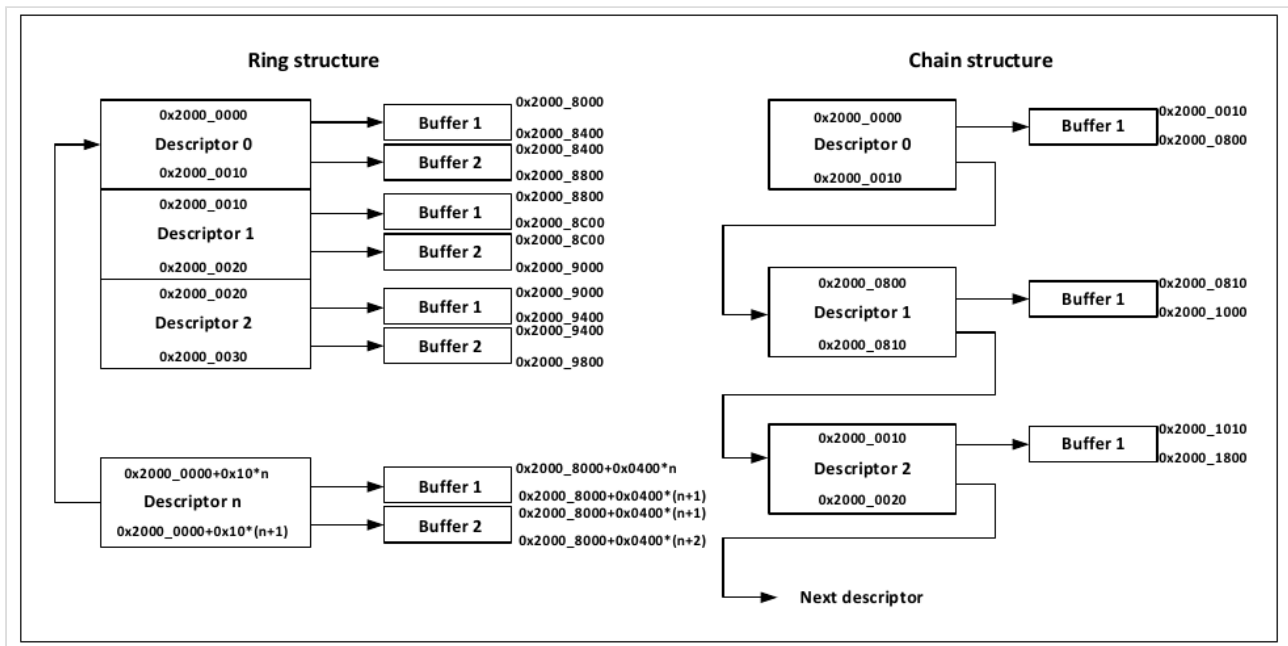


Figure 29-3: Ring structure and chain structure buffer allocation schemes

29.1.10.3 Transmit DMA Descriptor

Table 29-4 shows the structure of the transmit descriptor.

Table 29-4: Structure of the transmit descriptor

	31	30:26	25	23:20	19:18	17	16:0
TDes0	OWN (31)	CTRL (30:26)	TTSE (25)	Control (23:20)	Reserved (19:18)	TTSS (17)	Status (16:0)
TDes1	Reserved (31:29)	Buffer 2 byte count (28:16)		Reserved (15:13)	Buffer 1 byte count (12:0)		
TDes2	Buffer 1 address / timestamp low						
TDes3	Buffer 2 address / next descriptor address / timestamp high						

As can be seen from the table above, the transmit descriptor consists of four 32-bit words: TDes0, TDes1, TDes2, and TDes3. TDes0 is used for control and returning transmit status, TDes1 indicates the transmit length, TDes2 indicates the transmit buffer location or returns the low part of the transmit timestamp, and TDes3 indicates the address of the second transmit buffer (when TCH is not set) or the address of the next descriptor (when TCH is set). When timestamp is enabled, it returns the high part of the IEEE1588 timestamp after transmission. The descriptions of each 32-bit word are as follows.

Table 29-5: Bit definitions of TDes0

31	30	29	28	27	26	25	24	23:22	21	20	19:18	17	16
OWN	IC	LS	FS	DC	DP	TTE	Res	CIC	TER	TCH	Res	TSS	IHE
15	14	13	12	11	10	9	8	7	6:3	2	1	0	
ES	JT	FF	IPE	LCA	NC	LCO	EC	VF	CC	ED	UF	DB	

Table 29-5: TDes0 field descriptions

Bits	Name	Access	Description	Reset
31	OWN	R/W	Descriptor ownership bit. This bit indicates who owns this descriptor. 0: this descriptor is owned by the CPU, which can modify the descriptor's	-

Bits	Name	Access	Description	Reset
			value; 1: this descriptor is owned by the DMA, and the CPU has no permission to modify it. When this bit is 0, after the CPU completes operations on the descriptor and buffer, the CPU sets it to 1. When this bit is 1, after the DMA completes operations on the descriptor and buffer, the DMA automatically writes 0. This completes the handover of descriptor and buffer operations between user software and hardware.	
30	IC	R/W	Transmit completion interrupt enable bit. When set, after the current frame is transmitted, the transmit interrupt flag bit (ETH_DMASR:TS) will be set.	-
29	LS	R/W	Last segment indicator. When set, this descriptor's buffer contains the end portion of the frame.	-
28	FS	R/W	First segment indicator. When set, this descriptor's buffer contains the beginning portion of the frame.	-
27	DC	R/W	Disable automatic CRC calculation. When set, the DMA controller will not calculate the CRC32 checksum of the Ethernet frame, and no value will be appended to the end of the frame. This bit is only valid when the FS bit is set. Additionally, the DP bit setting takes priority over this bit.	-
26	DP	R/W	Disable automatic padding. When set, the DMA controller will not add automatic padding for Ethernet frames shorter than 64 bytes. When this bit is 0, the DMA will automatically add padding and CRC checksum for Ethernet frames shorter than 64 bytes, regardless of whether DC is set.	-
25	TTE	R/W	Timestamp transmit enable. When ETH_PTPTSCR:TSE is set, setting this bit enables the IEEE1588 timestamp function for the Ethernet frame indicated by the current descriptor. This bit is only valid when FS is set.	-
24	Reserved	-	Unused.	-
[23:22]	CIC	R/W	Checksum insertion control field. 00: disable checksum insertion; 01: enable IP header checksum calculation and insertion only; 10: reserved; 11: enable IP header checksum and payload checksum calculation and insertion, enable pseudo-header checksum calculation.	-
21	TER	R/W	Transmit descriptor end flag. When set by the user, it indicates to the DMA controller that the current transmit descriptor is the last one in the transmit descriptor array. The DMA controller will read the first descriptor in the transmit descriptor array next time.	-
20	TCH	R/W	Next descriptor address valid indicator. When set, the second address is the address of the next descriptor rather than the address of the next buffer. When set, the TBS2 field value has no effect. This bit is only valid when FS is set. The TER bit takes priority over this bit.	-
[19:18]	Reserved	-	Unused.	-
17	TSS	R	Transmit timestamp capture status. The DMA controller sets this bit to indicate that the transmit timestamp has been captured and stored in TDes6 and TDes7.	-
16	IHE	R	IP header error status. The DMA controller checks the IP header against the received data: for IPv4, the DMA controller checks whether the header length field is correct; for IPv6, the DMA controller checks whether the header is 40 bytes. Additionally, the IP protocol type must be consistent with the type/length field in the Ethernet frame.	-

Bits	Name	Access	Description	Reset
15	ES	R	Error summary. If an error is encountered during frame transmission, the DMA controller sets this bit. The ES bit is set when any of the following bits is set: UF [TDES0:1] data underflow error; IPE [TDES0:12] IP data error; FF [TDES0:13] frame flush; JT [TDES0:14] jabber timeout; IHE [TDES0:16] IP header error.	-
14	JT	R	Jabber timeout. When set, it indicates that a jabber timeout error occurred at the MAC transmit side. This bit is only set when the JD bit (ETH_MACCR:22) is not set.	-
13	FF	R	Frame flush. When set, it indicates that the DMA controller flushed the frame from the FIFO due to a CPU command.	-
12	IPE	R	IP header error indicator. The MAC compares the total packet length in the IPv4 or IPv6 header of received TCP/UDP/ICMP packets with the actual packet length; if they do not match, this bit is set.	-
11	LCA	R	Loss of carrier indicator. When set, it indicates that a carrier was lost during frame transmission, i.e., the CSR signal was in an invalid state. This bit only takes effect in half-duplex mode.	-
10	NC	R	No carrier indicator. This bit indicates that the physical layer carrier sense signal was invalid during frame transmission. This bit only takes effect in half-duplex mode.	-
9	LCO	R	Late collision indicator. This bit indicates that a collision occurred after the preamble was transmitted. This bit only takes effect in half-duplex mode.	-
8	EC	R	Excessive collision indicator. This bit indicates that more than 16 collisions occurred during frame transmission. If the RD bit of MACCR is set, this bit indicates only a single collision. This bit only takes effect in half-duplex mode.	-
7	VF	R	VLAN bit. This bit is set when transmitting a VLAN frame.	-
[6:3]	CC	R	Collision counter field. This field indicates the number of collisions that occurred during frame transmission. Invalid when EC is set. This field only takes effect in half-duplex mode.	-
2	ED	R	Excessive deferral indicator. When set, it indicates that when the DC bit of MACCR is set, frame transmission failed due to deferral exceeding 24288 bits. This bit only takes effect in half-duplex mode.	-
1	UF	R	Data underflow error. When the DMA controller finds the buffer empty while fetching data from the specified RAM during transmission, transmission enters a suspended state, and this bit and the relevant bits of the DMASR register are set.	-
0	DB	R	Deferral indicator. When set, it indicates that the transmission process failed due to carrier occupation. This bit only takes effect in half-duplex mode.	-

Table 29-6: Bit definitions of TDes1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TBS2											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				TBS1											

Table 29-6: TDes1 field descriptions

Bits	Name	Access	Description	Reset
[31:29]	Reserved	-	Unused.	-
[28:16]	TBS2	R/W	Size of transmit buffer 2.	-
[15:13]	Reserved	-	Unused.	-
[12:0]	TBS1	R/W	Size of transmit buffer 1.	-

Table 29-7: Bit definitions of TDes2

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TBAD1/TTSL															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TBAD1/TTSL															

Table 29-7: TDes2 field descriptions

Bits	Name	Access	Description	Reset
[31:0]	TBAD1/TTSL	R/W	The entire 32 bits of TDes2 are used as a single unit to store the address of transmit buffer 1. In IEEE1588 mode enabled, after transmission is completed, TDes2 is also used to store the timestamp returned by the MAC, and the MAC clears the OWN bit (TDes0:31). This field stores the low 32 bits of the timestamp.	-

Table 29-8: Bit definitions of TDes3

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TDAD2/TTSH															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TDAD2/TTSH															

Table 29-8: TDes3 field descriptions

Bits	Name	Access	Description	Reset
[31:0]	TDAD2/TTSH	R/W	This field is used to store the address of the next descriptor. When TCH is not set, this field stores the address of transmit buffer 2. In IEEE1588 mode enabled, after transmission is completed, TDes3 is used to store the timestamp returned by the MAC, and the MAC clears the OWN bit (TDes0:31). This field stores the high 32 bits of the timestamp.	-

29.1.10.4 Receive DMA Descriptor

Table 29-9: Structure of the receive descriptor

	31	30:0				
RDes0	OWN (31)	Status (30:0)				
RDes1	Control (31)	Reserved (30:29)	Buffer 2 byte count (28:16)	RER (15:14)	13	Reserved (12:0)
RDes2	Buffer 1 address, timestamp low					
RDes3	Buffer 2 address, next descriptor address, timestamp high					

As can be seen from the figure above, the receive descriptor also consists of four 32-bit words. The first 32-bit word mainly returns the receive status, the second 32-bit word contains the received data length, the third 32-bit word defines the receive buffer address or returns the low part of the

timestamp, and the fourth 32-bit word holds the second buffer address (when RCH is not set), the next descriptor address (when RCH is set), or returns the high part of the timestamp.

The meaning of each bit in the receive descriptor words is as follows:

Table 29-10: Bit definitions of RDes0

31	30	29:16														
OWN	AFM	FL														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
ES	DE	SAF	LE	OE	VLAN	FS	LS	IPHCE	LCO	PT	RWT	RE	DE	CE	PCE	

Table 29-10: RDes0 field descriptions

Bits	Name	Access	Description	Reset
31	OWN	R/W	Descriptor ownership bit. This bit indicates who owns this descriptor. 0: this descriptor is owned by the CPU, which can modify the descriptor's value; 1: this descriptor is owned by the DMA, and the CPU has no permission to modify it. When this bit is 0, after the CPU completes operations on the descriptor and buffer, the CPU sets it to 1. When this bit is 1, after the DMA completes operations on the descriptor and buffer, the DMA automatically writes 0. This completes the handover of descriptor and buffer operations between user software and hardware.	-
30	AFM	R	Destination address filter fail flag. If the received frame did not pass the MAC's destination address filter, this flag is set.	-
[29:16]	FL	R	Frame length field. This field is valid when the ES bit [RDes0:15] is 0. When the LS bit [RDes0:8] is 1, this field indicates the length of the frame received by the DMA controller, including CRC. When the LS bit is 0, this field indicates the cumulative length transferred to memory by the DMA controller so far, in bytes.	-
15	ES	R	Error summary. When the MAC detects any of the following errors, it sets this bit: CE [RDes0:1] CRC error; RE [RDes0:3] receive error; RWT [RDes0:4] watchdog timeout; IPHCE [RDes0:7] giant frame (note: when IPHCE is reported, it is necessary to distinguish whether it is a giant frame or an IP header error); OE [RDes0:11] overflow error; DE [RDes0:14] descriptor error.	-
14	DE	R	Descriptor error bit. When set, it indicates that the frame could not fit in the buffer indicated by the descriptor and was truncated, and the DMA does not own the next descriptor. This bit is only valid when the LS bit [RDes0:8] is set.	-
13	SAF	R	Source address filter fail flag. When set, it indicates that the frame did not pass the MAC's source address filter.	-
12	LE	R	Length error bit. When set, it indicates that the actual received frame length does not match the length indicated in the Ethernet type/length field. This bit is only valid when the FT bit [RDes0:5] is set.	-
11	OE	R	Overflow error bit. When set, it indicates that the received frame was corrupted due to receive FIFO overflow.	-
10	VLAN	R	VLAN tag bit. When set, it indicates that the received frame is a VLAN frame.	-
9	FS	R	First descriptor indicator. When set, this descriptor contains the beginning of the frame.	-
8	LS	R	Last descriptor indicator. When set, this descriptor contains the end of the frame.	-

Bits	Name	Access	Description	Reset
7	IPHCE	R	IP header checksum error flag. When set, it indicates an error in the IPv4 or IPv6 header. Possible causes: 1. The protocol indicated by the Ethernet frame type/length field does not match the actual IP version; 2. The IP header checksum is incorrect; 3. The length indicated in the IP header is incorrect.	-
6	LCO	R	Late collision indicator. When set, it indicates that a late collision occurred. This bit only takes effect in half-duplex mode.	-
5	FT	R	Frame type indicator. When 1, the received frame is an Ethernet type-encapsulated frame (RFC 894). When 0, the received frame is an IEEE 802.3 type-encapsulated frame (RFC 1042). When the frame length is less than 14 bytes, this bit is invalid. Note: The special meaning of FT is described in Table 29-11.	-
4	RWT	R	Receive watchdog timeout flag. When set, it indicates that a watchdog timeout occurred while receiving the current frame, and the current frame was truncated.	-
3	RE	R	Receive error flag. When set, it indicates that the RX_ERR signal was active while RX_DV was valid during frame reception.	-
2	DE	R	Dribble bit error. When set, it indicates that the frame received by the MAC has a length that is not an integer multiple of 8 bits, possibly due to missed clock cycles.	-
1	CE	R	CRC error. When set, it indicates that the received frame has a CRC checksum error. This bit is only valid when the LS bit [RDes0:8] is set.	-
0	PCE	R	Payload checksum error. When set, it indicates that the TCP/UDP/ICMP packet received by the MAC does not match the value indicated in its checksum field.	-

As can be seen, the MAC's receive process includes a checksum verification mechanism. In fact, the Ethernet frame layer (data link layer), network layer (IPv4/IPv6), and transport layer (TCP/UDP/SCTP) all specify their own data lengths and perform some form of verification on data correctness. Bits 0, 5, and 7 of RDes0 all mention data checksum verification, summarized in the table below.

Table 29-11: Relationship between RDes0:7/5/0 values and received frame status

RDes0:5 (FT)	RDes0:7 (IPHCE)	RDes0:0 (PCE)	Frame status
1	0	0	IP type frame, no IP header or payload checksum errors detected
1	0	1	IP type frame, payload checksum error
1	1	0	IP type frame, IP header checksum error
1	1	1	IP type frame, IP header checksum error, payload checksum error
0	0	0	IEEE 802.3 type-encapsulated frame (RFC 1042)
0	0	1	IP type frame, no IP header checksum error detected; payload checksum not checked due to unsupported
0	1	1	Not an IP type frame (e.g., ARP frame)
0	1	0	Reserved

Table 29-12: Bit definitions of RDes1

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DIC	Reserved		RBS2												
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RER	RCH	Res	RBS1												

Table 29-12: RDes1 field descriptions

Bits	Name	Access	Description	Reset
31	DIC	R/W	Disable receive completion interrupt setting.	-
[30:29]	Reserved	-	Unused.	-
[28:16]	RBS2	R/W	Receive buffer 2 size.	-
15	RER	R/W	End of receive descriptor list flag. Indicates that the current descriptor is the last descriptor. The DMA controller returns to the descriptor list base address register (ETH_DMARDLAR) to fetch the next descriptor.	-
14	RCH	R/W	Next receive descriptor address valid bit. When set, the last 32-bit word contains the address of the next receive descriptor; otherwise it is the address of the second buffer.	-
13	Reserved	-	Unused.	-
[12:0]	RBS1	R/W	Receive buffer 1 size.	-

Table 29-13: Bit definitions of RDes2

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RBAD1/RTSL															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RBAD1/RTSL															

Table 29-13: RDes2 field descriptions

Bits	Name	Access	Description	Reset
[31:0]	RBAD/ RTSL	R/W	The entire 32 bits of RDes2 are used as a single unit to store the receive buffer address. In IEEE1588 mode enabled, after reception is completed, RDes2 is also used to store the timestamp returned by the MAC, and the MAC clears the OWN bit [RDes0:31].	-

Table 29-14: Bit definitions of RDes3

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDAD2/RTSH															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDAD2/RTSH															

Table 29-14: RDes3 field descriptions

Bits	Name	Access	Description	Reset
[31:0]	RDAD/ RTSH	R/W	When RCH is set, this field stores the address of the next descriptor; otherwise it stores the address of the second buffer. In IEEE1588 mode enabled, after reception is completed, RDes3 is used to store the timestamp returned by the MAC, and the MAC clears the OWN bit [RDes0:31]. This field stores the high 32 bits of the timestamp.	-

29.1.10.5 Data Cache Alignment

Since the transmit/receive buffers and transmit/receive descriptors are both invoked by the DMA controller and physically reside in RAM space, and the DMA is 32-bit, the start address of the transmit/receive descriptor queues must be placed at 4-byte aligned positions. However, this alignment requirement is not mandatory for buffers.

For efficiency, it is most appropriate to receive an entire Ethernet frame in one buffer. Setting the buffer to 1518 bytes can accommodate the maximum standard Ethernet frame, including the source/destination address field, length/type field, data padding field, and CRC checksum field. Note that the maximum frame length of a tagged VLAN frame is 4 bytes longer than a standard Ethernet frame, i.e., 1522 bytes. If the user wants to achieve 4-byte alignment, a buffer can be 1524 bytes.

29.1.10.6 DMA Transmit/Receive Configuration

The DMA is responsible for automatically transferring and pushing transmit and receive data. However, if a fatal error is encountered, the DMA will stop operating and update the DMA status register. The user needs to re-initialize the DMA and manually start it to resume operation.

29.1.10.6.1 Transmit DMA Configuration

The steps for the DMA controller to establish the operation mechanism are as follows:

1. Set up the transmit buffer queue, fill the content to be sent into the transmit buffers. Set up the transmit descriptor queue, fill in all fields and bits of the transmit descriptors, set the OWN bit, and hand over management of the transmit descriptors to the DMA controller. Register the descriptor start address in the DMATDLAR register.
2. Set the ST bit (DMAOMR:13) to start the DMA.
3. In run mode, the DMA automatically reads the transmit descriptor content and pushes data to the transmit FIFO according to the address and length indicated. After completion, the DMA reads the immediately following next transmit descriptor according to the chained structure for the next transmission. When the DMA controller detects that the OWN bit is not set, resulting in no access to the transmit descriptor, or other normal errors, it terminates the transfer and sets the TBUS bit (DMASR:2) or other bits (for errors not caused by OWN being 0) and the NIS bit (DMASR:16).
4. A single frame is not allowed to span multiple descriptors; one frame must be clearly associated with one descriptor and one buffer.
5. If the MAC has IEEE1588 PTP mode enabled, after the DMA controller pushes data to the FIFO, the MAC writes the transmit timestamp to TDes2 and TDes3 and resets the OWN bit.
6. After transmitting a frame, if the transmit descriptor has enabled the transmit completion interrupt (TDes1:31 set), the DMA controller sets the transmit completion interrupt flag bit (DMASR:0) and then proceeds to fetch the next transmit descriptor.

The figure below shows the default transmit flow.

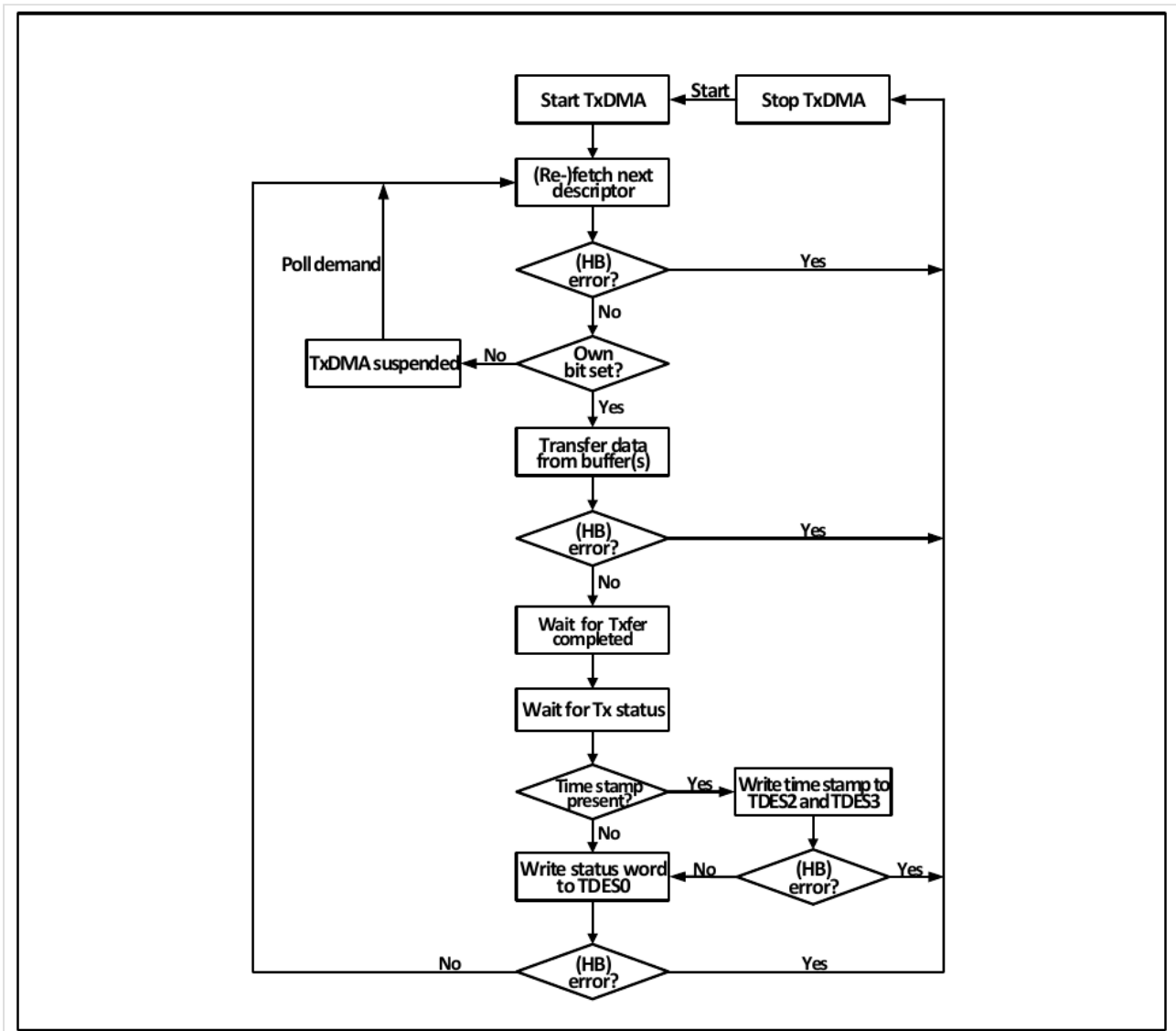


Figure 29-4: Transmit flow

29.1.10.6.2 Receive DMA Configuration

The steps to establish the DMA receive operation mechanism are as follows:

1. Set up the receive buffer queue and descriptor queue, configure all fields and bits of the receive descriptors. Register the descriptor start address in the DMARDLAR register. Set the OWN bit to hand over descriptor usage rights to the DMA controller.
2. Set the SR bit to start the receive process.
3. While the receive operation mechanism is running, the DMA controller fetches the next descriptor, checks the receive descriptor configuration, and when the FIFO receives the next frame, performs multiple checks such as filtering and identification on the frame content, transfers the frame data to the buffer specified by the descriptor, writes the descriptor status field, fetches the next receive descriptor, and reports the receive completion interrupt. If the frame does not pass filtering, it will be marked or discarded. If the frame has a checksum error, CRC error, or is too short, it will be marked. If the frame is too long, it may be truncated or a receive watchdog timeout error may be reported. If the DMA controller encounters a fatal error such as an unavailable receive descriptor, it will stop the receive process; the user should pay special attention to this.

4. The user must enable at least one receive completion interrupt. In the Ethernet interrupt handler, restore the used receive descriptors to standby state to ensure that the receive process can run continuously. The user can pass the pending data buffer address out in the interrupt handler, or handle some exceptional events that interrupt the receive process.
5. If PTP timestamp is enabled, when the DMA controller transfers data and writes the descriptor status field, it also writes the current timestamp into the last two words of the descriptor. The user should read out the timestamp promptly and restore the buffer address and next descriptor address originally written in the last two words.

The figure below shows the default receive operation mechanism:

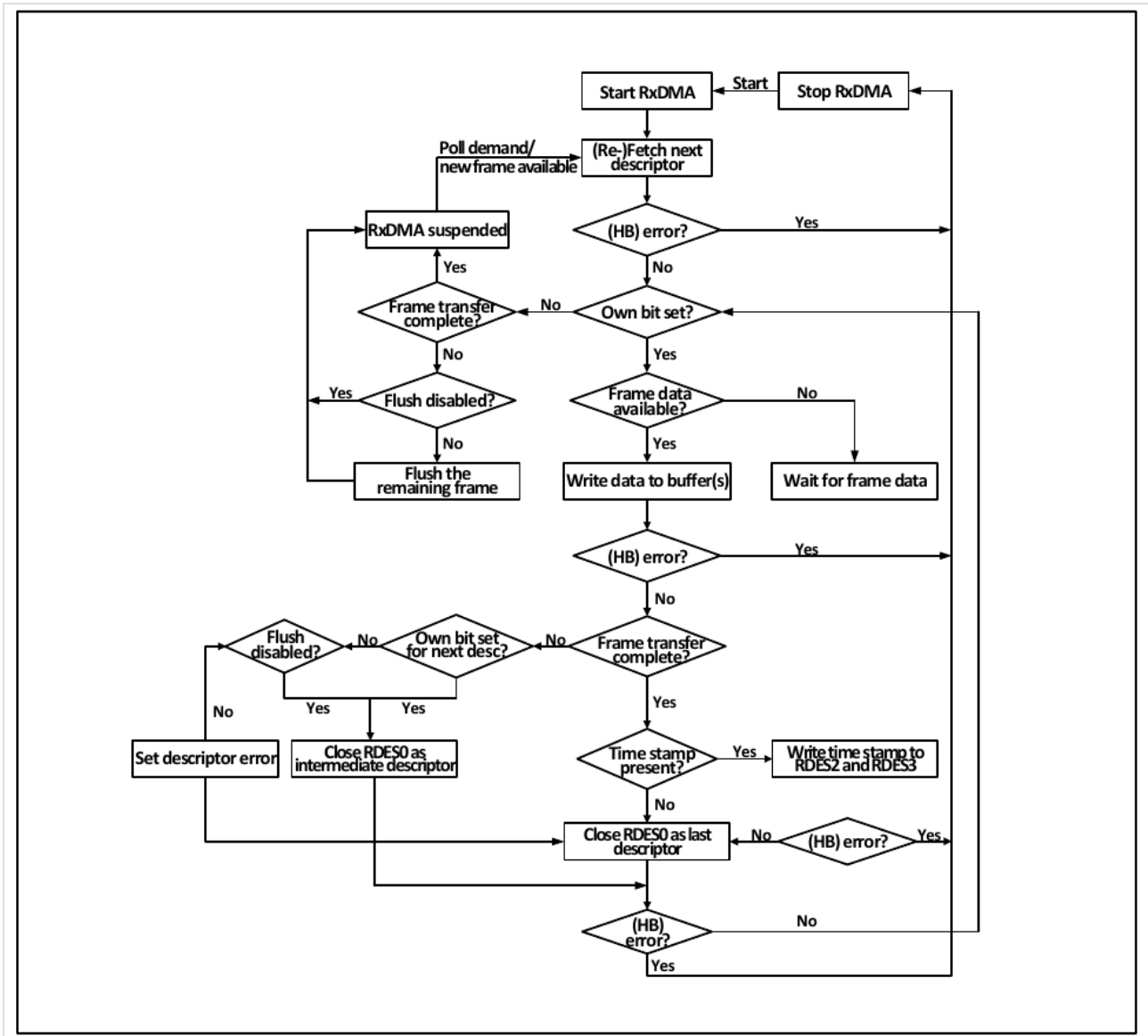


Figure 29-5: Receive flow

29.1.11 Interrupts

The Ethernet transceiver has two interrupt vectors: one for Ethernet wake-up events and the other for normal transmit/receive events. When a wake-up frame or magic frame is detected, the Ethernet wake-up event is triggered. Normal Ethernet transmit/receive interrupt events include DMA interrupts and ETH interrupts.

29.2 Register Description

MAC control related register address mapping.

The physical base address of the MAC control related registers is: 0x4002 8000

Table 29-15: Ethernet transceiver related register list

Name	Address	Description	Reset Value
R32_ETH_MACCR	0x40028000	MAC control register	0x00000000
R32_ETH_MACFFR	0x40028004	MAC frame filter register	0x00000000
R32_ETH_MACHTHR	0x40028008	Hash table register high	0x00000000
R32_ETH_MACHTLR	0x4002800C	Hash table register low	0x00000000
R32_ETH_MACMIAR	0x40028010	MII address register	0x00000000
R32_ETH_MACMIIDR	0x40028014	MII data register	0x00000000
R32_ETH_MACFCR	0x40028018	MAC flow control register	0x00000000
R32_ETH_MACVLAN	0x4002801C	VLAN tag register	0x00000000
R32_ETH_MACRWUFR	0x40028028	Remote wake-up frame filter register	0x00000000
R32_ETH_MACPMTCSR	0x4002802C	PMT control and status register	0x00000000
R32_ETH_MACSR	0x40028038	MAC interrupt status register	0x00000000
R32_ETH_MACIMR	0x4002803C	MAC interrupt mask register	0x00000000
R32_ETH_MACA0HR	0x40028040	MAC address 0 high 32 bits	0x8000FFFF
R32_ETH_MACA0LR	0x40028044	MAC address 0 low 32 bits	0xFFFFFFFF
R32_ETH_MACA1HR	0x40028048	MAC address 1 high 32 bits	0x0000FFFF
R32_ETH_MACA1LR	0x4002804C	MAC address 1 low 32 bits	0xFFFFFFFF
R32_ETH_MACA2HR	0x40028050	MAC address 2 high 32 bits	0x0000FFFF
R32_ETH_MACA2LR	0x40028054	MAC address 2 low 32 bits	0xFFFFFFFF
R32_ETH_MACA3HR	0x40028058	MAC address 3 high 32 bits	0x0000FFFF
R32_ETH_MACA3LR	0x4002805C	MAC address 3 low 32 bits	0xFFFFFFFF
R32_ETH_PHY_CR	0x40028080	PHY configuration register	0x40000001

MMC control related register address mapping. Note: addresses are non-contiguous.

The physical base address of the MMC control related registers is: 0x4002 8100

Table 29-16: MMC related register list

Name	Address	Description	Reset Value
R32_ETH_MMCCR	0x40028100	MMC control register	0x00000000
R32_ETH_MMCRIR	0x40028104	MMC receive interrupt register	0x00000000
R32_ETH_MMCTIR	0x40028108	MMC transmit interrupt register	0x00000000
R32_ETH_MMCRIMR	0x4002810C	MMC receive interrupt mask register	0x00000000
R32_ETH_MMCTIMR	0x40028110	MMC transmit interrupt mask register	0x00000000
R32_ETH_MMCTGFSCCR	0x4002814C		0x00000000

Name	Address	Description	Reset Value
		MMC transmit good frames single collision counter register	
R32_ETH_MMCTGFMSCCR	0x40028150	MMC transmit good frames more than single collision counter register	0x00000000
R32_ETH_MMCTGFCR	0x40028168	MMC transmit good frames counter register	0x00000000
R32_ETH_MMCRFCECR	0x40028194	MMC receive CRC error frame counter register	0x00000000
R32_ETH_MMCRFAECR	0x40028198	MMC receive alignment error frame counter register	0x00000000
R32_ETH_STATE	0x400281A0	Ethernet state machine register	0x00000000
R32_ETH_MMCRGUFCR	0x400281C4	MMC receive good unicast frame counter register	0x00000000

IEEE 1588 (PTP) related register address mapping.

The physical base address of the IEEE 1588 (PTP) related registers is: 0x4002 8700

Table 29-17: IEEE 1588 (PTP) related register list

Name	Address	Description	Reset Value
R32_ETH_PTPTSCR	0x40028700	PTP timestamp control register	0x00000000
R32_ETH_PTPSSIR	0x40028704	PTP sub-second increment register	0x00000000
R32_ETH_PTPTSHR	0x40028708	PTP timestamp register high	0x00000000
R32_ETH_PTPTSLR	0x4002870C	PTP timestamp register low	0x00000000
R32_ETH_PTPTSHUR	0x40028710	PTP timestamp update register high	0x00000000
R32_ETH_PTPTSLUR	0x40028714	PTP timestamp update register low	0x00000000
R32_ETH_PTPTSAR	0x40028718	PTP timestamp addend register	0x00000000
R32_ETH_PTPTTHR	0x4002871C	PTP target time register high 32 bits	0x00000000
R32_ETH_PTPTTLR	0x40028720	PTP target time register low 32 bits	0x00000000

DMA related register address mapping. Note: addresses are non-contiguous.

The physical base address of the DMA related registers is: 0x4002 9000

Table 29-18: DMA related register list

Name	Address	Description	Reset Value
R32_ETH_DMABMR	0x40029000	DMA bus mode register	0x00000001
R32_ETH_DMATPDR	0x40029004	DMA transmit poll demand register	0x00000000
R32_ETH_DMARPDR	0x40029008	DMA receive poll demand register	0x00000000
R32_ETH_DMARDLAR	0x4002900C	DMA receive descriptor list address register	0x00000000
R32_ETH_DMATDLAR	0x40029010	DMA transmit descriptor list address register	0x00000000
R32_ETH_DMASR	0x40029014	DMA status register	0x00000000
R32_ETH_DMAOMR	0x40029018	DMA operation mode register	0x00000000
R32_ETH_DMAIER	0x4002901C	DMA interrupt enable register	0x00000000
R32_ETH_DMAMFBOCR	0x40029020	DMA missed frame and buffer overflow counter register	0x00000000
R32_ETH_DMACTDR	0x40029048	DMA current transmit descriptor register	0x00000000

Name	Address	Description	Reset Value
R32_ETH_DMACHRDR	0x4002904C	DMA current receive descriptor register	0x00000000
R32_ETH_DMACHTBAR	0x40029050	DMA current transmit buffer address register	0x00000000
R32_ETH_DMACHRBAR	0x40029054	DMA current receive buffer address register	0x00000000

The offset addresses of the internal PHY registers are used in the SMI interface. All PHY related registers are configured through the R32_ETH_MACMIAR and R32_ETH_MACMIIDR registers.

Internal 10M/100M physical layer related register addresses.

Table 29-19: Internal 10M/100M physical layer related register list

Name	Offset Address	Description	Reset Value
R16_BMCR	0x00	Basic control register	0x3100
R16_BMSR	0x01	Basic status register	0x7849
R16_PHYIDR1	0x02	PHY identifier register 1	0x7371
R16_PHYIDR2	0x03	PHY identifier register 2	0x9011
R16_ETH_ANAR	0x04	Auto-negotiation advertisement register	0x01E1
R16_ETH_ANALPAR	0x05	Auto-negotiation link partner ability register	0x0001
R16_ANER	0x06	Auto-negotiation expansion register	0x0004
R16_PAGE_SEL	0x1F	Page selection register	0x0000

PAGE0 related register addresses.

Table 29-20: PAGE0 related register list

Name	Offset Address	Description	Reset Value
R16_PHY_STATUS1	0x1A	PHY status register	0x0020
R16_INTERRUPT_IND	0x1E	Interrupt indication register	0x0000

PAGE7 related register addresses.

Table 29-21: PAGE7 related register list

Name	Offset Address	Description	Reset Value
R16_INTERRUPT_MASK	0x13	Interrupt/WOL/LED function register	0x0030

29.2.1 MAC Control Related Registers

29.2.1.1 MAC Control Register (R32_ETH_MACCR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved							WD	JD	Reserved			IFG[2:0]		Reserved	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		FES	Reserved		LM	DM	IPCO	Reserved		APCS	Reserved		TE	RE	Reserved

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
23	WD	RW	Watchdog setting bit: 1: MAC disables the watchdog, can receive Ethernet frames up to 16384 bytes; 0: MAC enables the watchdog, can only receive Ethernet frames up to 2048 bytes; the excess is truncated.	0
22	JD	RW	Jabber setting bit: 1: MAC disables the Jabber timer, can transmit up to 16384 bytes of Ethernet frames; 0: If the user attempts to transmit Ethernet frames exceeding 2048 bytes, the MAC will disable the transmitter.	0
[21:20]	Reserved	RO	Reserved.	0
[19:17]	IFG[2:0]	RW	Inter-frame gap setting field. This sets the minimum time gap between two transmitted frames. 000: 96 bit times; 001: 88 bit times; 010: 80 bit times; 011: 72 bit times; 100: 64 bit times; 101: 56 bit times; 110: 48 bit times; 111: 40 bit times.	0
[16:15]	Reserved	RO	Reserved.	0
14	FES	RO	Fast Ethernet. This bit indicates the speed of the Fast Ethernet (MII) mode: 1: 100 Mbit/s; 0: 10 Mbit/s.	0
[13:12]	Reserved	RO	Reserved.	0
11	DM	RW	Duplex mode enable bit: Setting this bit enables full-duplex mode, allowing simultaneous transmit and receive.	0
10	IPCO	RW	IPv4 checksum offload enable bit: 1: Enables IPv4 checksum checking; the MAC controller performs checksum verification on TCP, UDP, and ICMP headers; 0: Disables the receive-side IPv4 checksum verification; the corresponding PCE and PHCE flags are always 0. See the receive descriptor bit definitions.	0
9	RD	RW	Retry disable.	0
8	Reserved	RO	Reserved.	0
7	APCS	RW	Automatic padding & CRC stripping enable bit: 1: MAC removes Pad/FCS fields from the frame only when the length/type field value is less than or equal to 1500 bytes. If the length/type field value is greater than or equal to 1501 bytes, the frame content is not modified; 0: MAC does not modify the frame content.	0
[6:4]	Reserved	RO	Reserved.	0
3	TE	RW	Transmit enable bit: 1: Enables the MAC transmitter;	0

Bits	Name	Access	Description	Reset
			0: After transmitting the current frame, the MAC disables the transmitter and does not transmit any more frames.	
2	RE	RW	Receive enable bit: 1: Enables the MAC receiver; 0: After receiving the current frame, the MAC disables the receiver and does not receive any more frames.	0
[1:0]	Reserved	RO	Reserved.	0

29.2.1.2 MAC Frame Filter Register (R32_ETH_MACFFR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RA	Reserved														
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					HPF	SAF	SAIF	PCF[1:0]		BFD	PAM	DAIF	HM	HU	PM

Bits	Name	Access	Description	Reset
31	RA	RW	Receive all bit: 1: MAC forwards all received frames to the receive queue, regardless of whether they pass the filter; 0: MAC only forwards frames that pass the filter to the receive queue.	0
[30:11]	Reserved	RO	Reserved.	0
10	HPF	RW	Hash or perfect filter selection bit: 1: When HM or HU is set, frames matching either the HASH filter or the perfect filter pass the address filter; 0: When HM or HU is set, only frames matching the HASH filter pass the address filter.	0
9	SAF	RW	Source address filter selection bit: 1: MAC directly discards frames that fail the source MAC address filter; 0: MAC marks frames that fail the source MAC address filter.	0
8	SAIF	RW	Source address filter inverse bit: 1: If the source address of the received frame matches the source address enabled in the MAC address register, it is considered to have failed the source address filter; 0: If the source address of the received frame does not match the source address enabled in the MAC address register, it is considered to have failed the source address filter.	0
[7:6]	PCF[1:0]	RW	Pass control frames control field. 00/01: MAC does not forward any control frames to the application; 10: MAC forwards all control frames to the application, including control frames that fail the address filter; 11: MAC only forwards control frames that pass the address filter.	0
5	BFD	RW	Broadcast frame receive control bit: 1: Discards all broadcast frames; 0: Receives all broadcast frames.	0
4	PAM	RW		0

Bits	Name	Access	Description	Reset
			Pass all multicast frames control bit: 1: All multicast frames pass the address filter; 0: Whether multicast frames pass the filter depends on the value of HM.	
3	DAIF	RW	Destination MAC address filter inverse control bit: 1: For multicast and unicast frames, the filter result is inverted before being applied; 0: The filter result is applied normally.	0
2	HM	RW	Multicast frame filter mode selection bit: 1: Performs HASH address filtering; 0: Performs perfect address filtering.	0
1	HU	RW	Unicast frame filter mode selection bit: 1: Performs HASH address filtering; 0: Performs perfect address filtering.	0
0	PM	RW	Promiscuous mode enable bit. All frames pass the address filter without marking the filter result.	0

29.2.1.3 Hash Table Registers (R32_ETH_MACHTHR, R32_ETH_MACHTLR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTH[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTH[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	HTH[31:0]	RW	Hash table high 32 bits.	0

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTL[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTL[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	HTL[31:0]	RW	Hash table low 32 bits.	0

29.2.1.4 MII Address Register (R32_ETH_MACMIAR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PA[4:0]				MR[4:0]				Reserved				CR[2:0]		MW	MB

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:11]	PA[4:0]	RW	Physical layer address field. The user writes the physical layer address to be accessed into this field.	0
[10:6]	MR[4:0]	RW	Physical layer register address field. The user writes the register address to be accessed into this field.	0
5	Reserved	RO	Reserved.	0
[4:2]	CR[2:0]	RW	Clock range setting field. The CR clock range options determine the HCLK frequency and are used to determine the MDC clock frequency. 000: HCLK (60–90 MHz), MDC clock (HCLK/42); 001: HCLK (90–120 MHz), MDC clock (HCLK/62); 010: HCLK (20–35 MHz), MDC clock (HCLK/16); 011: HCLK (35–62 MHz), MDC clock (HCLK/26); 100: HCLK (150–200 MHz), MDC clock (HCLK/102); 101: HCLK (120–150 MHz), MDC clock (HCLK/74); Others: Reserved.	0
1	MW	RW	Read/write setting bit: 1: Write operation to the physical layer; 0: Read operation to the physical layer.	0
0	MB	W1	MII busy flag bit: This bit is set by the user, indicating the command for hardware to start a read or write operation. During the read/write, the physical address, register address, and data fields should be kept unchanged. After the hardware clears this bit, the operation is complete.	0

29.2.1.5 MII Data Register (R32_ETH_MACMIIDR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MD[15:0]															

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	MD[15:0]	RW	MII operation data field. This field holds the data to be read from or written to the physical layer via the MII interface.	0

29.2.1.6 MAC Flow Control Register (R32_ETH_MACFCR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PT[15:0]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										UPFD		RFCE		TFCE	FCB

Bits	Name	Access	Description	Reset
[31:16]	PT[15:0]	RW	Pause time field. This field is used as the value for the pause time field, in units of the time to transmit 64 bytes on the current MII interface.	0
[15:4]	Reserved	RO	Reserved.	0
3	UPFD	RW	Unicast pause frame detect bit: 1: MAC also detects whether the pause frame matches the unicast address defined in MAC address register 0; 0: MAC only receives pause frames with the unique address defined by the protocol specification.	0
2	RFCE	RW	Receive flow control enable bit: 1: MAC parses pause frames and disables the transmitter for a period of time; 0: MAC does not parse pause frames.	0
1	TFCE	RW	Transmit flow control enable bit: 1: MAC enables transmit flow control and can send pause frames; 0: MAC disables transmit flow control and does not send pause frames.	0
0	FCB	W1	Flow control busy flag bit. Setting this bit sends a pause frame; it is cleared by hardware after the transmission is complete. When operating on the entire MACFCR register, ensure that the FCB bit is 0.	0

29.2.1.7 VLAN Tag Register (R32_ETH_MACVLANT)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Reserved															VLANT	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
VLANTI[15:0]																

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
16	VLANT	RW	Tag comparison control bit: 1: Only the [11:0] bits of bytes 15 and 16 of the VLAN frame are compared with the corresponding bits of the VLANTI field; 0: All 16 bits of bytes 15 and 16 of the VLAN frame are compared with the VLANTI field.	0
[15:0]	VLANTI[15:0]	RW	Tag comparison sample field. According to IEEE 802.1, bits [15:13] of the VLAN frame are user priority, [12] is the canonical format indicator, and bits [11:0] are the VLAN identifier field. If VLANTI is all zeros, the MAC no longer cares about bytes 15 and 16 of the VLAN frame; when bytes 13 and 14 are 0x8100 (note endianness), the frame is identified as a VLAN frame.	0

29.2.1.8 Remote Wake-Up Frame Filter Register (R32_ETH_MACRWUFR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RWUFR[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RWUFR[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	RWUFR[31:0]	RW	The wake-up frame filter register is actually eight separate registers. Eight consecutive read operations read all registers, and eight consecutive write operations write to all eight registers. The bit descriptions of these eight registers are shown in the table below.	0											

Table 29-22: Wake-up frame filter register contents

Filter register	Content
Filter register 0	Byte mask of filter 0
Filter register 1	Byte mask of filter 1
Filter register 2	Byte mask of filter 2
Filter register 3	Byte mask of filter 3
Filter register 4	Filter 3 command Filter 2 command Filter 1 command Filter 0 command
Filter register 5	Filter 3 offset Filter 2 offset Filter 1 offset Filter 0 offset
Filter register 6	Filter 1 CRC-16 Filter 0 CRC-16
Filter register 7	Filter 3 CRC-16 Filter 2 CRC-16

29.2.1.9 PMT Control and Status Register (R32_ETH_MACPMTCSR)**Offset address:** 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WFFRP		Reserved													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved					GU	Reserved	PHYR	WFR	MPR	Reserved	PHYE	WFE	MPE	PD	

Bits	Name	Access	Description	Reset
31	WFFRP	W	Wake-up frame filter register pointer reset. Setting this bit clears all MACRWUFR registers. This bit is automatically cleared after one clock cycle.	0
[30:10]	Reserved	RO	Reserved.	0
9	GU	RW	Global unicast bit. Setting this bit causes the MAC to treat all unicast frames that pass the filter as wake-up frames.	0
8	Reserved	RO	Reserved.	0
7	PHYR	RZ	PHY wake-up frame flag bit. When a PHY wake-up frame is received, this bit is set. Read clears automatically.	0
6	WFR	RZ	Wake-up frame received flag bit. When a wake-up frame is received, this bit is set. Read clears automatically.	0
5	MPR	RZ	Magic packet received flag bit. When a magic packet is received, this bit is set. Read clears automatically.	0
4	Reserved	RO	Reserved.	0
3	PHYE	WO		0

Bits	Name	Access	Description	Reset
			PHY PMEB enable bit. Setting this bit allows the PHY to generate a PMT event when a wake-up frame is received. This bit always reads as 0.	
2	WFE	RW	Wake-up frame enable bit. Setting this bit allows a PMT event to be generated upon receiving a wake-up frame.	0
1	MPE	RW	Magic packet enable bit. Setting this bit allows a PMT event to be generated upon receiving a magic packet.	0
0	PD	W	Power-down control bit. Setting this bit causes the MAC to enter power-down mode: all other frames are discarded until the PHY receives a wake-up frame or the MAC receives a wake-up frame or magic packet. After wake-up, the MAC automatically clears this bit. Before setting this bit, PHYE or WFE or MPE must be set.	0

29.2.1.10 MAC Interrupt Status Register (R32_ETH_MACSR)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				TSTS		Reserved		MMCTS	MMCRS	MMCS	PMTS	Reserved			

Bits	Name	Access	Description	Reset
[31:10]	Reserved	RO	Reserved.	0
9	TSTS	RZ	Timestamp trigger interrupt flag bit. When the PTP system clock reaches the set time, this flag bit is set.	0
[8:7]	Reserved	RO	Reserved.	0
6	MMCTS	R	MMC transmit interrupt flag bit. When any interrupt is generated in the MMCTIR register of the MMC register group, this bit is set. When the MMCTIR register in the MMC register group is all cleared, this bit is also cleared.	0
5	MMCRS	R	MMC receive interrupt flag bit. When any interrupt is generated in the MMCRIR register of the MMC register group, this bit is set. When the MMCRIR register in the MMC register group is all cleared, this bit is also cleared.	0
4	MMCS	R	MMC status flag bit. When MMCTS or MMCRS is set, this bit is set; when MMCTS and MMCRS are both cleared, this bit is cleared.	0
3	PMTS	R	PMT status flag bit. In power-down mode, if the MAC receives a magic packet or wake-up frame that wakes up the MAC, this bit is set; this bit is cleared after clearing the WFR and MPR bits.	0
[2:0]	Reserved	RO	Reserved.	0

29.2.1.11 MAC Interrupt Mask Register (R32_ETH_MACIMR)

Offset address: 0x3C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						TSTIM	Reserved			PMTIM	Reserved				

Bits	Name	Access	Description	Reset
[31:10]	Reserved	RO	Reserved.	0
9	TSTIM	RW	Timestamp interrupt mask bit. Setting this bit disables timestamp interrupts.	0
[8:4]	Reserved	RO	Reserved.	0
3	PMTIM	RW	PMT interrupt mask bit. Setting this bit masks PMT interrupts.	0
[2:0]	Reserved	RO	Reserved.	0

29.2.1.12 MAC Address 0 High 32 Bits (R32_ETH_MACA0HR)

Offset address: 0x40

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MO	Reserved														

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA0H[15:0]															

Bits	Name	Access	Description	Reset
31	MO	RO	Always 1.	1
[30:16]	Reserved	RO	Reserved.	0
[15:0]	MACA0H[15:0]	RW	High 16 bits of the MAC address, i.e., bits [47:32].	0xFFFF

29.2.1.13 MAC Address 0 Low 32 Bits (R32_ETH_MACA0LR)

Offset address: 0x44

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA0L[31:16]															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA0L[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	MACA0L[31:0]	RW	Low 32 bits of the MAC address, i.e., bits [31:0]. Generally, the address in MAC address register 0 is the MAC's own address.	0xFFFFFFFF

29.2.1.14 MAC Address 1 High 32 Bits (R32_ETH_MACA1HR)

Offset address: 0x48

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AE	SA	MBC[5:0]						Reserved							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA1H[15:0]															

Bits	Name	Access	Description	Reset
31	AE	RW	Address filter enable bit: 1: The address filter uses MAC address 1 for perfect filtering; 0: The address filter ignores MAC address 1.	0
30	SA	RW	Address role selection bit: 1: MAC address 1 is used to compare with the source address of received frames; 0: MAC address 1 is used to compare with the destination address of received frames.	0
[29:24]	MBC[5:0]	RW	Mask byte control field. Each bit of MBC corresponds to masking a byte of MAC address 1, used to prevent a byte of MAC address 1 from participating in the comparison with the source or destination address of received frames. R32_ETH_MACA1HR[29] set masks MACA1H[15:8]; R32_ETH_MACA1HR[28] set masks MACA1H[7:0]; R32_ETH_MACA1HR[27] set masks MACA1L[31:24]; R32_ETH_MACA1HR[26] set masks MACA1L[23:16]; R32_ETH_MACA1HR[25] set masks MACA1L[15:8]; R32_ETH_MACA1HR[24] set masks MACA1L[7:0].	0
[23:16]	Reserved	RO	Reserved.	0
[15:0]	MACA1H[15:0]	RW	High 16 bits of the MAC address, i.e., bits [47:32].	0xFFFF

29.2.1.15 MAC Address 1 Low 32 Bits (R32_ETH_MACA1LR)

Offset address: 0x4C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA1L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA1L[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	MACA1L[31:0]	RW	Low 32 bits of the MAC address, i.e., bits [31:0].	0xFFFFFFFF

29.2.1.16 MAC Address 2 High 32 Bits (R32_ETH_MACA2HR)

Offset address: 0x50

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AE	SA	MBC[5:0]						Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA2H[15:0]															

Bits	Name	Access	Description	Reset
31	AE	RW	Address filter enable bit: 1: The address filter uses MAC address 2 for perfect filtering; 0: The address filter ignores MAC address 2.	0
30	SA	RW	Address role selection bit: 1: MAC address 2 is used to compare with the source address of	0

Bits	Name	Access	Description	Reset
			received frames; 0: MAC address 2 is used to compare with the destination address of received frames.	
[29:24]	MBC[5:0]	RW	Mask byte control field. Each bit of MBC corresponds to masking a byte of MAC address 2, used to prevent a byte of MAC address 2 from participating in the comparison with the source or destination address of received frames. R32_ETH_MACA2HR[29] set masks MACA2H[15:8]; R32_ETH_MACA2HR[28] set masks MACA2H[7:0]; R32_ETH_MACA2HR[27] set masks MACA2L[31:24]; R32_ETH_MACA2HR[26] set masks MACA2L[23:16]; R32_ETH_MACA2HR[25] set masks MACA2L[15:8]; R32_ETH_MACA2HR[24] set masks MACA2L[7:0].	0
[23:16]	Reserved	RO	Reserved.	0
[15:0]	MACA2H[15:0]	RW	High 16 bits of the MAC address, i.e., bits [47:32].	0xFFFF

29.2.1.17 MAC Address 2 Low 32 Bits (R32_ETH_MACA2LR)

Offset address: 0x54

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA2L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA2L[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	MACA2L[31:0]	RW	Low 32 bits of the MAC address, i.e., bits [31:0].	0xFFFFFFFF

29.2.1.18 MAC Address 3 High 32 Bits (R32_ETH_MACA3HR)

Offset address: 0x58

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
AE	SA	MBC[5:0]						Reserved							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA3H[15:0]															

Bits	Name	Access	Description	Reset
31	AE	RW	Address filter enable bit: 1: The address filter uses MAC address 3 for perfect filtering; 0: The address filter ignores MAC address 3.	0
30	SA	RW	Address role selection bit: 1: MAC address 3 is used to compare with the source address of received frames; 0: MAC address 3 is used to compare with the destination address of received frames.	0
[29:24]	MBC[5:0]	RW	Mask byte control field. Each bit of MBC corresponds to masking a byte of MAC address 3, used to prevent a byte of MAC address 3	0

Bits	Name	Access	Description	Reset
			from participating in the comparison with the source or destination address of received frames. R32_ETH_MACA3HR[29] set masks MACA3H[15:8]; R32_ETH_MACA3HR[28] set masks MACA3H[7:0]; R32_ETH_MACA3HR[27] set masks MACA2L[31:24]; R32_ETH_MACA3HR[26] set masks MACA2L[23:16]; R32_ETH_MACA3HR[25] set masks MACA3L[15:8]; R32_ETH_MACA3HR[24] set masks MACA3L[7:0].	
[23:16]	Reserved	RO	Reserved.	0
[15:0]	MACA3H[15:0]	RW	High 16 bits of the MAC address, i.e., bits [47:32].	0xFFFF

29.2.1.19 MAC Address 3 Low 32 Bits (R32_ETH_MACA3LR)

Offset address: 0x5C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MACA3L[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MACA3L[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	MACA3L[31:0]	RW	Low 32 bits of the MAC address, i.e., bits [31:0].	0xFFFFFFFF

29.2.1.20 PHY Configuration Register (R32_ETH_PHY_CR)

Offset address: 0x80

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PHY_RSTN		PHY_PD		Reserved											
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				DUPIEX		SPEED		Reserved		PHYADDR_EN		Reserved		REPHYADDR[4:0]	

Bits	Name	Access	Description	Reset
31	PHY_RSTN	RW	Ethernet PHY global reset signal: 1: Normal operation mode; 0: Reset the Ethernet PHY.	0
30	PHY_PD	RW	In power-down mode, the MAC can access PHY registers through the MDIO interface: 1: Power-down mode; 0: Normal operation mode.	1
[29:11]	Reserved	RO	Reserved.	0
10	DUPIEX	RO	Link mode selection: 1: Link is full-duplex; 0: Link is half-duplex.	0
9	SPEED	RO	1: Link is 100M; 0: Link is 10M.	0
8	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
7	PHYADDR_EN	RW	Reconfigure PHY address: 1: bits[4:0] are used as the PHY address; 0: PHY address defaults to 1.	0
[6:5]	Reserved	RO	Reserved.	0
[4:0]	REPHYADDR[4:0]	RW	When bit 7 is 1, bits[4:0] are used as the PHY address.	0x01

Note: (1) When using Ethernet, the PHY reset signal and PD signal must be deasserted first.
(2) The bits of the R32_ETH_PHY_CR register always read as 0.

29.2.2 MMC Related Registers

29.2.2.1 MMC Control Register (R32_ETH_MMCCR)

Offset address: 0x0100

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											MCF		ROR	CSR	CR

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
3	MCF	RW	Counter freeze control bit. Setting this bit freezes all MMC counter values. Clearing it resumes counting. If ROR is set while frozen, reading any counter causes that counter to be cleared.	0
2	ROR	RW	Reset on read control bit. Setting this bit causes the counter value to be cleared after reading any counter.	0
1	CSR	RW	Counter stop rollover bit. Setting this bit causes the counter to stop at the maximum value without rolling over to zero.	0
0	CR	W	Counter reset control bit. Setting this bit resets all counters. This bit is automatically cleared after one system cycle.	0

29.2.2.2 MMC Receive Interrupt Register (R32_ETH_MMCRIR)

Offset address: 0x0104

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved													RGUFS		Reserved
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											RFCES		Reserved		

Bits	Name	Access	Description	Reset
[31:18]	Reserved	RO	Reserved.	0
17	RGUFS	RZ	Set when the number of received good frames reaches half.	0
[16:6]	Reserved	RO	Reserved.	0

Bits	Name	Access	Description	Reset
5	RFCES	RZ	Set when the number of received frames with CRC errors reaches half.	0
[4:0]	Reserved	RO	Reserved.	0

29.2.2.3 MMC Transmit Interrupt Register (R32_ETH_MMCTIR)

Offset address: 0x0108

31	30	29	28	27	26	25	24	23	22	21	20			19	18	17	16				
Reserved										TGFS		Reserved									
15				14				13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGFMSCS				TGFSCS			Reserved														

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
21	TGFS	RZ	Set when the number of transmitted frames reaches half.	0
[20:16]	Reserved	RO	Reserved.	0
15	TGFMSCS	RW	In half-duplex mode, set when the counter for frames encountering more than one collision reaches half.	0
14	TGFSCS	RW	In half-duplex mode, set when the counter for frames encountering only one collision reaches half.	0
[13:0]	Reserved	RO	Reserved.	0

29.2.2.4 MMC Receive Interrupt Mask Register (R32_ETH_MMCRIMR)

Offset address: 0x010C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16			
Reserved														RGUFM		Reserved		
15	14	13	12	11	10	9	8	7				6	5	4	3	2	1	0
Reserved								RFAEM			RFCEM		Reserved					

Bits	Name	Access	Description	Reset
[31:18]	Reserved	RO	Reserved.	0
17	RGUFM	RW	Received good frame half interrupt mask bit. Setting this bit masks the interrupt generated when the received good frame counter value reaches half.	0
[16:7]	Reserved	RO	Reserved.	0
6	RFAEM	RW	Received alignment error frame half interrupt mask bit. Setting this bit masks the interrupt generated when the received misaligned frame counter value reaches half.	0
5	RFCEM	RW	Received CRC error frame half interrupt mask bit. Setting this bit masks the interrupt generated when the received CRC error frame counter value reaches half.	0
[4:0]	Reserved	RO	Reserved.	0

29.2.2.5 MMC Transmit Interrupt Mask Register (R32_ETH_MMCTIMR)

Offset address: 0x0110

31	30	29	28	27	26	25	24	23	22	21	20		19	18	17	16					
Reserved										TGFM		Reserved									
15				14				13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGMSCM				TGFSCM				Reserved													

Bits	Name	Access	Description	Reset
[31:22]	Reserved	RO	Reserved.	0
21	TGFM	RW	Transmitted good frame half interrupt mask bit. Setting this bit masks the interrupt generated when the transmitted good frame counter value reaches half.	0
[20:16]	Reserved	RO	Reserved.	0
15	TGMSCM	RW	Transmitted good frame multiple collision flag bit.	0
14	TGFSCM	RW	Transmitted good frame single collision flag bit.	0
[13:0]	Reserved	RO	Reserved.	0

29.2.2.6 MMC Transmit Good Frames Single Collision Counter Register (R32_ETH_MMCTGFSCCR)

Offset address: 0x014C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TGFSCCR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGFSCCR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	TGFSCCR[31:0]	RO	This field counts the number of transmitted frames that encountered only one collision in half-duplex mode.	0

29.2.2.7 MMC Transmit Good Frames More Than Single Collision Counter Register (R32_ETH_MMCTGFMSCCR)

Offset address: 0x0150

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TGMSCCR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGMSCCR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	TGMSCCR[31:0]	RO	This field counts the number of transmitted frames that encountered more than one collision in half-duplex mode.	0

29.2.2.8 MMC Transmit Good Frame Counter Register (R32_ETH_MMCTGFCR)

Offset address: 0x0168

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TGFC[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TGFC[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	TGFC[31:0]	RO	Counter for correct frames transmitted by the MAC.	0											

29.2.2.9 MMC Receive CRC Error Frame Counter Register (R32_ETH_MMCRFCECR)

Offset address: 0x0194

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RFCECR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RFCECR[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	RFCECR[31:0]	RO	Counter for frames received by the MAC that have CRC errors.	0											

29.2.2.10 MMC Receive Alignment Error Frame Counter Register (R32_ETH_MMCRFAECR)

Offset address: 0x0198

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RFAECR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RFAECR[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	RFAECR[31:0]	RO	Counter for frames received by the MAC that have alignment errors.	0											

29.2.2.11 Ethernet State Machine Register (R32_ETH_STATE)

Offset address: 0x01A0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				ERX_STATE[2:0]				Reserved				ETX_STATE[2:0]			

Bits	Name	Access	Description	Reset
[31:11]	Reserved	RO	Reserved.	0
[10:8]	ERX_STATE[2:0]	RO		0

Bits	Name	Access	Description	Reset
			Ethernet receive state machine: 000: IDLE; 001: MAC_ADDR; 010: RCV_LEN_TYPE; 011: RCV_DATA; 100: RCV_FCS_PAD; 101: RCV_ZZZ; Others: Reserved.	
[7:3]	Reserved	RO	Reserved.	0
[2:0]	ETX_STATE[2:0]	RO	Ethernet transmit state machine: 000: SHED; 001: IDLE; 010: PREAM; 011: SEND_DATA; 100: SEND_PAD; 101: SEND_FCS; 110: SEND_JAM; 111: SEND_EEE.	0

29.2.2.12 MMC Receive Good Unicast Frame Counter Register (R32_ETH_MMCRGUFCR)

Offset address: 0x01C4

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RGUFCR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RGUFCR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	RGUFCR[31:0]	RO	Number of good unicast frames received by the MAC.	0

29.2.3 IEEE 1588 (PTP) Related Registers

29.2.3.1 PTP Timestamp Control Register (R32_ETH_PTPTSCR)

Offset address: 0x0700

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16						
Reserved																					
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
Reserved										TSARU		TSITE		TSSTU		TSSTI		TSFCU		TSE	

Bits	Name	Access	Description	Reset
[31:6]	Reserved	RO	Reserved.	0
5	TSARU	RW	Addend register update control bit. When set, the addend register value is added to the accumulator. Used in fine adjustment mode. After the accumulator increment is complete, this bit is automatically cleared. This bit can only be set when it is 0.	0

Bits	Name	Access	Description	Reset
4	TSITE	RW	Timestamp interrupt trigger enable bit. When set, an interrupt is generated when the PTP system time reaches the value set in the target time register.	0
3	TSSTU	RW	System time update control bit. When set, the PTP system time is incremented by the value in the update register. After the update is complete, this bit is automatically cleared.	0
2	TSSTI	RW	Timestamp initialization control bit. When set, the PTP system time is replaced with the value in the update register. After the update is complete, this bit is automatically cleared.	0
1	TSFCU	RW	Update mode selection control bit: 1: Fine adjustment mode; 0: Coarse adjustment mode.	0
0	TSE	RW	Timestamp enable bit: 1: After a transmit or receive is complete, a timestamp is added to the descriptor; 0: No timestamp is added to the descriptor.	0

29.2.3.2 PTP Sub-Second Increment Register (R32_ETH_PTPSSIR)

Offset address: 0x0704

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								STSSI[7:0]							

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
[7:0]	STSSI[7:0]	RW	Sub-second step value. In coarse adjustment mode, the PTP system time increments by this value every master clock cycle; in fine adjustment mode, the PTP system time increments by this value when the accumulator overflows.	0

29.2.3.3 PTP Timestamp Register High (R32_ETH_PTPTSHR)

Offset address: 0x0708

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STS[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STS[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	TSHR[31:0]	RO	PTP system time value, real-time value, in seconds.	0

29.2.3.4 PTP Timestamp Register Low (R32_ETH_PTPTSLR)

Offset address: 0x070C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
STPNS		STSS[30:16]													
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STSS[15:0]															
Bits	Name	Access	Description												Reset
31	STPNS	RO	System time positive/negative flag bit: 1: Current time is negative; 0: Current time is positive.												0
[30:0]	STSS[30:0]	RO	PTP system time value, real-time value, in sub-seconds, i.e., approximately 0.46 ns. When STSS overflows, STS increments by 1 second.												0

29.2.3.5 PTP Timestamp Update Register High (R32_ETH_PTPTSHUR)

Offset address: 0x0710

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TSUS[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSUS[15:0]															
Bits	Name	Access	Description												Reset
[31:0]	TSUS[31:0]	RW	Timestamp update seconds value, used to replace the PTP system time high bits or to represent the seconds added to or subtracted from the system time.												0

29.2.3.6 PTP Timestamp Update Register Low (R32_ETH_PTPTSLUR)

Offset address: 0x0714

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
TSUSS[31:16]																
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
TSUSS[15:0]																
Bits	Name	Access	Description													Reset
[31:0]	TSUSS[31:0]	RW	Timestamp update sub-seconds value, used to replace the PTP system time high bits or to represent the sub-seconds added to or subtracted from the system time.													0

29.2.3.7 PTP Timestamp Addend Register (R32_ETH_PTPTSAR)

Offset address: 0x0718

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TSA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSA[15:0]															
Bits	Name	Access	Description												Reset
[31:0]	TSA[31:0]	RW	Timestamp addend value. This register is only used in fine adjustment mode. The TSA value is added to the accumulator every system cycle. If the accumulator overflows, a system time update in fine adjustment mode is triggered.												0

29.2.3.8 PTP Target Time Register High (R32_ETH_PTPTHR)

Offset address: 0x071C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
TTSH[31:16]																
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
TTSH[15:0]																
Bits	Name	Access	Description													Reset
[31:0]	TTSH[31:0]	RW	Target time seconds value. If the PTP system time reaches or exceeds this value and the relevant interrupt is enabled, an interrupt is generated.													0

29.2.3.9 PTP Target Time Register Low (R32_ETH_PPTTLR)

Offset address: 0x0720

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
TTSL[31:16]																
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
TTSL[15:0]																
Bits	Name	Access	Description													Reset
[31:0]	TTSL[31:0]	RW	Target time sub-seconds value. If the PTP system time reaches or exceeds this value and the relevant interrupt is enabled, an interrupt is generated.													0

29.2.4 DMA Control Related Registers

29.2.4.1 DMA Bus Mode Register (R32_ETH_DMABMR)

Offset address: 0x1000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								DSL[4:0]				Reserved			SR

Bits	Name	Access	Description	Reset
[31:7]	Reserved	RO	Reserved.	0
[6:2]	DSL[4:0]	RW	Descriptor skip length: These bits define the skip distance between two descriptors that are not linked in a chain, in units of words (32 bits). The address skip is the address difference from the end of the current descriptor to the beginning of the next descriptor. When the DSL field is 0, in ring structure, the DMA considers the descriptors to be arranged contiguously and adjacent.	0
1	Reserved	RO	Reserved.	0
0	SR	RW	Software reset: When set to 1, the MAC DMA controller resets all internal registers and logic of the MAC subsystem. After the reset operations of the MAC internal different clock domain modules are complete, this bit is automatically cleared. Before rewriting MAC registers, this bit should be ensured to be 0.	1

29.2.4.2 DMA Transmit Poll Demand Register (R32_ETH_DMATPDR)

Offset address: 0x1004

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TPDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TPDR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	TPDR[31:0]	RW	Transmit poll demand. The user writes any value to this register to resume the suspended transmit process. After the transmit process is resumed, this register is automatically cleared.	0

29.2.4.3 DMA Receive Poll Demand Register (R32_ETH_DMARPDR)

Offset address: 0x1008

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RPDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RPDR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	RPDR[31:0]	RW	Receive poll demand. The receive process may be interrupted by various unexpected events; the user writes any value to this register to resume the receive process. After the receive process is resumed, this register is automatically cleared.	0

29.2.4.4 DMA Receive Descriptor List Address Register (R32_ETH_DMARDLAR)

Offset address: 0x100C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RDLAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDLAR[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	RDLAR[31:0]	RW	This register holds the address of the first receive DMA descriptor. Note that the descriptor must be 4-byte aligned.	0											

29.2.4.5 DMA Transmit Descriptor List Address Register (R32_ETH_DMATDLAR)

Offset address: 0x1010

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
TDLAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TDLAR[15:0]															
Bits	Name	Access	Description												Reset
[31:0]	TDLAR[31:0]	RW	This register holds the address of the first transmit DMA descriptor. Note that the descriptor must be 4-byte aligned.												0

29.2.4.6 DMA Status Register (R32_ETH_DMASR)

Offset address: 0x1014

31	30		29	28		27	26	25	24	23	22	21	20	19	18	17	16
Reserved		TSTS	Reserved		MMCS	EBS[2:0]		TPS[2:0]		RPS[2:0]		NIS					
15	14	13	12		11	10	9	8	7	6	5	4	3	2	1	0	
AIS	ERS	Reserved		PHYLINK_SR	ETS	RWTS	RPSS	RBUS	RS	TUS	ROS	TJTS	TBUS	TPSS	TS		

Bits	Name	Access	Description	Reset
[31:30]	Reserved	RO	Reserved.	0
29	TSTS	RO	Timestamp trigger status flag bit. When an interrupt event is generated in the PTP section, this bit is set; if the PTP interrupt is enabled, an interrupt is generated. After clearing all PTP section flag bits, this bit is automatically cleared.	0
28	PMTS	RO	PMT trigger status flag bit. When an interrupt event is generated in the PMT section, this bit is set; if the PMT interrupt is enabled, an interrupt is generated. After clearing all PMT section flag bits, this bit is automatically cleared.	0
27	MMCS	RO	MMC trigger status flag bit. When an interrupt event is generated in the MMC section, this bit is set; if the MMC interrupt is enabled, an interrupt is generated. After clearing all MMC section flag bits, this bit is automatically cleared.	0
26	Reserved	RO	Reserved.	0
[25:23]	EBS[2:0]	RO	Error status field. This field indicates the type of bus error. This field is only valid when DMASR[13] bit is set.	0

Bits	Name	Access	Description	Reset
			DMASR[23]: 1: Error during receive DMA data transfer; 0: Error during transmit DMA data transfer. DMASR[24]: 1: Error during write data transfer; 0: Error during read data transfer. DMASR[25]: 1: Error during data buffer access; 0: Error during descriptor access.	
[22:20]	TPS[2:0]	RO	Transmit process status field. This field indicates the current transmit DMA state. 000: Stopped, received reset or stop transmit command; 001: Running, fetching transmit descriptor; 010: Running, waiting for status information; 011: Running, reading data from transmit buffer and pushing into FIFO; 100, 101: Reserved; 110: Suspended, transmit descriptor unavailable or transmit buffer data underflow; 111: Running, closing transmit descriptor.	0
[19:17]	RPS[2:0]	RO	Receive process status field. This field indicates the current receive DMA state. 000: Stopped, received reset or stop transmit command; 001: Running, fetching receive descriptor; 010: Reserved; 011: Running, waiting for receive packet; 100: Suspended, receive descriptor unavailable; 101: Running, closing receive descriptor; 110: Reserved; 111: Running, pushing received data from FIFO into memory.	0
16	NIS	RW1	Normal interrupt summary bit. Under interrupts enabled in the DMAIER register, if any of the following bits is set, the NIS bit is also set. - DMASR[0]: Transmit interrupt; - DMASR[2]: Transmit buffer unavailable; - DMASR[6]: Receive interrupt; - DMASR[11]: Ethernet physical layer link status changed; - DMASR[14]: Early receive interrupt. The NIS bit is a sticky bit. When the corresponding interrupt status bit that caused NIS to be set is cleared, the NIS bit must also be cleared by writing 1.	0
15	AIS	RW1	Abnormal interrupt summary bit. Under interrupts enabled in the DMAIER register, if any of the following bits is set, the AIS bit is also set. - DMASR[1]: Transmit process stopped; - DMASR[3]: Transmit jabber timeout; - DMASR[4]: Receive FIFO overflow; - DMASR[5]: Transmit data underflow; - DMASR[7]: Receive buffer unavailable; - DMASR[8]: Receive process stopped; - DMASR[9]: Receive watchdog timeout; - DMASR[10]: Early transmit. The AIS bit is a sticky bit. When the corresponding interrupt status bit	0

Bits	Name	Access	Description	Reset
			that caused AIS to be set is cleared, the AIS bit must also be cleared by writing 1.	
14	ERS	RW1	Early receive status bit. When set, indicates that the DMA has filled the first buffer upon receiving a data frame, but the complete frame has not yet been received. After RS is set, the ERS bit is automatically cleared.	0
[13:12]	Reserved	RO	Reserved.	0
11	PHYLINK_SR	RW1Z	The PHY has generated an interrupt; the PHY's related registers need to be read to determine the interrupt source. This bit is cleared by writing 1.	0
10	ETS	RW1	Early transmit status bit. When set, indicates that the transmit frame has been fully pushed into the FIFO.	0
9	RWTS	RW1	Receive watchdog timeout flag bit. When set, indicates that the frame length has exceeded 2048 bytes.	0
8	RPSS	RW1	Receive process stopped status bit. When set, indicates that the receive process has stopped.	0
7	RBUS	RW1	Receive buffer unavailable status bit. When set, indicates that the receive descriptor is owned by the CPU, the DMA cannot acquire it, and the receive process has been suspended. The user needs to release the descriptor and write a value to the RPDR register to resume the receive process. The DMA will retry acquiring the descriptor on the next frame received.	0
6	RS	RW1	Receive complete status bit. When set, indicates that a frame has been received, and the frame information has been updated in the descriptor.	0
5	TUS	RW1	Transmit data underflow bit. When set, indicates that the transmit buffer underflowed during frame transmission. At this point, the transmit process is suspended and the data underflow error bit is set.	0
4	ROS	RW1	Receive status overflow bit. When set, indicates that the receive buffer has experienced a data overflow.	0
3	TJTS	RW1	Transmit jabber timeout bit. When set, indicates that the transmitter is too busy, transmission has stopped, and the descriptor's jabber timeout bit has been set.	0
2	TBUS	RW1	Transmit buffer unavailable status bit. When set, indicates that the transmit descriptor is occupied by the CPU, the DMA cannot acquire it, and the transmit process has been suspended. Bits [22:20] show the current transmit status. The application needs to release the transmit descriptor.	0
1	TPSS	RW1	Transmit process stopped status bit. When set, indicates that the transmit process has stopped.	0
0	TS	RW1	Transmit complete flag bit. When set, indicates that a frame has been transmitted and the descriptor ownership has been returned to the CPU.	0

29.2.4.7 DMA Operation Mode Register (R32_ETH_DMAOMR)

Offset address: 0x1018

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved					DTCEFD	Reserved					TSF	FTF	Reserved		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				ST	Reserved				FEF	FUGF	Reserved			SR	Reserved

Bits	Name	Access	Description	Reset
[31:27]	Reserved	RO	Reserved.	0
26	DTCEFD	RW	Do not discard TCP/IP checksum error frames control bit: 1: When a checksum error is found in protocols such as TCP/IP/ICMP/UDP, the frame is not discarded; 0: If the FEF bit is 0, the MAC discards all frames with errors.	0
[25:22]	Reserved	RO	Reserved.	0
21	TSF	RW	Transmit store-and-forward control bit: 1: The transmit process starts transmitting only after the complete frame is written into the FIFO; 0: The transmit process starts transmitting after the data written into the FIFO reaches a defined threshold.	0
20	FTF	RW	Flush transmit FIFO control bit. Setting this bit resets the transmit FIFO.	0
[19:14]	Reserved	RO	Reserved.	0
13	ST	RW	Start or stop transmit control bit: 1: Sets the transmit process to running state; 0: After transmitting the current frame, the transmit process enters stopped state.	0
[12:8]	Reserved	RO	Reserved.	0
7	FEF	RW	Forward error frames control bit: 1: All frames except overly short frames are forwarded to the DMA; 0: The receive FIFO discards frames with errors.	0
6	FUGF	RW	Forward undersized good frames control bit: 0: The receive FIFO discards all frames shorter than 64 bytes; 1: The receive FIFO forwards undersized frames.	0
[5:2]	Reserved	RO	Reserved.	0
1	SR	RW	Start or stop receive control bit: 1: Starts the receive process; the DMA fetches the receive descriptor from the current position or from the head of the descriptor list; 0: After forwarding the currently received frame, the receive DMA enters stopped mode; the next transfer starts from the current receive descriptor position.	0
0	Reserved	RO	Reserved.	0

29.2.4.8 DMA Interrupt Enable Register (R32_ETH_DMAIER)

Offset address: 0x101C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Reserved															NISE	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
AISE	ERS	Reserved	PHYLINK_IER	ETIE	RWTIE	RPSIE	RBUIE	RIE	Reserved	TUIE	ROIE	TJTIE	TBUIE	TPSIE	TIE	

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
16	NISE	RW	Normal interrupt enable bit. Enabling this bit enables the following interrupts. - DMASR[0]: Transmit interrupt; - DMASR[2]: Transmit buffer unavailable; - DMASR[6]: Receive interrupt; - DMASR[14]: Early receive interrupt.	0
15	AISE	RW	Abnormal interrupt. Enabling this bit enables the following interrupts. - DMASR[1]: Transmit process stopped; - DMASR[3]: Transmit jabber timeout; - DMASR[4]: Receive FIFO overflow; - DMASR[5]: Transmit data underflow; - DMASR[7]: Receive buffer unavailable; - DMASR[8]: Receive process stopped; - DMASR[9]: Receive watchdog timeout; - DMASR[10]: Early transmit.	0
14	ERS	RW	Early receive interrupt enable bit. Enabling this bit generates an early receive interrupt. NISE must be set. AISE must be set.	0
[13:12]	Reserved	RO	Reserved.	0
11	PHYLINK_IER	RW	Transceiver link interrupt enable bit. Enabling this bit generates a transceiver link interrupt.	0
10	ETIE	RW	Early transmit interrupt enable bit. Enabling this bit generates an early transmit interrupt. AISE must be set.	0
9	RWTIE	RW	Receive watchdog interrupt enable bit. Enabling this bit generates a receive watchdog timeout interrupt. AISE must be set.	0
8	RPSIE	RW	Receive process stopped interrupt enable bit. Enabling this bit generates a receive process stopped interrupt.	0
7	RBUIE	RW	Receive buffer unavailable interrupt enable bit. Enabling this bit generates a receive buffer unavailable interrupt. AISE must be set.	0
6	RIE	RW	Receive complete interrupt enable bit. Enabling this bit generates a receive complete interrupt. NISE must be set.	0
5	TUIE	RW	Transmit underflow interrupt enable. Enabling this bit generates a transmit underflow interrupt. AISE must be set.	0
4	ROIE	RW	Receive overflow interrupt. Enabling this bit generates a receive overflow interrupt. AISE must be set.	0
3	TJTIE	RW	Transmit jabber timeout interrupt enable. Enabling this bit generates a transmit jabber timeout interrupt. AISE must be set.	0
2	TBUIE	RW	Transmit buffer unavailable interrupt enable. Enabling this bit generates a transmit buffer unavailable interrupt. NISE must be set.	0

Bits	Name	Access	Description	Reset
1	TPSIE	RW	Transmit process stopped interrupt enable. Enabling this bit generates a transmit process stopped interrupt. AISE must be set.	0
0	TIE	RW	Transmit complete interrupt enable. Enabling this bit generates a transmit complete interrupt. NISE must be set.	0

29.2.4.9 DMA Missed Frame and Buffer Overflow Counter Register (R32_ETH_DMAMFBOCR)

Offset address: 0x1020

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved			OFOC	MFA[10:0]										OMFC	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MFC[15:0]															

Bits	Name	Access	Description	Reset
[31:29]	Reserved	RO	Reserved.	0
28	OFOC	RZ	FIFO overflow counter overflow bit.	0
[27:17]	MFA[10:0]	RZ	Frames missed by the application.	0
16	OMFC	RZ	Missed frame counter overflow bit.	0
[15:0]	MFC[15:0]	RZ	Missed frame counter. This field indicates the number of frames lost due to receive buffer being unavailable.	0

29.2.4.10 DMA Current Transmit Descriptor Register (R32_ETH_DMACHTDR)

Offset address: 0x1048

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTDAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTDAR[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	HTDAR[31:0]	RO	This register value points to the transmit descriptor currently in use. This register is updated in real-time by the DMA.	0											

29.2.4.11 DMA Current Receive Descriptor Register (R32_ETH_DMACHRDR)

Offset address: 0x104C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HRDAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HRDAR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	HRDAR[31:0]	RO	This register value points to the receive descriptor currently in use. This register is updated in real-time by the DMA.	0

29.2.4.12 DMA Current Transmit Buffer Address Register (R32_ETH_DMACHTBAR)

Offset address: 0x1050

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HTBAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HTBAR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	HTBAR[31:0]	RO	This register value points to the transmit buffer currently in use. This register is updated in real-time by the DMA.	0

29.2.4.13 DMA Current Receive Buffer Address Register (R32_ETH_DMACHRBAR)

Offset address: 0x1054

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
HRBAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HRBAR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	HRBAR[31:0]	RO	This register value points to the receive buffer currently in use. This register is updated in real-time by the DMA.	0

29.2.5 Internal 10M/100M Physical Layer Related Register Addresses

29.2.5.1 Basic Control Register (R16_BMCR)

Offset address: 0x00

Bits	Name	Access	Description	Reset
15	RST	RW/SC	1: PHY reset; 0: Normal operation.	0
14	Loopback	RW	1: Enable loopback; 0: Normal operation.	0
13	Speed Selection	RW	1: 100 Mb/s; 0: 10 Mb/s. After auto-negotiation, indicates the negotiated operating speed status.	1
12	Auto-Negotiation	RW	1: Enable auto-negotiation; 0: Disable auto-negotiation.	1
11	Power Down	RW		0

Bits	Name	Access	Description	Reset
			1: Power down; 0: Normal operation.	
10	Isolate	RW	1: MII/RMII interface is isolated from PHY, PHY still responds to MDC/MDIO; 0: Normal operation.	0
9	Restart Auto-Negotiation	RW/SC	1: Restart auto-negotiation; 0: Normal operation.	0
8	Duplex Mode	RW	1: Full-duplex; 0: Half-duplex.	1
7	Collision Test	RW	Collision test. 1: Collision detection enabled; 0: Normal operation.	0
[6:0]	Reserved	RO	Reserved.	0

29.2.5.2 Basic Status Register (R16_BMSR)

Offset address: 0x01

Bits	Name	Access	Description	Reset
15	100BASE-T4	RO	1: 100BASE-T4 capable; 0: 100BASE-T4 not capable.	0
14	100BASE-TX Full Duplex	RO	1: 100BASE-TX full-duplex capable; 0: 100BASE-TX full-duplex not capable.	1
13	100BASE-TX Half Duplex	RO	1: 100BASE-TX half-duplex capable; 0: 100BASE-TX half-duplex not capable.	1
12	10BASE-T Full Duplex	RO	1: 10BASE-T full-duplex capable; 0: 10BASE-T full-duplex not capable.	1
11	10BASE-T Half Duplex	RO	1: 10BASE-T half-duplex capable; 0: 10BASE-T half-duplex not capable.	1
[10:7]	Reserved	RO	Reserved.	0
6	MF Preamble Suppression	RO	Allows management frame suppression of preamble.	1
5	Auto-Negotiation Complete	RO	1: Auto-negotiation complete; 0: Auto-negotiation not complete.	0
4	Remote Fault	RO	1: Remote fault condition detected (read clears); 0: No remote fault condition detected.	0
3	Auto-Negotiation Ability	RO	1: PHY is able to perform auto-negotiation; 0: PHY is not able to perform auto-negotiation.	1
2	Link Status	RO	1: Valid link established; 0: No valid link established.	0
1	Jabber Detect	RO	1: Jabber condition detected; 0: No jabber condition detected.	0
0	Extended Capability	RO	1: Extended register capabilities; 0: No extended register capabilities.	1

29.2.5.3 Auto-Negotiation Advertisement Register (R16_ETH_ANAR)

Offset address: 0x04

Bits	Name	Access	Description	Reset
15	NP	RW	Next page bit: 1: Send protocol rule data page; 0: Send primary capability data page.	0
14	ACK	RO	1: Acknowledge receipt of link partner capability data word; 0: No acknowledge signal received.	0
13	RF	RW	1: Notify remote fault detection capability; 0: Do not notify remote fault detection capability.	0
12	Reserved	RO	Reserved.	0
11	ASYPAUSE	RW	1: Asymmetric flow control supported; 0: Asymmetric flow control not supported.	0
10	PAUSE	RW	1: Flow control supported; 0: Flow control not supported.	0
9	100Base-T4	RO	1: Local node supports 100Base-T4; 0: Local node does not support 100Base-T4.	0
8	100Base-TX-FD	RW	1: Local node supports 100Base-TX full-duplex; 0: Local node does not support 100Base-TX full-duplex.	1
7	100Base-TX	RW	1: Local node supports 100Base-TX; 0: Local node does not support 100Base-TX.	1
6	10Base-T-FD	RW	1: Local node supports 10Base-T full-duplex; 0: Local node does not support 10Base-T full-duplex.	1
5	10Base-T	RW	1: Local node supports 10Base-T half-duplex; 0: Local node does not support 10Base-T half-duplex.	1
[4:0]	SELECT[4:0]	RO	This node supports binary encoded selector. Currently only CSMA/CD 00001; no other protocol is supported.	1

29.2.5.4 Auto-Negotiation Link Partner Ability Register (R16_ETH_ANALPAR)

Offset address: 0x05

Bits	Name	Access	Description	Reset
15	NP	RW	Next page bit. 1: Send protocol detail data page; 0: Send basic capability data page.	0
14	ACK	RO	1: Acknowledge receipt of peer auto-negotiation capability register; 0: No acknowledge signal received.	0
13	RF	RW	1: Notify remote fault detection capability; 0: Do not notify remote fault detection capability.	0
12	Reserved	RO	Reserved.	0
11	ASYPAUSE	RO	1: Supports asymmetric pause; 0: Does not support asymmetric pause.	0

Bits	Name	Access	Description	Reset
			When auto-negotiation is enabled, this bit reflects the link partner's capability.	
10	PAUSE	RO	1: Supports symmetric pause; 0: Does not support symmetric pause.	0
9	100Base-T4	RO	1: PHY supports 100BASE-T4; 0: PHY does not support 100BASE-T4.	0
8	100Base-TX-FD	RO	1: PHY supports 100BASE-TX full-duplex; 0: PHY does not support 100BASE-TX full-duplex.	0
7	100Base-TX	RO	1: PHY supports 100BASE-TX; 0: PHY does not support 100BASE-TX.	0
6	10Base-T-FD	RO	1: PHY supports 10BASE-T full-duplex; 0: PHY does not support 10BASE-T full-duplex.	0
5	10Base-T	RO	1: PHY supports 10BASE-T; 0: PHY does not support 10BASE-T.	0
[4:0]	SELECT[4:0]	RO	PHY supports binary encoded selector. Currently only CSMA/CD 00001; no other protocol is supported.	0x01

29.2.5.5 Auto-Negotiation Expansion Register (R16_ANER)

Offset address: 0x06

Bits	Name	Access	Description	Reset
[15:7]	Reserved	RO	Reserved.	0
6	RX_NPLA	RO	Whether the next page sent by the peer is stored in the location determined by bit 5.	0
5	RX_NPSL	RO	1: Indicates whether the peer's next page is stored in register 8; 0: Indicates whether the peer's next page is stored in register 5.	0
4	Parallel Detection Fault	RZ	1: An error was detected during parallel detection; 0: No error was detected during parallel detection.	0
3	Link Partner Next Page Ability	RO	1: Link partner has next page capability; 0: Link partner does not have next page capability.	0
2	Local Next Page Ability	RO	1: Local device has next page capability; 0: Local device does not have next page capability.	1
1	Page Received	RZ	1: A new page was received; 0: No new page was received.	0
0	Link Partner Auto-Negotiation Ability	RO	If the local device auto-negotiation is enabled, this bit means: 1: Link partner has auto-negotiation capability; 0: Link partner does not have auto-negotiation capability.	0

29.2.5.6 Page Selection Register (R16_PAGE_SEL)

Offset address: 0x1F

Bits	Name	Access	Description	Reset
[15:8]	Reserved	RO	Reserved.	0
[7:0]	SEL_PAGE_ADDR[7:0]	RW	Select page address: 00000000–11111111.	0

29.2.6 PAGE0 Related Registers

29.2.6.1 PHY Status Register (R16_PHY_STATUS1)

Offset address: 0x1A

Bits	Name	Access	Description	Reset
15	incompatible_link	RO	Compatibility status of local device and peer device capability after auto-negotiation priority resolution: 1: Incompatible; 0: Compatible.	0
[14:7]	Reserved	RO	Reserved.	0
6	signal_status	RO	100M signal status: 1: Good signal status; 0: No signal.	0
5	mdi_mdix_status	RO	MDI/MDIX status: 1: MDI; 0: MDIX.	1
4	polarity_status	RO	Polarity status: 1: Polarity reversed; 0: Polarity normal.	0
[3:0]	phy_sepd_dup[3:0]	RO	PHY operating speed and full/half-duplex mode.	0

Table 29-23: phy_sepd_dup details

Bits	Name	Access	Description
3	phy_100bt	RO	Operating speed 100M.
2	phy10_indicate	RO	Operating speed 10M.
1	full_indicat	RO	Full-duplex.
0	hfull_indicat	RO	Half-duplex.

29.2.6.2 Interrupt Indication Register (R16_INTERRUPT_IND)

Offset address: 0x1E

Bits	Name	Access	Description	Reset
[15:12]	Reserved	RO	Reserved.	0
11	linkstatus_chg	RZ	Link status change indication.	0
[10:0]	Reserved	RO	Reserved.	0

29.2.7 PAGE7 Related Registers

29.2.7.1 Interrupt/WOL/LED Flag Register (R16_INTERRUPT_MASK)

PAGE7 offset address: 0x13

Bits	Name	Access	Description	Reset
[15:14]	Reserved	RO	Reserved.	0
13	int_linkchg	RW	Link change interrupt MASK.	0
[12:10]	Reserved	RO	Reserved.	0
9	led_enable	RW	LED function enable.	0
[8:0]	Reserved	RO	Reserved.	0x30

Chapter 30: Random Number Generator (RNG)

The Random Number Generator (RNG) is based on continuous analog noise and provides a 32-bit random number when the host reads data.

30.1 Main Features

- Can generate 32-bit random numbers
- Implements error management
- Can be individually disabled to reduce power consumption

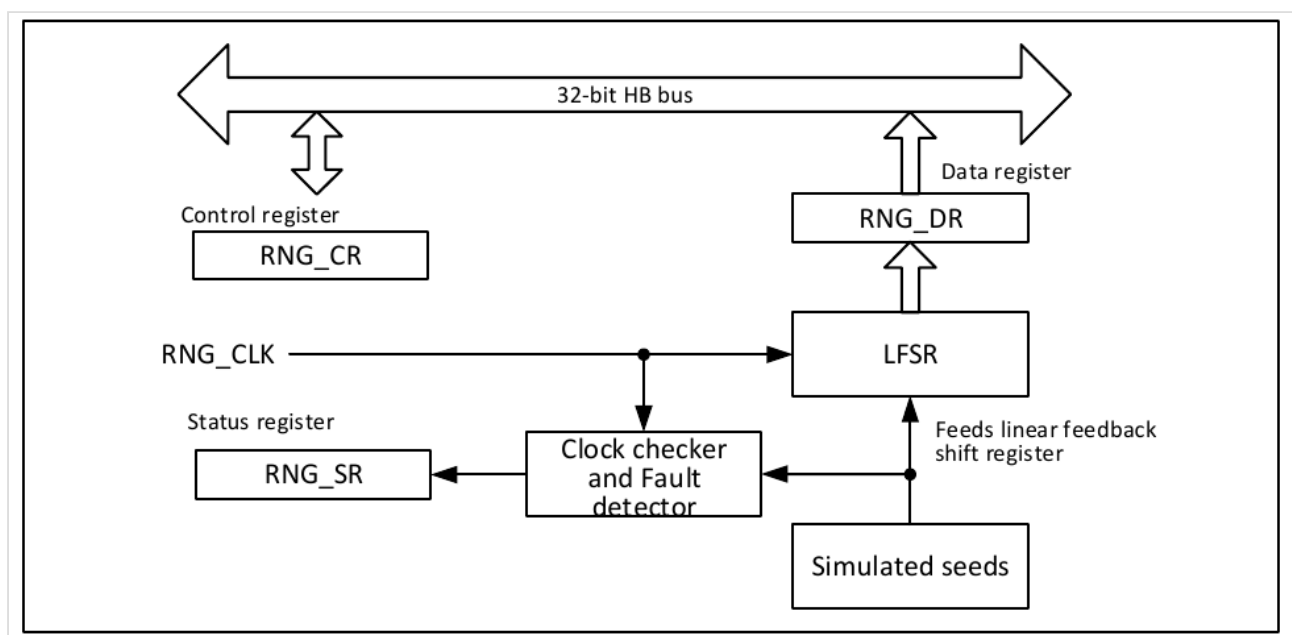


Figure 30-1: RNG block diagram

30.2 Functional Description

The random number generator is implemented using an analog circuit that generates seeds for the linear feedback shift register (RNG_LFSR) to produce 32-bit random numbers. The RNG_LFSR is clocked by a dedicated clock (PLL48CLK) at a constant frequency, so the quality of the random number is related to the HCLK clock. When a large number of seeds are introduced into the RNG_LFSR, the content of the RNG_LFSR is transferred to the data register (RNG_DR).

30.2.1 RNG Operation

The specific operation steps of the RNG are as follows:

1. If interrupts are enabled, the IE bit in the RNG_CR register must be set to 1 (this interrupt is generated when a random number is ready or an error occurs).
2. Enable random number generation by configuring the RNGEN bit of the RNG_CR register, which simultaneously activates the analog section, the RNG_LFSR, and the error detector.

3. If interrupts are enabled, each time an interrupt is generated, check that the SEIS and CEIS bits in the RNG_SR register are 0 to confirm that no error has occurred and that DRDY is 1 to confirm that the random number is ready. Then the contents of the RNG_DR register can be read.

30.2.2 Error Management

RNG errors include clock errors and seed errors. When a clock error occurs, the RNG can no longer generate random numbers. In this case, check whether the clock controller is correctly configured and can provide the RNG clock, then clear the CEIS bit. When the CECS bit is 0, the RNG can operate normally. When a clock error occurs, it does not affect the previous random number, which can still be used normally. When a seed error occurs, the value in the RNG_DR register cannot be used as a random number. To reuse the RNG, first clear the SEIS bit, then clear the RNGEN bit and set it to 1 again to re-initialize and restart the RNG.

30.3 Register Description

Table 30-1: RNG related register list

Name	Address	Description	Reset Value
R32_RNG_CR	0x40023C00	RNG control register	0x00000000
R32_RNG_SR	0x40023C04	RNG status register	0x00000000
R32_RNG_DR	0x40023C08	RNG data register	0x00000000

30.3.1 RNG Control Register (RNG_CR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved										IE		RNGEN		Reserved	

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
3	IE	RW	Interrupt enable control: 1: Enable RNG interrupt; 0: Disable RNG interrupt.	0
2	RNGEN	RW	Random number generator enable: 1: Enable random number generator; 0: Disable random number generator.	0
[1:0]	Reserved	RW	Reserved.	0

30.3.2 RNG Status Register (RNG_SR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved										SEIS	CEIS	Reserved		SECS	CECS	DRDY

Bits	Name	Access	Description	Reset
[31:7]	Reserved	RO	Reserved.	0
6	SEIS	RW	Seed error interrupt status (this bit is set simultaneously with SECS): 1: One of the following error sequences detected: - More than 64 consecutive identical bits; - More than 32 consecutive alternating 0s and 1s. 0: No error sequence detected.	0
5	CEIS	RW	Clock error interrupt status (this bit is set simultaneously with CECS): 1: PLL48CLK clock not correctly detected; 0: PLL48CLK clock correctly detected.	0
[4:3]	Reserved	RO	Reserved.	0
2	SECS	RO	Seed error current status: 1: One of the following error sequences detected: - More than 64 consecutive identical bits; - More than 32 consecutive alternating 0s and 1s. 0: No error sequence detected.	0
1	CECS	RO	Clock error current status: 1: PLL48CLK clock not correctly detected; 0: PLL48CLK clock correctly detected.	0
0	DRDY	RO	Data ready (this bit is cleared to 0 after reading the R32_RNG_DR register): 1: R32_RNG_DR register is valid, this random number is available; 0: R32_RNG_DR register is invalid, this random number is not available.	0

30.3.3 RNG Data Register (RNG_DR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
RNDATA[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RNDATA[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	RNDATA	RO	32-bit random number.	0

Chapter 31: Electronic Signature (ESIG)

The electronic signature contains chip identification information: flash memory area capacity and a unique device identifier. It is programmed by the manufacturer into the system storage area of the memory module at the factory, and can be read via SWD (SDI) or application code.

31.1 Functional Description

Flash memory area capacity: Indicates the size available for the user application on the current chip.

Unique device identifier: A 96-bit binary code that is unique to any given microcontroller. Users can only read it and cannot modify it. This unique identification information can be used as a security password for the microcontroller (product), encryption/decryption key, product serial number, etc., to improve system security mechanisms or indicate identity information.

All of the above content can be accessed by users in 8/16/32-bit reads.

31.2 Register Description

Table 31-1: ESIG related register list

Name	Address	Description	Reset Value
R16_ESIG_FLACAP	0x1FFFF7E0	Flash capacity register	0XXXXX
R32_ESIG_UNIID1	0x1FFFF7E8	UID register 1	0XXXXXXXXX
R32_ESIG_UNIID2	0x1FFFF7EC	UID register 2	0XXXXXXXXX
R32_ESIG_UNIID3	0x1FFFF7F0	UID register 3	0XXXXXXXXX

31.2.1 Flash Capacity Register (ESIG_FLACAP)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F_SIZE[15:0]															
Bits	Name	Access	Description	Reset											
[15:0]	F_SIZE[15:0]	RO	Flash capacity in Kbytes. Example: 0x0080 = 128 K bytes	X											

31.2.2 UID Register (ESIG_UNIID1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	U_ID[31:0]	RO	Bits 0-31 of the UID.	X											

31.2.3 UID Register (ESIG_UNIID2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID[63:48]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID[47:32]															
Bits	Name		Access		Description						Reset				
[31:0]	U_ID[63:32]		RO		Bits 32-63 of the UID.						X				

31.2.4 UID Register (ESIG_UNIID3)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
U_ID[95:80]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U_ID[79:64]															
Bits	Name		Access		Description						Reset				
[31:0]	U_ID[95:64]		RO		Bits 64-95 of the UID.						X				

Chapter 32: Flash and User Option Bytes (FLASH)

32.1 Flash Organization

The chip's internal flash memory organization is as follows:

Table 32-1: Flash Organization Structure

Block	Name	Address Range	Size (bytes)
Main Memory	Page 0	0x08000000 – 0x080000FF	256
	Page 1	0x08000100 – 0x080001FF	256
	Page 2	0x08000200 – 0x080002FF	256
	Page 3	0x08000300 – 0x080003FF	256
	Page 4	0x08000400 – 0x080004FF	256
	Page 5	0x08000500 – 0x080005FF	256
	Page 6	0x08000600 – 0x080006FF	256
	Page 7	0x08000700 – 0x080007FF	256
	...	...	...
	Page 3967	0x080F7F00 – 0x080F7FFF	256
Information Block	System Boot Code Storage	0x1FFF8000 – 0x1FFFEFFF	28K
	User Option Bytes	0x1FFFF800 – 0x1FFFF87F	128

Note:

1. The main memory area above is used for storing user application programs, and write protection is partitioned in units of 4K bytes (16 pages); except for the "manufacturer configuration word" area which is factory-locked and not accessible by the user, other areas can be operated by the user under certain conditions.
2. For chips whose lot number's fifth-from-last digit is greater than 0, the zero-wait-region supports both RVV instruction and DMA 64-bit accesses.

32.2 Flash Programming and Security

32.2.1 Two Programming/Erase Methods

- **Standard Programming:** This is the default programming method (compatible mode). In this mode, the CPU performs programming in single 2-byte operations, and performs erase and full-chip erase operations in single 4K-byte operations.
- **Fast Programming:** This method uses page operations (recommended). After unlocking via a specific sequence, it performs single 256-byte programming, 32K-byte erase, and full-chip erase operations.

Note: FLASH erase/write related functions can only be placed in the zero-wait-region FLASH.

32.2.2 Security — Preventing Illegal Access (Read, Write, Erase)

- Page write protection
- Read protection

When the chip is in read-protected state:

1. Main memory pages 0–15 (4K bytes) are automatically write-protected and are not controlled by the FLASH_WPR register; when read protection is released, all main memory pages are controlled by the FLASH_WPR register.
2. The system boot code area, SWD or SDI mode, and RAM area cannot erase or program the main memory, except for full-chip erase. The user option bytes area can be erased or programmed. If an attempt is made to release read protection (programming user option bytes), the chip will automatically erase the entire user area.

32.3 FLASH Enhanced Read Mode

The FLASH enhanced read mode is applicable when the user program runs in FLASH (the user code space exceeds the CODE size space configured by the user option byte RAM_CODE_MOD[1:0] bits). Enabling this mode can improve FLASH access efficiency. To enable this mode, set the EHMODO bit of the FLASH_CTLR register to 1; at this point, read the EHMODO bit — if the EHMODO bit is 1, the enhanced read mode has taken effect. To disable this mode, first clear the EHMODO bit to 0, then set RSENACT to 1. The access clock frequency can also be selected by configuring the SCKMOD bit of the FLASH_CTLR register.

Note: When using the FLASH enhanced read mode, please note the following:

1. Before performing any erase or programming operations on the FLASH (including releasing read protection and other user option byte programming), you must first exit the enhanced read mode, otherwise erase and programming operations will fail.
2. Before entering stop mode, you must first exit the enhanced read mode, otherwise the stop mode may be abnormal.
3. After power reset and system reset are completed, the chip automatically exits the enhanced read mode under hardware control.

32.4 Register Description

Table 32-2: FLASH Related Register List

Name	Address	Description	Reset Value
R32_FLASH_KEYR	0x40022004	FPEC key register	0xFFFFFFFF
R32_FLASH_OBKEYR	0x40022008	OBKEY register	0xFFFFFFFF
R32_FLASH_STATR	0x4002200C	Status register	0x00000000
R32_FLASH_CTLR	0x40022010	Control register	0x02008080
R32_FLASH_ADDR	0x40022014	Address register	0x00000000
R32_FLASH_OBR	0x4002201C	Option byte register	0xFFFFFFFF

Name	Address	Description	Reset Value
R32_FLASH_WPR	0x40022020	Write protection register	0xFFFFFFFF
R32_FLASH_MODEKEYR	0x40022024	Extended key register	0xFFFFFFFF

32.4.1 FPEC Key Register (FLASH_KEYR)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
KEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEYR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	KEYR[31:0]	WO	FPEC key, used to input the FPEC unlock key including: RDPRT key = 0x000000A5; KEY1 = 0x45670123; KEY2 = 0xCDEF89AB.	X

32.4.2 OBKEY Register (FLASH_OBKEYR)

Offset address: 0x08

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
OBKEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OBKEYR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	OBKEYR[31:0]	WO	Option byte key, used to input the option byte key to release OBWRE.	X

32.4.3 Status Register (FLASH_STATR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				EHMODS		EOP	Reserved		WRPRTERR	Reserved		WRBSY	BSY		

Bits	Name	Access	Description	Reset
[31:8]	Reserved	RO	Reserved.	0
7	EHMODS	RO	FLASH enhanced read mode enable status: 1: FLASH enhanced read mode enabled; 0: FLASH enhanced read mode not enabled.	0
6	Reserved	RO	Reserved.	0
5	EOP	RW1		0

Bits	Name	Access	Description	Reset
			Indicates end of operation, write 1 to clear. Hardware sets this bit on each successful erase or programming.	
4	WRPRTER	RW1	Indicates write protection error, write 1 to clear. Hardware sets this bit if programming is attempted at a write-protected address.	0
[3:2]	Reserved	RO	Reserved.	0
1	WRBSY	RO	This bit is used during fast page programming to indicate that programming data is being written. During page programming, when data is written, this bit is set to '1' and automatically cleared to '0' by hardware; if this bit is '0', the next data can be written.	0
0	BSY	RO	Indicates busy status: 1: Flash operation in progress; 0: Operation finished.	0

Note: When performing programming operations, ensure that the STRT bit of the FLASH_CTLR register is 0.

32.4.4 Control Register (FLASH_CTLR)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16				
Reserved		SCKMOD[1:0]		EHMOD		Reserved		RSENACT		PGSTRT		Reserved		BER32		Reserved		FTPG	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
FLOCK	Reserved	EOPIE	Reserved	ERRIE	Reserved	OBWRE	Reserved	LOCK	STRT	OBER	OBPG	Reserved	MER	PER	PG				

Bits	Name	Access	Description	Reset
[31:27]	Reserved	RO	Reserved.	0
[26:25]	SCKMOD[1:0]	RW	FLASH access clock configuration: 00: FLASH access clock frequency = HCLK; 01: FLASH access clock frequency = (HCLK/2); 10: FLASH access clock frequency = (HCLK/4); 11: Reserved. Note: FLASH access clock frequency does not exceed 100MHz; this bit is standard unlock + fast unlock.	01b
24	EHMOD	RW	FLASH enhanced read mode: This mode can improve access efficiency when the program runs in FLASH. 1: Enable FLASH enhanced read mode. 0: Disable FLASH enhanced read mode; must be used together with the RSENACT bit — the exit steps are: first clear the EHM0D bit to 0, then set RSENACT to 1. This bit is standard unlock + fast unlock.	0
23	Reserved	RO	Reserved.	0
22	RSENACT	WO	Exit enhanced read mode, automatically cleared by hardware; must be used together with the EHM0D bit — the exit steps are: first clear the EHM0D bit to 0, then set RSENACT to 1. This bit is standard unlock.	0
21	PGSTRT	RW0		0

Bits	Name	Access	Description	Reset
			Start. Set to 1 to start a page programming operation, automatically cleared by hardware. This bit is standard unlock.	
[20:19]	Reserved	RO	Reserved.	0
18	BER32	RW	Execute 32KB erase. This bit is standard unlock + fast unlock.	0
17	Reserved	RO	Reserved.	0
16	FTPG	RW	Execute fast page programming operation. This bit is standard unlock + fast unlock.	0
15	FLOCK	RW1	Fast programming lock. Can only write '1'. When this bit is '1', the fast programming/erase mode is not available. After detecting the correct unlock sequence, hardware clears this bit to '0'. Software sets it to 1 to re-lock. This bit is standard unlock.	1
[14:13]	Reserved	RO	Reserved.	0
12	EOPIE	RW	Operation complete interrupt control (EOP set in FLASH_STATR register): 1: Interrupt generation enabled; 0: Interrupt generation disabled. This bit is standard unlock.	0
11	Reserved	RO	Reserved.	0
10	ERRIE	RW	Error status interrupt control (PGERR/WRPRTERR set in FLASH_STATR register): 1: Interrupt generation enabled; 0: Interrupt generation disabled. This bit is standard unlock.	0
9	OBWRE	RW0	User option byte lock, software clears to 0: 1: Indicates that user option byte programming operations can be performed. Hardware sets this bit after the correct sequence is written to the FLASH_OBKEYR register; 0: Software clears to re-lock user option bytes. This bit is standard unlock.	0
8	Reserved	RO	Reserved.	0
7	LOCK	RW1	Lock. Can only write '1'. When this bit is '1', FPEC and FLASH_CTLR are locked and not writable. After detecting the correct unlock sequence, hardware clears this bit to '0'. After an unsuccessful unlock attempt, this bit will not change again until the next system reset. This bit is standard unlock.	1
6	STRT	RW1	Start. Set to 1 to start an erase operation, hardware automatically clears to 0 (BSY becomes '0'). This bit is standard unlock.	0
5	OBER	RW	Execute user option byte erase. This bit is standard unlock.	0
4	OBPG	RW	Execute user option byte programming. This bit is standard unlock.	0

Bits	Name	Access	Description	Reset
3	Reserved	RO	Reserved.	0
2	MER	RW	Execute full erase operation (erase the entire user area). This bit is standard unlock.	0
1	PER	RW	Execute sector (4KB) erase operation. This bit is standard unlock.	0
0	PG	RW	Execute standard programming operation. This bit is standard unlock.	0

32.4.5 Address Register (FLASH_ADDR)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FAR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FAR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	FAR[31:0]	WO	Flash address. For programming, this is the programming address; for erase, this is the starting address of the erase. This register cannot be written when the BSY bit in the FLASH_STATR register is '1'.	0

32.4.6 Option Byte Register (FLASH_OBR)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		RAM_CODE_MOD		Reserved		STANDYRST		STOPRST		IWDGSW		RDPRT		OBERR	

Bits	Name	Access	Description	Reset
[31:10]	Reserved	RO	Reserved.	0
9	Reserved	RO	Reserved.	x
8	RAM_CODE_MOD	RO	1: CODE-576KB + RAM-136KB 0: CODE-512KB + RAM-200KB	x
[7:5]	USER/Reserved	RO	Reserved.	x
4	STANDYRST	RO	System reset control in standby mode.	x
3	STOPRST	RO	System reset control in stop mode.	x
2	IWDGSW	RO	Independent watchdog (IWDG) hardware enable bit.	1
1	RDPRT	RO	Read protection status. 1: Indicates flash is currently read-protected; 0: Indicates flash read protection is not active.	1
0	OBERR	RO		0

Bits	Name	Access	Description	Reset
			Option byte error. 1: Indicates option byte and its complement do not match; 0: Indicates option byte and its complement match.	

Note: USER and RDPRT are loaded from the user option byte area after system reset.

32.4.7 Write Protection Register (FLASH_WPR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
WRP[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WRP[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	WRP[31:0]	RO	Flash write protection status. 1: Write protection not active; 0: Write protection active. Each bit represents the write protection status of 4K bytes (16 pages) of memory.	X

Note: WPR is loaded from the user option byte area after system reset.

32.4.8 Extended Key Register (FLASH_MODEKEYR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODEKEYR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODEKEYR[15:0]															

Bits	Name	Access	Description	Reset
[31:0]	MODEKEYR[31:0]	WO	Input the following sequence to unlock fast programming/erase mode: KEY1 = 0x45670123; KEY2 = 0xCDEF89AB.	X

32.5 Flash Operation Flow

32.5.1 Read Operation

Direct addressing is performed within the universal address space. Any 8/16/32-bit data read operation can access the flash module's content and obtain the corresponding data.

32.5.2 Unlocking Flash

After system reset, the flash controller (FPEC) and FLASH_CTLR register are locked and not accessible. The flash controller module can be unlocked by writing a sequence to the FLASH_KEYR register.

Unlock sequence:

1. Write KEY1 = 0x45670123 to the FLASH_KEYR register (step 1 must be KEY1);
2. Write KEY2 = 0xCDEF89AB to the FLASH_KEYR register (step 2 must be KEY2).

The above operations must be executed in order and consecutively; otherwise, they are considered erroneous and will lock the FPEC module and FLASH_CTLR register and generate a bus error until the next system reset.

The flash controller (FPEC) and FLASH_CTLR register can be re-locked by setting the "LOCK" bit of the FLASH_CTLR register to 1.

32.5.3 Main Memory Standard Programming

Standard programming writes 2 bytes at a time. When the PG bit of the FLASH_CTLR register is '1', each half-word (2 bytes) written to the flash address starts a programming operation. Writing any non-half-word data will cause the FPEC to generate a bus error. During programming, the BSY bit is '1'; when programming is complete, the BSY bit is '0' and the EOP bit is '1'.

Note: When the BSY bit is '1', writing to any register is prohibited.

图 32-1 FLASH 编程

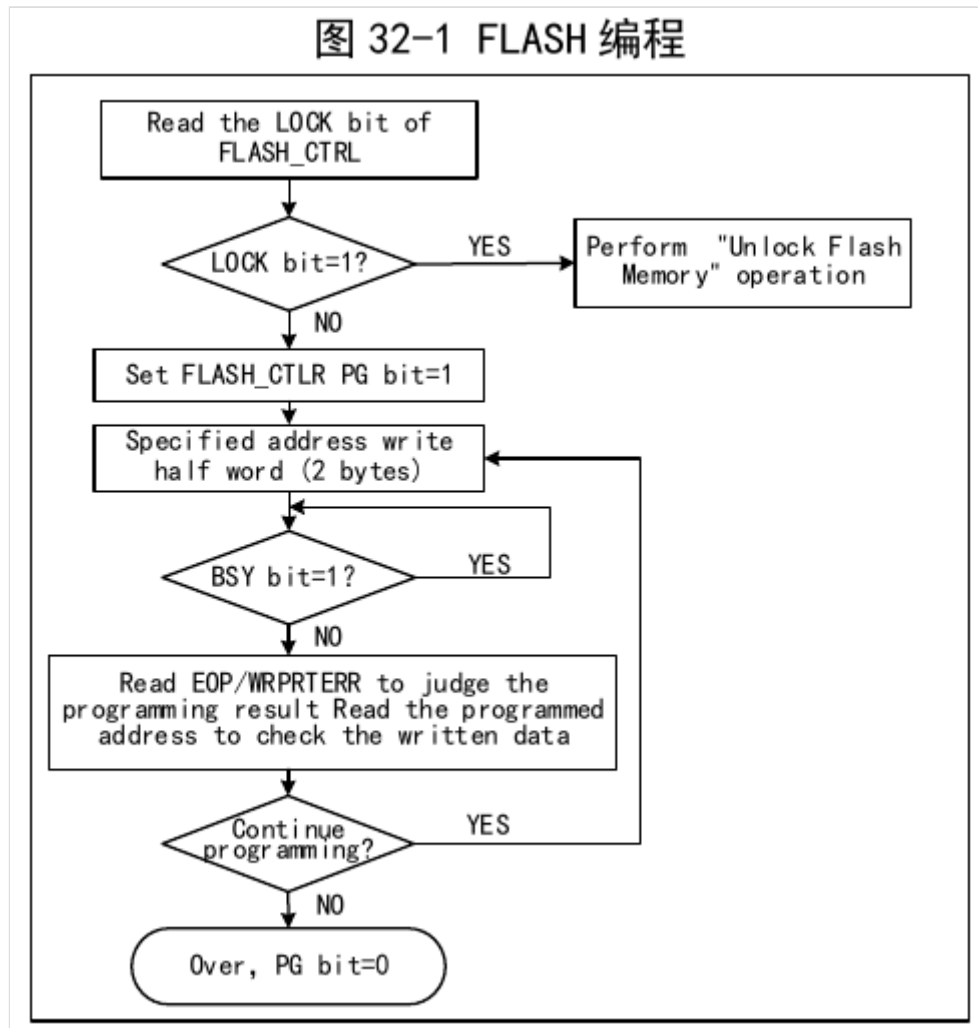


Figure 32-1: FLASH Programming

1. Check the LOCK bit of the FLASH_CTRL register; if it is 1, perform the "Unlock Flash" operation.
2. Set the PG bit of the FLASH_CTRL register to '1' to enable standard programming mode.
3. Write the half-word to be programmed to the specified flash address (even address).
4. Wait for the BSY bit to become '0' or the EOP bit of the FLASH_STATR register to become '1' to indicate programming is complete, then clear the EOP bit to 0.
5. Check the FLASH_STATR register for errors, or read the programmed address data for verification.
6. To continue programming, repeat steps 3–5; to finish programming, clear the PG bit to 0.

32.5.4 Main Memory Standard Erase

Flash can be erased by sector (4K bytes) or by full chip.

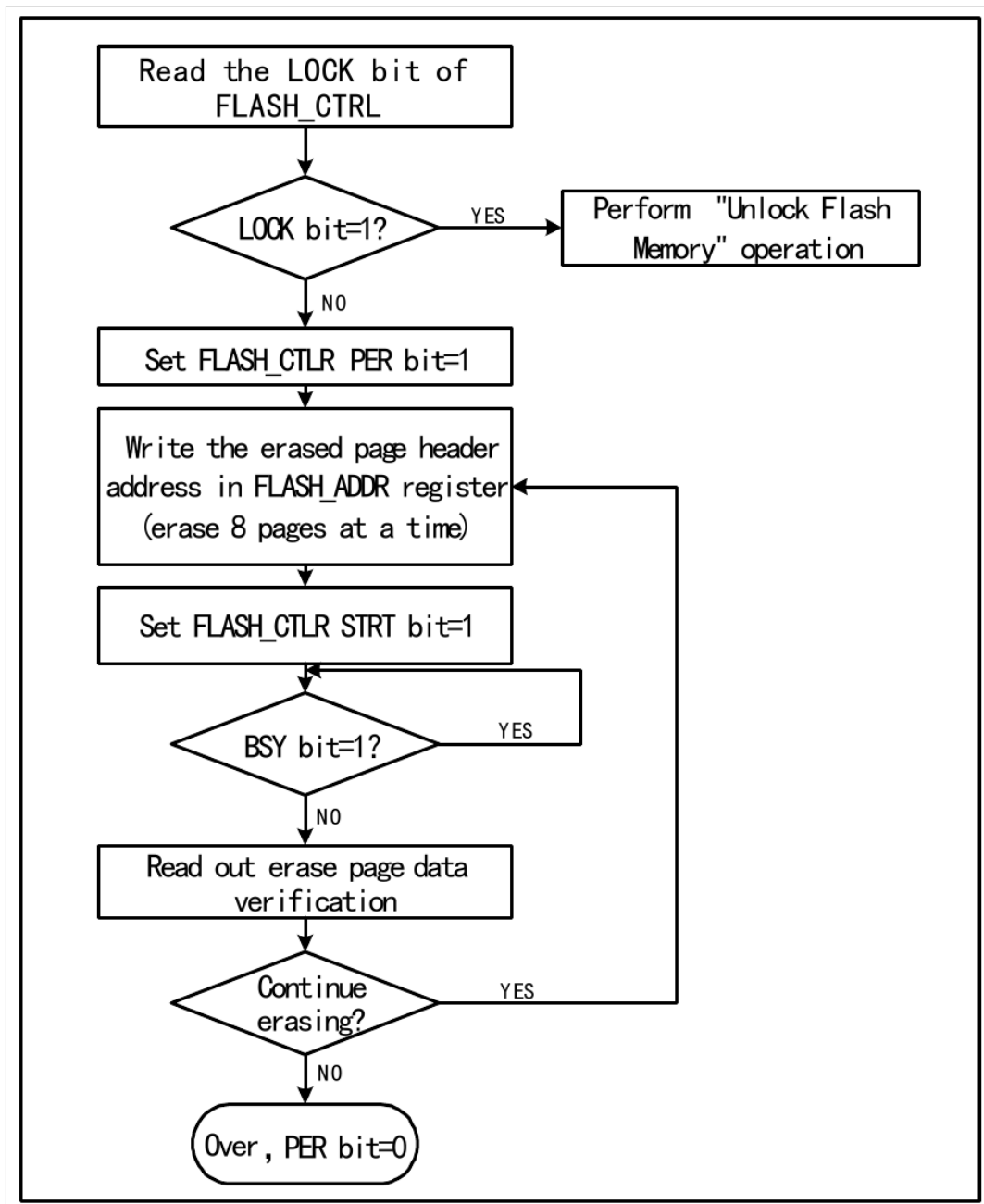


Figure 32-2: FLASH Page Erase

1. Check the LOCK bit of the FLASH_CTRL register; if it is 1, perform the "Unlock Flash" operation.
2. Set the PER bit of the FLASH_CTRL register to '1' to enable sector erase mode.
3. Write the page header address to be erased to the FLASH_ADDR register.
4. Set the STRT bit of the FLASH_CTRL register to '1' to start an erase operation.
5. Wait for the BSY bit to become '0' or the EOP bit of the FLASH_STATR register to become '1' to indicate erase is complete, then clear the EOP bit to 0.
6. Read the erased page data for verification.
7. To continue sector erase, repeat steps 3–5; to finish erasing, clear the PER bit to 0.

Note: After successful erase, word read = 0xe339e339, half-word read = 0xe339, even address byte read = 0x39, odd address read = 0xe3.

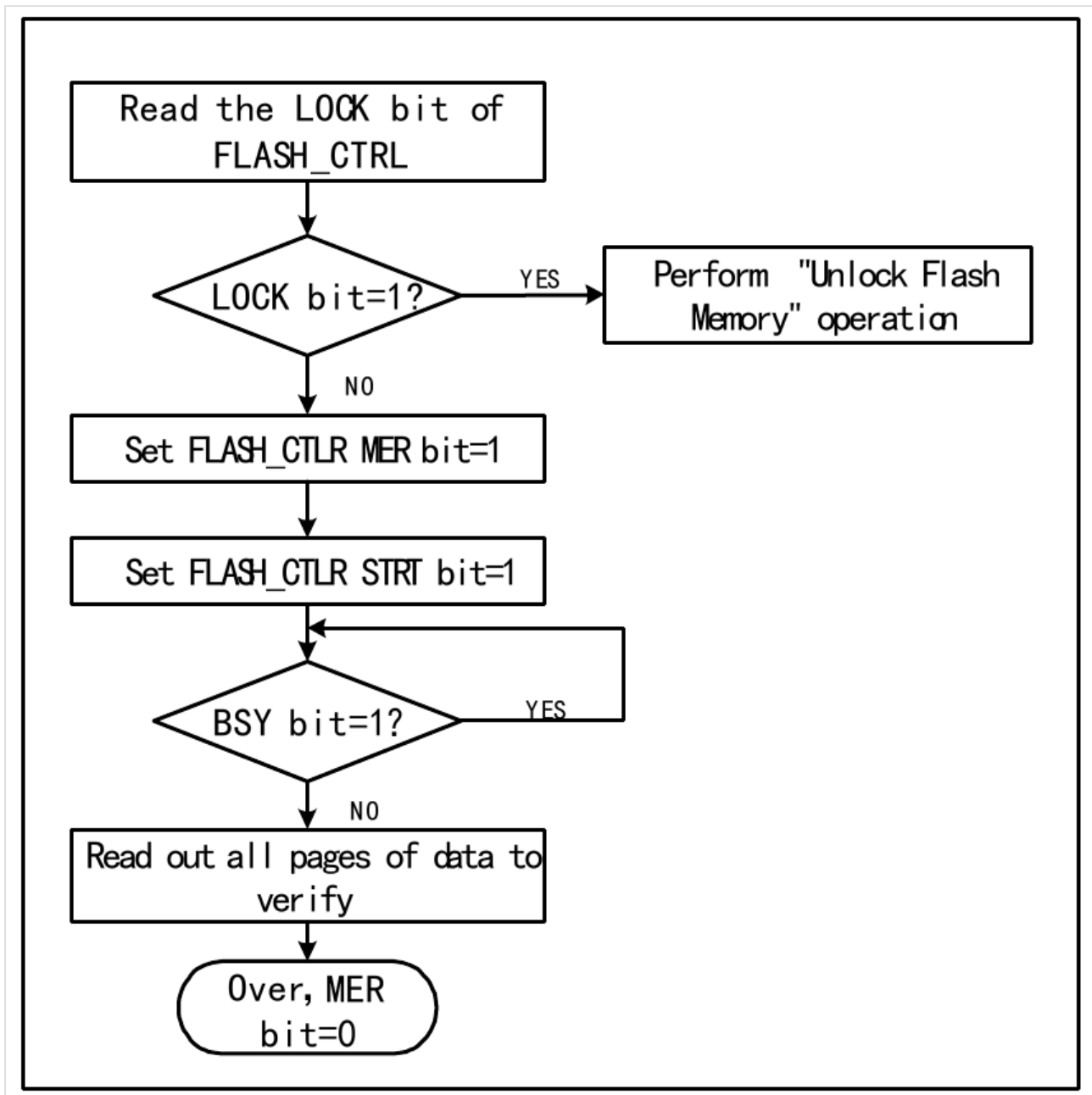


Figure 32-3: FLASH Full-Chip Erase

1. Check the LOCK bit of the FLASH_CTRL register; if it is 1, perform the "Unlock Flash" operation.
2. Set the MER bit of the FLASH_CTRL register to '1' to enable full-chip erase mode.
3. Set the STRT bit of the FLASH_CTRL register to '1' to start the erase operation.
4. Wait for the BSY bit to become '0' or the EOP bit of the FLASH_STATR register to become '1' to indicate erase is complete, then clear the EOP bit to 0.
5. Read the erased page data for verification.
6. Clear the MER bit to 0.

32.5.5 Fast Programming Mode Unlock

The fast programming mode can be unlocked by writing a sequence to the FLASH_MODEKEYR register. After unlocking, the FLOCK bit of the FLASH_CTRL register will be cleared to 0, indicating that fast erase and programming operations can be performed. It can be re-locked by software setting the "FLOCK" bit of the FLASH_CTRL register to 1.

Unlock sequence:

1. Write KEY1 = 0x45670123 to the FLASH_MODEKEYR register;
2. Write KEY2 = 0xCDEF89AB to the FLASH_MODEKEYR register.

The above operations must be executed in order and consecutively; otherwise, they are considered erroneous and will lock until the next system reset before they can be unlocked again.

Note: Fast programming operations require unlocking both the "LOCK" and "FLOCK" layers.

32.5.6 Main Memory Fast Programming

Fast programming is performed per page (256 bytes).

1. Check the LOCK bit of the FLASH_CTLR register; if it is '1', perform the "Unlock Flash" operation.
2. Check the FLOCK bit of the FLASH_CTLR register; if it is '1', perform the "Fast Programming Mode Unlock" operation.
3. Check the BSY bit of the FLASH_STATR register to confirm no other programming operations are in progress.
4. Set the FTPG bit of the FLASH_CTLR register to '1' to enable fast page programming mode.
5. Write data to the FLASH address using 32-bit mode, for example:

```
*(uint32_t*)0x80000000 = 0x12345678;
```
6. Wait for the WR_BSY bit of the FLASH_STATR register to become '0', then write the next data.
7. Repeat steps 5–6 a total of 64 times.
8. Set the PGSTRT bit of the FLASH_CTLR register to '1' to start fast page programming.
9. Wait for the BSY bit to become '0' or the EOP bit of the FLASH_STATR register to become '1' to indicate a fast page programming is complete, then clear the EOP bit to 0.
10. Check the FLASH_STATR register for errors, or read the programmed address data for verification.
11. To continue fast page programming, repeat steps 5–10; to finish programming, clear the FTPG bit to 0.

32.5.7 Main Memory Fast Erase

Fast erase is performed per block (32K bytes).

1. Check the LOCK bit of the FLASH_CTLR register; if it is 1, perform the "Unlock Flash" operation.
2. Check the FLOCK bit of the FLASH_CTLR register; if it is 1, perform the "Fast Programming Mode Unlock" operation.
3. Check the BSY bit of the FLASH_STATR register to confirm no other programming operations are in progress.
4. Set the BER32 bit of the FLASH_CTLR register to '1' to enable fast block erase (32K bytes) mode.
5. Write the starting address of the fast erase block to the FLASH_ADDR register.
6. Set the STRT bit of the FLASH_CTLR register to '1' to start a fast block erase (32K bytes) operation.
7. Wait for the BSY bit to become '0' or the EOP bit of the FLASH_STATR register to become '1' to indicate erase is complete, then clear the EOP bit to 0.
8. Check the FLASH_STATR register for errors, or read the erased page address data for verification.
9. To continue fast page erase, repeat steps 5–8; to finish erasing, clear the BER32 bit to 0.

Note: After successful erase, word read = 0xe339e339, half-word read = 0xe339, even address byte read = 0x39, odd address read = 0xe3.

32.6 User Option Bytes

User option bytes are burned into FLASH and are reloaded into the corresponding registers after system reset. The user can erase and program them freely. The user option byte information block has a total of 8 bytes (4 bytes for write protection, 1 byte for read protection, 1 byte for configuration options, 2 bytes for storing user data). Each bit has its complement bit for verification during the loading process. The option byte information structure and meaning are described below.

Table 32-3: 32-bit Option Byte Format Partition

[31:24]	[23:16]	[15:8]	[7:0]
Option Byte 1 Complement	Option Byte 1	Option Byte 0 Complement	Option Byte 0

Table 32-4: User Option Byte Information Structure

Address	Bits	[31:24]	[23:16]	[15:8]	[7:0]
0x1FFFF800		nUSER	USER	nRDPR	RDPR
0x1FFFF804		nData1	Data1	nData0	Data0
0x1FFFF808		nWRPR1	WRPR1	nWRPR0	WRPR0
0x1FFFF80C		nWRPR3	WRPR3	nWRPR2	WRPR2

Name/ Byte	Description	Reset Value
RDPR	Read protection control bit, configures whether the code in flash can be read out. 0xA5: If this byte is 0xA5 (nRDP must be 0x5A), indicates the current code is in non-read-protected state and can be read out; Other values: Indicates code read protection state, not readable. Pages 0–15 (4K) will be automatically write-protected, not controlled by WRPR0.	0xA5
USER bit 7	Reserved.	1
USER bit 6	RAM_CODE_MOD 1: CODE-576KB + RAM-136KB 0: CODE-512KB + RAM-200KB	x
USER bits [5:3]	Reserved.	111b
USER bit 2	STANDYRST — System reset control in standby mode: 1: Not enabled, entering standby mode does not reset system; 0: Enabled, entering standby mode generates system reset.	1
USER bit 1	STOPRST — System reset control in stop mode: 1: Not enabled, entering stop mode does not reset system; 0: Enabled, entering stop mode generates system reset.	1
USER bit 0	IWDGSW — Independent watchdog (IWDG) hardware enable bit: 1: IWDG function enabled by software, hardware enable disabled; 0: IWDG function enabled by hardware (determined by LSI clock).	1
	Store 2 bytes of user data.	0xFFFF

Name/ Byte	Description	Reset Value
Data0– Data1		
WRPR0 - WRPR3	Write protection control bits. Each bit controls the write protection status of one sector (4K bytes/sector) in main memory: 1: Write protection disabled; 0: Write protection enabled. The 4 bytes are used to protect a total of 992K bytes of main memory. WRPR0: Write protection control for sectors 0–7; WRPR1: Write protection control for sectors 8–15; WRPR2: Write protection control for sectors 16–23; WRPR3: Bits 0–6 provide write protection for sectors 24–30; bit 7 provides write protection for sectors 31–247.	0xFFFFFFFF

32.6.1 User Option Byte Unlock

The user option byte operation can be unlocked by writing a sequence to the FLASH_OBKEYR register. After unlocking, the OBWRE bit of the FLASH_CTLR register will be set to 1, indicating that user option byte erase and programming can be performed. It can be re-locked by software clearing the "OBWRE" bit of the FLASH_CTLR register to 0.

Unlock sequence:

1. Write KEY1 = 0x45670123 to the FLASH_OBKEYR register;
2. Write KEY2 = 0xCDEF89AB to the FLASH_OBKEYR register.

Note: User option byte operations require unlocking both the "LOCK" and "OBWRE" layers.

32.6.2 User Option Byte Programming

Only standard programming mode is supported, writing a half-word (2 bytes) at a time. In practice, when programming user option bytes, the FPEC only uses the low byte of the half-word and automatically calculates the high byte (the high byte is the complement of the low byte), then starts the programming operation. This ensures that the option byte and its complement are always correct.

1. Check the LOCK bit of the FLASH_CTLR register; if it is 1, perform the "Unlock Flash" operation.
2. Check the BSY bit of the FLASH_STATR register to confirm no other programming operations are in progress.
3. Check the OBWRE bit of the FLASH_CTLR register; if it is 0, perform the "User Option Byte Unlock" operation.
4. Set the OBPG bit of the FLASH_CTLR register to '1'.
5. Write the half-word (2 bytes) to be programmed to the specified address.
6. Wait for the BSY bit to become '0' or the EOP bit of the FLASH_STATR register to become '1' to indicate programming is complete, then clear the EOP bit to 0.
7. Read the programmed address data for verification.
8. To continue programming, repeat steps 5–7; to finish programming, clear the OBPG bit to 0.

Note: When modifying the "read protection" in the option bytes to "non-protected" state, a full-chip erase of the main memory area will be automatically executed. If options other than "read protection" are modified, no full-chip erase operation will occur.

32.6.3 User Option Byte Erase

Directly erase the entire 128-byte user option byte area.

1. Check the LOCK bit of the FLASH_CTLR register; if it is 1, perform the "Unlock Flash" operation.
2. Check the BSY bit of the FLASH_STATR register to confirm no programming operation is in progress.
3. Check the OBWRE bit of the FLASH_CTLR register; if it is 0, perform the "User Option Byte Unlock" operation.
4. Set the OBER bit of the FLASH_CTLR register to '1', then set the STRT bit of the FLASH_CTLR register to '1' to start user option byte erase.
5. Wait for the BSY bit to become '0' or the EOP bit of the FLASH_STATR register to become '1' to indicate erase is complete, then clear the EOP bit to 0.
6. Read the erased address data for verification.
7. Clear the OBER bit to 0 to finish.

Note: After successful erase, word read = 0xe339e339, half-word read = 0xe339, byte read = 0x39.

32.6.4 Releasing Read Protection

Whether the flash is read-protected is determined by the user option bytes. Reading the FLASH_OBR register, when the RDPRT bit is '1', it indicates the current flash is in read-protected state, and flash operations are subject to a series of security protections of the read-protected state. The process of releasing read protection is as follows:

1. Erase the entire user option byte area. At this point, the read protection field RDPR still has read protection in effect.
2. Program the user option bytes, writing the correct RDPR code 0xA5 to release the flash read protection. (This step will first cause the system to automatically perform a full-chip erase operation on the flash.)
3. Perform a power-on reset to reload the option bytes (including the new RDPR code); at this point, read protection is released.

32.6.5 Releasing Write Protection

Whether the flash is write-protected is determined by the user option bytes. Reading the FLASH_WPR register, each bit represents 4K bytes of flash space. When a bit is '1', it indicates non-write-protected state; when '0', it indicates write-protected. The process of releasing write protection is as follows:

1. Erase the entire user option byte area.
2. Write the correct RDPR code 0xA5 to allow read access;
3. Perform a system reset to reload the option bytes (including the new WRPR[3:0] bytes); write protection is released.

Chapter 33: Extended Configuration (EXTEN)

33.1 Extended Configuration

The system provides an EXTEN extended configuration unit (EXTEN_CTR register). This unit uses the HB clock and only performs reset actions during system reset. It mainly includes the following extended control bit functions:

1. **Internal voltage adjustment:** The ULLDOTRIM field selects the default value, which can be modified when adjusting performance and power consumption.
2. **Lock-up function monitoring:** When the LKUPEN field is enabled, the system's Lock-up condition monitoring is turned on. Once a Lock-up condition occurs, the system performs a software reset and sets the LKUPRST field to 1, which can be cleared by writing 1 after reading.
3. **VDD18 adjustment:** The voltage of VDD18 can be configured.

33.2 Register Description

Table 33-1: EXTEN related register list

Name	Address	Description	Reset Value
R32_EXTEN_CTR	0x40023800	Extended control configuration register	0x00000440
R32_FEATURE_SIGN	0x1FFFF7D0	Feature information indication register	0xE339XXXX

33.2.1 Extended Control Configuration Register (EXTEN_CTR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved		LDO18_ADJ		Reserved		ULLDOTRIM[1:0]		LKUPRST		LKUPEN		Reserved			

Bits	Name	Access	Description	Reset
[31:13]	Reserved	RO	Reserved.	0
12	LDO18_ADJ	RW	VDD18 adjustment bit. 1: 1.9V; 0: 1.8V.	0
[11:10]	Reserved	RO	Reserved.	0
[9:8]	ULLDOTRIM[1:0]	RW	Adjusts the ULLDO voltage value in low-power mode.	10b
7	LKUPRST	RW1	LOCKUP reset flag: 1: A LOCKUP occurred causing a system reset, write 1 to clear; 0: Normal.	0
6	LKUPEN	RW	LOCKUP monitoring function: 1: Enabled, the system performs a reset and sets LOCKUP_RST	1

Bits	Name	Access	Description	Reset
			when a lock-up occurs; 0: Not enabled.	
[5:0]	Reserved	RO	Reserved.	0

33.2.2 Feature Information Indication Register (FEATURE_SIGN)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								Reserved				ETH_LED_EN			

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0xE339
[15:8]	Reserved	RO	bit[15:8] reset value is the bitwise inversion of bit[7:0] reset value.	X
[7:1]	Reserved	RO	Reserved.	0x7F
0	ETH_LED_EN	RO	ETH LED function indication bit: 1: ETH LED function is enabled, providing LINK and ACT functions; 0: ETH LED function is not enabled, providing only LINK function.	X

Chapter 34: Pseudo-Static Random Access Memory (PSRAM)

The module description in this chapter applies only to the CH32V467 microcontroller product.

The CH32V467 chip, based on the CH32V407, also integrates a 4M or 8M byte extended PSRAM memory. It supports data storage through address and read/write instructions, and supports DMA.

34.1 Main Features

- 8-bit hardware width, supports 8-bit, 16-bit, or 32-bit random or block read/write
- Supports up to 300MHz clock rate, supports dual-edge, maximum 600MBps data transfer rate
- Supports write data masking
- Supports high-speed read operations via data strobe
- Supports register-configurable write and read initial latency
- Supports half-sleep mode and deep sleep mode

34.2 Register Description

34.2.1 PSRAM Related Registers

Table 34-1: PSRAM Related Register List

Name	Address	Description	Reset Value
R32_PSRAM_CTLR	0x40024400	PSRAM control register	0x00000300
R32_PSRAM_TIMING	0x40024404	PSRAM timing register	0x070C0014
R32_PSRAM_DATA	0x40024408	PSRAM data register	0x00000000
R32_PSRAM_ADDR	0x4002440C	PSRAM address register	0x00000000
R32_PSRAM_CMD0_CFG	0x40024410	PSRAM command word 0 configuration register	0xA0208000
R32_PSRAM_CMD1_CFG	0x40024414	PSRAM command word 1 configuration register	0xB0FFC040
R32_PSRAM_DMACR	0x40024418	PSRAM DMA control register	0x00000000
R32_PSRAM_DATA_NUM	0x4002441C	PSRAM data transfer quantity register	0x00000000
R32_DMA_MEM_ADDR	0x40024420	DMA transfer memory start address register	0x00000000
R32_DMA_PSRAM_ADDR	0x40024424	DMA transfer PSRAM start address register	0x00000000
R32_PSRAM_ISR	0x40024428	PSRAM interrupt status register	0x00000000
R32_PSRAM_IER	0x4002442C	PSRAM interrupt error enable register	0x00000000
R32_PSRAM_STATUS	0x40024430	PSRAM controller state machine register	0x00000000
R32_PSRAM_CECFG	0x40024434	PSRAM CE pulse configuration register	0x000004B0
R32_PSRAM_LATENCY	0x40024438	PSRAM read/write latency register	0x00050505

34.2.1.1 PSRAM Control Register (R32_PSRAM_CTLR)

Offset address: 0x00

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Reserved															EXIT_LPMD	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved					CAP_CFG[2:0]			Reserved			DFIFO	CFIFO	Reserved			

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
16	EXIT_LPMD	RW	PSRAM exit current low-power state enable: 1: Make PSRAM exit current low-power state; 0: No behavior.	0
[15:11]	Reserved	RW	Reserved.	0
[10:8]	CAP_CFG[2:0]	RW	PSRAM capacity configuration: 001: 32Mb; 011: 64Mb; 101: 128Mb; 111: 256Mb; Others: Reserved.	0x3
[7:6]	Reserved	RW	Reserved.	0
5	DFIFO	RW	DMA write operation FIFO flush method selection: 1: DMA write operation simultaneously clears data in the HB slave interface read cache; 0: DMA write operation does not clear data in the HB slave interface read cache.	0
4	CFIFO	RW	Read FIFO flush method selection: 1: Writing to any PSRAM address will flush the read FIFO; 0: Read FIFO is flushed only when the write address matches the address in the read buffer.	0
[3:0]	Reserved	RW	Reserved.	0

34.2.1.2 PSRAM Timing Register (R32_PSRAM_TIMING)

Offset address: 0x04

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
TXLPD[7:0]									TCPH[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reserved										TRC[7:0]						

Bits	Name	Access	Description	Reset
[31:24]	TXLPD[7:0]	RW	PSRAM_CE_n pulse width sustained when exiting low-power (half-sleep and deep sleep): $T_{pulse} = TXLPD[7:0] * HSI \text{ clock period.}$	0x07
[23:16]	TCPH[7:0]	RW	Minimum interval between two consecutive bursts. $t_{CPH} = TCPH[7:0] * PSRAM \text{ clock period.}$	0x0C

Bits	Name	Access	Description	Reset
			Example: If PSRAM_CLK is 333M, to satisfy the 32ns tCPH time, this field should be written with a minimum of 12; If PSRAM_CLK is 166M, to satisfy the 22ns tCPH time, this field should be written with a minimum of 4.	
[15:8]	Reserved	RW	Reserved.	0x00
[7:0]	TRC[7:0]	RW	Minimum read/write cycle time: $T = TRC[7:0] * PSRAM \text{ clock period.}$ Example: To satisfy the 60ns tRC_read and tRC_write time, if PSRAM_CLK is 333M, this should be written with a minimum of 0x14; If PSRAM_CLK is 166M, this should be written with a minimum of 0x0A.	0x14

34.2.1.3 PSRAM Data Register (R16_PSRAM_DATA)

Offset address: 0x08

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RW_HDATA[7:0]								RW_LDATA[7:0]							

Bits	Name	Access	Description	Reset
[15:8]	RW_HDATA[7:0]	RW	High byte data written or read out.	0
[7:0]	RW_LDATA[7:0]	RW	Low byte data written or read out.	0

34.2.1.4 PSRAM Address Register (R32_PSRAM_ADDR)

Offset address: 0x0C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								MR_ADDR0[7:0]							

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMD[7:0]								Reserved				MW		BUSY	

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
[23:16]	MR_ADDR0[7:0]	RW	Register 0 address to operate on.	0
[15:8]	CMD[7:0]	RW	Command to execute: Read register: 0x40; Write register: 0xC0.	0
[7:2]	Reserved	RO	Reserved.	0
1	MW	RW	1: Informs PSRAM that a write operation to the PSRAM register is about to start; 0: Informs PSRAM to start a read operation, placing data into the R32_PSRAM_DATA data register.	0
0	BUSY	RW	This bit should be read as logic 0 before writing to R32_PSRAM_DATA and R32_PSRAM_ADDR. During PSRAM register access, this bit is set to 1 by the application to indicate that a read or write access is in progress. During a write operation to the PSRAM register, R32_PSRAM_DATA must remain valid until the	0

Bits	Name	Access	Description	Reset
			controller clears this bit. During a read operation to the PSRAM register, R32_PSRAM_DATA remains invalid until the controller clears this bit.	

34.2.1.5 PSRAM Command Word 0 Configuration Register (R32_PSRAM_CMD0_CFG)

Offset address: 0x10

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
SW_LB[7:0]								SR_LB[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SW[7:0]								SR[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	SW_LB[7:0]	RW	Configurable synchronous write (linear burst) command.	0xA0
[23:16]	SR_LB[7:0]	RW	Configurable synchronous read (linear burst) command.	0x20
[15:8]	SW[7:0]	RW	Configurable synchronous write command.	0x80
[7:0]	SR[7:0]	RW	Configurable synchronous read command.	0

34.2.1.6 PSRAM Command Word 1 Configuration Register (R32_PSRAM_CMD1_CFG)

Offset address: 0x14

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
REFRESH[7:0]								GLRST[7:0]							
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MREGW[7:0]								MREGR[7:0]							

Bits	Name	Access	Description	Reset
[31:24]	REFRESH[7:0]	RW	Configurable refresh command.	0xB0
[23:16]	GLRST[7:0]	RW	Configurable global reset command.	0xFF
[15:8]	MREGW[7:0]	RW	Configurable register write command.	0xC0
[7:0]	MREGR[7:0]	RW	Configurable register read command.	0x40

34.2.1.7 PSRAM DMA Control Register (R32_PSRAM_DMCCR)

Offset address: 0x18

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved								TOWBL_TOUT[3:0]				DMA_ONEBL[3:0]			
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				MEMORY_SIZE[1:0]				Reserved		DAT_DIR		Reserved		DMAEN	

Bits	Name	Access	Description	Reset
[31:24]	Reserved	RO	Reserved.	0
[23:20]	TOWBL_TOUT[3:0]	RW	Pause time between two bursts, in HCLK units: 0000: Transfer without pause; 0001: Pause 1 cycle; 0010: Pause 2 cycles; ... 1110: Pause 14 cycles; 1111: Pause 15 cycles.	0
[19:16]	DMA_ONEBL[3:0]	RW	DMA single burst length: 0000: 2; 0001: 4; 0010: 6; 0011: 8; Others: 16.	0
[15:12]	Reserved	RO	Reserved.	0
[11:10]	MEMORY_SIZE[1:0]	RW	Memory data width: 00: Reserved; 01: 16-bit; 10: 32-bit; 11: Reserved.	0
[9:5]	Reserved	RO	Reserved.	0
4	DAT_DIR	RW	Data transfer direction: 1: MEM to PSRAM; 0: PSRAM to MEM. This bit is set and cleared by software. Note: When DMA enable is turned off, this bit must be cleared simultaneously.	0
[3:1]	Reserved	RO	Reserved.	0
0	DMAEN	RW	DMA enable: 1: Enable; 0: Disable.	0

34.2.1.8 PSRAM Data Transfer Quantity Register (R32_PSRAM_DATA_NUM)

Offset address: 0x1C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA_NUM[15:0]															

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:0]	DATA_NUM[15:0]	RW	Data transfer quantity: Written when the DMAEN bit of the R32_PSRAM_DMACR register is 0; read-only when the DMAEN bit is 1, indicating the remaining number of transfers.	0

34.2.1.9 DMA Transfer Memory Start Address Register (R32_DMA_MEM_ADDR)

Offset address: 0x20

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMA_MEM_ADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMA_MEM_ADDR[15:0]															
Bits	Name			Access	Description										Reset
[31:0]	DMA_MEM_ADDR[31:0]			RW	DMA transfer memory start address (single-byte aligned).										0

34.2.1.10 DMA Transfer PSRAM Start Address Register (R32_DMA_PSRAM_ADDR)

Offset address: 0x24

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
DMA_PSRAM_ADDR[31:16]															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMA_PSRAM_ADDR[15:0]															
Bits	Name	Access	Description	Reset											
[31:0]	DMA_PSRAM_ADDR[31:0]	RW	DMA transfer PSRAM start address (4-byte aligned in 32-bit mode, 2-byte aligned in 16-bit mode).	0											

34.2.1.11 PSRAM Interrupt Status Register (R32_PSRAM_ISR)

Offset address: 0x28

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16		
Reserved																	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
Reserved											AEIF	CENEIF		DMATF		Reserved	

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
3	AEIF	RO	Address error interrupt flag, address exceeded MEM capacity.	0
2	CENEIF	RW1Z	CEM error interrupt flag: 1: CEM error interrupt generated; 0: No CEM error interrupt generated.	0
1	DMATF	RW1Z	DMA bus transfer complete flag: 1: DMA bus transfer complete; 0: DMA bus transfer not complete.	0
0	Reserved	RO	Reserved.	0

34.2.1.12 PSRAM Interrupt Error Enable Register (R32_PSRAM_IER)

Offset address: 0x2C

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						AEIEN		CENEIEN		DMATEN		Reserved			

Bits	Name	Access	Description	Reset
[31:4]	Reserved	RO	Reserved.	0
3	AEIEN	RW	Address error interrupt enable: 1: Enable; 0: Disable.	0
2	CENEIEN	RW	CEM error interrupt enable: 1: Enable; 0: Disable.	0
1	DMATEN	RW	DMA bus transfer complete enable: 1: Enable; 0: Disable.	0
0	Reserved	RO	Reserved.	0

34.2.1.13 PSRAM Controller State Machine Register (R32_PSRAM_STATUS)

Offset address: 0x30

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved											PSRAM_STATE[4:0]				

Bits	Name	Access	Description	Reset
[31:5]	Reserved	RO	Reserved.	0
[4:0]	PSRAM_STATE[4:0]	RO	PSRAM controller state machine: 0000: Idle state; 0001: Send command address state; 0010: Write register state; 0011: Read register state; 0100: Write latency state; 0101: Reserved; 0110: Write data state; 0111: Read latency state; 1111: Wait for minimum CEM completion state.	0

34.2.1.14 PSRAM CE Pulse Configuration Register (R32_PSRAM_CECFG)

Offset address: 0x34

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved														CEPWEN	
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CELPW[11:0]									

Bits	Name	Access	Description	Reset
[31:17]	Reserved	RO	Reserved.	0
16	CEPWEN	RW	CE pulse width detection function enable: 1: Disable CE pulse width detection function; 0: Enable CE pulse width detection function.	0
[15:12]	Reserved	RO	Reserved.	0
[11:0]	CELPW[11:0]	RW	Maximum duration of PSRAM_CE_n low pulse (low pulse width), excluding exiting half-sleep: $T = \text{CELPW}[11:0] * \text{PSRAM clock period.}$	0x4B0

34.2.1.15 PSRAM Read/Write Latency Register (R32_PSRAM_LATENCY)

Offset address: 0x38

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WLATENCY[7:0]								RLATENCY[7:0]							

Bits	Name	Access	Description	Reset
[31:16]	Reserved	RO	Reserved.	0
[15:8]	WLATENCY[7:0]	RW	Write latency, the delay from issuing a write command to write data being valid.	0x05
[7:0]	RLATENCY[7:0]	RW	Read latency, the delay from issuing a read command to read data being valid.	0x05

Chapter 35: Debug Support (DBG)

35.1 Main Features

This register allows configuring the MCU in debug state. It includes:

- Support for Independent Watchdog (IWDG) counter
- Support for Window Watchdog (WWDG) counter
- Support for timer counters
- Support for I2C SMBus timeout control
- Support for CAN communication

35.2 Register Description

35.2.1 Debug MCU Configuration Register (DBGMCU_CR)

Address: 0x7C0 (CSR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved										CAN1_STOP		TIM8_STOP	TIM7_STOP	TIM6_STOP	TIM5_STOP
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIM4_STOP	TIM3_STOP	TIM2_STOP	TIM1_STOP	Reserved	I2C_SMBUS_TIMEOUT	WWDG_STOP	IWDG_STOP	Reserved	STANDBY	STOP	SLEEP				

Bits	Name	Access	Description	Reset
[31:21]	Reserved	RO	Reserved.	0
20	CAN1_STOP	RW	CAN debug stop bit. When the core enters debug state, CAN1 stops running. 1: The CAN receive register does not continue to receive data; 0: CAN still operates normally.	0
19	TIM8_STOP	RW	Timer 8 debug stop bit. When the core enters debug state, the counter stops working. 1: Timer 8 counter stops working; 0: Timer 8 counter still operates normally.	0
18	TIM7_STOP	RW	Timer 7 debug stop bit. When the core enters debug state, the counter stops working. 1: Timer 7 counter stops working; 0: Timer 7 counter still operates normally.	0
17	TIM6_STOP	RW	Timer 6 debug stop bit. When the core enters debug state, the counter stops working. 1: Timer 6 counter stops working; 0: Timer 6 counter still operates normally.	0
16	TIM5_STOP	RW	Timer 5 debug stop bit. When the core enters debug state, the counter stops working.	0

Bits	Name	Access	Description	Reset
			1: Timer 5 counter stops working; 0: Timer 5 counter still operates normally.	
15	TIM4_STOP	RW	Timer 4 debug stop bit. When the core enters debug state, the counter stops working. 1: Timer 4 counter stops working; 0: Timer 4 counter still operates normally.	0
14	TIM3_STOP	RW	Timer 3 debug stop bit. When the core enters debug state, the counter stops working. 1: Timer 3 counter stops working; 0: Timer 3 counter still operates normally.	0
13	TIM2_STOP	RW	Timer 2 debug stop bit. When the core enters debug state, the counter stops working. 1: Timer 2 counter stops working; 0: Timer 2 counter still operates normally.	0
12	TIM1_STOP	RW	Timer 1 debug stop bit. When the core enters debug state, the counter stops working. 1: Timer 1 counter stops working; 0: Timer 1 counter still operates normally.	0
11	Reserved	RO	Reserved.	0
10	I2C_SMBUS_TIMEOUT	RW	SMBUS timeout mode debug stop bit. When the core enters debug state, the SMBUS timeout mode is stopped. 1: Freezes the SMBUS timeout control; 0: Operates the same as normal mode.	0
9	WWDG_STOP	RW	Window watchdog debug stop bit. When the core enters debug state, the debug window watchdog stops working. 1: Window watchdog counter stops working; 0: Window watchdog counter still operates normally.	0
8	IWDG_STOP	RW	Independent watchdog debug stop bit. When the core enters debug state, the watchdog stops working. 1: Watchdog counter stops working; 0: Watchdog counter still operates normally.	0
[7:3]	Reserved	RW	Reserved.	0
2	STANDBY	RW	Debug standby mode bit. 1: (FCLK on, HCLK on) The digital circuit section is not powered down, and FCLK and HCLK clocks are provided by the internal RC oscillator. In addition, the microcontroller exits STANDBY mode by generating a system reset, which is the same as a reset; 0: (FCLK off, HCLK off) The entire digital circuit section is powered down. From the software point of view, exiting STANDBY mode is the same as a reset (except that some status bits indicate that the microcontroller has just exited from the STANDBY state).	0
1	STOP	RW	Debug stop mode bit. 1: (FCLK on, HCLK on) In stop mode, FCLK and HCLK clocks are provided by the internal RC oscillator. When exiting stop mode, the software must reconfigure the clock system to start the PLL, crystal oscillator, etc. (same as when this bit is	0

Bits	Name	Access	Description	Reset
			set to 0); 0: (FCLK off, HCLK off) In stop mode, the clock controller disables all clocks (including HCLK and FCLK). When exiting STOP mode, the clock configuration is the same as after a reset (the microcontroller is clocked by the 20MHz internal RC oscillator (HSI)). Therefore, the software must reconfigure the clock control system to start the PLL, crystal oscillator, etc.	
0	SLEEP	RW	Debug sleep mode bit. 1: (FCLK on, HCLK on) In sleep mode, both FCLK and HCLK clocks are provided by the previously configured system clock; 0: (FCLK on, HCLK off) In sleep mode, FCLK is provided by the previously configured system clock, while HCLK is turned off. Since sleep mode does not reset the configured clock system, the software does not need to reconfigure the clock system when exiting sleep mode.	0